ETSI TR 133 929-1 V19.0.0 (2026-10)



TECHNICAL REPORT

LTE;
5G;
Lawful Interception (LI) implementation guidance;
LI for IMS based services
(3GPP TR 33.929-1 version 19.0.0 Release 19)



Reference
DTR/TSGS-0333929-1vj00

Keywords
5G,LTE

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part of this document may be reproduced in any form, by any means and in any media, without the prior written authorization of ETSI and except as expressly permitted below.
By way of exception and when the document is a normative deliverable (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)), ETSI authorizes to reproduce and incorporate into products, services and technical documentation only those extracts (e.g. templates) that are strictly necessary for the technical implementation of the normative deliverable, to ensure compliance with the latter.
Nothing in this notice shall be construed as limiting any mandatory exceptions to copyright provided by applicable law.

© ETSI 2026.
All rights reserved.

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)) may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Legal Notice

This Technical Report (TR) has been produced by ETSI 3rd Generation Partnership Project (3GPP).

The present document may refer to technical specifications or reports using their 3GPP identities. These shall be interpreted as being references to the corresponding ETSI deliverables.

The cross reference between 3GPP and ETSI identities can be found at 3GPP to ETSI numbering cross-referencing.

Modal verbs terminology

In the present document "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Contents

Intellectual Property Rights 2
Legal Notice 2
Modal verbs terminology 2
Foreword 10
Introduction 10
1 Scope 11
2 References 11
3 Definitions of terms, symbols and abbreviations 12
3.1 Terms 12
3.2 Symbols 12
3.3 Abbreviations 12
4 Illustration of LI for IMS-based services 14
4.1 General 14
4.2 Intra-CSP domain IMS sessions 20
4.2.1 General 20
4.2.2 Originating sessions 20
4.2.2.1 Party A (target) calls Party B 20
4.2.2.2 Party A (target) calls Party B, redirected to Party C 21
4.2.3 Terminating sessions 22
4.2.3.1 Party A calls Party B (target) 22
4.2.3.2 Party A calls Party B, redirected to Party C (target) 22
4.2.4 Redirecting scenarios 23
4.2.4.1 Party A calls Party B (target), redirected to Party C 23
4.3 Inter-CSP domain IMS sessions 24
4.3.1 General 24
4.3.2 Inter-CSP domain originations 24
4.3.2.1 Introduction 24
4.3.2.2 Party A (CS domain) calls Party B (target) 24
4.3.2.3 Party A (non-local ID target, CS domain) calls Party B 25
4.3.2.4 Party A (non-local ID target, IP domain) calls Party B 26
4.3.3 Inter-CSP domain terminations 26
4.3.3.1 Introduction 26
4.3.3.2 Party A (target) calls Party B (CS domain) 26
4.3.3.3 Party A calls Party B (non-local ID target, CS domain) 27
4.3.3.4 Party A calls Party B (non-local ID target, IP domain) 28
4.3.4 Inter-CSP domain redirections 29
4.3.4.1 Introduction 29
4.3.4.2 Party A calls Party B (target) redirected to Party C (CS domain) 29
4.3.4.3 Party A calls Party B (target) redirected to Party C (IP domain) 30
4.3.4.4 Party A calls Party B redirected to Party C (non-local ID target, CS domain) 31
4.3.4.5 Party A calls Party B redirected to Party C (non-local ID target, IP domain) 31
4.3.4.6 Special case – multiple targets, inter-CSP domains at both ends of the session 32
4.4 LBO roaming scenarios 33
4.4.1 General 33
4.4.2 Roaming at the origination end 33
4.4.2.1 Introduction 33
4.4.2.2 Roaming Party A (target in HPLMN) calls Party B 33
4.4.2.3 Roaming Party A (target in VPLMN) calls Party B 34
4.4.2.4 Roaming Party A (target in VPLMN and HPLMN) calls Party B 35
4.4.2.5 Roaming Party A calls Party B (non-local ID target in VPLMN) 35
4.4.2.6 Roaming Party A (target) calls Party B (non-local ID target in VPLMN) 36
4.4.3 Roaming at the terminating end 37
4.4.3.1 Introduction 37

4.4.3.2 Party A calls roaming Party B (target in HPLMN) 37
4.4.3.3 Party A calls roaming Party B (target in VPLMN) 38
4.4.3.4 Party A calls roaming Party B (target in VPLMN and HPLMN) 38
4.4.3.5 Party A (non-local ID target in VPLMN) calls roaming Party B 39
4.4.3.6 Party A (non-local ID target in VPLMN) calls roaming Party B (target) 40
4.4.4 Roaming with redirections 40
4.4.4.1 Introduction 40
4.4.4.2 Party A calls Party B (target) redirected to roaming Party C 41
4.4.4.3 Party A calls roaming Party B (target) redirected to Party C 41
4.4.4.4 Party A calls roaming Party B (target) redirected to Party C (roaming) 42
4.4.4.5 Party A calls roaming Party B (target) redirected to Party C (CS domain) 43
4.4.4.6 Party A calls roaming Party B (target) redirected to Party C (IP domain) 44
4.4.4.7 Party A (non-local ID target in VPLMN) calls Party B redirected to roaming Party C 45
4.4.4.8 Party A (non-local ID target in VPLMN) calls Party B (non-local ID target in VPLMN) redirected to roaming Party C (target) 45
4.5 IMS based conferencing 46
4.5.1 General 46
4.5.2 Ad-hoc conferencing 47
4.5.2.1 Introduction 47
4.5.2.2 Party A (target) initiates a conference with Party B and Party C – case 1 47
4.5.2.3 Party A (target) initiates a conference with Party B and Party C – Case 2 49
4.5.2.4 Party A (target) initiates a conference with Party B and Party C - Alternative 50
4.5.2.5 Party A initiates a conference with Party B and Party C (target) 50
4.5.2.6 Party A initiates a conference with Party B (non-local ID target) and Party C 51
4.5.3 Group conferencing 52
4.5.3.1 Introduction 52
4.5.3.2 Party A, Party B and Party C in a group conference (conference URI target) – case 1 53
4.5.3.3 Party A, Party B and Party C in a group conference (conference URI target) – case 2 54
4.5.3.4 Party A (target), Party B, Party C in a group conference 55
4.5.3.5 Party A (target), Party B, Party C (target) in a group conference (conference URI (target) 56
4.6 IMS session hold 58
4.6.1 General 58
4.6.2 Single Party on hold 58
4.6.2.1 Introduction 58
4.6.2.2 Party A (target) in session with Party C with Party B on hold 59
4.6.2.3 Party A (target) in session with Party C with Party B on hold - alternative 60
4.6.2.4 Party A in session with Party C with Party B (non-local ID target) on hold – case 1 60
4.6.2.5 Party A in session with Party C with Party B (target) on hold – case 2 61
4.6.2.6 Party A in session with Party C with Party B (target in other CSP domain) on hold – case 3 62
4.6.3 Conference on hold 63
4.6.3.1 Introduction 63
4.6.3.2 Party A (target) in session with Party D, with conference on hold – case 1 64
4.6.3.3 Party A (target) in session with Party D, with conference on hold – case 1 (alternative) 66
4.6.3.4 Party A (target) in session with Party D, with conference on hold – case 2 66
4.6.3.5 Party A (target) in session with Party D, with conference on hold – case 2 (alternative) 67
4.6.3.6 Party A in session with Party D, with a Party C (non-local ID target) on the held conference leg 68
4.6.3.7 Party A in session with Party D, with a Party C (target) on the held conference leg 70
4.6.3.8 Party A in session with Party D, with a Party C (in a different CSP domain, target) on the held conference leg 71
4.6.4 Group conference on hold 72
4.6.4.1 Introduction 72
4.6.4.2 Party A (target) in communication with Party D, with group conference on hold 73
4.6.4.3 Party A (target) in communication with Party D, with group conference on hold (alternative) 74
4.6.4.4 Party A places a group conference on hold, conference URI target 74
4.6.4.5 Party A and Party C place a group conference on hold, conference URI target 75
4.6.4.6 Party A places a group conference with Party B and Party C (target) on hold 76
4.6.4.7 Special case – multiple targets 77
4.7 Special media on target terminating sessions 78
4.7.1 General 78
4.7.2 Special media with intra-office terminating sessions 78
4.7.3 Special media with inter-office terminating sessions 79
4.8 Home routed roaming architecture 80

4.8.1 Introduction....80
4.8.2 Topology overviews....81
4.8.2.1 N9HR LI....81
4.8.2.2 S8HR LI, EPC with CUPS....82
4.8.2.3 S8HR LI, EPC without CUPS....83
4.8.3 HR LI principles....84
4.8.3.1 N9HR LI....84
4.8.3.2 S8HR LI, EPC with CUPS....85
4.8.3.3 S8HR LI, EPC without CUPS....86
4.8.4 HR LI process....87
4.8.4.1 N9HR LI....87
4.8.4.1.1 Initial configuration - applicable to all inbound roamers....87
4.8.4.1.2 Intercept activation - applicable to target UE with IRI + CC....88
4.8.4.1.3 Intercept de-activation - applicable to target UE with IRI + CC....89
4.8.4.2 S8HR LI, EPC with CUPS....90
4.8.4.2.1 Initial configuration - applicable to all inbound roamers....90
4.8.4.2.2 Intercept activation - applicable to target UE with IRI + CC....92
4.8.4.2.3 Intercept de-activation - applicable to target UE with IRI + CC....93
4.8.4.3 S8HR LI, EPC without CUPS....94
4.8.4.3.1 Initial configuration - applicable to all inbound roamers....94
4.8.4.3.2 Intercept activation - applicable to target UE with IRI + CC....95
4.8.4.3.3 Intercept de-activation - applicable to target UE with IRI + CC....96
4.8.5 Correlation principles....97
4.8.5.1 Correlating the IRI....97
4.8.5.2 Correlating the IRI and CC....97
4.8.6 UE location reporting....97
4.9 Illustrations of IMS CC-POI triggering scenarios....98
4.9.1 Symmetric media streams....98
4.9.1.1 General....98
4.9.1.2 SDP offer in the forward message – originating session....99
4.9.1.3 SDP offer in the response message – originating session....100
4.9.1.4 SDP offer in the forward message – terminating session....102
4.9.1.5 SDP offer in the response message – terminating session....103
4.9.1.6 SDP offer in the forward message – session to target non-local ID (IP)....105
4.9.1.7 SDP offer in the forward message – session from target non-local ID (IP)....106
4.9.1.8 SDP offer in the forward message – session to target non-local ID (CS)....108
4.9.1.9 SDP offer in the forward message – session from target non-local ID (CS)....109
4.9.2 Asymmetric media streams....110
4.9.2.1 General....110
4.9.2.2 SDP offer in the forward message – originating session (core)....110
4.9.2.3 SDP offer in the response message – originating session (core)....111
4.9.2.4 SDP offer in the forward message – terminating session (core)....112
4.9.2.5 SDP offer in the response message – terminating session (core)....113
4.9.3 Illustrations of IMS CC-POI triggering and xCC generation scenarios....114
4.9.3.1 General....114
4.9.3.2 Symmetric media streams....115
4.9.3.2.1 General....115
4.9.3.2.2 Interception at the core network session – originating session....115
4.9.3.2.3 Interception at the core network side – terminating session....116
4.9.3.2.4 Interception at the access side – originating session....117
4.9.3.2.5 Interception at the access side – terminating session....117
4.9.3.3 Asymmetric media streams....118
4.9.3.3.1 General....118
4.9.3.3.2 Interception at the core network session – originating session....118
4.9.3.3.3 Interception at the core network side – terminating session....119
4.9.3.3.4 Interception at both ends of IMS media function – target originating....120
4.9.3.3.5 Interception at both ends of IMS media function – terminating to target....120
4.10 Illustration of LI for IMS data channel....121
4.10.1 IMS data channel background....121
4.10.1.1 General overview....121
4.10.1.2 General principles....122
4.10.1.3 IMS Network and IMS UE capabilities with respect to IMS data channel....123

4.10.1.3.1 IMS network capabilities 123
4.10.1.3.2 IMS UE capabilities 124
4.10.1.3.3 IMS UE perspective 124
4.10.1.3.4 IMS network perspective 124
4.10.1.3.5 Summary 124
4.10.1.4 IMS network functions and interfaces using generic topology views 125
4.10.1.4.1 General 125
4.10.1.4.2 IMS network has IMS DC capabilities 126
4.10.1.4.2.1 General 126
4.10.1.4.2.2 Scenario 1 - originating UE is the local UE, remote UE has IMS DC capabilities 126
4.10.1.4.2.3 Scenario 2 - originating UE is the local UE, remote UE does not have IMS DC capabilities 128
4.10.1.4.2.4 Scenario 3 - originating IMS UE with IMS DC capabilities is the remote UE 129
4.10.1.4.2.4.1 General 129
4.10.1.4.2.4.2 IMS US has subscribed to IMS DC 129
4.10.1.4.2.4.3 IMS UE has not subscribed to IMS DC 129
4.10.1.4.2.5 Scenario 4 - originating IMS UE without the IMS DC capabilities 130
4.10.1.4.3 IMS network without IMS DC capabilities, with limited IMS DC support 131
4.10.1.4.3.1 General 131
4.10.1.4.3.2 Scenario 1 – as a remote network, registered IMS UE has IMS DC capabilities 132
4.10.1.4.3.3 Scenario 2 – as a remote network, registered IMS UE does not have IMS DC capabilities 133
4.10.1.4.4 IMS network does not have IMS DC capabilities 134
4.10.1.4.4.1 General 134
4.10.1.4.4.2 Scenario 1 – as a remote network, locally registered IMS UE has IMS DC capabilities 134
4.10.1.4.4.3 Scenario 2 – as a remote network, locally registered IMS UE has no IMS DC capabilities 135
4.10.1.4.5 Special cases when redirecting network does not support IMS DC capabilities 135
4.10.1.4.5.1 General 135
4.10.1.4.5.2 Scenario 1 – redirecting network has no IMS DC capabilities, redirected-to UE is the remote UE 135
4.10.1.4.5.3 Scenario 2 – redirecting network has no IMS DC capabilities, originating UE is the remote UE 136
4.10.1.4.6 Inter-network redirecting cases 137
4.10.1.4.6.1 General 137
4.10.1.4.6.2 Three IMS UEs in three different networks 137
4.10.1.4.6.3 Only two networks involved in a redirection case 137
4.10.1.5 MF operating modes 138
4.10.1.6 IMS data channels 139
4.10.1.6.1 General 139
4.10.1.6.2 Bootstrap IMS data channel 140
4.10.1.6.3 P2A and A2P IMS data channel 140
4.10.1.6.4 P2A2P IMS data channel 142
4.10.1.6.5 P2P IMS data channel 144
4.10.1.6.6 Media transcoding on P2A2P IMS DC 146
4.10.1.6.7 Media transcoding on P2P IMS data channel 148
4.10.2 LI for IMS DC sessions 149
4.10.2.1 Conventions used 149
4.10.2.2 Illustration principles 150
4.10.2.3 LI in the local end of the IMS DC session 150
4.10.2.3.1 General 150
4.10.2.3.2 IRI-POI for IMS DC 151
4.10.2.3.3 Media interception for bootstrap IMS DC 152
4.10.2.3.4 Media interception for P2A and A2P IMS DC 152
4.10.2.3.4.1 General 152
4.10.2.3.4.2 MF operating in HTTP proxy mode 152
4.10.2.3.4.3 MF operating in DC application proxy mode 153
4.10.2.3.4.4 MF operating in UDP proxy mode 153
4.10.2.3.5 Media interception for P2A2P IMS DC 154
4.10.2.3.5.1 General 154
4.10.2.3.5.2 MF operating in HTTP proxy mode 155
4.10.2.3.5.3 MF operating in DC application proxy mode 155
4.10.2.3.5.4 MF operating in UDP proxy mode 156
4.10.2.3.6 Media interception for P2P IMS DC 157

4.10.2.3.6.1 General 157
4.10.2.3.6.2 MF operating in DC application proxy mode 157
4.10.2.3.6.3 MF operating in UDP proxy mode 158
4.10.2.3.6.4 Without MF 158
4.10.2.3.7 Media interception with transcoded media transfer to the Remote UE 159
4.10.2.3.7.1 General 159
4.10.2.3.7.2 Transcoded media with P2A2P IMS DC with MF acting in an HTTP proxy mode 159
4.10.2.3.7.3 Transcoded media with P2A2P IMS DC with MF operating in DC application proxy mode 160
4.10.2.3.7.4 Transcoded media with P2A2P IMS DC with MF operating in UDP proxy mode 161
4.10.2.3.7.5 Transcoded media with P2P IMS DC with MF operating in DC application proxy mode 162
4.10.2.4 LI at the remote end of IMS DC session 163
4.10.2.4.1 General 163
4.10.2.4.2 Remote network and remote UE have IMS DC capabilities 164
4.10.2.4.2.1 General 164
4.10.2.4.2.2 IRI-POI for IMS DC 164
4.10.2.4.2.3 Media interception for bootstrap IMS DC 165
4.10.2.4.2.4 Media interception for P2A and A2P IMS DC 166
4.10.2.4.2.5 Media interception for P2A2P IMS DC 167
4.10.2.4.2.6 Media interception for P2P IMS DC 168
4.10.2.4.2.7 MF operating in an UDP proxy mode 169
4.10.2.4.3 Either the remote network or remote UE does not have IMS DC capabilities 169
4.10.2.4.3.1 General 169
4.10.2.4.3.2 LI with transcoded media at the remote end (for P2A2P IMS DC from local UE's perspective) 170
4.10.2.4.3.3 LI with transcoded media at the remote end (for P2P IMS DC from local UE's perspective) 170
4.10.2.4.4 Remote network does not have IMS DC capabilities but has limited IMS DC support 171
4.10.3 LI Network topologies 173
4.10.3.1 Few conventions used 173
4.10.3.2 Network topologies from local network's perspective 174
4.10.3.2.1 General 174
4.10.3.2.2 Scenario 1: With IMS DCs to remote UE – DC-AS is in the CSP domain 174
4.10.3.2.3 Scenario 2: With IMS DCs to remote UE – DC-AS is in the third party domain 175
4.10.3.2.4 Scenario 3: Transcoded IMS DC media to remote UE - DC-AS in the CSP domain 176
4.10.3.2.5 Scenario 4: Transcoded IMS DC media to the remote UE - DC-AS in the third party domain 178
4.10.3.2.6 Scenario 5: Terminating party (Remote UE) is a target non-local ID in the originating network with IMS DCs to remote UE. 179
4.10.3.2.7 Scenario 6: Terminating party (Remote UE) is a target non-local ID in the originating network with transcoded IMS DC media to the remote UE 180
4.10.3.3 Network topologies from remote network's perspective with IMS DC capabilities at the remote network 181
4.10.3.3.1 General 181
4.10.3.3.2 Scenario 1: Terminating party as a remote UE is the target 181
4.10.3.3.3 Scenario 2: Redirected-to party as the remote UE 182
4.10.3.3.4 Scenario 3: Originating party is a target non-local ID in the terminating network 183
4.10.3.3.5 Scenario 4: Originating party is a target non-local ID in the redirected-to network 184
4.10.3.3.6 Scenario 5: Redirecting party is a target non-local ID in the redirected-to network 185
4.10.3.4 Network topologies from remote network's perspective (either remote network or remote UE does not have IMS DC capabilities) 185
4.10.3.4.1 General 185
4.10.3.4.2 Scenario 1: Terminating party as a remote UE is the target with transcoded media 185
4.10.3.4.3 Scenario 2: Redirected-to party as the remote UE with transcoded media 187
4.10.3.4.4 Scenario 3: Originating party is a target non-local ID in the terminating network with transcoded media 188
4.10.3.4.5 Scenario 6: Originating party is a target non-local ID in the redirected-to network 188
4.10.3.4.6 Scenario 5: Redirecting party is a target non-local ID in the redirected-to network 189
4.10.3.5 Network topologies for special redirection cases 190
4.10.3.5.1 General 190
4.10.3.5.2 Scenario 1: Redirecting party is the target 190
4.10.3.5.3 Scenario 2: Originating party is a target non-local ID in the redirecting network 191
4.10.3.5.4 Scenario 3: Redirecting party is a target non-local ID in the redirected-to network 192

4.10.3.5.5 Scenario 4: Redirecting party is the target with transcoded IMS DC media to remote UE 192
4.10.3.5.6 Scenario 5: Redirected-to party is a target non-local ID in the redirecting network with transcoded IMS DC media to the remote UE 193
4.10.3.5.7 Scenario 6: Originating party is a target non-local ID in the redirecting network with transcoded IMS DC media to the remote UE 194
4.10.3.6 Topology view for two IMS DC sessions in parallel 194
4.10.3.6.1 General 194
4.10.3.6.2 Originating party is the target 194
4.10.3.6.3 Terminating party is the target 196

5 Example call flows of LI for IMS-based services 197
5.1 General remarks 197
5.2 Originating IMS sessions from the target – non-roaming 197
5.2.1 Introduction 197
5.2.2 Party_A (target) calls Party_B 198
5.2.3 Party_A (target) dials a special number 199
5.3 Terminating IMS sessions to the target – non-roaming 199
5.3.1 Introduction 199
5.3.2 Party_A calls Party_B (target) 200
5.4 Redirected IMS sessions – no roaming 200
5.4.1 Introduction 200
5.4.2 Intra-CSP redirection unconditional – Party_A calls Party_B (target) redirected to Party_C 201
5.4.3 Intra-CSP redirection due to no answer - Party_A calls Party_B (target) redirected to Party_C – flow 1 of 2 202
5.4.4 Intra-CSP redirection due to no answer - Party_A calls Party_B (target) redirected to Party_C – flow 2 of 2 203
5.4.5 Inter-CSP redirection unconditional - Party_A calls Party_B (target) redirected to Party_C 204
5.5 Originating IMS sessions from the target – roaming 204
5.5.1 Introduction 204
5.5.2 Party_A (outbound roaming target) calls Party_B 205
5.5.3 Party_A (outbound roaming target) dials a special number 206
5.5.4 CC Unavailable in home CSP due to optimal media routing 207
5.5.5 Party_A (inbound roaming target) calls Party_B 208
5.5.6 Party_A (inbound roaming target) dials a special number 209
5.6 Terminating IMS sessions to the target – roaming 209
5.6.1 Introduction 209
5.6.2 Party_A calls Party_B (outbound roaming target) 210
5.6.3 Party_A calls Party_B (outbound roaming target), redirected due to no answer 211
5.6.4 Party_A calls Party_B (inbound roaming target) 212
5.6.5 Party_A calls Party_B (inbound roaming target), redirected due to no answer 213
5.7 Ad-hoc IMS conference established by the target 213
5.7.1 Introduction 213
5.7.2 Party_A (target) creates the conference 214
5.7.3 Party_C (conferee) joins the conference 215
5.7.4 Party_B joins the conference 216
5.7.5 Party C drops out of the conference 219
5.7.6 Reconfiguration from conference to two-party session 220
5.7.7 Party_A (target) places conference on hold 221
5.7.8 Party_A (target) retrieves conference from hold 222
5.8 Home routed roaming architecture 222
5.8.1 General 222
5.8.2 Call origination (IRI + CC) 223
5.8.3 Call Origination (IRI only) 223
5.8.4 Call termination (IRI + CC) 224
5.8.5 Mid-call interception 225
5.8.5.1 Before the call is completely answered 225
5.8.5.2 After the call is completely answered 226
5.8.6 BBIFF relocation scenarios 227
5.8.6.1 BBIFF-U relocation 227
5.8.6.2 BBIFF-C relocation 228

Annex A: Change history 230

History 231

Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services (present document).
- Part 2: LI for IMS based STIR/SHAKEN. See TR 33.929-2 [10].
- Part 3: LI for messaging services. See TR 33.929-3 [11].
- Part 4: LI for data in 5G core. See TR 33.929-4 [12].
- Part 5: LI for ID association caching. See TR 33.929-5 [13].
- Part 6: LI for IMS based RCS. See TR 33.929-6 [14].
- Part 7: LI location acquisition capabilities. See TR 33.929-7 [15].
- Part 8: LI for MMS. See TR 33.929-8 [16].
- Part 9: LI for PTC. See TR 33.929-9 [17].

In Part 1 (present document), the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2, the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3, the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4, the illustrations of LI for data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5, the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6, the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7, the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8, the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9, the illustrations of LI for PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3] and TS 33.128 [4].

The present document covers the LI for IMS based services that focus on illustrating the LI aspects of session-based IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (Local Breakout and Home-routed) scenarios. The IMS based STIR/SHAKEN related LI reporting are illustrated in TR 33.929-2 [10].

The LI for IMS-based services is agnostic to the packet core network.

The present document also provides guidance on the LI aspects of IMS Data Channel (IMS DC) capabilities defined in TS 23.228 [8] and TS 24.186 [18].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TS 24.147: "Conferencing using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[6] 3GPP TS 23.501: "System architecture for the 5G System (5GS)".

[7] 3GPP TS 24.174: "Support of multi-device and multi-identity in the IP Multimedia Subsystem (IMS); Stage 3".

[8] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS), Stage 2".

[9] 3GPP TR 33.928: "ADMF logic for provisioning Lawful Interception (LI)".

[10] 3GPP TR 33.929-2: "Lawful Interception (LI) implementation guidance; LI for IMS based STIR/SHAKEN".

[11] 3GPP TR 33.929-3: "Lawful Interception (LI) implementation guidance; LI for messaging services".

[12] 3GPP TR 33.929-4: "Lawful Interception (LI) implementation guidance; LI for data in 5G core".

[13] 3GPP TR 33.929-5: "Lawful Interception (LI) implementation guidance; LI for ID association caching ".

[14] 3GPP TR 33.929-6: "Lawful Interception (LI) implementation guidance; LI for IMS based RCS".

[15] 3GPP TR 33.929-7: "Lawful Interception (LI) implementation guidance; LI location acquisition capabilities".

[16] 3GPP TR 33.929-8: "Lawful Interception (LI) implementation guidance; LI for MMS".

[17] 3GPP TR 33.929-9: "Lawful Interception (LI) implementation guidance; LI for PTC".

[18] 3GPP TS 24.186: "IMS Data Channel applications; Protocol specification".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

3GPP	3rd Generation Partnership Program
A2P	Application to Person
ADC	Application Data Channel
ADMF	Administration Function
AMF	Access and Mobility Management Function
APN	Access Pointe Name
AS	Application Server
BBIFF	Bearer Binding Intercept and Forward Function
BBIFF-C	Bearer Binding Intercept and Forward Function Control plane
BBIFF-U	Bearer Binding Intercept and Forward Function User plane
BDC	Bootstrap Data Channel
BGCF	Border Gateway Control Function
CAN	Core Access Network
CC	Content of Communication
CSCF	Call Session Control Function
CSP	Communication Service Provider
CUPS	Control and User Plane Separation
DC	Data Channel
DCSF	Data Channel Signalling Function
DNN	Data Network Name
eCNAM	Enhanced Calling NAMe
EPC	Evolved Packet Core
E-CSCF	Emergency Call Session Control Function
HPLMN	Home PLMN
HSS	Home Subscriber Server
HTTP	Hypertext Transfer Protocol
H-SMF	Home SMF
H-UPF	Home UPF
IBCF	Interrogating Border Control Function
I-CSCF	Interrogating CSCF

ID	Identity
IMPI	IP Multimedia Private Identity
IMPU	IP Multimedia Public Identity
IMS	IP Multimedia Subsystem
IMS-AGW	IMS Media Gateway
IMS DC	IMS Data Channel
IP	Internet Protocol
IRI	Intercept Related Information
LALS	Lawful Access Location Services
LBO	Local Break Out
LEA	Law Enforcement Agency
LEMF	Law Enforcement Monitoring Facility
LI	Lawful Interception
LICF	Lawful Interception Control Function
LI_HI1	LI_Handover Interface 1
LI_HI2	LI_Handover Interface 2
LI_HI3	LI_Handover Interface 3
LIPF	LI Provisioning Function
LI_T1	Lawful Interception Triggering Interface 1
LI_T3	Lawful Interception Triggering Interface 3
LI_X1	Lawful Interception Internal Interface 1
LI_X2	Lawful Interception Internal Interface 2
LI_X3	Lawful Interception Internal Interface 3
LMISF	LI Mirror IMS State Function
LMISF-CC	LMISF for the handling of CC
LMISF-IRI	LMISF for the handling of IRI
LTE	Long Term Evolution
MCPTT	Mission Critical Push To Talk
MDF	Mediation and Delivery Function
MDF2	Mediation and Delivery Function 2
MDF3	Mediation and Delivery Function 3
MF	Media Function
MGCF	Media Gateway Control Function
MMS	Multimedia Message Service
MMTel	Multimedia Telephony
MRFC	Media Resource Function Controller
MRFP	Multimedia Resource Function Processor
N9HR	N9 Home Routed
NEF	Network Exposure Function
NF	Network Function
NAS	Non-Access Stratum
NR	New Radio
PASSporT	Personal ASSertion Token
P2A	Person to Application
P2A2P	Person to Application and Application to Person
P2P	Person to Person
P-CSCF	Proxy Call Session Control Function
PDN	Packet Data Network
PEI	Permanent Equipment Identifier
PGW	PDN Gateway
PGW-C	PDN Gateway Control Plane
PGW-U	PDN Gateway User Plane
PLMN	Public Land Mobile Network
PoC	Push to talk over Cellular
POI	Point Of Interception
PTC	Push to Talk over Cellular
RCD	Rich Call Data
RCS	Rich Communication Suite
RTP	Real-time Transport Protocol
S8HR	S8 Home Routed
S-CSCF	Serving Call Session Control Function
SCTP	Stream Control Transmission Protocol

SDP	Session Description Protocol
SHAKEN	Signature-based Handling of Asserted information using toKENs
SMF	Session Management Function
SGW	Serving Gateway
SGW-C	Serving Gateway Control Plane
SGW-U	Serving Gateway User Plane
SMS	Short Messaging Service
SUPI	Subscriber Permanent Identifier
SIP	Session Initiation Protocol
STIR	Secure Telephony Identity Revisited
TCP	Transmission Control Protocol
TF	Triggering Function
TLS	Transport Layer Security
TrGW	Transit Gateway
UDM	Unified Data Management
UDP	User Datagram Protocol
UE	User Equipment
UPF	User Plane Function
VPLMN	Visited PLMN
V-SMF	Visted SMF
V-UPF	Visited UPF
xCC	LI_X3 Communications Content
xIRI	LI_X2 Intercept Related Information

4 Illustration of LI for IMS-based services

4.1 General

The subsequent clauses contain a series of IMS topology diagrams that illustrate two aspects:

- Network functions involved in an IMS session setup.
- Network functions involved in providing the LI functions.

In principle, an IMS-based SIP session establishment is independent of the IP connectivity access network. Therefore, within the diagrams, the method used by the IMS UE to access the IMS network is not shown. These illustrations can be applicable to LTE, NR, non-3GPP access network and fixed line access.

The diagrams shown in the subsequent clauses are based on the LI functional architecture defined in TS 33.127 [3] and a simplified generic form is shown in figure 4.1-1 below.

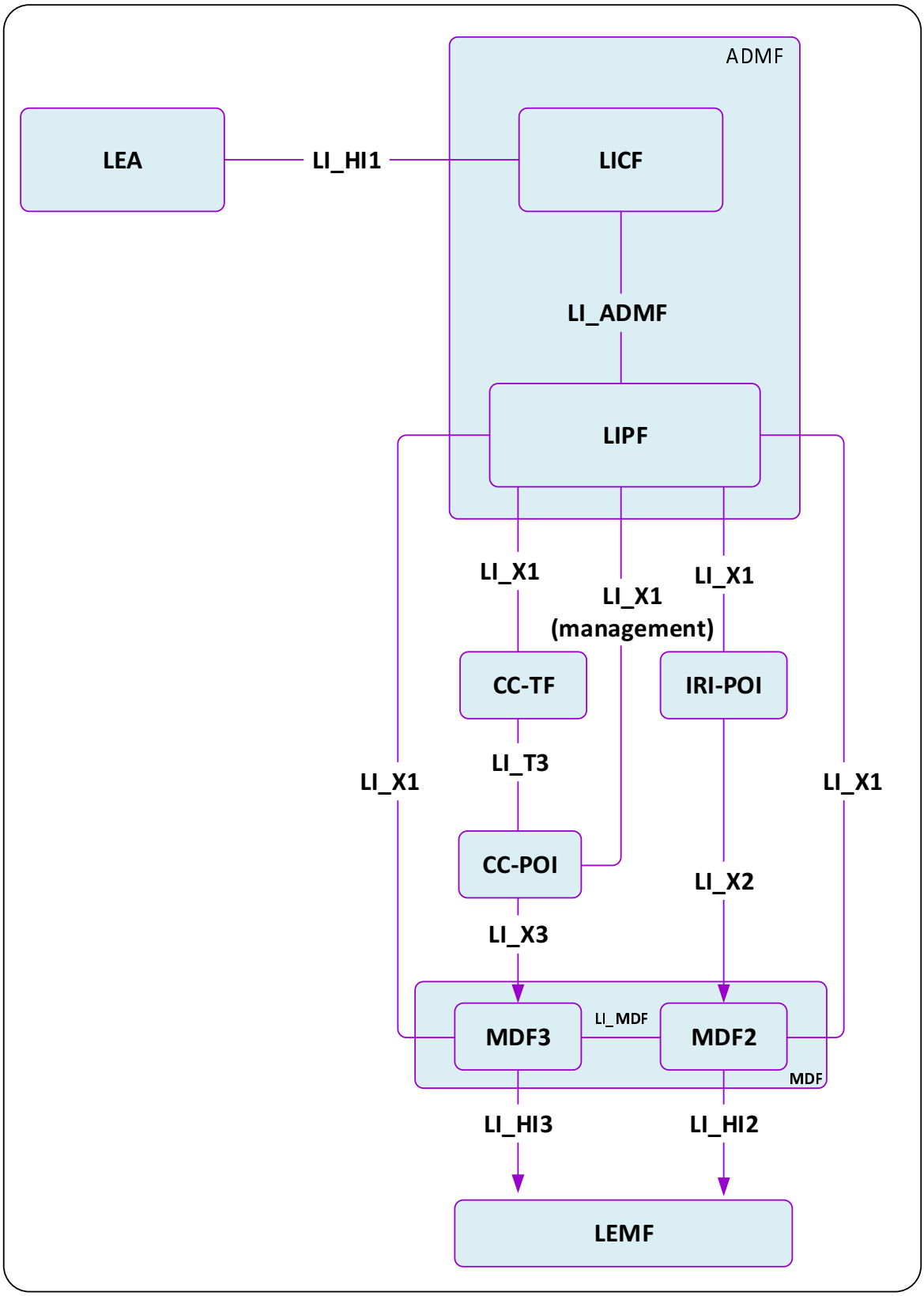


Figure 4.1-1: Generic LI architecture for IMS

The LICF present in the ADMF receives the warrant from an LEA, derives the intercept information from the warrant and provides it to the LIPF. The LIPF present in the ADMF provisions IRI-POI, CC-TF, MDF2 and MDF3 over the LI_X1 interfaces.

The CC-TF sends the CC intercept trigger to the CC-POI over the LI_T3 interface. The IRI-POI generates the xIRI and delivers it to the MDF2 over the LI_X2 interface. The CC-POI generates the xCC and delivers it to the MDF3 over theLI_X3 interface.

The MDF2 generates IRI messages from the received xIRI and delivers them to the LEMF over the LI_HI2 interface. The MDF3 generates the CC from the received xCC and delivers it to the LEMF over the LI_HI3 interface.

The network configuration and IMS service scenarios including the roaming scenarios determine the network functions that provide the IRI-POI, CC-TF and CC-POI functions. The IRI-POI functions are provided by the network functions that handle the SIP messages and the triggered CC-POI functions are provided by the network functions that handle the media. The CC-TF functions are also provided by the network functions that handle the SIP messages and manage the network functions that handle the media. The network functions that provide the CC-TF functions can be different from the network functions that provide the IRI-POI functions.

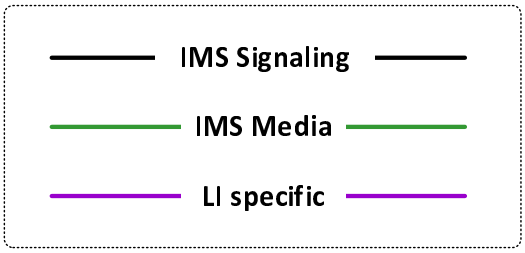
TS 33.127 [3] clause 7.4.6.2 provides several options based on the network topology, session scenario etc. The architecture scenarios shown in the subsequent clauses are examples of default implementation models.

The series of diagrams shown in the subsequent clauses do not include the following LI functions and LI interfaces shown in figure 4.1-1:

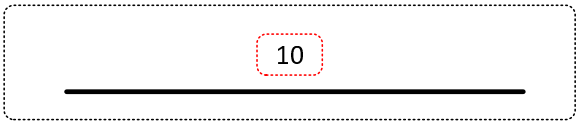
- LEA.
- LICF.
- LIPF.
- LI_HI1.
- LI_X1.
- LI_X1 (management).

The above LI functions and the interfaces apply to all diagrams and hence, can be presumed that they are common to all.

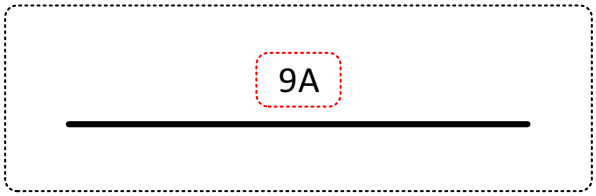
NOTE 1: The dark lines show IMS signalling (not limited to SIP messages), green lines show IMS media, and purple line show LI specific interfaces or information flow.



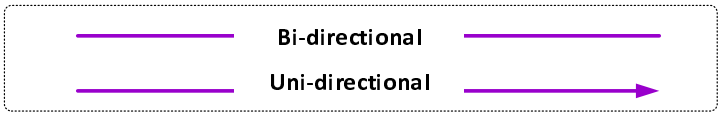
NOTE 2: The sequential numerical number next to the dark line is used to show the progress of the end-to-end IMS session setup. For example, the following is the 10th step in the progress of an end-to-end IMS session setup.



NOTE 3: A numerical sequence number illustrates a SIP message and an alpha-numeric sequence number illustrates a non-SIP message (e.g. H.248). In the following, 9A is a non-SIP step and progressively, it comes after step 9.



NOTE 4: A line without an arrow indicates a bi-directional information flow (see below for a comparison).



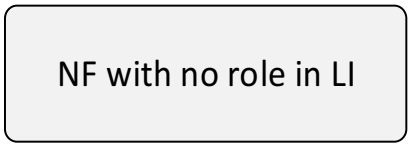
NOTE 5: A network function that has an LI function is shown as an orange coloured box with a thick boundary (see below).



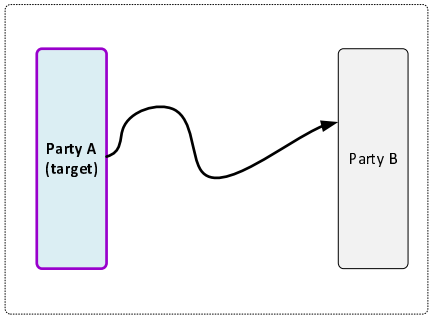
NOTE 6: An LI specific function is shown as a blue coloured box with a thick purple boundary (see below).



NOTE 7: A network function that does not have a role to play in providing the LI function for the scenario is shown as a thin lined white or grey box (see below).



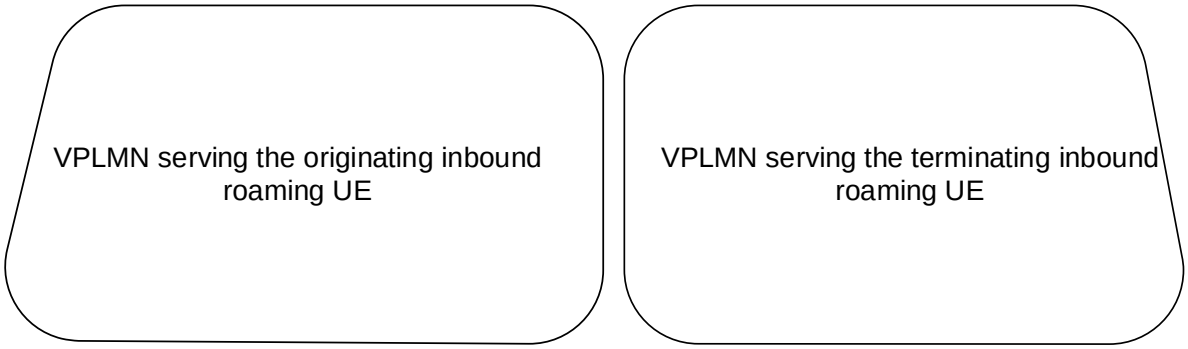
NOTE 8: Each IMS session involves at least two parties. Party A is always an originating party. In a session involving two parties, Party B is the called party. The party that is also a target is shown as a blue coloured box with purpled boundary.



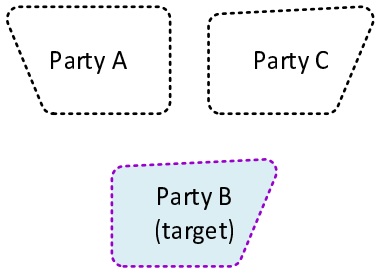
NOTE 9: The functions handling the session related to a party are grouped and accordingly recognized. In the following, the functions (not shown) within the dashed line below represent the A-side of the session.



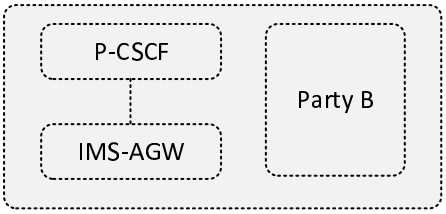
NOTE 10: The functions within a VPLMN handling the IMS sessions of inbound roaming users are grouped and accordingly recognized using a slanted boundary as shown below.



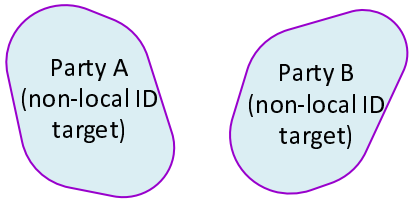
NOTE 11: An outbound roaming UE is shown with a dotted/slanted boundary. In the following Party A (originating), Party B (terminating) and Party C (terminating) are outbound roaming UEs. Party B is an outbound roaming target.



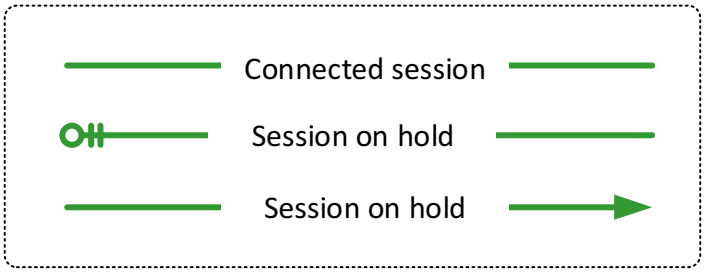
NOTE 12: The functions and the UE that would have normally been involved in the session but are not within the PLMN due to the redirecting feature (or roaming) are shown with dotted boundary. In the following the Party B, the P-CSCF and the IMS-AGW are such UEs and functions.



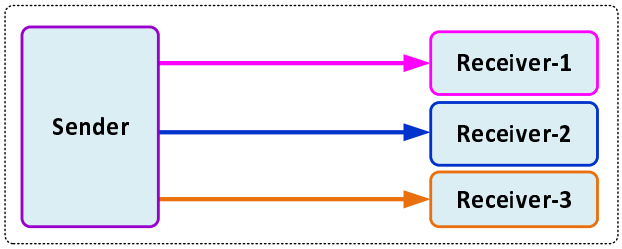
NOTE 13: A non-local ID target is shown with slanted solid boundary as shown below. The direction of the slant indicates whether a session is incoming from a non-local ID target (left) or whether the session is to the non-local ID target (right).



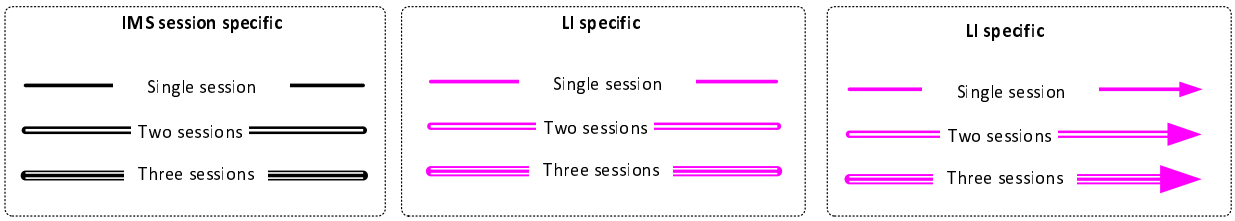
NOTE 14: A session on hold is shown with a special symbol at the end of the media line (preferred in the TR) indicating that the party placing the session on hold is not receiving any media streams. Alternatively, a session on hold may also be shown with a unidirectional media flow line.



NOTE 15: Different colours (e.g. pink, blue, gold) are used to show the associated grouping of the LI specific information flow when one or more of those information flows have multiple recipients (see below).



NOTE 16: Double line or triple lines are used to show that the information flows are associated with multiple IMS sessions (see below).



A function present in two groups may be provided by a single network function. For example, when an incoming session to Party B is redirected (intra-CSP domain) to Party C, the HSS/UDM shown in the B-side of the session and the C-side of the session can be the same.

4.2 Intra-CSP domain IMS sessions

4.2.1 General

Clause 4.2 illustrates the IMS LI for various intra-CSP domain IMS session scenarios, i.e. all parties involved in the session are served by the same CSP domain.

4.2.2 Originating sessions

4.2.2.1 Party A (target) calls Party B

Figure 4.2.2.1-1 shows a scenario where Party A calls Party B and Party A is the target.

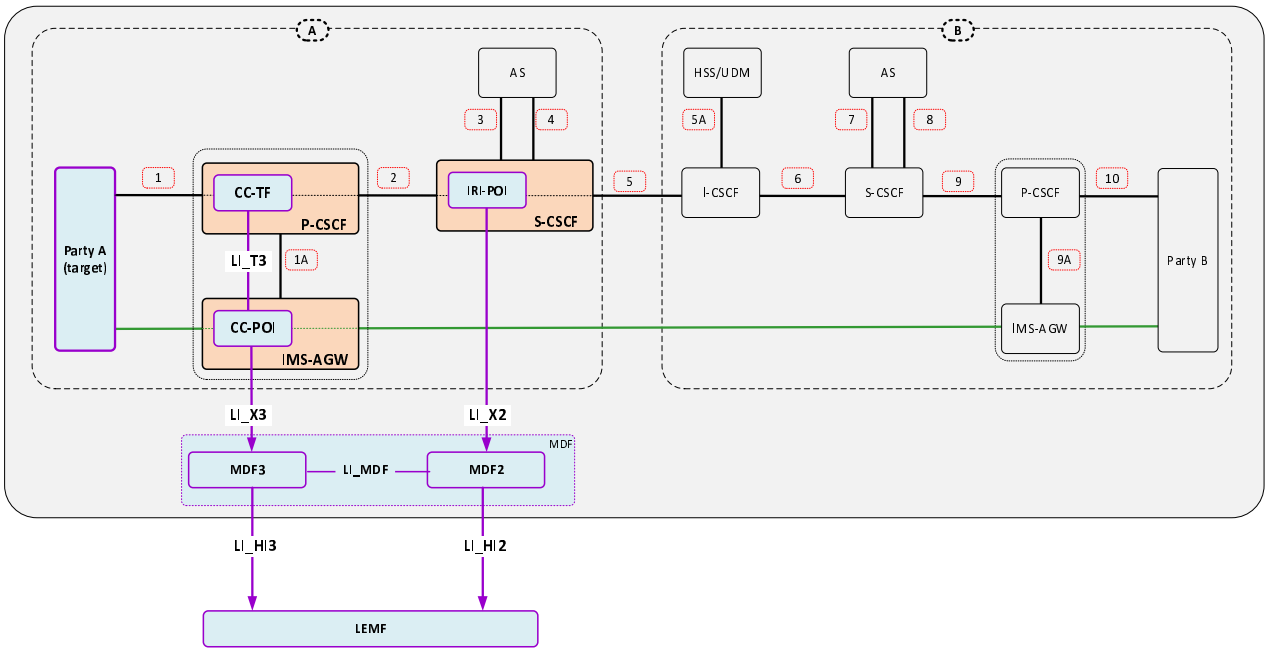


Figure 4.2.2.1-1: Party A (target) calls Party B

As shown in figure 4.2.2.1-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.2.2.2 Party A (target) calls Party B, redirected to Party C

Figure 4.2.2.2-1 shows a scenario where Party A calls Party B while Party B has set up to redirect all incoming sessions to Party C. In this scenario, Party A is the target.

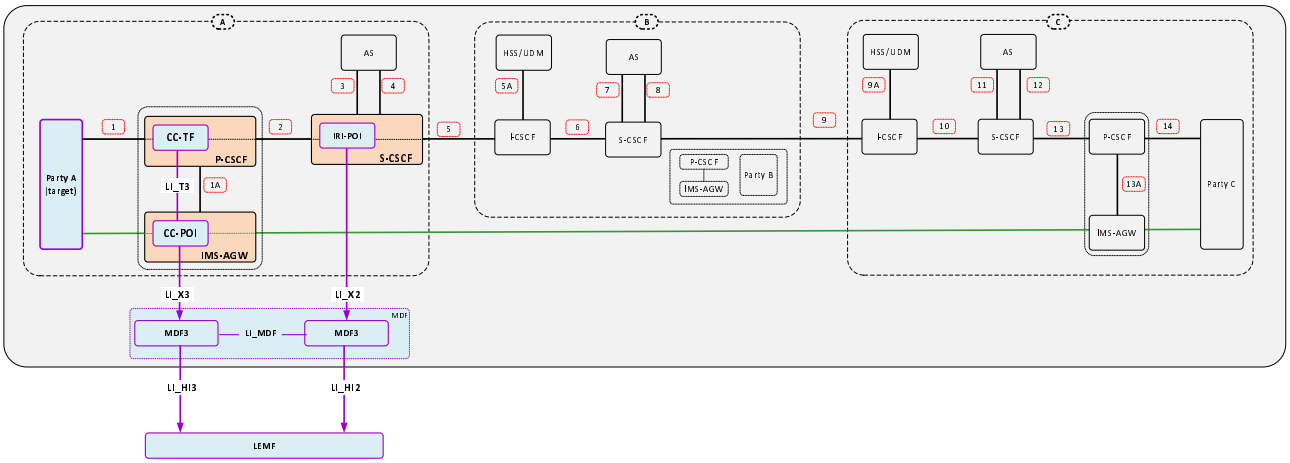


Figure 4.2.2.2-1: Party A (target) calls Party B, redirected to Party C

As shown in figure 4.2.2.2-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions. This illustration shows that when the originating party is a target, only the network

functions involved in the originating party side of the call may have an LI impact, independently of how the call is further progressed within the network.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.2.3 Terminating sessions

4.2.3.1 Party A calls Party B (target)

Figure 4.2.3.1-1 shows a scenario where Party A calls Party B and Party B is the target.

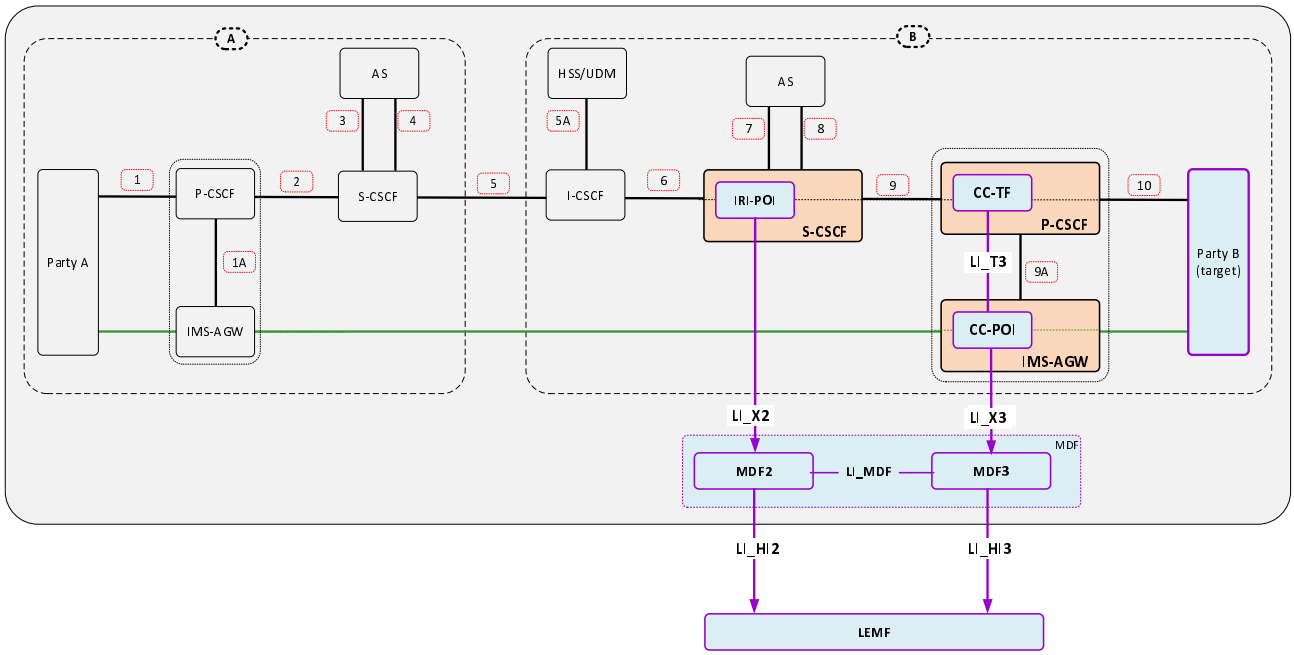


Figure 4.2.3.1-1: Party A calls Party B (target)

As shown in figure 4.2.3.1-1, only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.2.3.2 Party A calls Party B, redirected to Party C (target)

Figure 4.2.3.2-1 shows a scenario where Party A calls Party B while Party B has set up to redirect all incoming sessions to Party C. In this scenario, Party C is the target.

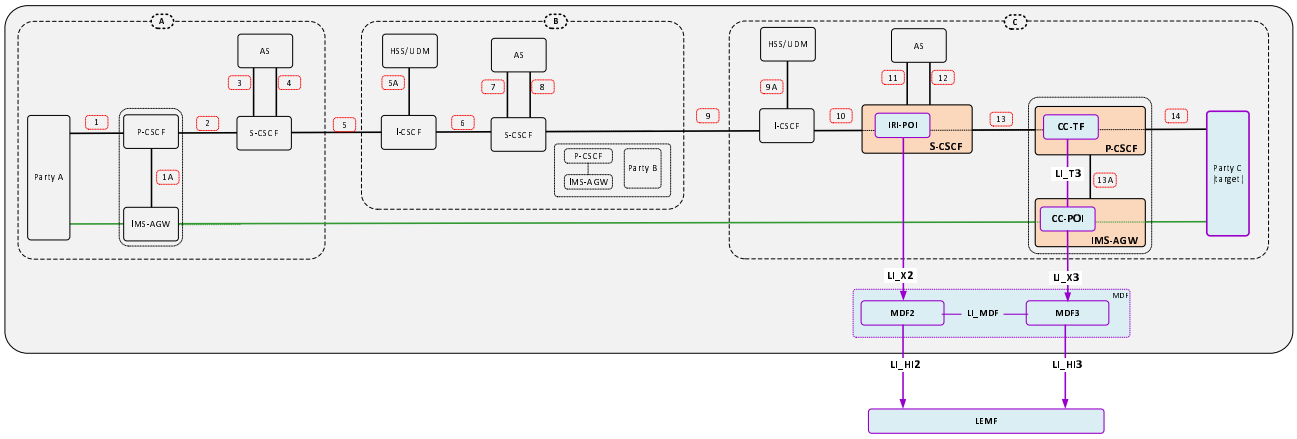


Figure 4.2.3.2-1: Party A calls Party B, redirected to Party C (target)

As shown in figure 4.2.3.2-1, only the network functions that handle the C-side (target side) of the session are involved in handling the LI functions. This illustration shows that when the terminating party is a target, only the network functions involved in the terminating party side of the call may have an LI impact and that is independent of how the call was previously progressed within the network.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.2.4 Redirecting scenarios

4.2.4.1 Party A calls Party B (target), redirected to Party C

Figure 4.2.4.1-1 shows a scenario where Party A calls Party B and Party B has setup to redirect all incoming sessions to Party C. In this scenario, Party B is the target.

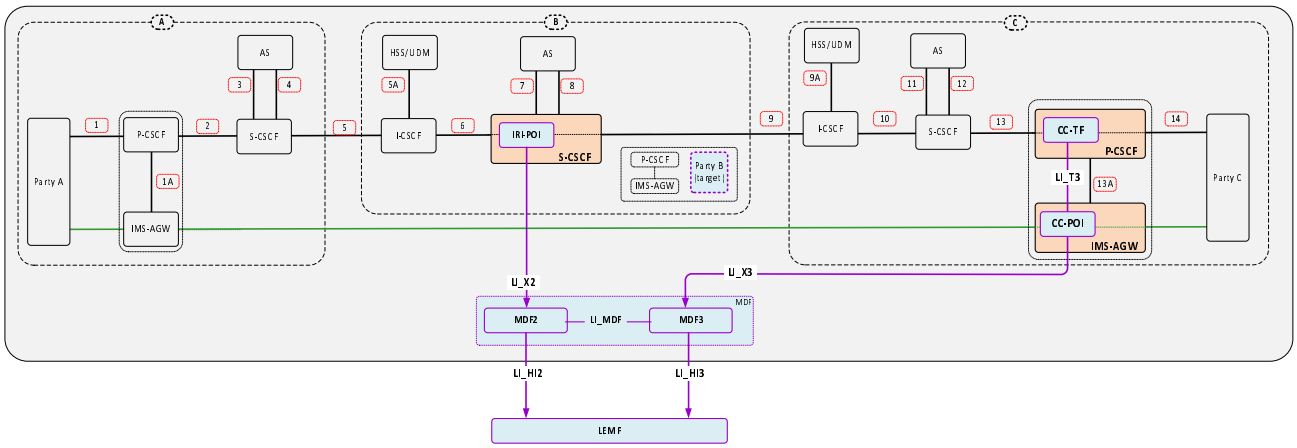


Figure 4.2.4.1-1: Party A calls Party B (target), redirected to Party C

As shown in figure 4.2.4.1-1, the network functions that handle the B-side (target side) and the C-side (redirected-to party side) of the session are involved in handling the LI functions. The network functions that handle the B-side (target side) are not involved in the handling the IMS media related to the session.

The IRI-POI present in the S-CSCF (B-side) accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF (C-side) triggers the CC-POI present in the IMS-AGW (C-side) for the xCC. The CC-POI present in the IMS-AGW (C-side) generates the xCC. The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.3 Inter-CSP domain IMS sessions

4.3.1 General

Clause 4.3 illustrates the IMS LI for various inter-CSP domain IMS session scenarios, i.e. one of the parties involved in the session is in another CSP domain.

NOTE: CSP domain of interest means the CSP network where the interception is done. A local target means that the target is served within the CSP domain of interest. A non-local ID target means that the target is a subscriber of another CSP domain.

4.3.2 Inter-CSP domain originations

4.3.2.1 Introduction

In these scenarios, the session is originated in another CSP domain. That CSP domain may be CS-based or IP-based, the latter can be an IMS-based as well. The session may also get redirected one or more times before terminating to the CSP domain of interest.

NOTE: For the cases of redirection happening before the sessions hit the CSP domain of interest, different domain types may be involved as well. For example, a session originating from a CS domain may get redirected to an IP domain before terminating at the CSP domain of interest.

The illustrations show that as far as lawful interception is concerned, it makes no difference whether the call is originated from another IMS domain, another non-IMS-based IP domain or a CS domain, unless the originating party or one of the redirecting parties in the other CSP domain is a non-local ID target in the CSP domain of interest.

4.3.2.2 Party A (CS domain) calls Party B (target)

Figure 4.3.2.2-1 shows a scenario where Party A from a CS domain (another CSP) calls Party B (target).

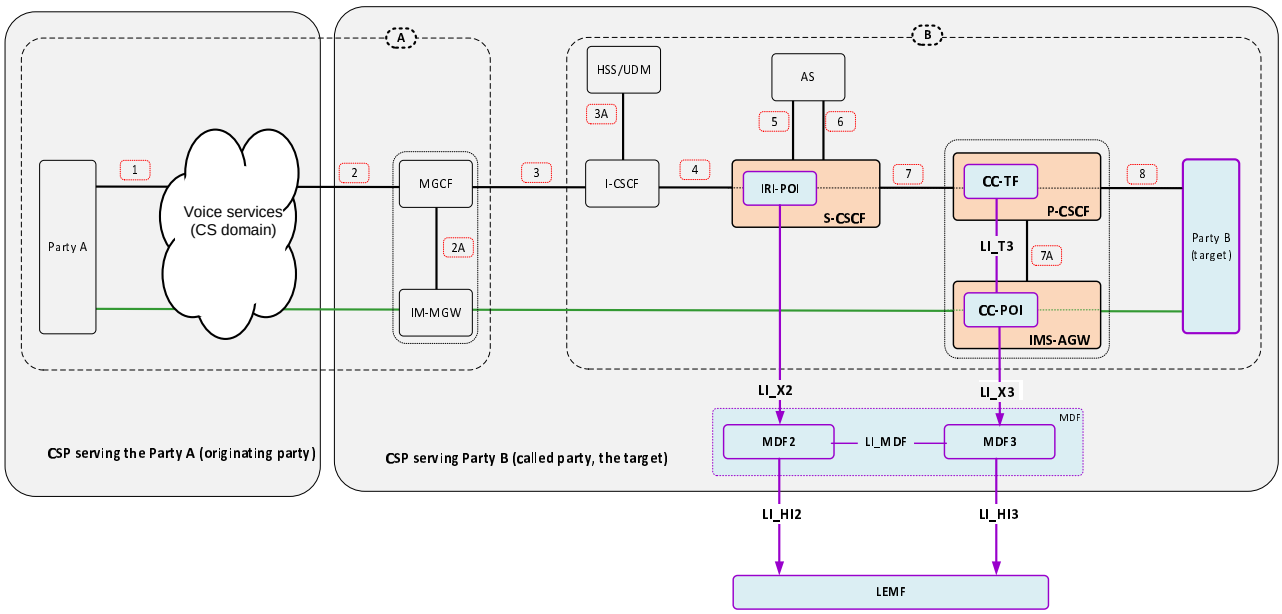


Figure 4.3.2.2-1: Party A from CS domain calls Party B (target)

As shown in figure 4.3.2.2-1, only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

NOTE: For sessions originating from an IP domain (e.g. another CSP's IMS domain), the architecture diagram shown in figure 4.3.2.2-1 applies, with IBCF/TrGW instead of MGCF/IM-MGW, handling the A-side of the session within the CSP domain of interest. No separate diagram is given since there is no LI function on the A-side of the session for the scenario.

4.3.2.3 Party A (non-local ID target, CS domain) calls Party B

Figure 4.3.2.3-1 shows a scenario where Party A from a CS domain (another CSP) calls Party B. In this example, Party A is a non-local ID target in the CSP domain of interest.

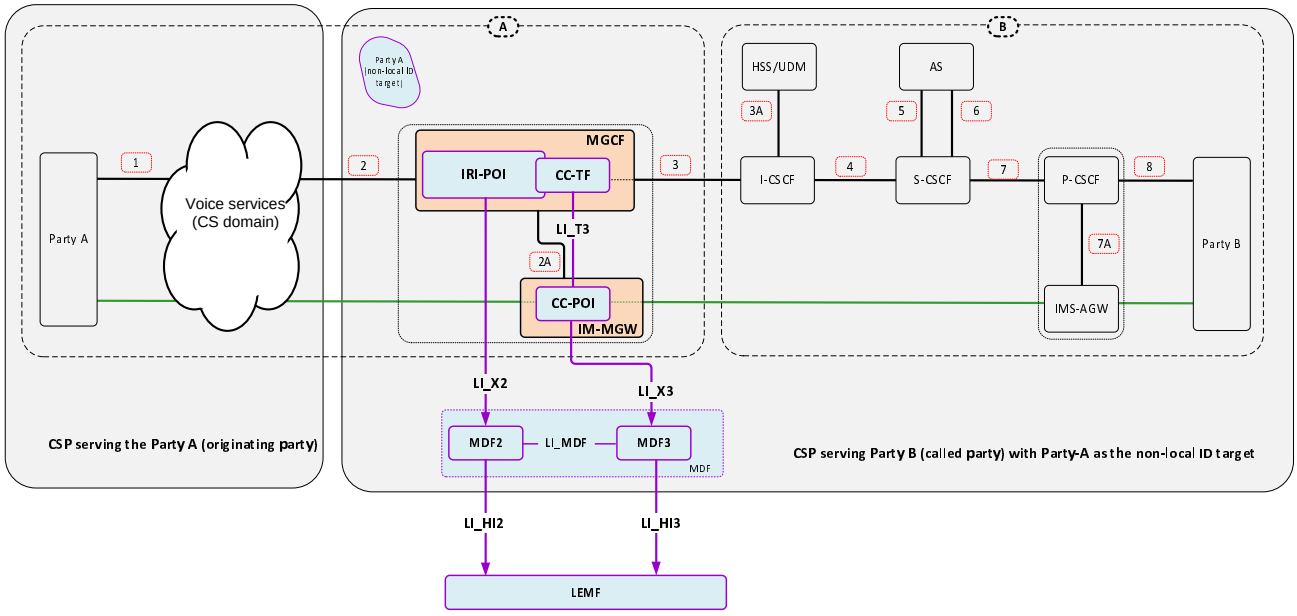


Figure 4.3.2.3-1: Party A (non-local ID target) from CS domain calls Party B

As shown in figure 4.3.2.3-1, only the network functions that handle the A-side (non-local ID target side) of the session are involved in handling the LI functions.

The IRI-POI present in the MGCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF triggers the CC-POI present in the IM-MGW for the xCC. The CC-POI present in the IM-MGW generates the xCC.

NOTE 1: Since only the egress signalling at the MGCF is SIP-based, this architecture assumes that the interception is done using the SIP messages seen at the step 3.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

NOTE 2: The session may experience one or more redirections within the CS domain before terminating to the CSP domain of interest. From LI architectural perspective, the diagram shown in figure 4.3.2.3-1 applies even if one of the redirecting parties within the CS domain is a non-local ID target.

4.3.2.4 Party A (non-local ID target, IP domain) calls Party B

Figure 4.3.2.4-1 shows a scenario where Party A from an IP domain (another CSP) calls Party B. In this example, Party A is a non-local ID target.

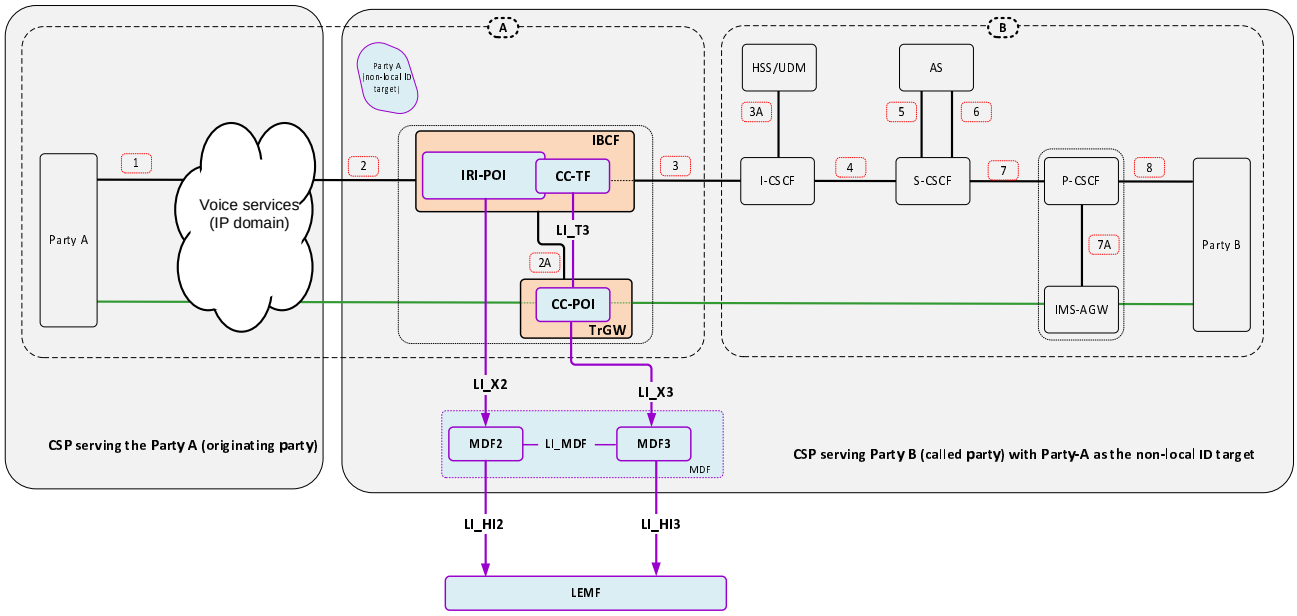


Figure 4.3.2.4-1: Party A (non-local ID target) from IP domain calls Party B

As shown in figure 4.3.2.4-1, only the network functions that handle the A-side (non-local ID target side) of the session are involved in handling the LI functions.

The IRI-POI present in the IBCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

NOTE: The session may experience one or more redirections within the IP domain before terminating to the CSP domain. From LI architectural perspective, the diagram shown in figure 4.3.2.4-1 applies even if one of the redirecting parties within the IP domain is a non-local ID target.

4.3.3 Inter-CSP domain terminations

4.3.3.1 Introduction

In these scenarios, the session is terminated in another CSP domain. That CSP domain may be CS-based or IP-based, the latter can be an IMS-based as well.

The illustrations show that as far as lawful interception is concerned, it makes no difference whether a call is terminated to another IMS domain, another non-IMS-based IP domain or a CS domain unless the terminating party is a non-local ID target in the CSP domain of interest.

4.3.3.2 Party A (target) calls Party B (CS domain)

Figure 4.3.3.2-1 shows a scenario where Party A (target) calls Party B in a CS domain (another CSP).

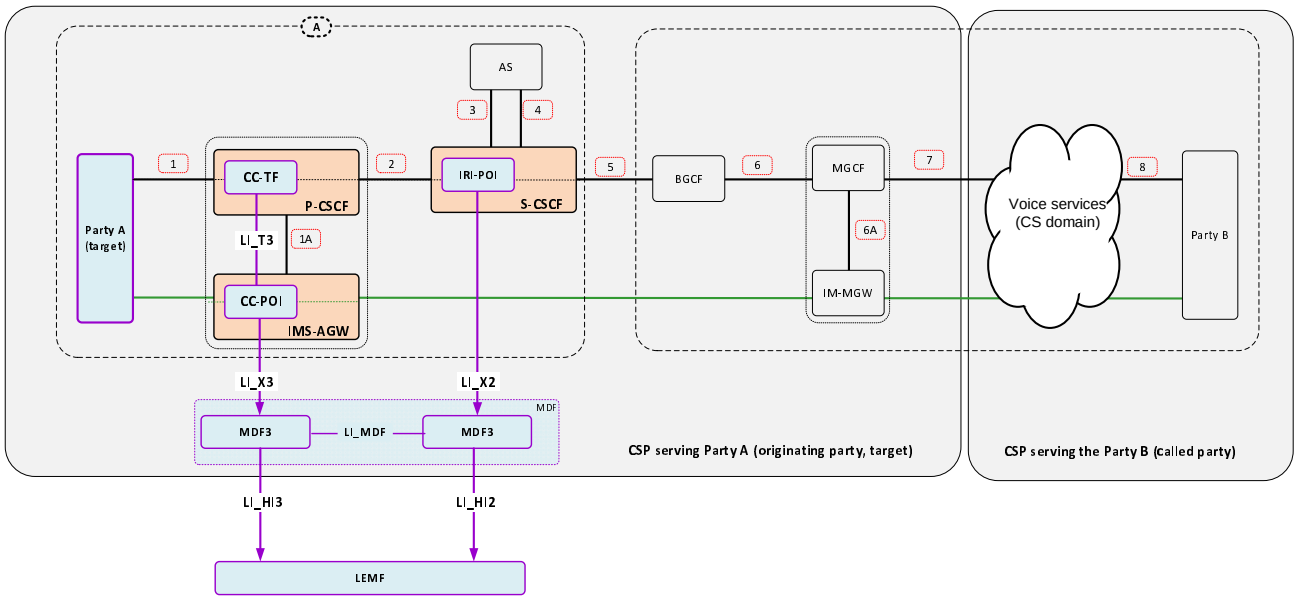


Figure 4.3.3.2-1: Party A (target) calls Party B in a CS domain

As shown in figure 4.3.3.2-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

NOTE: For sessions terminating to an IP domain (e.g. another CSP's IMS domain), the architecture diagram shown in figure 4.3.3.2-1 applies, with IBCF/TrGW, instead of MGCF/IM-MGW, handling the B-side of the session within the CSP domain of interest. No separate diagram is given since there is no LI function on the B-side of the session for the scenario.

4.3.3.3 Party A calls Party B (non-local ID target, CS domain)

Figure 4.3.3.3-1 shows a scenario where Party A calls Party B in a CS domain (another CSP). In this example, Party B is a non-local ID target.

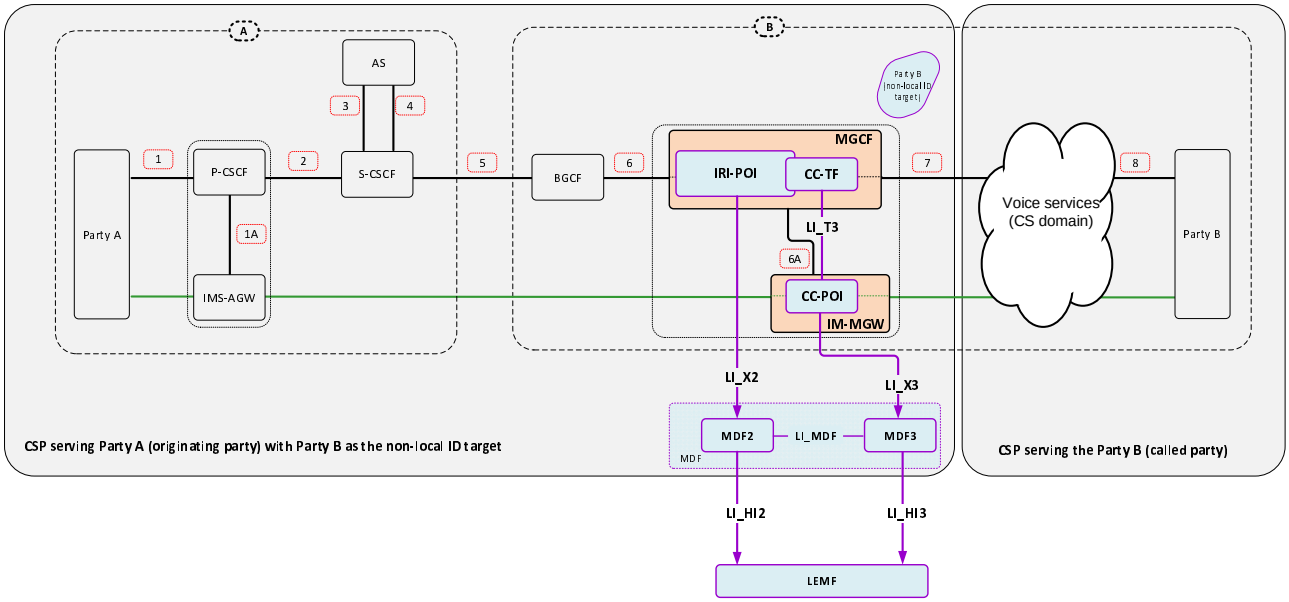


Figure 4.3.3.3-1: Party A calls Party B (non-local ID target) in a CS domain

As shown in figure 4.3.3.3-1, only the network functions that handle the B-side (non-local ID target side) of the session are involved in handling the LI functions.

The IRI-POI present in the MGCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF triggers the CC-POI present in the IM-MGW for the xCC. The CC-POI present in the IM-MGW generates the xCC.

NOTE: Since only the ingress signalling at the MGCF is SIP-based, this architecture assumes that the interception is done using the SIP messages seen at the step 6.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.3.3.4 Party A calls Party B (non-local ID target, IP domain)

Figure 4.3.3.4-1 shows a scenario where Party A calls Party B in IP domain (another CSP). In this example, Party B is a non-local ID target.

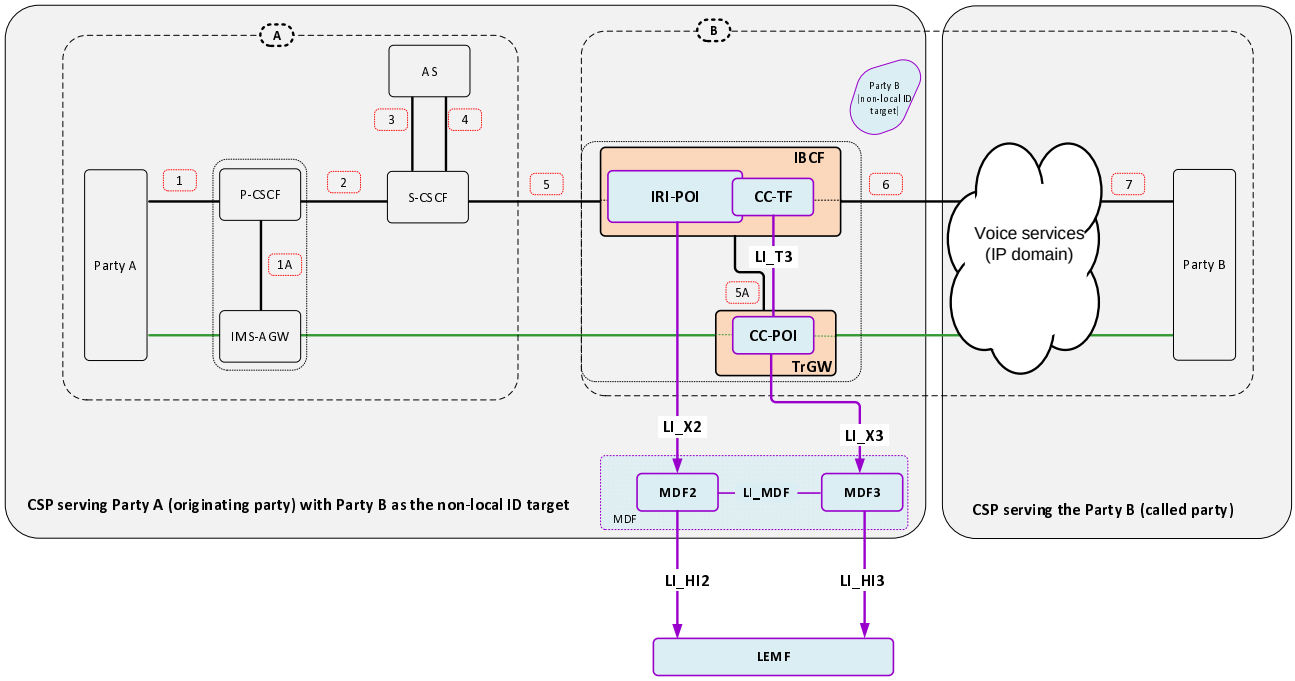


Figure 4.3.3.4-1: Party A calls Party B (non-local ID target) in IP domain

As shown in figure 4.3.3.4-1, only the network functions that handle the B-side (non-local ID target side) of the session are involved in handling the LI functions.

The IRI-POI present in the IBCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.3.4 Inter-CSP domain redirections

4.3.4.1 Introduction

In these scenarios, the session is redirected to another CSP domain. That CSP domain may be CS-based or IP-based, the latter can be an IMS-based as well.

4.3.4.2 Party A calls Party B (target) redirected to Party C (CS domain)

Figure 4.3.4.2-1 shows a scenario where Party A calls Party B and Party B has setup to redirect all incoming sessions to Party C. In this scenario, Party B is the target and Party C is in a CS domain (another CSP).

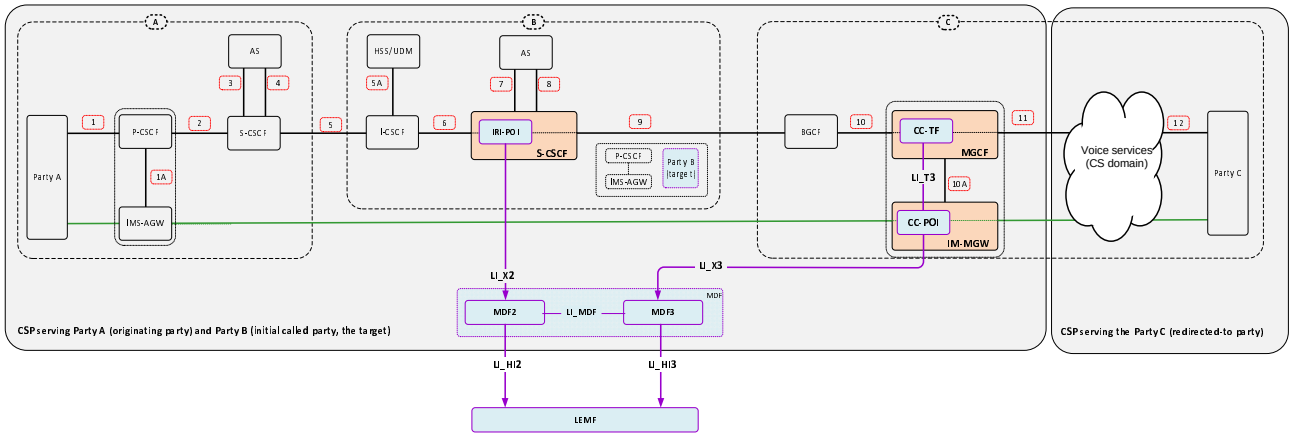


Figure 4.3.4.2-1: Party A calls Party B (target) redirected to Party C in a CS domain

As shown in figure 4.3.4.2-1, the network functions that handle the B-side (target side) and the C-side (redirected-to party side) of the session are involved in handling the LI functions. The network functions that handle the B-side (target side) are not involved in the handling the IMS media related to the session.

The IRI-POI present in the S-CSCF (B-side) accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF (C-side) triggers the CC-POI present in the IM-MGW (C-side) for the xCC. The CC-POI present in the IM-MGW (C-side) generates the xCC.

NOTE: Since only the ingress signalling at the MGCF is SIP-based, this architecture assumes that the for the CC intercept trigger the MGCF would use the SIP messages seen at the step 10.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.3.4.3 Party A calls Party B (target) redirected to Party C (IP domain)

Figure 4.3.4.3-1 shows a scenario where Party A calls Party B and Party B has setup to redirect all incoming sessions to Party C. In this scenario, Party B is the target and Party C is in an IP domain (another CSP).

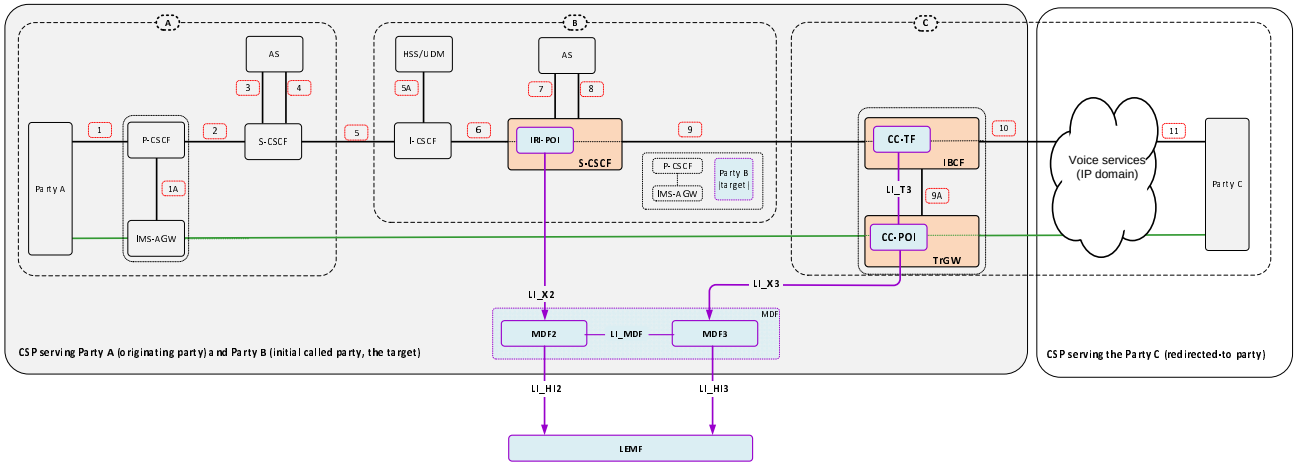


Figure 4.3.4.3-1: Party A calls Party B (target) redirected to Party C in an IP domain

As shown in figure 4.3.4.3-1, the network functions that handle the B-side (target side) and the C-side (redirected-to party side) of the session are involved in handling the LI functions. The network functions that handle the B-side (target side) are not involved in the handling the IMS media related to the session.

The IRI-POI present in the S-CSCF (B-side) accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF (C-side) triggers the CC-POI present in the TrGW (C-side) for the xCC. The CC-POI present in the TrGW (C-side) generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.3.4.4 Party A calls Party B redirected to Party C (non-local ID target, CS domain)

Figure 4.3.4.4-1 shows a scenario where Party A calls Party B and Party B has setup to redirect all incoming sessions to Party C. In this scenario, Party C is the non-local ID target and is in a CS domain.

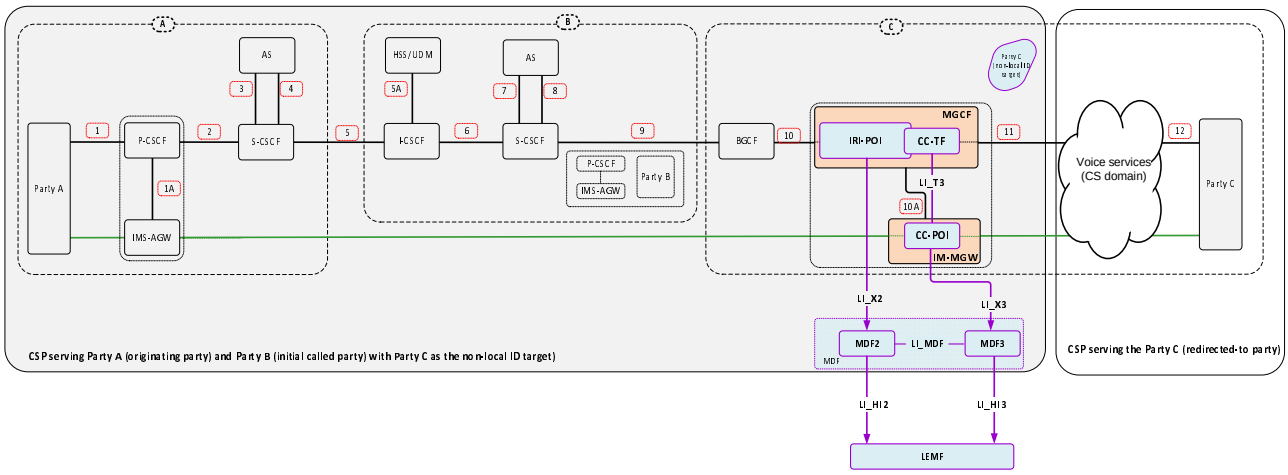


Figure 4.3.4.4-1: Party A calls Party B redirected to Party C (non-local ID target) in a CS domain

As shown in figure 4.3.4.4-1, only the network functions that handle the C-side (non-local ID target side) of the session are involved in handling the LI functions.

The IRI-POI present in the MGCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF triggers the CC-POI present in the IM-MGW for the xCC. The CC-POI present in the IM-MGW generates the xCC.

NOTE: Since only the ingress signalling at the MGCF his SIP-based, this architecture assumes that the interception is done using the SIP messages seen at the step 10.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.3.4.5 Party A calls Party B redirected to Party C (non-local ID target, IP domain)

Figure 4.3.4.5-1 shows a scenario where Party A calls Party B and Party B has setup to redirect all incoming sessions to Party C. In this scenario, Party C is the non-local ID target and is in an IP domain.

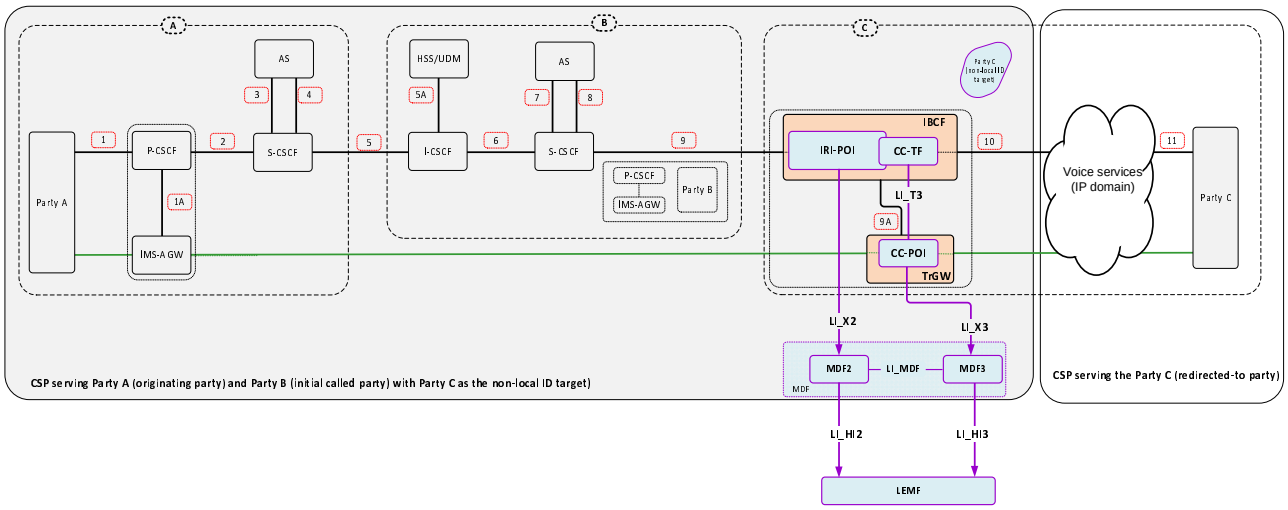


Figure 4.3.4.5-1: Party A calls Party B redirected to Party C (non-local ID target) in IP domain

As shown in figure 4.3.4.5-1, only the network functions that handle the C-side (non-local ID target side) of the session are involved in handling the LI functions.

The IRI-POI present in the IBCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.3.4.6 Special case – multiple targets, inter-CSP domains at both ends of the session

Figure 4.3.4.6-1 shows a scenario where Party A calls Party B and Party B has setup to redirect all incoming sessions to Party C. In this scenario, Party B is a local target in the CSP domain of interest whereas Party A and Party C are the non-local ID targets. Party A is in an IP domain and Party C is in a CS domain.

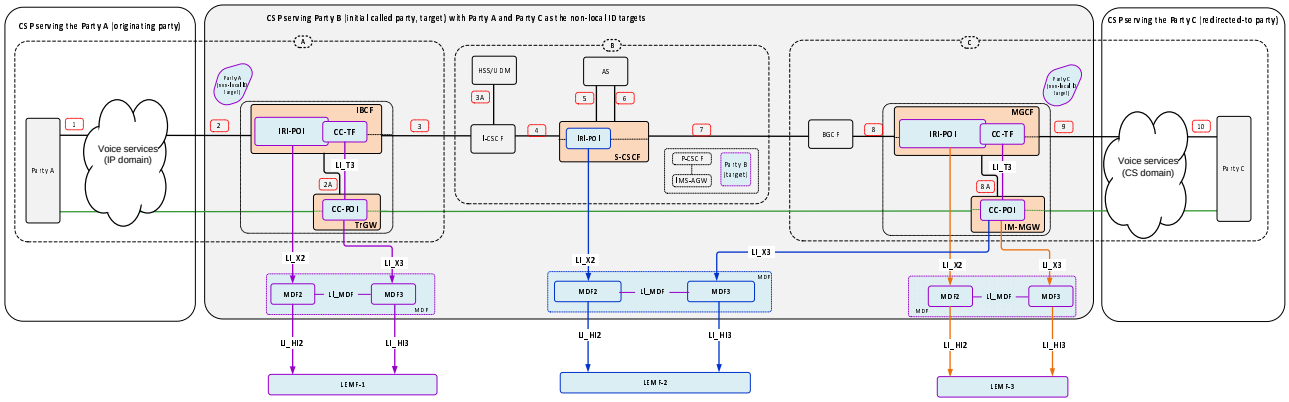


Figure 4.3.4.6-1: Party A (non-local ID target) calls Party B (target) redirected to Party C (non-local ID target)

As shown in figure 4.3.4.6-1, there are three active intercepts in this scenario. The network functions that handle the A-side (originating non-local ID target), B-side (local target) and C-side (terminating non-local ID target) of the session are involved in handling the LI functions. For CC interception for Party B and for Party C are done at the C-side of the session.

Party A's interception

The IRI-POI present in the IBCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-1. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-1.

Party B's interception

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF (C-side) triggers the CC-POI present in the IM-MGW for the xCC. The CC-POI present in the IM-MGW generates the xCC.

NOTE 1: Since only the ingress signalling at the MGCF is SIP-based, this architecture assumes that for the CC intercept trigger, the MGCF would use the SIP messages seen at the step 8.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-2. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-2.

Party C’s interception

For Party C’s interception, the IRI-POI present in the MGCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF triggers the CC-POI present in the IM-MGW for the xCC. The CC-POI present in the IM-MGW generates the xCC.

NOTE 2: Since only the ingress signalling at the MGCF is SIP-based, this architecture assumes that for the CC intercept trigger, the MGCF would use the SIP messages seen at the step 8.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-3. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-3.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4 LBO roaming scenarios

4.4.1 General

Clause 4.4 illustrates the IMS LI for various roaming scenarios with local breakout (LBO) as the roaming architecture.

The VPLMN may have an active intercept on the inbound roaming party, the HPLMN may have an active intercept on the outbound roaming party or both HPLMN and VPLMN have the two separate independent active intercepts on the roaming party.

4.4.2 Roaming at the origination end

4.4.2.1 Introduction

In these scenarios, the session is originated in the VPLMN, i.e. originating party is roaming.

4.4.2.2 Roaming Party A (target in HPLMN) calls Party B

Figure 4.4.2.2-1 shows a scenario where Party A (calling party) is roaming and Party B (called party) is not roaming. In the scenario illustrated, the HPLMN has an active intercept on Party A, who is outbound roaming.

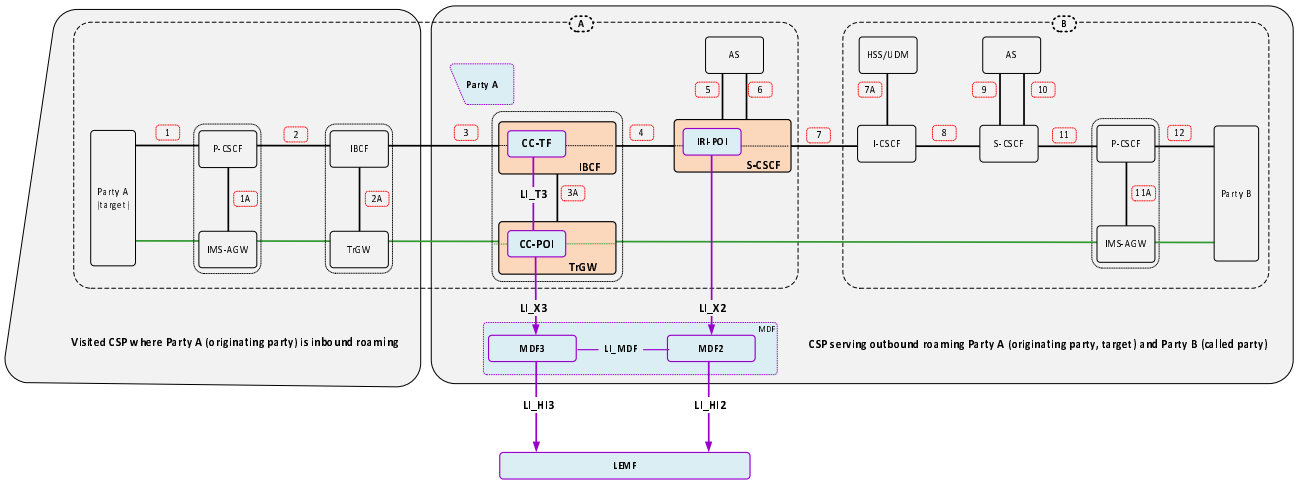


Figure 4.4.2.2-1: Party A (roaming, target in HPLMN) calls Party B

As shown in figure 4.4.2.2-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.2.3 Roaming Party A (target in VPLMN) calls Party B

Figure 4.4.2.3-1 shows a scenario where Party A (calling party) is roaming and Party B (called party) is not roaming. In the scenario illustrated, the VPLMN has an active intercept on Party A, who is inbound roaming.

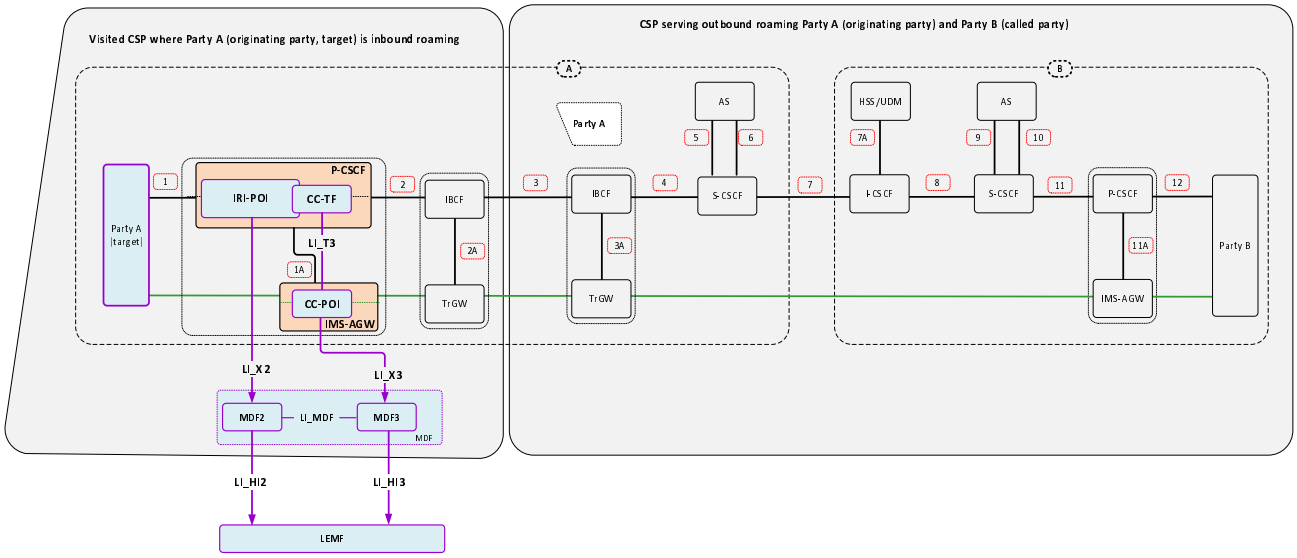


Figure 4.4.2.3-1: Party A (roaming, target in VPLMN) calls Party B

As shown in figure 4.4.2.3-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions. Note that the network functions in the VPLMN are considered to be the A-side of the session since the Party A is originating the session in the VPLMN.

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.2.4 Roaming Party A (target in VPLMN and HPLMN) calls Party B

Figure 4.4.2.4-1 shows a scenario where Party A (calling party) is roaming and Party B (called party) is not roaming. In the illustrated scenario, HPLMN has an active intercept on Party A, who is outbound roaming and VPLMN also has an active intercept on Party A who is inbound roaming.

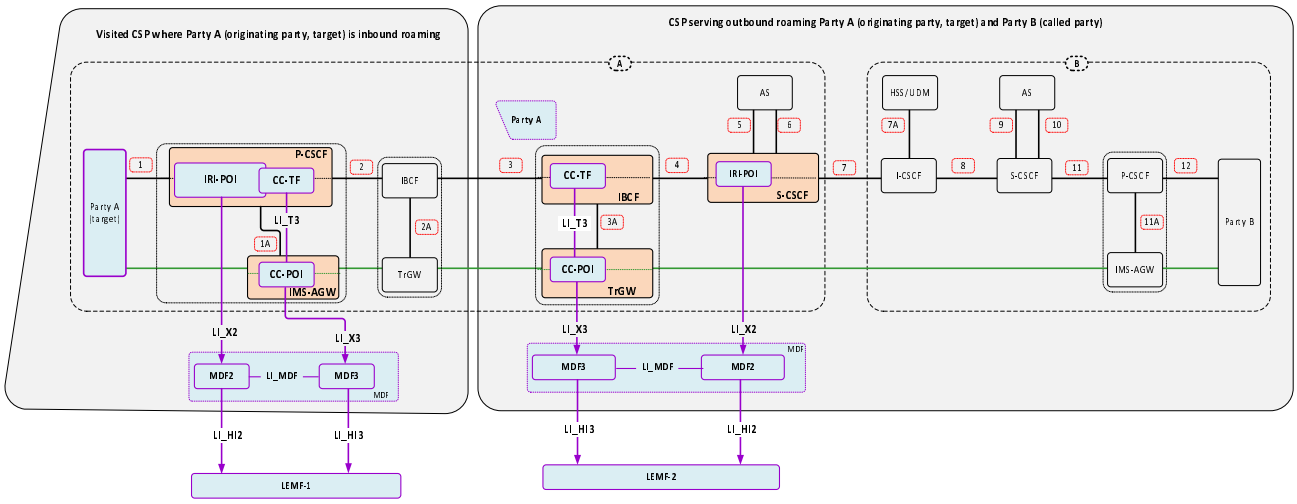


Figure 4.4.2.4-1: Party A (roaming, target in VPLMN, target in HPLMN) calls Party B

As shown in figure 4.4.2.4-1, there are two intercepts active on Party A. Only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions. Note that the network functions in the VPLMN are considered the A-side of the session since the Party A is originating the session in the VPLMN.

Interception in the VPLMN

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-1.

Interception in the HPLMN

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-2.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.2.5 Roaming Party A calls Party B (non-local ID target in VPLMN)

Figure 4.4.2.5-1 shows a scenario where Party A (calling party) is roaming and Party B (called party) is not roaming. In the scenario illustrated, the VPLMN has an active intercept on Party B as a non-local ID target.

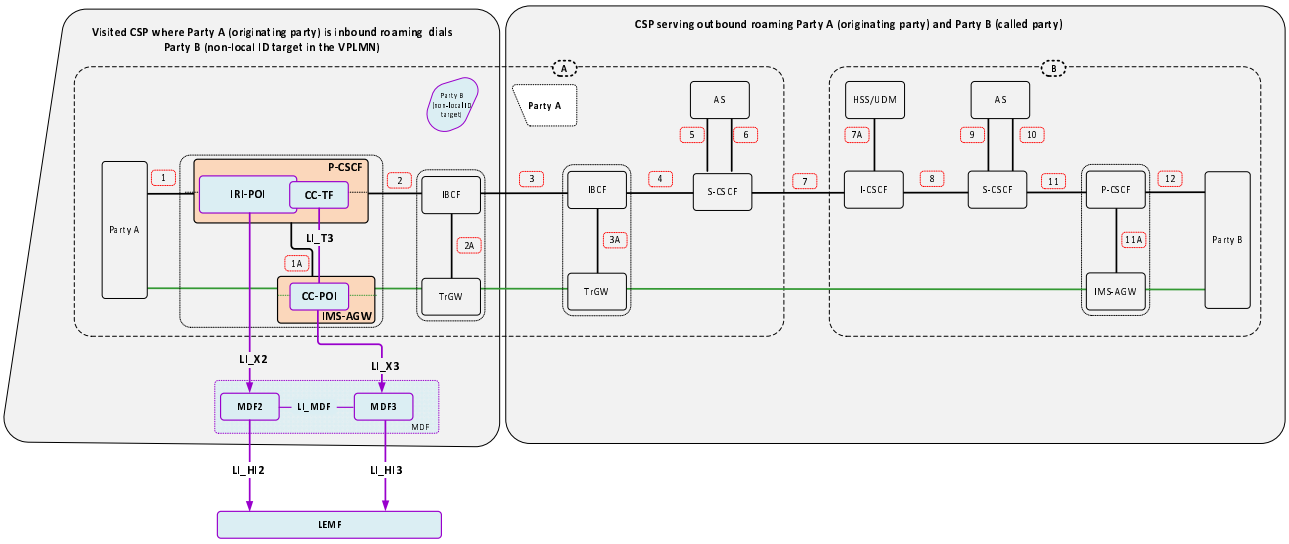


Figure 4.4.2.5-1: Party A (roaming) calls Party B (non-local ID target in VPLMN)

As shown in figure 4.4.2.5-1, the network functions in the VPLMN are the A-side of the session since the Party A is originating the session in the VPLMN. Therefore, even though the target is Party B, the network functions involved in handling the LI functions are part of the A-side of the session.

From a network topology perspective, this is same as the topology described in clause 4.4.2.3. Here, the called party ID is the target identity.

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.2.6 Roaming Party A (target) calls Party B (non-local ID target in VPLMN)

Figure 4.4.2.6-1 shows a scenario where Party A (calling party) is roaming and Party B (called party) is not roaming. In the scenario illustrated, the VPLMN has an active intercept on Party A. The VPLMN also has an active intercept on Party B as non-local ID target.

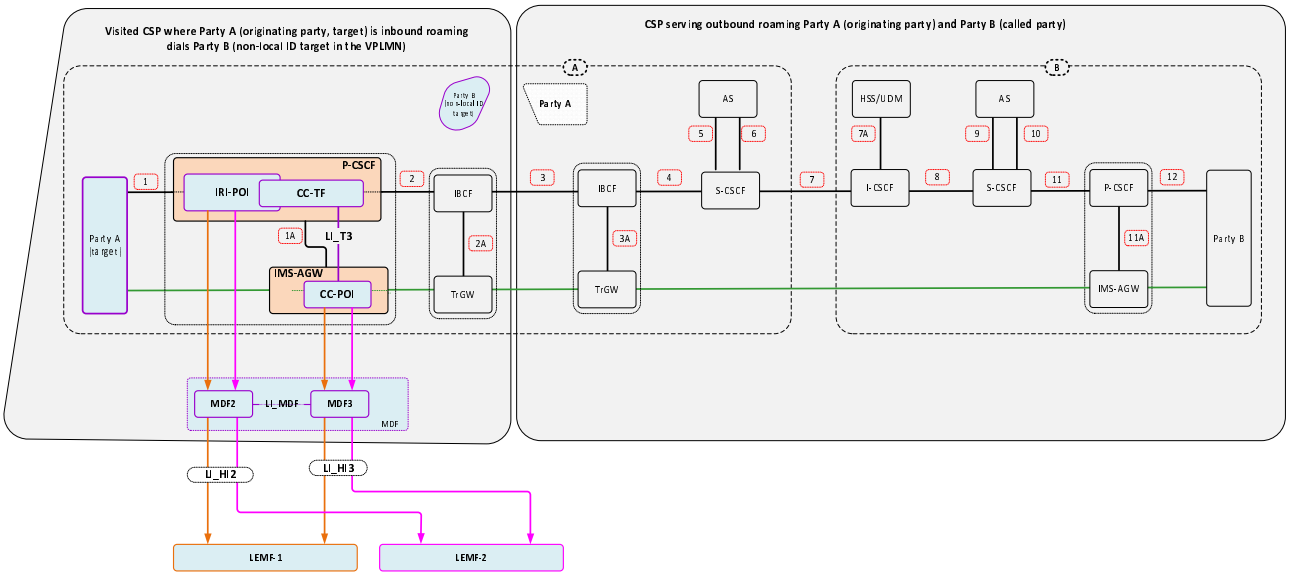


Figure 4.4.2.6-1: Party A (roaming, target) calls Party B (non-local ID target in VPLMN)

As shown in figure 4.4.2.6-1, the network functions in the VPLMN are the A-side of the session since the Party A is originating the session in the VPLMN. Therefore, even though the target is Party B, the network functions involved in handling the LI functions for Party B as a non-local ID are part of the A-side of the session. Since the Party A is also a target for another intercept, the same network functions handle the LI functions for that intercept as well.

From a network topology perspective, this is same as the topology described in clause 4.4.2.3. Here, both calling party ID as well as the called party ID are the target identities.

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-1 (active intercept on Party A) and LEMF-2 (active intercept on Party B as non-local ID target). The MDF3 generates the CC from the received xCC and delivers it to the LEMF-1 (active intercept on Party A) and LEMF-2 (active intercept on Party B as non-local ID target).

NOTE: To illustrate the point that two separate identities are used in filtering the target identity, two separate lines shown for LI_X2 and LI_X3 from POIs to the MDF2 and two separate LI_HI2 and LI_H3 lines are shown from MDFs to LEMFs.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.3 Roaming at the terminating end

4.4.3.1 Introduction

In these scenarios, the session is terminated in the VPLMN, i.e. the terminating party is roaming.

4.4.3.2 Party A calls roaming Party B (target in HPLMN)

Figure 4.4.3.2-1 shows a scenario where Party B (called party) is roaming and Party A (calling party) is not roaming. In the scenario illustrated, the HPLMN has an active intercept on Party B, who is outbound roaming.

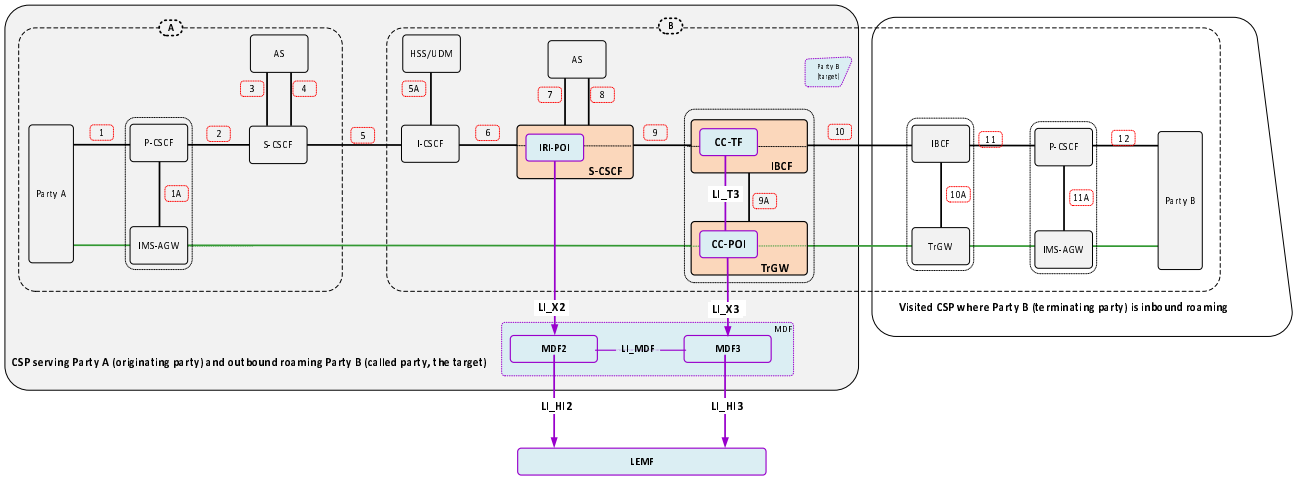


Figure 4.4.3.2-1: Party A calls Party B (roaming, target in HPLMN)

As shown in figure 4.4.3.2-1, only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.3.3 Party A calls roaming Party B (target in VPLMN)

Figure 4.4.3.3-1 shows a scenario where Party B (called party) is roaming and Party A (calling party) is not roaming. In the scenario illustrated, the VPLMN has an active intercept on Party B, who is inbound roaming.

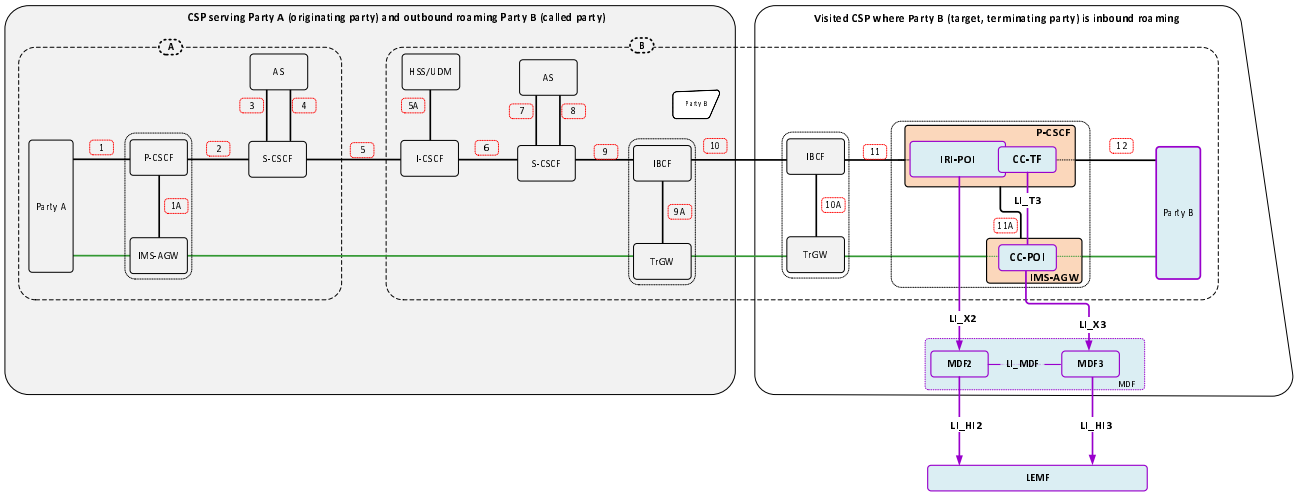


Figure 4.4.3.3-1: Party A calls Party B (roaming, target in VPLMN)

As shown in figure 4.4.3.3-1, only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions. Note that the network functions in the VPLMN are considered to be the B-side of the session since session is terminated to the Party B in the VPLMN.

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.3.4 Party A calls roaming Party B (target in VPLMN and HPLMN)

Figure 4.4.3.4-1 shows a scenario where Party B (called party) is roaming and Party A (calling party) is not roaming. In the illustrated scenario, HPLMN has an active intercept on Party B, who is outbound roaming and VPLMN also has an active intercept on Party B who is inbound roaming.

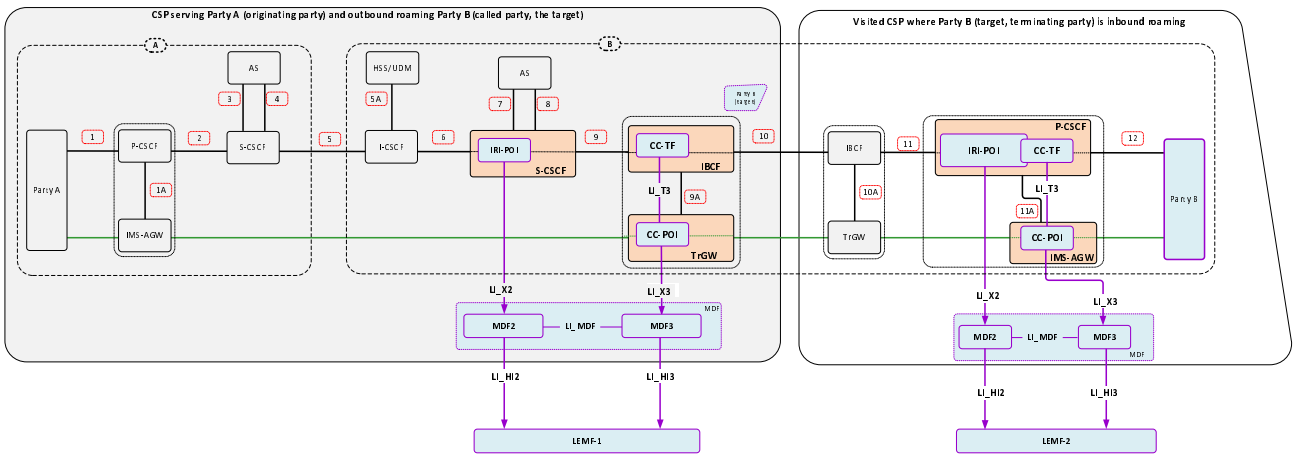


Figure 4.4.3.4-1: Party A calls Party B (roaming, target in VPLMN, target in HPLMN)

As shown in figure 4.4.3.4-1, there are two intercepts active on Party B.- one in HPLMN and one in VPLMN, Only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions. Note that the network functions in the VPLMN are considered to be the B-side of the session since the session is terminated to the Party A in the VPLMN.

Interception in the HPLMN

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-1. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-1.

Interception in the VPLMN

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-2. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-2.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.3.5 Party A (non-local ID target in VPLMN) calls roaming Party B

Figure 4.4.3.5-1 shows a scenario where Party B (called party) is roaming and Party A (calling party) is not roaming. In the scenario illustrated, the VPLMN has an active intercept on Party A as a non-local ID target.

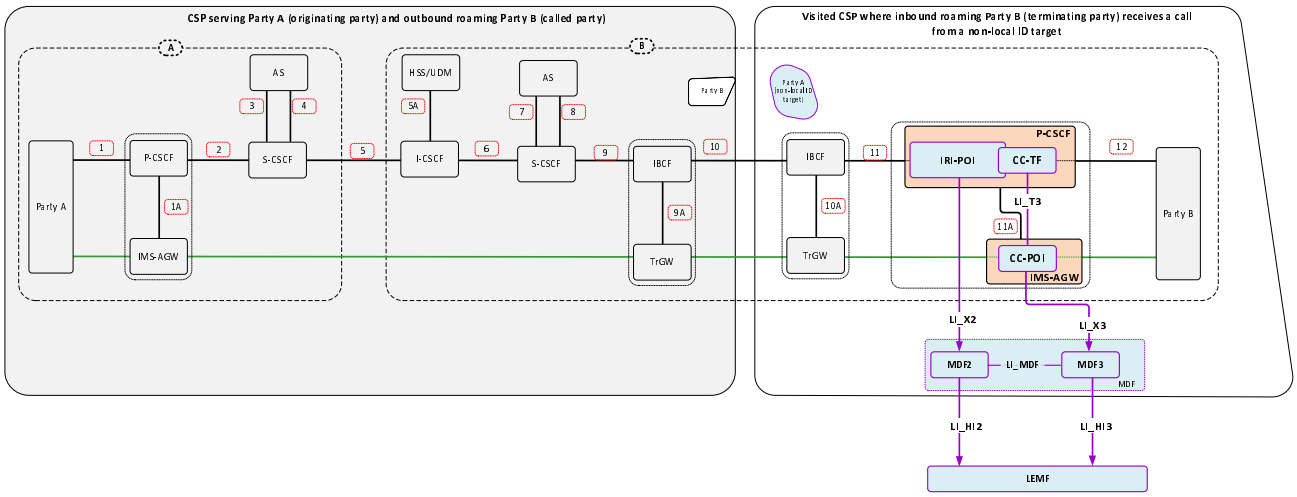


Figure 4.4.3.5-1: Party A (non-local ID target in VPLMN) calls Party B (roaming)

As shown in figure 4.4.3.5-1, the network functions in the VPLMN are the B-side of the session since the Party B is a terminating party of the session. Therefore, even though the target is Party A, the network functions involved in handling the LI functions are part of the B-side of the session.

From a network topology perspective, this is same as the topology described in clause 4.4.3.3. Here, the calling party ID is the target identity.

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.3.6 Party A (non-local ID target in VPLMN) calls roaming Party B (target)

Figure 4.4.3.6-1 shows a scenario where Party B (called party) is roaming and Party A (calling party) is not roaming. In the scenario illustrated, the VPLMN has an active intercept on Party B. The VPLMN also has an active intercept on Party A, a non-local ID target.

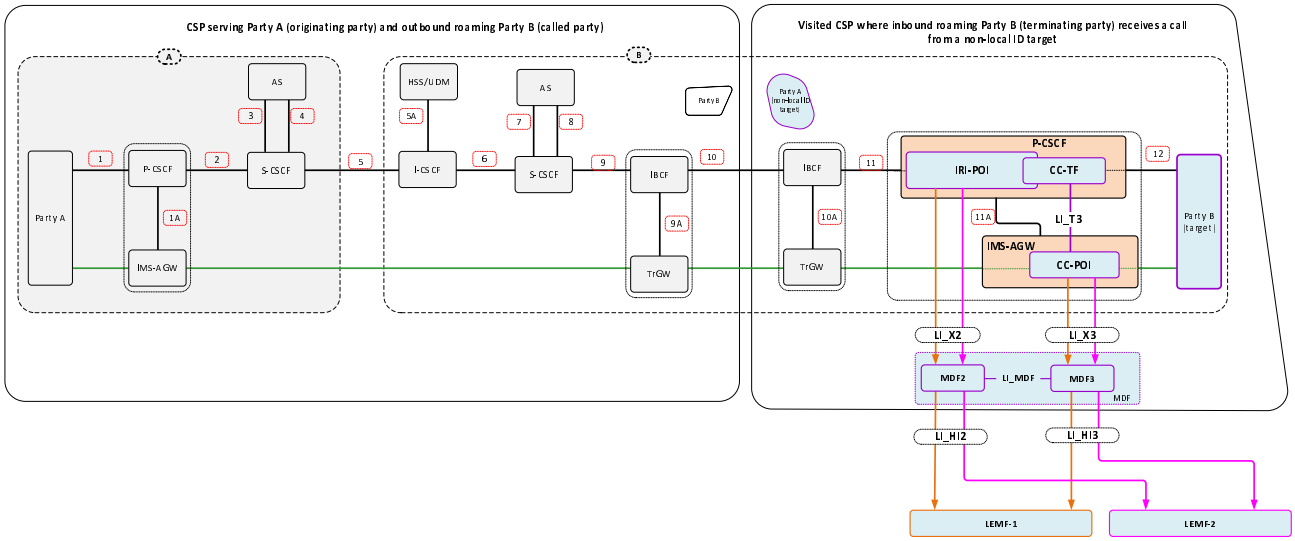


Figure 4.4.3.6-1: Party A (roaming, target) calls Party B (non-local ID target in VPLMN)

As shown in figure 4.4.3.6-1, the network functions in the VPLMN are the B side of the session since the Party B is terminating party of the session. Therefore, even though the target is Party A, the network functions involved in handling the LI functions for Party A as a non-local ID are part of the B-side of the session. Since the Party B is also a target for another intercept, the same network functions handle the LI functions for that intercept as well.

From a network topology perspective, this is same as the topology described in clause 4.4.3.3. Here, both calling party ID as well as the called party ID are the target identities.

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-1 (active intercept on Party B) and LEMF-2 (active intercept on Party A as non-local ID target). The MDF3 generates the CC from the received xCC and delivers it to the LEMF-1 (active intercept on Party B) and LEMF-2 (active intercept on Party A as non-local ID target).

NOTE: To illustrate the point that two separate identities are used in filtering the target identity, two separate lines are shown for LI_X2 and LI_X3 from POIs to the MDF2 and two separate LI_HI2 and LI_H3 lines are shown from MDFs to LEMFs.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.4 Roaming with redirections

4.4.4.1 Introduction

In these scenarios, the session is redirected and either the redirecting party, or the redirected to party, or both are roaming.

NOTE: For redirecting scenarios other than the do-not-answer case, the VPLMN where the redirecting party is inbound roaming has no role to play in the LI even if that VPLMN has an active intercept on that redirecting party, since none of the network functions of that VPLMN are involved in handling the redirected session.

Special cases, where the redirecting party is roaming and redirected-to-party in another CSP domain are also included.

4.4.4.2 Party A calls Party B (target) redirected to roaming Party C

Figure 4.4.4.2-1 shows a scenario Party C (redirected-to party) is roaming. Party A (calling party) and Party B (initial called party, redirecting party) are not roaming. Party B has setup to redirect all incoming calls to Party C. Party B is the target.

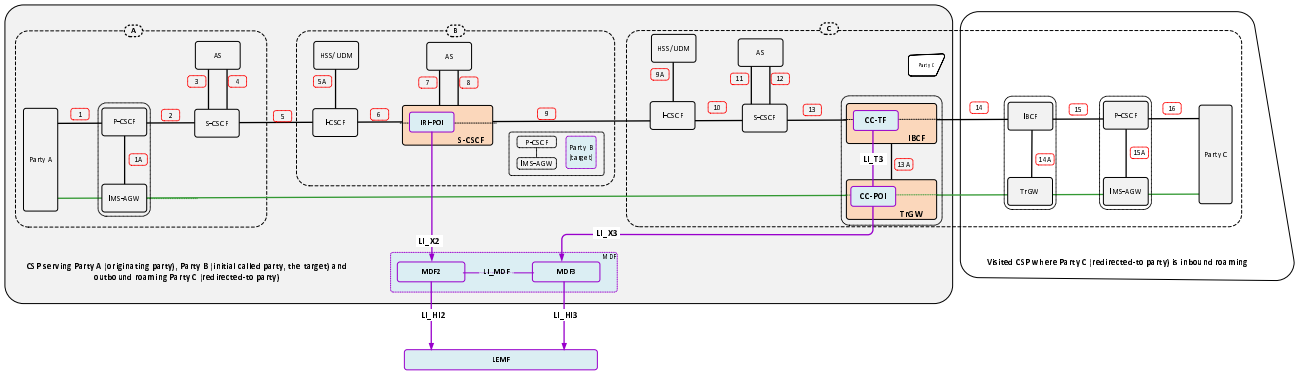


Figure 4.4.4.2-1: Party A calls Party B (non-roaming, target) redirected to Party C (roaming)

As shown in figure 4.4.4.2-1, only the network functions that handle the B-side (target side) and C-side (redirected-to party side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.4.3 Party A calls roaming Party B (target) redirected to Party C

Figure 4.4.4.3-1 shows a scenario Party B (the initial called party, redirecting party) is roaming. Party A (calling party) and Party C (redirected-to party) are not roaming. Party B has setup to redirect all incoming calls to Party C. Party B is the target.

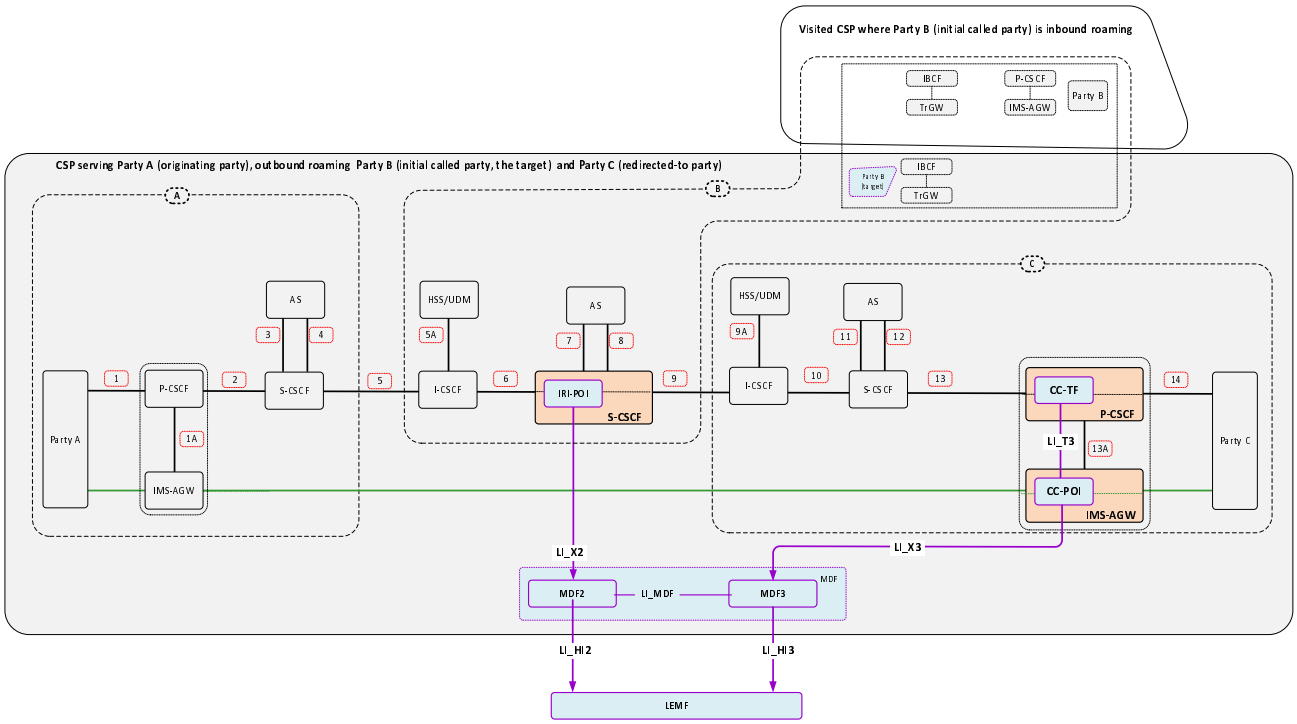


Figure 4.4.4.3-1: Party A calls Party B (roaming, target) redirected to Party C (non-roaming)

As shown in figure 4.4.4.3-1, only the network functions that handle the B-side (target side) and C-side (redirected-to party side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.4.4 Party A calls roaming Party B (target) redirected to Party C (roaming)

Figure 4.4.4.4-1 shows a scenario where Party B (the initial called party, redirecting party) and Party C (redirected-to party) are roaming. Party A (calling party) is not roaming. Party B has setup to redirect all incoming calls to Party C. Party B is the target. In this example, Party B and Party C are roaming in two different VPLMNs.

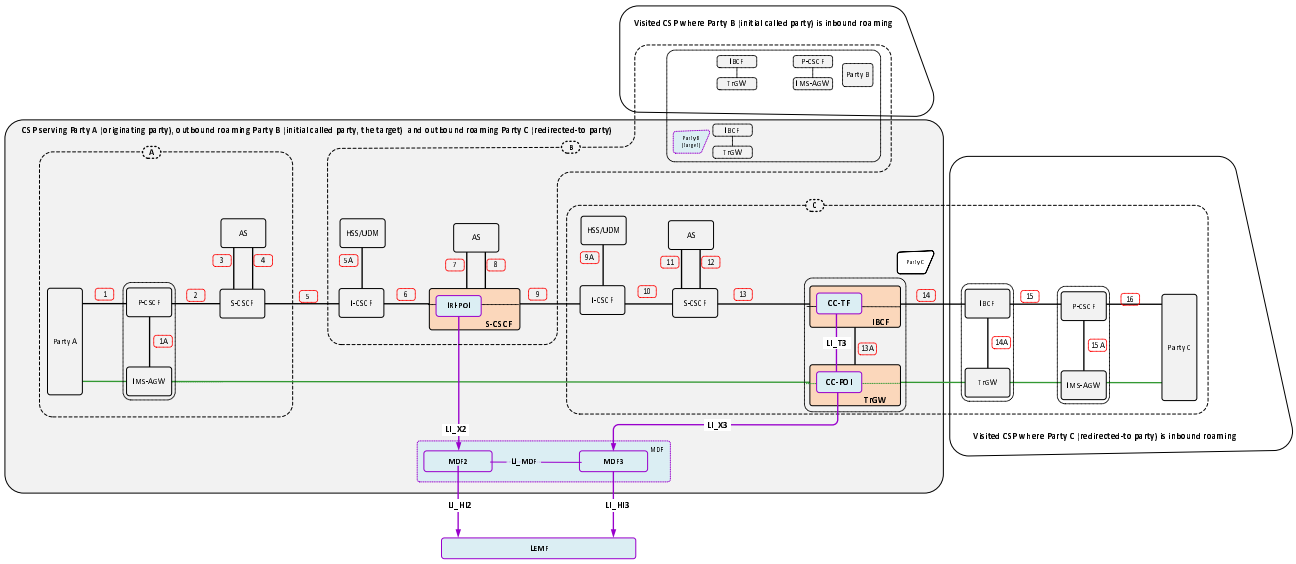


Figure 4.4.4.4-1: Party A calls Party B (roaming, target) redirected to Party C (roaming)

As shown in figure 4.4.4.4-1, only the network functions that handle the B-side (target side) and C-side (redirected-to party side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.4.5 Party A calls roaming Party B (target) redirected to Party C (CS domain)

Figure 4.4.4.5-1 shows a scenario where Party B (the initial called party, redirecting party) is roaming and Party C (redirected-to party) in a CS domain (i.e. another CSP domain). Party A (calling party) is not roaming. Party B has setup to redirect all incoming sessions to Party C. Party B is the target.

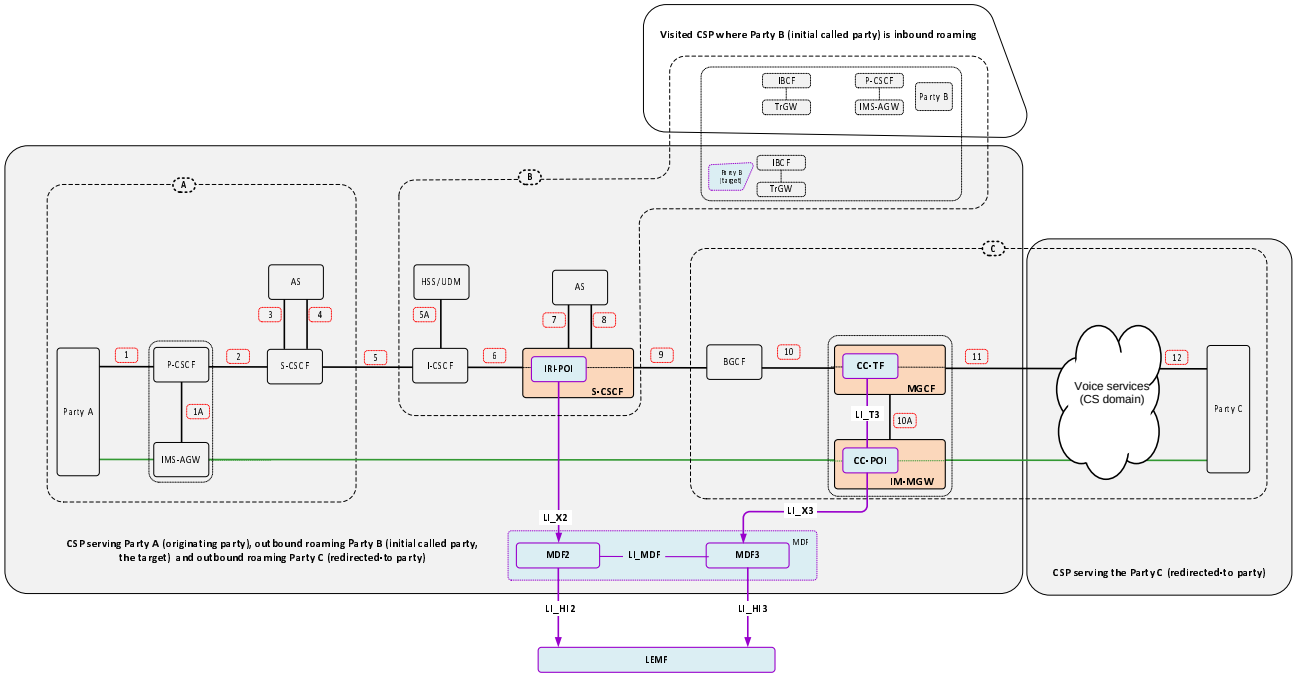


Figure 4.4.4.5-1: Party A calls Party B (roaming, target) redirected to Party C (CS domain)

As shown in figure 4.4.4.5-1, only the network functions that handle the B-side (target side) and C-side (redirected-to party side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF triggers the CC-POI present in the IM-MGW for the xCC. The CC-POI present in the IM-MGW generates the xCC.

NOTE: Since only the ingress signalling at the MGCF is SIP-based, this architecture assumes that the interception is done using the SIP messages seen at the step 10.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.4.6 Party A calls roaming Party B (target) redirected to Party C (IP domain)

Figure 4.4.4.6-1 shows a scenario where Party B (the initial called party, redirecting party) is roaming and Party C (redirected-to party) is an IP domain (i.e. another CSP domain). Party A (calling party) is not roaming. Party B has setup to redirect all incoming sessions to Party C. Party B is the target.

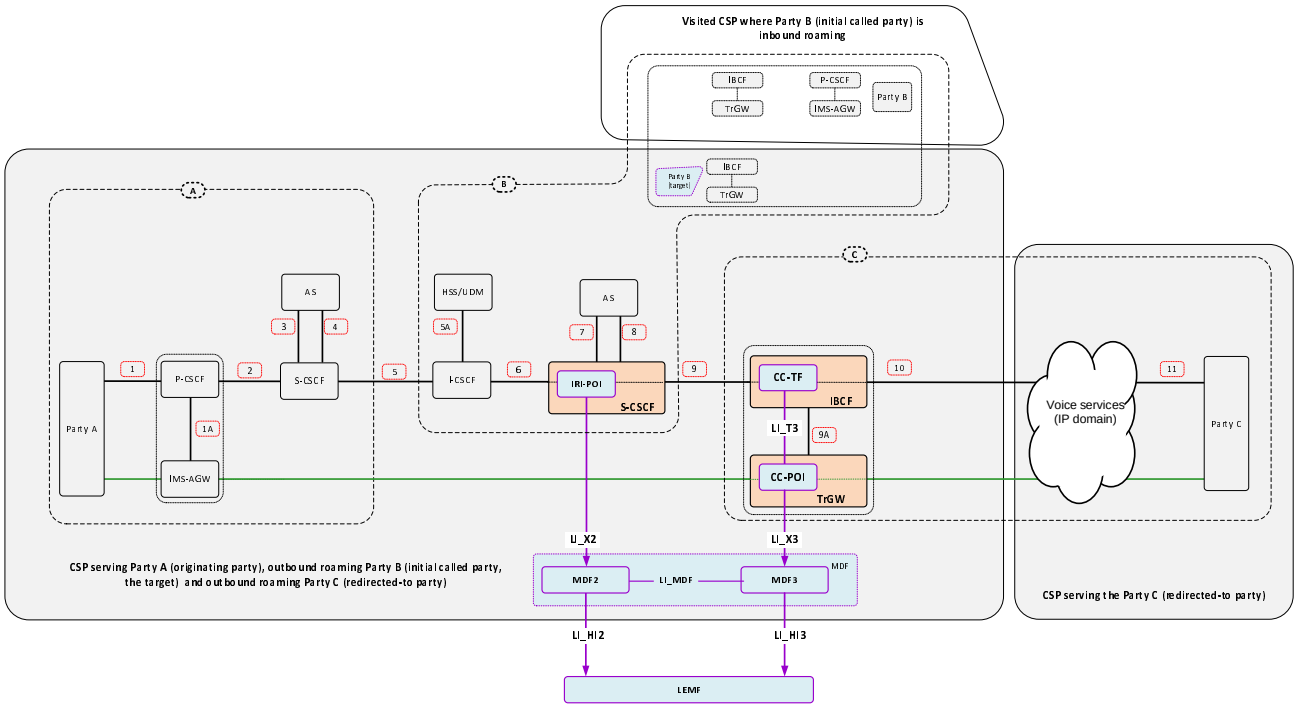


Figure 4.4.4.6-1: Party A calls Party B (roaming, target) redirected to Party C (IP domain)

As shown in figure 4.4.4.6-1, only the network functions that handle the B-side (target side) and C-side (redirected-to party side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.4.7 Party A (non-local ID target in VPLMN) calls Party B redirected to roaming Party C

Figure 4.4.4.7-1 shows a scenario where Party C (redirected-to party) is roaming and Party A (calling party) and Party B (redirecting party) are not roaming. In the scenario illustrated, the VPLMN has an active intercept on Party A as a non-local ID target.

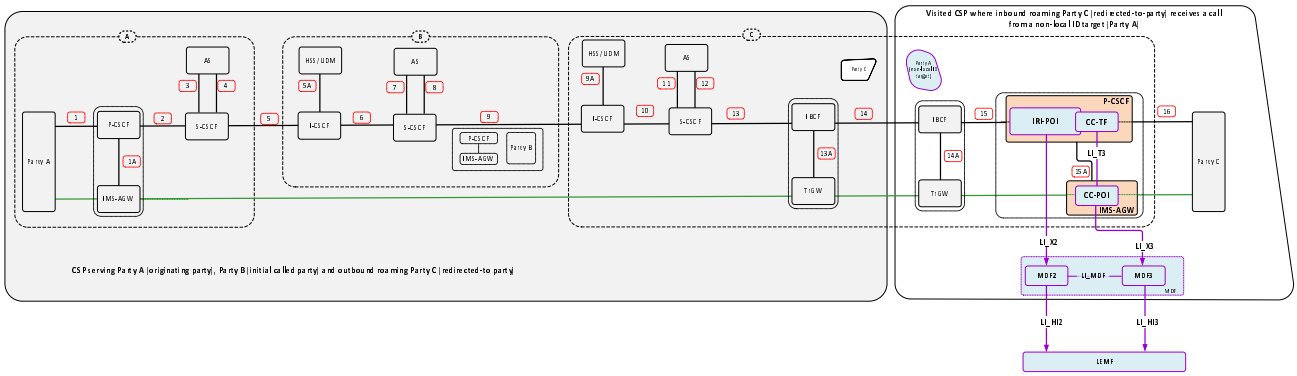


Figure 4.4.4.7-1: Party A (non-local ID target in VPLMN) calls Party B redirected to Party C (roaming)

As shown in figure 4.4.4.7-1, the network functions in the VPLMN are the C-side of the session since the Party C is a redirected-to party of the session. Therefore, even though the target is Party A, the network functions involved in handling the LI functions are part of the C-side of the session.

From a network topology perspective, this is same as the topology described in clause 4.4.3.3. Here, the session is redirected and the calling party ID is the target identity.

NOTE: In the VPLMN, the Party B (the redirecting party) can also be non-local ID target and in that case, the network topology will be same as shown in figure 4.4.4.7-1 with redirecting party ID as the target identity.

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.4.4.8 Party A (non-local ID target in VPLMN) calls Party B (non-local ID target in VPLMN) redirected to roaming Party C (target)

Figure 4.4.4.8-1 shows a scenario where Party C (redirected-to party) is roaming and Party A (calling party) and Party B (redirecting party) are not roaming. In the scenario illustrated, the VPLMN has three active intercepts on Party A as a non-local ID target, Party B as a non-local ID target and Party C as a target.

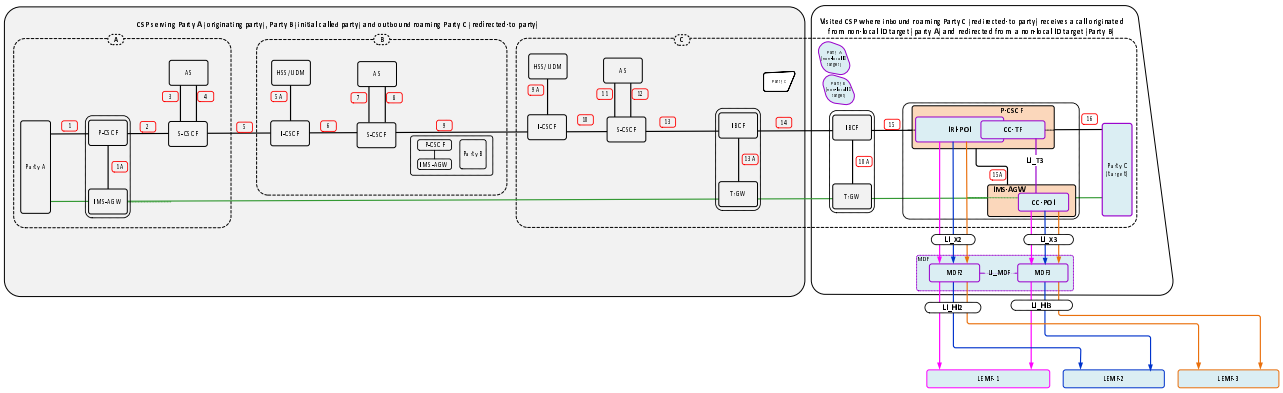


Figure 4.4.4.8-1: Party A (roaming, target) calls Party B (non-local ID target in VPLMN)

As shown in figure 4.4.4.8-1, the network functions in the VPLMN are the C-side of the session since the Party C is redirecting party of the session. Therefore, even though the targets are Party A and Party B, the network functions involved in handling the LI functions for Party A as a non-local ID and Party B as non-local ID targets are part of the B-side of the session. Since the Party C is also a target for another intercept, the same network functions handle the LI functions for that intercept as well.

From a network topology perspective, this is same as the topology described in clause 4.4.3.3. Here, calling party ID, redirecting party ID and called party ID are the target identities.

The IRI-POI present in the P-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-1 (active intercept on Party C) and LEMF-2 (active intercept on Party A as non-local ID target) and LEMF-3 (active intercept on Party Bas non-local ID target). The MDF3 generates the CC from the received xCC and delivers it to the LEMF-1 (active intercept on Party C) and LEMF-2 (active intercept on Party A as non-local ID target) and LEMF-3 (active intercept on Party V as non-local ID target).

NOTE: To illustrate the point that three separate identities are used in filtering the target identity, three separate lines are shown for LI_X2 and LI_X3 from the POIs to the MDF2 and three separate LI_HI2 and LI_H3 lines are shown from the MDFs to the LEMFs.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5 IMS based conferencing

4.5.1 General

Clause 4.5 illustrates the IMS LI for various conferencing scenarios.

The 3GPP TS 24.147 [5] defines procedures for invoking the conference, mainly, based on the network configuration, signalling flow and the conferencing types (e.g. ad-hoc conferencing, group conferencing).

A user connected to a conference is referred to as conference participant. In an ad-hoc conference, one of the conference participants initiates the conference. In a group conferencing, each conference participant basically has the same role.

In the presented illustrations, a conference participant, or a conference URI, can be the target. The possibility of the conference server associated with the conference URI present in a different network (i.e. different CSP domain) is also considered in the illustrations.

4.5.2 Ad-hoc conferencing

4.5.2.1 Introduction

In the illustrations presented here, a session that involves two parties is extended to a conferencing session by one of the parties involved in that session and one of the parties involved in the conference session is a target. A typical example of an ad-hoc conferencing is a three-way conferencing.

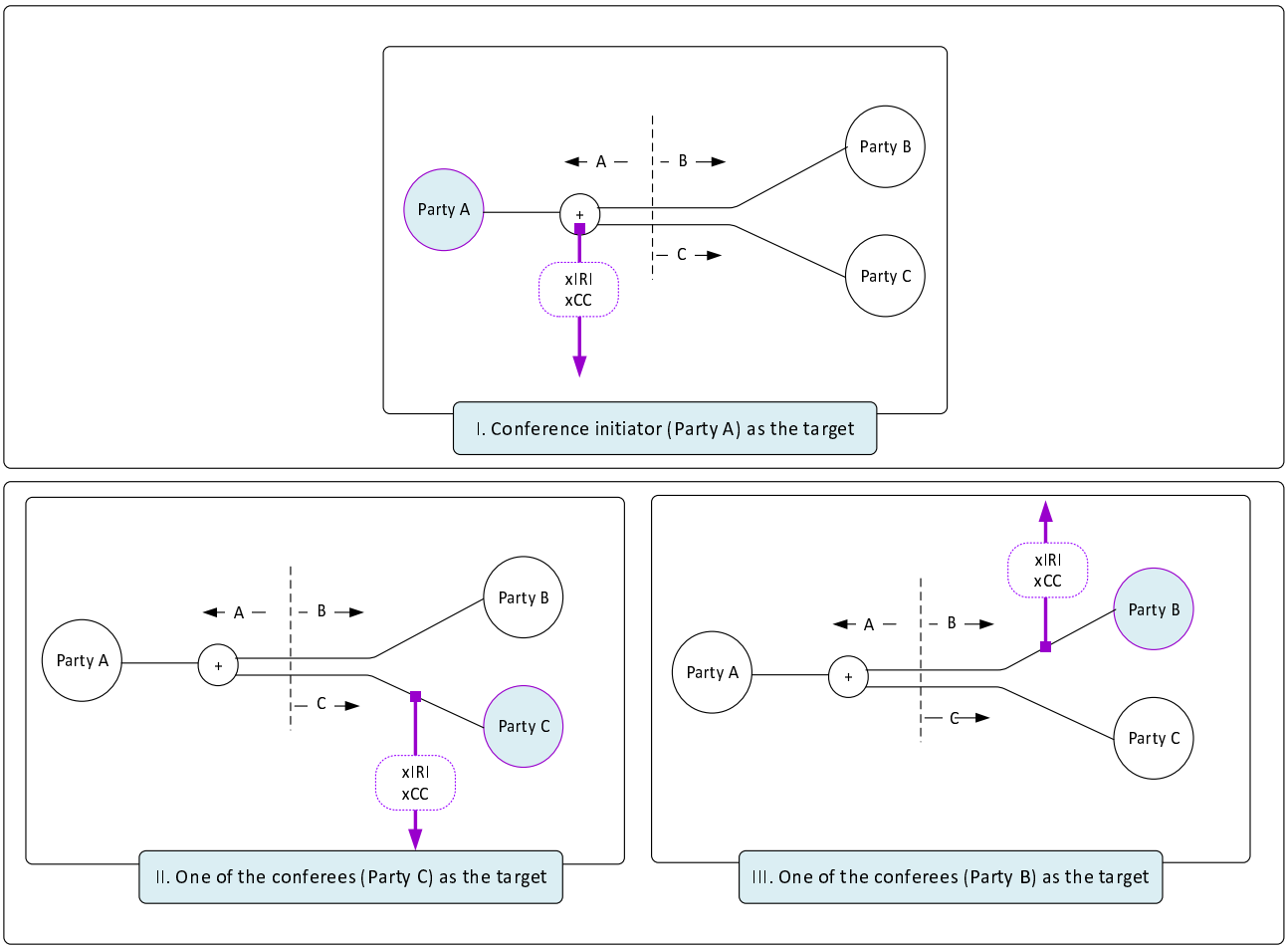


Figure 4.5.2.1-1: Ad-hoc conferencing scenarios

In each of the illustrations, Party A is conference initiator. Party B and Party C are the other two conference participants (i.e. conferees). Any of the three can be the target.

NOTE: TS 24.147 [5], defines different methods that can be used to establish an ad-hoc conferencing session between the parties involved in the conference.

The illustrations assume that the conference server is present in the same CSP domain as that of the conference initiator. Furthermore, the illustrations assume that the Application Server (AS), in the role of a conference focus, is different from the AS that is used in the regular voice sessions.

4.5.2.2 Party A (target) initiates a conference with Party B and Party C – case 1

Figure 4.5.2.2-1 shows the scenario where Party A (target) is in an IMS-based conference session with Party B and Party C.

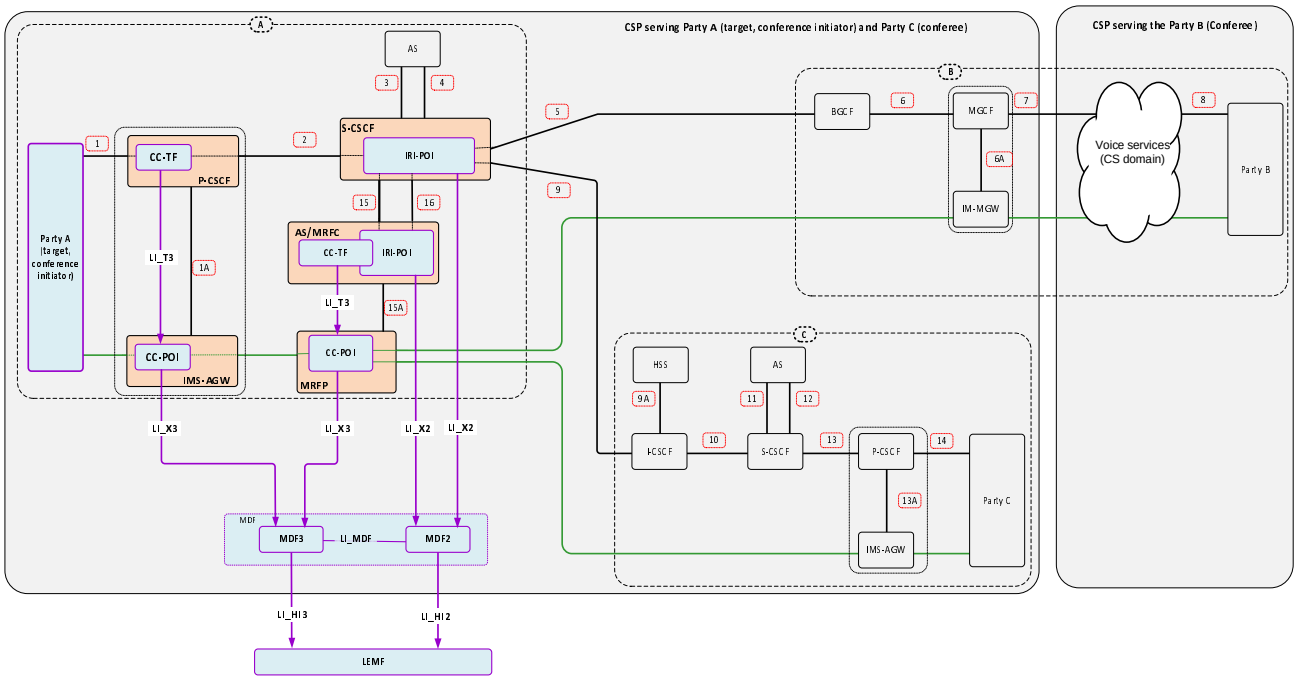


Figure 4.5.2.2-1: Party A (target) in a conference session with Party B and Party C – case 1

As shown in figure 4.5.2.2-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions. Party A and Party C are in the same CSP domain. Party B is in a CS domain.

The session progressive steps 1 to 4 are presumed to be applicable to the multiple sessions depicted in figure 4.5.2.2-1. For example, for the scenario where the Party A establishes a session to Party B, and then to Party C and then invokes the conference to merge the two sessions, the session progressive steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8 as shown in the scenario illustrated in clause 4.3.3.2.
- The IMS session setup to Party C follows the steps, 1 to 4 and the steps 9 to 14.
- The conference invoking (before inviting Party B and Party C into the conference) follows the steps 1 to 4 and the steps 15 to 16.

Before the conference:

- When Party A is in an independent session with either of the two parties (Party B or Party C), the IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI.
- The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

During the conference:

- The IRI-POI present in the S-CSCF continues to access the SIP messages and generate the required xIRI. The IMS-AGW continues to generate the xCC.
- In addition, the IRI-POI present in the AS/MRFC (conference focus), generates the xIRI related to the conference session. The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC related to the conference session. The MRFP generates the xCC for the conference session.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The xCC from the CC-POI present in the IMS-AGW includes the communication content of Party A (sent and received). The xCC from the CC-POI present in the MRFP includes the communication content from the conference. When all three parties are connected to the conference, the xCC from IMS-AGW and xCC from the MRFP provide the

same information. Either way, the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C.

NOTE 1: The Party A may invite additional parties into the conference. The network topology providing the LI functions for such scenarios is same as the one shown in figure 4.5.2.2-1 with the other parties connected to the conference as conference participants.

NOTE 2: The use of the method inferred by the topology illustrated in figure 4.5.2.2-1 may be applicable when the delivery of communication content of held conference session is required.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5.2.3 Party A (target) initiates a conference with Party B and Party C – Case 2

Figure 4.5.2.3-1 shows an alternative illustration for the scenario where Party A (target) is in an IMS-based conference session with Party B and Party C. In this alternative, the CC-POI functions are limited to the IMS-AGW. This scenario can only be used when the communication content of held conference session is not required.

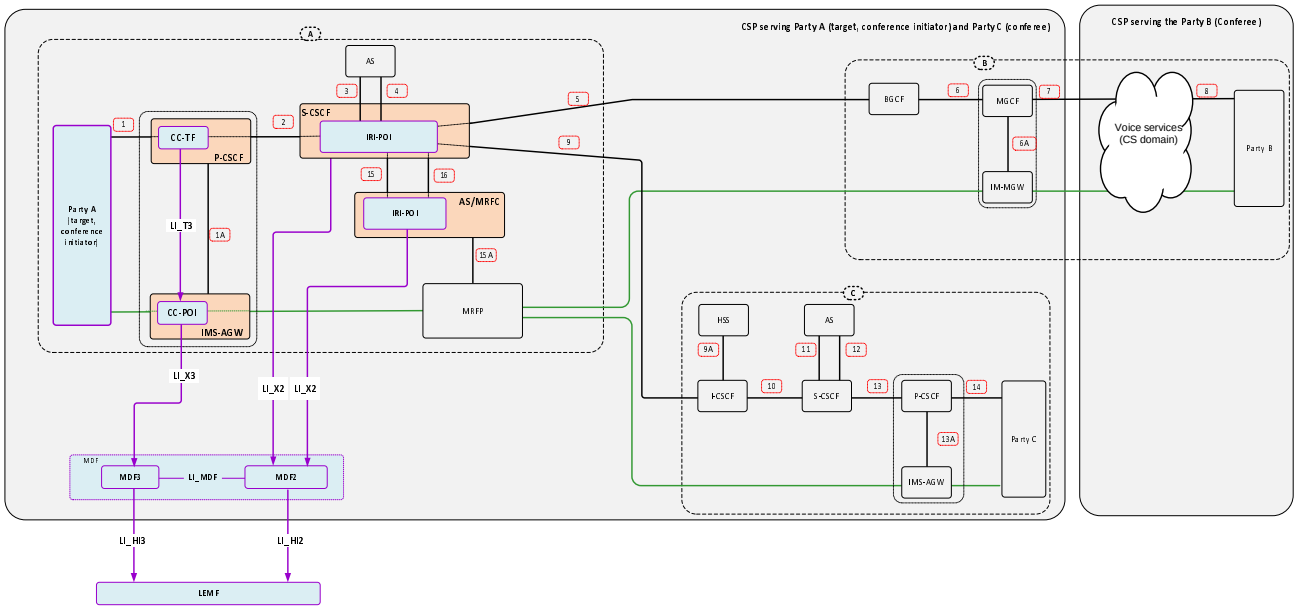


Figure 4.5.2.3-1: Party A (target) in a conference session with Party B and Party C – case 2

The illustration shown in figure 4.5.2.3-1 differs from the illustration shown in figure 4.5.2.2-1 in the way the CC interception is performed. In this illustration, the CC-POI functions are provided only by the IMS-AGW even during the conference session.

The IMS session establishment procedures are same as described in clause 4.5.2.2.

The CC-TF present in the P-CSCF provides the trigger for the CC-POI present in the IMS-AGW for the xCC and the CC-POI present in the IMS-AGW generates the xCC. An MRFP based CC-POI is not used in this alternative.

The generation of xIRI is same as shown in figure 4.5.2.2-1 and as described in clause 4.5.2.2.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The xCC received from the CC-POI present in the IMS-AGW includes the communication content of Party A (sent and received). Therefore, the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5.2.4 Party A (target) initiates a conference with Party B and Party C - Alternative

Figure 4.5.2.4-1 shows another alternative illustration to depict the scenario where Party A (target) is in an IMS-based conference session with Party B and Party C. In this alternative, the CC-POI functions are limited to the MRFP.

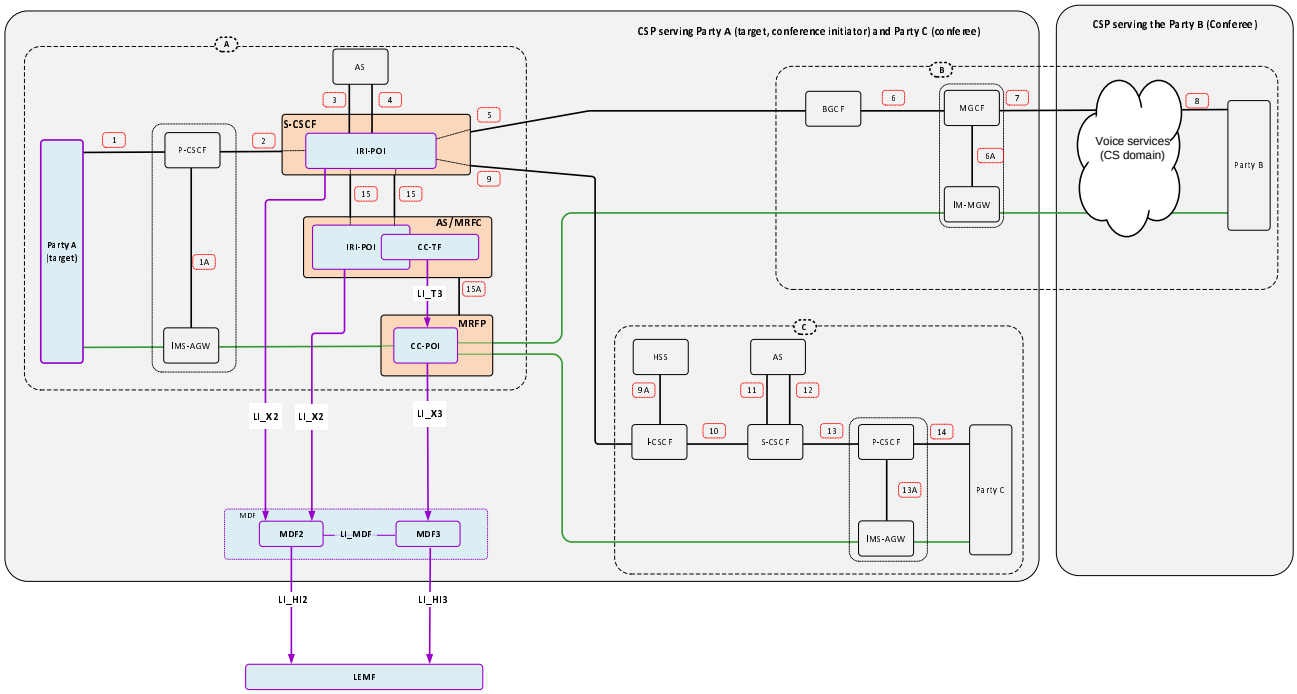


Figure 4.5.2.4-1: Party A (target) in a conference session with Party B and Party C – alternative

Figure 4.5.2.4-1 shows an alternative to figure 4.5.2.2-1 as far as the CC interception is concerned. In this alternative illustration, the CC-POI functions are provided only by the MRFP during the conference session.

NOTE 1: The use of this method, however, requires additional processing when a conference is placed on hold and if the communication content of a held conference session is not to be delivered.

The IMS session establishment procedures are same as described in clause 4.5.2.2.

The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC and the MRFP generates the xCC. An IMS-AGW based CC-POI is not used in this alternative.

NOTE 2: The method used to detect and stop the CC-POI present in the IMS-AGW from generating the xCC is outside the scope of this TR.

The generation of xIRI is same as shown in figure 4.5.2.2-1 and as described in clause 4.5.2.2.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers the same to the LEMF.

The xCC from the CC-POI present in the MRFP includes the communication content from the conference. In the illustration, the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5.2.5 Party A initiates a conference with Party B and Party C (target)

Figure 4.5.2.5-1 shows the scenario where Party A is in an IMS-based conference session with Party B and Party C. In this illustration, Party C is the target.

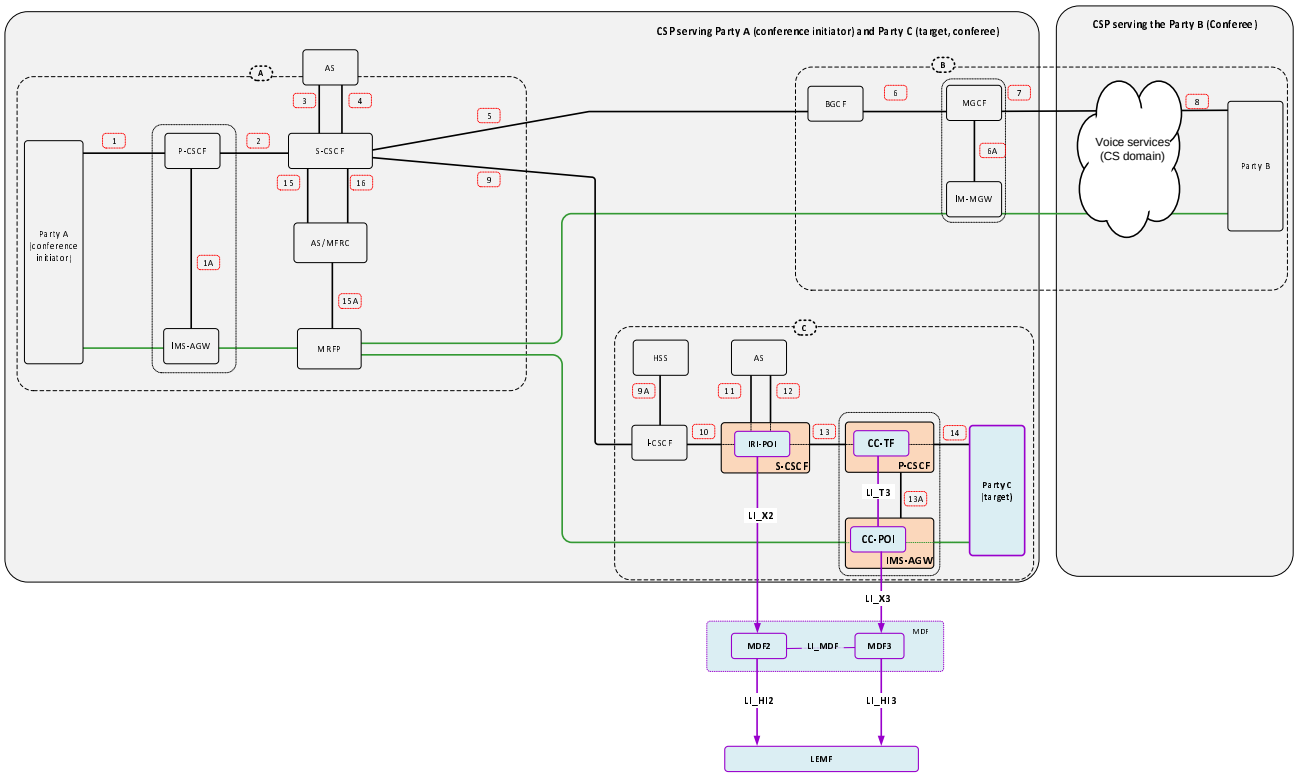


Figure 4.5.2.5-1: Party A in a conference session with Party B and Party C (target)

As shown in figure 4.5.2.5-1, only the network functions that handle the C-side (target side) of the session are involved in handling the LI functions. Party A and Party C are in the same CSP domain. Party B is in a CS domain.

The session progressive steps 1 to 4 are presumed to be applicable to the multiple sessions depicted in figure 4.5.2.5-1. For example, for the scenario where the Party A establishes a session to Party B, and then to Party C and then invokes the conference to merge the two sessions, the session progressive steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8 as shown in the scenario illustrated in clause 4.3.3.2. Party B is in a CS domain.
- The IMS session setup to Party C (target) follows the steps, 1 to 4 and the steps 9 to 14.
- The conference invoking (before inviting Party B and Party C into the conference) follows the steps 1 to 4 and the steps 15 to 16.

The IRI-POI present in the S-CSCF (that serves Party C) accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The CC delivered to the LEMF includes the communication content of Party C (sent and received). When all three parties are in conference (as shown), the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5.2.6 Party A initiates a conference with Party B (non-local ID target) and Party C

Figure 4.5.2.6-1 shows the scenario where Party A is in an IMS-based conference session with Party B and Party C. In this illustration, Party B is a non-local ID target.

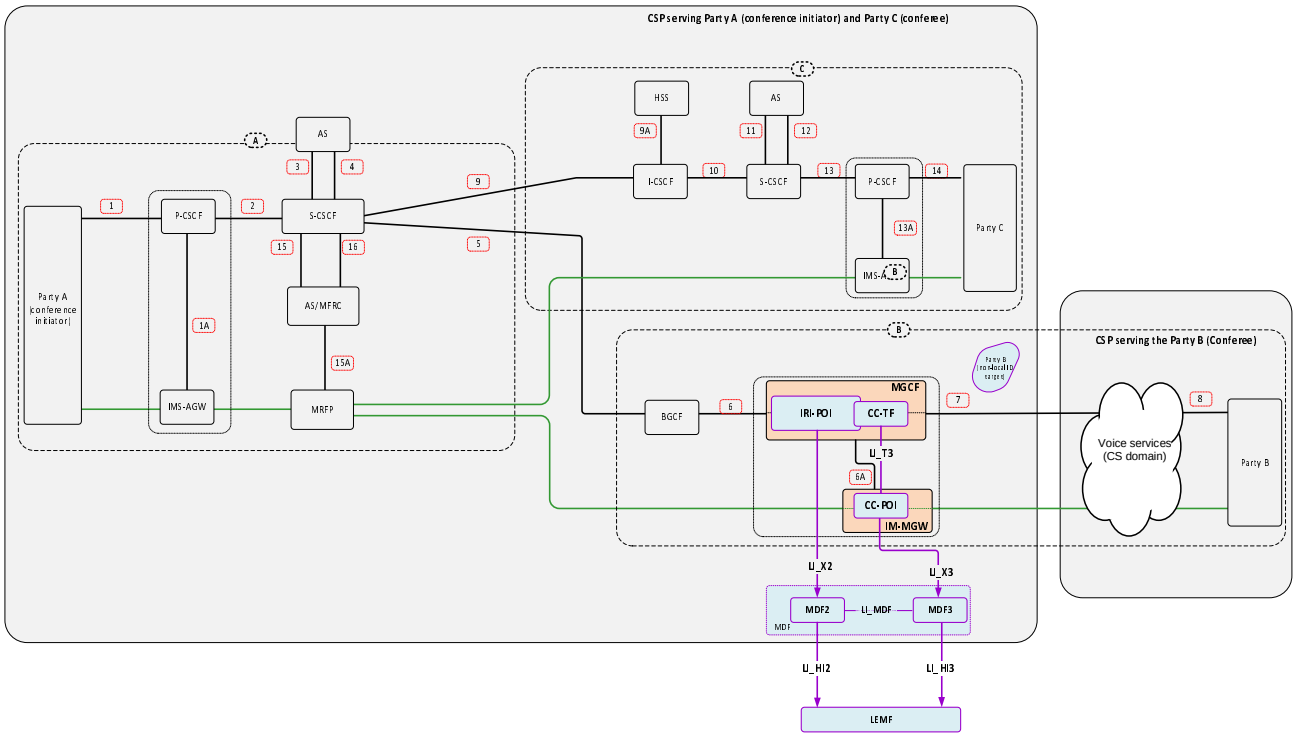


Figure 4.5.2.6-1: Party A in a conference session with Party B (non-local ID target) and Party C

As shown in figure 4.5.2.6-1, only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions. Party A and Party C are in the same CSP domain. Party B is in a CS domain.

The session progressive steps 1 to 4 are presumed to be applicable to the multiple sessions depicted in figure 4.5.2.6-1. For example, for the scenario where the Party A establishes a session to Party B, and then to Party C and then invokes the conference to merge the two sessions, the session progressive steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8 as shown in the scenario illustrated in clause 4.3.3.3. Party B is in a CS domain and is a non-local ID target in the CSP domain serving Party A.
- The IMS session setup to Party C follows the steps, 1 to 4 and the steps 9 to 14.
- The conference invoking (before inviting Party B and Party C into the conference) follows the steps 1 to 4 and the steps 15 to 16. The IRI-POI present in the MGCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF triggers the CC-POI present in the IM-MGW for the xCC. The CC-POI present in the IM-MGW generates the xCC.

NOTE: Since only the ingress signalling at the MGCF is SIP-based, this architecture assumes that the interception is done using the SIP messages seen at the step 6.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The CC delivered to the LEMF includes the communication content of Party B (sent and received). When all three parties are in conference (as shown), the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5.3 Group conferencing

4.5.3.1 Introduction

In the scenarios presented here, the conference participants join a pre-defined conference either independently, or due to an invitation from the conference server or one of the other conference participants. These scenarios apply to group conferencing such as meet-me conferencing.

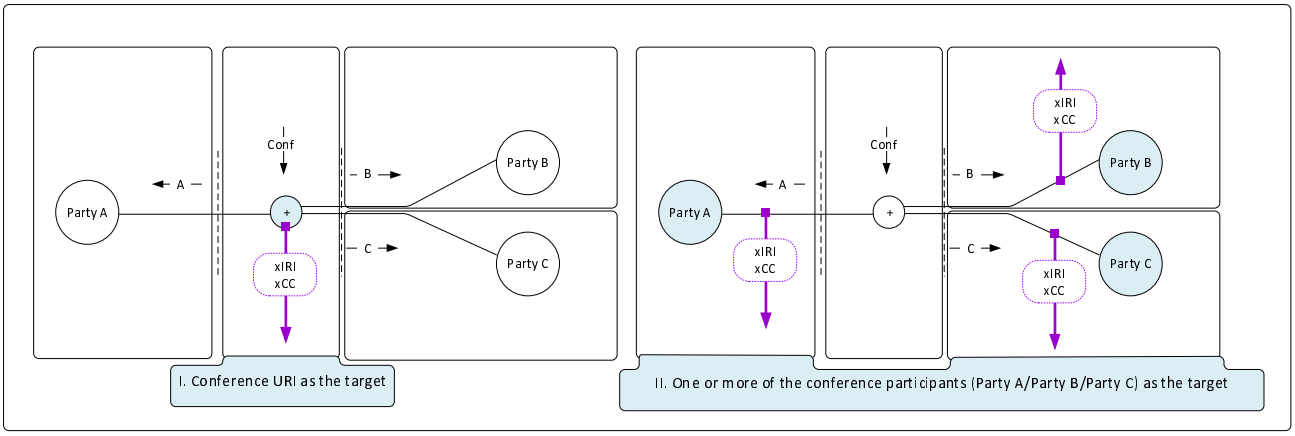


Figure 4.5.3.1-1: Group Conferencing LI scenarios

The illustrations show that either the conference URI, or one or more of the conference participants can be the target.

The conference URI may be provided in a separate CSP domain that does not serve any of the conference participants or provided by a CSP domain that serves one of the conference participants. The SIP signalling flow is presumed to be the same for either of the two cases, which of-course, depends how the conference session is setup.

NOTE: The present document assumes that for group conferencing, the conference server is served by a different CSP domain.

4.5.3.2 Party A, Party B and Party C in a group conference (conference URI target) – case 1

In this illustration, Party A dials into the conference URI, Party B dials into the conference and Party C dials into the conference. Party A and Party B are served by IP domains and Party C is served by a CS domain. The conference server is in a different CSP domain.

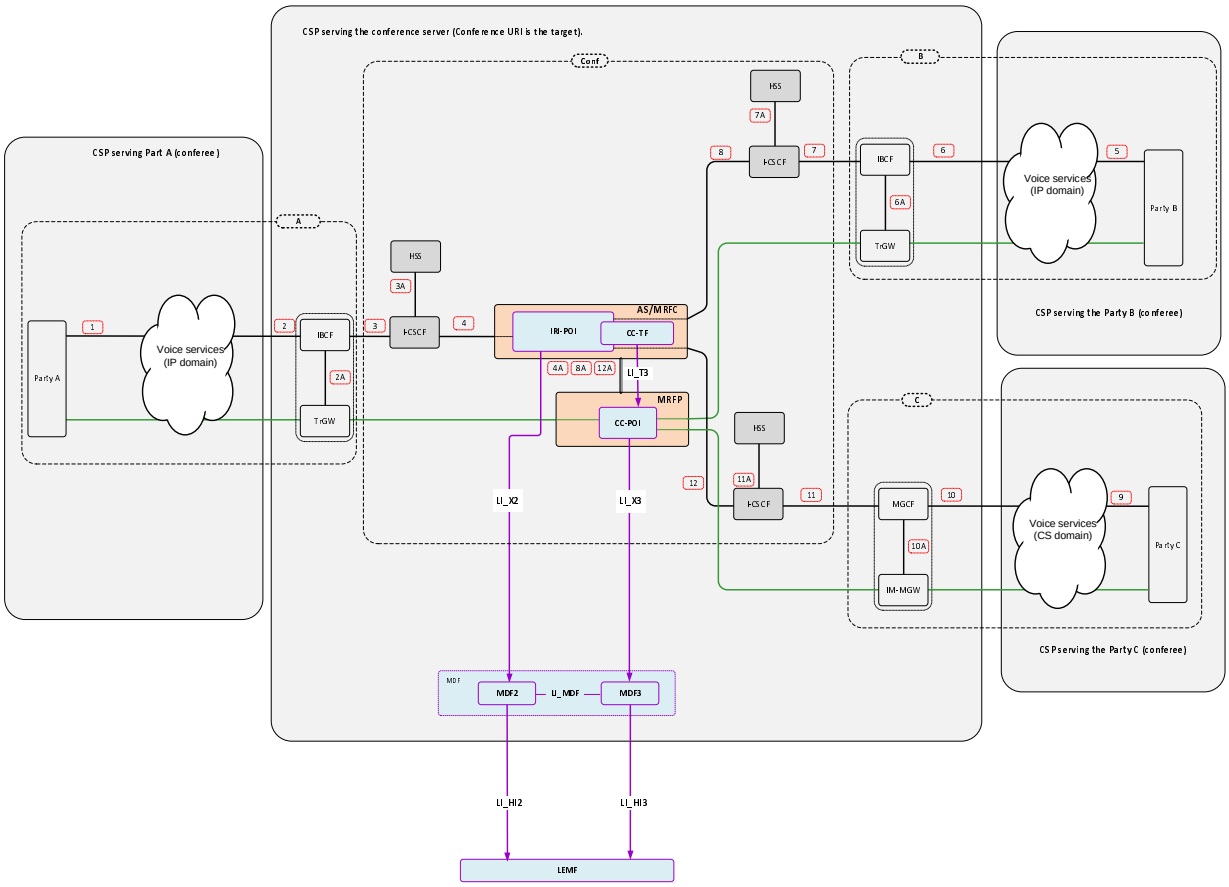


Figure 4.5.3.2-1: Party A, Party B and Party C in group conference (conference URI target) - case 1

NOTE: The same I-CSCF and HSS are used in all of the three session establishments shown above. The diagram shows them separately, for the simplicity of the drawing.

As shown in figure 4.5.3.2-1, the interception is in the CSP domain that serves the conference URI, the target. Party A, Party B and Party C are in different IP domains.

For session progressive steps, the illustration assumes that Party A, Party B and Party C are dialling into the conference individually. The steps should be read as follows:

- Party A dialling into the conference URI (target) follows the steps 1 to 4A.
- Party B dialling into the conference URI (target) follows the steps 5 to 8A.
- Party C dialling into the conference URI (target) follows the steps 9 to 12A.The IRI-POI present in the AS/MRFC accesses the SIP messages and generates the required xIRI. The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC. The CC-POI present in the MRFP delivers the xCC to the MDF3.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF. When all three parties are in the conference, the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5.3.3 Party A, Party B and Party C in a group conference (conference URI target) – case 2

In this illustration, Party A dials into the conference URI, then through the conference server, invites Party B and Party C into the conference. Since Party A invites the other two parties (via the conference server), in this illustration, Party A may also be referred to as conference initiator. The illustration shows the three CSP domains that serves the three conference participants are also IMS domains. The conference server is in a different CSP domain.

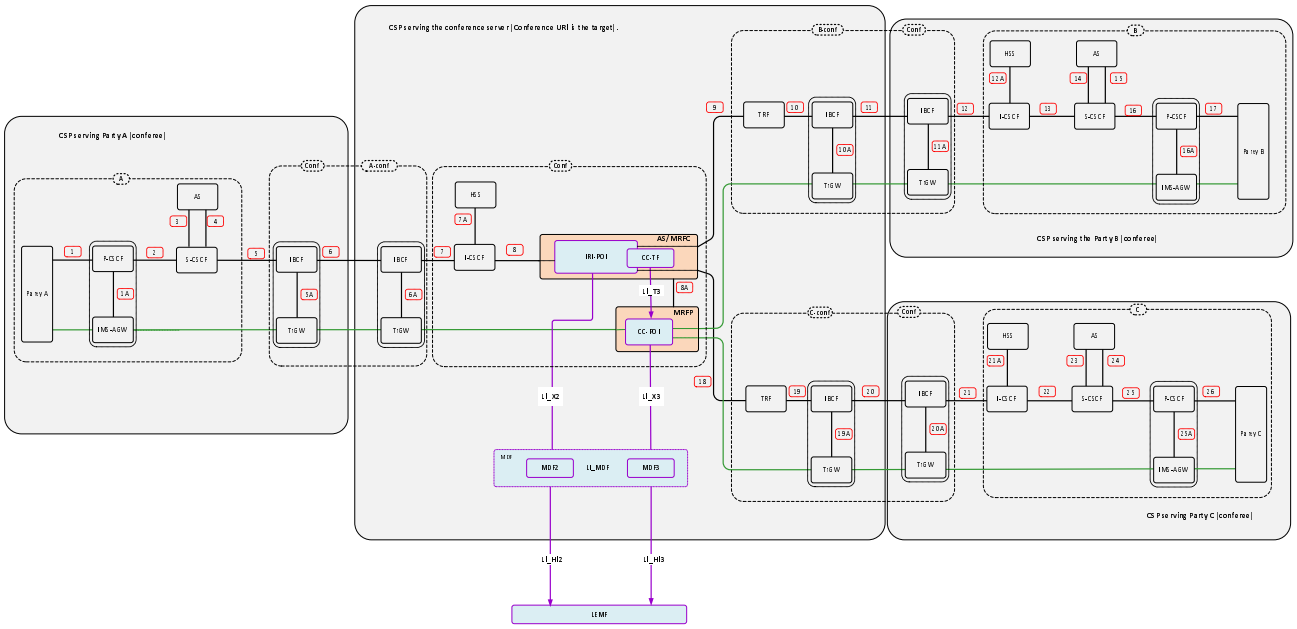


Figure 4.5.3.3-1: Party A, Party B and Party C in group conference (conference URI target) - case 2

As shown in figure 4.5.3.3-1, the interception is in the CSP domain that serves the conference URI, the target. Party A, Party B and Party C are in different IMS domains.

For session progressive steps, the illustration assumes that Party A dials into the conference, then through the conference invites the Party B and Party C. The steps should be read as follows:

- Party A dialling into the conference URI (target) follows the steps 1 to 8A.
- Conference URI invitation to Party B follows the steps 9 to 17.
- Conference URI invitation to Party C follows the steps 18 to 26. The IRI-POI present in the AS/MRFC accesses the SIP messages and generates the required xIRI. The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC. The CC-POI present in the MRFP delivers the xCC to the MDF3.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF. When all three parties are in conference, the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5.3.4 Party A (target), Party B, Party C in a group conference

In this illustration, Party A dials into the conference URI, then through the conference server invites Party B and Party C into the conference. In this scenario, the conference server is in a different CSP domain. The Party A is a target in the respective CSP domain.

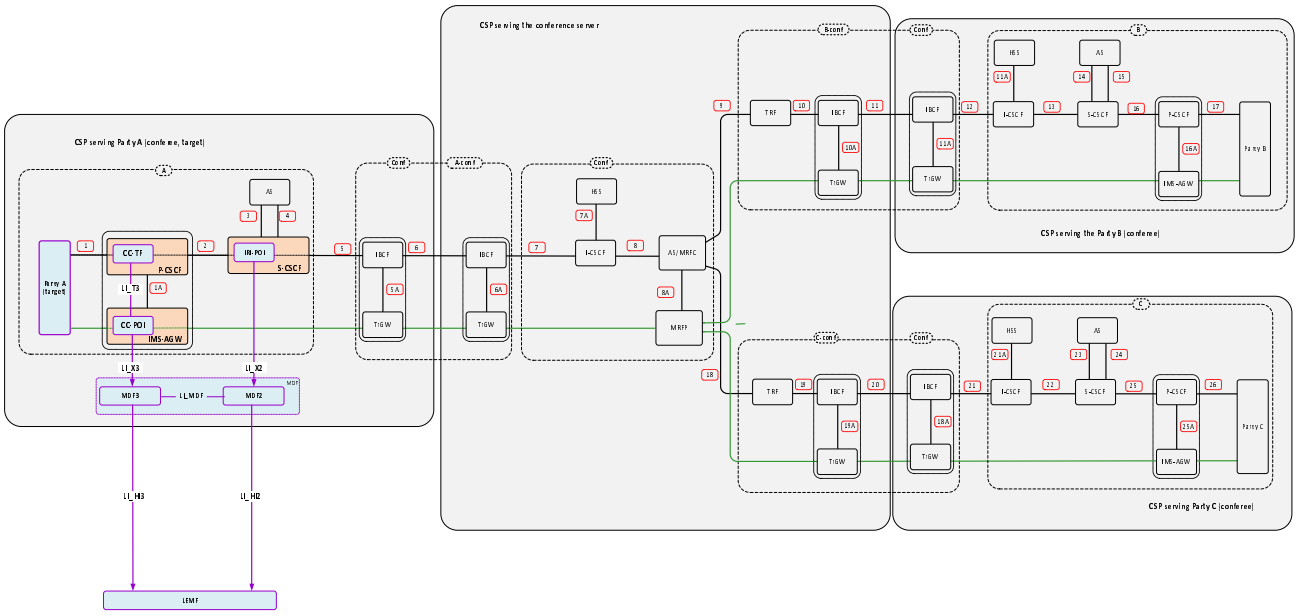


Figure 4.5.3.4-1: Party A (target), Party B, Party C in a group conference

As shown in figure 4.5.3.4-1, the interception is on the A-side of the session within the CSP domain that serves the conference participant (target). The Party A, Party B and Party C are in different IMS domains.

NOTE: If Party B or Party C were to be the targets, the interception would occur in the respective CSP domains in the same way as shown in this clause.

For session progressive steps, the illustration assumes that Party A dials into the conference, then through the conference invites Party B and Party C. The steps should be read as follows:

- Party A (target) dialling into the conference URI follows the steps 1 to 8A.
- Conference URI invitation to Party B follows the steps 9 to 17.
- Conference URI invitation to Party C follows the steps 18 to 26. The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF. The xCC received from IMS-AGW includes the communication content of Party A (sent and received). Therefore, CC delivered to the LEMF includes the communication content of Party A, Party B and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.5.3.5 Party A (target), Party B, Party C (target) in a group conference (conference URI (target)

In this illustration, Party A dials into the conference URI, then through the conference invites Party B and Party C into the conference. Since Party A invites the other two parties, in this illustration, Party A may also be referred to as conference initiator. In this scenario, the conference server is in a different CSP domain. The Party A and the conference URI are the targets in the respective CSP domains.

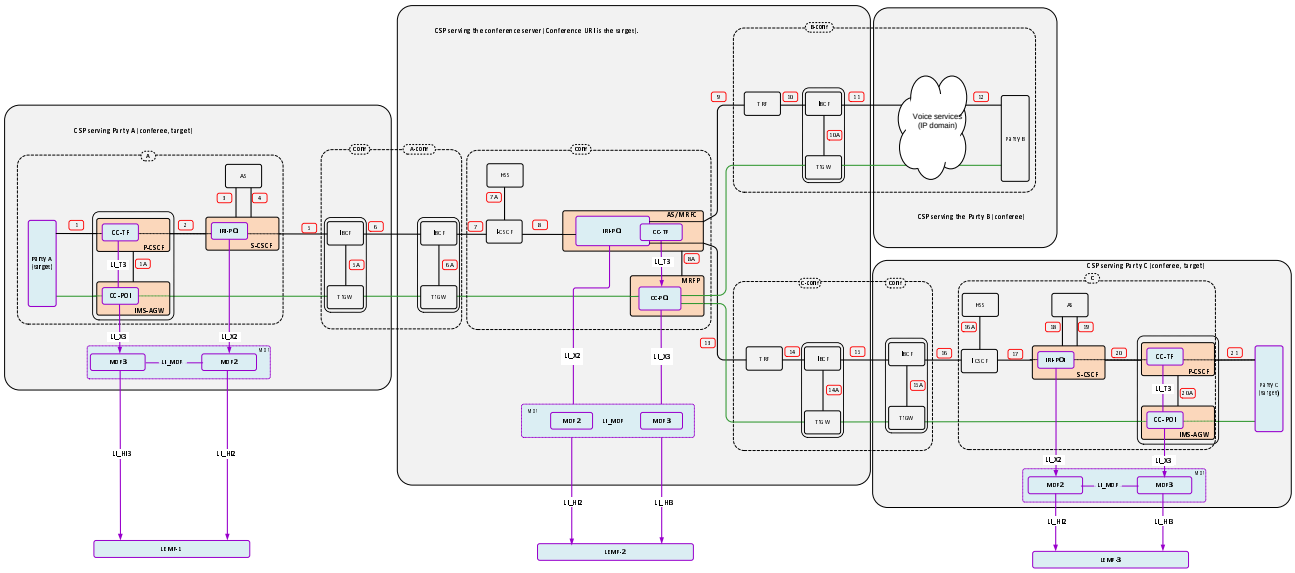


Figure 4.5.3.5-1: Party A (target), Party B, Party C (target) in a group conference (conference URI target)

As shown in figure 4.5.3.5-1, in the CSP domain that serves Party A (target), the interception is on the A-side of the session. In the CSP domain that serves the conference server, the interception is within that CSP domain. In the CSP domain that serves Party C (target), the interception is on the C-side of the session. The Party B is in another IP domain.

For session progressive steps, the illustration assumes that Party A dials into the conference, then through the conference invites Party B and Party C. The steps should be read as follows:

- Party A dialling into the conference URI (target) follows the steps 1 to 8A.
- Conference URI invitation to Party B follows the steps 9 to 12.
- Conference URI invitation to Party C follows the steps 13 to 21.

CSP domain that serves Party A (target)

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.
- The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-1. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-1. The xCC received from the CC-POI present in the IMS-AGW includes the communication content of Party A (sent and received). Therefore, CC delivered to the LEMF-1 includes the communication content of Party A, Party B and Party C.

CSP domain that serves conference URI (target)

- The IRI-POI present in the AS/MRFC accesses the SIP messages and generates the required xIRI. The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC. The CC-POI present in the MRFP generates the xCC.
- The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-2. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-2. When all three parties are in the conference, the CC delivered to the LEMF-2 includes the communication content of Party A, Party B and Party C.

CSP domain that serves Party C (target)

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

- The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-3. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-3. The xCC received from the CC-POI present in the IMS-AGW includes the communication content of Party C (sent and received). Therefore, CC delivered to the LEMF-3 includes the communication content of Party A, Party B and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6 IMS session hold

4.6.1 General

Clause 4.6 illustrates the IMS LI for various session leg hold scenarios. In all illustrations, the party placing the session on hold is presumed to be involved in multiple independent IMS sessions. The illustrations show a single party hold scenario, a conference hold scenario, a session leg involving a group conference hold scenario.

The party placing the session on hold can be a target, or the party or the conference (in the case of group conferencing) placed on hold can be the target in the respective CSP domains.

In case of group conferencing, the possibility of the conference server present in a different CSP domain is also considered in the illustrations.

4.6.2 Single Party on hold

4.6.2.1 Introduction

In the scenarios presented here, three IMS users are involved in two independent IMS sessions with one session being placed on hold by of one of the IMS users.

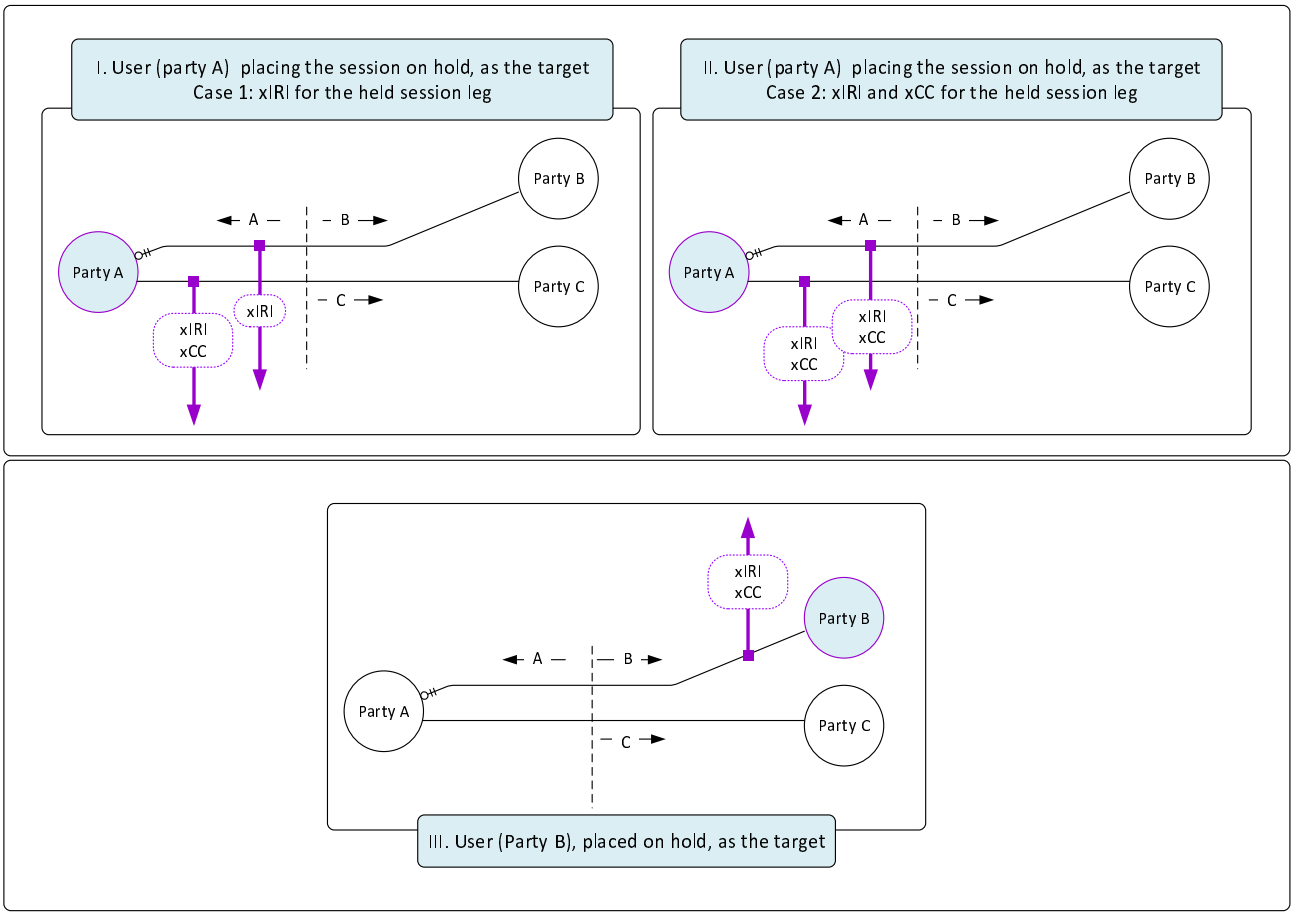


Figure 4.6.2.1-1: Single Party on hold scenarios

In one set of the scenarios, the user who places the session on hold is the target and in another set of scenarios, the user who is placed on hold is the target.

4.6.2.2 Party A (target) in session with Party C with Party B on hold

Figure 4.6.2.2-1 shows a scenario where Party A (target) is in an IMS session with Party C with Party B on hold.

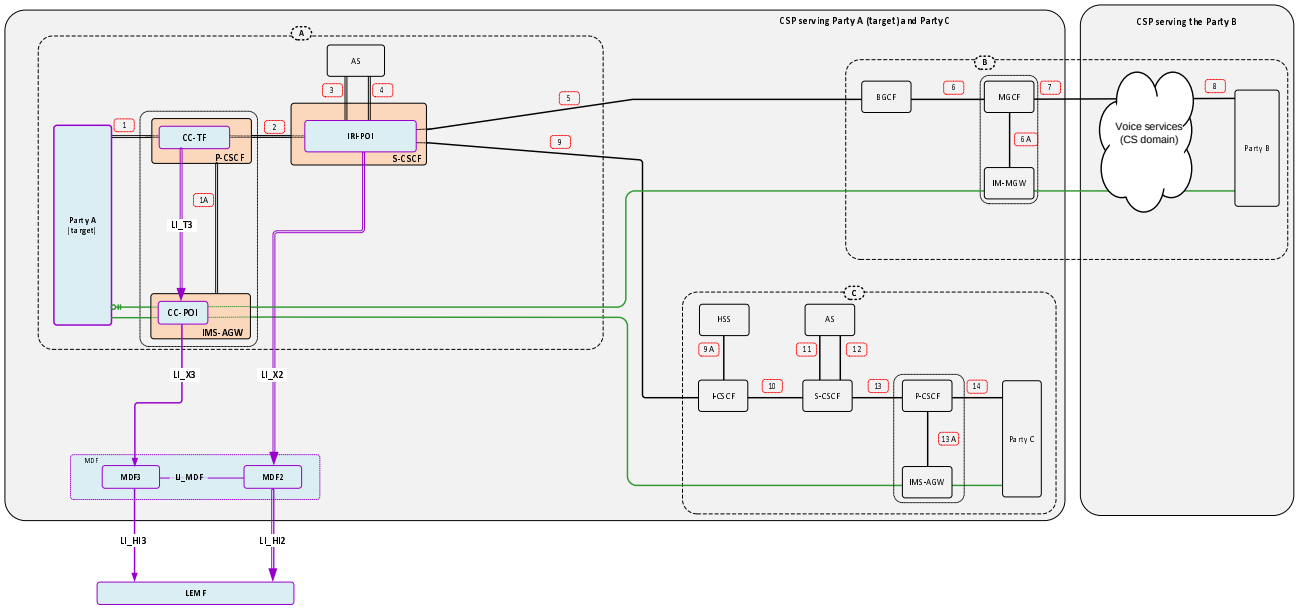


Figure 4.6.2.2-1: Party A (target) is in a session with Party C with Party B on hold

As shown in figure 4.6.2.2-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions. Party B is in a CS domain. Party C in the same CSP domain as that of Party A.

The session progressive steps 1 to 4 are presumed to be applicable to the two sessions depicted in figure 4.6.2.2-1, Party A to Party B and Party A to Party C. In this illustration, Party A establishes a session to Party B, places the session on hold and then establishes a session to Party C. The steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8 as shown in the scenario illustrated in clause 4.3.3.2. Party B is in a CS domain.

- After placing the session to Party B on hold, the IMS session setup to Party C follows the steps, 1 to 4 and the steps 9 to 14. Party C is in the same CSP domain as that of Party A.

The initial IMS session setup to Party B followed the steps 1 to 8 as shown in the scenario.

Held session

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI for the held session. The xCC for the held session is not generated.

Active session with Party C

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The IRI messages delivered to the LEMF include the xIRI received for both sessions. The IRI messages associated with the two sessions are correlated individually.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.2.3 Party A (target) in session with Party C with Party B on hold - alternative

Figure 4.6.2.3-1 shows an alternative to the scenario shown in figure 4.6.2.2-1, where Party A (target) is in an IMS session with Party C with Party B on hold.

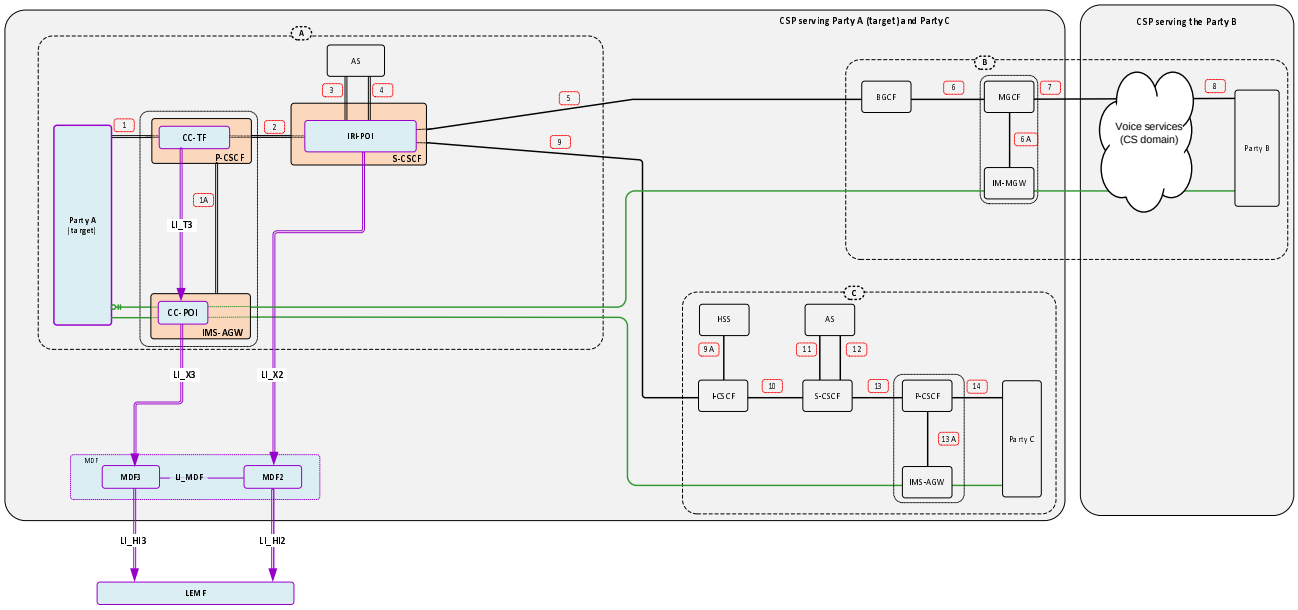


Figure 4.6.2.3-1: Party A (target) is in a session with Party C with Party B on hold - alternative

In this alternative, the xCC generation at the CC-POI present in the IMS-AGW is not stopped for the held session leg. Other than that, all the descriptions described in clause 4.6.2.2 apply to the scenario illustrated in figure 4.6.2.3-1.

4.6.2.4 Party A in session with Party C with Party B (non-local ID target) on hold – case 1

Figure 4.6.2.4-1 shows a scenario where Party A is in an IMS session with Party C with Party B on hold. Party B is in the CS domain and is a non-local ID target in the CSP domain of Party A and Party C. The purpose of this illustration is to show the LI interactions with a held non-local ID target.

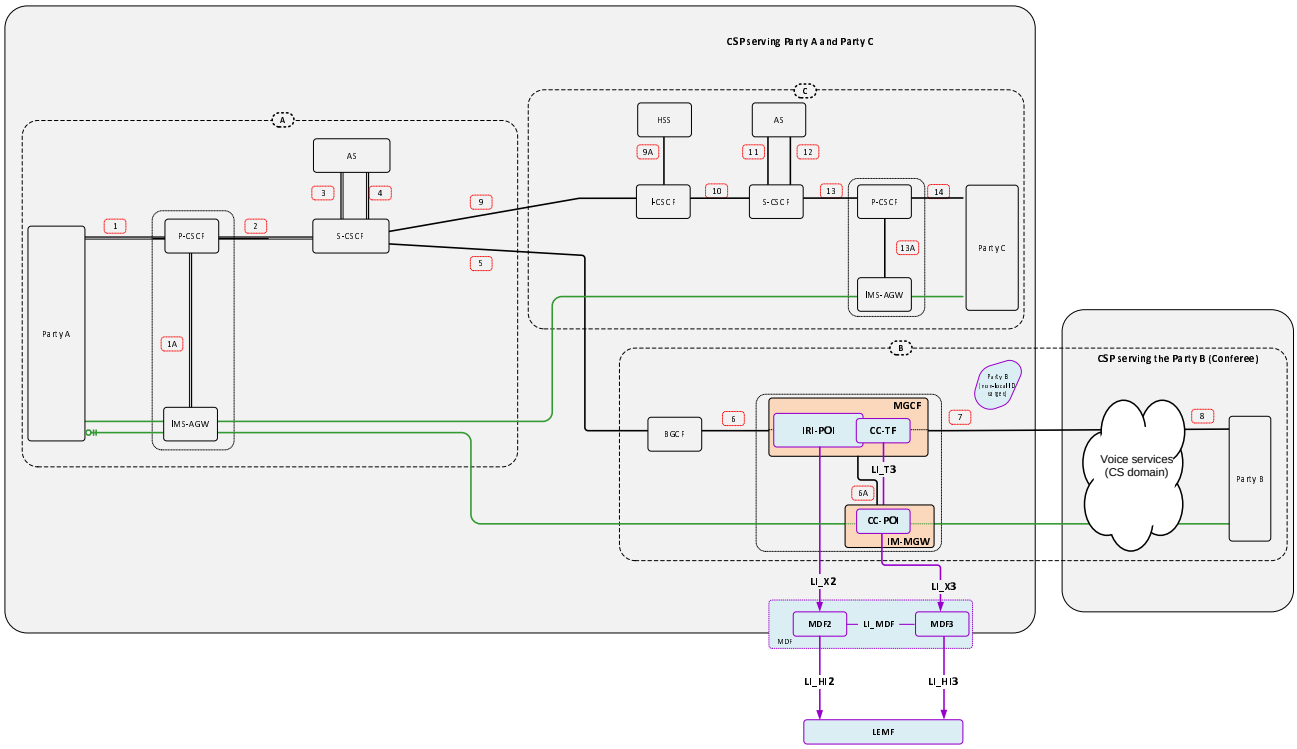


Figure 4.6.2.4-1: Party A is in a session with Party C with Party B (non-local ID target) on hold

As shown in figure 4.6.2.4-1, only the network functions that handle the B-side (non-local ID target side) of the session are involved in handling the LI functions.

The session progressive steps 1 to 4 are presumed to be applicable to the two sessions depicted in figure 4.6.2.4-1, Party A to Party B and Party A to Party C. The steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8 as shown in the scenario illustrated in clause 4.3.3.2. Party B is in a CS domain. Party B is a non-local ID target.
- After placing the session to Party B on hold, the IMS session setup to Party C follows the steps, 1 to 4 and the steps 9 to 14. Party C is in the same CSP domain as that of Party A.

The IRI-POI present in the MGCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the MGCF triggers the CC-POI present in the IM-MGW for the xCC.

NOTE 1: Since only the ingress signalling at the MGCF is SIP-based, this architecture assumes that the interception is done using the SIP messages seen at the step 6.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE 2: Even if Party A is not in an active communication with Party B, from the perspective of LI functions provided in the B-side of the call, the Party B is in active communication with Party A.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.2.5 Party A in session with Party C with Party B (target) on hold – case 2

Figure 4.6.2.5-1 shows a scenario where Party A is in an IMS session with Party C with Party B on hold. Party B is the target in this illustration.

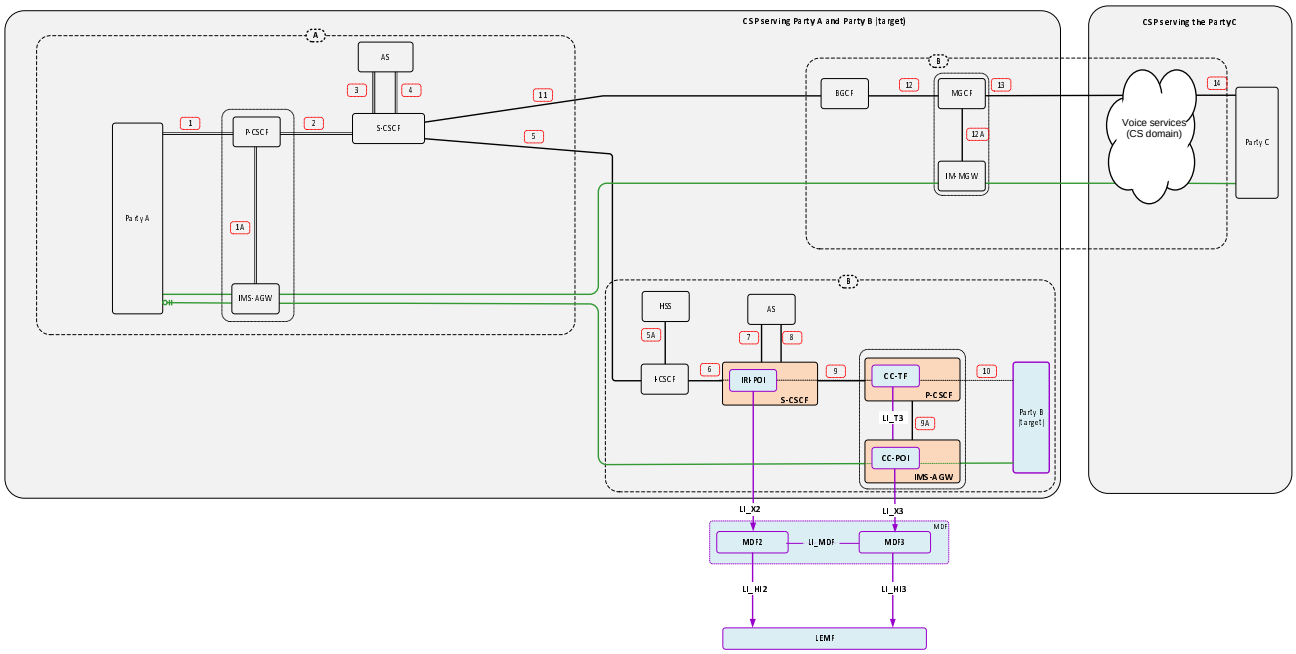


Figure 4.6.2.5-1: Party A is in a session with Party C with Party B (target) on hold

As shown in figure 4.6.2.5-1, only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions.

The session progressive steps 1 to 4 are presumed to be applicable to the two sessions depicted in figure 4.6.2.5-1, Party A to Party B and Party A to Party C. The steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 10 as shown in the scenario illustrated in clause 4.2.3.1. Party B is in the same CSP domain as that of Party A. The Party B is the target in this illustration.
- After placing the session to Party B on hold, the IMS session setup to Party C follows the steps, 1 to 4 and the steps 11 to 14. Party C is in a CS domain.

The IRI-POI present in the S-CSCF (serving Party B) accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE: Even if Party A is not in an active communication with Party B, from the perspective of LI functions provided on the B-side of the call, the Party B is in active communication with Party A.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.2.6 Party A in session with Party C with Party B (target in other CSP domain) on hold – case 3

Figure 4.6.2.6-1 shows a scenario where Party A is in an IMS session with Party C with Party B on hold. The Party B is in another IMS domain. For this illustration, the CSP domain of interest is the CSP that serves Party B. The purpose of this illustration is to show the LI functions when a Party is placed on hold in a different CSP domain.

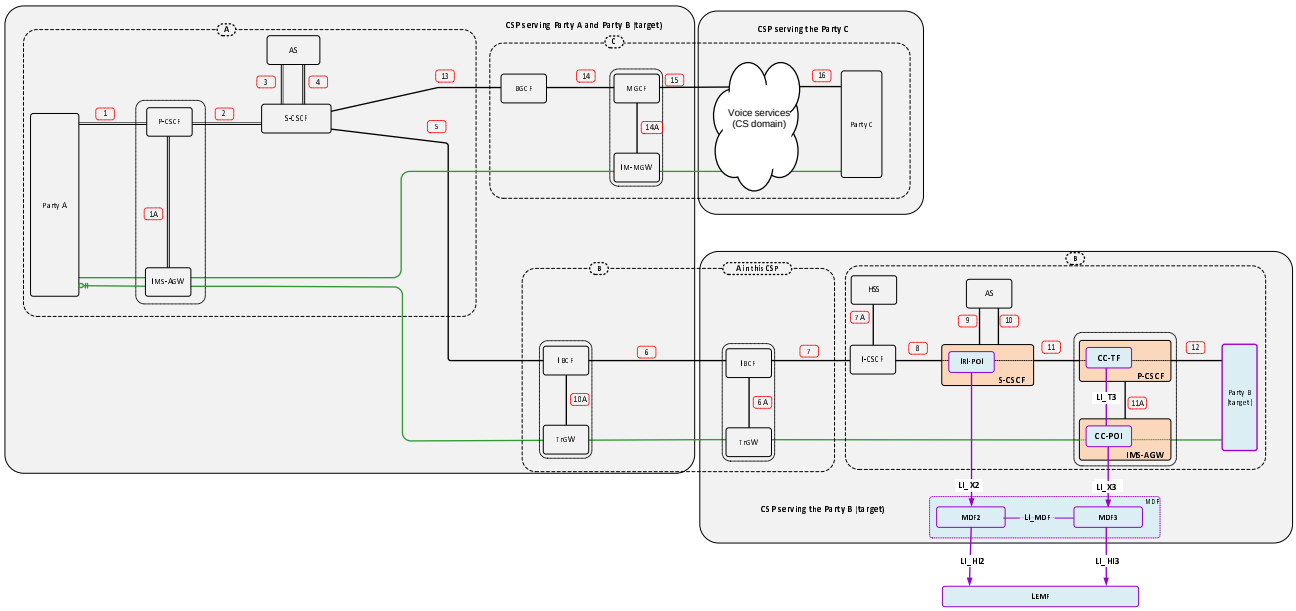


Figure 4.6.2.6-1: Party B (target) is placed on hold by Party A of a different CSP domain

As shown in figure 4.6.2.6-1, only the network functions that handle the B-side of the session are involved in handling the LI functions.

The session progressive steps 1 to 4 are presumed to be applicable to the two sessions depicted in figure 4.6.2.6-1, Party A to Party B and Party A to Party C. The steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 12. Party B is in a different IMS domain. Party B is the target.
- After placing the session to Party B on hold, the IMS session setup to Party C follows the steps, 1 to 4 and the steps 13 to 16. Party C is in a CS domain.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE: Even if Party A is not in an active communication with Party B, from the perspective of LI functions provided in the CSP domain of Party B, the Party B is in active communication with Party A.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.3 Conference on hold

4.6.3.1 Introduction

In the scenarios presented here, a conference is on hold by the conference initiator while active on a different communication session.

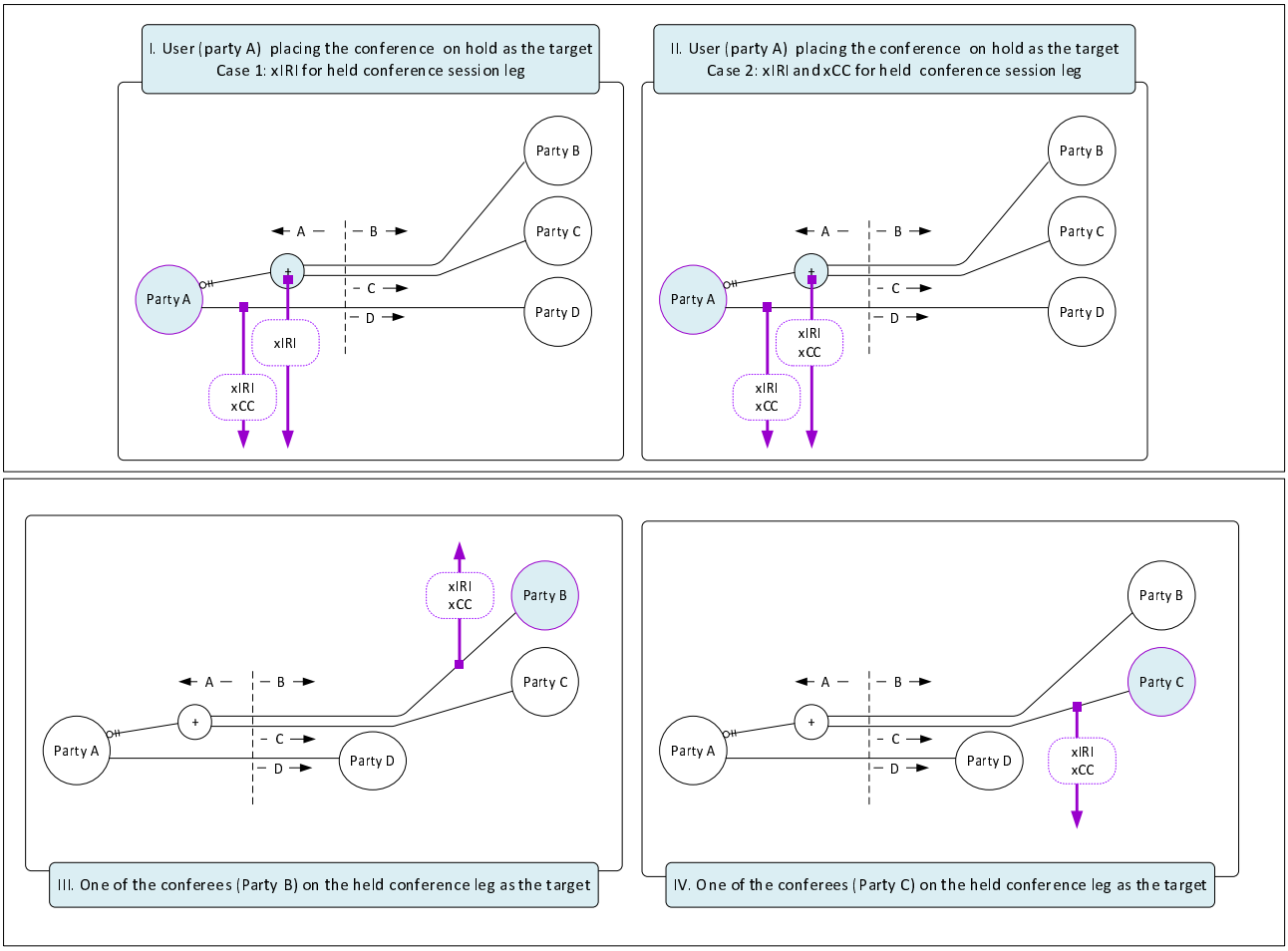


Figure 4.6.3.1-1: Conference on hold scenarios

In one scenario, the user who places the conference on hold is the target; and in another scenario, the target is part of the conference (as a conferee) that is placed on hold.

The illustrations assume that the conference server is present in the same CSP domain that serves the conference initiator. The illustrations also assume that the Application Server (AS), which is the conference focus, is different from the AS that is used in the regular voice sessions.

4.6.3.2 Party A (target) in session with Party D, with conference on hold – case 1

Figure 4.6.3.2-1 shows a scenario where Party A (target) is in an IMS session with Party D with conference (with party B and Party C) on hold.

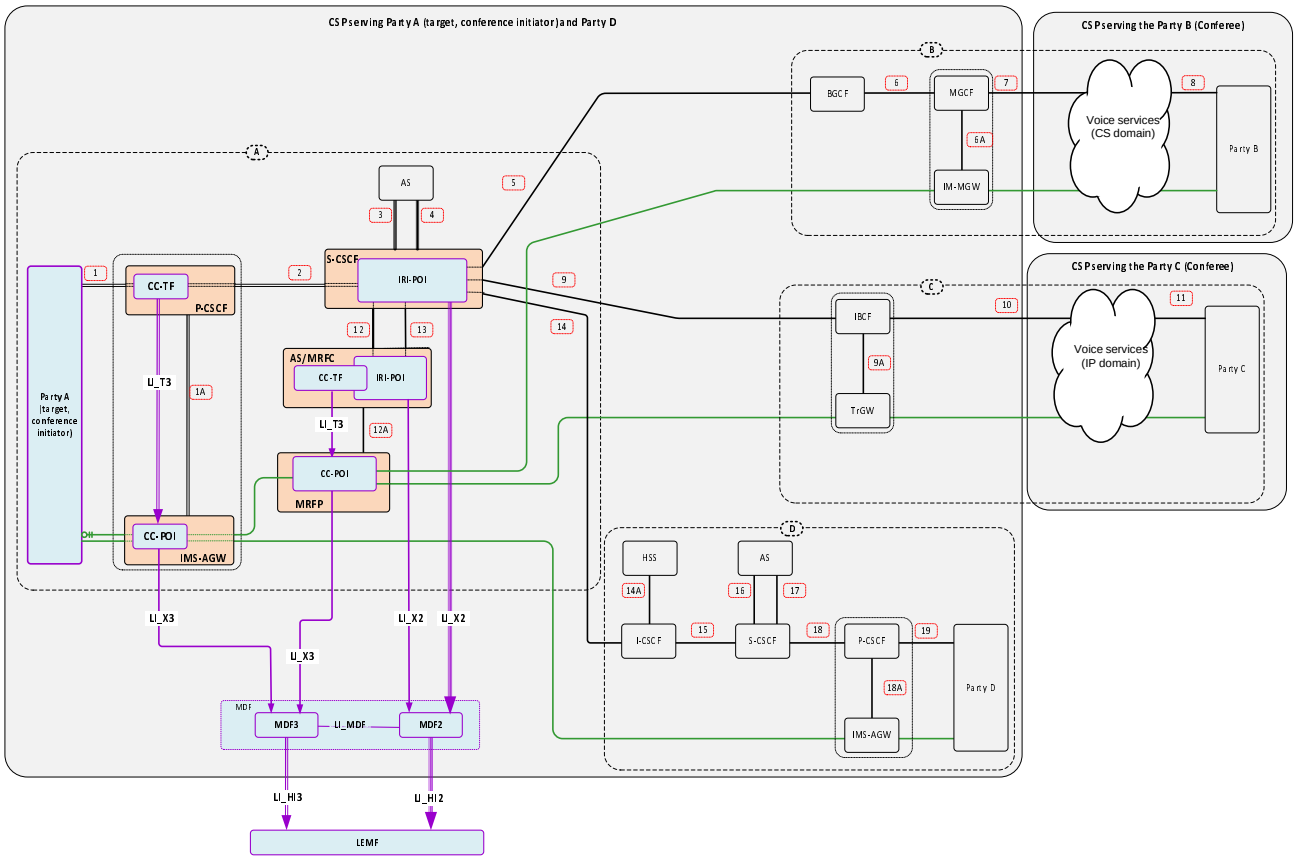


Figure 4.6.3.2-1: Party A (target) is in a session with Party D with conference on hold – case 1

As shown in figure 4.6.3.2-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions. Party B is in a CS domain. Party C is in an IP domain. Party D is in the same CSP domain as that of Party A.

The session progressive steps 1 to 4 are presumed to be applicable to the multiple sessions depicted in figure 4.6.3.2-1. For example, for the scenario where the Party A establishes a session to Party B, and then to Party C, and then invokes the conference to merge the two sessions, and then places the conference on hold, and then establishes a session to Party D, session progressive steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8.
- The IMS session setup to Party C follows the steps, 1 to 4 and the steps 9 to 11.
- The conference invoking (before inviting Party B and Party C into the conference) follows the steps 1 to 4 and the steps 12 to 13.
- After placing the conference session on hold, the IMS session setup to Party D follows the steps, 1 to 4 and then 14 to 19.

Held conference leg

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the xIRI. The IRI-POI present in the AS/MRFC accesses the SIP messages and generates the required xIRI for the held conference. The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC. The CC-POI present in the MRFP generates the xCC.

Active session with Party D

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

Two separate copies of CC are delivered to the LEMF: one that includes the Party A communication content with Party D and the other includes communication content of held conference. The IRI messages delivered to the LEMF include the xIRI received for both sessions. The IRI messages and CC delivered to the LEMF for the two sessions are correlated individually.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.3.3 Party A (target) in session with Party D, with conference on hold – case 1 (alternative)

Figure 4.6.3.3-1 shows an alternative to the scenario shown in figure 4.6.3.2-1, where Party A (target) is in an IMS session with Party D with conference (with party B and Party C) on hold.

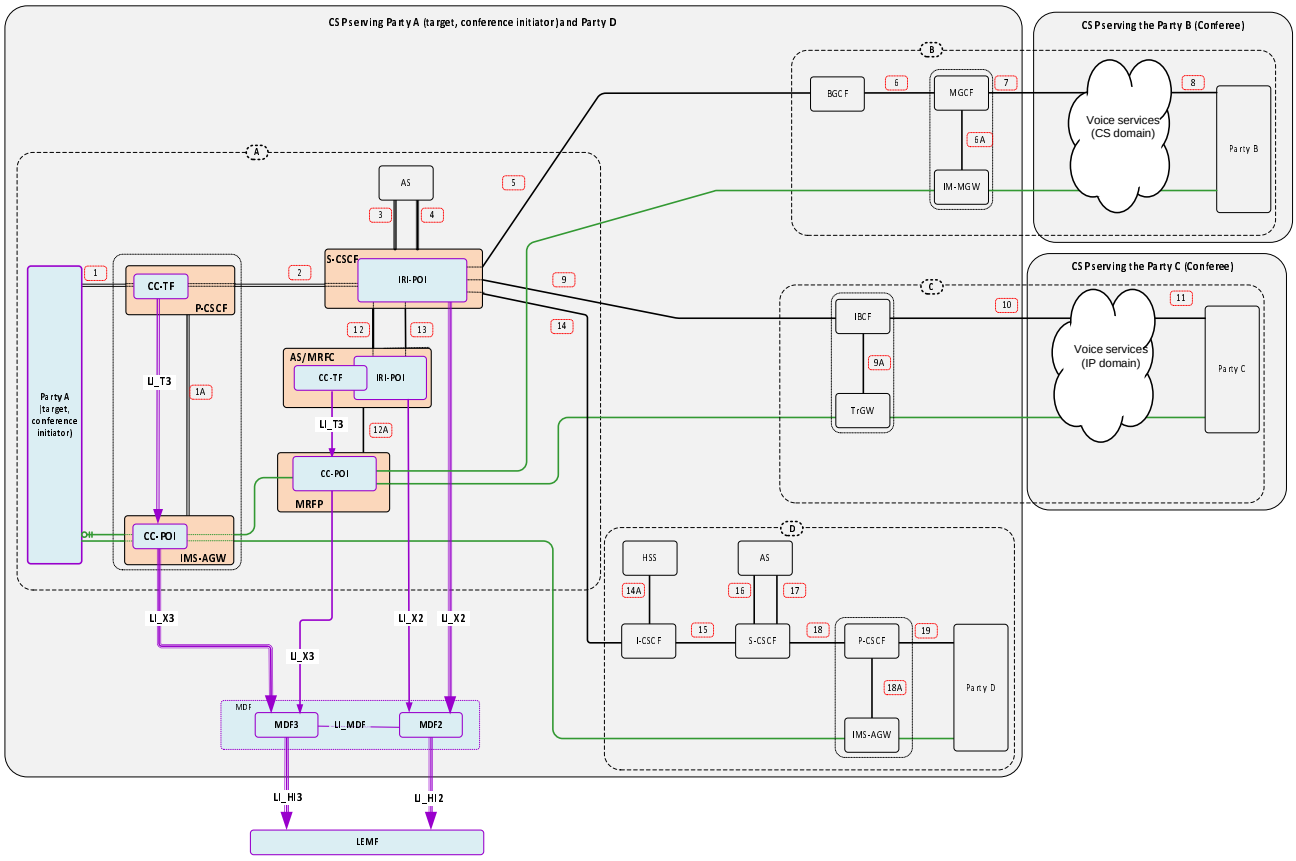


Figure 4.6.3.3-1: Party A (target) is in a session with Party D with conference on hold – case 1 (alternative)

In this alternative, the xCC generation at the CC-POI present in the IMS-AGW is not stopped for the held conference leg. Other than that, all the descriptions described in clause 4.6.3.2 apply to the scenario illustrated in figure 4.6.3.3-1.

4.6.3.4 Party A (target) in session with Party D, with conference on hold – case 2

Figure 4.6.3.4-1 shows an alternative illustration where Party A (target) is in an IMS session with Party D and with a conference (with Party B and Party C) on hold. This illustration is applicable to the case where the communication content of a held conference session is not to be delivered.

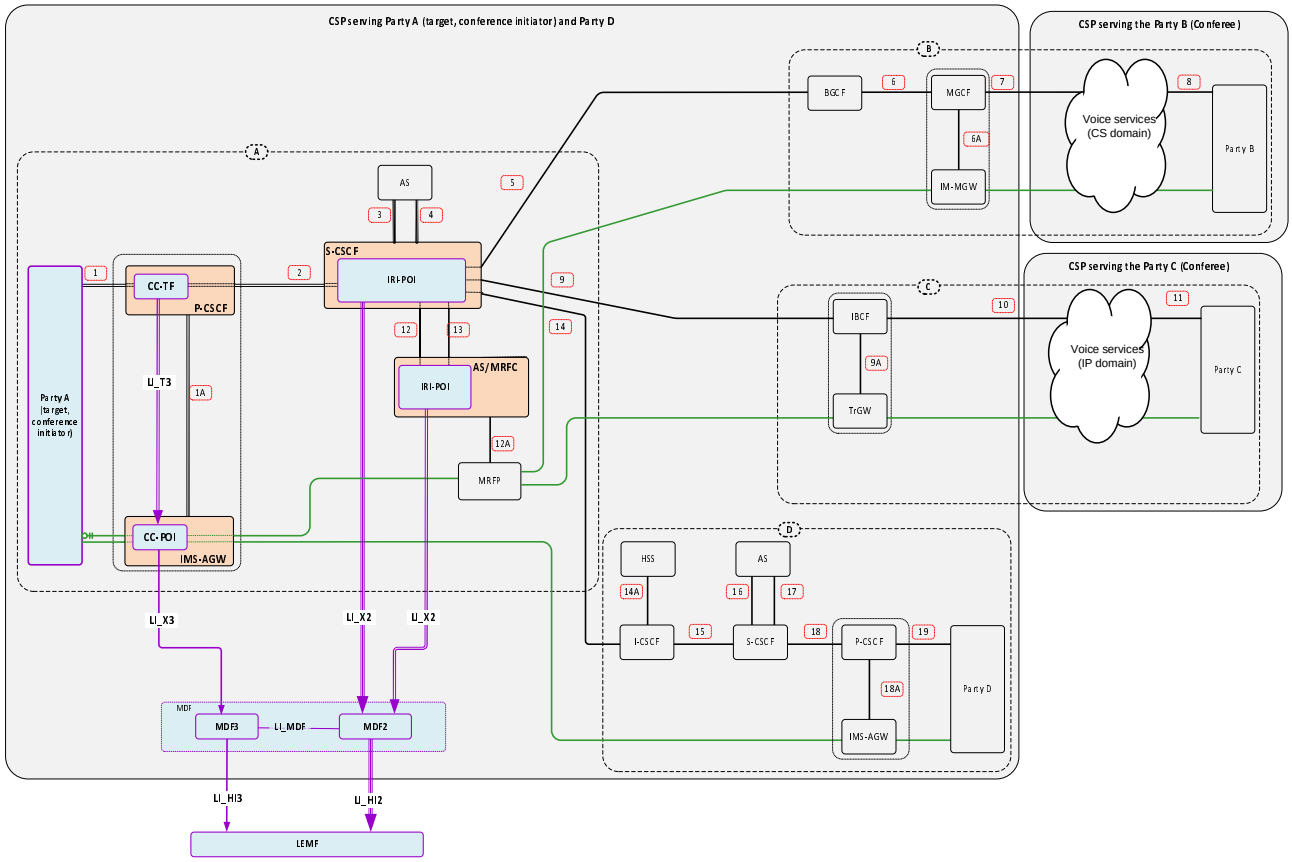


Figure 4.6.3.4-1: Party A (target) is in a session with Party D with conference on hold – case 2

This illustration is applicable to the scenario where communication content of a held conference session is not to be delivered to the LEMF.

The IMS session establishment procedures are same as described in clause 4.6.3.2.

Held conference leg

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the xIRI. The IRI-POI present in the AS/MRFC accesses the SIP messages and generates the required xIRI for the held conference. No xCC generation takes place for a held conference session.

Active session with Party D

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

The IRI messages delivered to the LEMF include the xIRI received for both sessions. The IRI messages and when applicable the CC delivered to the LEMF for the two sessions are correlated individually.

4.6.3.5 Party A (target) in session with Party D, with conference on hold – case 2 (alternative)

Figure 4.6.3.5-1 shows an alternative to the scenario shown in figure 4.6.3.4-1, where Party A (target) is in an IMS session with Party D and with a conference (with Party B and Party C) on hold. This illustration is applicable to the case where the communication content of a held conference session is not to be delivered to the LEMF.

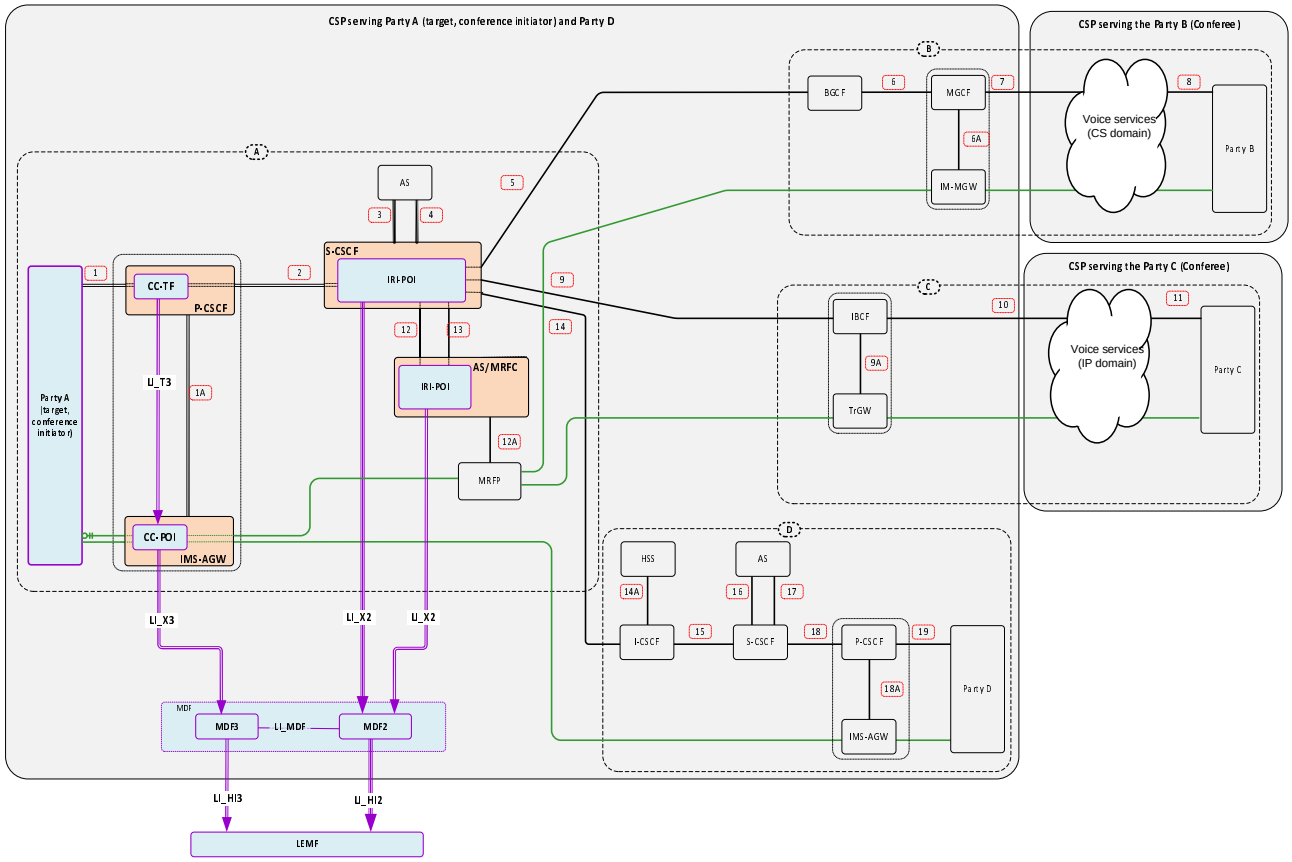


Figure 4.6.3.5-1: Party A (target) is in a session with Party D with conference on hold – case 2 (alternative)

In this alternative, the xCC generation at the CC-POI present in the IMS-AGW is not stopped for the held conference leg. Other than that, all the descriptions described in clause 4.6.3.4 apply to the scenario illustrated in figure 4.6.3.5-1.

4.6.3.6 Party A in session with Party D, with a Party C (non-local ID target) on the held conference leg

Figure 4.6.3.6-1 shows a scenario where Party A is in an IMS session with Party D and with a conference on hold. In this illustration, Party C one of the conferees in the held conference leg is a non-local ID target.

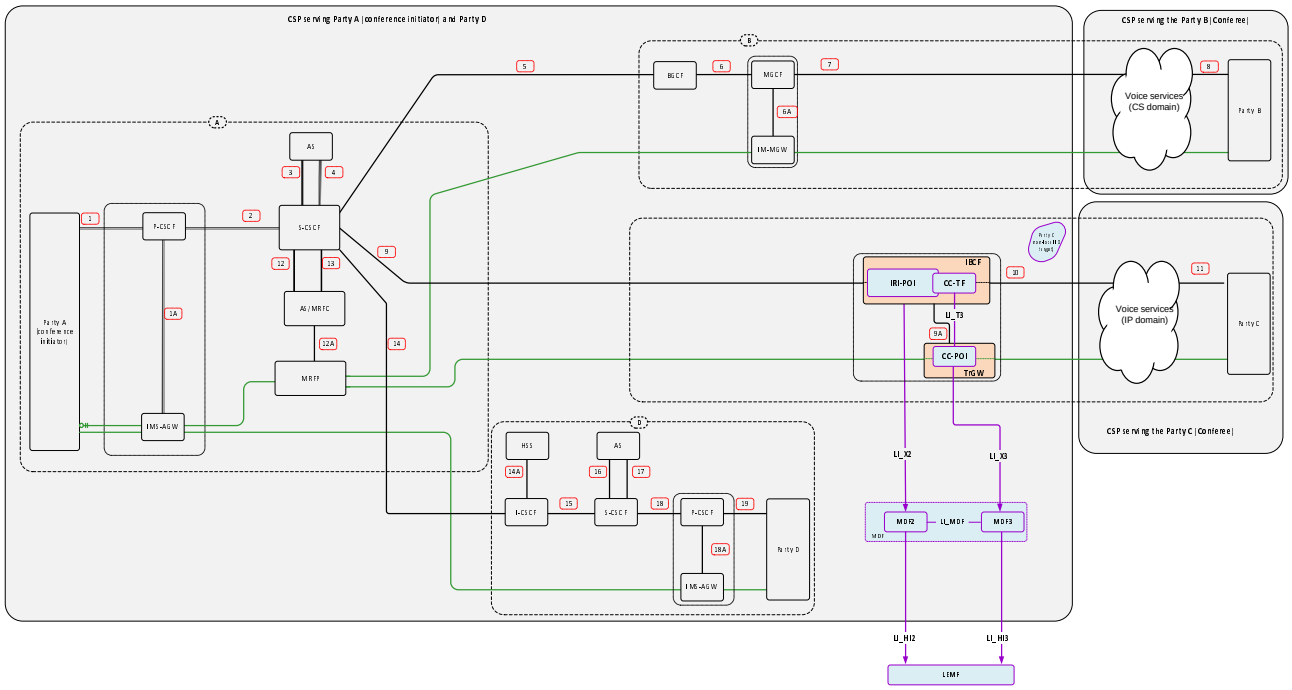


Figure 4.6.3.6-1: Party A is in a session with Party D, with Party C (non-local ID target) on held conference leg

As shown in figure 4.6.3.6-1, only the network functions that handle the C-side (target side) of the session are involved in handling the LI functions. Party B is in a CS domain. Party C is in an IP domain. Party D is in the same CSP domain as that of Party A. Party C is a non-local ID target.

NOTE 1: If Party B was a non-local ID target, the LI functions are provided by the MGCF and MGW, similar to the way IBCF and TrGW provide the LI functions as illustrated in this clause.

The session progressive steps 1 to 4 are presumed to be applicable to the multiple sessions depicted in figure 4.6.3.6-1. For example, for the scenario where the Party A establishes a session to Party B, and then to Party C, and then invokes the conference to merge the two sessions, and then places the conference on hold, and then establishes a session to Party D, the session progressive steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8.
- The IMS session setup to Party C follows the steps, 1 to 4 and the steps 9 to 11.
- The conference invoking (before inviting Party B and Party C into the conference) follows the steps 1 to 4 and the steps 12 to 13.
- After placing the conference session on hold, the IMS session setup to Party D follows the steps, 1 to 4 and then 14 to 19.

The IRI-POI present in the IBCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the IBCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE 2: Even if Party A is not in an active communication with Party B or Party C, from the perspective of LI functions provided in the C-side of the call, the Party C is in active communication with the conference.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.3.7 Party A in session with Party D, with a Party C (target) on the held conference leg

Figure 4.6.3.7-1 shows a scenario where Party A is in an IMS session with Party D with conference on hold. In this illustration, Party C one of the conferees in the held conference leg is the target.

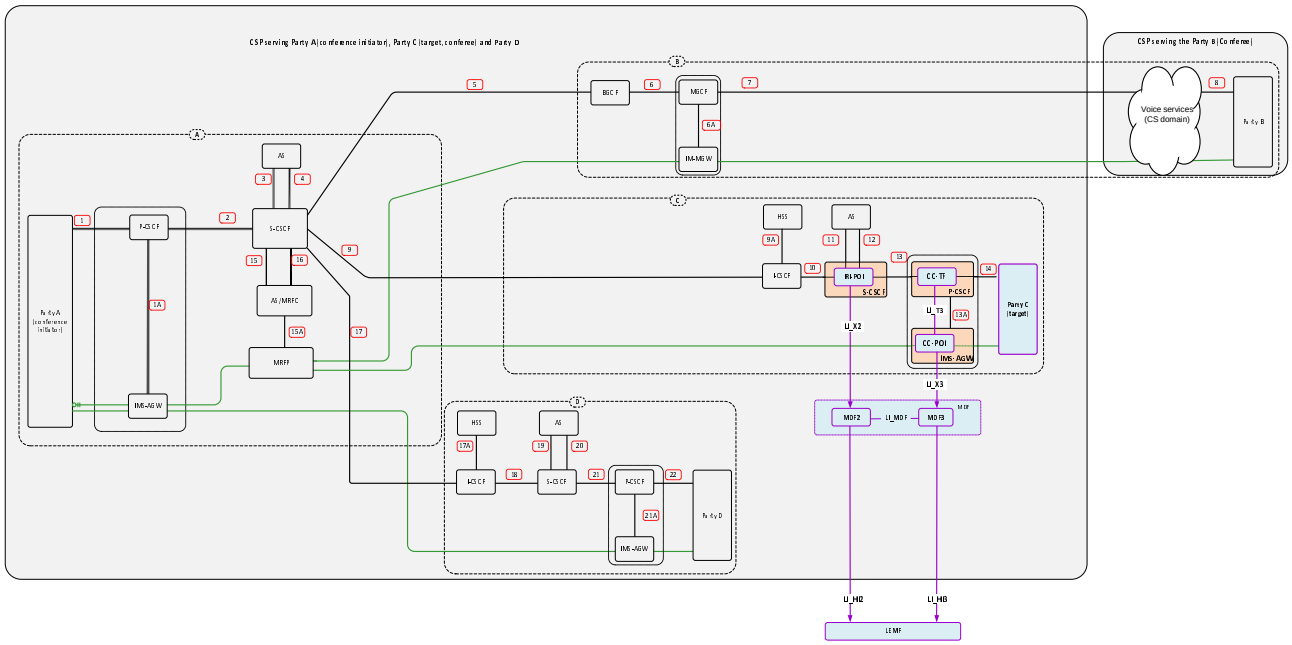


Figure 4.6.3.7-1: Party A is in a session with Party D, with Party C (target) on the held conference leg

As shown in figure 4.6.3.7-1, only the network functions that handle the C-side (target side) of the session are involved in handling the LI functions. Party B is in a CS domain. Party C is in the same CSP domain as that of Party A and Party D. Party C is the target.

The session progressive steps 1 to 4 are presumed to be applicable to the multiple sessions depicted in figure 4.6.3.7-1. For example, for the scenario where the Party A establishes a session to Party B, and then to Party C, and then invokes the conference to merge the two sessions, and then places the conference on hold, and then establishes a session to Party D, the session progressive steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8.
- The IMS session setup to Party C follows the steps, 1 to 4 and the steps 9 to 14.
- The conference invoking (before inviting Party B and Party C into the conference) follows the steps 1 to 4 and the steps 15 to 16.
- After placing the conference session on hold, the IMS session setup to Party D follows the steps, 1 to 4 and then 17 to 22.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the _P-CSCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE: Even if Party A is not in an active communication with Party B or Party C, from the perspective of LI functions provided in the C-side of the call, the Party C is in active communication with the conference.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.3.8 Party A in session with Party D, with a Party C (in a different CSP domain, target) on the held conference leg

Figure 4.6.3.8-1 shows a scenario where Party A is in an IMS session with Party D and with a conference on hold. In this illustration, Party C one of the conferees in the held conference leg is the target.

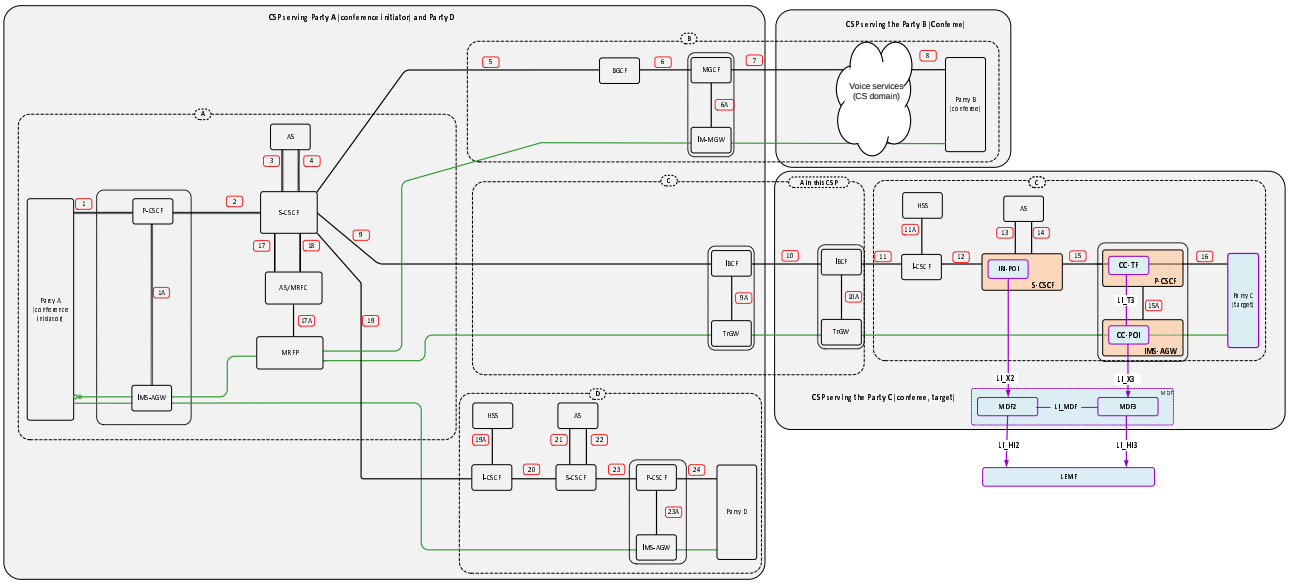


Figure 4.6.3.8-1: Party A is in a session with Party D, with Party C (different CSP domain, target) on the held conference leg

As shown in figure 4.6.3.8-1, only the network functions that handle the C-side (target side) of the session are involved in handling the LI functions. Party B is in a CS domain. Party C is in a different CSP from Party A.

The session progressive steps 1 to 4 are presumed to be applicable to the multiple sessions depicted in figure 4.6.3.8-1. For example, for the scenario where the Party A establishes a session to Party B, and then to Party C, and then invokes the conference to merge the two sessions, and then places the conference on hold, and then establishes a session to Party D, the session progressive steps should be read as follows:

- The initial IMS session setup to Party B followed the steps 1 to 8.
- The IMS session setup to Party C follows the steps, 1 to 4 and the steps 9 to 16.
- The conference invoking (before inviting Party B and Party C into the conference) follows the steps 1 to 4 and the steps 17 to 18.
- After placing the conference session on hold, the IMS session setup to Party D follows the steps, 1 to 4 and then 19 to 24.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the TrGW for the xCC. The CC-POI present in the TrGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE: Even if Party A is not in an active communication with Party B or Party C, from the perspective of LI functions provided in the C-side of the call, the Party C is in active communication with the conference.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.4 Group conference on hold

4.6.4.1 Introduction

The hold scenarios related to the group conferencing are illustrated in the subsequent clauses.

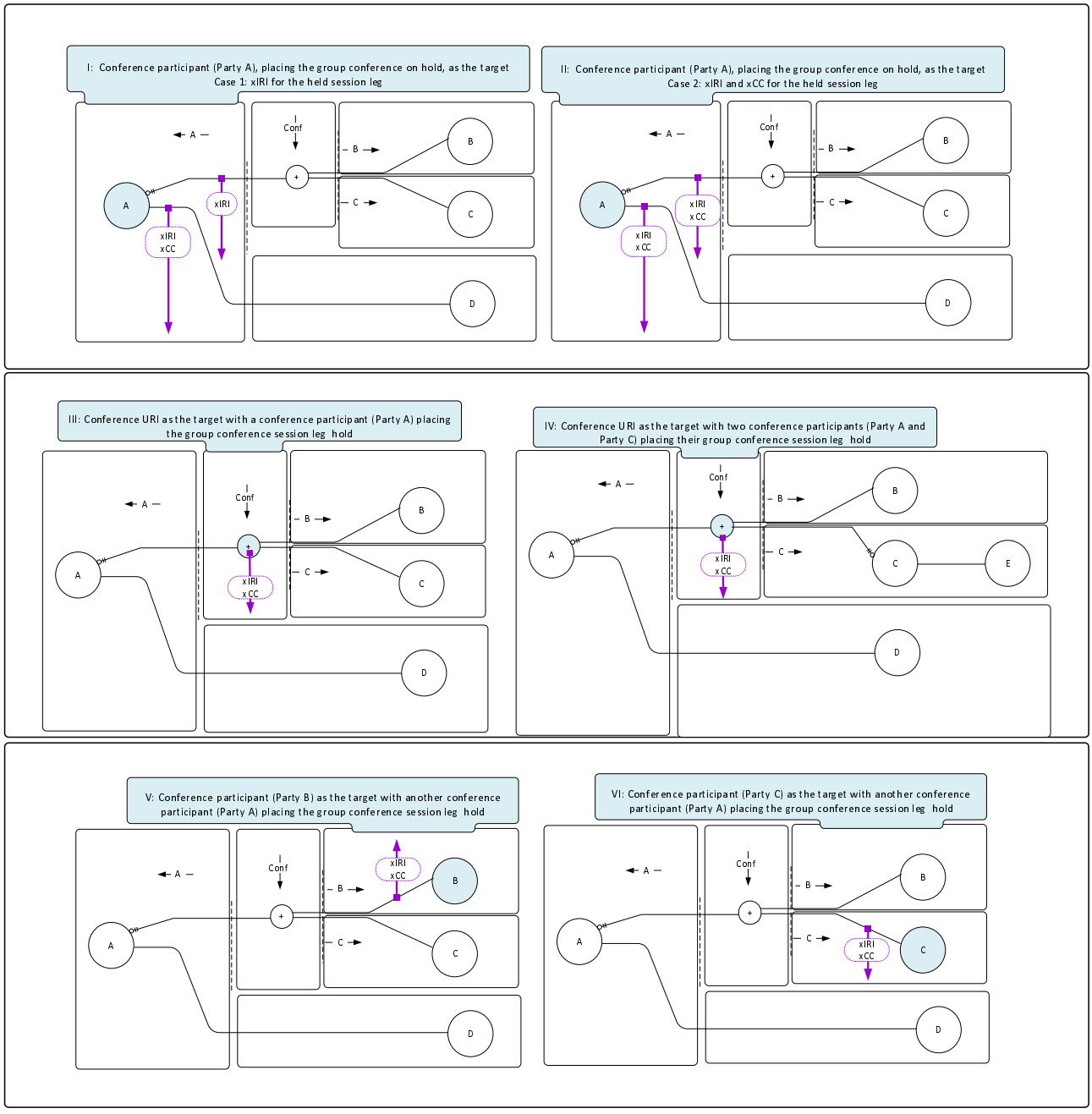


Figure 4.6.4.1-1: Group conferencing hold scenarios

The party placing the group conference on hold, or the conference URI, or the other conference parties can be the target.

The conference URI may be provided in a separate CSP domain that does not serve any of the conference participants or provided by a CSP domain that serves one of the conference participants. The SIP signalling flow is presumed to be the same for either of the two cases, which of-course, depends how the conference session is setup.

NOTE: The present document assumes that for group conferencing, the conference server is served by a different CSP domain.

4.6.4.2 Party A (target) in communication with Party D, with group conference on hold

Figure 4.6.4.2-1 shows a scenario where Party A (target) is in an IMS session with Party D with group conference on hold.

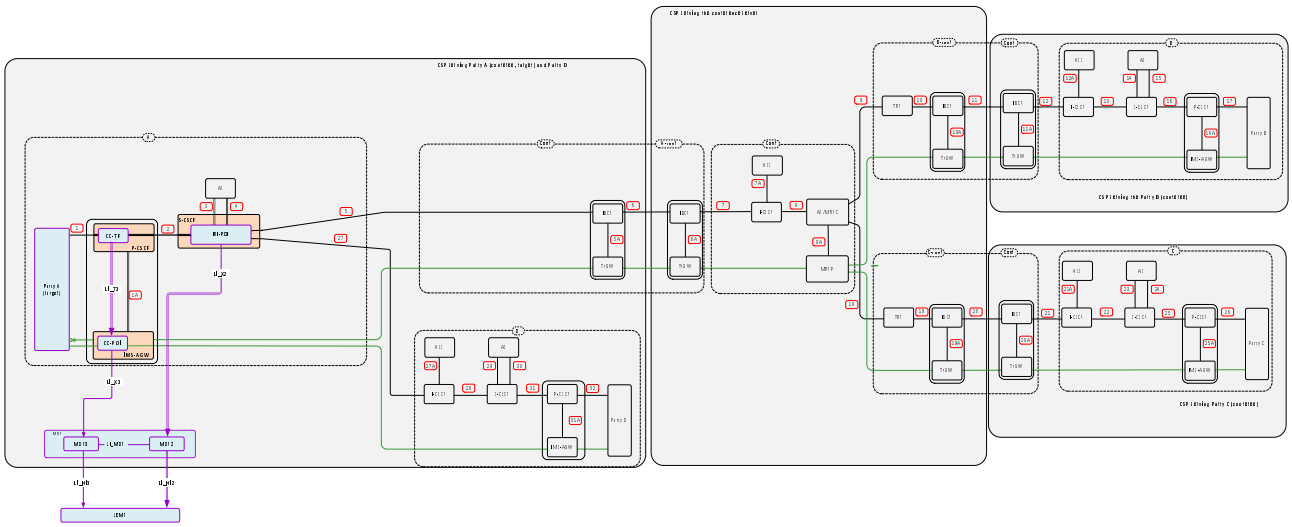


Figure 4.6.4.2-1: Party A (target) is in a session with Party D with group conference on hold

As shown in figure 4.6.4.2-1, only the network functions that handle the A-side (target side) of the session are involved in handling the LI functions. The conference server is in a different CSP domain. Party B and Party C are in different IP domains. Party D is in the same CSP domain as that of Party A.

For session progressive steps, the illustration assumes that the Party A dials into the conference, then through the conference invites the Party B and Party C, and then places that session leg on hold before establishing the session to Party D. The steps should be read as follows:

- Party A dialling into the conference URI (target) follows the steps 1 to 8A.
- Conference URI invitation to Party B follows the steps 9 to 17.
- Conference URI invitation to Party C follows the steps 18 to 26.
- After placing the group conference on hold, Party A's IMS session establishment to Party D follows the steps 27 to 32.

Held session

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI for the held session. The xCC for the held session is not generated.

Active session with Party D

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF. The IRI messages and CC delivered to the LEMF for the two sessions are correlated individually.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.4.3 Party A (target) in communication with Party D, with group conference on hold (alternative)

Figure 4.6.4.3-1 shows an alternative to the scenario shown in figure 4.6.4.2-1, where Party A (target) is in an IMS session with Party D with group conference on hold.

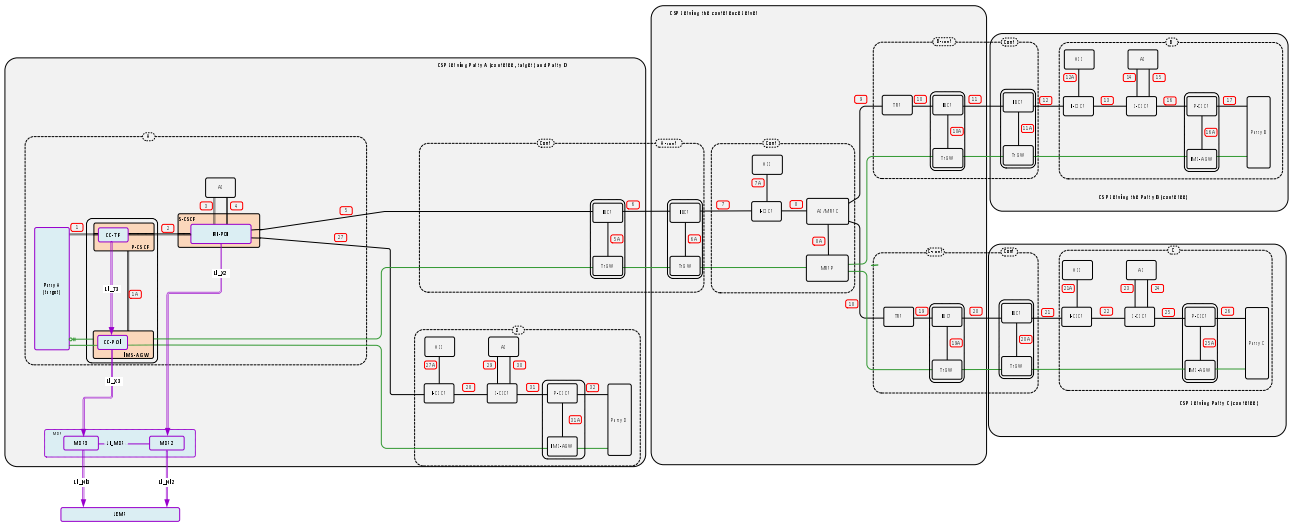


Figure 4.6.4.3-1: Party A (target) is in a session with Party D with group conference on hold (alternative)

In this alternative, the xCC generation at the CC-POI present in the IMS-AGW is not stopped for the held session leg. Other than that, all the descriptions described in clause 4.6.4.2 apply to the scenario illustrated in figure 4.6.4.3-1.

4.6.4.4 Party A places a group conference on hold, conference URI target

Figure 4.6.4.4-1 shows a scenario where Party A is in an IMS session with Party D with group conference on hold. In this illustration, conference URI is the target.

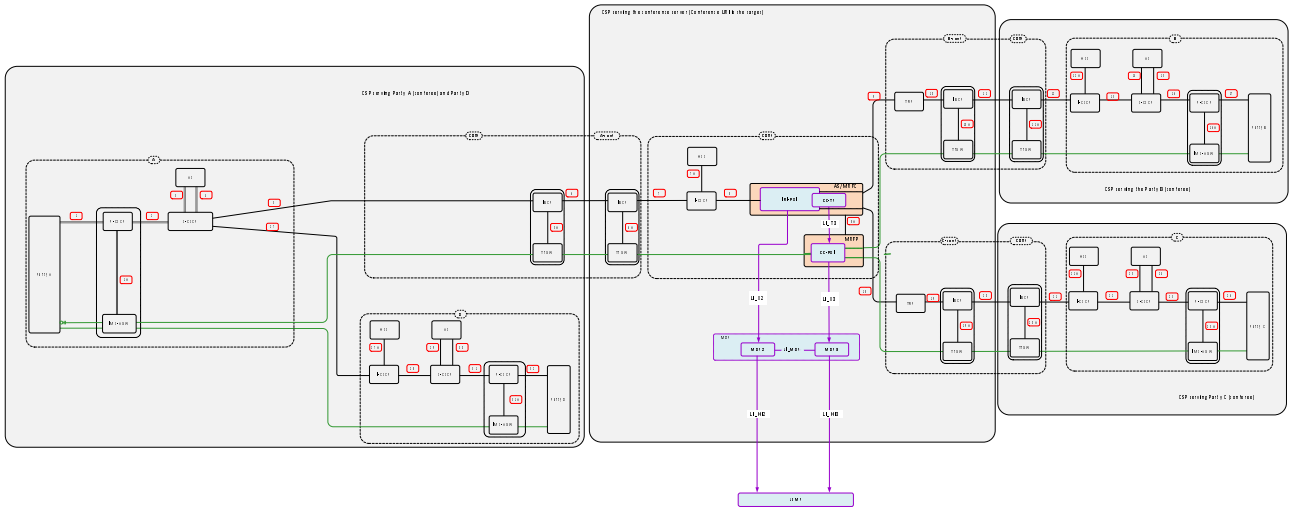


Figure 4.6.4.4-1: Party A is in a session with Party D, with group conference on hold (conference URI target)

As shown in figure 4.6.4.4-1 the interception is in the CSP domain that serves the conference URI, the target. Party A, Party B, Party C are different IMS domains. Party D is in the same CSP domain as that of Party A.

For session progressive steps, the illustration assumes that the Party A dials into the conference, then through the conference invites the Party B and Party C, and then places that session leg on hold before establishing the session to Party D. The steps should be read as follows:

- Party A dialling into the conference URI (target) follows the steps 1 to 8A.
- Conference URI invitation to Party B follows the steps 9 to 17.
- Conference URI invitation to Party C follows the steps 18 to 26.
- After placing the group conference on hold, Party A's IMS session establishment to Party D follows the steps 27 to 32.

The IRI-POI present in the AS/MRFC accesses the SIP messages and generates the required xIRI. The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE: From the perspective of CC-POI present in the MRFP, the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C, however, in reality, that CC does not include any communication content from Party A, since group conference is placed on hold by the Party A.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.4.5 Party A and Party C place a group conference on hold, conference URI target

Figure 4.6.4.5-1 shows a scenario where Party A is in an IMS session with Party D, with a group conference on hold. Also, Party C is on an IMS session with Party E with a group conference on hold. In this illustration, the conference URI is the target.

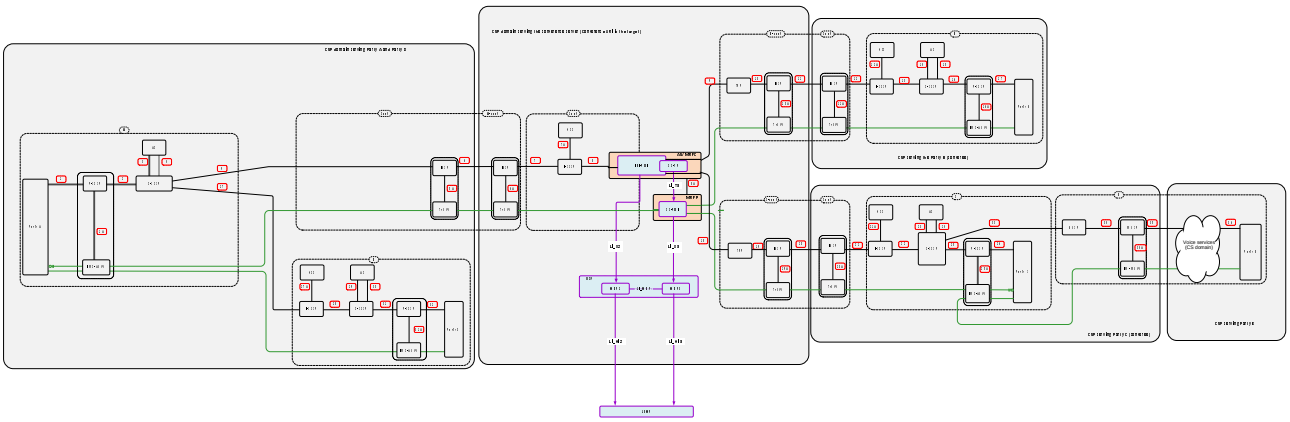


Figure 4.6.4.5-1: Party A and Party C have placed group conference on hold (conference URI target)

As shown in figure 4.6.4.5-1 the interception is in the CSP domain that serves the conference URI, the target. Party A, Party B, Party C are different IMS domains. Party D is in the same CSP domain as that of Party A. Party E is in another IP domain.

For session progressive steps, the illustration assumes that the Party A dials into the conference, then through the conference invites the Party B and Party C, and then places that session leg on hold before establishing the session to Party D. The Party C also places the group session leg on hold before establishing a session to Party E. The steps should be read as follows:

- Party A dialling into the conference URI (target) follows the steps 1 to 8A.
- Conference URI invitation to Party B follows the steps 9 to 17.
- Conference URI invitation to Party C follows the steps 18 to 26.
- After placing the group conference on hold, Party A's IMS session establishment to Party D follows the steps 27 to 32.
- After placing the group conference on hold, Party C's IMS session establishment to Party E follows the steps 33 to 36.

The IRI-POI present in the AS/MRFC accesses the SIP messages and generates the required xIRI. The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE: From the perspective of CC-POI present in the MRFP, the CC delivered to the LEMF includes the communication content of Party A, Party B and Party C, however, in reality, that CC does not include any communication content from Party A or Party C, since the group conference is placed on hold by the Party A and by the Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.4.6 Party A places a group conference with Party B and Party C (target) on hold

Figure 4.6.4.6-1 shows a scenario where Party A is in an IMS session with Party D with a group conference on hold. Also, Party C, another conference participant, is the target in this illustration.

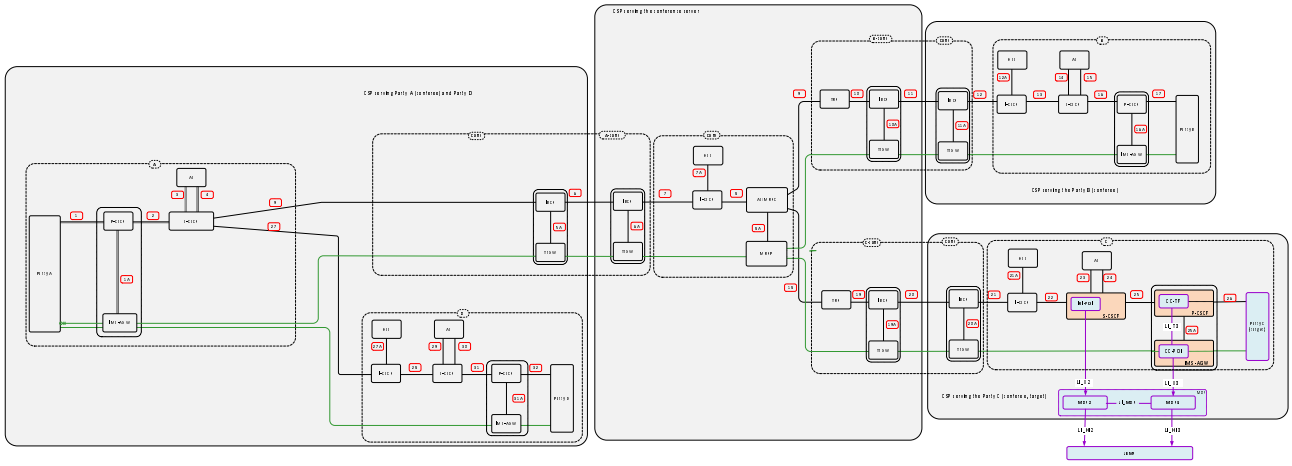


Figure 4.6.4.6-1: Party A placed group conference on hold with Party C as the target

As shown in figure 4.6.4.6-1, the interception is in the CSP domain that serves the Party C, the target. Party A, Party B and Party C are in different IMS domains. Party D is in the same CSP domain as that of Party A. The conference server is in a different CSP domain.

NOTE 1: The topology will be similar if the Party B is the target.

For session progressive steps, the illustration assumes that the Party A dials into the conference, then through the conference invites the Party B and Party C, and then places that session leg on hold before establishing the session to Party D. The steps should be read as follows:

- Party A dialling into the conference URI (target) follows the steps 1 to 8A.
- Conference URI invitation to Party B follows the steps 9 to 17.
- Conference URI invitation to Party C follows the steps 18 to 26.
- After placing the group conference on hold, Party A's IMS session establishment to Party D follows the steps 27 to 32.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

NOTE 2: From the perspective of CC-POI present in the IMS-AGW, the CC delivered to the LEMF includes the communication content of Party C, receives and transmits. In the illustration, that includes the communication content coming out of the group conference (Party B) and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.6.4.7 Special case – multiple targets

Figure 4.6.4.7-1 shows a scenario where Party A is in an IMS session with Party D with a group conference on hold. In this illustration, Party A, Party C and the conference URI are the targets in the respective CSP domains.

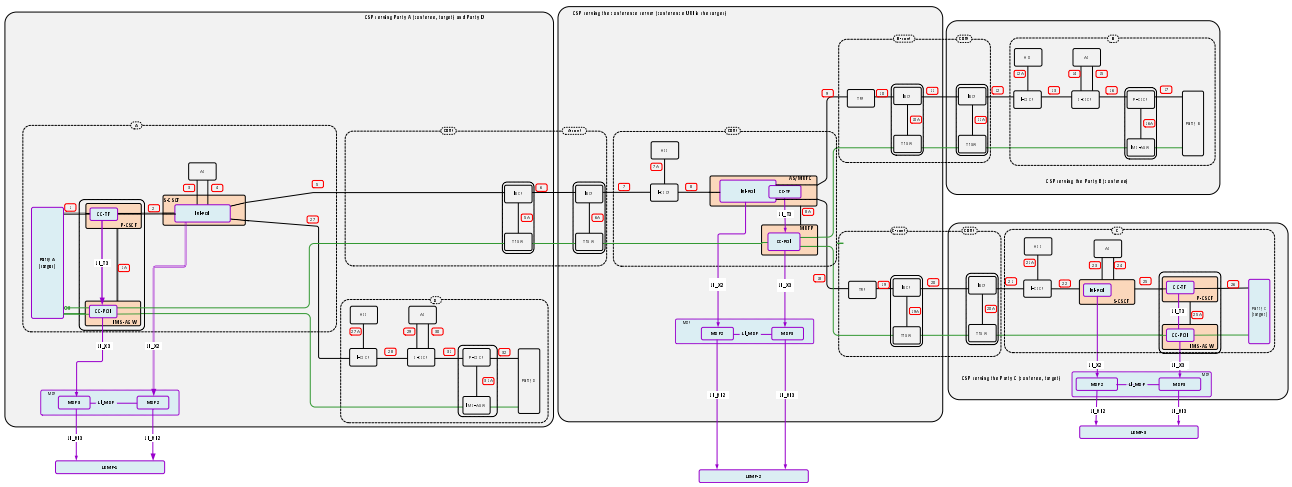


Figure 4.6.4.7-1: Party A (target) placed group conference on hold (conference URI and Party C are targets as well)

As shown in figure 4.6.4.7-1, the interception is in the CSP domains that serve the targets Party A, the conference URI and Party C. The conference server is in a different CSP domain. Party B and Party C are in different IP domains. Party D is in the same CSP domain as that of Party A.

NOTE 1: The topology will be similar if the Party B is the target instead of Party C.

For session progressive steps, the illustration assumes that the Party A dials into the conference, then through the conference invites the Party B and Party C, and then places that session leg on hold before establishing the session to Party D. The steps should be read as follows:

- Party A dialling into the conference URI (target) follows the steps 1 to 8A.
- Conference URI invitation to Party B follows the steps 9 to 17.
- Conference URI invitation to Party C follows the steps 18 to 26.
- After placing the group conference on hold, Party A's IMS session establishment to Party D follows the steps 27 to 32.

CSP domain that serves Party A:

Held session

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI for the held session. The xCC for the held session is not generated.

Active session with Party D

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC. The CC-POI present in the IMS-AGW generates the xCC.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-1. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-1. The IRI messages and CC delivered to the LEMF for the two sessions are correlated individually.

CSP domain that serves conference URI:

- The IRI-POI present in the AS/MRFC accesses the SIP messages and generates the required xIRI. The CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC.

- The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-2.

NOTE 2: From the perspective of CC-POI present in the MRFP, the CC delivered to the LEMF-2 includes the communication content of Party A, Party B and Party C, however, in reality, that CC does not include any communication content from Party A, since group conference is placed on hold by the Party A.

CSP domain that serves Party C:

- The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. The CC-TF present in the P-CSCF triggers the CC-POI present in the IMS-AGW for the xCC.

- The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF-3. The MDF3 generates the CC from the received xCC and delivers it to the LEMF-3.

NOTE 3: From the perspective of CC-POI present in the IMS-AGW, the CC delivered to the LEMF-3 includes the communication content of Party C receives and transmits. In the illustration, that includes the communication content coming out of the group conference (Party B) and Party C.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.7 Special media on target terminating sessions

4.7.1 General

The subsequent clauses illustrate the scenarios where special media (e.g. early media with special tones, music, announcement) are applied to the IMS terminating sessions to a target prior to the answer. Once the session is answered, the two-way communication path is established between the originating party and the terminating party (target).

In the scenarios illustrated, it is assumed that such special media is applied at the MRFP. Therefore, in the illustrated scenarios, the CC interception is provided at the MRFP prior to the call answer.

4.7.2 Special media with intra-office terminating sessions

Figure 4.7.2-1 shows a scenario where Party A calls Party B (target), however, prior to the answer, the media path is routed to an MRFP to provide the special media to the originating party.

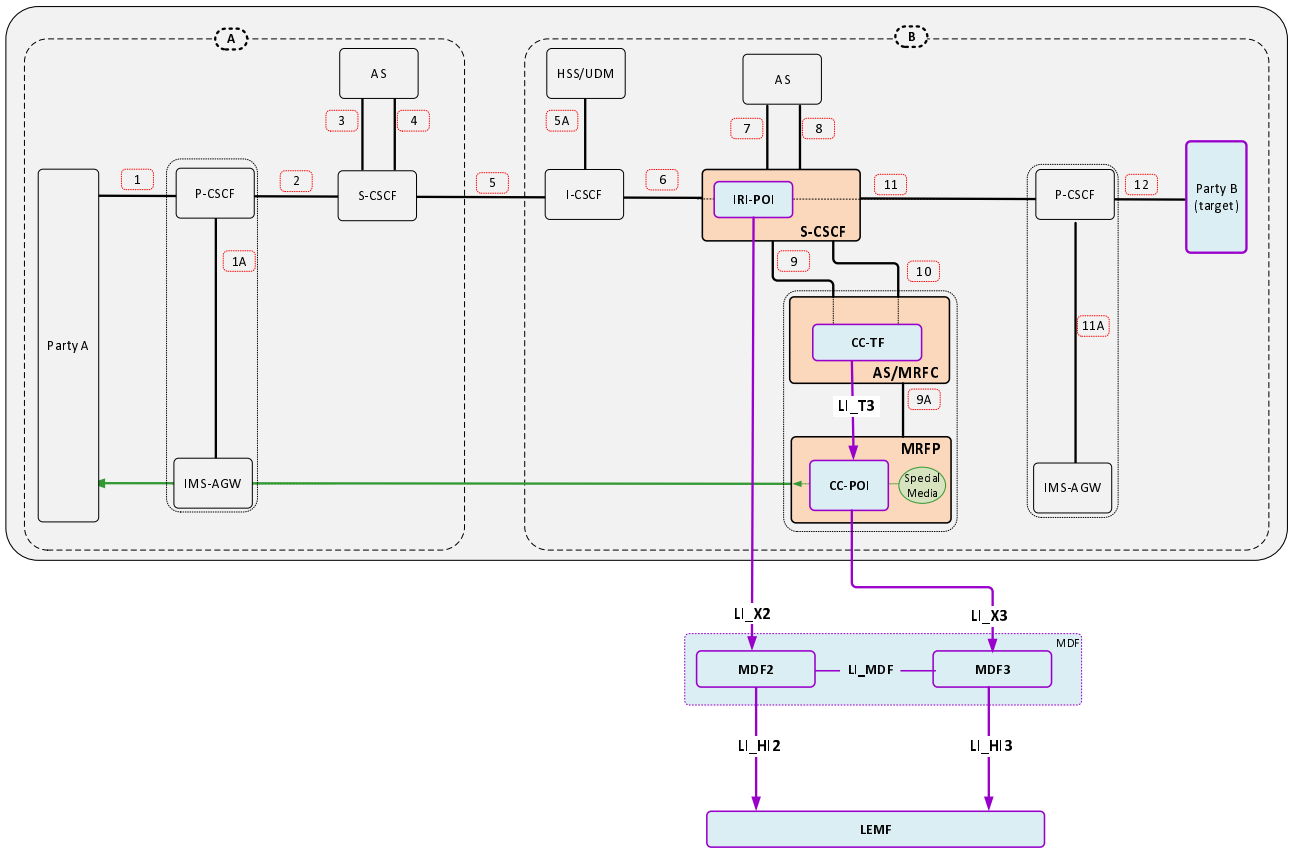


Figure 4.7.2-1: Party A calls Party B (target) with special media prior to answer

As shown in figure 4.7.2-1, only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. Prior to the answer, the CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC. The CC-POI present in the MRFP generates the xCC.

NOTE: Figure 4.7.2-1 shows that a separate AS is handling the special media. However, it is possible to have an architecture that uses only one AS in the signalling path.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

When the session is answered by the Party B (target), the media path is reconfigured with CC-TF present in the P-CSCF triggering the CC-POI present in the IMS-AGW for the CC interception. The IRI-POI present in the S-CSCF continues to provide the IRI interception. The architecture diagram showing these are illustrated in figure 4.2.3.1-1, clause 4.2.3.

It is also possible that the session may get redirected due to no answer. The redirection may be intra-office, or inter-office, or to an outbound roaming party. Figure 4.2.4.1-1 (clause 4.2.4) shows an architecture diagram for intra-office redirected sessions, figures 4.3.4.2-1 and 4.3.4.3-1 (clause 4.3.4) show the architecture diagrams for inter-office redirected sessions and figure 4.4.4.3-1 (clause 4.4.4) shows an architecture diagram when the redirection is to an outbound roaming party.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.7.3 Special media with inter-office terminating sessions

Figure 4.7.3-1 shows a scenario where Party A from a CS domain (another CSP) calls Party B (target), however, prior to the answer, the media path is routed to an MRFP to provide the special media to the originating party.

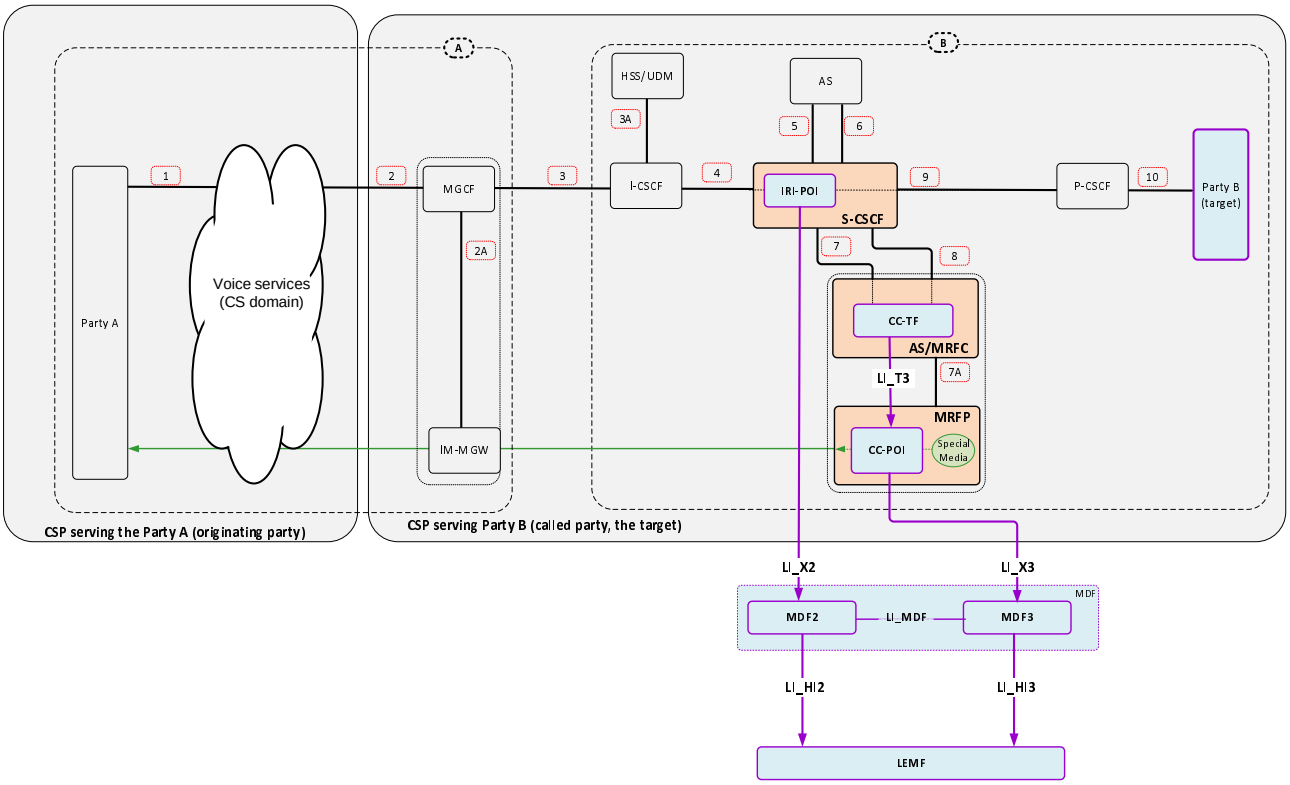


Figure 4.7.3-1: Inter-office terminating sessions to Party B (target) with special media prior to answer

As shown in figure 4.7.3-1, only the network functions that handle the B-side (target side) of the session are involved in handling the LI functions.

The IRI-POI present in the S-CSCF accesses the SIP messages and generates the required xIRI. Prior to the answer, the CC-TF present in the AS/MRFC triggers the CC-POI present in the MRFP for the xCC. The CC-POI present in the MRFP generates the xCC.

NOTE: Figure 4.7.3-1 shows that a separate AS is handling the special media. However, it is possible to have an architecture that uses only one AS in the signalling path.

The MDF2 generates the IRI messages from the xIRI and delivers them to the LEMF. The MDF3 generates the CC from the received xCC and delivers it to the LEMF.

When the session is answered by the Party B (target), the media path is reconfigured with CC-TF present in the P-CSCF triggering the CC-POI present in the IMS-AGW for the CC interception. The IRI-POI present in the S-CSCF continues to provide the IRI interception. The architecture diagram showing these are illustrated in figure 4.3.2.2-1, clause 4.3.2.

As described in clause 4.7.2, it is also possible that the session may get redirected due to no answer. The redirection may be intra-office, or inter-office, or to an outbound roaming party. The architecture diagrams that illustrate the IRI and CC interception are similar to the diagrams shown in figure 4.2.4.1-1 (clause 4.2.4), 4.3.4.2-1 and 4.3.4.3-1 (clause 4.3.4) and 4.4.4.3-1 (clause 4.4.4) except that the sessions are originated from another CSP domain.

The details of the above LI functions and the interfaces are described in TS 33.127 [3] and TS 33.128 [4].

4.8 Home routed roaming architecture

4.8.1 Introduction

This clause gives an overview of the principles of the LI functions to be provided within the visited CSP domain for voice communications involving a target when the home routed roaming architecture is used.

With home routed roaming architecture, all the IMS related functions reside in the home CSP domain. A part of the packet core network also resides in the home CSP domain.

With 5GC as the packet core network, the SMF/UPF reside in both home CSP domain (HPLMN) and the visited CSP domain (VPLMN). With EPC as the packet core network, the PGW resides in the home CSP domain (i.e. in the HPLMN) and the SGW resides in the visited CSP domain (i.e. in the VPLMN).

The reference point (aka interface) used between the VPLMN and the HPLMN to transport the packets that carry the IMS signalling messages and the IMS media is referenced in naming the LI functions provided in the visited CSP domain with home routed roaming architecture:

- N9HR LI: The reference point N9 is used between the UPF (in the VPLMN) and the UPF (in the HPLMN).
- S8HR LI: The reference point S8 is used between the SGW (in the VPLMN) and the PGW (in the HPLMN).
- S8UHR LI: The reference point S8-U is used between the SGW-U (in the VPLMN) and the PGW-U (in the HPLMN).

N9HR LI refers to the LI functions provided in the visited CSP domain for voice services involving a target with 5GC based packet core network. Similarly, S8HR LI refers to the LI functions provided in the visited CSP domain for voice services involving a target with an EPC based packet core network.

In the EPC, the SGW and the PGW may also be deployed with Control and User Plane Separated (CUPS) with the SGW-C and PGW-C providing the control plane functions and with the SGW-U and PGW-U providing the user plane functions. The reference point between the SGW-U and the PGW-U is referred to as S8-U. Accordingly, S8UHR refers to the LI functions provided in the visited CSP domain for voice services involving a target with EPC based packet core network deployed with the CUPS model.

NOTE: As an alternate to the introduction of a new term S8UHR LI, the same can be named S8HR LI when the EPC is deployed with CUPS. This alternate approach is preferred in the TR.

4.8.2 Topology overviews

4.8.2.1 N9HR LI

The overall network configuration of a 5G system with a home-routed architecture with the LI aspects in the VPLMN is shown in figure 4.8.2.1-1 using the service-based representation (as shown in TS 23.501 [6]) with the use of point-to-point LI system.

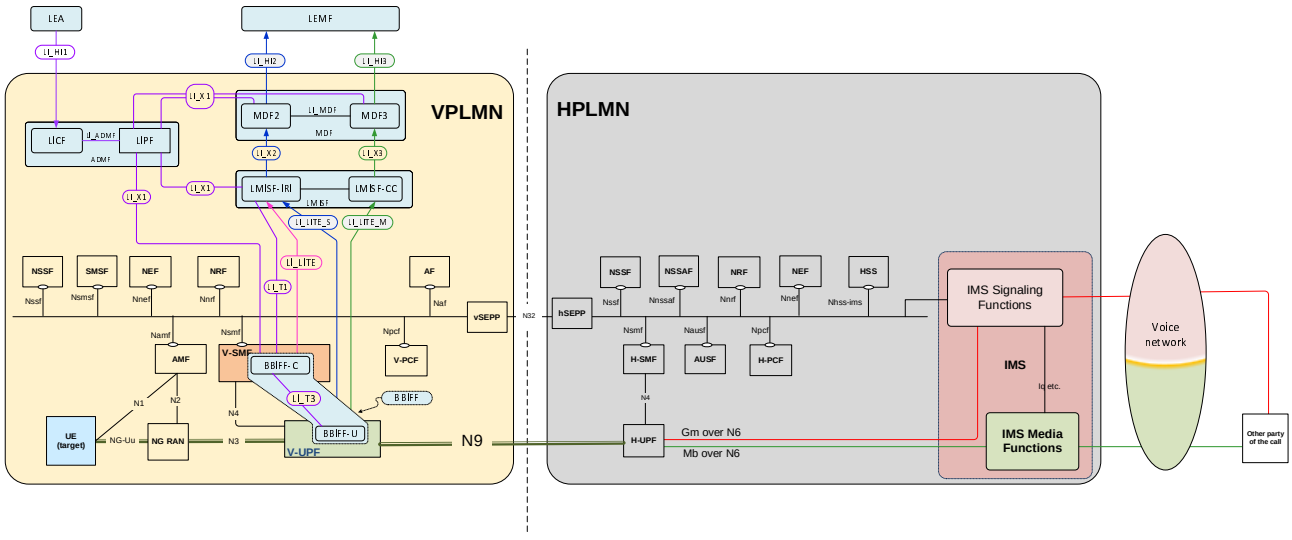


Figure 4.8.2.1-1: Topology view of N9HR LI system

As specified in TS 33.127 [2], the N9HR LI introduces new LI specifics functions referred to as LMISF-IRI and LMISF-CC (together referred to as LMISF) and BBIFF-C and BBIFF-U (together referred to as BBIFF). The BBIFF-C is present in the V-SMF and the BBIFF-U is present in the V-UPF.

For N9HR LI:

- BBIFF-C present in the V-SMF notifies the LMISF-IRI whenever a PDU session is established, modified or deleted. Such notifications include the UE location. This is done for all inbound roaming UEs with N9HR.
- BBIFF-U present in the V-UPF delivers the User Plane (UP) packets that relate to the IMS signalling to the LMISF-IRI. This is done for all inbound roaming UEs with N9HR.
- LMISF-IRI provides the IRI-POI functions for the IMS signalling related xIRI. This is done for target communications.
- When the interception requires CC, the LMISF-IRI triggers the BBIFF-C present in the V-SMF over LI_T1 interface. The BBIFF-C present in the V-SMF passes the trigger to BBIFF-U present in the V-UPF over LI_T3 interface.
- BBIFF-U present in the V-UPF delivers the UP packets that relate to the IMS media to the LMISF-CC. This is done for target communications that requires CC interception.
- LMLISF-CC provides the CC-POI functions for the IMS media related xCC. This is done for target communications that requires CC interception.

The LIPF present in the ADMF configures and provisions MDF2, MDF3, LIMISF-IRI and BBIFF-C for the N9HR LI related intercept data.

4.8.2.2 S8HR LI, EPC with CUPS

This clause refers when the EPC is deployed with CUPS. The overall network configuration of an EPC system with a home-routed architecture with the LI aspects in the VPLMN is shown in figure 4.8.2.2-1.

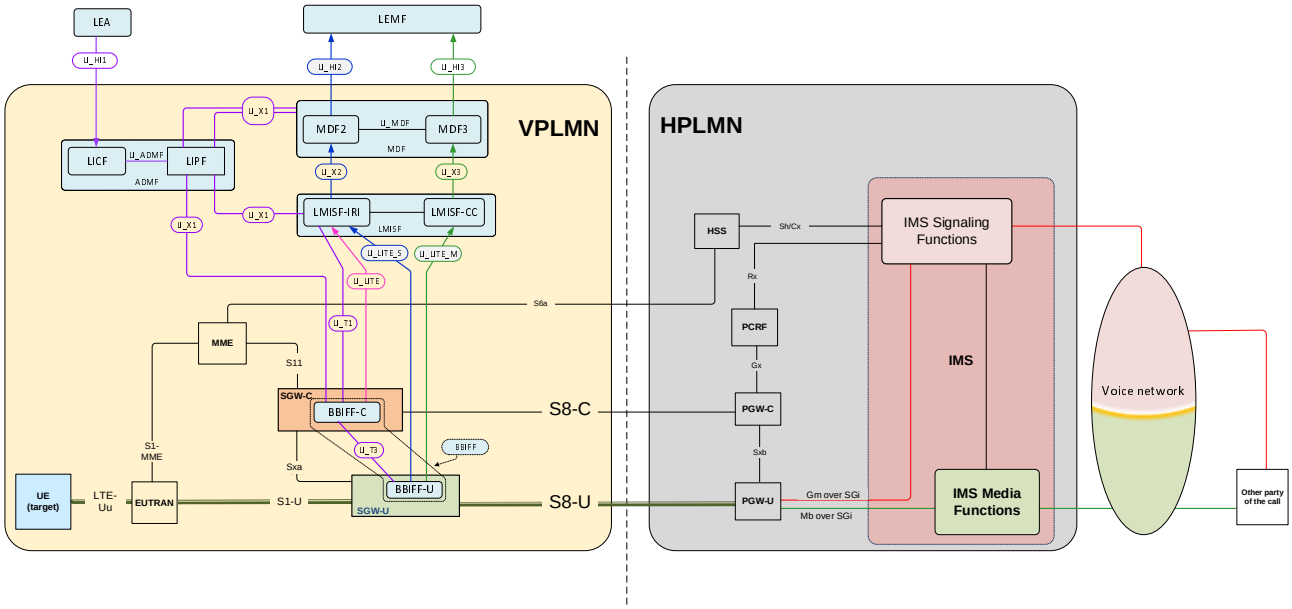


Figure 4.8.2.2-1: Topology view of S8HR LI (with CUPS)

As specified in TS 33.127 [2], the S8HR LI introduces new LI specific functions referred to as LMISF-IRI and LMISF-CC (together referred to as LMISF) and BBIFF-C and BBIFF-U (together referred to as BBIFF). The BBIFF-C is present in the SGW-C and the BBIFF-U is present in the SGW-U.

For S8HR LI where EPC is deployed with CUPS:

- BBIFF-C present in the SGW-C notifies the LMISF-IRI whenever an IMS bearer within a PDN connection is created, modified, or deleted with the UE location. This is done for all inbound roaming UEs with S8HR.
- BBIFF-U present in the SGW-U delivers the User Plane (UP) packets that relate to the IMS signalling to the LMISF-IRI. This is done for all inbound roaming UEs with S8HR.
- LMISF-IRI provides the IRI-POI functions for the IMS signalling related xIRI. This is done for target communications.

- When the interception requires CC, the LMISF-IRI triggers the BBIFF-C present in the SGW-C over LI_T1 interface. The BBIFF-C present in the SGW-C passes the trigger to BBIFF-U present in the SGW-U over LI_T3 interface.
- BBIFF-U present in the SGW-U delivers the UP packets that relate to the IMS media bearer to the LMISF-CC. This is done for target communications that requires CC interception.
- LMISF-CC provides the CC-POI functions for the IMS media related xCC. This is done for target communications that requires CC interception.

The LIPF present in the ADMF configures and provisions MDF2, MDF3, LIMISF-IRI and BBIFF-C for the S8HR LI related intercept data.

4.8.2.3 S8HR LI, EPC without CUPS

This clause refers when the EPC is deployed without the CUPS. The overall network configuration of an EPC system with a home-routed architecture with the LI aspects in the VPLMN is shown in figure 4.8.2.3-1.

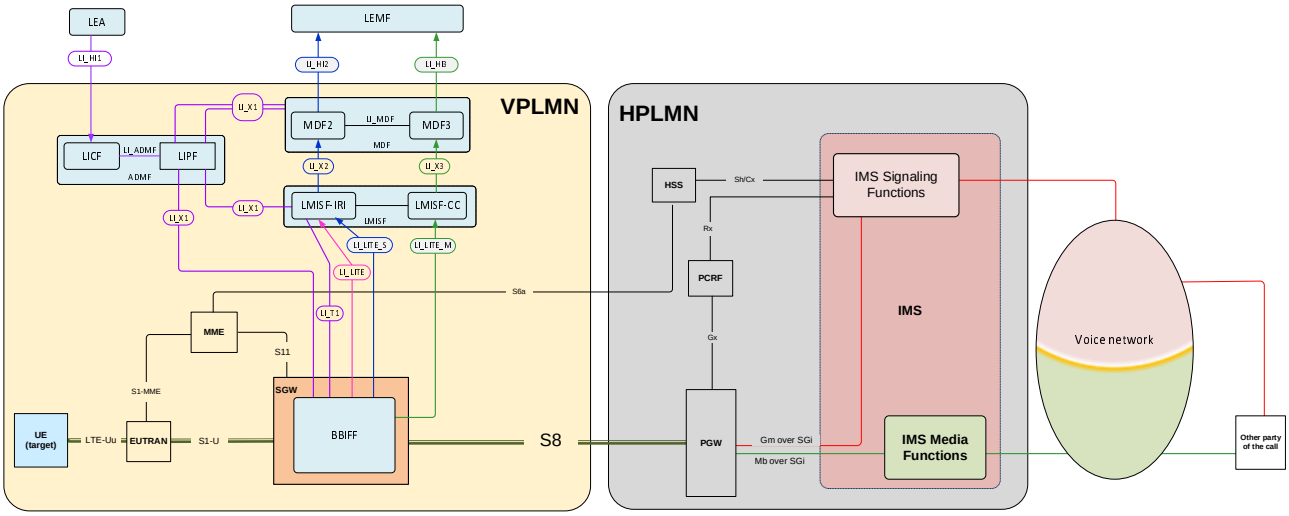


Figure 4.8.2.3-1: Topology view of S8HR LI (EPC without CUPS)

As specified in TS 33.127 [2], the S8HR LI introduces new LI specifics function referred to as LMISF-IRI and LMISF-CC (together referred to as LMISF) and BBIFF in the SGW.

For S8HR LI where EPC is deployed without CUPS:

- BBIFF present in the SGW notifies the LMISF-IRI whenever an IMS bearer within a PDN connection is created, modified, or deleted with the UE location. This is done for all inbound roaming UEs with S8HR.
- BBIFF present in the SGW delivers the User Plane (UP) packets that relate to the IMS signalling to the LMISF-IRI. This is done for all inbound roaming UEs with S8HR.
- LMISF-IRI provides the IRI-POI functions for the IMS signalling related xIRI. This is done for target communications.
- When the interception requires CC, the LMISF-IRI triggers the BBIFF present in the SGW over LI_T1 interface.
- BBIFF present in the SGW delivers the UP packets that relate to the IMS media bearer to the LMISF-CC. This is done for target communications that requires CC interception.
- LMISF-CC provides the CC-POI functions for the IMS media related xCC. This is done for target communications that requires CC interception.

The LIPF present in the ADMF configures and provisions MDF2, MDF3, LIMISF-IRI and BBIFF for the S8HR LI related intercept data.

4.8.3 HR LI principles

4.8.3.1 N9HR LI

Within the 5G packet core system in relation to N9HR, the IMS signalling packets and the IMS media packets are transported as User Plane (UP) packets over different QoS flows within the same PDU session established between the UE and the H-UPF. A PDU session has a GTP tunnel between the NG-RAN and the V-UPF and then between the V-UPF and the H-UPF. The QoS Flow Identifier (QFI) present within the GTP-U extension header identifies the QoS flow over which the UP packets are transported. In other words, the QFI present in the GTP-U extension header can be used to identify a particular type of UP packet (e.g. IMS signalling, IMS media).

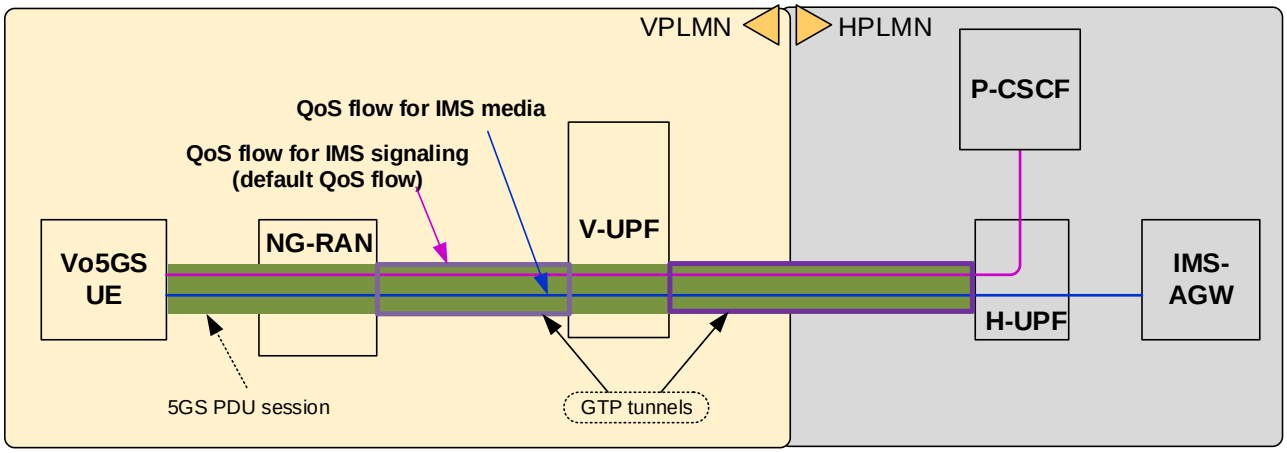


Figure 4.8.3.1-1: IMS signalling and media related QoS flows within the same PDU session

NOTE 1: At the control plane, the 5QI values identify the QoS characteristics of a QoS flow. According to the GSMA guidelines, the 5QI value 5 of is used for IMS signalling and 5QI value of 1 is used for IMS voice media.

NOTE 2: NG-RAN inserts the QFI into the GTP-U extension header for the upstream UP packets and the H-UPF insert the QFI into the GTP-U extension header for the downstream UP packets.

NOTE 3: The UP plane packets associated with a particular QoS flow can be associated with one or more IMS session legs (e.g. IMS hold).

The process of providing the LI functions is described in TS 33.127 [2]. This clause depicts the same using the flow diagrams observing the following points:

- IMS signalling messages are between the UE and the IMS Signalling Functions in the HPLMN and use default QoS flow of the PDU session.
- IMS media is between the UE and the IMS Media Functions in the HPLMN and use dedicated QoS flow of the PDU session.
- PDU session establishment (default QoS flow) is requested by the UE during the initial registration phase.
- The PDU session is modified to add IMS media related QoS flows. The PDU session modification is initiated by the HPLMN for IMS media related QoS flow.
- H-SMF in the HPLMN is aware of the QFI used for a QoS flow and provides the same to H-UPF and V-SMF during the PDU session establishment/modification steps. H-SMF also provides the 5QI value that relates to a QoS flow to the V-SMF during the PDU session establishment/modification steps.
- V-SMF provides the QFI, 5QI to the NG-RAN during the PDU session establishment/modification steps.
- V-SMF can determine QFI to QoS flow associations based on the 5QI values received from the H-SMF (i.e. 5 for IMS signalling related QoS flow and 1 for IMS voice media related QoS flow, see NOTE 1).
- Neither the QFI nor the 5QI is passed to the V-UPF by the V-SMF and hence, the V-UPF is not aware of any relationship between the QFI value and the packet type.

- V-UPF, however, can be made to recognise and hence, isolate the UP packets based on a QFI value (present in the GTP-U extension header). N9HR LI is based on this principle.

- When a SMF/UPF or just the UPF relocation occurs, the new SMF can determine about the relocation based on the information that it receives during the PDU session establishment procedures (upCnxState IE is present).

During the initial PDU session establishment (the same time, default QoS flow is created), the BBIFF-C present in the V-SMF includes that QFI value (associated to the default QoS flow which is used for the IMS signalling) in the trigger sent to the BBIFF-U present in the V-UPF to identify and capture the UP packets with that QFI. The captured UP packets at the BBIFF-U present in the V-UPF are the IMS signalling UP packets.

When the PDU session is modified to add IMS media related QoS flow, the BBIFF-C present in the V-SMF includes the QFI value (associated to the QoS flow used for the IMS voice media) in the trigger sent to the BBIFF-U present in the V-UPF to identify and capture the UP packets with that QFI. The captured UP packets at the BBIFF-U present in the V-UPF are the IMS voice media related UP packets.

4.8.3.2 S8HR LI, EPC with CUPS

When the EPC system is deployed with the CUPS, the IMS signalling packets and IMS media packets are transported as User Plane (UP) packets over different bearers of the PDN connection established between the UE and the PGW-U. Each bearer within a PDN connection has a GTP tunnel between the eNB (EUTRAN Access Node) and the SGW-U and then between the SGW-U and the PGW-U. The UP packets transported within a bearer have the same QoS characteristics. The IMS signalling packets are transported over the default bearer (aka IMS signalling bearer) and IMS media packets are transported over dedicated bearers (aka IMS media bearer).

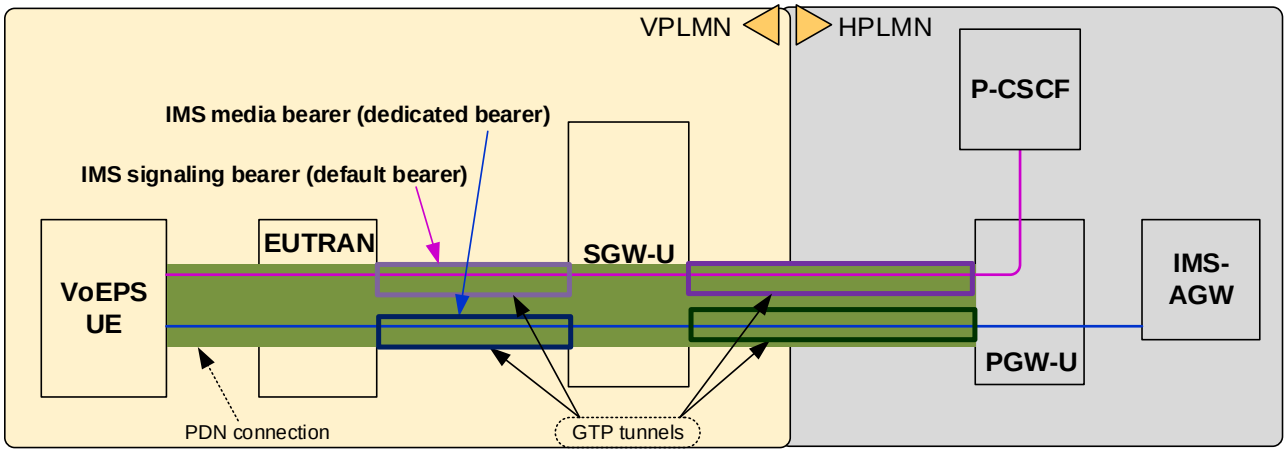


Figure 4.8.3.2-1: IMS signalling and media related bearers within a PDN connection (with CUPS)

NOTE 1: At the control plane, the QCI values identify the QoS characteristics of a bearer. According to the GSMA guidelines, the QCI value 5 of is used for IMS signalling and QCI value of 1 is used for IMS voice media.

NOTE 2: The UP plane packets of a bearer can be associated with one or more IMS session legs (e.g. IMS hold).

The process of providing the LI functions is described in TS 33.127 [2]. This clause depicts the same using the flow diagrams observing the following points:

- The IMS signalling messages are between the UE and the IMS Signalling Functions in HPLMN and are transported over the default bearer of the PDU connection. The bearer is also referred to as IMS signalling bearer.

- The IMS media packets are between the UE and the IMS Media Functions in the HPLMN and are transported over a dedicated bearer of the PDN connection. This dedicated bearer is also referred to as IMS media bearer.

- The PGW-C in HPLMN is aware of the QCI used for a bearer and provides the same to the SGW-C during the bearer activation steps.

- The SGW-C can determine whether a bearer is used for IMS signalling or IMS media based on the QCI value received from the PGW-C (i.e. 5 for IMS signalling bearer and 1 for IMS voice media bearer, see NOTE 1).

- Within the SGW-U, the IMS signalling packets and the IMS voice media packets are on separate bearers of the PDN connection. Therefore, based on the bearer information, the SGW-U can be made to recognise and hence, isolate the UP packets. S8HR LI is based on this principle.

- The IMS signalling bearer ID is linked to the IMS media bearer. This helps to associate the IMS media with the IMS signalling.

- When a SGW relocation occurs, the new SGW-C can determine that a relocation has occurred based on the timing at which it receives the eNB tunnel ID used for the default bearer during the PDN connection establishment procedure.

When the default bearer is created, the BBIFF-C present in the SGW-C instructs the BBIFF-U present in the SGW-U to capture the UP packets from that bearer. The captured UP packets at the BBIFF-U are the IMS signalling UP packets.

When the dedicated bearer for IMS voice media is created, the BBIFF-C present in the SGW-C instructs the BBIFF-U present in the SGW-U to capture the UP packets from that bearer. The captured UP packets at the BBIFF-U present in the SGW-U are the IMS voice media related UP packets.

4.8.3.3 S8HR LI, EPC without CUPS

When the EPC system is deployed without CUPS, the IMS signalling packets and the IMS media packets are transported as User Plane (UP) packets over different bearers of the PDN connection established between the UE and the PGW. Each bearer within a PDN connection has a GTP tunnel between the eNB (EUTRAN Access Node) and the SGW and then between the SGW and the PGW. The UP packets transported within a bearer have the same QoS characteristics. The IMS signalling packets are transported over the default bearer (aka IMS signalling bearer) and the IMS media packets are transported over dedicated bearers (aka IMS media bearer).

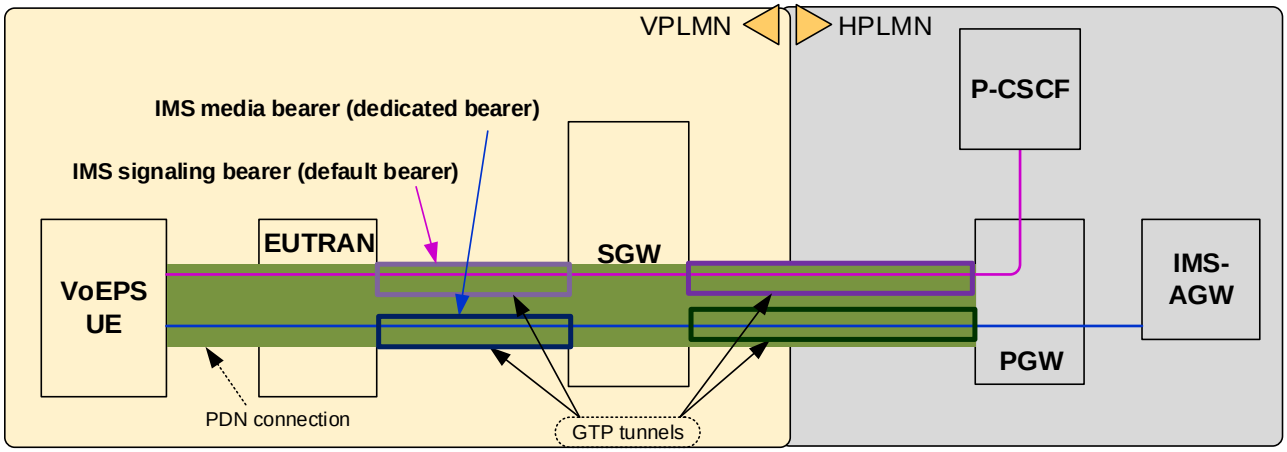


Figure 4.8.3.3-1: IMS signalling and media related Bearers within a PDN connection (without CUPS)

NOTE 1: At the control plane, the QCI values identify the QoS characteristics of a bearer. According to the GSMA guidelines, the QCI value 5 of is used for IMS signalling and QCI value of 1 is used for IMS voice media.

NOTE 2: The UP plane packets of a bearer can be associated with one or more IMS session legs (e.g. IMS hold).

The process of providing the LI functions is described in TS 33.127 [2]. This clause depicts the same using the flow diagrams observing the following points:

- The IMS signalling messages are between the UE and the IMS Signalling Functions in HPLMN and are transported over the default bearer of the PDU connection. The bearer is also referred to as IMS signalling bearer.

- The IMS media packets are between the UE and the IMS Media Functions in the HPLMN and are transported over a dedicated bearer in the PDN connection. This dedicated bearer is also referred to as IMS media bearer.

- The PGW in HPLMN is aware of the QCI used for a bearer and provides the same to SGW during the bearer activation steps.

- The SGW can determine whether a bearer is used for IMS signalling or IMS media based on the QCI value received from the PGW (i.e. 5 for IMS signalling bearer and 1 for IMS voice media bearer, see NOTE 1).
- Within the SGW, the IMS signalling packets and the IMS voice media packets are on separate bearers of the PDN connection. Therefore, based on the bearer information, the SGW can recognise and hence, isolate the UP packets. S8HR LI is based on this principle.
- The IMS signalling bearer ID is linked to the IMS media bearer. This helps to associate the IMS media with the IMS signalling.
- When a SGW relocation occurs, the new SGW can determine that a relocation has occurred based on the timing at which it receives the eNB tunnel ID used for the default bearer during the PDN connection establishment procedure.

BBIFF present in the SGW delivers the UP packets from the IMS signalling bearer to the LMISF-IRI for all inbound roamers and the UP packets from the IMS media bearer to the LMISF-CC for only for the target UEs that require CC interception.

4.8.4 HR LI process

4.8.4.1 N9HR LI

4.8.4.1.1 Initial configuration - applicable to all inbound roamers

The N9HR LI process applicable to all inbound roamers with home-routed roaming is illustrated in figure 4.8.4.1.1-1.

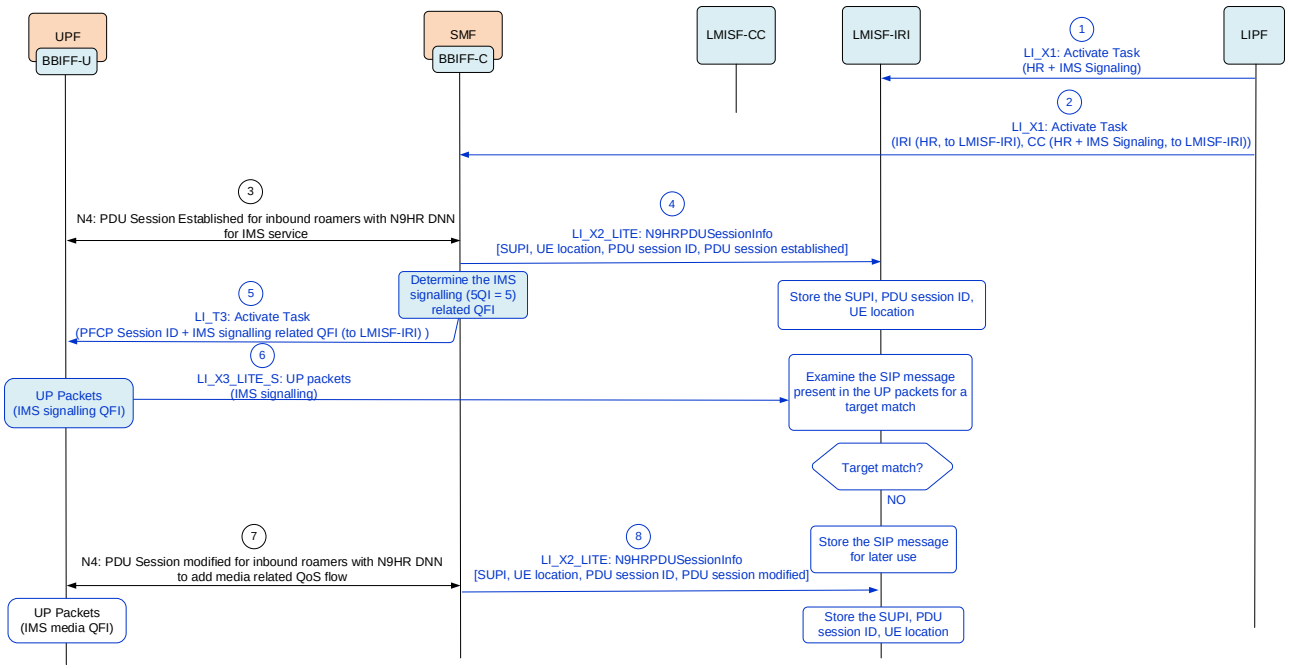


Figure 4.8.4.1.1-1: N9HR LI process – initial configuration (applicable to all inbound roamers)

The following describe the steps shown in figure 4.8.4.1.1-1:

1. LIPF initiates the N9HR LI process by provisioning the LMISF-IRI (over the LI_X1 reference point) with "HR" + "IMSSignaling"as the target identities. However, there is no DID specified in the Activate Task as the LMISF-IRI is not supposed to forward any of the received notifications.
2. LIPF provisions the BBIFF-C present in the SMF (over the LI_X1 reference point) with the "HR" as the target identity for xIRI and with "HR" + "IMSSignaling" as the target identities for xCC that corresponds to the IMS signalling. LMISF-IRI is the DID for both.

3. The BBIFF-C present in the SMF detects that PDU sessions are being established for the inbound roaming UEs with home-routed roaming (i.e. for N9HR DNN).

4. The BBIFF-C notifies the LMISF-IRI about the PDU session establishments over the LI_X2_LITE reference point. The notifications include the SUPI, PDU session ID and the UE location information. LMISF stores the SUPI, PDU session ID and the UE location information.

5. Based on the 5QI value of 5, the BBIFF-C present in the SMF determines that the QoS flow is for IMS signalling. The BBIFF-C sends a trigger over LI_T3 to the BBIFF-U present in the UPF to capture and deliver the UP packets associated with the QoS flow. The trigger includes the PFCP session ID and the QFI. The DID is the LMISF-IRI.

6. The BBIFF-U present in the UPF captures and delivers the UP packets from the PDU session that have the indicated QFI to the LMISF-IRI over the LI_X3_LITE_S reference point. The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. When there is no target match, the LMISF-IRI extracts and stores the SIP message.

7. The BBIFF-C present in the SMF detects that PDU sessions are modified for inbound roaming UEs with home-routed roaming.

8. The BBIFF-C present in the SMF notifies the LMISF-IRI about the PDU session modifications over the LI_X2_LITE reference point. The notifications include the SUPI, PDU session ID and the UE location information. LMISF stores the SUPI, PDU session ID and the UE location information.

NOTE 1: The UP packets for IMS media will have a different QFI than the one received in step 5 and hence, the BBIFF-U won't deliver those packets to the LMISF-IRI.

NOTE 2: The BBIFF-C would notify the LMISF-IRI whenever a PDU session is established, modified or deleted independent of the associated QoS flows using the xIRI: N9HRPDUSessionInfo message.

4.8.4.1.2 Intercept activation - applicable to target UE with IRI + CC

The N9HR LI process applicable to a target UE with home-routed roaming is illustrated in figure 4.8.4.1.2-1.

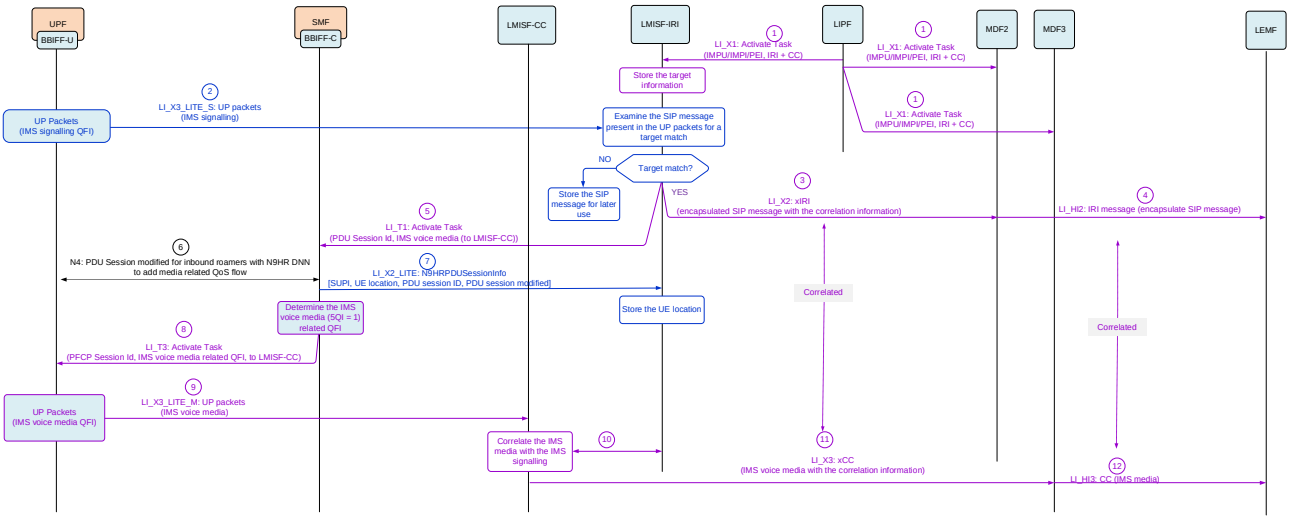


Figure 4.8.4.1.2-1: N9HR LI process – intercept activation (applicable to target UE with IRI + CC)

The following describe the steps shown in figure 4.8.4.1.2-1:

1. LIPF provisions the LMISF-IRI, MDF2 and MDF3 for IMS based voice interception with IRI + CC. The target identity is IMPU, IMPI or PEI as they are the identities supported with the IMS LI.

2. As per the procedures illustrated in flow of figure 4.8.4.1.1-1, the BBIFF-U present in the UPF captures and delivers the UP packets from the PDU session to the LMISF-IRI over LI_X3_LITE_S reference point according to the trigger previously received from the LMISF-IRI. The UP packets are associated with the IMS signalling related QoS flow.

3. The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. With a target match, the LMISF-IRI in the role of IRI-POI for IMS signalling related messages, generates the xIRI that contains the encapsulated SIP message and delivers the same to the MDF2.

4. The MDF2 delivers the IRI containing the encapsulated SIP message to the LEMF.

5. This interception requires IMS voice media interception and therefore, the LMISF-IRI sends a trigger to the BBIFF-C (over LI_T1) present in the SMF for the interception of IMS voice media (see NOTE 1). The trigger includes the SUPI and the PDU session ID previously received from the BBIFF-C. The DID included in the trigger is LMISF-CC.

6. The BBIFF-C present in the SMF detects that the PDU session is modified.

7. The BBIFF-C present in the SMF notifies the LMISF-IRI about the PDU session modification over the LI_X2_LITE reference point. The notification includes the SUPI, PDU session ID and the UE location information. The LMISF stores the PDU session ID and the UE location information.

8. The BBIFF-C present in the SMF determines that the PDU session modification (detected in step 6) is for IMS voice media related QoS flow (5QI = 1) and is associated with the PDU session ID indicated within the trigger received from the LMISF-IRI. The BBIFF-C sends a trigger over LI_T3 to the BBIFF-U present in the UPF to capture and deliver the UP packets associated with the QoS flow. The trigger includes the PFCP session ID and the QFI. The DID is LMISF-CC.

9. The BBIFF-U captures and delivers the UP packets from the PDU session that have the indicated QFI to the LMISF-CC over the LI_X3_LITE_M reference point.

10. The LMISF-CC interacts with the LMISF-IRI (this reference point is not standardized) to obtain the correlation information to be used in the xCC.

11. The LMISF-CC generates and delivers the xCC to the MDF3.

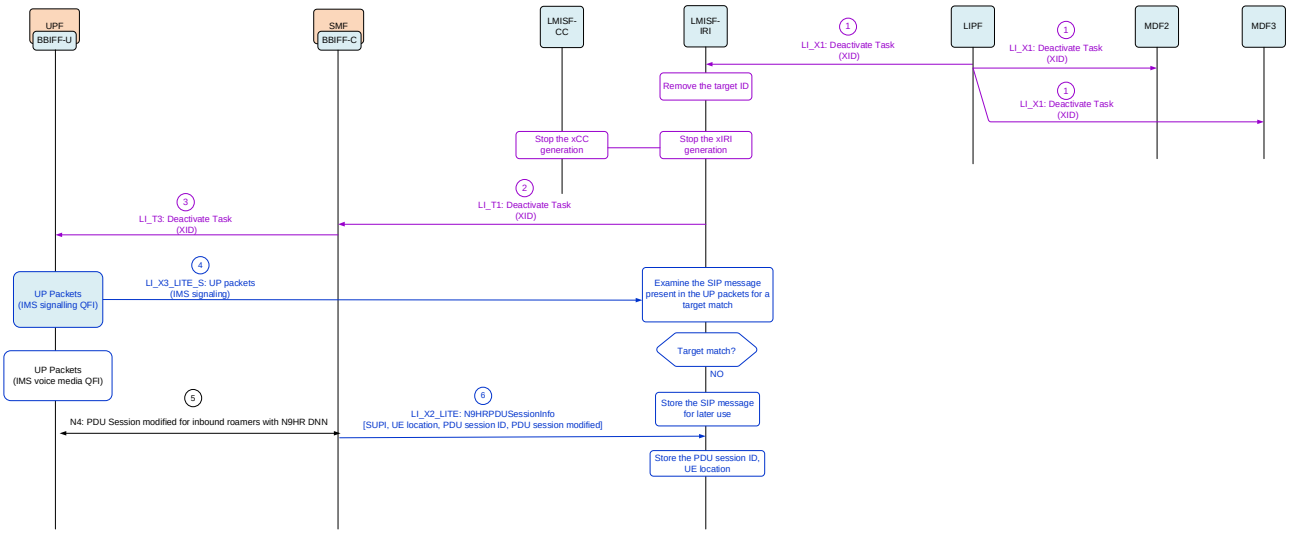
12. The MDF3 forwards the CC to the LEMF.

NOTE 1: At step 5, the LMISF-IRI and the BBIFF-C present in the SMF do not know the QFI to be associated with the IMS voice media related QoS flow. That information, the SMF in the VPLMN would receive from the SMF present in the HPLMN when the PDU session is modified (step 6).

NOTE 2: A QoS flow may carry the packets of one or more IMS session legs. In this case, it is the responsibility of the LMISF-IRI and the LMISF-CC to determine and use proper correlation information.

4.8.4.1.3 Intercept de-activation - applicable to target UE with IRI + CC

When an interception is stopped the N9HR LI process applicable to the same is illustrated in 4.8.4.1.3-1.



4.8.4.1.3-1: N9HR LI process – interception deactivation (applicable to target UE with IRI + CC)

The following describe the steps shown in 4.8.4.1.3-1:

1. LIPF de-provisions (i.e. deactivates) a previously provisioned intercept at the LMISF-IRI, the MDF2 and the MDF3.

2. The LMISF-IRI removes the intercept information (i.e. removes the target ID), stops the generation of IMS session related xIRI, interacts with the LMISF-CC to stop the generation of IMS session related xCC and sends a trigger to the BBIFF-C (over LI_T1) present in the SMF to deactivate the interception of IMS voice related media. The trigger includes the XID associated with the Task.

3. The BBIFF-C present in the SMF sends a trigger over LI_T3 to the BBIFF-U present in the UPF to deactivate the delivery of UP packets associated with the IMS media related QFI to the LMISF-CC. The trigger includes the XID associated with the Task.

4. The BBIFF-U present in the UPF continues to capture and deliver the UP packets from the PDU session that have the QFI (which was not deactivated by the BBIFF-C) to the LMISF-IRI over the LI_X3_LITE_S reference point. These are the IMS signalling packets. The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. When there is no target match, the LMISF-IRI extracts and stores the SIP message. This is normal process before any interception is activated (see figure 4.8.4.1.1-1). However, the BBIFF-U does not deliver the UP packets from the PDU session that have the QFI which was deactivated by the BBIFF-C. Accordingly, the IMS voice media related packets are not delivered to the LMISF-CC.

5. The BBIFF-C present in the SMF detects that the PDU session is modified for inbound roaming UEs with home-routed roaming.

6. The BBIFF-C present in the SMF notifies the LMISF-IRI about the PDU session modifications over the LI_X2_LITE reference point. The notifications include the PDU session ID and the UE location information. The LMISF stores the PDU session ID and the UE location information.

4.8.4.2 S8HR LI, EPC with CUPS

4.8.4.2.1 Initial configuration - applicable to all inbound roamers

The S8HR LI process applicable to all inbound roamers with home-routed roaming is illustrated in figure 4.8.4.2.1-1.

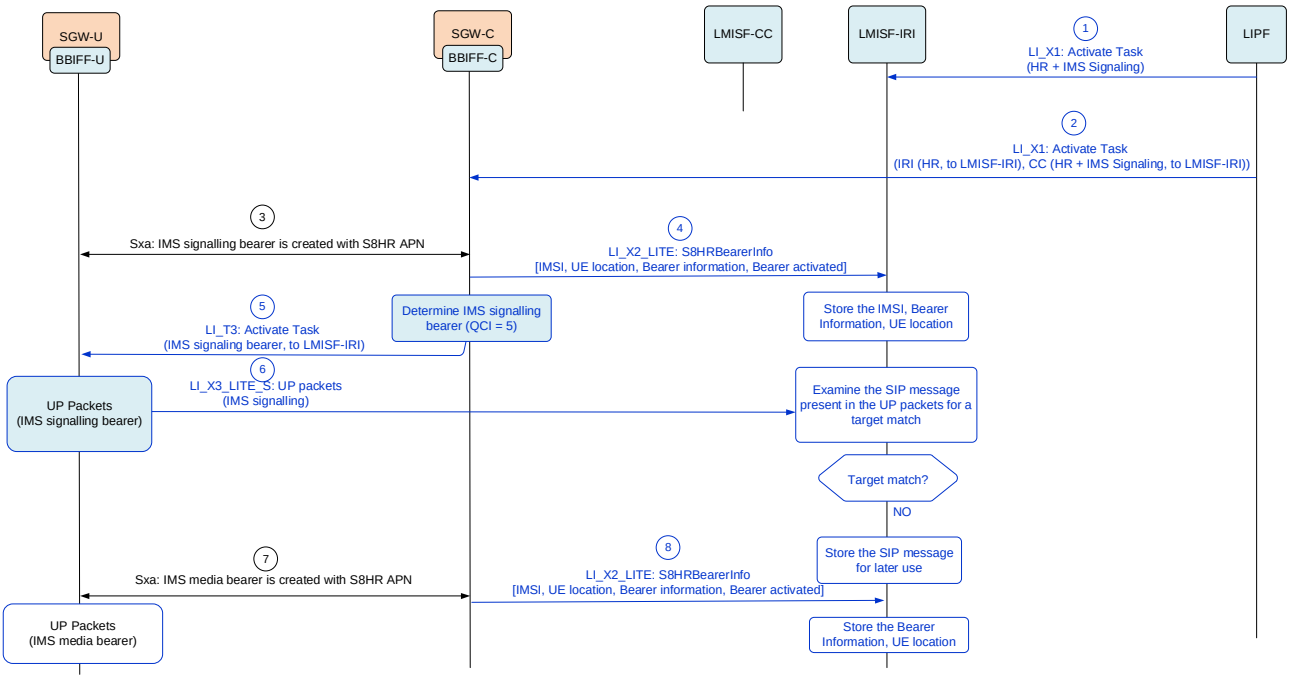


Figure 4.8.4.2.1-1: S8HR LI (EPC with CUPS) process applicable to all inbound roamers

The following describe the steps shown in figure 4.8.4.2.1-1:

1. The LIPF initiates the S8HR LI process by provisioning the LMISF-IRI with "HR" + "IMSSignaling" as the target identities. However, there is no DID specified in the Activate Task as the LMISF-IRI is not supposed to forward any of the received notifications.

2. The LIPF provisions the BBIFF-C present in the SGW-C with the "HR" as the target identity for xIRI and with "HR" + "IMSSignaling" as the target identities for xCC that corresponds to the IMS signalling. . LMISF-IRI is the DID for both.

3. The BBIFF-C present in the SGW-C detects that an IMS signalling bearer is created (see also the step 5), for the inbound roaming UEs with home-routed roaming (i.e. for S8HR APN).

4. The BBIFF-C present in the SGW-C notifies the LMISF-IRI about the IMS signalling bearer information over the LI_X2_LITE reference point. The notifications include the IMSI, bearer information (e.g. bearer ID) and the UE location information. LMISF stores the IMSI, bearer information and the UE location information.

5. Based on the QCI value of 5, the BBIFF-C present in the SGW-C determines that the bearer is created for IMS signalling. The BBIFF-C sends a trigger over LI_T3 to the BBIFF-U present in the SGW-U to capture and deliver the UP packets associated with the IMS signalling bearer. The DID is LMISF-IRI

6. The BBIFF-U present in the SGW-U captures and delivers the UP packets from the indicated bearer (i.e. IMS signalling bearer) to the LMISF-IRI over the LI_X3_LITE_S reference point. The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. When there is no target match, the LMISF-IRI extracts and stores the SIP message.

7. The BBIFF-C present in the SGW-C detects that a dedicated bearer (for IMS media) is created for inbound roaming UE with home-routed roaming (i.e. for S8HR APN).

8. The BBIFF-C present in the SGW-C notifies the LMISF-IRI about the IMS media bearer information over the LI_X2_LITE reference point. The notifications include the IMSI, bearer information (e.g. bearer ID, the linked bearer ID) and the UE location information. LMISF stores the IMSI, bearer information and the UE location information.

NOTE 1: The UP packets for IMS media bearer are not delivered to the LMISF-IRI when there is no active interception for involved UE.

NOTE 2: The BBIFF-U would notify the LMISF-IRI whenever an IMS bearer is created, modified or deleted using the xIRI: S8HRBearerInfo message.

4.8.4.2.2 Intercept activation - applicable to target UE with IRI + CC

The S8HR LI process applicable to target UE with home-routed roaming is illustrated in figure 4.8.4.2.2-1.

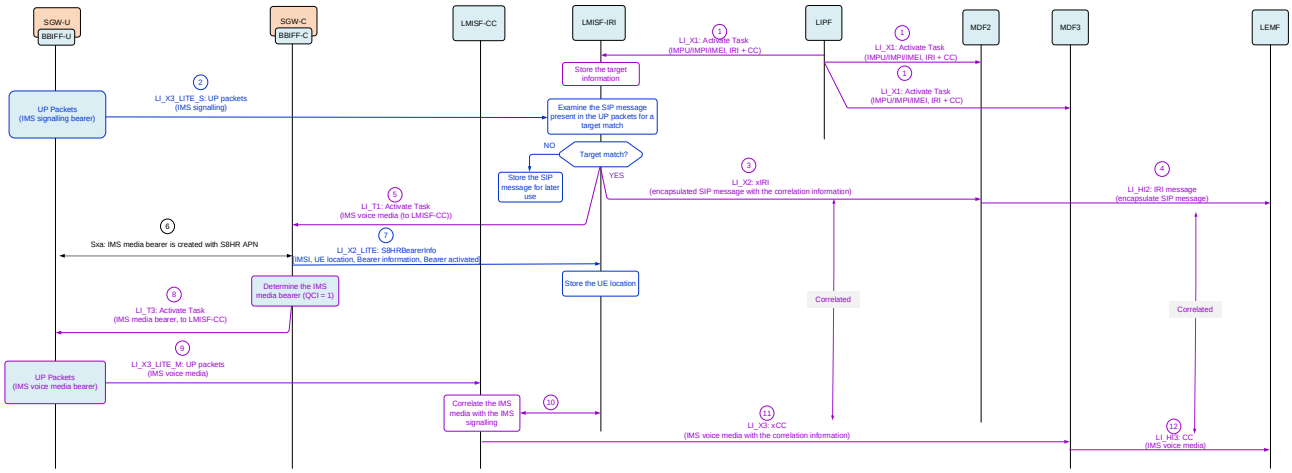


Figure 4.8.4.2.2-1: S8HR LI (EPC with CUPS) process – intercept activation (applicable to target UE with IRI + CC

The following describe the steps shown in figure 4.8.4.2.2-1:

1. The LIPF provisions the LMISF-IRI, the MDF2 and the MDF3 for IMS based voice interception with IRI + CC. The target identity is IMPU, IMPI or the IMEI as they are the identities supported with the IMS LI.

2. As per the procedures illustrated in flow of figure 4.8.4.2.1-1, the BBIFF-U present in the SGW-U captures and delivers the UP packets from the IMS signalling bearer (default bearer) to the LMISF-IRI over LI_X3_LITE_S reference point according to the trigger previously received from the BBIFF-C.

3. The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. With a target match, the LMISF-IRI in the role of IRI-POI for IMS signalling related messages, generates the xIRI that contains the encapsulated SIP message and delivers the same to MDF2.

4. The MDF2 delivers the IRI containing the encapsulated SIP message to the LEMF.

5. This interception requires IMS voice media interception and therefore, the LMISF-IRI sends a trigger to the BBIFF-C (over LI_T1) present in the SGW-C for the interception of IMS voice media. The trigger includes the IMSI, bearer ID of the IMS signalling bearer previously received from the BBIFF-C. The DID included in the trigger is LMISF-CC.

6. The BBIFF-C present in the SGW-C detects that IMS media bearer is created.

7. The BBIFF-C present in the SGW-C notifies the LMISF-IRI about the IMS media bearer creation over the LI_X2_LITE reference point. The notification includes the IMSI, bearer information (bearer ID, the linked bearer ID of the IMS signalling bearer and the UE location information. LMISF stores the bearer information and the UE location information.

8. The BBIFF-C present in the SGW-C determines that the bearer created (detected in step 6) is for IMS voice media (QCI = 1) and is linked to the IMS signalling bearer based on the bearer ID indicated within the trigger received from the LMISF-IRI. The BBIFF-C sends a trigger over LI_T3 to the BBIFF-U present in the SGW-U to capture and deliver the UP packets from the IMS voice media bearer. The DID is LMISF-CC.

9. The BBIFF-U captures and delivers the UP packets from the IMS voice media bearer to the LMISF-CC over the LI_X3_LITE_M reference point.

10. The LMISF-CC interacts with the LMISF-IRI (this reference point is not standardized) to obtain the correlation information to be used in the xCC.

11. The LMISF-CC generates and delivers the xCC to the MDF3.

12. The MDF3 forwards the CC to the LEMF.

NOTE: A bearer within the EPC may carry the packets of one or more IMS session legs. In this case, it is the responsibility of LMISF-IRI and LMISF-CC to determine and use proper correlation information.

4.8.4.2.3 Intercept de-activation - applicable to target UE with IRI + CC

When an interception is stopped the S8HR LI process applicable to the same is illustrated in figure 4.8.4.2.3-1.

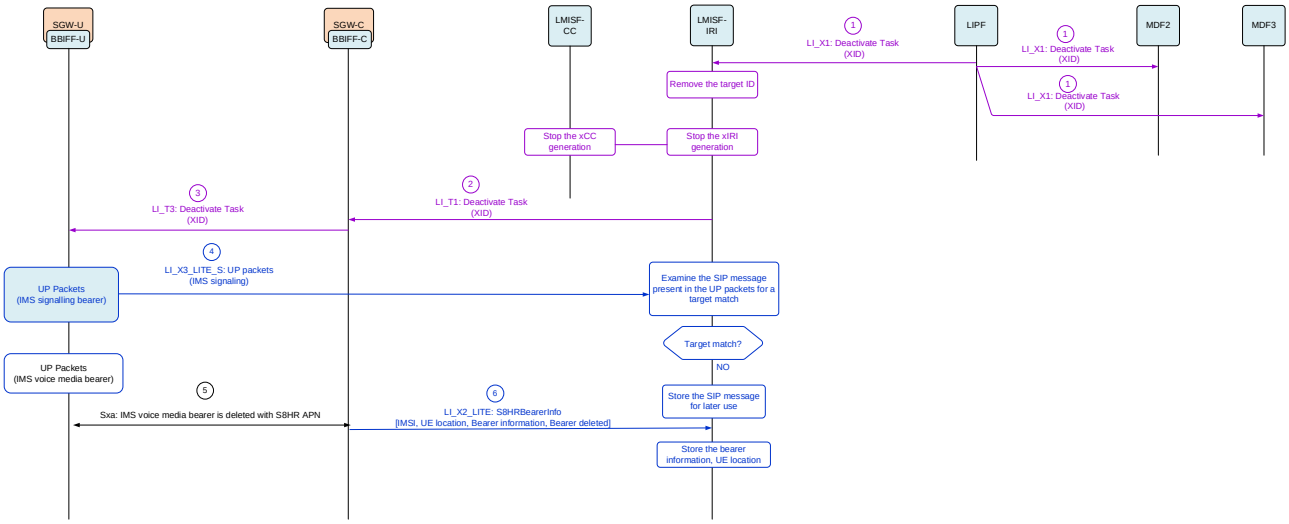


Figure 4.8.4.2.3-1: S8HR LI (EPC with CUPS) process – interception deactivation (applicable to target UE with IRI + CC)

The following describe the steps shown in figure 4.8.4.2.3-1:

1. LIPF de-provisions (i.e. deactivates) a previously provisioned intercept at the LMISF-IRI, MDF2 and MDF3.

2. The LMISF-IRI removes the intercept information (i.e. removes the target ID), stops the generation of IMS session related xIRI, interacts with the LMISF-CC to stop the generation of IMS session related xCC and sends a trigger to the BBIFF-C (over LI_T1) present in the SGW-C to deactivate the interception of IMS voice media. The trigger includes the XID associated with the Task.

3. The BBIFF-C present in the SGW-C sends a trigger over LI_T3 to the BBIFF-U present in the SGW-U to deactivate the delivery of UP packets associated with the IMS voice media bearer to the LMISF-CC. The trigger includes the XID associated with the Task.

4. The BBIFF-U present in the SGW-C continues to capture and deliver the UP packets from the IMS signalling bearer to the LMISF-IRI over the LI_X2_LITE reference point The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. When there is no target match, the LMISF-IRI extracts and stores the SIP message. This is normal process before any interception is activated (see figure 4.8.4.2.1-1). However, the BBIFF-U does not deliver the UP packets from the IMS media bearer.

5. The BBIFF-C present in the SGW-C detects that IMS media bearer is deleted for inbound roaming UEs with home-routed roaming.

6. The BBIFF-C present in the SGW-C notifies the LMISF-IRI about the bearer deletion notification over the LI_X2_LITE reference point. The notifications include the IMSI, linked IMS signalling bearer ID, UE location information. LMISF-IRI stores the IMSI, bearer information and the UE location information.

4.8.4.3 S8HR LI, EPC without CUPS

4.8.4.3.1 Initial configuration - applicable to all inbound roamers

The S8HR LI process applicable to all inbound roamers with home-routed roaming is illustrated in figure 4.8.4.3.1-1.

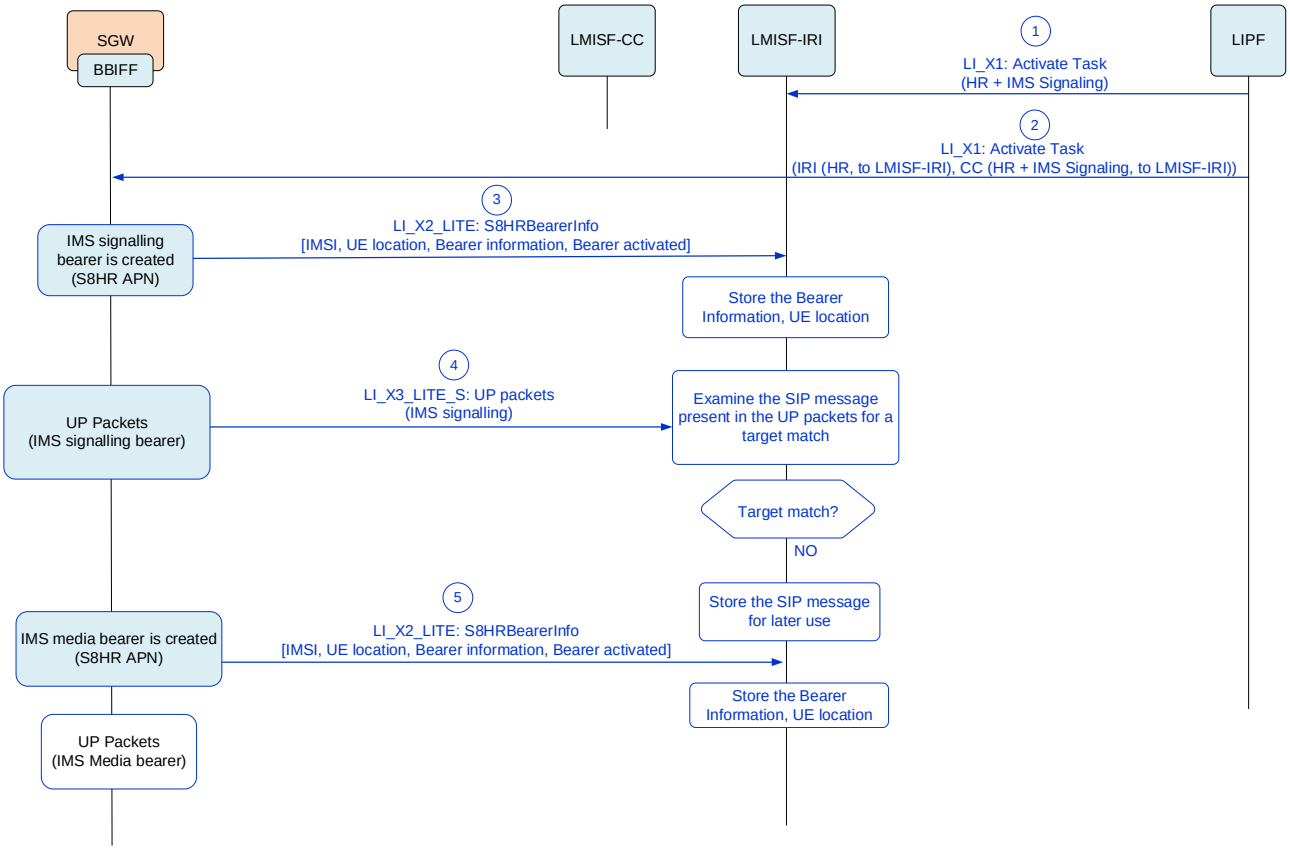


Figure 4.8.4.3.1-1: S8HR LI (EPC without CUPS) process applicable to all inbound roamers

The following describe the steps shown in figure 4.8.4.3.1-1:

1. The LIPF initiates the S8HR LI process by provisioning the LMISF-IRI with "HR" + "IMSSignaling"as the target identities. However, there is no DID specified in the Activate Task as the LMISF-IRI is not supposed to forward of any of the received notifications.

2. The LIPF provisions the BBIFF present in the SGW with "HR" as the target identity for xIRI and with "HR" + "IMSSignaling" as the target identities for xCC that corresponds to the IMS signalling. LMISF-IRI is the DID for both.

3. The BBIFF present in the SGW detects that an IMS signalling bearer is created (see also the step 5), for the inbound roaming UEs with home-routed roaming (i.e. for S8HR APN).

4. The BBIFF present in the SGW notifies the LMISF-IRI about the IMS signalling bearer information over the LI_X2_LITE reference point. The notifications include the IMSI, bearer information (e.g. bearer ID) and the UE location information. The LMISF stores the bearer information and the UE location information.

5. Based on the QCI value of 5, the BBIFF present in the SGW determines that the bearer is created for IMS signalling.

6. The BBIFF present in the SGW captures and delivers the UP packets from the IMS signalling bearer to the LMISF-IRI over the LI_X3_LITE_S reference point. The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. When there is no target match, the LMISF-IRI extracts and stores the SIP message.

7. The BBIFF present in the SGW detects that IMS media bearer is created for inbound roaming UE with home-routed roaming (i.e. for S8HR APN).

8. The BBIFF present in the SGW notifies the LMISF-IRI about the IMS media bearer information over the LI_X2_LITE reference point. The notifications include the IMSI, bearer information (e.g. bearer ID, bearer ID of the IMS signalling bearer) and the UE location information. The LMISF-IRI stores the bearer information and the UE location information.

NOTE 1: The UP packets for the IMS media bearer are not delivered to the LMISF-IRI when there is no active interception for involved UE.

NOTE 2: The BBIFF would notify the LMISF-IRI whenever an IMS bearer is created, modified or deleted using xIRI: S8HRBearerInfo message.

4.8.4.3.2 Intercept activation - applicable to target UE with IRI + CC

The S8HR LI process applicable to target UE with home-routed roaming is illustrated in figure 4.8.4.3.2-1.

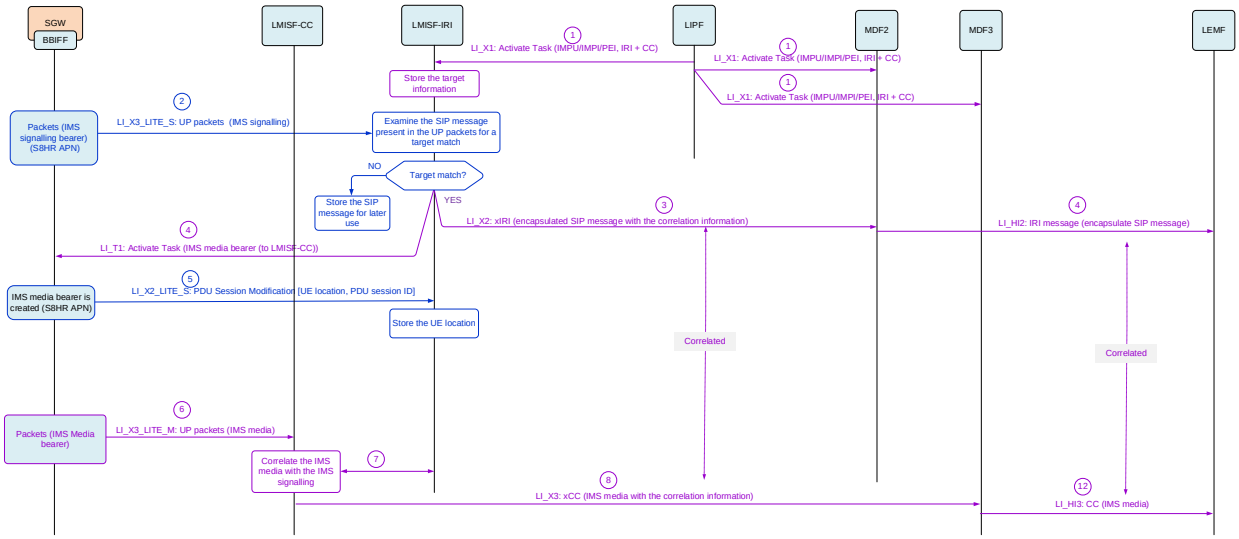


Figure 4.8.4.3.2-1: S8HR LI (EPC without CUPS) process – applicable to target UE with IRI + CC

The following describe the steps shown in figure 4.8.4.3.2-1:

1. The LIPF provisions the LMISF-IRI, the MDF2 and the MDF3 for IMS based voice interception with IRI + CC. The target identity is IMPU, IMPI or the IMEI as they are the identities supported with the IMS LI.

2. As per the procedures illustrated in flow of figure 4.8.4.3.1-1, the BBIFF present in the SGW captures and delivers the UP packets from the IMS signalling bearer (default bearer) to the LMISF-IRI over LI_X3_LITE_S reference.

3. The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. With a target match, the LMISF-IRI in the role of IRI-POI for IMS signalling related messages, generates the xIRI that contains the encapsulated SIP message and delivers the same to MDF2.

4. The MDF2 delivers the IRI containing the encapsulated SIP message to the LEMF.

5. This interception requires IMS voice media interception and therefore, the LMISF-IRI sends a trigger to the BBIFF (over LI_T1) present in the SGW for the interception of IMS voice media. The trigger includes the IMSI, bearer ID of the IMS signalling bearer previously received from the BBIFF. The DID included in the trigger is LMISF-CC.

6. The BBIFF present in the SGW detects that IMS media bearer is created.

7. The BBIFF present in the SGW notifies the LMISF-IRI about the IMS media bearer creation over the LI_X2_LITE reference point. The notification includes the IMSI, bearer information (bearer ID, the linked

bearer ID of the IMS signalling bearer and the UE location information). LMISF stores the bearer information and the UE location information.

8. The BBIFF present in the SGW determines that the bearer created (detected in step 6) is for IMS voice media (QCI = 1) and is linked to the IMS signalling bearer based on the bearer ID indicated within the trigger received from the LMISF-IRI.

9. The BBIFF captures and delivers the UP packets from the IMS voice media bearer to the LMISF-CC over the LI_X3_LITE_M reference point.

10. The LMISF-CC interacts with the LMISF-IRI (this reference point is not standardized) to obtain the correlation information to be used in the xCC.

11. The LMISF-CC generates and delivers the xCC to the MDF3.

12. The MDF3 forwards the CC to the LEMF.

NOTE: A bearer within the EPC may carry the packets of one or more IMS session legs. In this case, it is the responsibility of LMISF-IRI and LMISF-CC to determine and use proper correlation information.

4.8.4.3.3 Intercept de-activation - applicable to target UE with IRI + CC

When an interception is stopped the S8HR LI process applicable to the same is illustrated in figure 4.8.4.3.3-1.

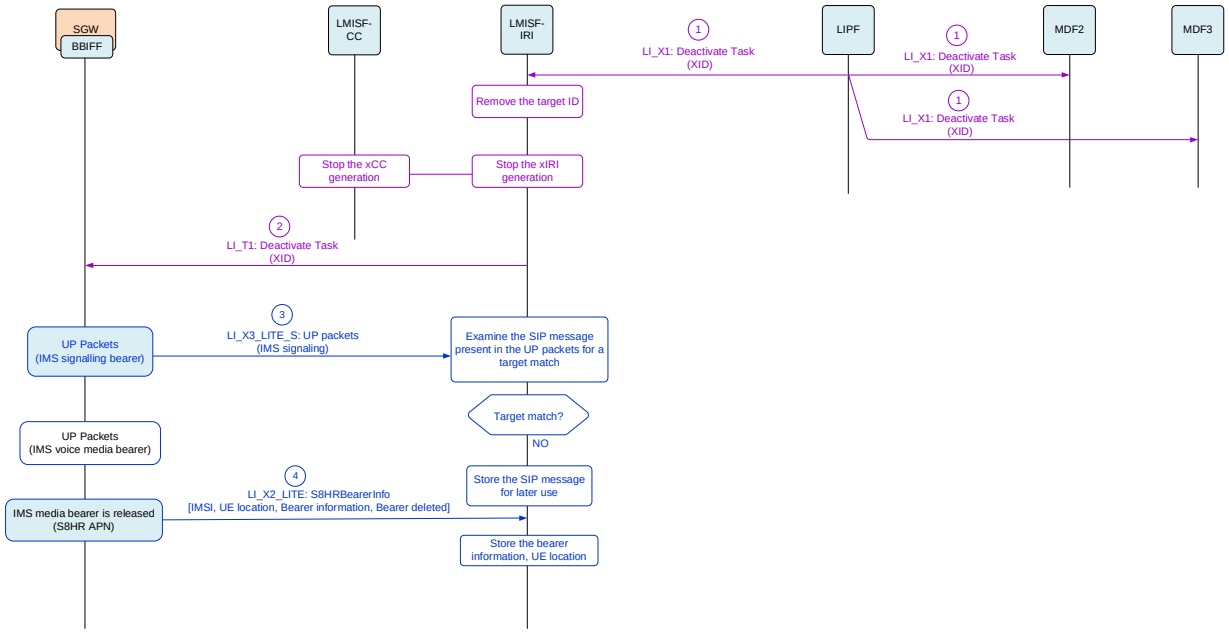


Figure 4.8.4.3.3-1: S8HR LI (EPC without CUPS) process – interception deactivation (applicable to target UE with IRI + CC)

The following describe the steps shown in figure 4.8.4.3.3-1:

1. The LIPF de-provisions (i.e. deactivates) a previously provisioned intercept at the LMISF-IRI, the MDF2 and the MDF3.

2. The LMISF-IRI removes the intercept information (i.e. removes the target ID), stops the generation of IMS session related xIRI, interacts with the LMISF-CC to stop the generation of IMS session related xCC and sends a trigger to the BBIFF (over LI_T1) present in the SGW to deactivate the interception of IMS media. The trigger includes the XID associated with the Task.

3. The BBIFF present in the SGW continues to capture and deliver the UP packets from the IMS signalling bearer to the LMISF-IRI over the LI_X3_LITE_S reference point. The LMISF-IRI examines the received UP packets and examines the SIP message included within those packets for any target match. When there is no target

match, the LMISF-IRI extracts and stores the SIP message. This is normal process before any interception is activated (see figure 4.8.4.3.1-1). However, the BBIFF does not deliver the UP packets from the IMS media bearer.

4. The BBIFF present in the SGW detects that IMS media bearer is deleted for inbound roaming UEs with home-routed roaming and notifies the same to the LMISF-IRI over the LI_X2_LITE reference point. The notification includes the IMSI linked IMS signalling bearer ID, UE location information. The LMISF-IRI stores the bearer information and the UE location information.

4.8.5 Correlation principles

4.8.5.1 Correlating the IRI

With N9HR LI and S8HR LI, a target is identified using IMPU, IMPI, PEI or the IMEI.

The LMISF-IRI examines the IMS signalling packets that it receives from the BBIFF-C over the LI_LITE_S reference point. While performing the check for a target match, the LMISF-IRI verifies the identities present in the SIP INVITE to the provisioned target identities.

Not all SIP messages carry those identities. However, the LMISF-IRI by maintaining the IMS call state is able to determine the subsequent SIP messages (i.e. after SIP INVITE) that correspond to the same target, to the same IMS session. In other words, the SIP messages that represent an IMS session can be identified within the LMISF-IRI.

LMISF-IRI includes the correlation information (e.g. the same Correlation Number) in all the xIRI that it sends to the MDF2.

When a target is involved in multiple IMS sessions, the LMISF-IRI will have the logic to associate and correlate the SIP messages that are related to an IMS session. For example, the SIP messages that have the same Call Identity value can be treated as the SIP messages of a particular IMS session and hence, when reported to the LEMF (via MDF2) can have the same correlation information (e.g. Correlation Number).

4.8.5.2 Correlating the IRI and CC

When an IMS session is established, the media information is exchanged between the two end points of the media stream (e.g. the target's UE and the IMS-AGW in the HPLMN) through the SDP offer and answer process. The combination of the IP address of the end point (e.g. UE and IMS AGW) and the UDP port numbers used to transport the RTP and RTCP are part of this SDP offer and answer along with other things like codec information. The media packets (i.e. RTP streams) exchanged between the two end points of the media use those IP addresses and the port numbers (assigned for RTP).

The LMISF-IRI will also examine the SIP messages that carry the SDP offer and SDP answer to determine the media information related to an IMS session. The LMISF-IRI will provide the SDP information and the correlation information to the LMISF-CC.

One method that can be used to establish the correlation between the IRI and the CC is to use the IP addresses and the UDP port numbers exchanged within the SDP offer and answer process and compare them with the IP addresses and UDP port numbers of the media packets to establish an association between the IMS session leg and the media.

In other words, the IP address and UDP port numbers associated with a media packet when compared with the IP address and UDP port numbers exchanged in the SDP offer and answer, one can determine to which IMS session a media packet corresponds to. Once that determination is made, a correlation can be made between the CC and the IRI.

That is exactly what will be done at the LMISF-CC: When the UP packets that represent the IMS media packets are received, the LMISF-CC will examine the UP media packets to determine to which IMS session, the media packets are related to. Once determined, the LMISF-CC will deliver the xCC to the MDF3 along with the correlation information previously received from the LMISF-IRI against the IMS session.

4.8.6 UE location reporting

Within the 5GC, the AMF sends the UE location information to the SMF whenever a PDU session is established, modified or deleted (e.g. Nsmf_PDUSession_CreateSMContext Request). The BBIFF-C present in the SMF notifies the

LMISF-IRI whenever the PDU session is established, modified, or deleted. Within those notifications, the BBIFF-C includes the UE location that the SMF receives from the AMF.

Within the EPC with CUPS, the MME sends the UE location to the SGW-C whenever an IMS signalling bearer or an IMS media bearer within a PDN connection is created, modified, or deleted (e.g. Create Session Request). The BBIFF-C present in the SGW-C notifies the LMISF-IRI whenever the IMS signalling bearer (i.e. default bearer) or media bearer (i.e. dedicated bearer linked to the IMS signalling bearer) is created, modified, or deleted. Within those notifications, the BBIFF-C includes the UE location that the SGW-C receives from the MME.

Within the EPC without CUPS, the MME sends the UE location to the SGW whenever an IMS signalling bearer or an IMS media bearer within a PDN connection is created, modified, or deleted (e.g. Create Session Request). The BBIFF present in the SGW notifies the LMISF-IRI whenever the IMS signalling bearer (i.e. default bearer) or media bearer (i.e. dedicated bearer linked to the IMS signalling bearer) is created, modified, or deleted. Within those notifications, the BBIFF includes the UE location that the SGW receives from the MME.

In summary, the LMISF-IRI receives the UE location information in the notifications received from the BBIFF-C (or BBIFF, EPC without the CUPS) over the LI_LITE reference point. The LMISF-IRI stores the received UE location along with the timestamp (i.e. the time at which that location was determined).

The LMISF-IRI includes the received UE location and the UE location timestamp in the appropriate xIRIs sent to the MDF2 over the LI_X2 reference point.

NOTE: The UE location reported in the xIRI that encapsulates the SIP INVITE may not be the most recent one.

The MDF2 delivers the UE Location to the LEMF (when required) as it is done for the non-roaming scenario or in a roaming with LBO scenario.

4.9 Illustrations of IMS CC-POI triggering scenarios

4.9.1 Symmetric media streams

4.9.1.1 General

This clause illustrates the LI_T3 scenarios when the media streams in an IMS session are symmetric where the source IP address and UDP port number of the IP layer of a media stream sent from a media node are the same as the destination IP address and IP address of the IP layer for that media stream that it has given to the remote end.

In this case, an implementation may choose to use only one LI_T3 ActivationTask trigger (as illustrated here) with the local IP address and UDP port number of the IMS media function that intercepts the media. An LI_T3 ModifyTask trigger may still be required to send the Remote SDP information to the CC-POI.

If an implementation uses two LI_T3 ActivationTask triggers even for symmetric media stream case, then the illustrations shown in clause 4.9.2 apply.

The following descriptions apply to all scenarios illustrated in the subsequent sub-clauses:

- The Remote SDP within the ActivateTask or ModifyTask includes the SDP information received from the originating end or terminating end of the session (depending on the scenario) in the SDP offer or answer (depending on the scenario).

- The Local SDP within the ActivateTask or ModifyTask includes the SDP information sent to the originating end or terminating end of the session (depending on the scenario) in the SDP offer or answer (depending on the scenario). The Trigger Scope value of Bidirectional indicates that the ActivateTask trigger applies for the interception of incoming and outgoing media. The target identifiers present in the ActivateTask or ModifyTask indicate the destination IP address and UDP port number for the incoming media streams. The same are the source IP address and UDP port numbers for the media streams in the reverse direction (i.e. outgoing).

- The Payload Direction Assignment gives the direction of media from the perspective of the target. When the value is ToTarget, the media destined to the IP address and UDP port number provided as the target identifiers in the ActivateTask or ModifyTask are to the target. In the reverse direction, the media would be from the target. Likewise, when the value is FromTarget, the media destined to the IP address and UDP port number provided as

the target identifiers in the ActivateTask or ModifyTask are from the target. In the reverse direction, the media would be to the target.

- When the SDP offer is received on the side where the media interception is done, the ActivateTask includes both Local SDP and the Remote SDP.
- When the SDP offer is sent from the side where the media interception is done, the Remote SDP can only be provided when an SDP answer is received. Therefore, in this case, a second trigger using ModifyTask is sent to the CC-POI.

Other than the Remote SDP, all other fields sent in the ModifyTask are same as the fields sent in the ActivateTask and have the same inference.

4.9.1.2 SDP offer in the forward message – originating session

Figure 4.9.1.2-1 illustrates the LI_T3 trigger sent from a CC-TF present in the P-CSCF to the CC-POI present in the IMS-AGW as per the 2 options described in TS 33.128 [4] for an IMS originating session from a target when the SDP offer is included in a forward SIP message (e.g. SIP INVITE).

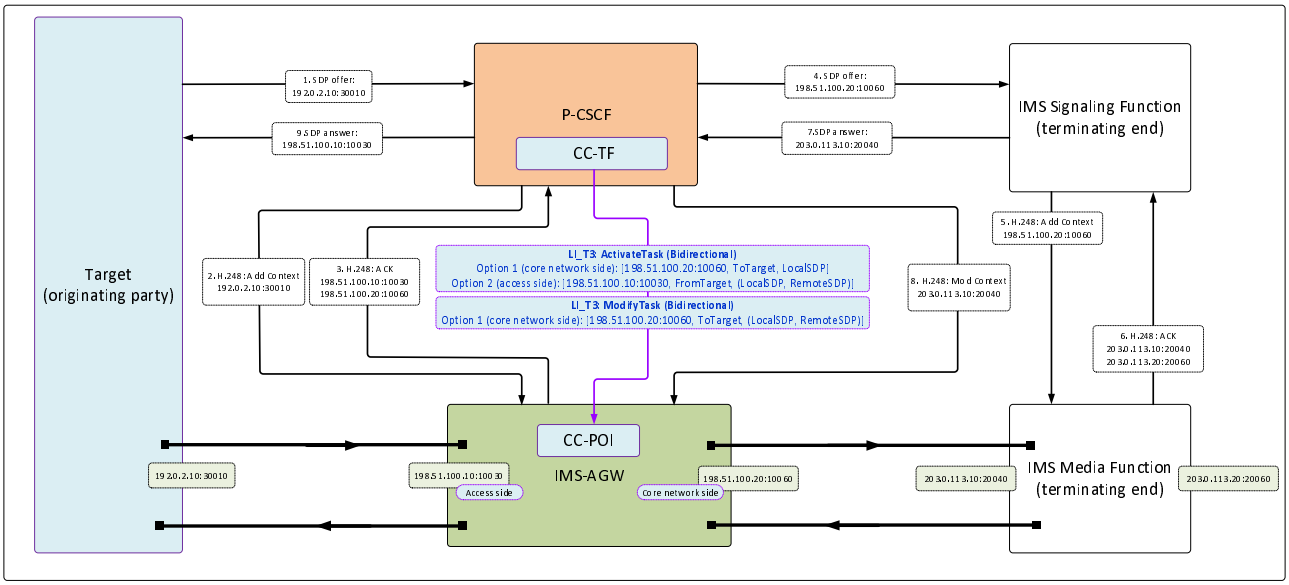


Figure 4.9.1.2-1: ActivateTask for target originated sessions (SDP offer in forward SIP)

The ActivateTask that can be sent after step 3 has the following information.

When the media interception is performed at the core network side:

- Target Identifiers:
  - IP address 198.51.100.20.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Trigger scope: Bidirectional.

When the media interception is performed at the access side:

- Target Identifiers:
  - IP address 198.51.100.10.
  - UDP port number 10030.

- Payload Direction Assignment: FromTarget.
- Remote SDP.
- Local SDP.
- Trigger Scope: Bidirectional.

The IP Address 198.51.100.20 and the UDP port number 10060 are also present in the SDP offer sent toward the terminating end of the session.

The IP Address 198.51.100.10 and the UDP port number 10030 are also present in the SDP answer sent toward the originating end of the session (i.e. target UE).

The media streams received at the IP address 198.51.100.20 and UDP port 10060 are to the target (core network side media interception) and the media streams received at the IP address 198.51.100.10 and the UDP port 10030 (access side media interception) are from the target.

When the media interception is done at the core network side, a ModifyTask is sent after step 8 with the following information:

- Target Identifiers:
  - IP address 198.51.100.20.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is done at the access side, the ModifyTask is not required since the Remote SDP is already provided to the CC-POI in the ActivateTask message.

4.9.1.3 SDP offer in the response message – originating session

Figure 4.9.1.3-1 illustrates the LI_T3 trigger sent from a CC-TF present in the P-CSCF to the CC-POI present in the IMS-AGW as per the 2 options described TS 33.128 [4] for an IMS originating session from a target when the SDP offer is included in a response SIP message (e.g. SIP 200 OK).

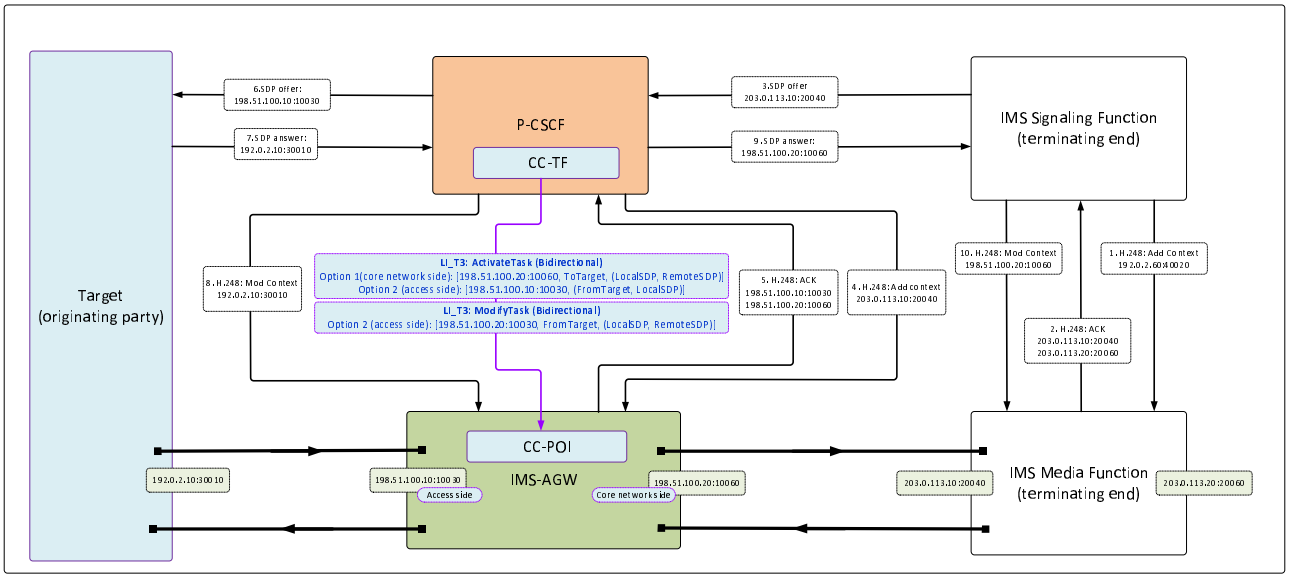


Figure 4.9.1.3-1: ActivateTask for target originated sessions (SDP offer in response SIP)

The ActivateTask that can be sent after step 5 has the following information.

When the media interception is performed at the core network side:

- Target Identifiers:
  - IP address 198.51.100.20.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is performed at the access side:

- Target Identifiers:
  - IP address 198.51.100.10.
  - UDP port number 10030.
- Payload Direction Assignment: FromTarget.
- Local SDP.
- Trigger Scope: Bidirectional.

When the media interception is done at the access side, a ModifyTask is sent after step 8 with the following information:

- Target Identifiers:
  - IP address 198.51.100.10.
  - UDP port number 10030.
- Payload Direction Assignment: FromTarget.

- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is done at the core network side, the ModifyTask is not required since the Remote SDP is already provided to the CC-POI in the ActivateTask message.

The IP Address 198.51.100.20 and the UDP port number 10060 are also present in the SDP answer sent toward the terminating end of the session.

The IP Address 198.51.100.10 and the UDP port number 10030 are also present in the SDP offer sent toward the originating end of the session (i.e. target UE).

The media streams received at the IP address 198.51.100.20 and UDP port 10060 (core network side media interception) are to the target and the media streams received at the IP address 198.51.100.10 and the UDP port 10030 (access side media interception) are from the target.

4.9.1.4 SDP offer in the forward message – terminating session

Figure 4.9.1.4-1 illustrates the LI_T3 trigger sent from a CC-TF present in the P-CSCF to the CC-POI present in the IMS-AGW as per the 2 options described in TS 33.128 [4] for an IMS terminating session from a target when the SDP offer is included in a forward SIP message (e.g. SIP INVITE).

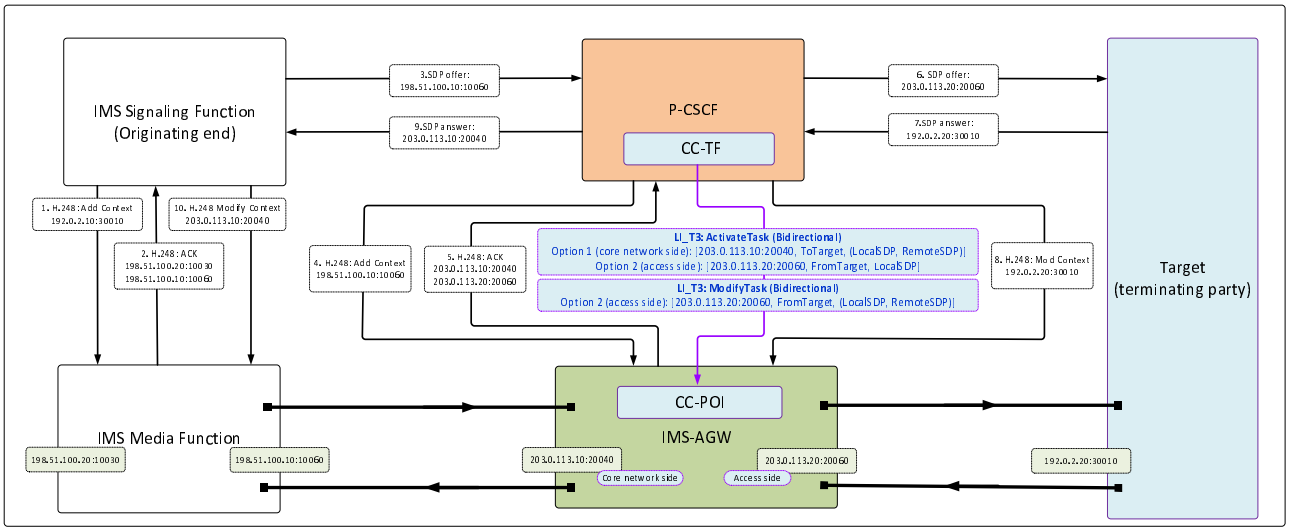


Figure 4.9.1.4-1: ActivateTask for target terminating sessions (SDP offer in forward SIP)

The ActivateTask that can be sent after step 5 has the following information.

When the media interception is performed at the core network side:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is performed at the access side:

- Target Identifiers:
  - IP address 203.0.113.20.
  - UDP port number 20060.
- Payload Direction Assignment: FromTarget.
- Local SDP.
- Trigger scope: Bidirectional.

The IP Address 203.0.113.10 and the UDP port number 20040 are also present in the SDP answer sent toward the originating end of the session.

The IP Address 203.0.113.20 and the UDP port number 20060 are also present in the SDP offer sent toward the terminating end of the session (i.e. target UE).

The media streams received at the IP address 203.0.113.10 and UDP port 20040 (core network side media interception) are to the target and media streams received at the IP address 203.0.1113.20 and the UDP port 20060 (access side media interception) are from the target.

When the media interception is done at the access side, a ModifyTask is sent after step 8 with the following information:

- Target Identifiers:
  - IP address 203.0.113.20.
  - UDP port number 20060.
- Payload Direction Assignment: FromTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is done at the core network side, the ModifyTask is not required since the Remote SDP is already provided to the CC-POI in the ActivateTask message.

4.9.1.5 SDP offer in the response message – terminating session

Figure 4.9.1.5-1 illustrates the LI_T3 trigger sent from a CC-TF present in the P-CSCF to the CC-POI present in the IMS-AGW as per the 2 options described in TS 33.128 [4] for an IMS terminating session from a target when the SDP offer is included in a response SIP message (e.g. SIP 200 OK).

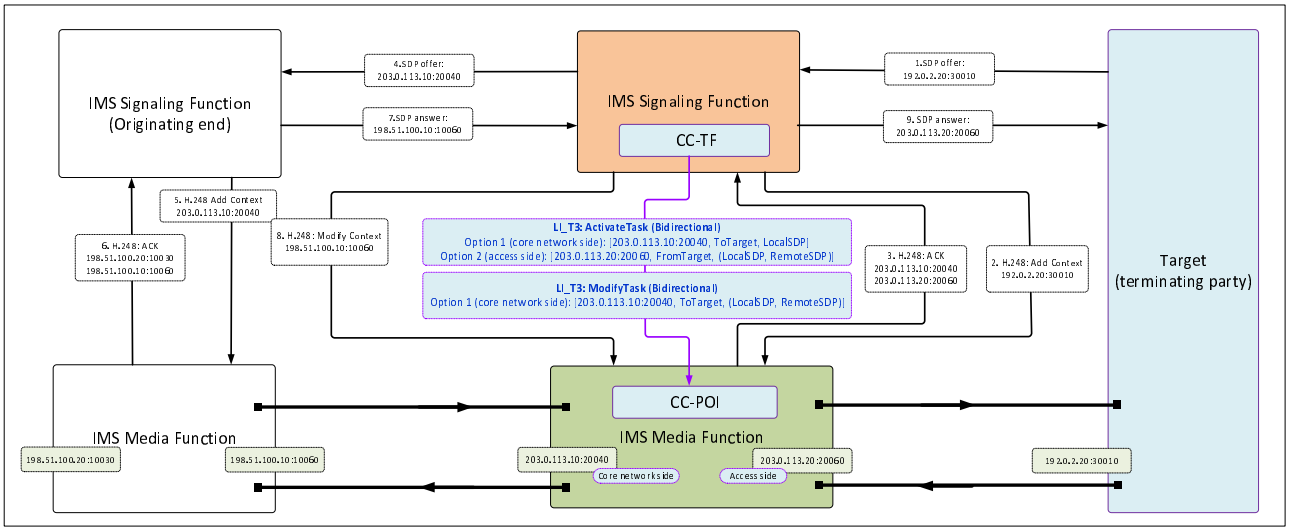


Figure 4.9.1.5-1: ActivateTask for target terminating sessions (SDP offer in response SIP

The ActivateTask that can be sent after step 3 has the following information.

When the media interception is performed at the core network side:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Trigger Scope: Bidirectional.

When the media interception is performed at the access side:

- Target Identifiers:
  - IP address 203.0.113.20.
  - UDP port number 20060.
- Payload Direction Assignment: FromTarget.
- Local SDP.
- Remote SDP.
- Trigger Scope: Bidirectional.

The IP Address 203.0.113.10 and the UDP port number 20040 are also present in the SDP offer sent toward the originating end of the session.

The IP Address 203.0.113.20 and the UDP port number 20060 are also present in the SDP answer sent toward the terminating end of the session (i.e. target UE).

The media streams received at the IP address 203.0.113.10 and UDP port 20040 (core network side media interception) are to the target and media streams received at the IP address 203.0.1113.20 and the UDP port 20060 (access side media interception) are from the target.

When the media interception is done at the core network side, a ModifyTask is sent after step 8 with the following information:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is done at the access side, the ModifyTask is not required since the Remote SDP is already provided to the CC-POI in the ActivateTask message.

4.9.1.6 SDP offer in the forward message – session to target non-local ID (IP)

Figure 4.9.1.6-1 illustrates the LI_T3 trigger sent from a CC-TF present in the IBCF to the CC-POI present in the TrGW as described in TS 33.128 [4] for an IMS session to a target non-local ID (in the IP domain) when the SDP offer is included in a forward SIP message (e.g. SIP INVITE).

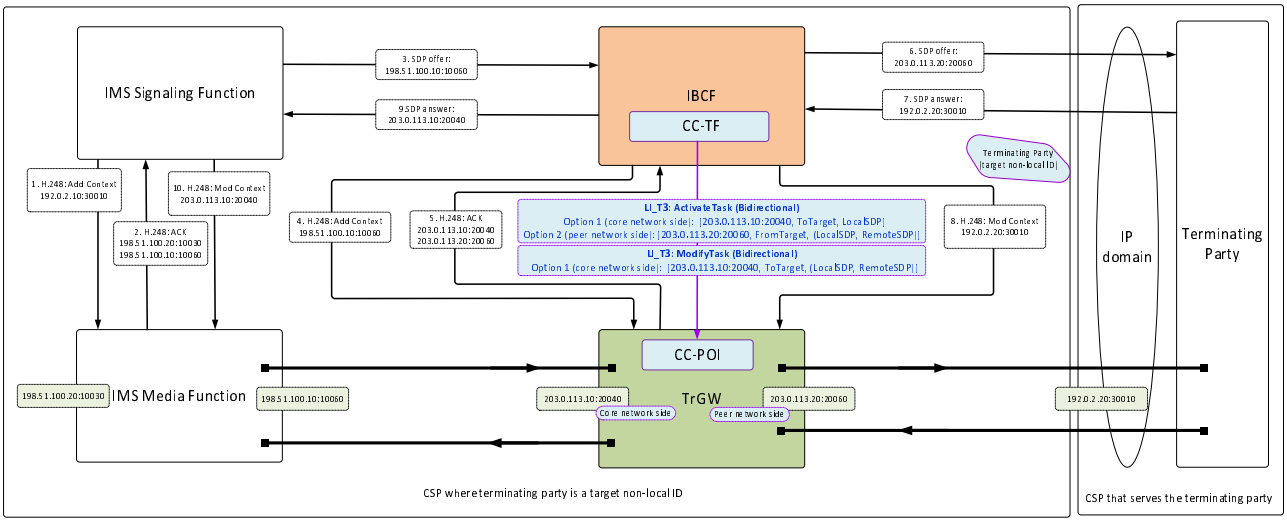


Figure 4.9.1.6-1: ActivateTask for sessions to target non-local ID in IP domain (SDP offer in forward SIP)

The ActivateTask that can be sent after step 5 has the following information.

When the media interception is performed at the core network side:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is performed at the peer network side:

- Target Identifiers:
  - IP address 203.0.113.20.
  - UDP port number 20060.
- Payload Direction Assignment: FromTarget.
- Local SDP.
- Trigger scope: Bidirectional.

The IP Address 203.0.113.10 and the UDP port number 20040 are also present in the SDP answer sent toward the originating end of the session.

The IP Address 203.0.113.20 and the UDP port number 20060 are also present in the SDP offer sent toward the terminating end of the session (i.e. toward the network that has the target non-local ID).

The media streams received at the IP address 203.0.113.10 and UDP port 20040 (core network side media interception) are to the target and media streams received at the IP address 203.0.1113.20 and the UDP port 20060 (peer network side media interception) are from the target.

When the media interception is done at the peer network side, a ModifyTask is sent after step 8 with the following information:

- Target Identifiers:
  - IP address 203.0.113.20.
  - UDP port number 20060.
- Payload Direction Assignment: FromTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is done at the core network side, the ModifyTask is not required since the Remote SDP is already provided to the CC-POI in the ActivateTask message.

4.9.1.7 SDP offer in the forward message – session from target non-local ID (IP)

Figure 4.9.1.7-1 illustrates the LI_T3 trigger sent from a CC-TF present in the IBCF to the CC-POI present in the TrGW as described in TS 33.128 [4] for an IMS session from a target non-local ID (IP domain) when the SDP offer is included in a forward SIP message (e.g. SIP INVITE).

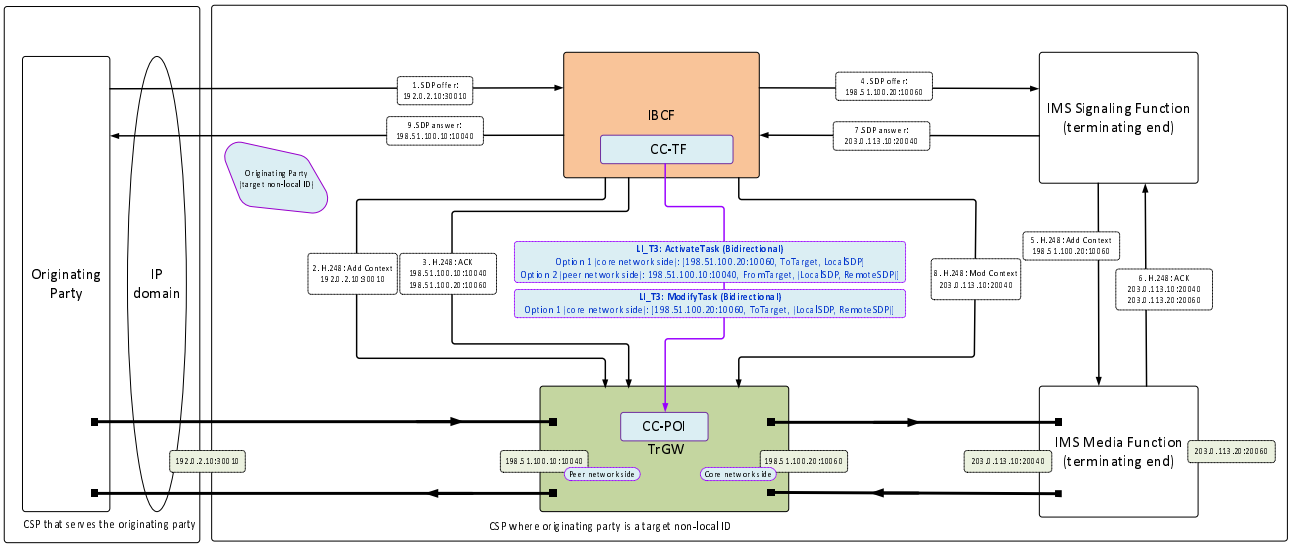


Figure 4.9.1.7-1: ActivateTask for sessions from target non-local ID in IP domain (SDP offer in forward SIP)

The ActivateTask that can be sent after step 3 has the following information.

When the media interception is performed at the core network side:

- Target Identifiers:
  - IP address 198.51.100.20.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Trigger scope: Bidirectional.

When the media interception is performed at the peer network side:

- Target Identifiers:
  - IP address 198.51.100.10.
  - UDP port number 10040.
- Payload Direction Assignment: FromTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

The IP Address 198.51.100.20 and the UDP port number 10060 are also present in the SDP offer sent toward the terminating end of the session.

The IP Address 198.51.100.10 and the UDP port number 10040 are also present in the SDP answer sent toward the originating end of the session.

The media streams received at the IP address 198.51.100.20 and UDP port 10060 (core network side media interception) are to the target and media streams received at the IP address 198.51.100.10 and the UDP port 10040 (peer network side media interception) are from the target.

When the media interception is done at the core network side, a ModifyTask is sent after step 8 with the following information:

- Target Identifiers:
  - IP address 198.51.100.20.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

When the media interception is done at the peer network side, the ModifyTask is not required since the Remote SDP is already provided to the CC-POI in the ActivateTask message.

4.9.1.8 SDP offer in the forward message – session to target non-local ID (CS)

Figure 4.9.1.8-1 illustrates the LI_T3 trigger sent from a CC-TF present in the MGCF to the CC-POI present in the IM-MGW as described in TS 33.128 [4] for an IMS session to a target non-local ID (CS domain) when the SDP offer is included in a forward SIP message (e.g. SIP INVITE).

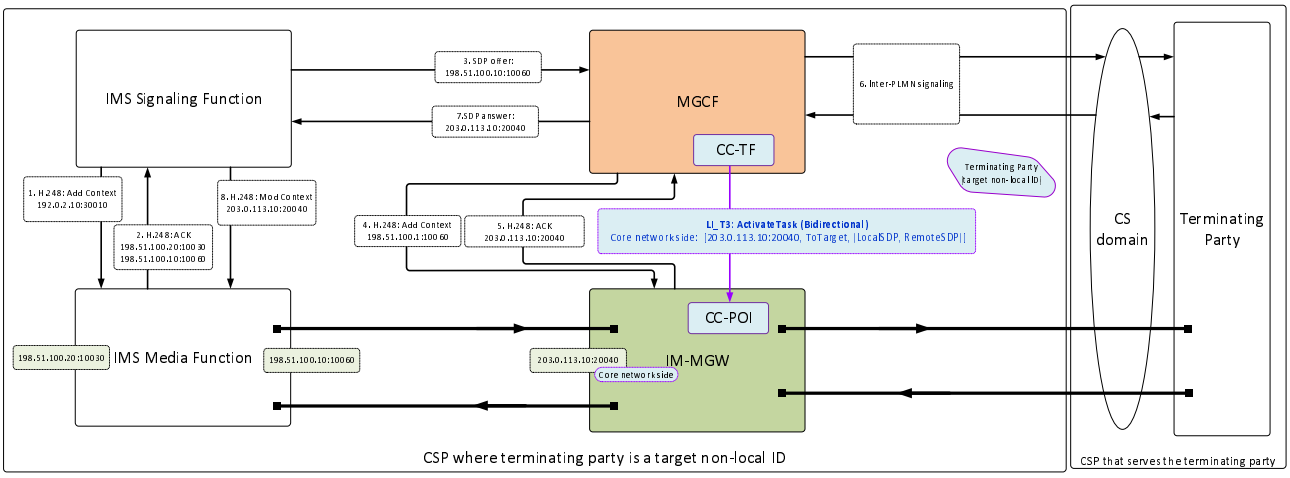


Figure 4.9.1.8-1: ActivateTask for sessions to target non-local ID in CS domain (SDP offer in forward SIP)

The ActivateTask that can be sent after step 5 has the following information.

The media interception is performed at the core network side:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Remote SDP.
- Trigger scope: Bidirectional.

The IP Address 203.0.113.10 and the UDP port number 20040 are also present in the SDP answer sent toward the originating end of the session.

The inter-PLMN signalling CS-based (e.g. ISUP). The details of this inter-PLMN signalling are not shown in figure 4.9.1.8-1.

The media streams received at the IP address 203.0.113.10 and UDP port 20040 are to the target.

4.9.1.9 SDP offer in the forward message – session from target non-local ID (CS)

Figure 4.9.1.9-1 illustrates the LI_T3 trigger sent from a CC-TF present in the MGCF to the CC-POI present in the IM-MGW as described in TS 33.128 [4] for an IMS session from a target non-local ID (CS domain) when the SDP offer is included in a forward SIP message (e.g. SIP INVITE).

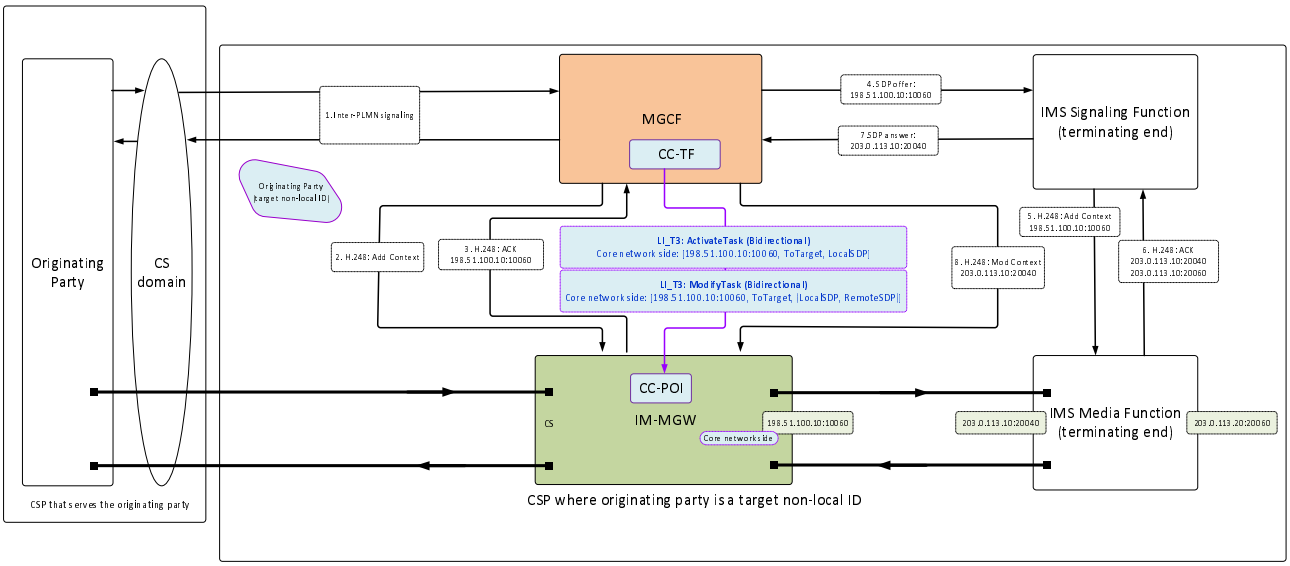


Figure 4.9.1.9-1: ActivateTask for sessions from target non-local ID in CS domain (SDP offer in forward SIP)

The ActivateTask that can be sent after step 3 has the following information.

The media interception is performed at the core network side:

- Target Identifiers:
  - IP address 198.51.100.10.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Trigger scope: Bidirectional.

The IP Address 198.51.100.10 and the UDP port number 10060 are also present in the SDP offer sent toward the terminating end of the session.

The inter-PLMN signalling is CS-based (e.g. ISUP). The details of this inter-PLMN signalling are not shown in figure 4.9.1.9-1.

The media streams received at the IP address 198.51.100.10 and UDP port 10060 are to the target.

A ModifyTask is sent after step 8 with the following information:

- Target Identifiers:
  - IP address 198.51.100.10.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.

- Remote SDP.
- Trigger scope: Bidirectional.

4.9.2 Asymmetric media streams

4.9.2.1 General

This clause illustrates the LI_T3 scenarios when the media streams in an IMS session are asymmetric, i.e. where the source IP address and UDP port number of the IP layer of a media stream sent from a media node is different from the destination IP address and UDP port number of the IP layer that it has given to the remote end.

In this case, a separate LI_T3 ActivationTask trigger is used to intercept the media for each of the media directions. For a simple session involving two parties (as illustrated here), two LI_T3 ActivationTask triggers are used.

Due to the possibility of a NAT (Network Address Translation) presence on the access side, it is presumed that symmetric media streams are used on the access side. Therefore, the illustrations shown in clause 4.9.1 apply for the cases where the media interception is done on the access side.

The following descriptions apply to all of the scenarios illustrated in the subsequent sub-clauses:

- The Remote SDP within the ActivateTask includes the SDP information received from the originating end or terminating end of the session (depending on the scenario) in the SDP offer or answer (depending on the scenario).
- The Local SDP within the ActivateTask includes the SDP information sent to the originating end or terminating end of the session (depending on the scenario) in the SDP offer or answer (depending on the scenario).
- The Trigger Scope value of Unidirectional indicates that the ActivateTask trigger applies for the interception of media in one direction. A separate ActivateTask trigger is sent to intercept the media in the opposite direction. The target identifiers present in the ActivateTask indicate the destination IP address and the UDP port number of the intercepted IMS media stream.
- The Payload Direction Assignment gives the direction of the media from the perspective of the target. When the value is ToTarget, the media destined to the IP address and UDP port number provided as the target identifiers in the ActivateTask are to the target. Likewise, when the value is FromTarget, the media destined to the IP address and UDP port number provided as the target identifiers in the ActivateTask are from the target.
- When the SDP offer is received on the side where the media interception is done, the two ActivateTask triggers are sent at the same time with one having the Local SDP and the other having the Remote SDP.
- When the SDP offer is sent from the side where the media interception is done, the Remote SDP can only be provided when an SDP answer is received. Therefore, in this case, the second trigger using ActivateTask is sent when the SDP offer is received.
- The Local SDP is used in the ActivateTask trigger where the local IP address and UDP port numbers are the target identifiers. The Remote SDP is included in the ActivateTask when the remote IP address and UDP port number are the target identifiers.

If the local IP address and UDP port number are sent in an SDP offer, then the remote IP address and UDP port number are received in the SDP answer. Likewise, if the remote IP address and UDP port number are received in an SDP offer, then the local IP address and UDP port number are sent in the SDP answer.

4.9.2.2 SDP offer in the forward message – originating session (core)

Figure 4.9.2.2-1 illustrates the LI_T3 triggers sent from a CC-TF present in the P-CSCF to the CC-POI present in the IMS-AGW when the media interception is done at the core network side of the IMS-AGW as per the two options described in TS 33.128 [4] for an IMS originating session from a target when the SDP offer is included in a forward SIP message (e.g. SIP INVITE).

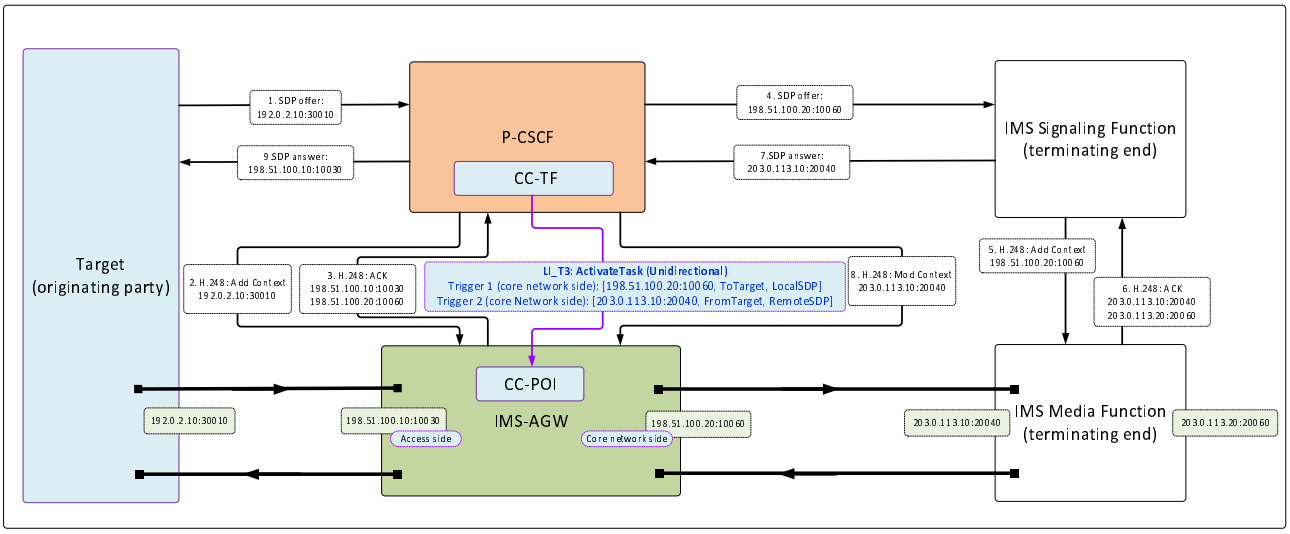


Figure 4.9.2.2-1: ActivateTasks for target originated sessions (SDP offer in forward SIP, core)

The first ActivateTask that can be sent after step 3 has the following information:

- Target Identifiers:
  - IP address 198.51.100.20.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Trigger Scope: Unidirectional.

The second ActivateTask that can be sent after the step 8 Ack (not shown) has the following information:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: FromTarget.
- Remote SDP.
- Trigger Scope: Unidirectional.

The IP Address 198.51.100.20 and the UDP port number 10060 are also present in the SDP offer sent toward the terminating end of the session.

The IP Address 203.0.113.10 and the UDP port number 20040 are also present in the SDP answer received from the terminating end of the session.

The media streams received at the IP address 198.51.100.20 and UDP port 10060 are to the target and the media streams sent to the IP address 203.0.113.10 and the UDP port 20040 are from the target.

4.9.2.3 SDP offer in the response message – originating session (core)

Figure 4.9.2.3-1 illustrates the LI_T3 triggers sent from a CC-TF present in the P-CSCF to the CC-POI present in the IMS-AGW when the media interception is done at the core network side of the IMS-AGW as per the two options described in TS 33.128 [4] for an IMS originating session from a target when the SDP offer is included in a response SIP message (e.g. SIP 200 OK).

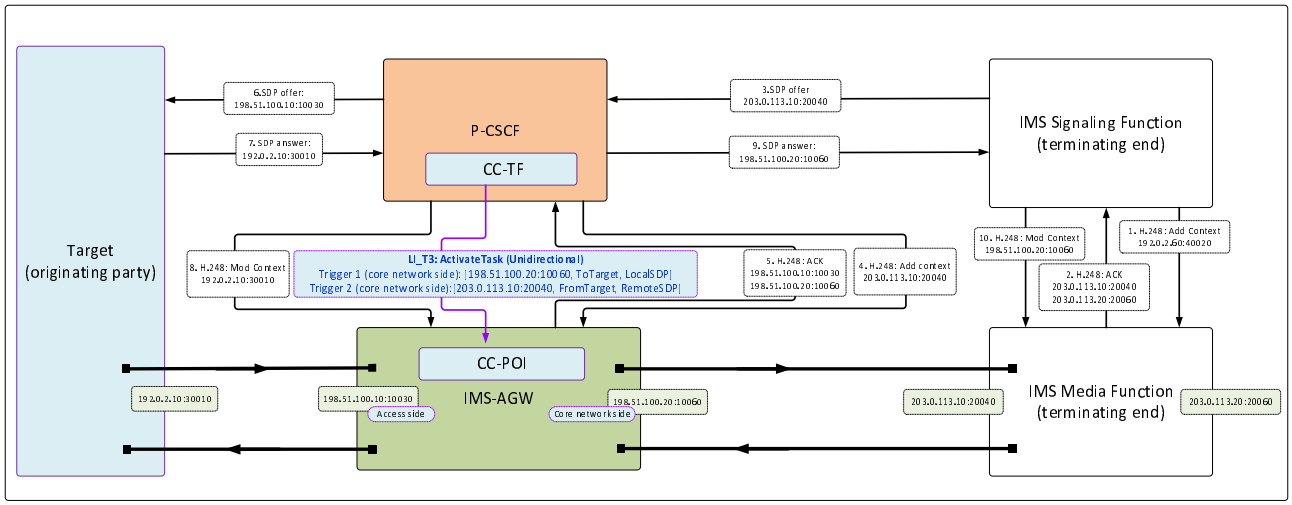


Figure 4.9.2.3-1: ActivateTask for target originated sessions (SDP offer in response SIP) (core)

The first ActivateTask that can be sent after step 5 has the following information:

- Target Identifiers:
  - IP address 198.51.100.20.
  - UDP port number 10060.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Trigger Scope: Unidirectional.

The second ActivateTask that can be sent after step 5 has the following information:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: FromTarget.
- Remote SDP.
- Trigger Scope: Unidirectional.

The IP Address 198.51.100.20 and the UDP port number 10060 are also present in the SDP answer sent toward the terminating end of the session.

The IP Address 203.0.113.10 and the UDP port number 20040 are also present in the SDP offer received from the terminating end of the session.

The media streams received at the IP address 198.51.100.20 and UDP port 10060 are to the target and the media streams sent to the IP address 203.0.113.10 and the UDP port 20040 are from the target.

4.9.2.4 SDP offer in the forward message – terminating session (core)

Figure 4.9.2.4-1 illustrates the LI_T3 triggers sent from a CC-TF present in the P-CSCF to the CC-POI present in the IMS-AGW when the media interception is done at the core network side of the IMS-AGW as per the two options described in TS 33.128 [4] for an IMS terminating session to a target when the SDP offer is included in a forward SIP message (e.g. SIP INVITE).

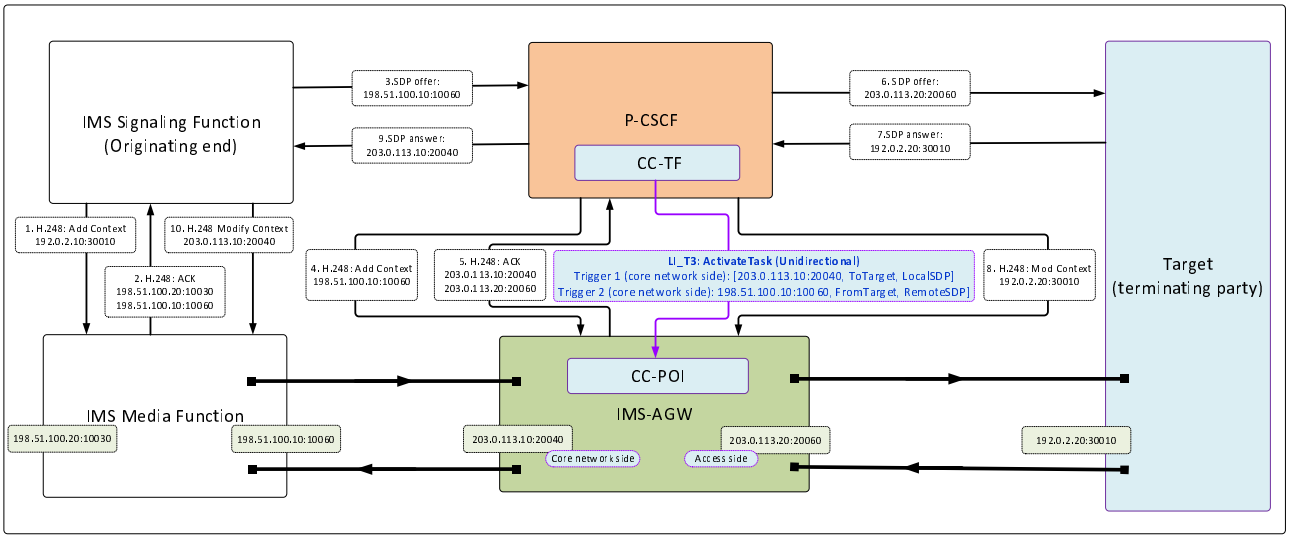


Figure 4.9.2.4-1: ActivateTask for target terminating sessions (SDP offer in forward SIP) (core)

The first ActivateTask that can be sent after step 5 has the following information:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Trigger Scope: Unidirectional.

The second ActivateTask that can be sent after step 5 has the following information:

- Target Identifiers:
  - IP address 198.51.100.10.
  - UDP port number 10060.
- Payload Direction Assignment: FromTarget.
- Remote SDP.
- Trigger Scope: Unidirectional.

The IP Address 203.0.113.10 and the UDP port number 20040 are also present in the SDP answer sent toward the originating end of the session.

The IP Address 198.51.100.10 and the UDP port number 10060 are also present in the SDP offer received from the originating end of the session.

The media streams received at the IP address 203.0.113.10 and UDP port 20040 are to the target and media streams sent to the IP address 198.51.100.10 and the UDP port 10060 are from the target.

4.9.2.5 SDP offer in the response message – terminating session (core)

Figure 4.9.2.5-1 illustrates the LI_T3 triggers sent from a CC-TF present in the P-CSCF to the CC-POI present in the IMS-AGW when the media interception is done at the core network side of the IMS-AGW as per the two options described in TS 33.128 [4] for an IMS terminating session to a target when the SDP offer is included in a response SIP message (e.g. SIP 200 OK).

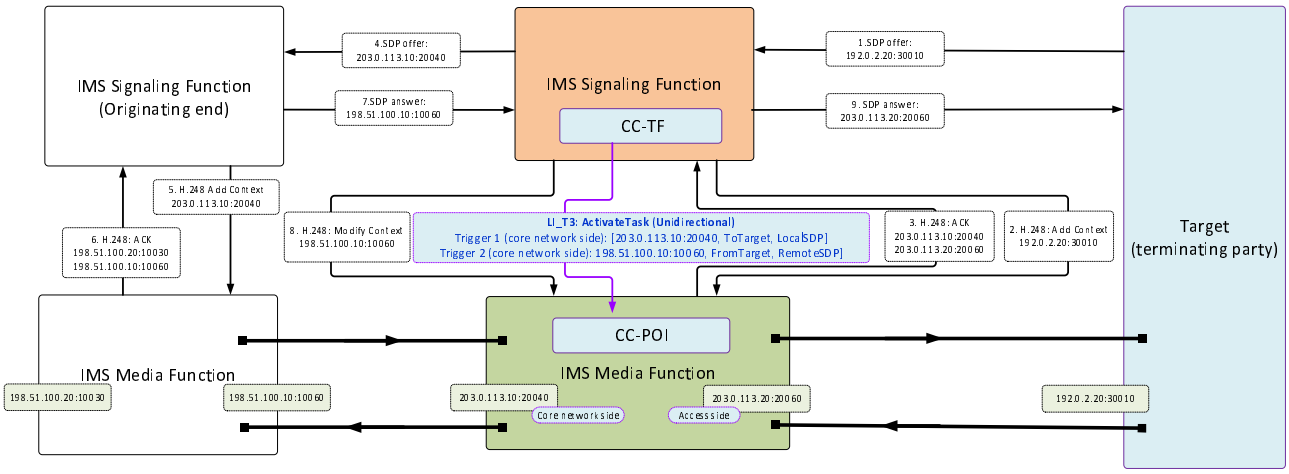


Figure 4.9.2.5-1: ActivateTask for target terminating sessions (SDP offer in response SIP) (core)

The first ActivateTask that can be sent after step 3 has the following information:

- Target Identifiers:
  - IP address 203.0.113.10.
  - UDP port number 20040.
- Payload Direction Assignment: ToTarget.
- Local SDP.
- Trigger Scope: Unidirectional.

The second ActivateTask that can be sent after step 8 Ack (not shown) has the following information:

- Target Identifiers:
  - IP address 198.51.100.10.
  - UDP port number 10060.
- Payload Direction Assignment: FromTarget.
- Remote SDP.
- Trigger Scope: Unidirectional.

The IP Address 203.0.113.10 and the UDP port number 20040 are also present in the SDP offer sent toward the originating end of the session.

The IP Address 198.51.100.10 and the UDP port number 10060 are also present in the SDP answer received from the originating end of the session.

The media streams received at the IP address 203.0.113.10 and UDP port 20040 are to the target and media streams sent to the IP address 198.51.100.10 and the UDP port 10060 are from the target.

4.9.3 Illustrations of IMS CC-POI triggering and xCC generation scenarios

4.9.3.1 General

This clause provides a few diagrams to illustrate the relationship between the parameters a CC-POI receives in the LI_T3 trigger (see clauses 4.9.1 and 4.9.2) and the xCC that it generates.

The following apply to all the diagrams presented in this clause:

- The Context ID is shown as C and C = ? infers that the value of the Context ID value is not known yet.

- The Termination Points are shown as T and T = ? infers that the value of the Termination Point is not known yet.
- The Local Descriptor is shown as L and L = ? infers that the details of the Local Descriptor (always assigned by the CC-POI) are not known yet.
- The Remote Descriptor is shown as R and R = ? infers that the details (always received from the remote end) are not know yet.
- The information passed on in the LI_T3 ActivateTask and LI_T3 ModifyTask are as defined in TS 33.128 [4].
- The xCC includes the entire IP packets that pass through the Termination Point (i.e. IP layer, UDP layer and RTP media streams).

In the illustrations, the SDP offer is always sent in the forward direction (i.e. from the originating end to the termination end). The illustrations of target originating session and target terminating session show the differences in the timing aspects of triggers for the SDP offer sent versus the SDP offer received scenarios.

The LI_T3 triggering aspects when the SDP offer is sent from the terminating end to the originating end are illustrated in clause 4.9.1 and 4.9.2. From the xCC generation point of view, that does not make much of a difference.

The IMS media can be symmetric or asymmetric when the interception is done at the core network side (IMS-AGW, TrGW, IM-MGW, MRFP) or peer network side (TrGW). The IMS media is presumed to be symmetric when the interception is done at the access side (IMS-AGW) due to the possibility of NAT presence.

4.9.3.2 Symmetric media streams

4.9.3.2.1 General

Based on the triggering illustrations shown in clause 4.9.1, this clause provides a few illustrations on how the CC-POI uses the information that it receives from the CC-TF in the LI_T3 triggering messages in constructing the IP packets that include the IMS media.

In the following illustrations, IMS media interception may be done either at the core network side or at access side of the IMS Media Function.

4.9.3.2.2 Interception at the core network session – originating session

Figure 4.9.3.2.2-1 illustrates the case where the SDP offer is received from the originating end of the session (target) and the media interception is done at the core network side of the IMS Media Function. As shown, the CC-TF sends two triggers, the second one to provide the Remote SDP information, since the same is not available at the CC-TF when the first trigger is sent.

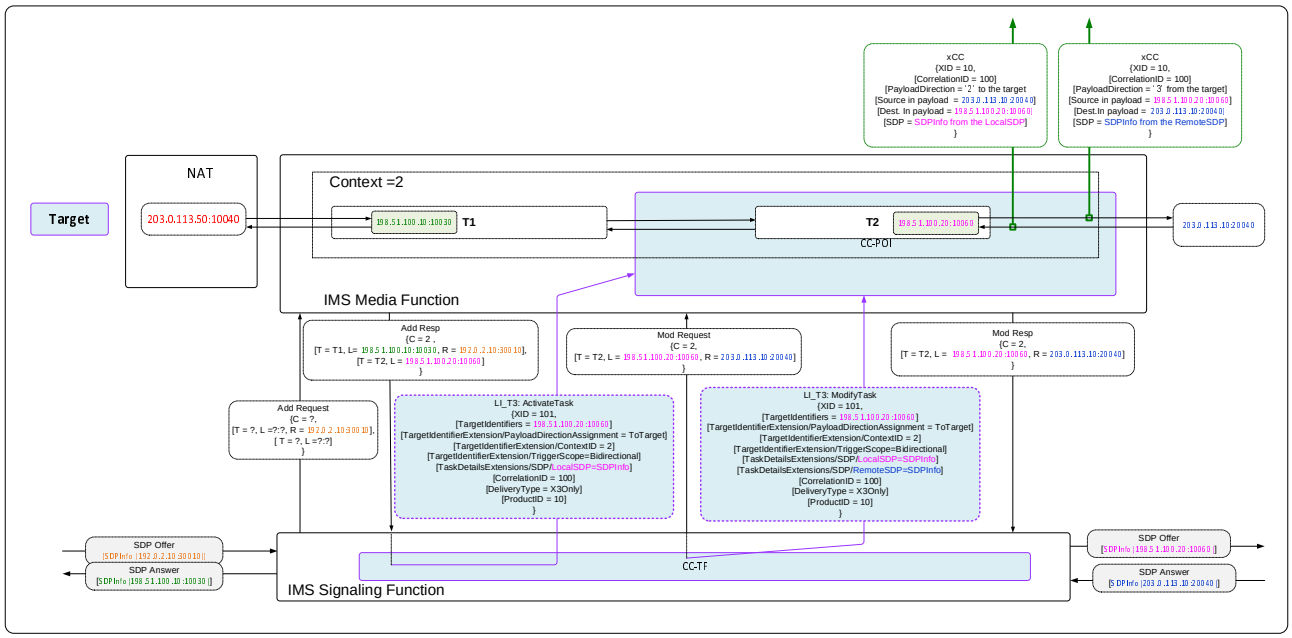


Figure 4.9.3.2.2-1: Media interception at the core network side (SDP offer is sent)

As shown in figure 4.9.3.2.2-1, when the LI_T3 ActivateTask is sent the Remote SDP is not known to the IMS Signalling Function. For that reason, a second trigger LI_T3 ModifyTask is sent later.

The SDP information included in the xCC is always the SDP information from the media destination perspective. For xCC generated for the incoming media, it would be based on the Local SDP that the CC-POI receives in the LI_T3 ActivateTask. For xCC generated for the outgoing media, it would be based on the Remote SDP that the CC-POI receives in the LI_T3 ModifyTask.

4.9.3.2.3 Interception at the core network side – terminating session

Figure 4.9.3.2.3-1 illustrates the case where the SDP offer is received from the originating end of the session and the media interception is done at the core network side of the IMS Media Function. The target is on the terminating side of the session. As shown, CC-TF sends only one trigger that includes both the Local SDP and the Remote SDP since the remote SDP information is already received in the SDP offer when the CC-TF sends the first trigger.

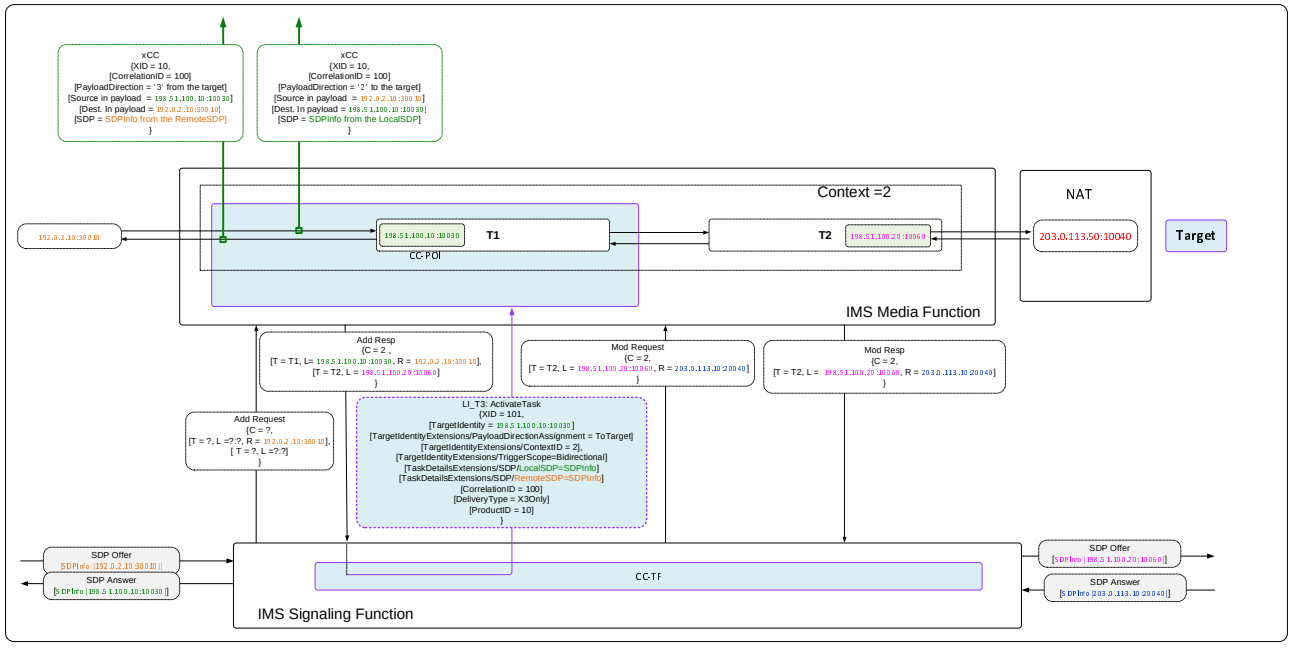


Figure 4.9.3.2.3-1: Media interception at the core network side (SDP offer is received)

The SDP information included in the xCC is always the SDP information from the media destination perspective. For xCC generated for the incoming media, it would be based on the Local SDP that the CC-POI receives in the LI_T3 ActivateTask. For xCC generated for the outgoing media, it would be based on the Remote SDP that the CC-POI receives in the same LI_T3 ActivateTask.

4.9.3.2.4 Interception at the access side – originating session

Figure 4.9.3.2.4-1 illustrates the case where the SDP offer is received from the originating end of the session (target) and the media interception is done at the access side of the IMS Media Function. As shown, the CC-TF sends only one trigger that includes both the Local SDP and the Remote SDP since the remote SDP information is already received in the SDP offer when the CC-TF sends the first trigger.

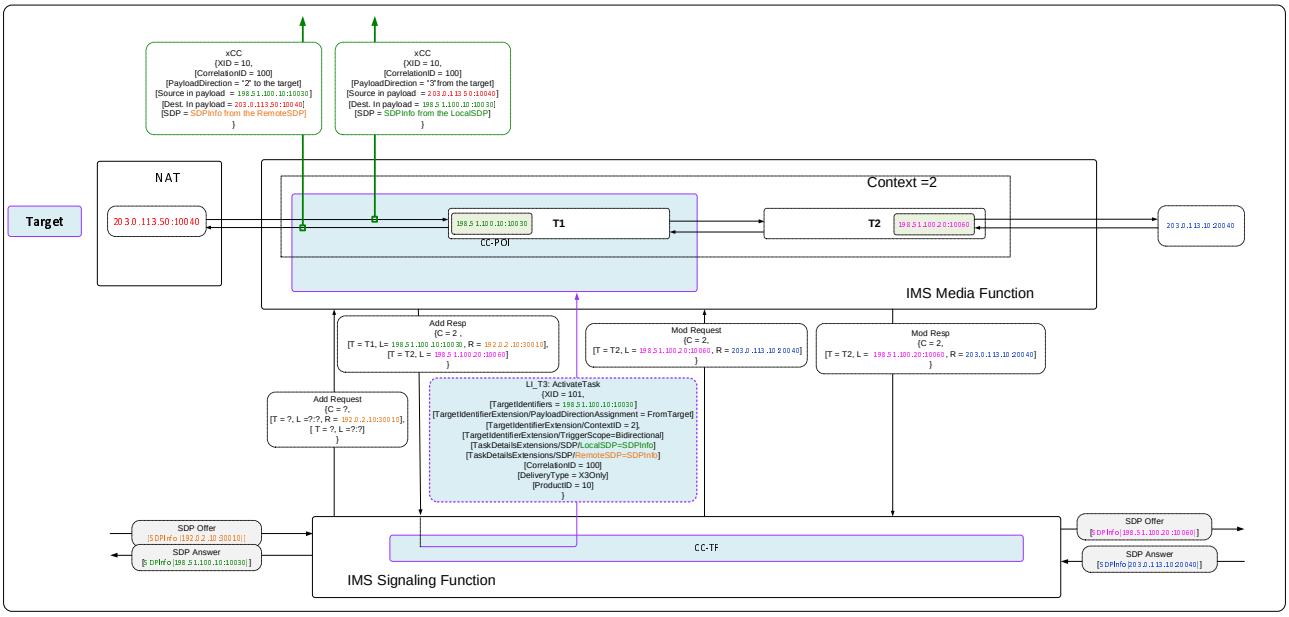


Figure 4.9.3.2.4-1: Media interception at the access side (SDP offer is received)

The SDP information included in the xCC is always the SDP information from the media destination perspective. For xCC generated for the incoming media, it would be based on the Local SDP that the CC-POI receives in the LI_T3 ActivateTask. For xCC generated for the outgoing media, it would be based on the Remote SDP that the CC-POI receives in the same LI_T3 ActivateTask.

In this case, due to the presence of a NAT, the IP address and UDP port number included in the SDP information of outgoing IP packets that include the media are different from the destination IP address and UDP port number of the same IP packets. The IMS Media Function always waits for the incoming IP packets to extract the source IP address and UDP port number and uses the same as the destination IP address and UDP port number in the outgoing IP packets. If NAT is present, then illustration shown in figure 4.9.3.2.3-1 can apply here as well.

4.9.3.2.5 Interception at the access side – terminating session

Figure 4.9.3.2.5-1 illustrates the case where the SDP offer is received from the originating end of the session and the media interception is done at the access side of the IMS Media Function. The target is on the terminating side of the session. As shown, the CC-TF sends two triggers, the second one to provide the Remote SDP information, since the same is not available at the CC-TF when the first trigger is sent.

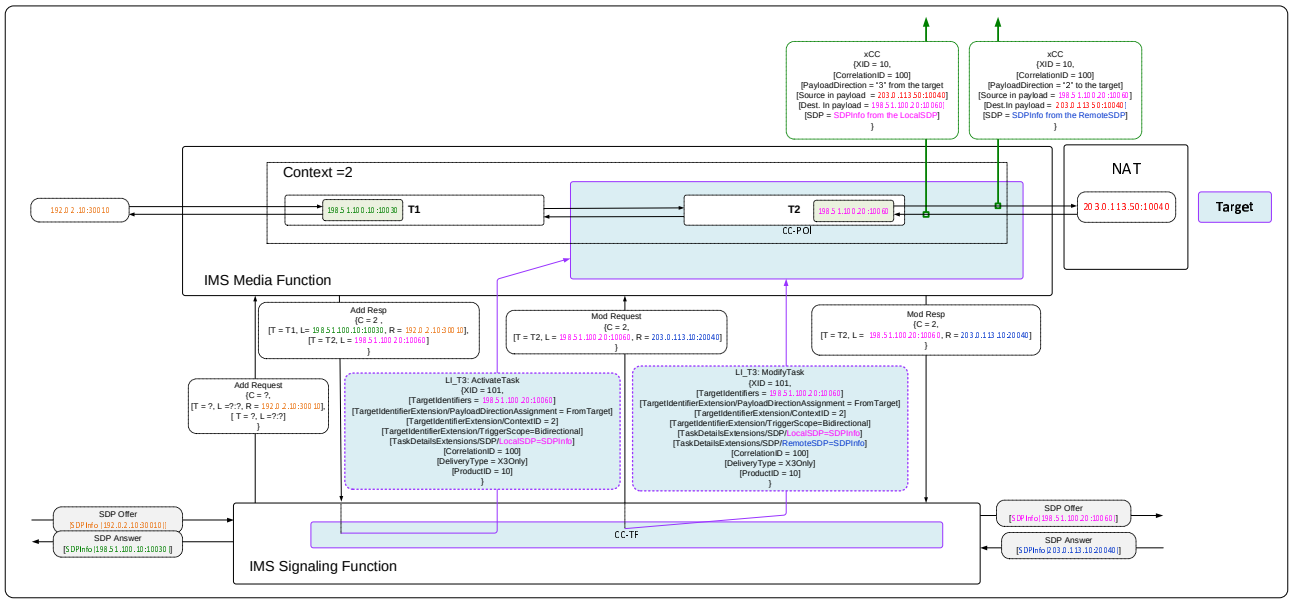


Figure 4.9.3.2.5-1: Media interception at the access side (SDP offer is sent)

As shown in figure 4.9.3.2.5-1, when the LI_T3 ActivateTask is sent the Remote SDP is not known to the IMS Signalling Function. For that reason, a second trigger LI_T3 ModifyTask is sent later.

The SDP information included in the xCC is always the SDP information from the media destination perspective. For xCC generated for the incoming media, it would be based on the Local SDP that the CC-POI receives in the LI_T3 ActivateTask. For xCC generated for the outgoing media, it would be based on the Remote SDP that the CC-POI receives in the LI_T3 ModifyTask.

In this case, due to the presence of a NAT, the IP address and UDP port number included in the SDP information of outgoing IP packets that include the media are different from the destination IP address and UDP port number of the same IP packets. The IMS Media Function always waits for the incoming IP packets to extract the source IP address and UDP port number and uses the same as the destination IP address and UDP port number in the outgoing IP packets. If NAT is present, then illustration shown in figure 4.9.3.2.2-1 can apply here as well.

4.9.3.3 Asymmetric media streams

4.9.3.3.1 General

Based on the triggering illustrations shown in clause 4.9.2, this clause provides a few illustrations on how the CC-POI uses the information that it receives from the CC-TF in the LI_T3 triggering messages in constructing the IP packets that include the IMS media.

In the following illustrations, IMS media interception is done either at the core network side of the IMS Media Function or at both sides of the IMS Media Function.

4.9.3.3.2 Interception at the core network session – originating session

Figure 4.9.3.3.2-1 illustrates the case where the SDP offer is received from the originating end of the session (target) and the media interception is done at the core network side of the IMS Media Function. As shown, the CC-TF sends two triggers, one for each direction of the media flow, with IP address and UDP port number included as the target identifiers within the trigger are the destination IP address and UDP port number for the media flow.

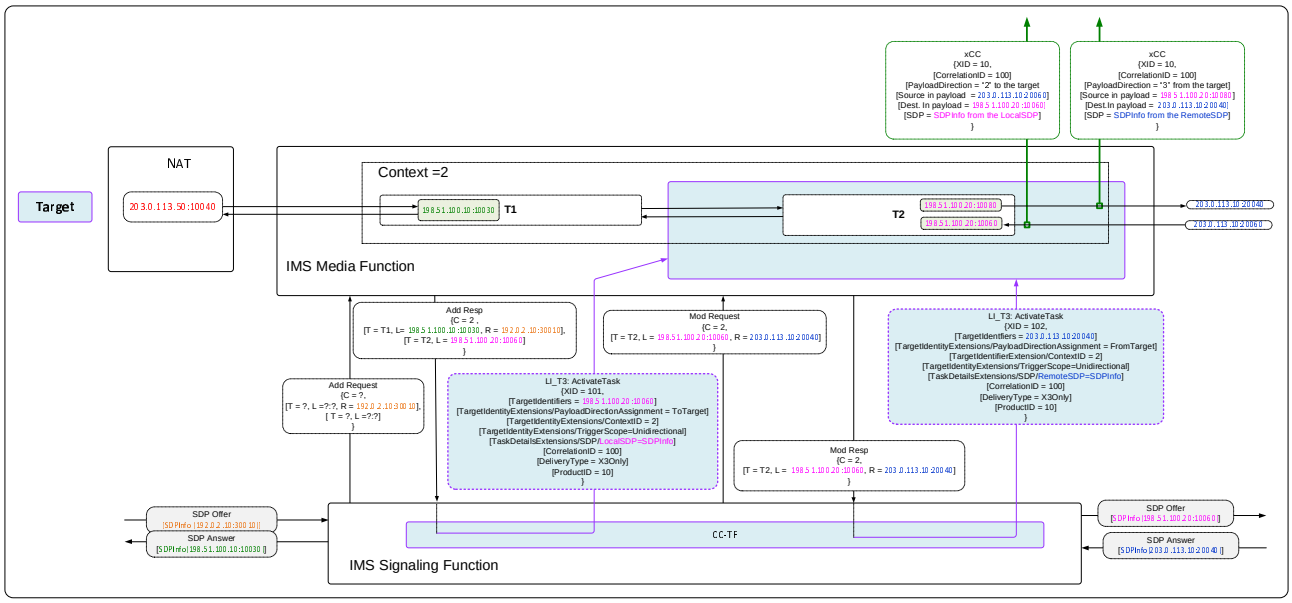


Figure 4.9.3.3.2-1: Media interception at the core network side (SDP offer is sent)

The SDP information included in the xCC is always the SDP information from the media destination perspective. For the xCC generated for the incoming media, it would be based on the Local SDP that the CC-POI receives in the LI_T3 ActivateTask. For xCC generated for the outgoing media, it would be based on the Remote SDP that the CC-POI receives in the second LI_T3 ActivateTask.

4.9.3.3.3 Interception at the core network side – terminating session

Figure 4.9.3.3.3-1 illustrates the case where the SDP offer is received from the originating end of the session and the media interception is done at the core network side of the IMS Media Function. The target is on the terminating side of the session. As shown, the CC-TF sends two triggers, one for each direction of the media flow, with IP address and UDP port number included as the target identifiers within the trigger are the destination IP address and UDP port number for the media flow.

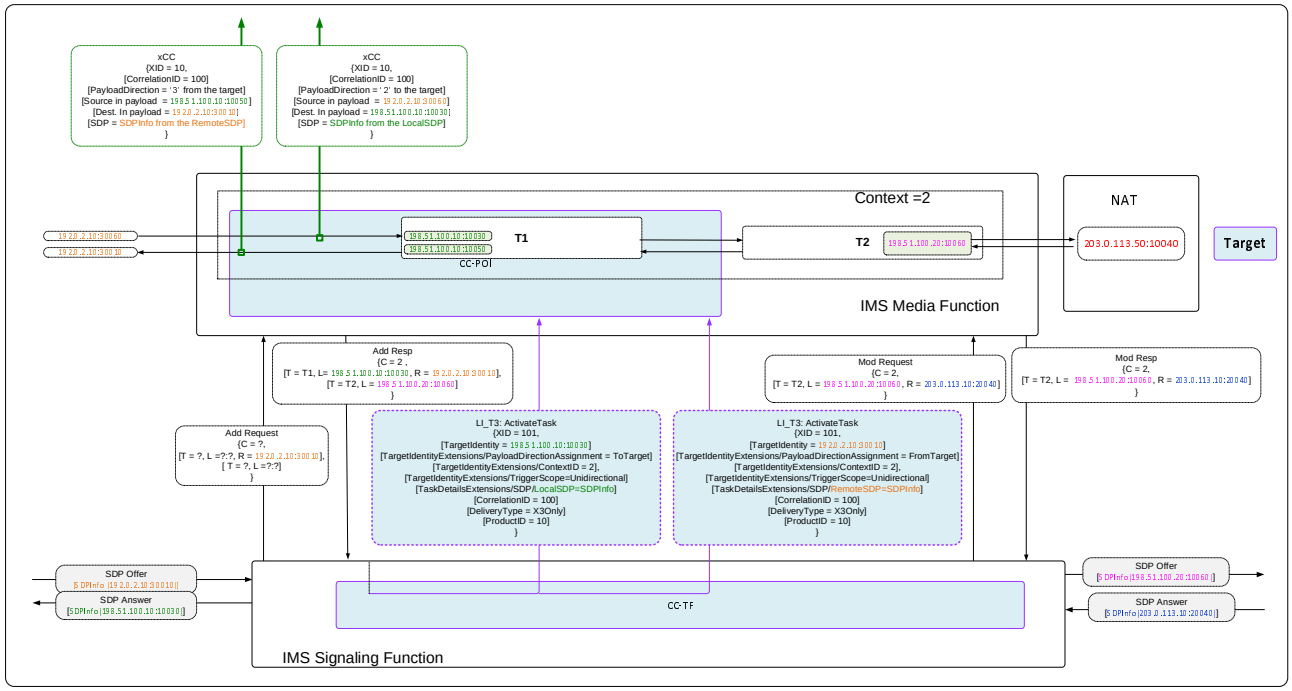


Figure 4.9.3.3.3-1: Media interception at the core network side (SDP offer is received)

In the flow-diagram shown in figure 4.9.3.3.3-1, both LI_T3 ActivateTask triggers are sent at the same time, i.e. when the H.248 Add Response is received from the IMS media function since both Remote SDP (received in the SDP offer) and Local SDP (that would be sent in the SDP answer) are available at the CC-TF.

The SDP information included in the xCC is always the SDP information from the media destination perspective. For the xCC generated for the incoming media, it would be based on the Local SDP that the CC-POI receives in the LI_T3 ActivateTask. For xCC generated for the outgoing media, it would be based on the Remote SDP that the CC-POI receives in the same LI_T3 ActivateTask.

4.9.3.3.4 Interception at both ends of IMS media function – target originating

Figure 4.9.3.3.4-1 illustrates the case where media interception is done on both ends of the IMS media function – each end intercepting the media in two different directions. As shown, CC-TF sends two triggers, one for each direction of the media flow, with IP address and UDP port number included as the target identifiers within the trigger are the destination IP address and UDP port number for the media flow.

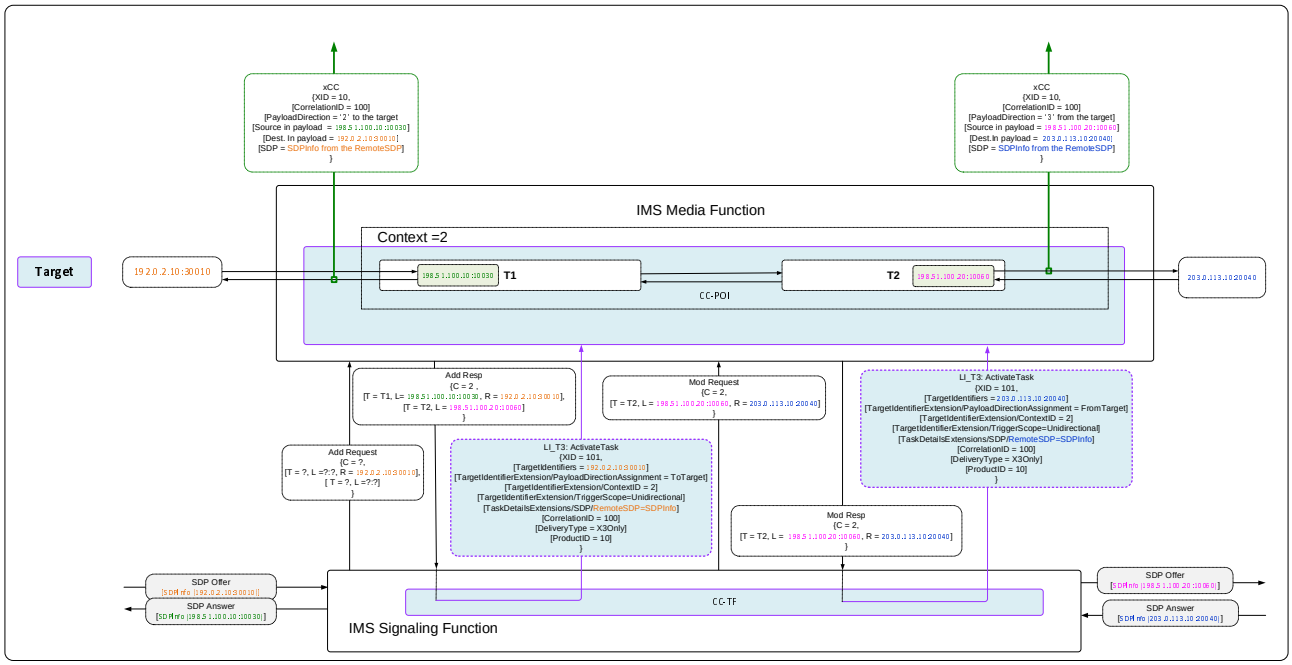


Figure 4.9.3.3.4-1: Media interception at both sides of IMS media function side (SDP offer is received)

In this illustration, the session is originated from a target. The SDP information included in the xCC is always the SDP information from the media destination perspective. In this case, the on both sides, the intercepted media are outgoing media streams. Therefore, both the LI_T3 ActivateTask triggers include the Remote SDP information.

4.9.3.3.5 Interception at both ends of IMS media function – terminating to target

Figure 4.9.3.3.5-1 illustrates the case where media interception is done on both ends of the IMS media function – each end intercepting the media in two different directions. As shown, CC-TF sends two triggers, one for each direction of the media flow, with IP address and UDP port number included as the target identifiers within the trigger are the destination IP address and UDP port number for the media flow.

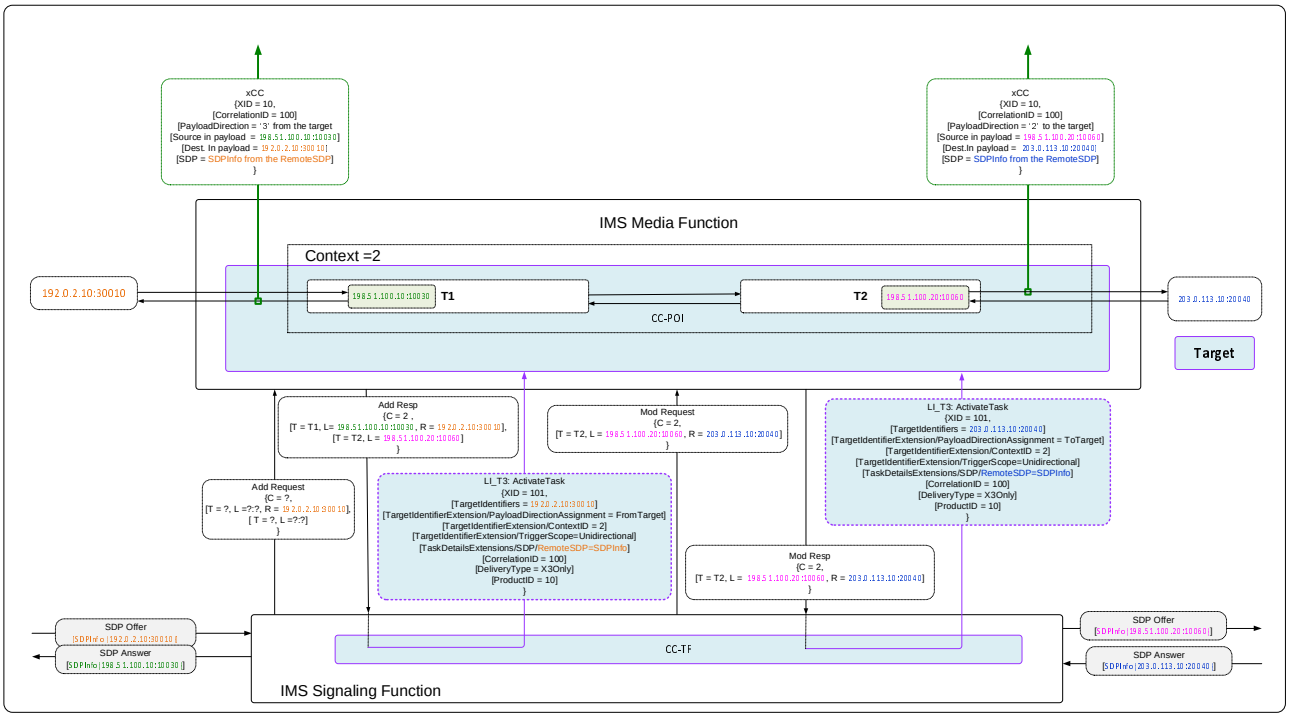


Figure 4.9.3.3.5-1: Media interception at both sides of IMS media function side (SDP offer is sent)

In this illustration, the session is terminated to a target. The SDP information included in the xCC is always the SDP information from the media destination perspective. In this case, the on both sides, the intercepted media are outgoing media streams. Therefore, both LI_T3 ActivateTask triggers include the Remote SDP information.

4.10 Illustration of LI for IMS data channel

4.10.1 IMS data channel background

4.10.1.1 General overview

The IMS Data Channel (IMS DC) architecture details are specified in TS 23.228 [8] and TS 24.186 [18]. These background clauses give a preview mainly keeping LI in mind even though LI specific guidance is provided in latter clauses.

IMS DC is an application specific bearer channel established between the two UEs present on an IMS session that allows the two users to enhance their IMS session experience through the sharing of application specific data (e.g. screensharing, live video streaming). The IMS DC that provides the control and signalling functions is referred to as Bootstrap Data Channel (BDC). The IMS DC used to exchange application specific media information is referred to as Application Data Channel (ADC).

NOTE: The terms IMS Bootstrap Data Channel, Bootstrap IMS Data Channel, BDC refer to the same Data Channel. Likewise, IMS Application Data Channel, Application IMS DC and ADC all refer to the same Data Channel.

The IMS DC specific functions that handle the IMS DC related functions include DCSF, DC-AS and MF. The DC-AS may also be in a third-party domain. The IMS DC session setup is done using the SIP signalling messages.

The Bootstrap Data Channel always terminates at the DCSF. IMS DC media passes through the MF. Figure 4.10.1.1-1 illustrates an overview of IMS DC Functions at a high-level.

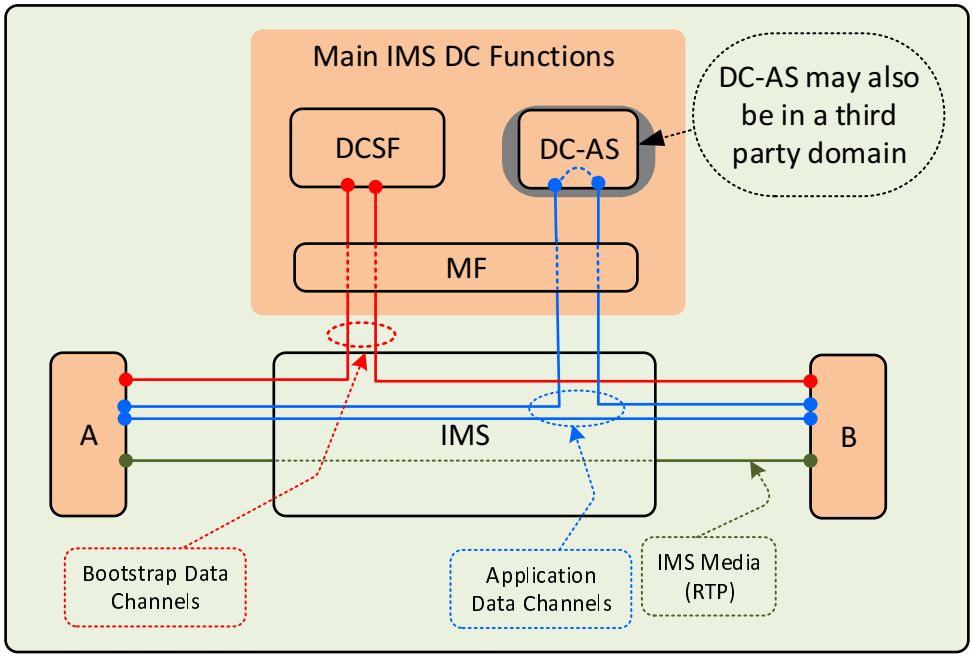


Figure 4.10.1.1-1 IMS DC Functions

In the illustration shown in figure 4.10.1.1-1, both A and B have IMS DC capabilities.

The Application IMS DC can be a P2A, A2P where the application specific media is exchanged between the UE and the DC-AS. The Application IMS DC can also be a P2A2P and P2P where the application specific media is exchanged between the two UEs with the former passing via the DC-AS. The MF may be configured to operate in multiple ways and the details of the same are described in the subsequent clauses. A reader familiar with the IMS DC capabilities defined in TS 23.228 [8] can go to clause 4.10.2 and 4.10.3 for the LI specific guidance in reference to IMS DC.

4.10.1.2 General principles

When a network provides the IMS DC capabilities, the registered UE that has subscribed for IMS may initiate the IMS DC setup with the other UE present on the same IMS session. In principle, either of the two UEs present on an IMS session may initiate IMS DC setup, independently, when the respective network where the UE has registered supports the IMS DC capabilities and the UEs have subscribed for the IMS DC. As a matter of fact, there can be two independent IMS DC setup can happen between the two UEs present on the same IMS session. In addition, the two UEs of an IMS session can be within the same network (i.e. intra-network IMS session) or in separate networks (i.e. inter-network IMS session). The terminating UE may also be a redirected-to UE (in a redirected IMS session).

The IMS session originating party can initiate the IMS DC either as a part of the initial IMS session setup or after the session is answered provided the originating IMS network supports the IMS DC capabilities. The terminating UE of an IMS session can only initiate the IMS DC setup after the session is answered provided the terminating network supports the IMS DC capabilities. In the present document, the UE that initiates the IMS DC setup is referred to as Local UE and the other UE is referred to as Remote UE. Furthermore, in the present document, the network where the Local UE is registered is referred to as Local Network and the network where the Remote UE is registered is referred to as Remote Network.

The IMS DC for both Local UE and Remote UE terminate (or pass through) at the IMS DC specific functions.

In the illustrations shown in figure 4.10.1.2-1, the IMS DC specific functions (DCSF, DC-AS and MF) are within the box shown as IMS DC. In the figure, the two cases where either the IMS session originating party (shown as A) or IMS session terminating party (shown as B) may initiate the IMS DC setup.

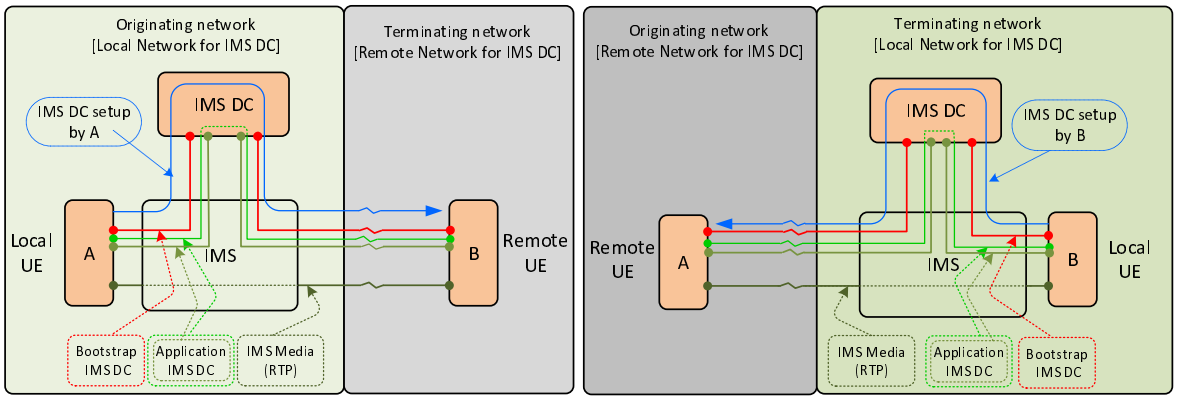


Figure 4.10.1.2-1: A generic overview of IMS DC setup principles

In the illustration shown in figure 4.10.1.2-1, the Remote UE has IMS DC capabilities and therefore, Bootstrap IMS DC and Application IMS DC can be setup to that Remote UE at the Local Network. The Application IMS DC shown in the above figure may be a P2A2P or P2P IMS DC which would be between the two UEs. The P2A or A2P IMS DC terminate at the IMS DC Functions.

When the Remote UE does not have the IMS DC capabilities, a Bootstrap IMS DC cannot be setup to that Remote UE and any application specific data (e.g. user-to-user, i.e. P2A2P, or P2P) can be transferred to and from that Remote UE only in a transcoded form. This is illustrated in figure 4.10.1.2-2 below.

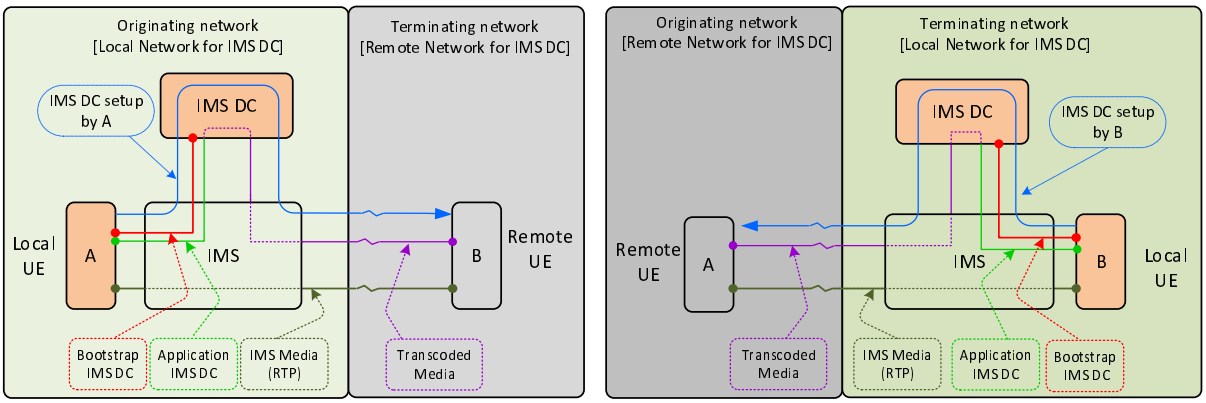


Figure 4.10.1.2-2: A generic overview of IMS DC setup principles with transcoded media to the Remote UE

In addition to the UE capabilities of IMS DC, the IMS DC related capabilities of the Remote Network also will have an influence on how the IMS DC media passes through in that network.

The subsequent clauses elaborate these points.

4.10.1.3 IMS Network and IMS UE capabilities with respect to IMS data channel

4.10.1.3.1 IMS network capabilities

With respect to the IMS DC, the following is a summary of the IMS network capabilities:

- The network has IMS DC capabilities.
- The network does not have any IMS DC capabilities.

An IMS network that does not have IMS DC capabilities may have limited IMS DC support (see NOTE) by allowing the locally registered IMS DC capable UEs to have IMS DC (BDC and ADC) setup by the other UE present on the IMS session, of-course, when the peer network where the other UE is registered has IMS DC capabilities.

The IMS network that does not have IMS DC capabilities presumed not to have DCSF, MF and DC-AS. The IMS network that does not IMS DC capabilities but has the limited IMS DC support may or may not have the DCSF and MF.

NOTE: The phrase "IMS network with limited IMS DC support" which is not defined in TS 23.228 [8] is introduced in the present document in referring to an IMS network that does not have IMS DC capabilities but allows its locally registered IMS capable UEs to have IMS DCs setup by the other UE of that IMS session present in peer network that has IMS DC capabilities. The present document uses the phrase for the illustration purposes.

4.10.1.3.2 IMS UE capabilities

With respect to IMS DC, the following is a summary of the IMS UE capabilities:

- IMS UE is IMD DC capable and subscribed to the IMS DC in the registered home network.
- IMS UE is IMD DC capable but not has subscribed to the IMS DC in the registered home network.
- IMS UE not IMS DC capable.

4.10.1.3.3 IMS UE perspective

From an IMS UE perspective:

- IMS DC capable UE can subscribe to IMS DC if and only if the registered home IMS network has IMS DC capabilities.

4.10.1.3.4 IMS network perspective

From an IMS network perspective:

- IMS network with IMS DC capabilities can be a Local Network for an IMS DC, if and only if the locally registered IMS DC capable UE has subscribed to IMS DC.
- IMS network with IMS DC capabilities can also be a Remote Network in all scenarios including the case when the locally registered IMS UE with IMS DC capabilities has subscribed to IMS DC.
- IMS network without the IMS DC capabilities can only be a Remote Network.

Note that the last case is true even when the IMS network that does not have the IMS DC capabilities but has limited IMS DC support.

4.10.1.3.5 Summary

Tables 4.10.1.3.5-1 and 4.10.1.3.5-2 summarize the applicability of IMS DCs on the above points.

Table 4.10.1.3.5-1: Applicable IMS DCs when the IMS network has IMS DC capabilities

IMS UE	IMS Network with IMS DC capabilities							
	As a Local Network				As a Remote Network			
	BDC	P2A/A2P	P2A2P	P2P	BDC	P2A/A2P	P2A2P	P2P
IMS DC capable subscribed to IMS DC	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
IMS DC capable, not subscribed to IMS DC	n/a	n/a	n/a	n/a	Yes	Yes	Yes	Yes
IMS DC not capable	n/a	n/a	n/a	n/a	No	No	Yes with transcoded media	Yes with transcoded media

Table 4.10.1.3.5-2: Applicable IMS DCs when the IMS network does not have IMS DC capabilities

IMS UE	IMS Network without the IMS DC capabilities							
	As a Local Network				As a Remote Network			
	Bootstrap	P2A/A2P	P2A2P	P2P	Bootstrap	P2A/A2P	P2A2P	P2P
IMS DC capable (option 1, NOTE)	n/a	n/a	n/a	n/a	Yes	Yes	Yes	Yes
IMS DC capable (option 2)	n/a	n/a	n/a	n/a	No	No	Yes with transcoded media	Yes with transcoded media
IMS DC not capable	n/a	n/a	n/a	n/a	No	No	Yes with transcoded media	Yes with transcoded media

NOTE: This option 1 is the case when the IMS network that does not have IMS DC capabilities has limited IMS DC support (see NOTE in clause 4.10.1.3.1).

4.10.1.4 IMS network functions and interfaces using generic topology views

4.10.1.4.1 General

The subsequent clauses illustrate the overall network topology views showing the IMS DC specific NFs and the interfaces without showing different IMS DC types. The illustrations are shown from the originating network (or originating end of the session for intra-network case) perspective. As described in clause 4.10.1.2, the originating party, the terminating party or the redirected-to party may setup the IMS DC, independently and parallelly.

The following conventions are used within the drawings when not self-explanatory.

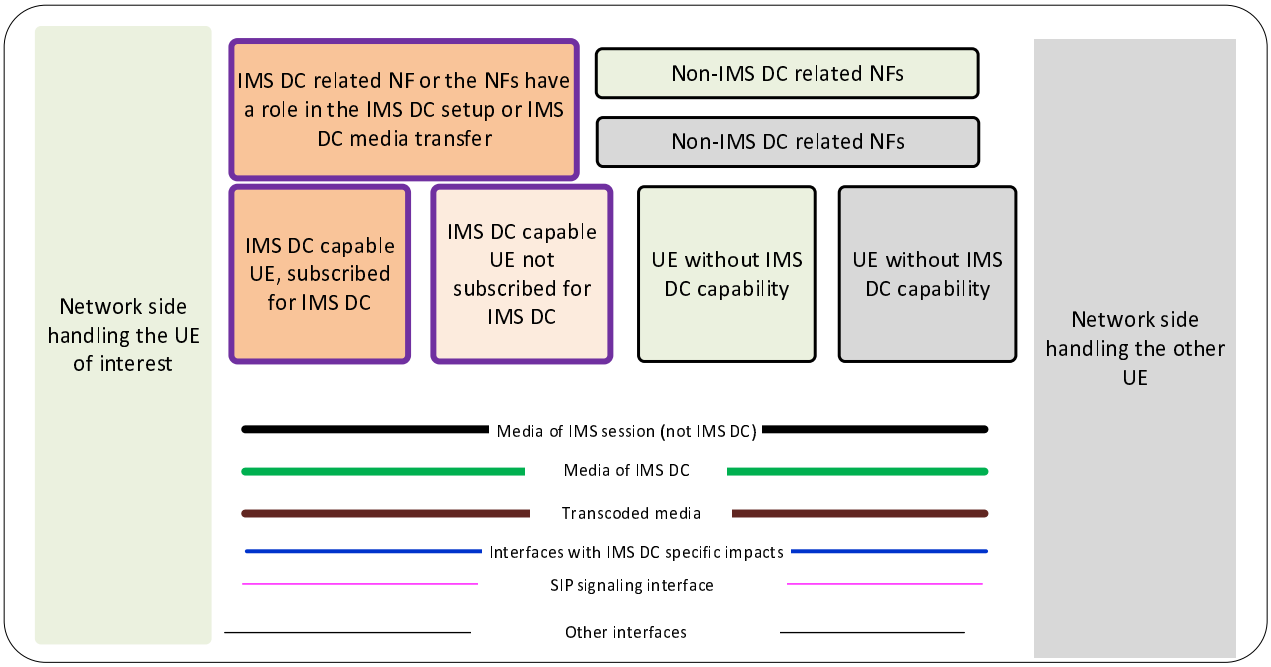


Figure 4.10.1.4.1-1: Conventions when not self-explanatory

4.10.1.4.2 IMS network has IMS DC capabilities

4.10.1.4.2.1 General

Clause 4.10.1.4.2 illustrates the scenarios when the IMS network has IMS DC capabilities.

NOTE: At the remote end of the IMS DC session, when the Remote UE is IMS DC capable and the Remote Network has the IMS DC capabilities, the insertion of the MF into the IMS DC media path is an option (irrespective of whether the Remote UE has subscribed to IMS DC or not) according to TS 23.228 [8]. However, the illustrations shown in the present document assume that the MF is always inserted as a guidance to provide the LI capabilities at the remote end of the IMS DC session.

When the locally registered UE has IMS DC capabilities and subscribed to IMS DC, the IMS network that has IMS DC capabilities can become a Local Network. When the locally registered UE does not have the IMS DC capabilities or has not subscribed IMS DC, then that IMS network can only be a Remote network for that IMS UE.

In all of the illustrations shown in the subsequent clauses, the IMS UEs present on an IMS session (and hence, on an IMS DC session) can be in the same network or in different networks. Furthermore, the terminating UE may also be a redirected-to UE of a redirected IMS session. Other applications such as MMTEL Enabler Server that may also be present are not shown.

4.10.1.4.2.2 Scenario 1 - originating UE is the local UE, remote UE has IMS DC capabilities

In the following illustrations, the UE-A is the originating party with the IMS DC capabilities and has subscribed to IMS DC. In all illustrations, the UE-A initiates the IMS DC setup. The DC-AS one of the IMS DC NFs may also be present in a third party domain.

In the illustration shown in figure 4.10.1.4.2.2-1, UE-A and UE-B are in the same network and have an IMS session. UE-A sets up the IMS session with UE-B. The UE-B is IMS DC capable, may or may not have subscribed for the IMS DC. The illustration assumes that MF is inserted into the IMS DC media path at the terminating end of the session.

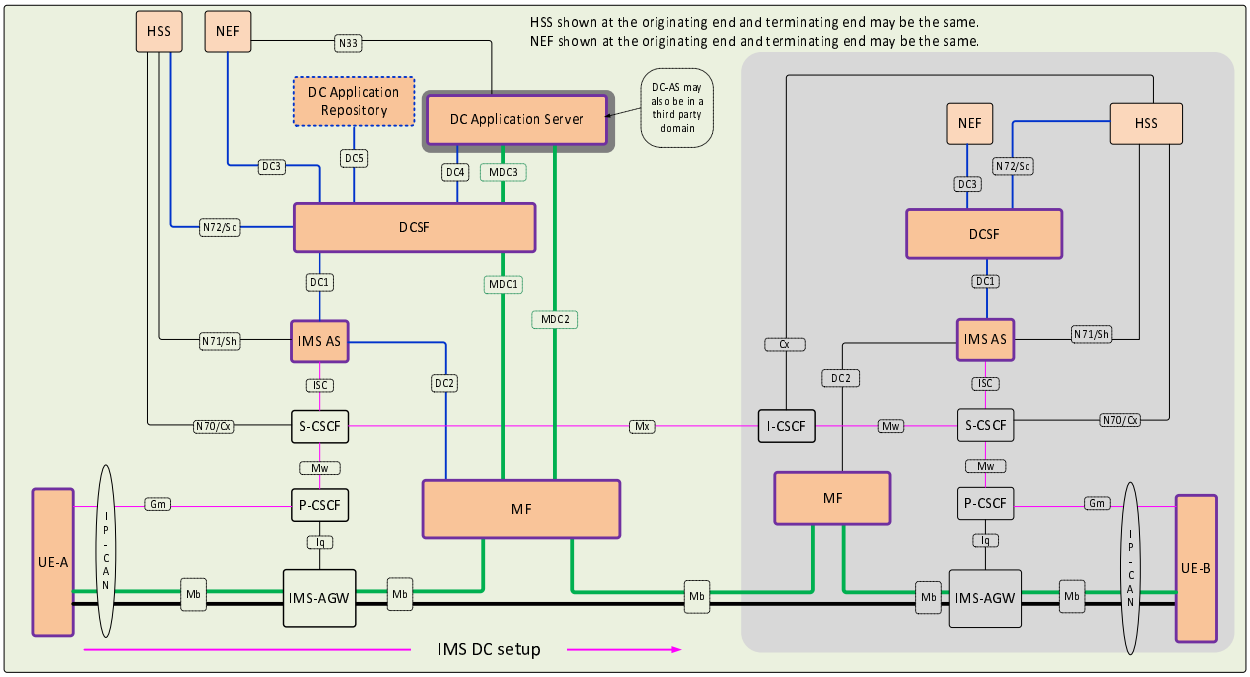


Figure 4.10.1.4.2.2-1: IMS network with IMS DC capabilities (intra-network case)

In the illustration shown in figure 4.10.1.4.2.2-2, UE-A and UE-B who are on an IMS session are in different networks. UE-A sets up the IMS session with UE-B and therefore, the originating network is the Local Network and the terminating network is the Remote Network. In this illustration, the UE-B is IMS DC capable, may or may not have subscribed for the IMS DC. In this illustration, MF is inserted into the IMS DC media path in the terminating network.

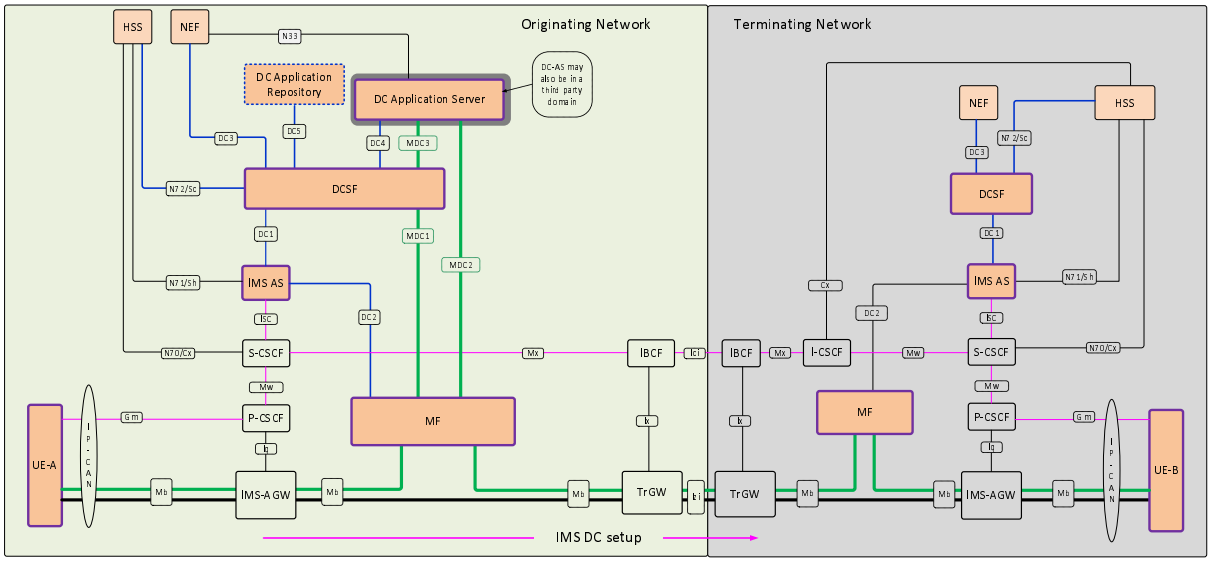


Figure 4.10.1.4.2.2-2: IMS network with IMS DC capabilities (inter-network case)

In the illustration shown in figure 4.10.1.4.2.2-3, an IMS session from UE-A to UE-B has been redirected to UE-C. UE-A sets up an IMS session to UE-C. The UE-C is IMS DC capable, may or may not have subscribed for the IMS DC. In this illustration, UE-C is IMS DC capable and MF is inserted into the IMS DC media path at the redirected-to end of the session.

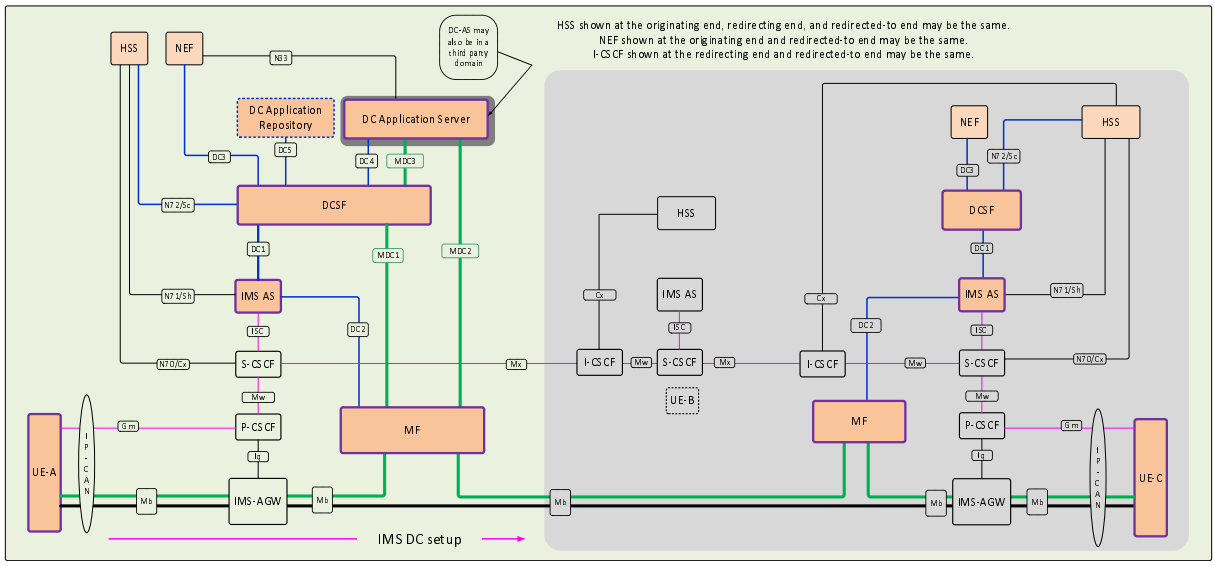


Figure 4.10.1.4.2.2-3: IMS network with IMS DC capabilities (intra-network redirecting case)

When the IMS UEs involved in a redirection case, several illustrations showing different inter-network possibilities are shown in clauses 4.10.1.4.5 and 4.10.1.4.6.

4.10.1.4.2.3 Scenario 2 - originating UE is the local UE, remote UE does not have IMS DC capabilities

In the illustration shown in figure 4.10.1.4.2.3-1, UE-A and UE-B have an IMS session. The UE-A is the originating party with the IMS DC capabilities and has subscribed to IMS DC. The UE-A sets up the IMS DC session with UE-B. In the illustration UE-B is not IMS DC capable and hence, the IMS DC media for a P2A2P DC and P2P DC is transferred in a transcoded form to the UE-B. Media transcoding may happen either at the DC-AS or at the MF depending on the type of IMS DC.

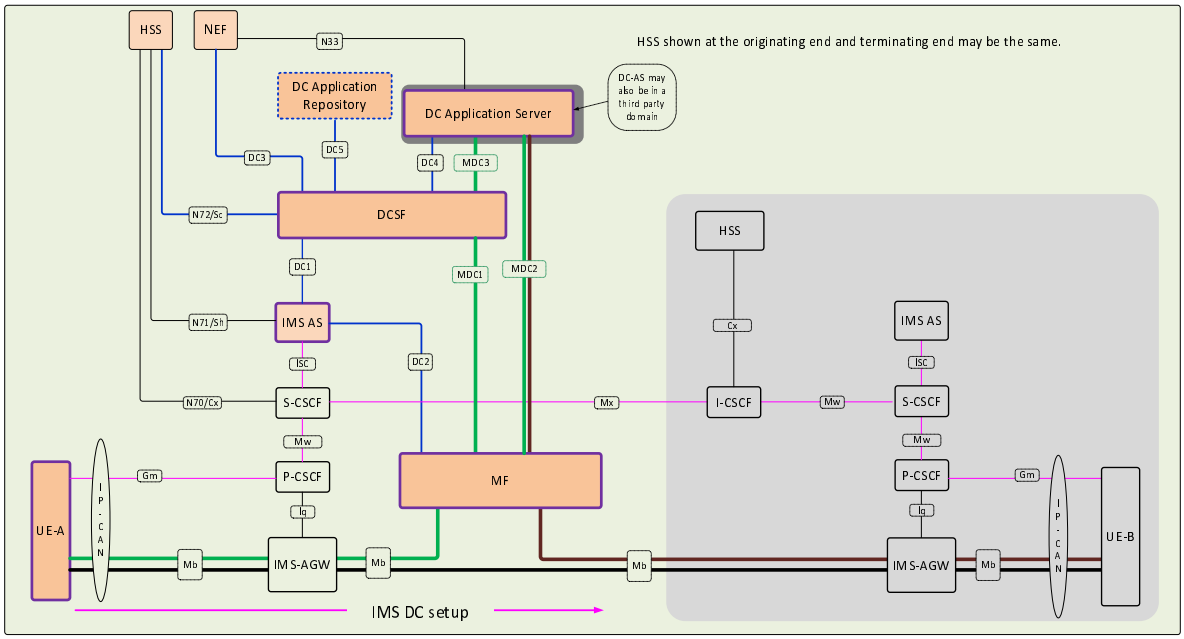


Figure 4.10.1.4.2.3-1: IMS network with IMS DC capabilities - remote UE does not have IMS DC capabilities

The illustration shown above is an intra-network scenario. The terminating party may also be in a different network. In addition, the terminating party may also be a redirected-to party in case the initial IMS session had a redirection.

4.10.1.4.2.4 Scenario 3 - originating IMS UE with IMS DC capabilities is the remote UE

4.10.1.4.2.4.1 General

The originating IMS UE with IMS DC capabilities can also be a Remote UE when the other IMS DC capable UE initiates the IMS DC session. The originating UE with the IMS DC capabilities may or may not have subscribed IMS DC. In the latter case, that originating UE can only be a Remote UE. From a topology point of view, the two scenarios look the same when MF is always inserted to the IMS DC media path.

4.10.1.4.2.4.2 IMS US has subscribed to IMS DC

In the illustration shown in figure 4.10.1.4.2.4.2-1, UE-A and UE-B have an IMS session. The UE-A is the originating party with the IMS DC capabilities. In this illustration, the UE-B (with IMS DC capabilities and subscribed for IMS DC) initiates the IMS DC setup. Therefore, in this illustration UE-A is a Remote UE. The illustration assumes that MF is inserted into the IMS media path at the originating end.

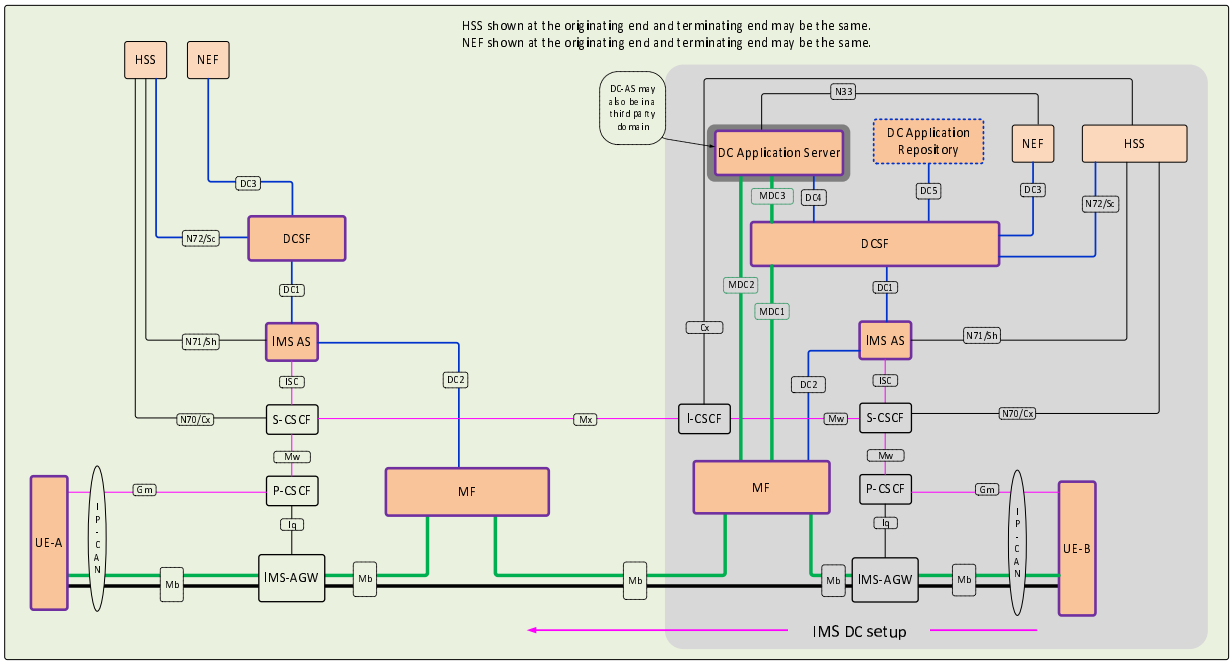


Figure 4.10.1.4.2.4.2-1: IMS network with IMS DC capabilities, originating UE subscribed to the IMS DC happens to be a Remote UE

4.10.1.4.2.4.3 IMS UE has not subscribed to IMS DC

In the illustration shown in figure 4.10.1.4.2.4.3-1, UE-A and UE-B have an IMS session. The UE-A is the originating party with the IMS DC capabilities but has not subscribed for IMS DC. Since UE-A has not subscribed for the IMS-DC, UE-A cannot initiate the IMS DC setup. Therefore, in this illustration UE-A is a Remote UE and other IMS UE (UE-B) present on the IMS session (subscribed to the IMS DC) has initiated the IMS DC setup. The illustration assumes that MF is inserted into the IMS media path at the originating end.

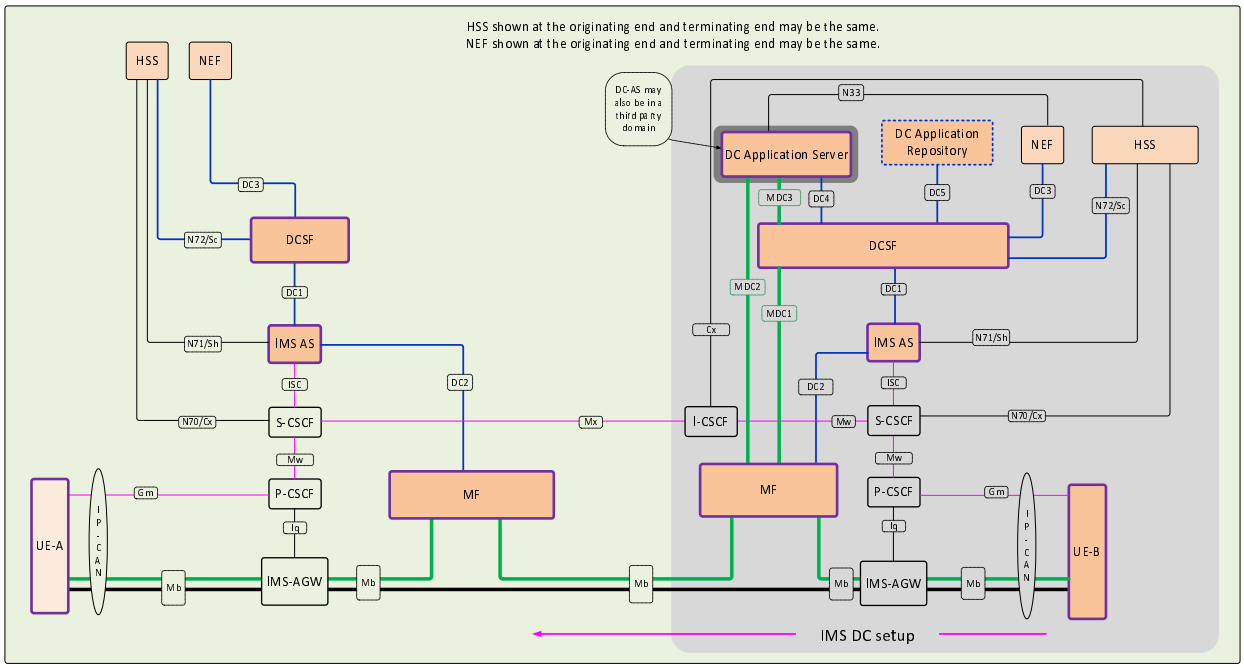


Figure 4.10.1.4.2.4.3-1: IMS network with IMS DC capabilities, originating UE has not subscribed for IMS DC

The illustrations show the intra-network scenarios. The terminating party may also be in a different network. In addition, the terminating party may also be a redirected-to party in case the initial IMS session had a redirection.

4.10.1.4.2.5 Scenario 4 - originating IMS UE without the IMS DC capabilities

In the illustration shown in figure 4.10.1.4.2.5-1, UE-A and UE-B have an IMS session. The UE-A is the originating party without the IMS DC capabilities. Since UE-A does not IMS-DC capabilities, UE-A can only be a Remote UE and can only receive the IMS DC media in a transcoded form when IMS DC capable other party (UE-B) present on the IMS session initiates the IMS DC setup. Media transcoding may happen either at the DC-AS or at the MF depending on the type of IMS DC. Such transcoded IMS DC media transfer is only applicable to P2A2P and P2P DCs.

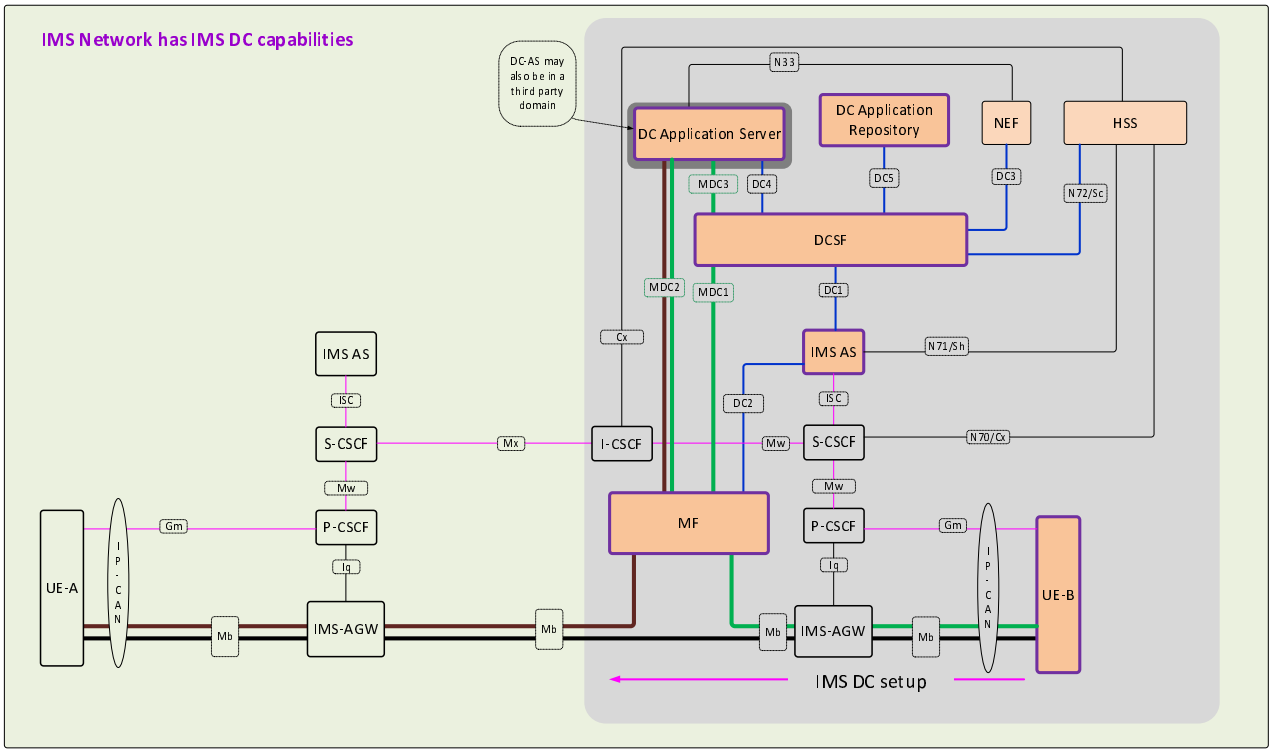


Figure 4.10.1.4.2.5-1: IMS network with IMS DC capabilities, originating UE does not have IMS DC capabilities

The terminating party may also be in a different network. In addition, the terminating party may also be a redirected-to party in case the initial IMS session had a redirection.

4.10.1.4.3 IMS network without IMS DC capabilities, with limited IMS DC support

4.10.1.4.3.1 General

The IMS network that does not have IMS DC capabilities, may have limited IMS DC support. In this case, the IMS network allows its locally registered IMS DC capable UEs to have IMS DCs (including BDC and P2A, A2P) setup by the other IMS UE on the same IMS session present in peer IMS network that has IMS DC capabilities.

The following aspects are considered to be part of IMS network that does not have IMS DC capabilities but has limited IMS DC support:

- DC-AS is not present.
- DCSF and MF may or may not be present (See NOTE).
- When the IMS network has DCSF and MF:
  - The subscription to IMS DC from an IMS DC capable UE is not supported.
  - Locally registered IMS DC capable UE cannot initiate an IMS DC setup.
- IMS DC setup request, if received from another network, is not blocked.
- Locally registered IMS DC capable UE can have Bootstrap IMS DC terminated at the peer network that has IMS DC capabilities.
- Locally registered IMS DC capable UE can have P2A, A2P IMS DCs with the DC-AS present in another network that has IMS DC capabilities.
- Locally registered IMS DC capable UE can have P2A2P IMS DC via the DC-AS present in another network that has the IMS DC capabilities with another IMS DC capable UE present in that peer network.

- Locally registered IMS DC capable UE can have P2P IMS DC with another IMS DC capable IUE present in the peer network that has IMS DC capabilities.

NOTE: The present document makes this assumption from the point of view of LI. Furthermore, the illustrations assume that when the IMS Network has the MF and it would insert the same to the IMS DC media path.

Based on the above aspects, an IMS network that has limited IMS DC support can only be a Remote Network for an IMS DC. In other words, any discussion of IMS DC may apply only for inter-network scenarios. As illustrated in clause 4.1.10.4.2, an IMS UE that is not capable of IMS DC can only receive or send IMS DC media in a transcoded form.

Since to support the IMS DC setup initiated by one of the IMS UEs present on the IMS session, the respective IMS network where the UE has registered has to have to IMS DC capabilities, the illustrations shown in the subsequent clauses do not apply for intra-network IMS sessions. The terminating party may also be a redirected-to party in case the initial IMS session had a redirection.

4.10.1.4.3.2 Scenario 1 – as a remote network, registered IMS UE has IMS DC capabilities

In the illustration shown in figure 4.10.1.4.3.2-1 UE-A is the originating party with IMS DC capabilities. Since the originating network does not have IMS DC capabilities, the UE-A can only be a Remote UE for an IMS DC setup by UE-B (subscribed to the IMS DC in the respective network) In this illustration the originating network that has the limited IMS DC support has DCSF and MF and it inserts the MF into the IMS DC media path.

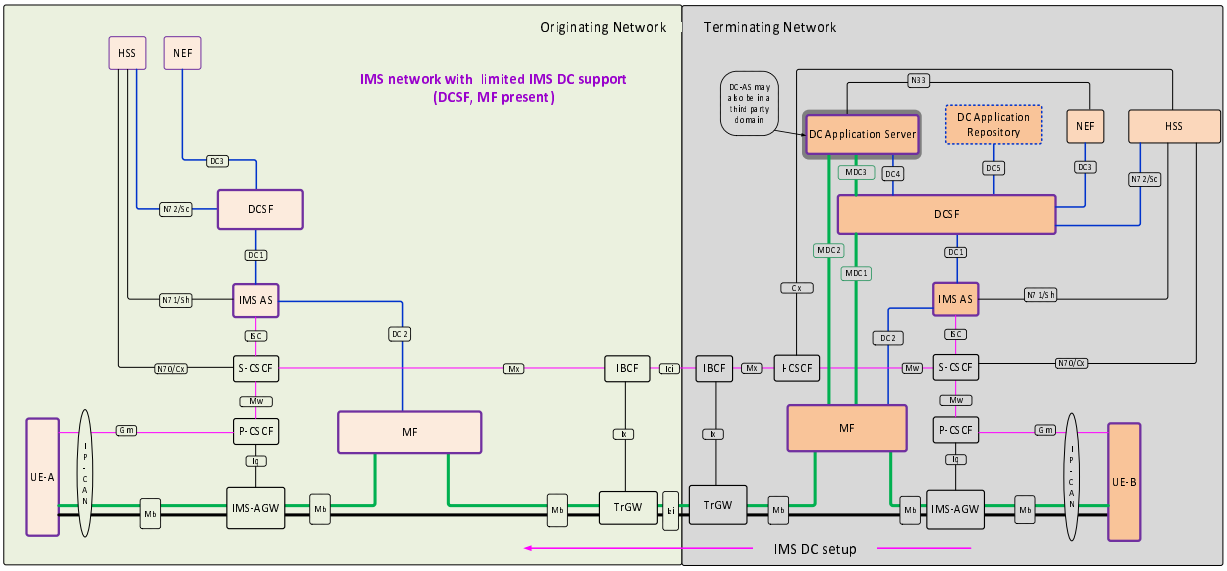


Figure 4.10.1.4.3.2-1: IMS network with limited IMS DC support, IMS UE has IMS DC capabilities, possibility 1

In the illustration shown in figure 4.10.1.4.3.2-2, UE-A is the originating party with IMS DC capabilities. Since the originating network does not have IMS DC capabilities, the UE-A can only be a Remote UE for an IMS DC setup by UE-B (subscribed to the IMS DC in the respective network). In this illustration the originating network that has limited IMS DC support has not deployed a DCSF or MF.

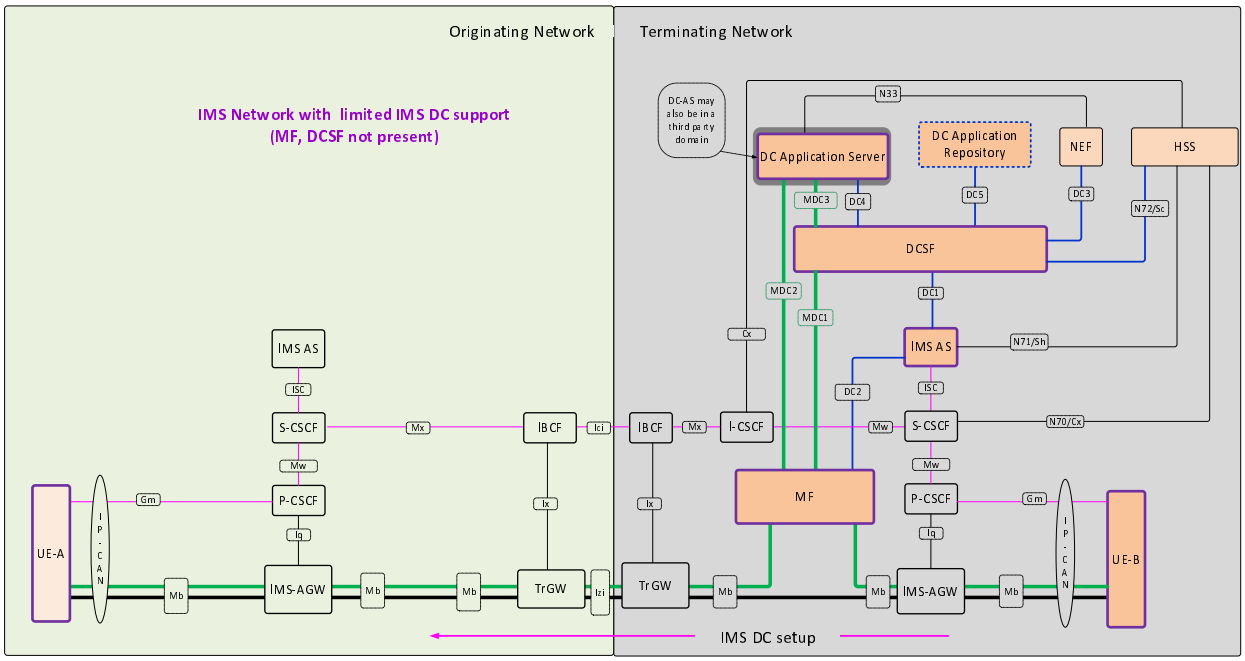


Figure 4.10.1.4.3.2-2: IMS network with limited IMS DC support, IMS UE has IMS DC capabilities, possibility 2

NOTE: Since there is no MF in the originating network, if UE-A is a target, the interception of IMS DC related media can be a challenge in the originating network. The possibility of this topology is still being studied.

4.10.1.4.3.3 Scenario 2 – as a remote network, registered IMS UE does not have IMS DC capabilities

In the illustration shown in figure 4.10.1.4.3.3-1 UE-A is the originating party without the IMS DC capabilities. Since the UE-A does not have IMS DC capabilities, UE-A can only receive the IMS DC media in a transcoded form when IMS DC capable UE-B that has subscribed to IMS DC (in the terminating network) initiates the IMS DC setup. Media transcoding may happen either at the DC-AS or at the MF (in the terminating network) depending on the type of IMS DC.

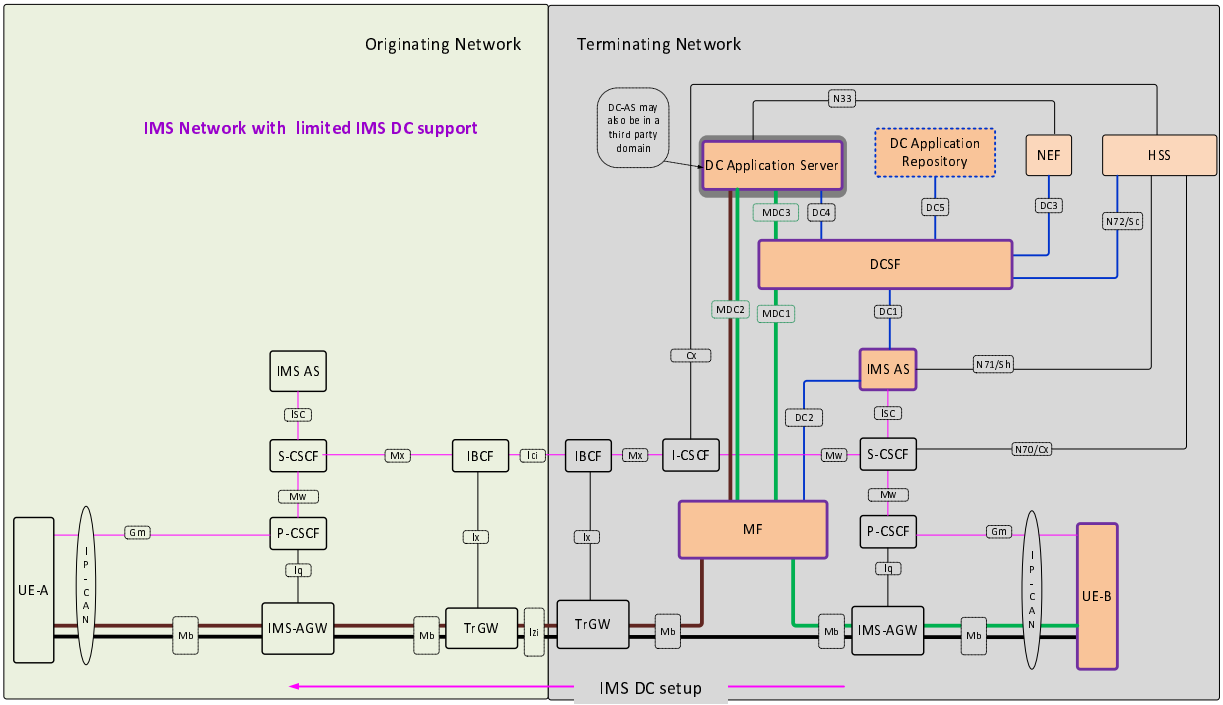


Figure 4.10.1.4.3.3-1: IMS network with limited IMS DC support – scenario 2

From the originating network's perspective, the transcoded media is same as the media of any other IMS media (i.e. not IMS DC).

4.10.1.4.4 IMS network does not have IMS DC capabilities

4.10.1.4.4.1 General

This clause describes the case when the IMS network does not have IMS DC capabilities.

When the IMS network does not have IMS DC capabilities, that network presumed to have no IMS DC specific NFs. Even if the locally registered IMS UE has IMS DC capabilities, that IMS UE cannot have a Bootstrap IMS DC channel when the other IMS UE present on the IMS session in the peer network initiates the IMS DC session setup. The network where the UE has registered upon receiving an SDP offer with IMS DC related attributes would downgrade the SDP to a regular IMS bearer. The locally registered IMS UEs may still have the P2A2P and P2P IMS DC media in a transcoded form.

Since to support the IMS DC setup initiated by one of the IMS UEs present on the IMS session, the respective IMS network where the UE has registered has to have to IMS DC capabilities, the illustrations shown in the subsequent clauses do not apply for intra-network IMS sessions. The terminating party may also be a redirected-to party in case the initial IMS session had a redirection.

The subsequent clauses illustrate a few cases that may show different aspects.

4.10.1.4.4.2 Scenario 1 – as a remote network, locally registered IMS UE has IMS DC capabilities

In the illustration shown in figure 4.10.1.4.4.2-1, UE-A is the originating party with the IMS DC capabilities. Since the originating network does not have the IMS DC capabilities and does not have the limited IMS DC support (as discussed in clause 4.10.1.4.3), the UE-A can only be a Remote UE and can only receive the IMS DC media in a transcoded form when IMS DC capable UE-B that has subscribed to IMS DC (in the terminating network) initiates the IMS DC setup. Media transcoding may happen either at the DC-AS or at the MF (in the terminating network) depending on the type of IMS DC.

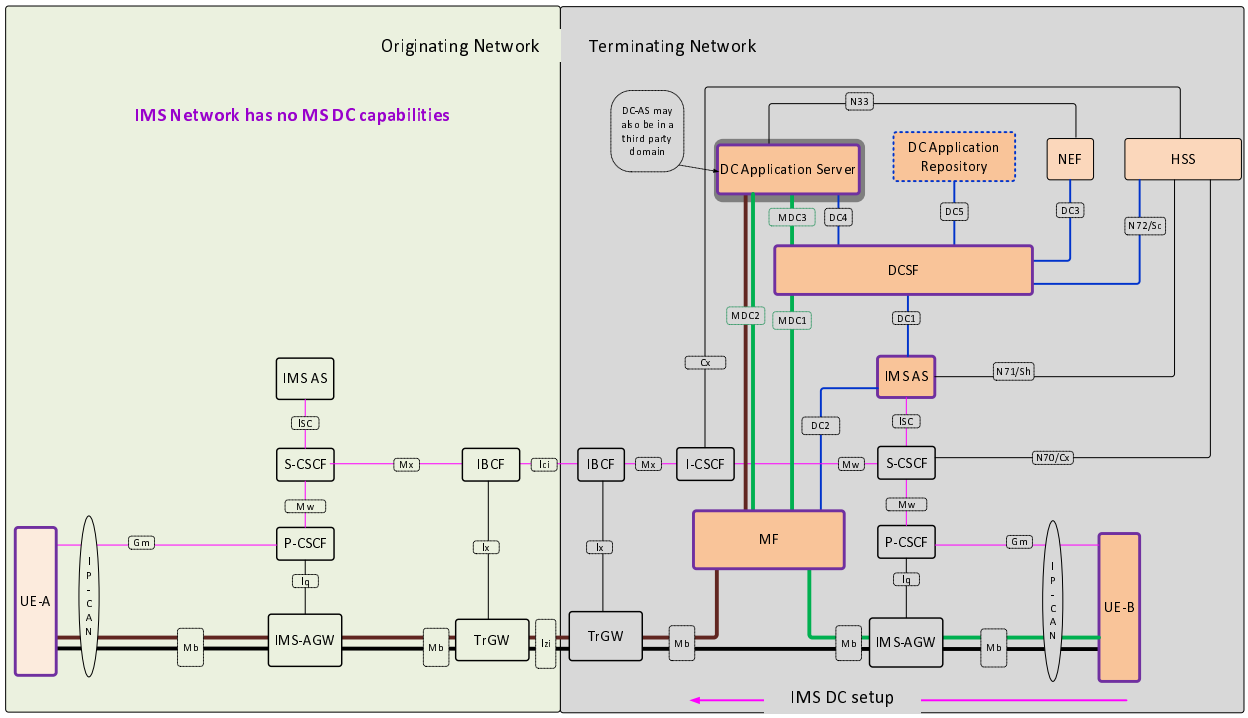


Figure 4.10.1.4.4.2-1: IMS network without IMS DC capabilities – scenario 1

From the originating network's perspective, the transcoded media is same as the media of any other IMS media (i.e. not IMS DC).

4.10.1.4.4.3 Scenario 2 – as a remote network, locally registered IMS UE has no IMS DC capabilities

In the illustration shown in figure 4.10.1.4.4.3-1, UE-A is the originating party without the IMS DC capabilities. In this case, the originating network also does not have the IMS DC capabilities and does not have the limited IMS DC support. Either of the two cases (no UE not IMS capable, the network has no IMS DC capabilities) make that the UE-A can only be a Remote UE and can only receive the IMS DC media in a transcoded form when IMS DC capable UE-B that has subscribed to IMS DC (in the terminating network) initiates the IMS DC setup. Media transcoding may happen either at the DC-AS or at the MF (in the terminating network) depending on the type of IMS DC.

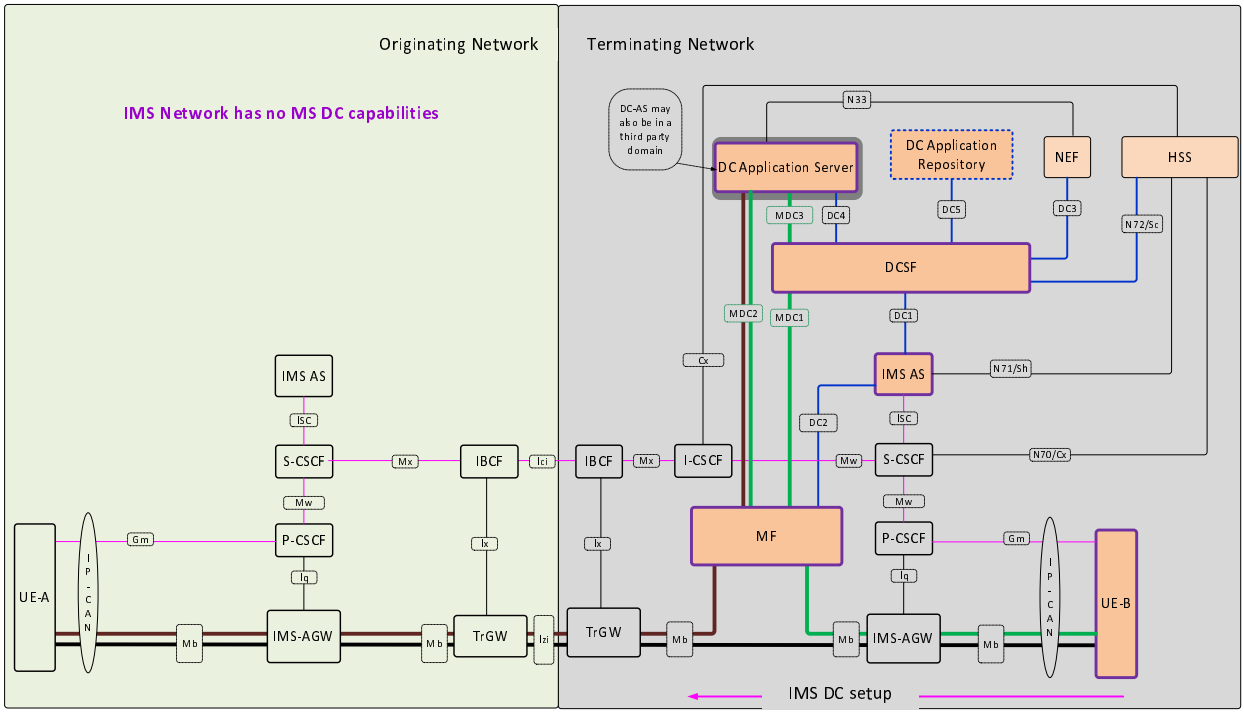


Figure 4.10.1.4.4.3-1: IMS network without IMS DC capabilities – scenario 2

From the originating network's perspective, the transcoded media is same as the media of any other IMS media (i.e. not IMS DC). From a topology point of view, this case same as the one shown in the figure 4.10.1.4.3.4-1 (clause 4.10.1.4.3.4).

4.10.1.4.5 Special cases when redirecting network does not support IMS DC capabilities

4.10.1.4.5.1 General

Clause 4.10.1.4.5 illustrates that when the IMS network does not have IMS DC capabilities (and does not have limited IMS DC support) a locally registered IMS UE can only be a Remote UE for an IMS DC setup by the other party, and can only receive the IMS DC media in a transcoded form. The illustrations shown in the subsequent clauses show as a consequence of a redirecting IMS network not having IMS DC capabilities.

In this case, when the originating UE initiates the IMS DC setup, the redirected-to UE even if IMS DC capable can only receive and send the IMS media in a transcoded form. This is true even if the redirected-to IMS network has IMS DC capabilities or limited IMS DC support. Likewise, when the redirected-to party initiates the IMS DC, the originating party even if IMS DC capable can only receive and send the IMS media in a transcoded form. This is true even if the originating IMS network has IMS DC capabilities or limited IMS DC support.

4.10.1.4.5.2 Scenario 1 – redirecting network has no IMS DC capabilities, redirected-to UE is the remote UE

The illustration shown in figure 4.10.1.4.5.2-1 is a redirecting case where the redirecting network does not have IMS DC capabilities (and does not have limited IMS DC support). Accordingly, any IMS DC setup initiation by the originating party would be downgraded to regular IMS bearer by the redirecting network. Therefore, even if UE-C has IMS DC capabilities and even if the redirected-to network supports IMS DC capabilities, the UE-C as a Remote UE can

only receive the IMS DC media in a transcoded form. Media transcoding may happen either at the DC-AS or at the MF (in the terminating network) depending on the type of IMS DC.

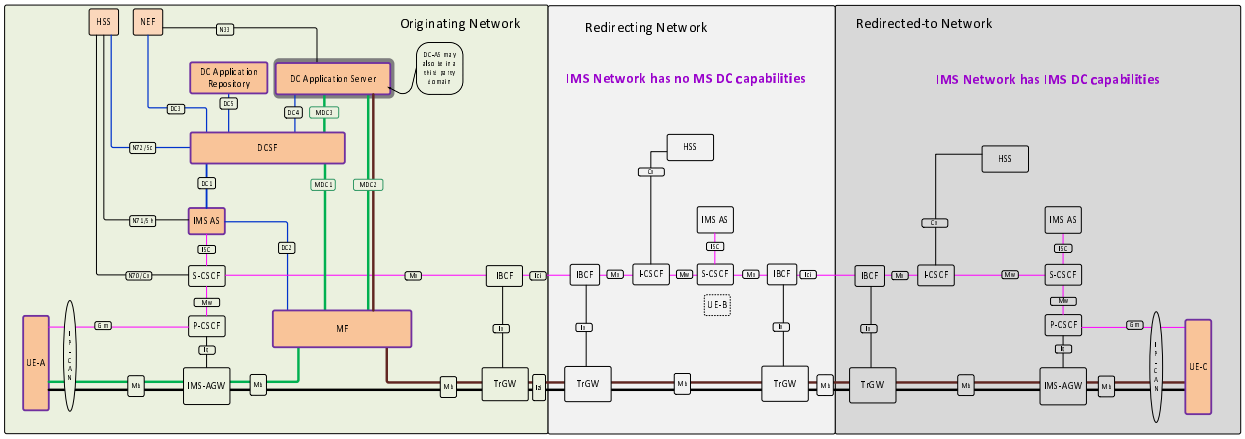


Figure 4.10.1.4.5.2-1: IMS network (redirecting network) without IMS DC capabilities, redirected-to UE is the remote UE

From the perspective of redirecting network and redirected-to network, the transcoded media is same as the media of any other IMS media (i.e. not IMS DC).

4.10.1.4.5.3 Scenario 2 – redirecting network has no IMS DC capabilities, originating UE is the remote UE

The illustration shown in figure 4.10.1.4.5.3-1 is a redirecting case where the redirecting network does not have IMS DC capabilities. Accordingly, any IMS DC setup initiation by the redirected-to party would be downgraded to regular IMS bearer by the redirecting network. Therefore, even if UE-A has IMS DC capabilities and even if the originating network supports IMS DC capabilities (either full or limited), the UE-A as a Remote UE can only receive the IMS DC media in a transcoded form. Media transcoding may happen either at the DC-AS or at the MF (in the terminating network) depending on the type of IMS DC.

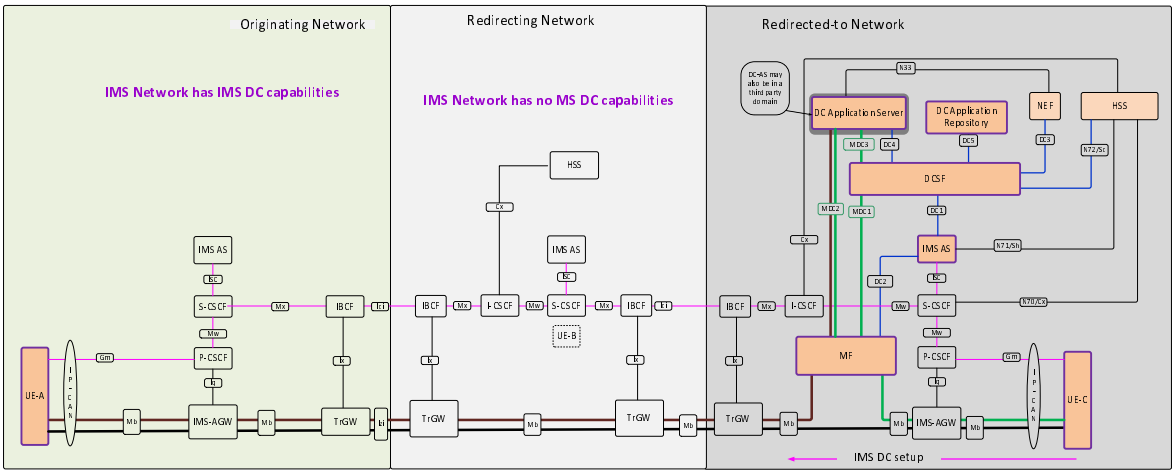


Figure 4.10.1.4.5.3-1: IMS network (redirecting network) without IMS DC capabilities, originating UE is the remote UE

From the perspective of originating network and redirecting network, the transcoded media is same as the media of any other IMS media (i.e. not IMS DC).

4.10.1.4.6 Inter-network redirecting cases

4.10.1.4.6.1 General

The subsequent clauses illustrate the IMS DC related topologies when multiple networks are involved in a redirecting case. In all the illustrations, an IMS session from UE-A to UE-B has been redirected to UE-C.

In the illustrations, the UE-A sets up an IMS DC session to UE-C. When the UE-C is IMS DC capable, it may or may not have subscribed for the IMS DC. In all the illustrations, the MF is inserted into the IMS DC media path at the redirected-to end of the session.

4.10.1.4.6.2 Three IMS UEs in three different networks

Figure 4.10.1.4.6.2-1 illustrates the case where the UE-C is IMS DC capable (irrespective of whether subscribed to IMS DC or not). The redirected-to network (as a Remote Network of IMS DC session) inserts the MF into the MF media path. The illustrations also assume that the redirecting network has the IMS DC capabilities so as not to block IMS DC setup requests. In this illustration, the IMS DC media path passes through the Ingress TrGW and Egress TrGW present in the redirecting network.

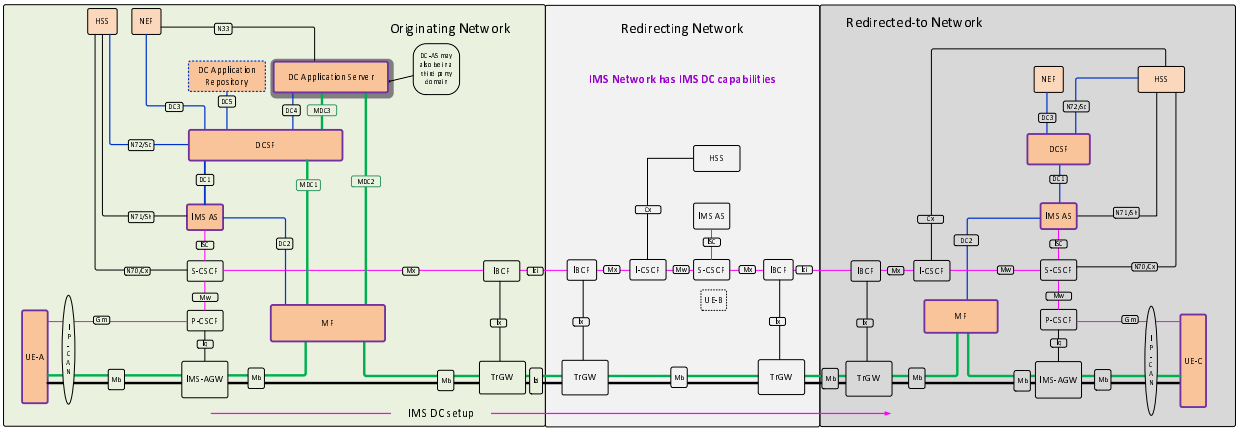


Figure 4.10.1.4.6.2-1: Inter-network redirection case – redirecting case, all UEs in different networks

The redirected-to party when subscribed to IMS DC in the redirected-to network can also initiate the IMS DC with the originating party if the redirecting network allows the IMS DC channel setup to happen. Furthermore, when the Remote UEs of IMS DC does not support IMS DC capabilities, the IMS DC media can only be received or sent by that UE (as a Remote UE) in a transcoded form.

4.10.1.4.6.3 Only two networks involved in a redirection case

In general, either two or all three parties involved in an IMS session redirection may also be in the same network. When two parties are in the same network, all permutations among the two of the three parties being in the same network is possible – UE-A and UE-B in the same network, or UE-A and UE-C in the network, or UE-B and UE-C in the same network. In such cases (i.e. two networks being involved), the IMS DC media traverse through the TrGWs.

The illustration shown in figure 4.10.1.4.6.3-1 illustrates the case where the redirecting party happens to be in the originating network.

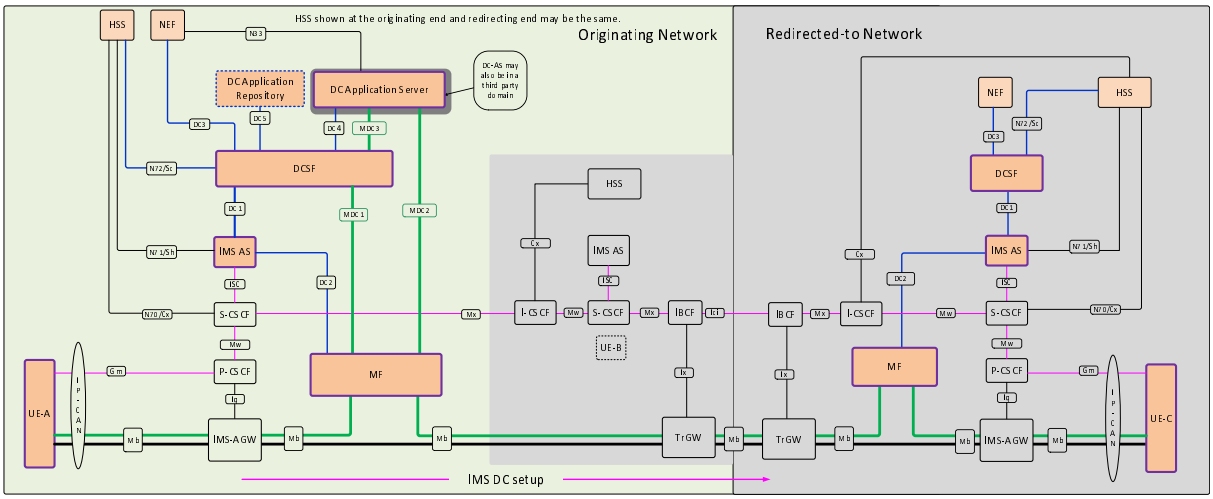


Figure 4.10.1.4.6.3-1: Inter-network redirection case – two UEs in the same network (UE-A and UE-B)

The illustration shown in figure 4.10.1.4.6.3-2 illustrates the case where the redirecting party happens to be in the terminating network.

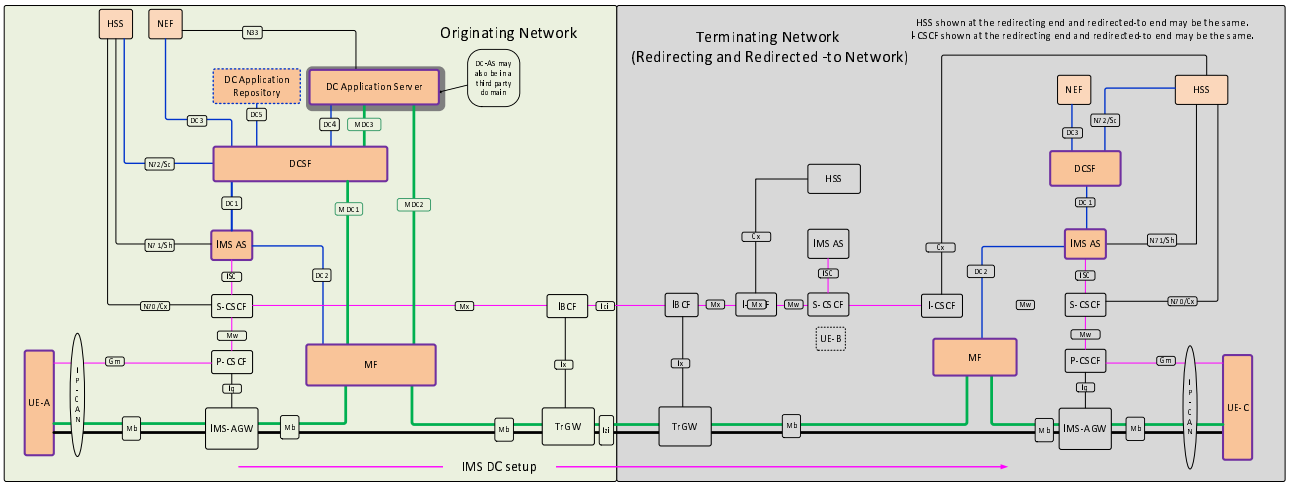


Figure 4.10.1.4.6.3-2: Inter-network redirection case – two UEs in the same network (UE-B and UE-C)

4.10.1.5 MF operating modes

The TS 23.228 [8] defines the following operating modes for the MF:

- HTTP Proxy.
- DC Application Proxy.
- UDP Proxy.

The present document introduces another operating mode for the MF - RTP Proxy. The MF operates in an RTP Proxy mode for the media transcoded at the DC-AS. The MF may also transcode the media that does not go through the DC-AS.

Figure 4.10.1.5-1 illustrates the protocol stack views of MF acting as an HTTP Proxy, DC Application Proxy and UDP Proxy.

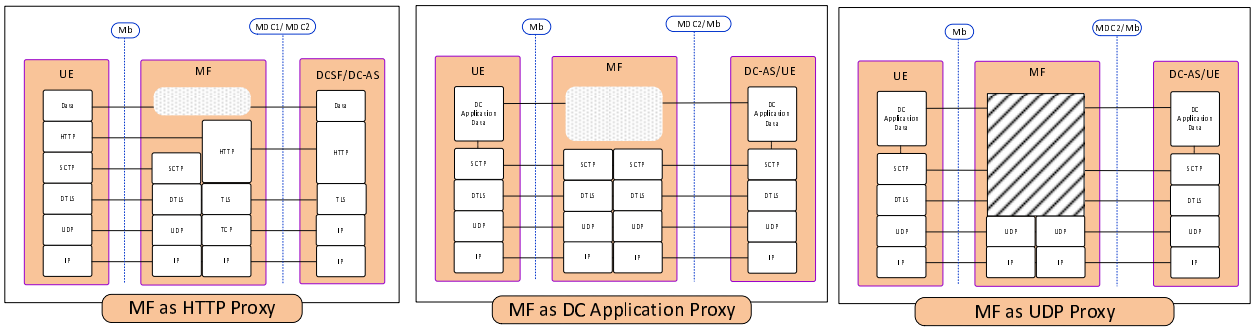


Figure 4.10.1.5-1: MF operating modes

In the above illustration, the UE is IMD DC capable. When two UEs are shown, one of the UEs is the Remote UE from an IMS DC session perspective.

When the Remote UE is not IMS DC capable that UE can send and receive the IMS DC related media only in a transcoded form. The media may be transcoded at the DC-AS or at the MF present in the Local Network. The transcoded media also applies to UE that may be IMS DC capable but registered at a network that does not have IMS DC capabilities.

Figure 4.10.1.5-2 shows the protocol stacks at the MF present in the Local Network when transcoding is involved.

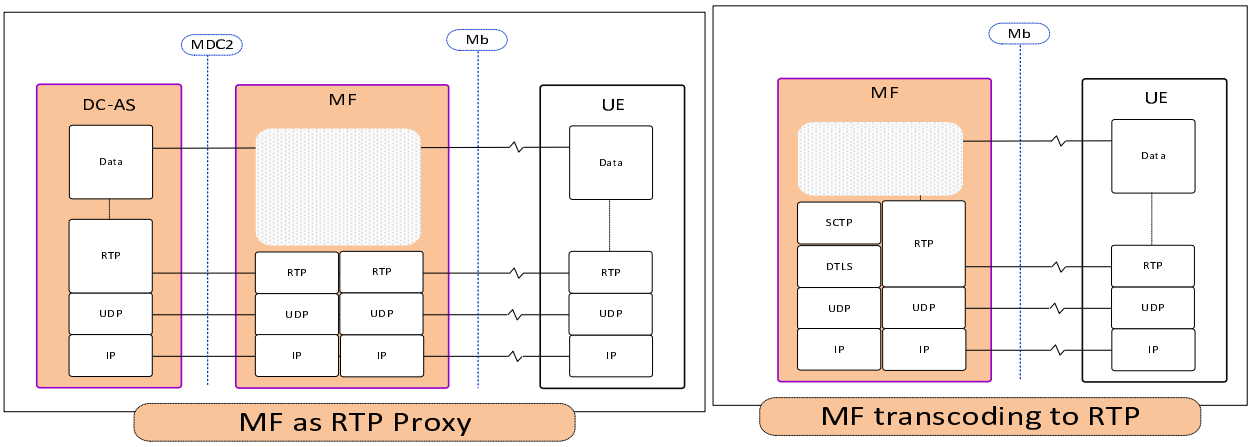


Figure 4.10.1.5-2: MF operating modes for transcoded media

The transcoded media applies only to P2A2P and P2P IMS DC media.

4.10.1.6 IMS data channels

4.10.1.6.1 General

The TS 23.228 [8] defines the following IMS DC types:

- Bootstrap IMS DC.
- P2A or A2P IMS DC.
- P2A2P IMS DC.
- P2P IMS DC.

As illustrated in clause 4.10.1.4, the Bootstrap IMS DC can be setup by an IMS DC capable UE subscribed for the IMS for self and to an IMS DC capable Remote UE which may be present in the same network or may be present in the peer network that has either the IMS DC capabilities or IMS network that has limited IMS DC support. The P2A and A2P IMS DC is between the IMS DC capable UE and the DC-AS. Similar to the Bootstrap IMS DC, the P2A and A2P IMS DC can also be setup to the IMS DC capable Remote UE that may be present in the same network or may be present in the peer network that has either the IMS DC capabilities or limited IMS DC support. The P2A2P IMS DC that passes

through the DC-AS is established between the two IMS UEs. When the Remote UE is not IMS DC capable or the Remote Network does not have IMS DC capabilities, the P2A2P media can only be sent to, or received from, the Remote UE in a transcoded form. The P2P IMS DC is between the two IMS UEs. According, to TS 23.228 [8], a P2P IMS DC in the Local Network may or may not pass through an MF. The present document assumes that the P2P IMS DC always passes through an MF in the Local Network in presenting the LI capabilities for the P2P IMS DC session.

The subsequent clauses illustrate the above IMS DCs using the protocol stacks as a guide.

4.10.1.6.2 Bootstrap IMS data channel

Bootstrap IMS FC is between the IMS DC capable UE and the DCSF. The IMS DC capable UE can be Local UE or the Remote UE. In the latter case, when the Remote UE is in a peer network, it is assumed that the peer network allows its locally registered IMS DC capable UE to have the Bootstrap IMS DC.

Figure 4.10.1.6.2-1 illustrates a high-level view of Bootstrap IMS DC between the Local UE and the DCSF and between the DCSF and the Remote UE. The presence or absence of an MF at the remote end of the session is a deployment choice.

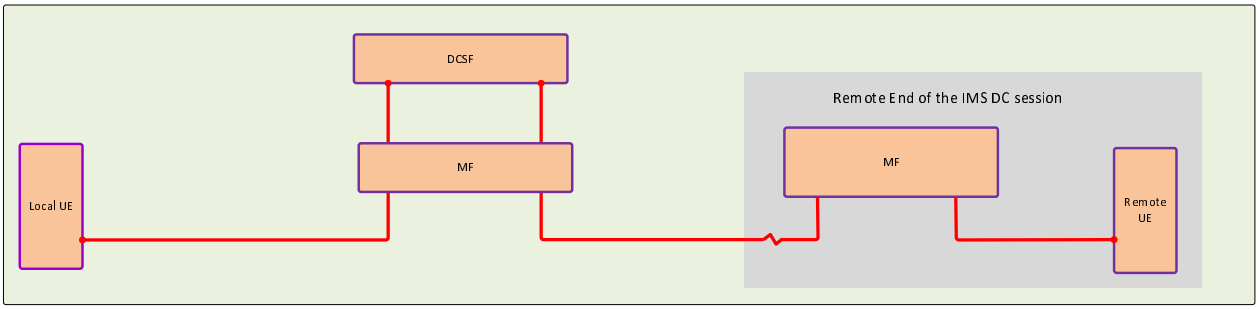


Figure 4.10.1.6.2-1: Bootstrap IMS DC – high level view

Figure 4.10.1.6.2-2 illustrates the Bootstrap IMS DC between the Local UE and the DCSF and between the DCSF and the Remote UE at a protocol level. The MF at the Local UE's end is operating in HTTP Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

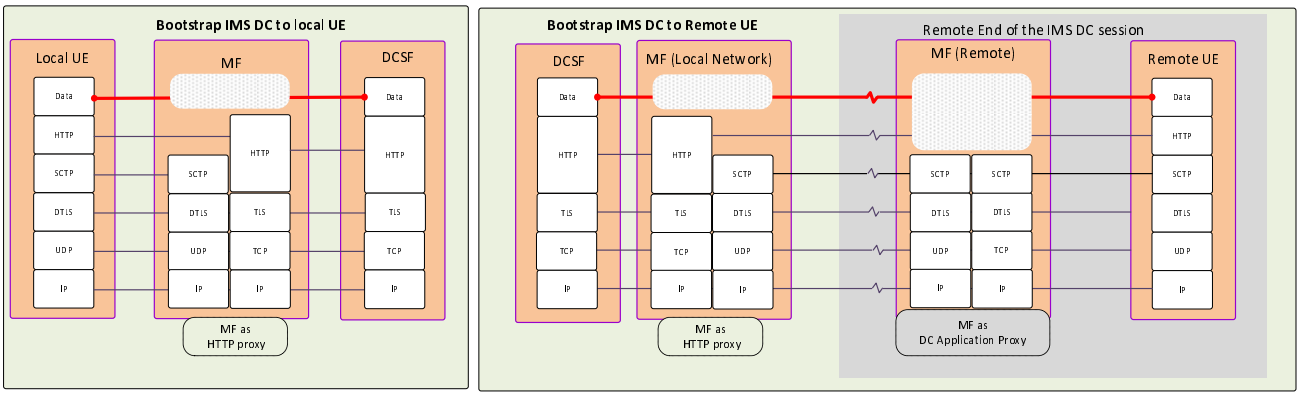


Figure 4.10.1.6.2-2: Bootstrap IMS DC to local UE and remote UE – protocol view

The Remote UE may be in the same network as the Local UE or maybe in a Remote Network. The Remote UE may also be a redirected-to party. The Remote UE may or may not have subscribed to the IMS DC in the respective IMS network where the UE has registered. It is also possible either the terminating party or the redirected-to party is the Local UE and the originating party is the Remote UE.

4.10.1.6.3 P2A and A2P IMS data channel

Figure 4.10.1.6.3-1 illustrates a high-level view of P2A/A2P IMS DC between the Local UE and the DC-AS and between the DC-AS and the Remote UE. The DC-AS may also be present in a third party domain. The presence or absence of an MF at the remote end of the session is a deployment choice.

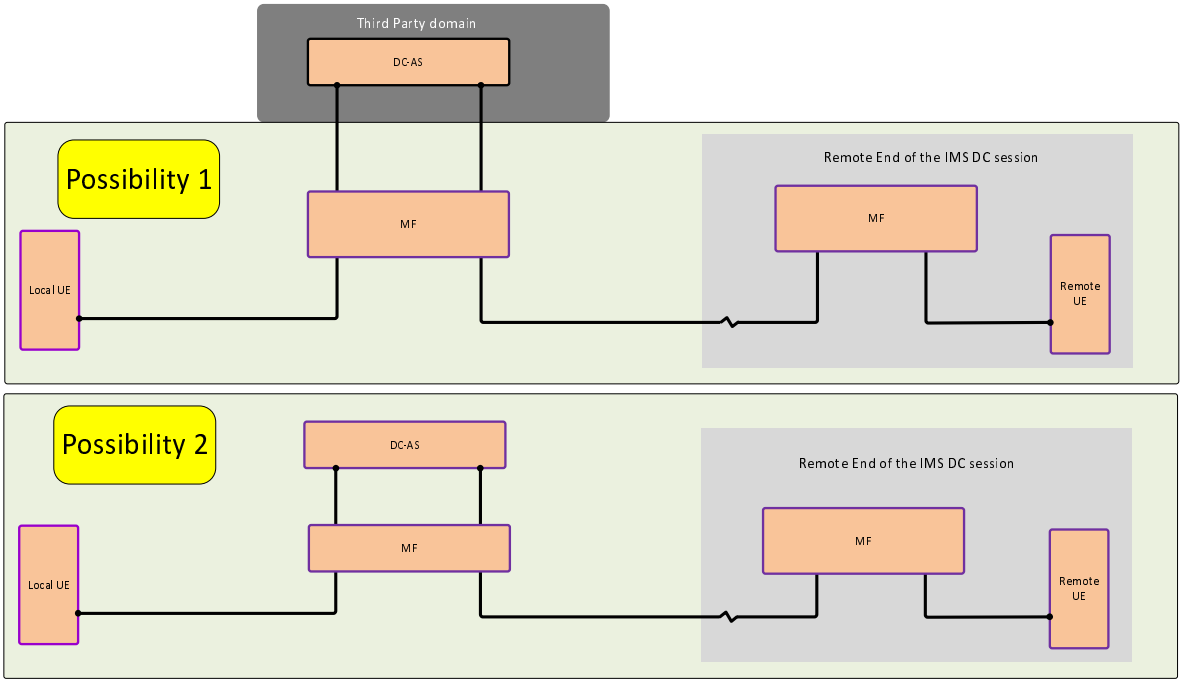


Figure 4.10.1.6.3-1: P2A and A2P IMS DC – high level view

Figure 4.10.1.6.3-2 illustrates the P2A/A2P IMS DC between the Local UE and the DC-AS and between the DC-AS and the Remote UE when the MF at the Local UE's end is operating in HTTP Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

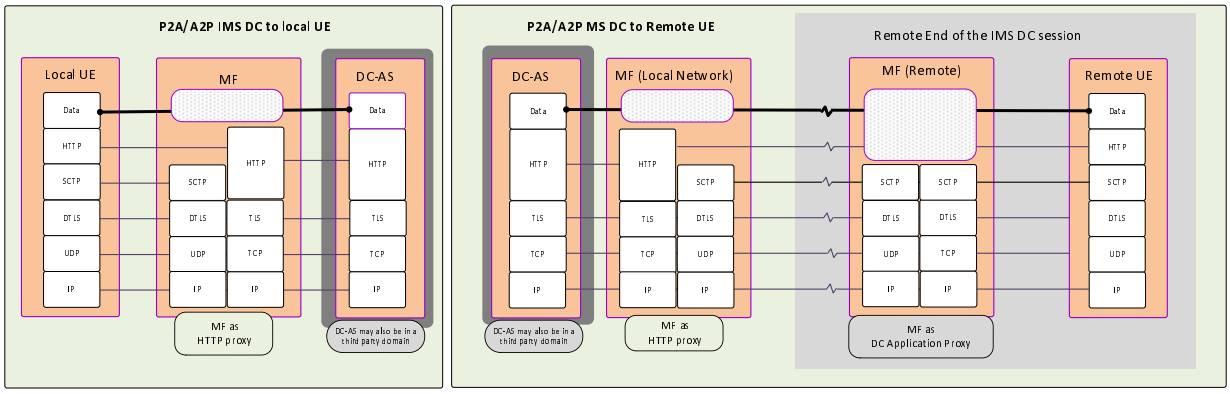


Figure 4.10.1.6.3-2: P2A and A2P IMS DC to local UE and remote UE with (MF at the local UE's end is operating in HTTP Proxy mode – protocol view

Figure 4.10.1.6.3-3 illustrates the P2A/A2P IMS DC between the Local UE and the DC-AS and between the DC-AS and the Remote UE when the MF at the Local UE's end is operating in DC Application Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

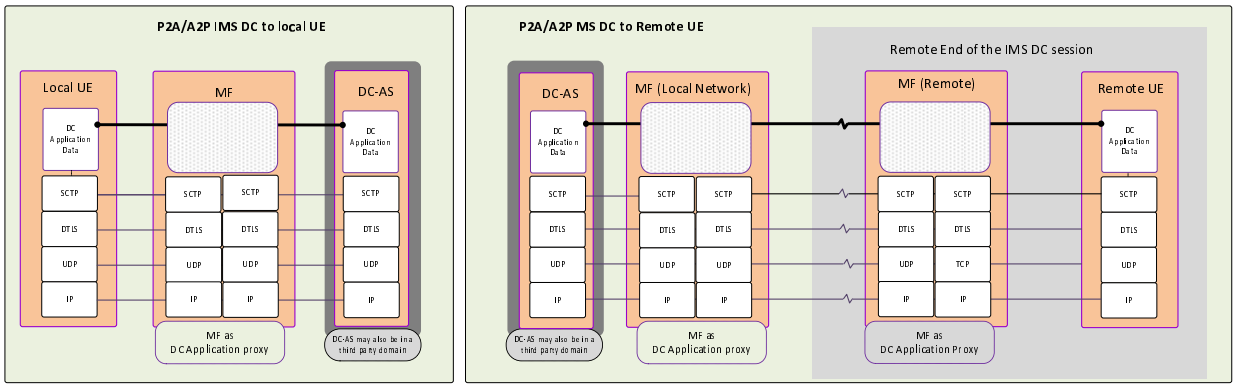


Figure 4.10.1.6.3-3: P2A and A2P IMS DC to local UE and remote UE with (MF at the local UE's end is operating in DC Application Proxy mode – protocol view

Figure 4.10.1.6.3-4 illustrates the P2A/A2P IMS DC between the Local UE and the DC-AS and between the DC-AS and the Remote UE when the MF at the Local UE's end is operating in UDP Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

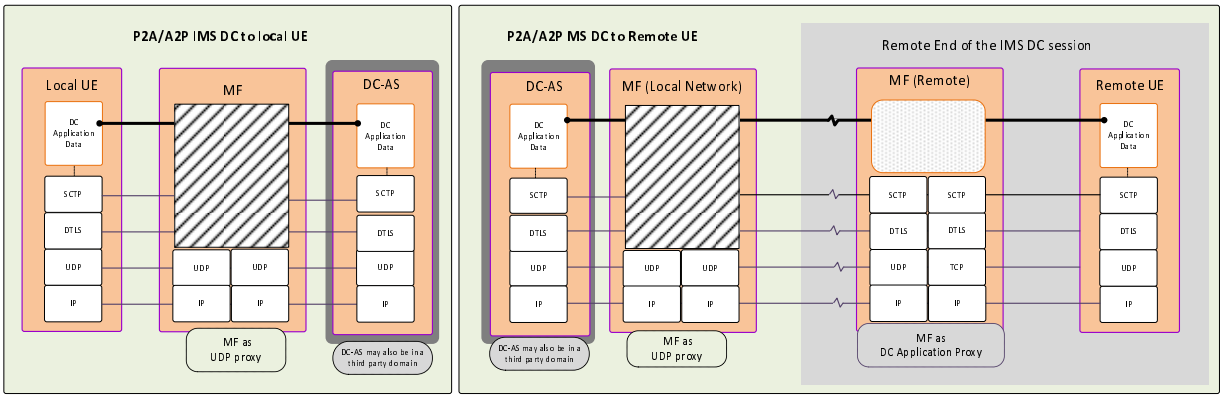


Figure 4.10.1.6.3-4: P2A and A2P IMS DC to local UE and remote UE with (MF at the local UE's end is operating in UDP Proxy mode – protocol view

In all three cases, the Remote UE may be in the same network as the Local UE or maybe in a Remote Network. The Remote UE may also be a redirected-to party. The Remote UE may or may not have subscribed to the IMS DC in the respective IMS network where the UE has registered to. It is also possible either the terminating party or the redirected-to party is the Local UE and the originating party is the Remote UE.

When the DC-AS is in a third party domain, the present document discourages the use of MF in UDP Proxy mode when the CSPs have LI obligations to meet.

4.10.1.6.4 P2A2P IMS data channel

Figure 4.10.1.6.4-1 illustrates high level view of P2A2P IMS DC between the Local UE and the Remote UE via the DC-AS present in the Local Network. The DC-AS may also be present in a third party domain. The presence or absence of an MF at the remote end of the session is a deployment choice.

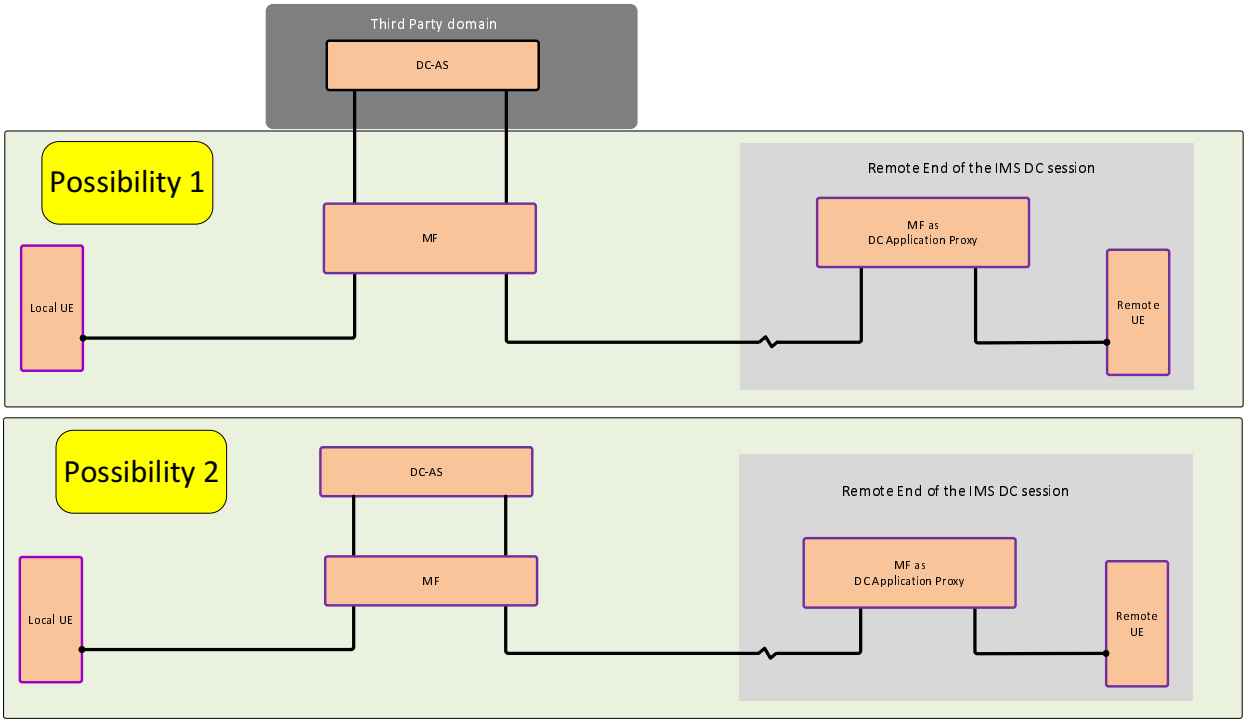


Figure 4.10.1.6.4-1: P2A2P IMS DC – high level view

Figure 4.10.1.6.4-2 illustrates the P2A2P IMS DC between the Local UE and the Remote UE via the DC-AS present in the Local Network (or third party domain) when the MF at the Local UE's end is operating in HTTP Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

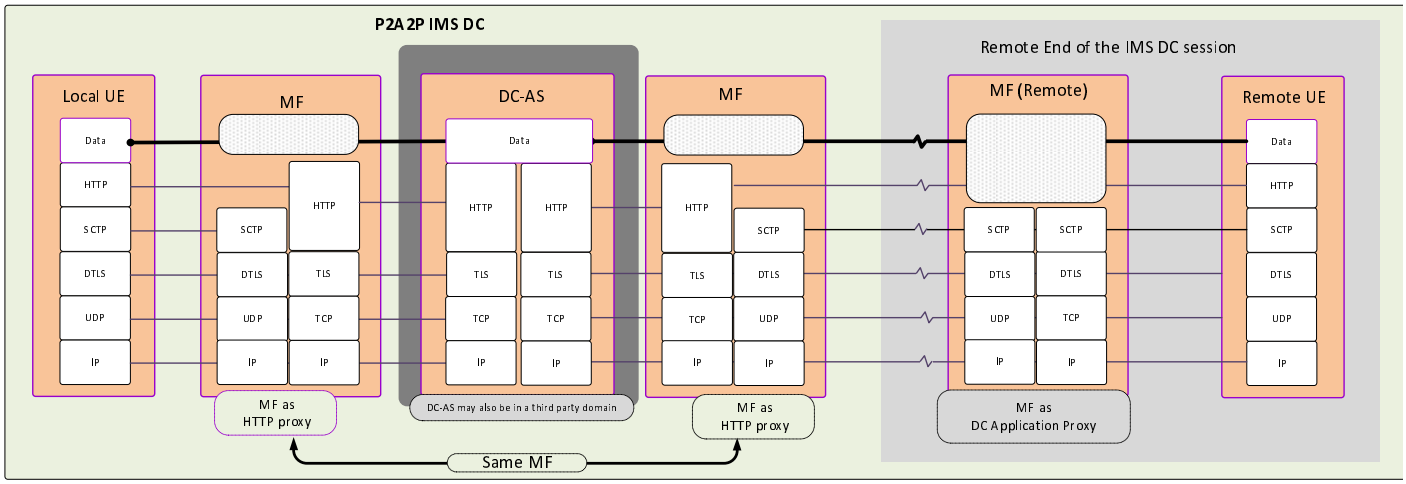


Figure 4.10.1.6.4-2: P2A2P IMS DC to local UE and remote UE with (MF at the local UE's end is operating in HTTP Proxy mode – protocol view

Figure 4.10.1.6.4-3 illustrates the P2A2P IMS DC between the Local UE and the Remote UE via the DC-AS present in the Local Network (or third party domain) when the MF at the Local UE's end is operating in DC Application Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

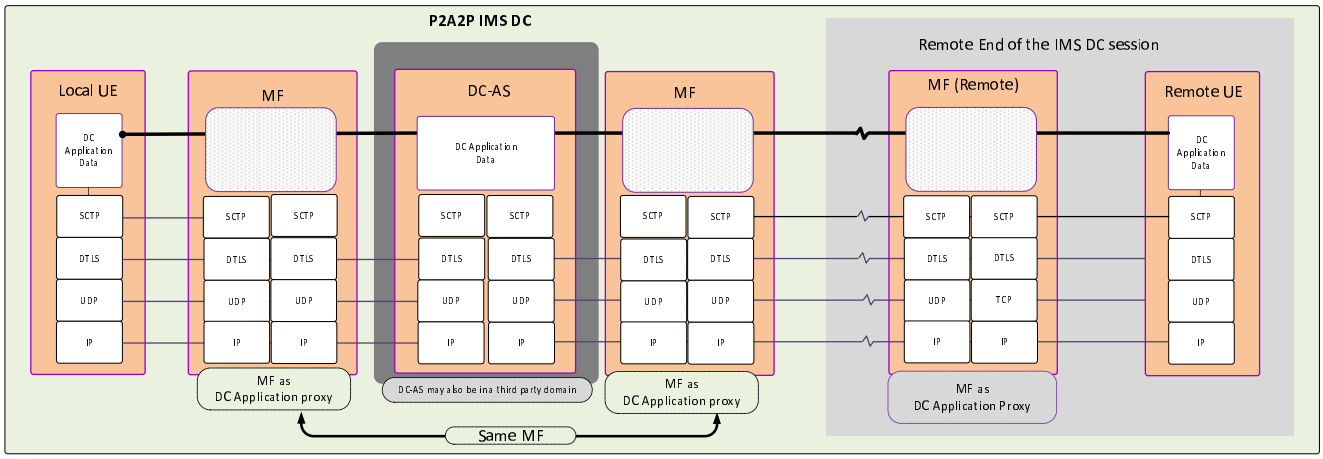


Figure 4.10.1.6.4-3: P2A2P IMS DC to local UE and remote UE with (MF at the local UE's end is operating in DC Application Proxy mode – protocol view

Figure 4.10.1.6.4-4 illustrates the P2A2P IMS DC between the Local UE and the Remote UE via the DC-AS present in the Local Network (or third party domain) when the MF at the Local UE's end is operating UDP Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

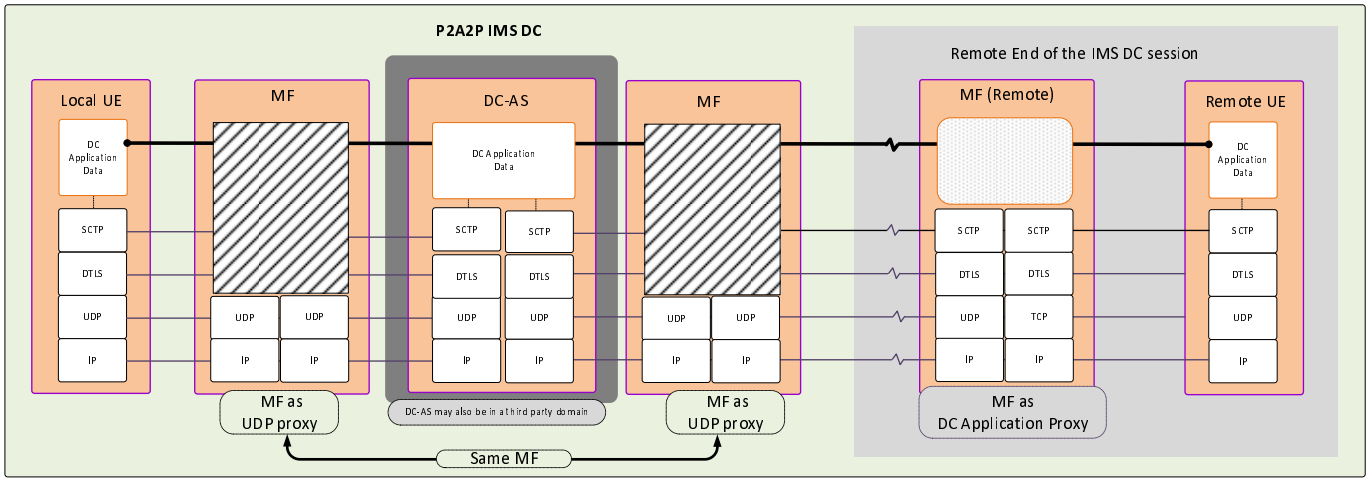


Figure 4.10.1.6.4-4: P2A2P IMS DC to local UE and remote UE with (MF at the local UE's end is operating in UDP Proxy mode – protocol view

In all three cases, the Remote UE may be in the same network as the Local UE or maybe in a Remote Network. The Remote UE may also be a redirected-to party. The Remote UE may or may not have subscribed to the IMS DC in the respective IMS network where the UE has registered to. It is also possible either the terminating party or the redirected-to party is the Local UE and the originating party is the Remote UE.

When the DC-AS is in a third party domain, the present document discourages the use of MF in UDP Proxy mode when the CSPs have LI obligations to meet.

4.10.1.6.5 P2P IMS data channel

Figure 4.10.1.6.5-1 illustrates high level view of P2P IMS DC between the Local UE and the Remote UE. The presence or absence of an MF at the remote end of the session is a deployment choice. The possibility of P2P IMS DC without passing through an MF is not discouraged in the present document when CSPs have LI obligations.

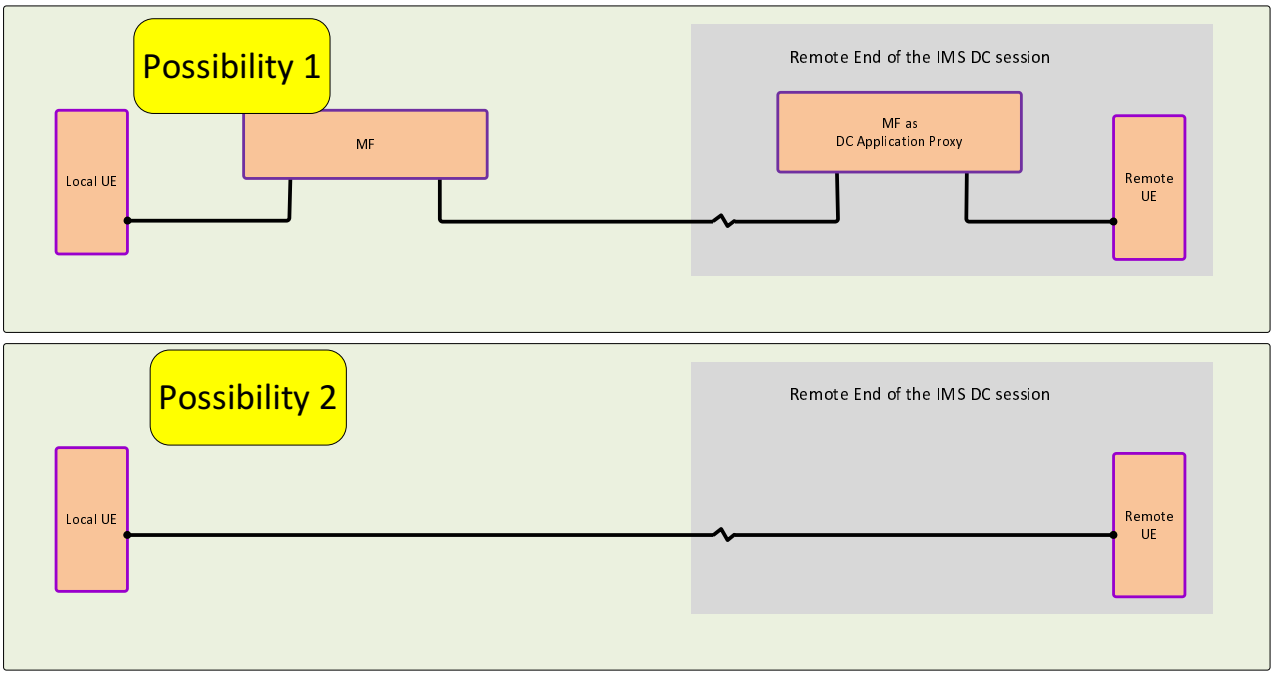


Figure 4.10.1.6.5-1: P2P IMS DC – high-level view

Figure 4.10.1.6.5-2 illustrates the P2P IMS DC between the Local UE and the Remote UE when the MF at the Local UE's end is operating in DC Application Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

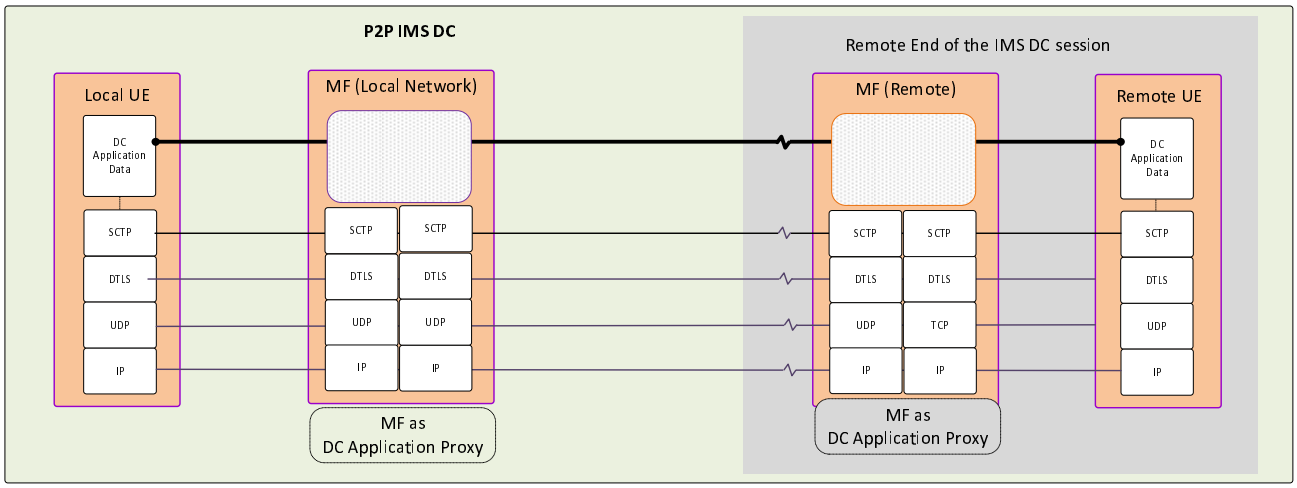


Figure 4.10.1.6.5-2: P2P IMS DC between local UE and remote UE with (MF at the local UE's end is operating in DC Application Proxy mode – protocol view

Figure 4.10.1.6.5-3 illustrates the P2P IMS DC between the Local UE and the Remote UE when the MF at the Local UE's end is operating in UDP Proxy mode. The MF (when present) at the Remote UE's end is operating in DC Application Proxy mode.

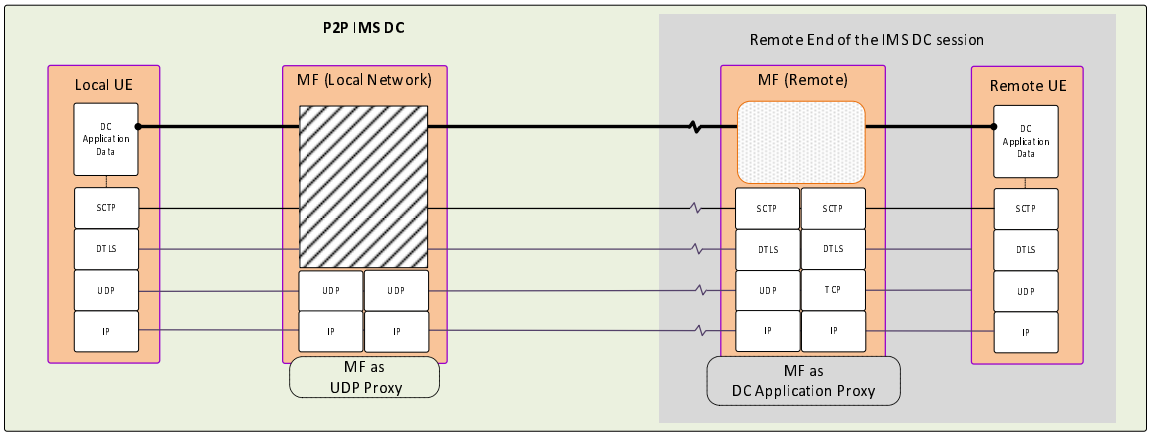


Figure 4.10.1.6.5-3: P2P IMS DC between local UE and remote UE (MF at the local UE's end is operating in UD Proxy mode) – protocol view

Figure 4.10.1.6.5-4 illustrates the P2P IMS DC between the Local UE and the Remote UE with DC media not passing through any MF. The presence of MF when the Remote UE is in a peer network is independent of the presence of MF in the Local Network.

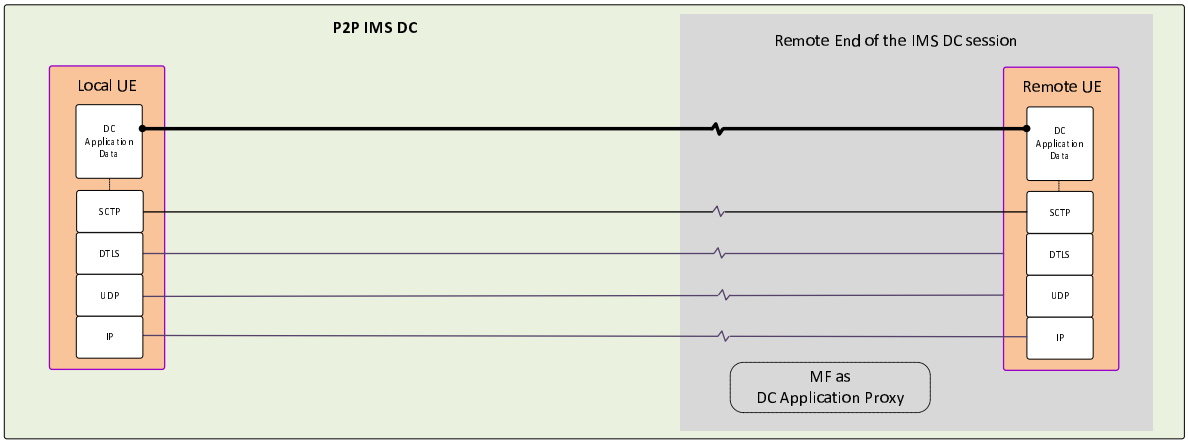


Figure 4.10.1.6.5-4: P2P IMS DC between local UE and remote UE (no MF) – protocol view

In all three cases, the Remote UE may be in the same network as the Local UE or maybe in a Remote Network. The Remote UE may also be a redirected-to party. The Remote UE may or may not have subscribed to the IMS DC in the respective IMS network where the UE has registered to. It is also possible either the terminating party or the redirected-to party is the Local UE and the originating party is the Remote UE.

The present document discourages the deployment of a P2P IMS DC without the MF or with the MF in UDP Proxy mode when the CSPs have LI obligations to meet.

4.10.1.6.6 Media transcoding on P2A2P IMS DC

Figure 4.10.1.6.6-1 illustrates high level view media transcoding for P2A2P IMS DC between the Local UE and the Remote UE. The media transcoding is done at the DC-AS.

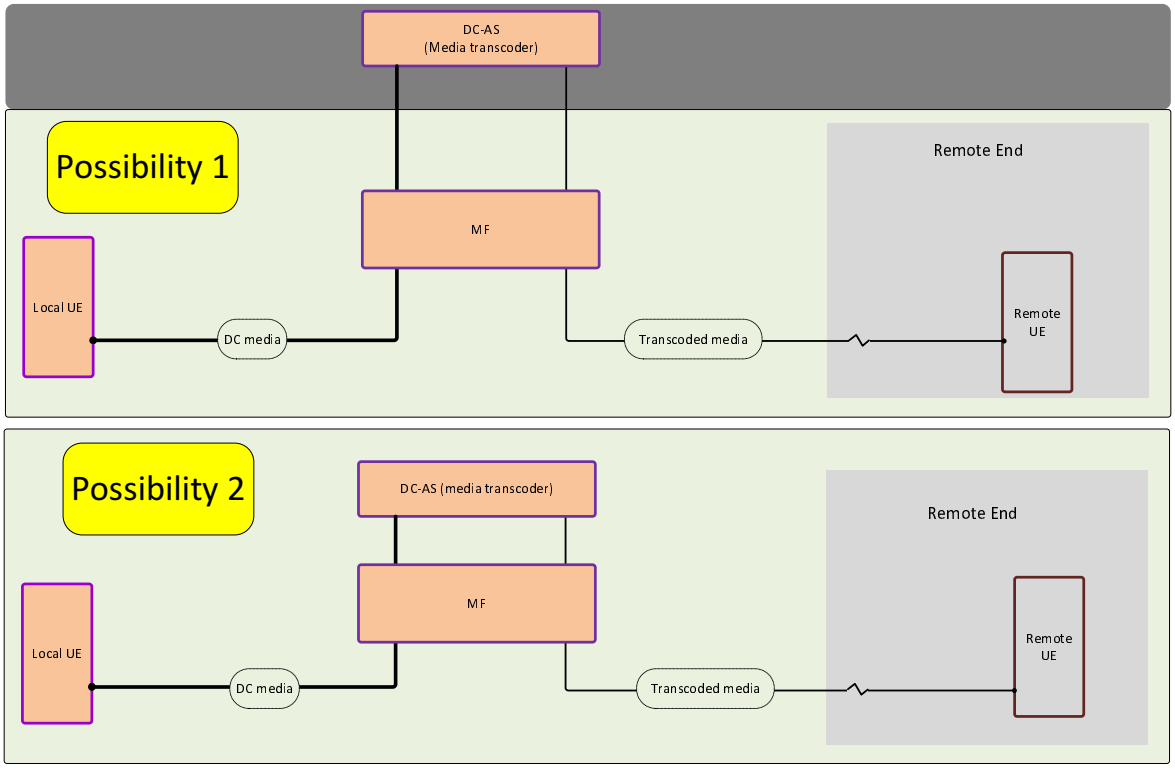


Figure 4.10.1.6.6-1: P2A2P IMS DC media transcoding – high level view

Figure 4.10.1.6.6-2 illustrates the media transcoding for P2A2P IMS DC between the Local UE and the Remote UE via the DC-AS present in the Local Network (or third party domain) when the MF at the Local UE's end is operating in HTTP Proxy mode. Media transcoding is done at the DC-AS. For the transcoded media, the MF present in the Local Network operates in a RTP Proxy mode.

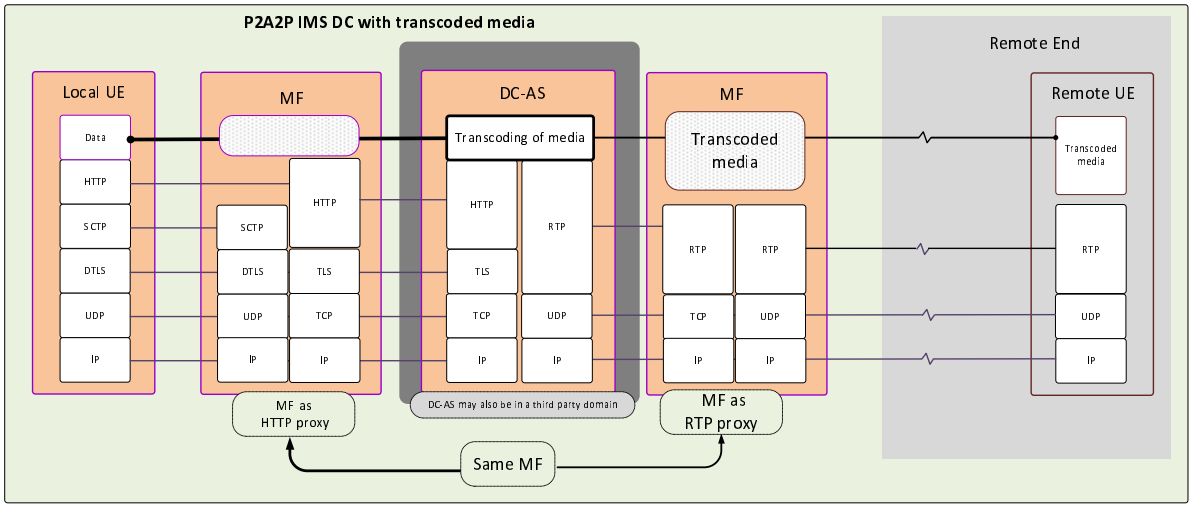


Figure 4.10.1.6.6-2: P2A2P IMS DC media transcoding (MF at the local UE's end is operating in HTTP Proxy mode and RTP Proxy) – protocol view

Figure 4.10.1.6.6-3 illustrates the media transcoding for P2A2P IMS DC between the Local UE and the Remote UE via the DC-AS present in the Local Network (or third party domain) when the MF at the Local UE's end is operating in DC Application Proxy mode. Media transcoding is done at the DC-AS. For the transcoded media, the MF present in the Local Network operates in a RTP Proxy mode.

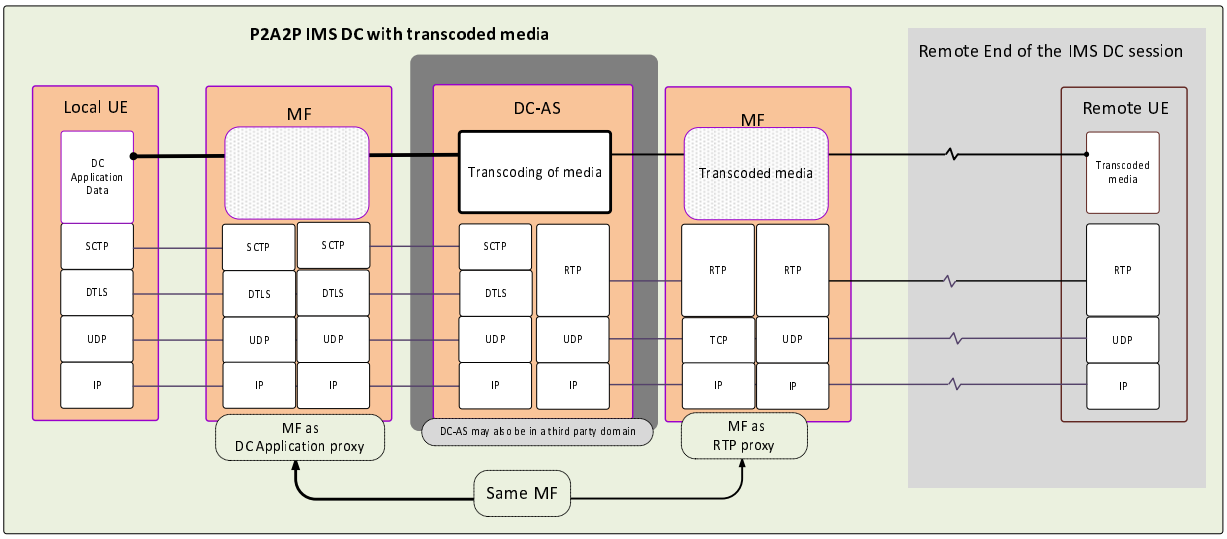


Figure 4.10.1.6.6-3: P2A2P IMS DC media transcoding (MF at the local UE's end is operating in DC Application Proxy mode and RTP Proxy) – protocol view

Figure 4.10.1.6.6-4 illustrates the media transcoding for P2A2P IMS DC between the Local UE and the Remote UE via the DC-AS present in the Local Network (or third party domain) when the MF at the Local UE's end is operating in UDP Proxy mode. Media transcoding is done at the DC-AS. For the transcoded media, the MF present in the Local Network operates in a RTP Proxy mode.

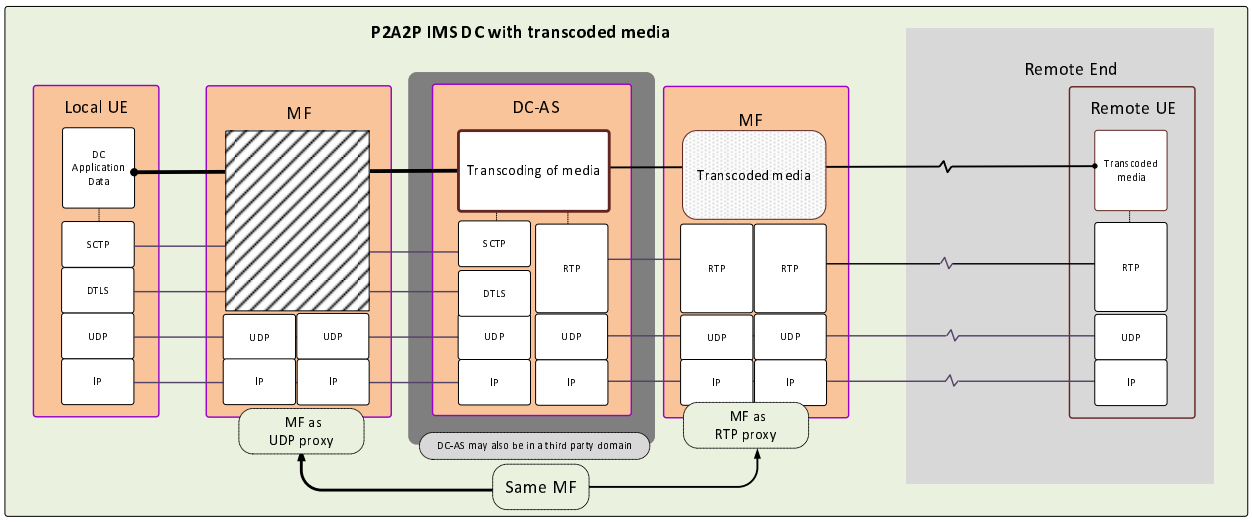


Figure 4.10.1.6.6-4: P2A2P IMS DC media transcoding (MF at the local UE's end is operating in UDP Proxy mode and RTP Proxy) – protocol view

In all three cases, the Remote UE may be in the same network as the Local UE or maybe in a Remote Network. The Remote UE may also be a redirected-to party. It is also possible either the terminating party or the redirected-to party is the Local UE and the originating party is the Remote UE.

When the DC-AS is in a third party domain, the present document discourages the use of MF in UDP Proxy mode when the CSPs have LI obligations to meet.

4.10.1.6.7 Media transcoding on P2P IMS data channel

Figure 4.10.1.6.7-1 illustrates a high level view of the media transcoding for P2P IMS DC between the Local UE and the Remote UE with MF providing the transcoding.

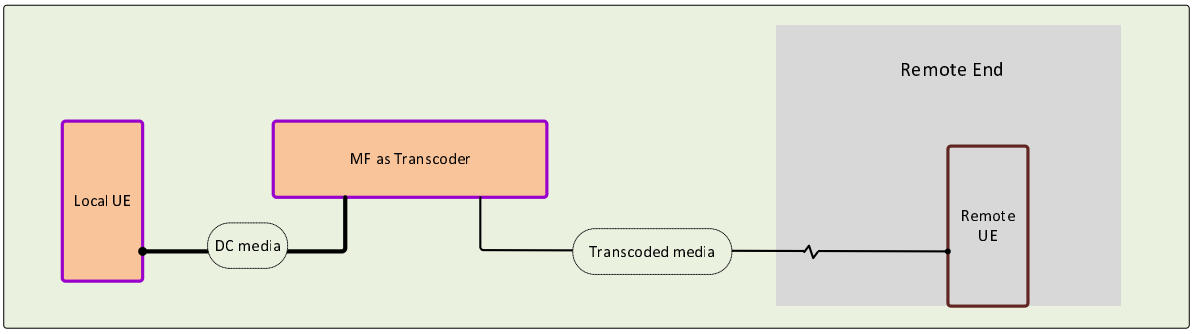


Figure 4.10.1.6.7-1: P2P media transcoding at MF – high level view

In the illustration shown in figure 4.10.1.6.7-2, the MF in the local network that provides the media transcoding is operating in a DC Application Proxy mode before the transcoding comes into act.

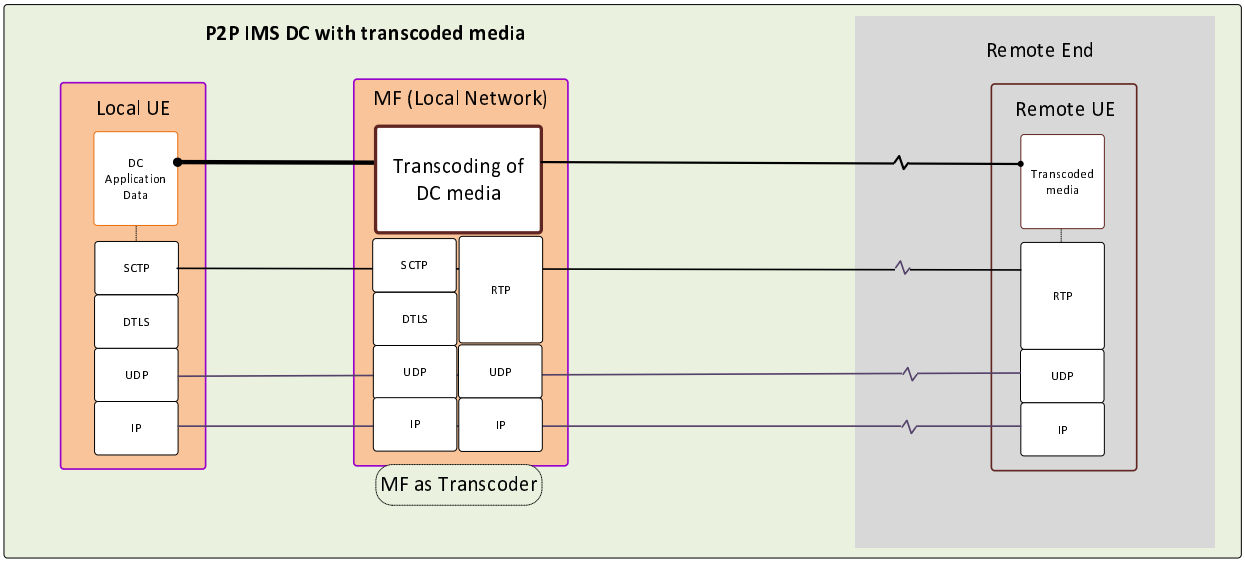


Figure 4.10.1.6.7-2: P2P media transcoding at MF – protocol view

The Remote UE may be in the same network as the Local UE or maybe in a Remote Network. The Remote UE may also be a redirected-to party. It is also possible either the terminating party or the redirected-to party is the Local UE and the originating party is the Remote UE.

4.10.2 LI for IMS DC sessions

4.10.2.1 Conventions used

Figure 4.10.2.1-1 shows some of the conventions used within the LI related drawings when the same are not self-explanatory:

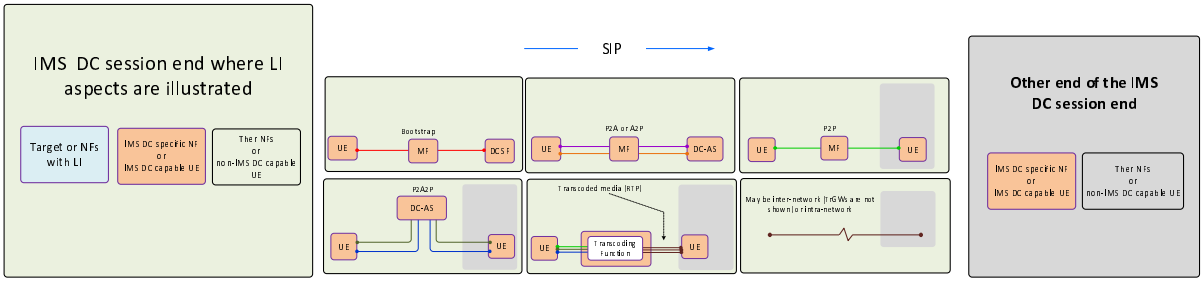


Figure 4.10.2.1-1: Conventions used when not self-explanatory

The SIP flow direction shown is for the initial IMS session setup which can also be flow direction when the originating party initiates the IMS DC session setup.

Figure 4.10.2.2-2 shows some of the conventions used for the IMS DC media flow at a NF, when the same are not self-explanatory.

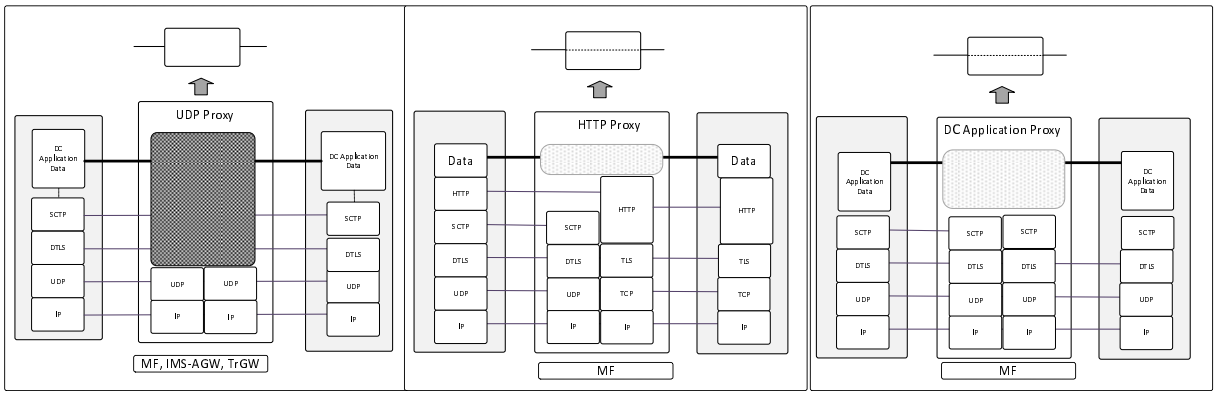


Figure 4.10.2.1-2: Conventions used for the IMS DC media flow through a NF when not self-explanatory

4.10.2.2 Illustration principles

The present document illustrates the LI for IMS DC sessions from the perspective of local end of IMS DC session and remote end of IMS DC session. In an inter-network scenario, the local end of the IMS DC session is handled by the Local Network and remote end of the IMS DC session is handled by the Remote Network. The present document includes the following illustrations:

- LI in the local end of the IMS DC session.
- LI in the remote end of the IMS DC session.

For LI at the local end, the Local UE is the target and for LI at the remote end, the Remote UE is the target. A target for IMS DC can also be a target for the IMS-based services. Therefore, the LI functions for IMS DC sessions and LI for other IMS-based services may happen at the same time. Within the topology views, the present document shows some aspects of IMS LI using the default deployment options defined in TS 33.127 [3]. In this aspect, even though the TS 33.127 [3] and TS 33.128 [4] specify that target non-local ID is not applicable to IMS DC sessions, a few examples are shown to address the case of target non-local ID to illustrate its influence of the same on the IMS DC sessions.

For the IMS DC sessions, the illustrations show that LI at the local end of the IMS DC session is independent of how the remote end handles the IMS DC sessions. Furthermore, the illustrations show that the LI at the remote end of the IMS DC session is independent of the MF operating mode at the local end of the IMS DC session.

Either the originating party or the terminating party of an IMS session may initiate the IMS DC session. The IMS DC session may also be initiated by the redirected-to party when the IMS session gets redirected. When both parties present in an IMS DC session are IMS DC capable and subscribed to IMS DC in the respective registered networks, both parties can initiate the IMS DC, independently and in parallel. The LI functions provided for such independent IMS DC sessions are handled independently, in the sense, IRIs and CCs are correlated independently.

4.10.2.3 LI in the local end of the IMS DC session

4.10.2.3.1 General

LI at the local end of the IMS DC session refers to the case where the target UE registered for IMS and subscribed for the IMS DC sets up the IMS DC. The case applies for intra-network and inter-network scenarios. As the illustrations show, the LI aspects in the local end of IMS DC session is independent of how the IMS DC is handled at the remote end of the IMS DC session.

As illustrated in clause 4.10.1, either the originating party or the terminating party can initiate the IMS DC setup and hence, can become the Local UE. It is possible that a redirected-to UE can also initiate the IMS DC session and hence, become the Local UE.

As defined in TS 33.127 [3] and TS 33.128 [4], the IRI-POI functions for the IMS DC setup related signalling information are provided by the DCSF.

As defined in TS 33.127 [3] and TS 33.128 [4], the CC-POI for IMS DC related media is provided by the MF or at the DC-AS depending on the IMS DC types and the operating mode of MF. For CC-POI present in the MF, CC-TF is present in the IMS AS. For the CC-POI present in the DC-AS, the CC-TF is in the DCSF. The DC-AS may also be present in a third party domain.

As explained in clause 4.10.1.6, the illustrations shown in the present document are based on the following deployment considerations to be made by the CSPs that have LI obligations to deliver the intercepted IMS DC media in an unencrypted form:

- The option of not having an MF on the P2P IMS DC media path is prohibited. This is true for all Local UEs.
- The option of MF operating in an UDP Proxy mode is prohibited for P2P IMS media.
- The option of MF operating in an UDP Proxy mode is prohibited for P2A, A2P, P2A2P IMS DC media when the DC-AS is in the third party domain.

In all the illustrations, originating party is shown as the Local UE. However, it has to be noted that on an IMS session, the other party can initiate the IMS DC session as well, independently, and parallelly. For example, originating party as a target can initiate the IMS DC and at the same time, can become a Remote UE for another IMS DC session setup by the peer party (i.e. terminating party or redirected-to party).

In an inter-network scenario, the Remote UE can be a target non-local ID for IMS-based services. TS 33.128 [4] specifies that the concept of target non-local ID does not apply to IMS DC.

4.10.2.3.2 IRI-POI for IMS DC

As defined in the TS 33.127 [3] and TS 33.128 [4], the IRI-POI functions for IMS DC are provided at the DCSF. Figure 4.10.2.3.2-1 illustrates topology view DCSF providing the IRI-POI functions when the Local UE (target) sets up the IMS DC.

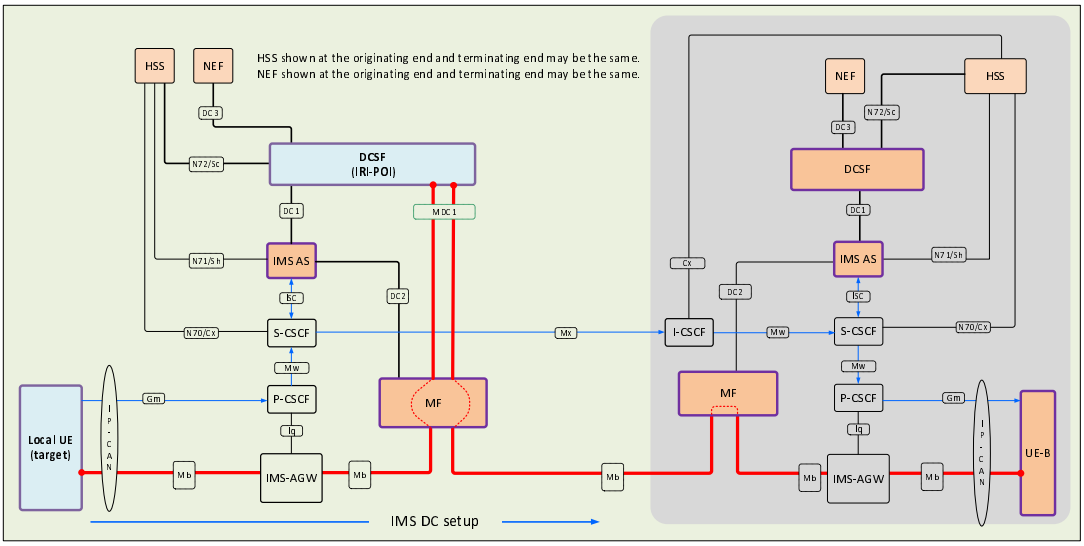


Figure 4.10.2.3.2-1: IRI-POI for IMS DC is at DCSF

The IRI-POI present in the DCSF generates the xIRI records when it detects of DSCF receiving IMS DC setup related event notifications from the IMS-AS over the reference point DC1. The details of xIRI records are defined in TS 33.128 [4].

4.10.2.3.3 Media interception for bootstrap IMS DC

TS 33.127 [3] and TS 33.128 [4] do not define the need to intercept the media of a Boot-strap IMS DC. Therefore, the following illustration can be used in case a requirement is added to provide the media interception for Boot-strap IMS DC.

Figure 4.10.2.3.3-1 illustrates the topology view and protocol stack level view for the media interception for the Boot-strap IMS DC. As shown, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

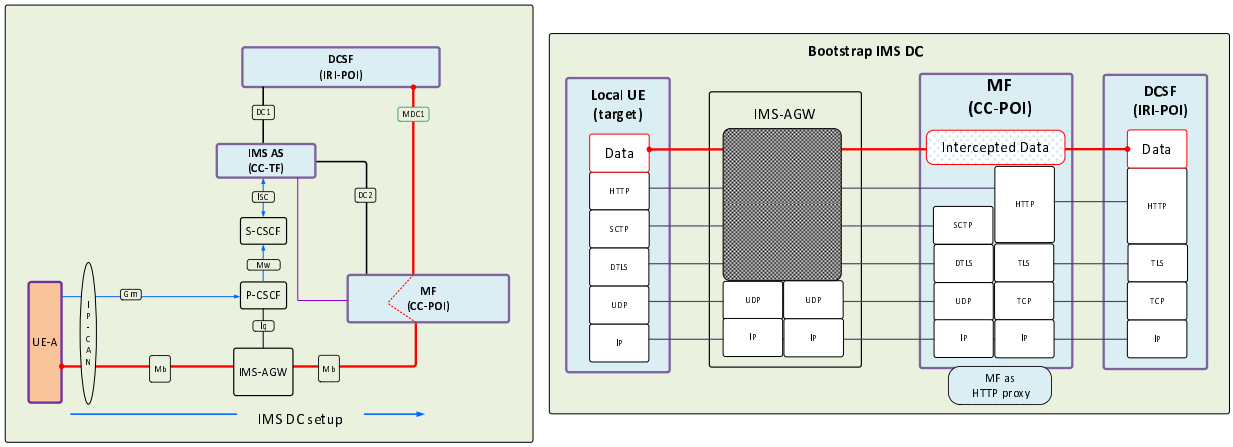


Figure 4.10.2.3.3-1: CC-POI for boot-strap IMS DC at MF

As shown in figure 4.10.2.3.3-1, MF is operating in HTTP Proxy mode.

4.10.2.3.4 Media interception for P2A and A2P IMS DC

4.10.2.3.4.1 General

The media interception of P2A and A2P IMS DC is done by the CC-POI present in the MF or DC-AS depending on the operating mode of MF. When MF provides the CC-POI functions, the CC-TF is in the IMS-AS and when DC-AS provides the CC-POI, the CC-TF is in the DCSF.

4.10.2.3.4.2 MF operating in HTTP proxy mode

Figure 4.10.2.3.4.2-1 illustrates the topology view and protocol stack level view for the media interception for the P2A and A2P IMS DC when the MF is operating in the HTTP Proxy mode. In this case, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

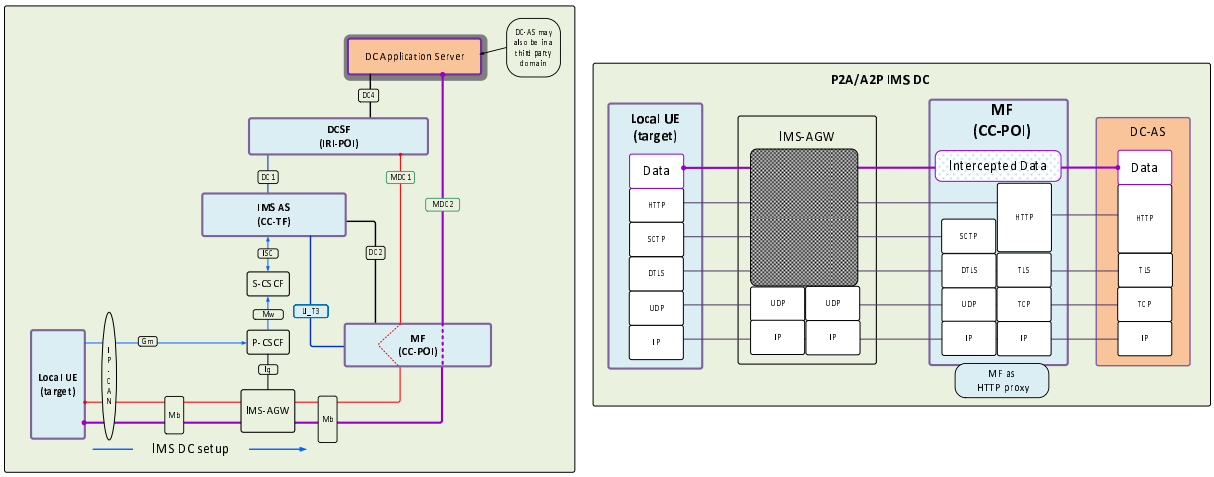


Figure 4.10.2.3.4.2-1: CC-POI for P2A, A2P IMS DC when MF is HTTP Proxy

In this case, DC-AS may also be in a third party domain.

4.10.2.3.4.3 MF operating in DC application proxy mode

Figure 4.10.2.3.4.3-1 illustrates the topology view and protocol stack level view for the media interception for the P2A and A2P IMS DC when the MF is operating in the DC Application Proxy mode. In this case, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

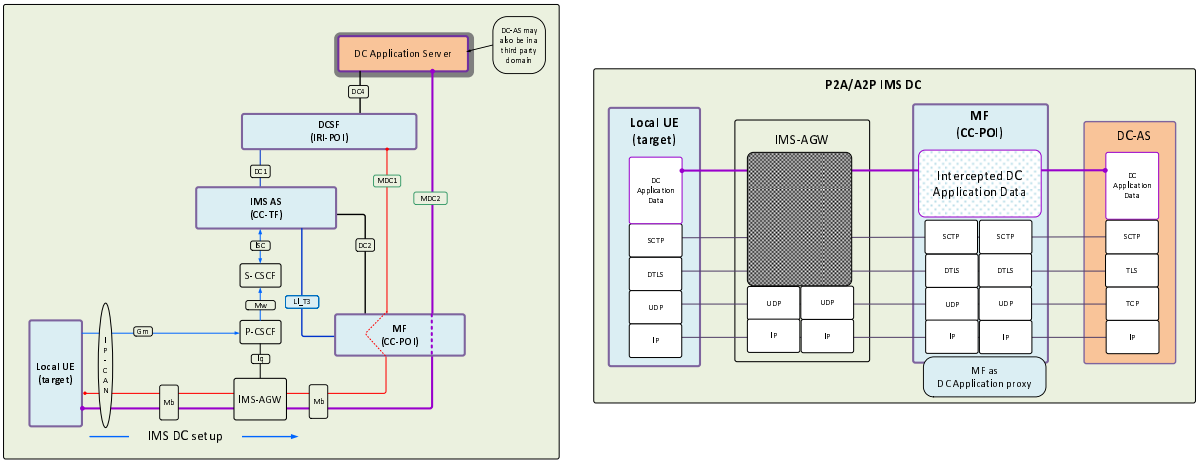


Figure 4.10.2.3.4.3-1: CC-POI for P2A, A2P IMS DC when MF is DC Application Proxy

In this case, DC-AS may also be in a third party domain.

4.10.2.3.4.4 MF operating in UDP proxy mode

Figure 4.10.2.3.4.4-1 illustrates the topology view and protocol stack level view for the media interception for the P2A and A2P IMS DC when the MF is operating in the UDP Proxy mode. In this case, the CC-POI functions are provided by the DC-AS and the trigger for the media interception is provided by the CC-TF present in the DCSF. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

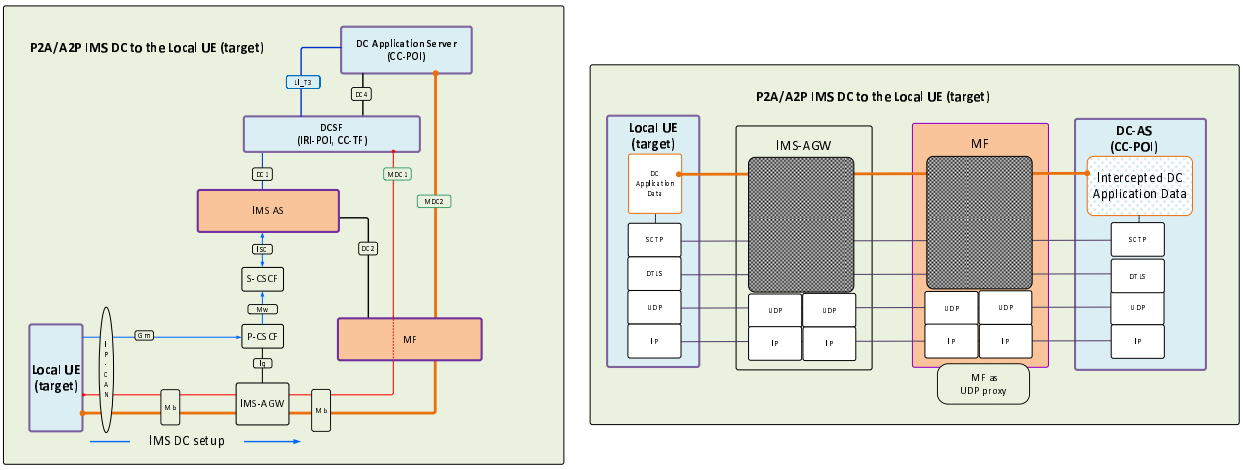


Figure 4.10.2.3.4.4-1: CC-POI for P2A, A2P IMS DC when MF is UDP Proxy

In this case, DC-AS cannot be in third party domain. In other words, when the DC-AS is in a third party domain, the use of MF in an UDP Proxy mode (as shown in figure 4.10.2.3.4.4-2) should be prohibited.

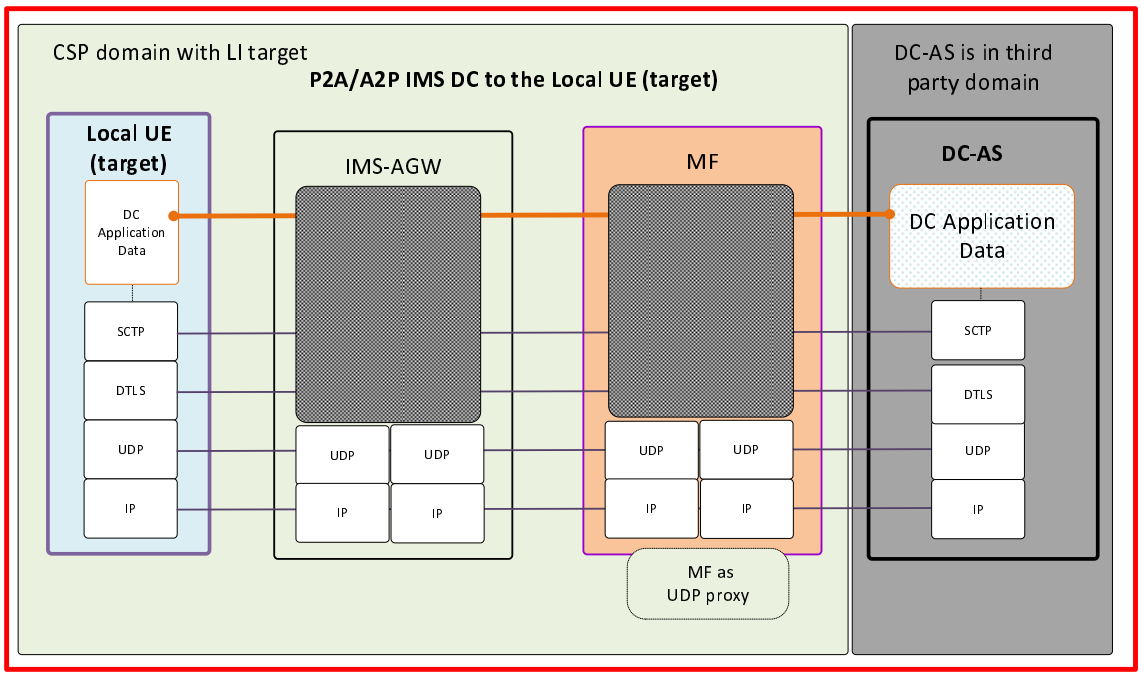


Figure 4.10.2.3.4.4-2: Unavailable NF for P2A, A2P (also true for P2A2P) MS DC media interception with MF as UDP Proxy when DC-AS is in third party

As can be observed in the illustration shown in figure 4.10.2.3.4.4-2, P2A, A2P (also true for P2A2P) IMS DC media is not available in an unencrypted for interception in the CSP domain. Any possibility of third party provider having the LI is outside the scope of the present document.

4.10.2.3.5 Media interception for P2A2P IMS DC

4.10.2.3.5.1 General

The media interception of P2A2P IMS DC is done by the CC-POI present in the MF or DC-AS depending on the operating mode of MF. When MF provides the CC-POI functions, the CC-TF is in the IMS-AS and when DC-AS provides the CC-POI, the CC-TF is in the DCSF.

In all illustrations the Remote UE is in the same network as the Local UE (target). However, Remote UE may also be in a different network in which case, TrGWs would be present at the egress end of the originating network and ingress end

of the terminating network. The Remote UE may also be a redirected-to network in case the IMS session redirection happens. In that case, additional TrGWs may be present in the redirecting network.

4.10.2.3.5.2 MF operating in HTTP proxy mode

Figure 4.10.2.3.5.2-1 illustrates the topology view and protocol stack level view for the media interception for the P2A2P IMS DC when the MF is operating in the HTTP Proxy mode. In this case, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

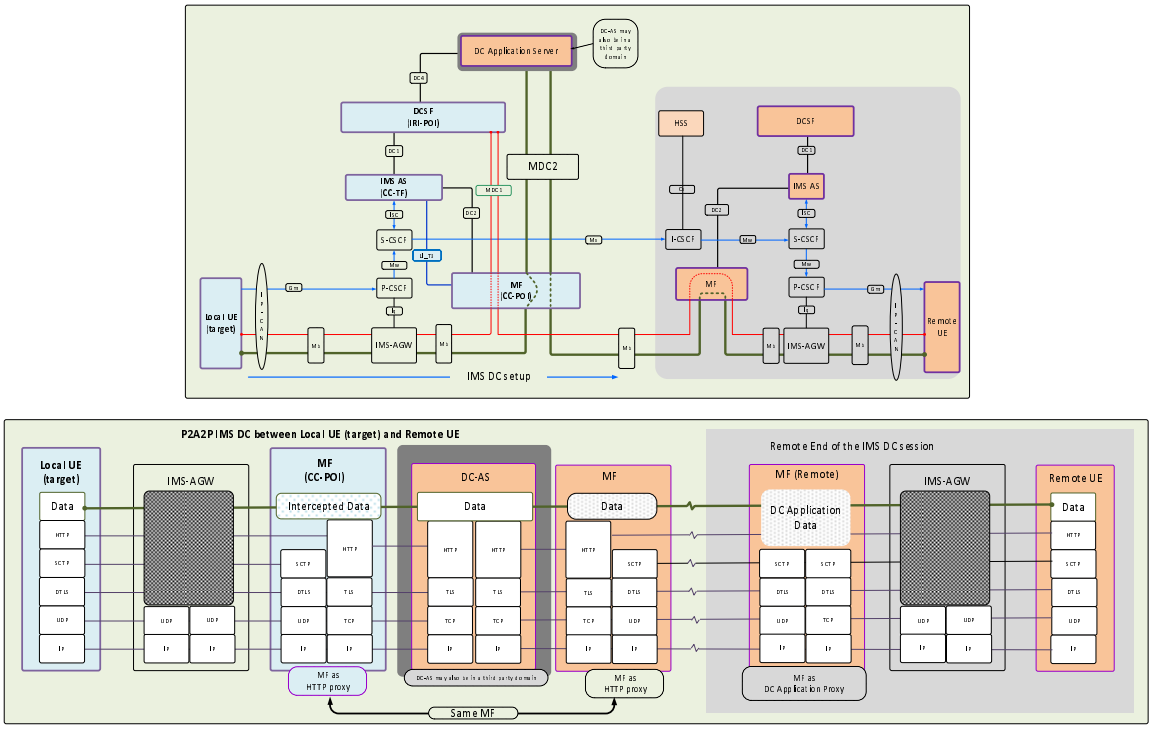


Figure 4.10.2.3.5.2-1: CC-POI for P2A2P IMS DC when MF is HTTP Proxy

In this case, DC-AS may also be in a third party domain.

4.10.2.3.5.3 MF operating in DC application proxy mode

Figure 4.10.2.3.5.3-1 illustrates the topology view and protocol stack level view for the media interception for the P2A2P IMS DC when the MF is operating in the DC Application Proxy mode. In this case, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

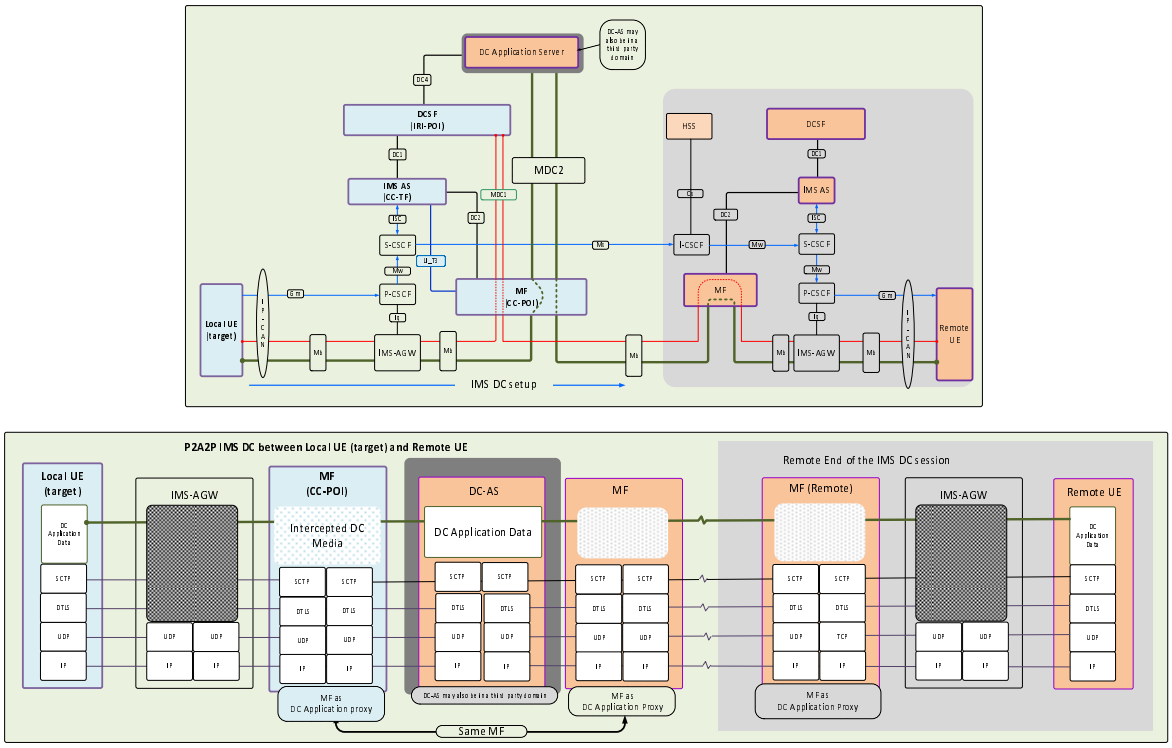


Figure 4.10.2.3.5.3-1: CC-POI for P2A2P IMS DC when MF is DC Application Proxy

In this case, DC-AS may also be in a third party domain.

4.10.2.3.5.4 MF operating in UDP proxy mode

Figure 4.10.2.3.5.4-1 illustrates the topology view and protocol stack level view for the media interception for the P2A2P IMS DC when the MF is operating in the UDP Proxy mode. In this case, the CC-POI functions are provided by the DC-AS and the trigger for the media interception is provided by the CC-TF present in the DCSF. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

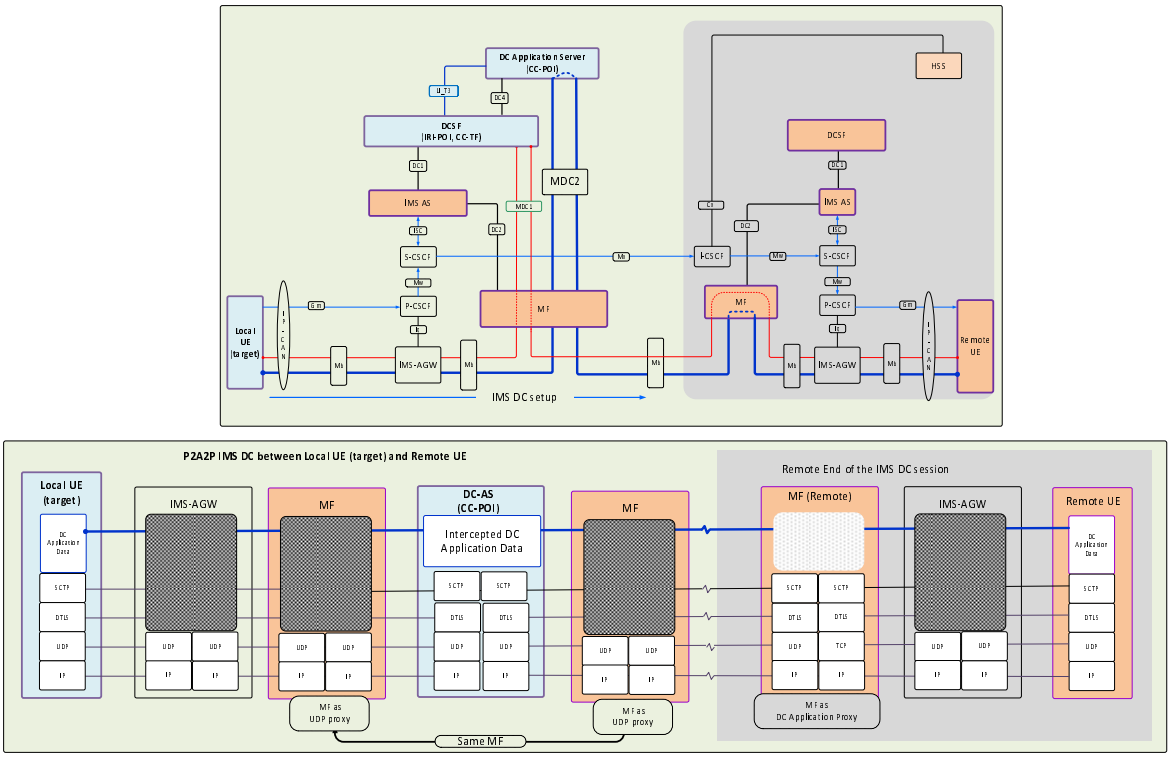


Figure 4.10.2.3.5.4-1: CC-POI for P2A2P IMS DC when MF is UDP Proxy

In this case, DC-AS cannot be in third party domain. In other words, when the DC-AS is in a third party domain, the use of MF in an UDP Proxy mode should be prohibited (see figure 4.10.2.3.4.4-2).

4.10.2.3.6 Media interception for P2P IMS DC

4.10.2.3.6.1 General

The media interception of P2P IMS DC is done by the CC-POI present in the MF with CC-TF at IMS AS.

In all illustration the Remote UE is in the same network as the Local UE (target). However, Remote UE may also be in a different network in which case, TrGWs would be present at the egress end of the originating network and ingress end of the terminating network. The Remote UE may also be a redirected-to network in case the IMS session redirection happens. In that case, additional TrGWs may be present in the redirecting network.

4.10.2.3.6.2 MF operating in DC application proxy mode

Figure 4.10.2.3.6.2-1 illustrates the topology view and protocol stack level view for the media interception for the P2P IMS DC when the MF is operating in the DC Application Proxy mode. In this case, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

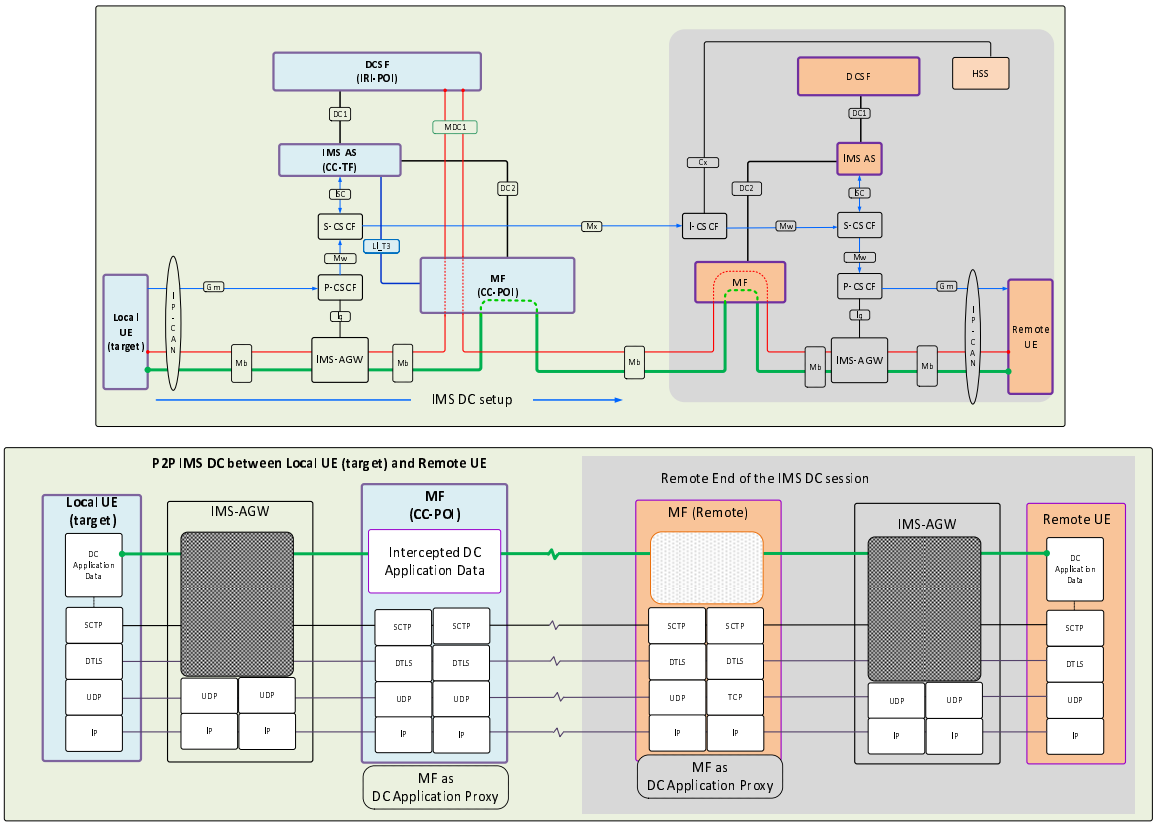


Figure 4.10.2.3.6.2-1: CC-POI for P2P IMS DC when MF is DC Application Proxy

4.10.2.3.6.3 MF operating in UDP proxy mode

Even though TS 23.228 [8] allows a P2P IMS DC with MF operating in a UDP Proxy mode, since the IMS DC media is not available in an unencrypted form which is necessary to perform the media interception, the use of MF in UDP Proxy mode (as shown in figure 4.10.2.3.6.3-1) should be prohibited for P2P IMS DC.

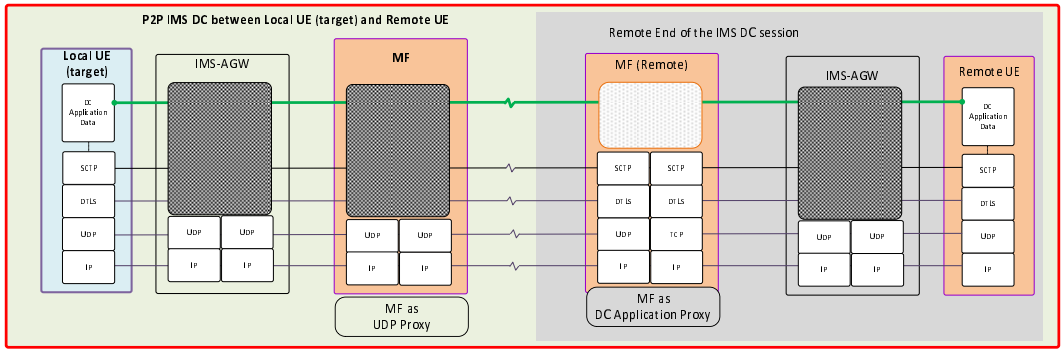


Figure 4.10.2.3.6.3-1: Unavailable NF for P2P IMS DC media interception with MF as UDP Proxy

As can be observed in the illustration shown in figure 4.10.2.3.6.3-1, P2P IMS DC media is not available in an unencrypted for interception in the CSP domain when MF is operating in UDP proxy mode.

4.10.2.3.6.4 Without MF

Even though TS 23.228 [8] allows a P2P IMS DC without the presence of an MF on the media path, since the IMS DC media is not available in an unencrypted form which is necessary to perform the media interception, the P2P IMS DC without an MF on the media path should be prohibited.

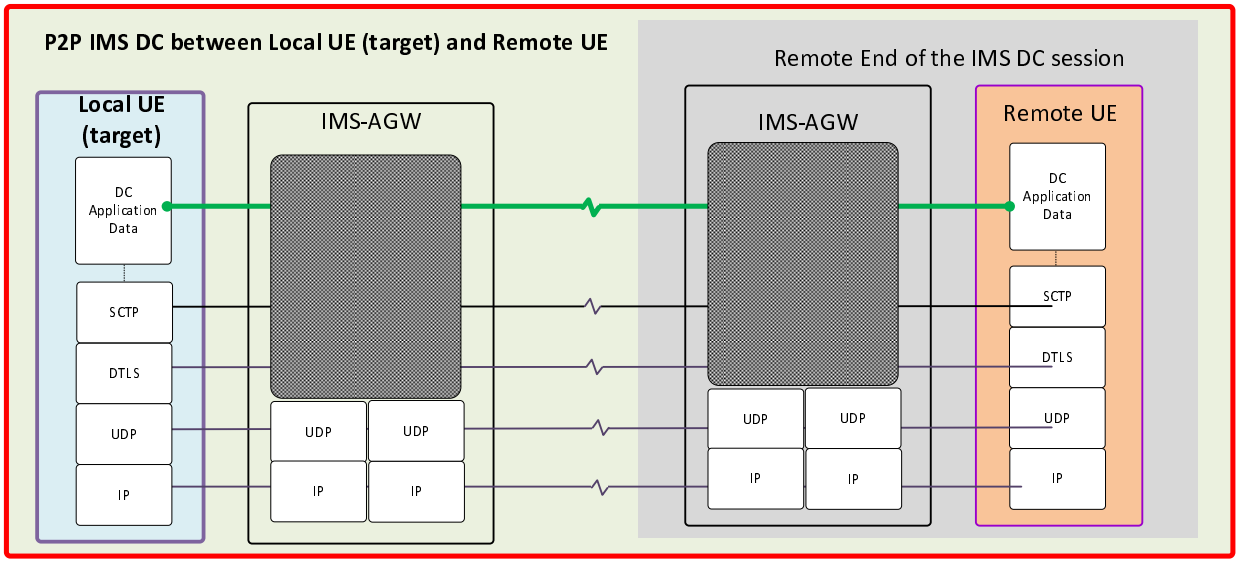


Figure 4.10.2.3.6.4-1: Unavailable NF for P2P IMS DC media interception without the MF

As can be observed in the illustration shown in figure 4.10.2.3.6.4-1, P2P IMS DC media is not available in an unencrypted for interception in the CSP domain without an MF.

4.10.2.3.7 Media interception with transcoded media transfer to the Remote UE

4.10.2.3.7.1 General

When the Remote UE is not IMS DC capable or when the Remote Network (in an inter-network scenario) does not have IMS DC capabilities, or the redirecting network does not have IMS DC capabilities for a redirected case, the P2A2P and P2P IMS DC media may be sent to, or received from, the Remote UE in a transcoded form. When the Local UE is the target, the media interception is done in the same way as in the case of non-transcoded media transfer to the Remote UE. This is because the IMS DC from the Local UE's (target's) perspective is the same. These are illustrated in the subsequent clauses.

In all illustrations the Remote UE is in the same network as the Local UE (target). Therefore, transcoding can happen only when the Remote UE is not IMS DC capable. However, Remote UE may also be in a different network in which case, a TrGWs would be present at the egress end of the originating network and ingress end of the terminating network. The Remote UE may also be a redirected-to network in case the IMS session redirection happens. In that case, additional TrGWs may be present in the redirecting network. In the inter-network cases, the transcoded media transfer may happen to the Remote UE even if the Remote UE is IMS DC capable, but the network where the Remote UE registered to does not support the IMS DC capabilities or in the case of a redirecting case, the redirecting network does not support the IMS DC capabilities.

In all the cases, no Bootstrap IMS DC is setup to the Remote UE.

For P2A2P IMS DC, the media transcoding is done at the DC-AS whereas for P2P IMS DC, the media transcoding is done at the MF.

4.10.2.3.7.2 Transcoded media with P2A2P IMS DC with MF acting in an HTTP proxy mode

Figure 4.10.2.3.7.2-1 illustrates the topology view and protocol stack level view for the media interception for the P2A2P IMS DC with transcoded media transfer to the Remote UE when the MF is operating in the HTTP Proxy mode. In this case, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

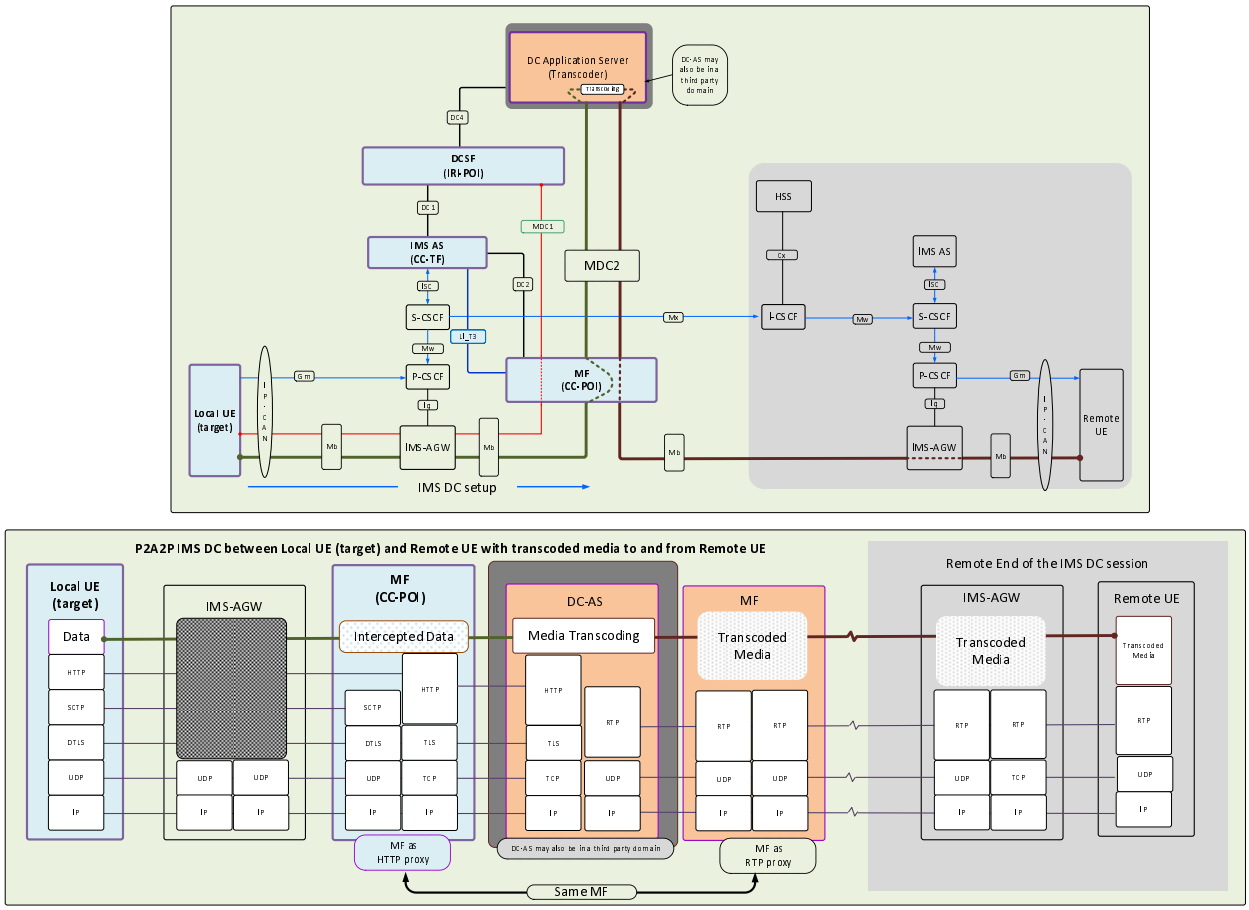


Figure 4.10.2.3.7.2-1: CC-POI for P2A2P IMS DC with transcoded media transfer to the remote UE when MF is HTTP Proxy

In this case, DC-AS may also be in a third party domain.

4.10.2.3.7.3 Transcoded media with P2A2P IMS DC with MF operating in DC application proxy mode

Figure 4.10.2.3.7.3-1 illustrates the topology view and protocol stack level view for the media interception for the P2A2P IMS DC with transcoded media transfer to the Remote UE when the MF is operating in the DC Application Proxy mode. In this case, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

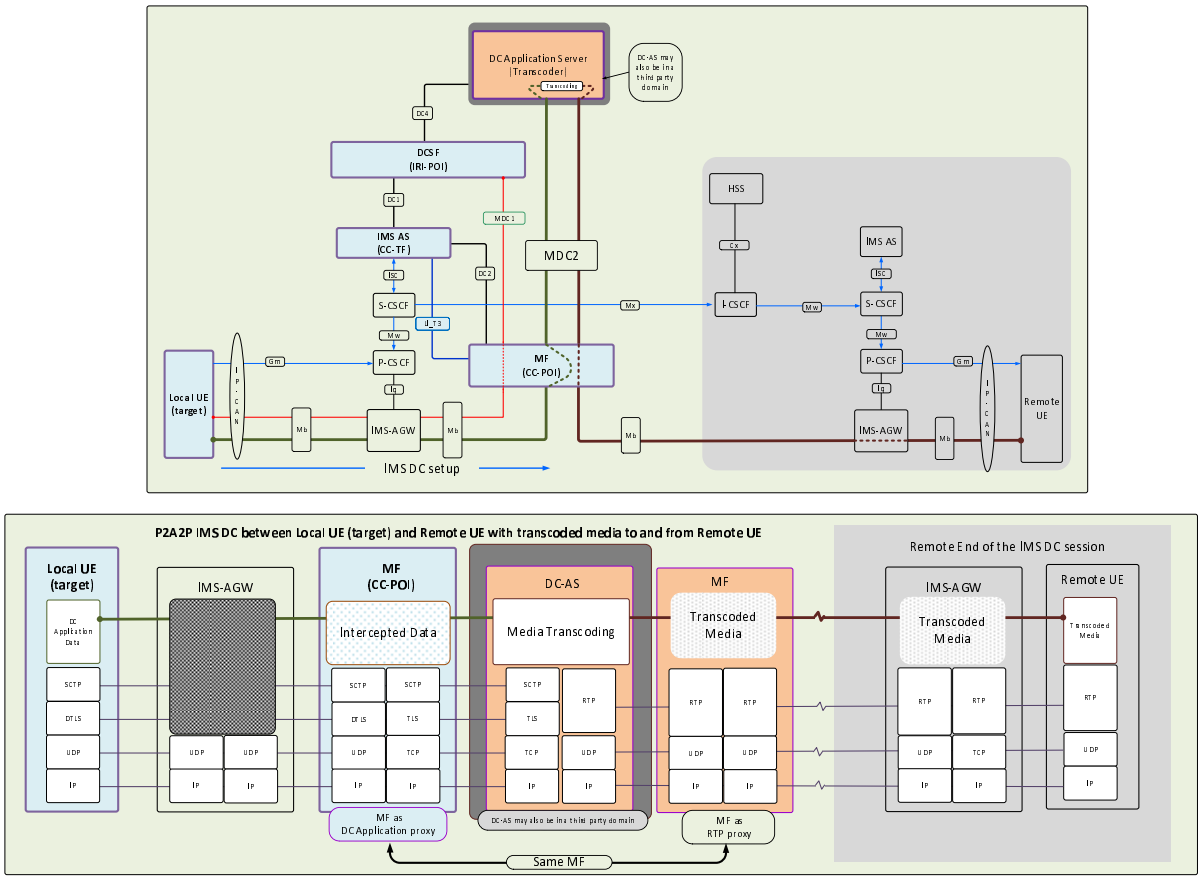


Figure 4.10.2.3.7.3-1: CC-POI for P2A2P IMS DC with transcoded media transfer to the remote UE when MF is DC Application Proxy

In this case, DC-AS may also be in a third party domain.

4.10.2.3.7.4 Transcoded media with P2A2P IMS DC with MF operating in UDP proxy mode

Figure 4.10.2.3.7.4-1 illustrates the topology view and protocol stack level view for the media interception for the P2A2P IMS DC with transcoded media transfer to the Remote UE when the MF is operating in the UDP Proxy mode. In this case, the CC-POI functions are provided by the DC-AS and the trigger for the media interception is provided by the CC-TF present in the DCSF. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

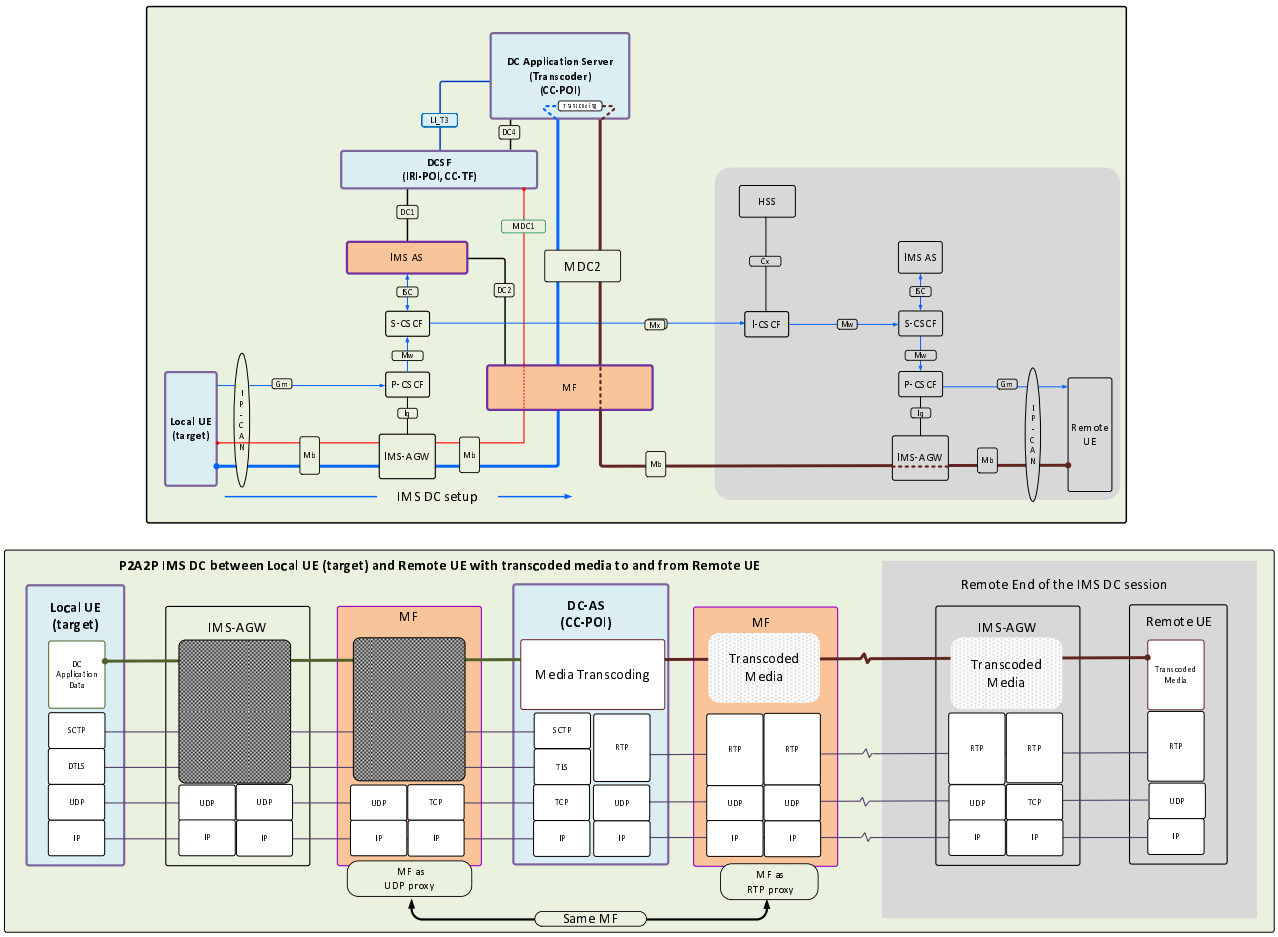


Figure 4.10.2.3.7.4-1: CC-POI for P2A2P IMS DC with transcoded media transfer to the remote UE MF is UDP Proxy

In this case, DC-AS cannot be in third party domain. In other words, when the DC-AS is in a third party domain, the use of MF in an UDP Proxy mode should be prohibited.

4.10.2.3.7.5 Transcoded media with P2P IMS DC with MF operating in DC application proxy mode

Figure 4.10.2.3.7.5-1 illustrates the topology view and protocol stack level view for the media interception for the P2P IMS DC with transcoded media transfer to the Remote UE when the MF is operating in the DC Application Proxy mode. In this case, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. The illustration also shows that IRI-POI is at the DCSF. In this illustration, Local UE is the target.

The topology view shows only the relevant IMS DC related NFs.

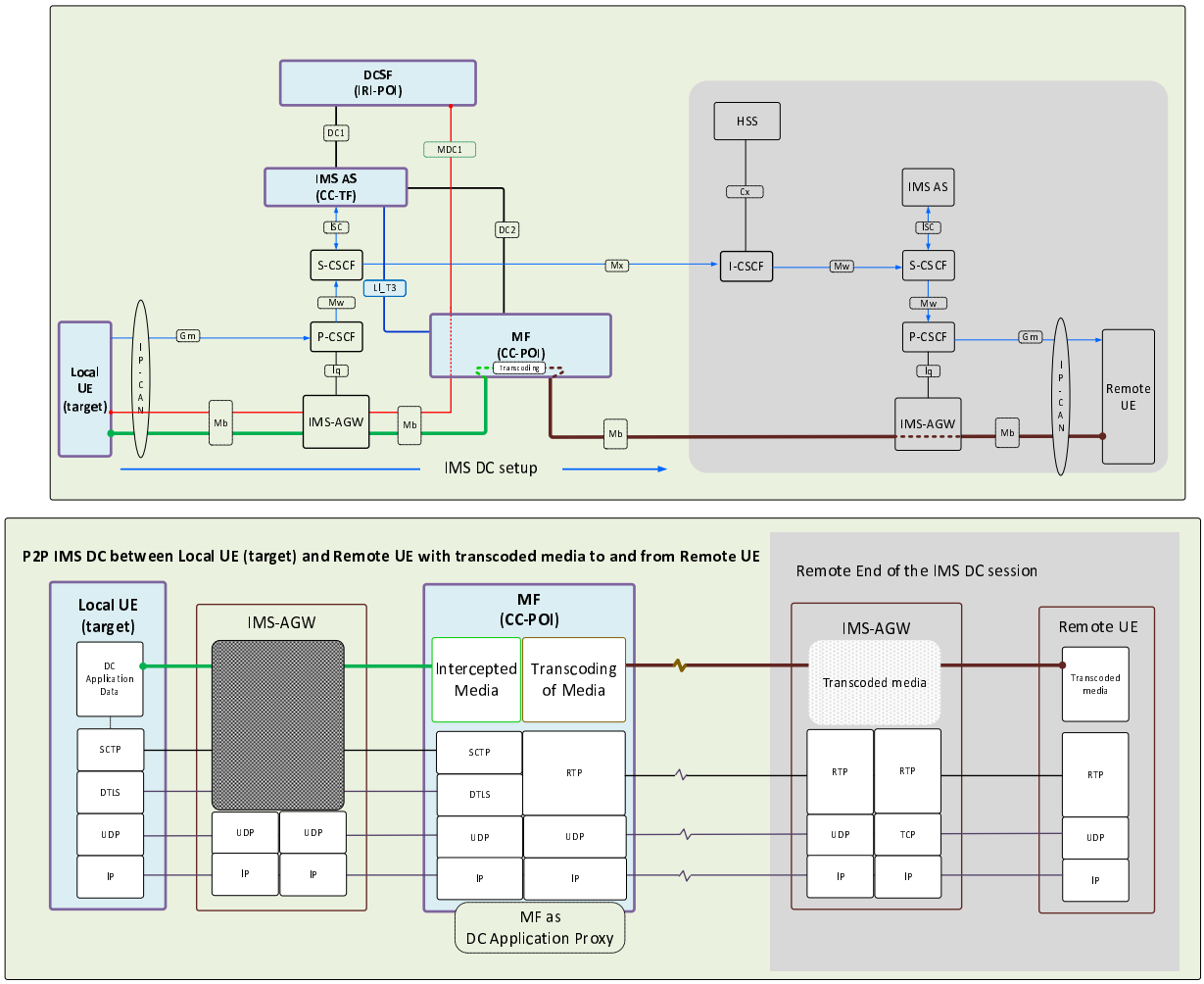


Figure 4.10.2.3.7.5-1: CC-POI for P2P IMS DC P2A2P IMS DC with transcoded media transfer to the remote when MF is DC Application Proxy

The P2P IMS DC with transcoded media transfer to the Remote UE cannot happen unless MF is operating in a DC Application Proxy mode.

4.10.2.4 LI at the remote end of IMS DC session

4.10.2.4.1 General

LI at the remote end of the IMS DC session refers to the case where the other IMS UE present on the IMS session initiates the IMS DC setup.

The case applies for intra-network and inter-network scenarios. In the inter-network scenario, the Remote Network may or may not have IMS DC capabilities. The Remote UE may or may not have IMS DC capabilities. When the Remote Network has IMS DC capabilities, the IMS DC capable UE may or may not have subscribed to IMS DC. The Remote Network that does not have IMS DC capabilities, may have limited support to allow its locally registered IMS DC capable UEs to have Bootstrap and other Application DCs setup by the other IMS UE present in the peer network.

The subsequent clauses illustrate the following two cases:

- Both Remote Network and Remote UE have IMS DC capabilities. In an intra-network case, by definition, the remote end of the IMS network for the IMS DC session has the IMS DC capabilities. It is assumed that the MF (at the remote end of IMS DC session) is inserted into the IMS DC media path and DCSF (at the remote end) is notified.

- Either Remote Network or the Remote UE does not have IMS DC capabilities. In the intra-network case, it is the Remote UE that does not have the IMS DC capabilities. It is assumed that the Remote UE in this case, can receive or send P2A2P or P2PIMS DC media only in a transcoded form.

The following scenarios are not illustrated in the subsequent clauses:

- When the network has the MF and DCSF, optional deployment case of not inserting the MF into the IMS DC media path.
- When the network does not have MF or DCSF, but allowing its locally IMS registered UEs to have Bootstrap and other application DCs setup by the other IMS UE present in the peer network.

This optionality of not inserting the MF into the IMS DC media path and not notifying the DCSF also includes the case where the network with IMS DC capabilities does not insert the MF into the IMS DC media path as a deployment choice, or the case where the IMS DC capable UE has not subscribed for IMS DC. In the latter case, if the network inserts the MF into the IMS DC path and notifies the DCSF, then the illustrations shown in the subsequent clauses can apply.

If network that does not have IMS DC capabilities with limited IMS DC support (allows its locally registered UEs to have bootstrap and other application DCs setup by the other UE present in the peer network) has MF and DCSF and inserts MF into the IMS DC media path and notifies the DCSF, then the illustrations shown in the subsequent clauses can apply.

As the illustrations show, the LI aspects in the remote end of IMS DC session is independent of which operating mode the MF is used at local end of the IMS DC session. The MF in the Remote end of the IMS DC session always operates in the DC Application Proxy mode. The possibility of MF operating in an UDP Proxy mode is not considered.

As illustrated in clause 4.10.1, either the originating party or the terminating party of an IMS session can initiate the IMS DC setup and hence, either the terminating party or the originating party can become the Remote UE. It is also possible that a redirected-to UE can also be a Remote UE for an IMS DC session.

In all the illustrations, terminating party is shown as the Remote UE. However, as stated above on an IMS session, either the terminating party or the redirected-to party can also initiate the IMS DC session, independently, and parallelly. In other words, the originating party can also become the Remote UE. As a matter of fact, both originating party and terminating party (or redirected-to party) can become the Local UE and the Remote UE at the same time.

4.10.2.4.2 Remote network and remote UE have IMS DC capabilities

4.10.2.4.2.1 General

When the Remote Network and Remote UE have IMS DC capabilities, the present document assumes that the MF (in Remote Network or at the remote end of the IMS DC session) is inserted into the IMS DC media path and DCSF (in Remote Network or at the remote end of the IMS DC session) is notified. The MF always operates in DC Application Proxy mode.

When the Remote UE is the target, the media interception for the IMS DCs happen at CC-POI present in the MF with CC-TF at IMS AS.

NOTE: According to TS 23.228 [8], even when the Remote Network has IMS DC capabilities, the insertion of MF into the IMS DC media path is optional or when an IMS DC capable Remote UE has not subscribed to IMS DC, the insertion of MF into the IMS DC media path is optional. From an illustration perspective, all these cases fall into category of further study as described in clause 4.10.2.4.3.

4.10.2.4.2.2 IRI-POI for IMS DC

As defined in the TS 33.127 [3] and TS 33.128 [4], the IRI-POI functions for IMS DC are provided at the DCSF. Figure 4.10.2.4.2.2-1 illustrates topology view DCSF providing the IRI-POI functions for the Remote UE.

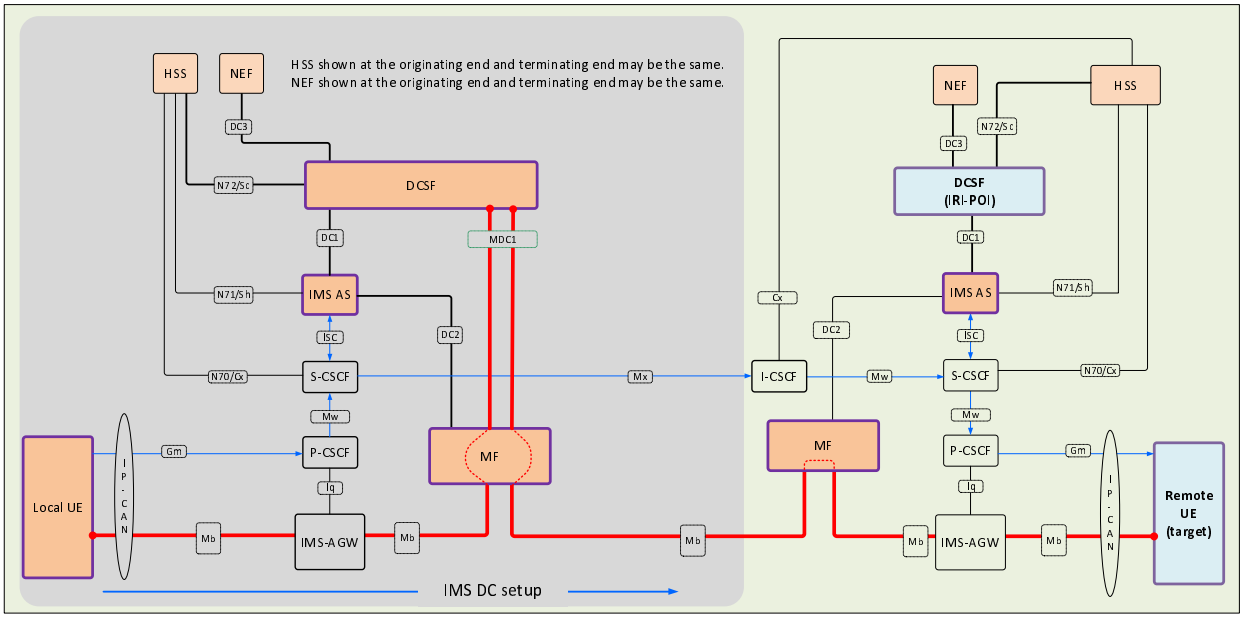


Figure 4.10.2.4.2.2-1: IRI-POI for IMS DC is at DCSF

The IRI-POI present in the DCSF generates the xIRI records when it detects of DSCF receiving IMS DC setup related event notifications from the IMS-AS over the reference point DC1. The details of xIRI records are defined in TS 33.128 [4].

4.10.2.4.2.3 Media interception for bootstrap IMS DC

TS 33.127 [3] and TS 33.128 [4] do not define the need to intercept the media of a Boot-strap IMS DC. Therefore, the following illustration can be used in case a requirement is added to provide the media interception for Boot-strap IMS DC.

Figure 4.10.2.4.2.3-1 illustrates the topology view and protocol stack level view for the media interception for the Boot-strap IMS DC. As shown, the CC-POI functions are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS. In this illustration, Remote UE is the target.

The topology view shows only the relevant IMS DC related NFs.

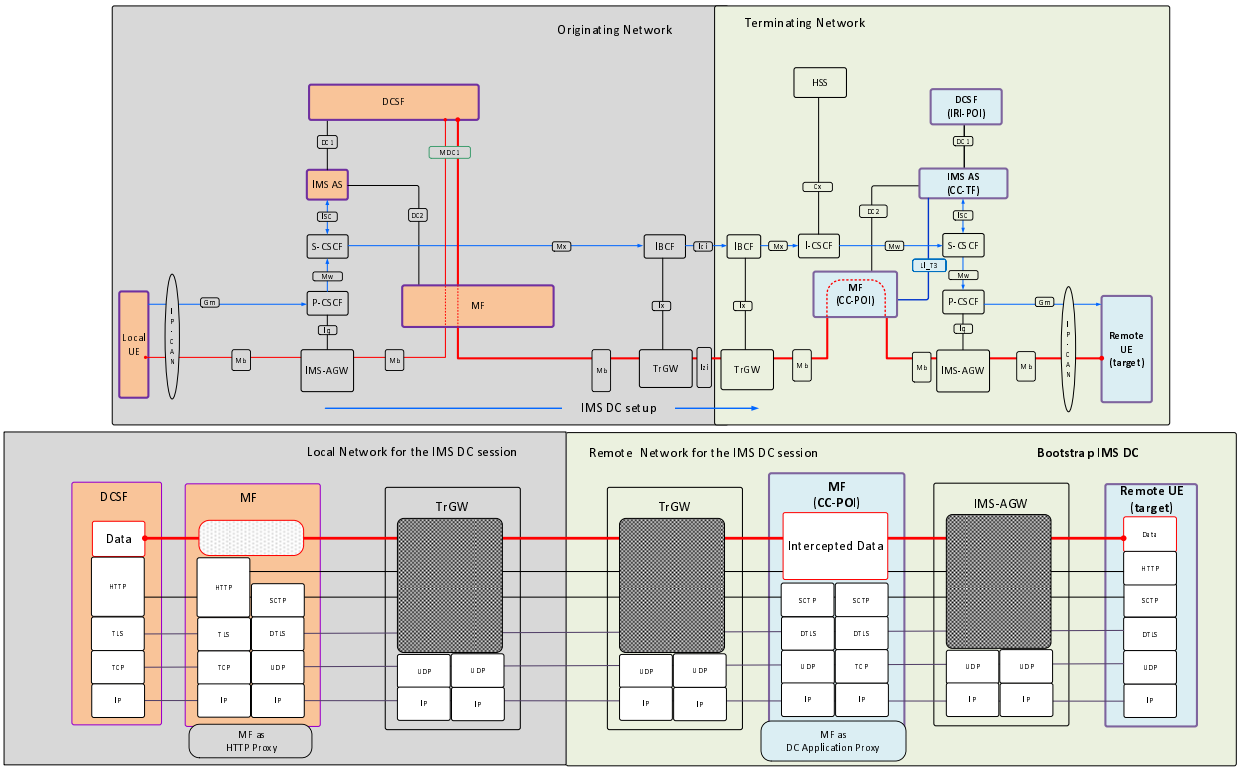


Figure 4.10.2.4.2.3-1: CC-POI for boot-strap IMS DC at MF

As shown in figure 4.10.2.4.2.3-1, MF in the Remote Network is operating in DC Application Proxy mode.

4.10.2.4.2.4 Media interception for P2A and A2P IMS DC

Figure 4.10.2.4.2.4-1 illustrates the topology view and protocol stack level view for the media interception for the P2A and A2P IMS DC in the Remote Network when the Remote UE is the target.

The CC-POI functions for the P2A or A2P media are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS.

The topology view shows Bootstrap IMS DC to the Remote UE even though it has nothing to do with the interception of P2A or A2P IMS DC media.

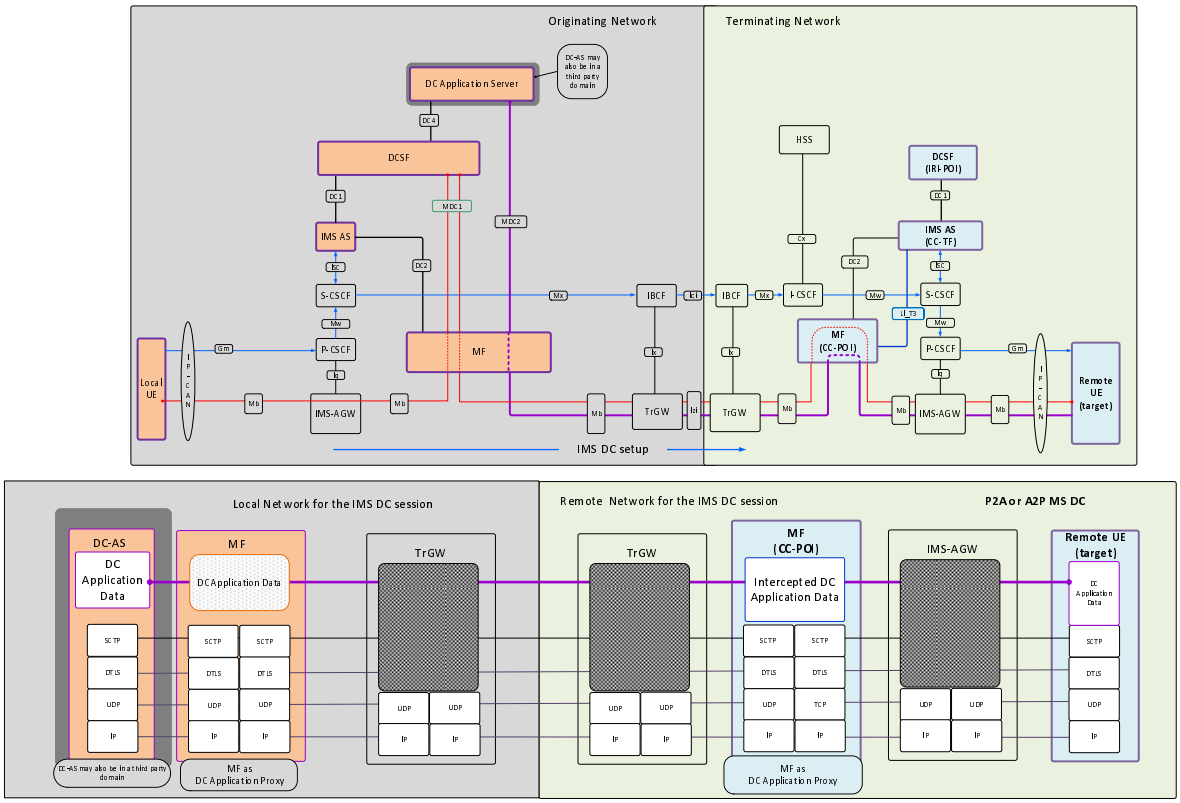


Figure 4.10.2.4.2.4-1: CC-POI for P2A, A2P IMS DC at MF

In the above illustration, the MF at the local end of the IMS DC session is operating in a DC Application Proxy mode. However, since the MF in the Remote end of the IMS DC session is always operating in DC Application Proxy mode, the same illustration can apply even if the MF at the local end is operating in an UDP proxy mode or HTTP proxy mode.

4.10.2.4.2.5 Media interception for P2A2P IMS DC

Figure 4.10.2.4.2.5-1 illustrates the topology view and protocol stack level view for the media interception for the P2A2P IMS DC in the Remote Network when the Remote UE is the target.

The CC-POI functions for the P2A2P media are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS.

The topology view shows Bootstrap IMS DC to the Remote UE even though it has nothing to do with the interception of P2A2P IMS DC media.

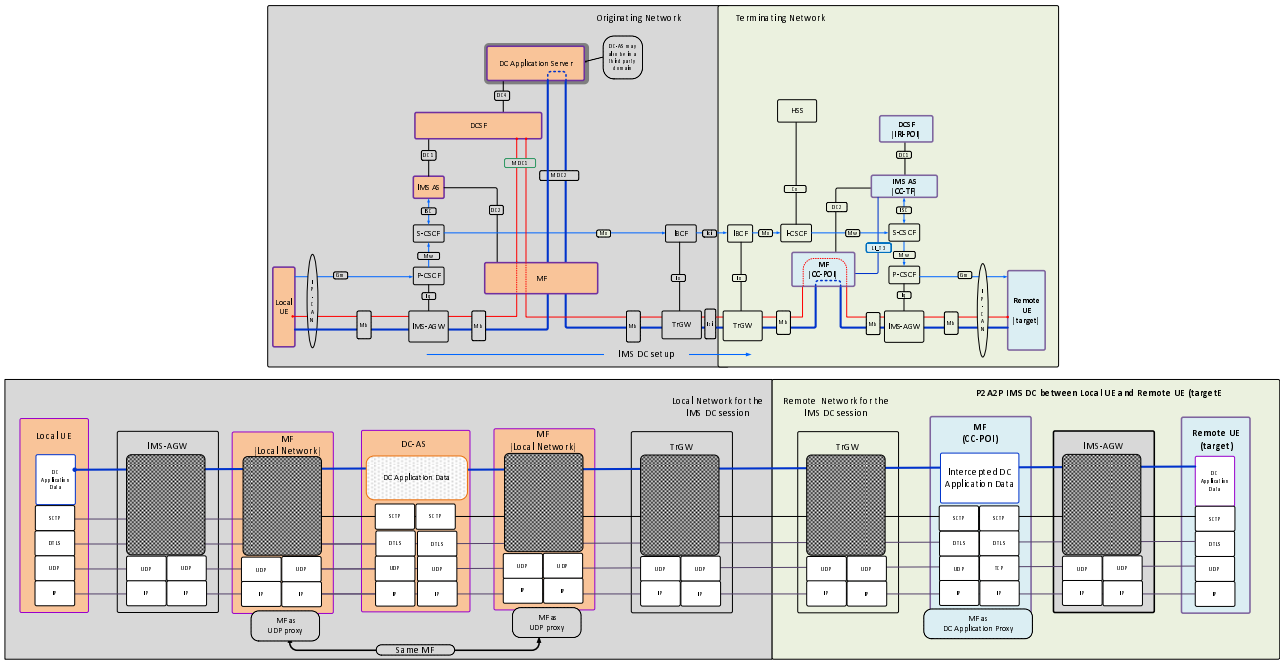


Figure 4.10.2.4.2.5-1: CC-POI for P2A2P IMS DC at MF

In the above illustration, the MF at the local end of the IMS DC session is operating in an UDP Proxy mode. However, since the MF in the Remote end of the IMS DC session is always operating in DC Application Proxy mode, the same illustration can apply even if the MF at the local end is operating in an DC Application Proxy mode or HTTP proxy mode.

4.10.2.4.2.6 Media interception for P2P IMS DC

Figure 4.10.2.4.2.6-1 illustrates the topology view and protocol stack level view for the media interception for the P2P IMS DC in the Remote Network when the Remote UE is the target.

The CC-POI functions for the P2P media are provided by the MF and the trigger for the media interception is provided by the CC-TF present in the IMS-AS.

The topology view shows Bootstrap IMS DC to the Remote UE even though it has nothing to do with the interception of P2P IMS DC media.

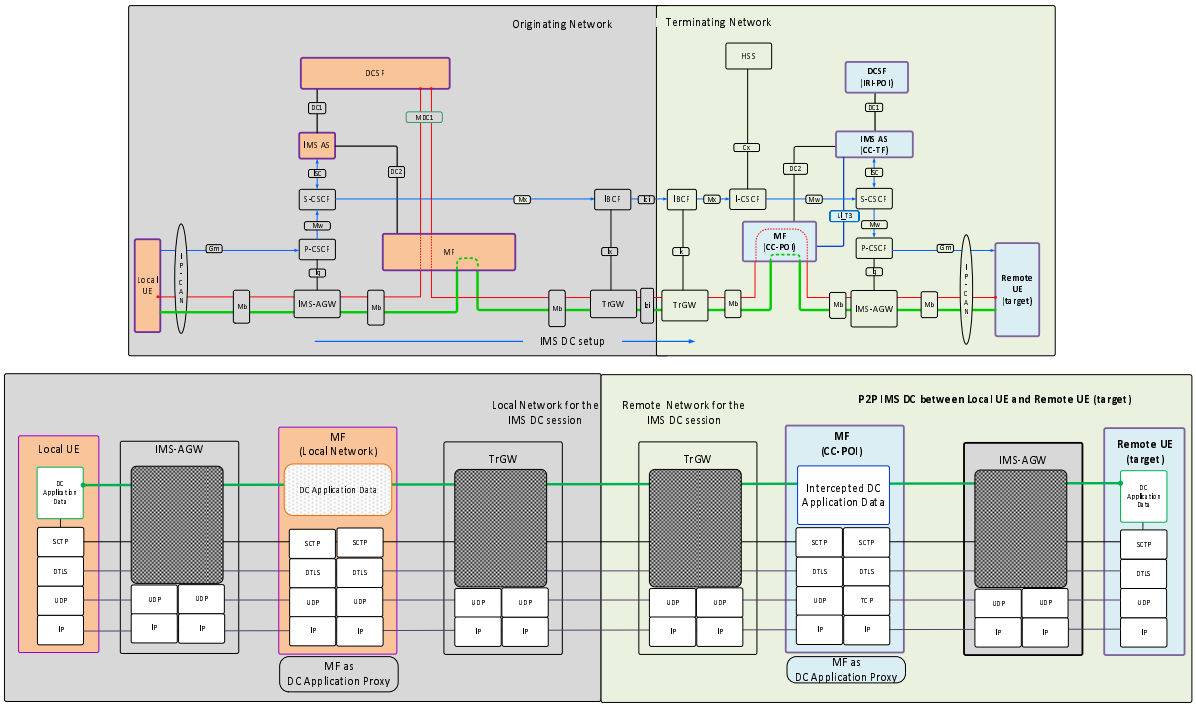


Figure 4.10.2.4.2.6-1: CC-POI for P2P IMS DC at MF

In the above illustration, the MF at the local end of the IMS DC session is operating in an DC Application Proxy mode.

4.10.2.4.2.7 MF operating in an UDP proxy mode

Since the IMS DC media is not available in an unencrypted form (as shown in figure 4.10.2.4.2.7-1) which is necessary to perform the media interception, the use of MF in an UDP Proxy mode in the Remote Network (or at the remote end of IMS DC session) has to be prohibited.

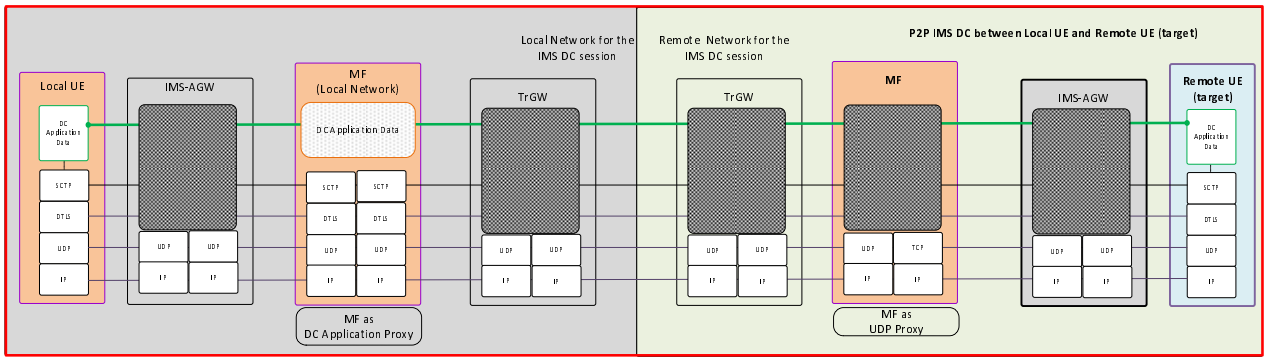


Figure 4.10.2.4.2.7-1: Unavailable NF for IMS DC media interception when MF operates in UDP Proxy mode in the Remote Network

As can be observed in the illustration shown in figure 4.10.2.4.2.7-1, P2P IMS DC media is not available in an unencrypted for interception in the CSP domain (as Remote Network) when MF is operating in an UDP proxy mode.

4.10.2.4.3 Either the remote network or remote UE does not have IMS DC capabilities

4.10.2.4.3.1 General

When the Remote UE is not IMS DC capable or when the Remote Network (in an inter-network scenario) does not have IMS DC capabilities, or the redirecting network does not have IMS DC capabilities for a redirected case, the P2A2P and P2P IMS DC media may be sent to, or received from, the Remote UE in a transcoded form. The media in the Remote Network is in RTP form and when the Remote UE is the target, the regular IMS LI would be used to provide the media interception for the transcoded media.

In all the cases, no Bootstrap IMS DC is setup to the Remote UE (target).

As the illustration show, the LI for transcoded media in the remote network is independent of where and how the media transcoding is done at the local end of the IMS DC session.

4.10.2.4.3.2 LI with transcoded media at the remote end (for P2A2P IMS DC from local UE's perspective)

Figure 4.10.2.4.3.2-1 illustrates the topology view and protocol stack level view for the media interception for the P2A2P IMS DC with transcoded media transfer to the Remote UE. In this case, the CC-POI functions are provided by the IMS-AGW and the trigger for the media interception is provided by the CC-TF present in the P-CSCF.

The topology view shows only the relevant IMS DC related NFs.

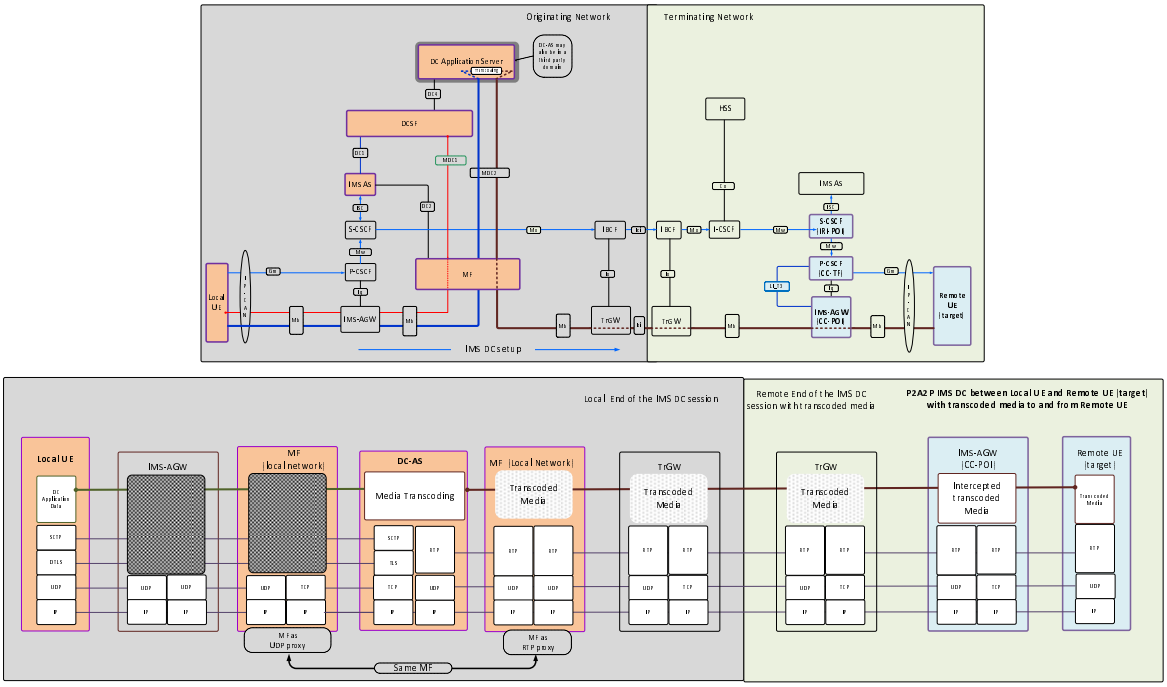


Figure 4.10.2.4.3.2-1: CC-POI for transcoded media from a P2A2P IMS DC

As shown in figure 4.10.2.4.3.2-1, the S-CSCF in the Remote Network provides the IRI-POI functions and IMS-AGW provides the CC-POI functions with CC-TF at the P-CSCF.

4.10.2.4.3.3 LI with transcoded media at the remote end (for P2P IMS DC from local UE's perspective)

Figure 4.10.2.4.3.3-1 illustrates the topology view and protocol stack level view for the media interception for the P2P IMS DC with transcoded media transfer to the Remote UE. In this case, the CC-POI functions are provided by the IMS-AGW and the trigger for the media interception is provided by the CC-TF present in the P-CSCF.

The topology view shows only the relevant IMS DC related NFs.

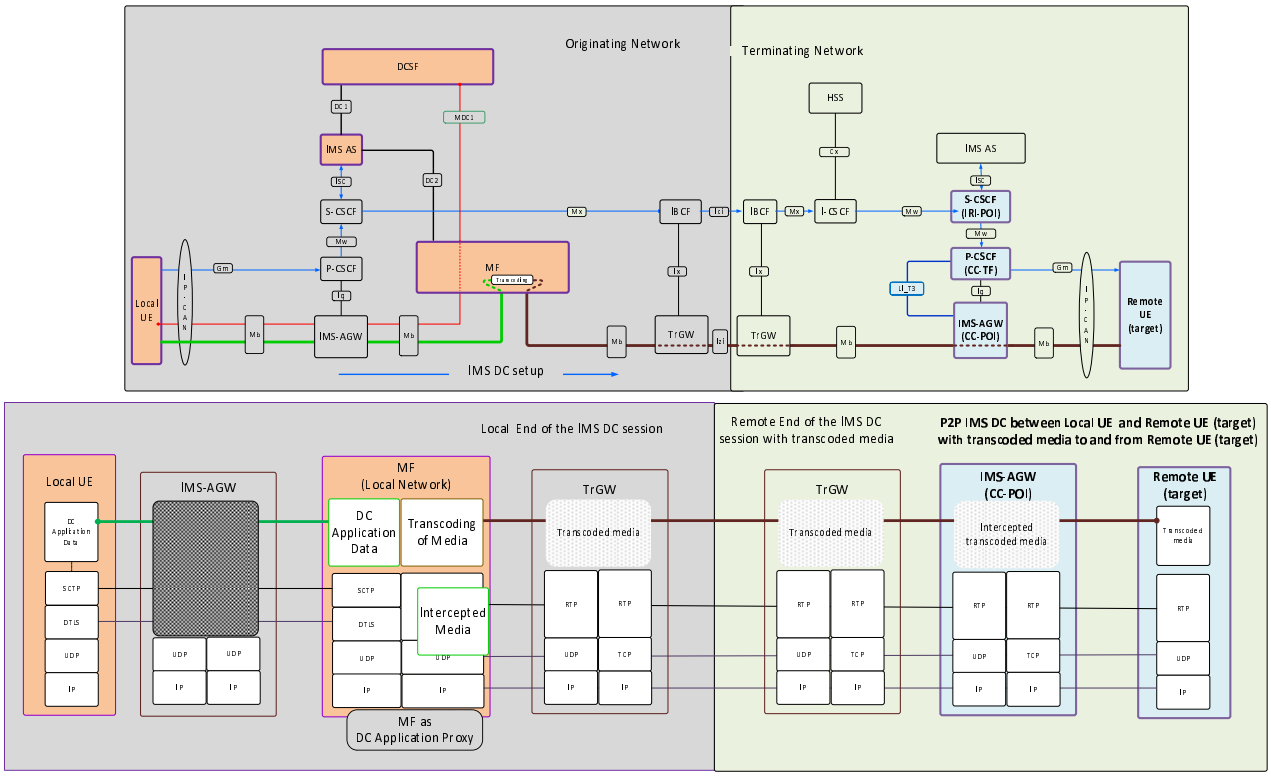


Figure 4.10.2.4.3.3-1: CC-POI for P2P IMS DC P2A2P IMS DC with transcoded media transfer to the remote when MF is DC Application Proxy

As shown in figure 4.10.2.4.3.3-1, the S-CSCF in the Remote Network provides the IRI-POI functions and IMS-AGW provides the CC-POI functions with CC-TF at the P-CSCF.

4.10.2.4.4 Remote network does not have IMS DC capabilities but has limited IMS DC support

The Remote Network that does not have IMS DC capabilities but has limited IMS DC support may or may not have the DCSF and MF. When it does have it, it may insert the MF into the IMS DC media path and may notify the DCSF about the IMS DC setup, or it may not. Three possibilities are shown below.

Figure 4.10.2.4.4-1 illustrates one possibility when the IMS network that has limited IMS DC support is presumed to have the MF and DCSF. When MF is inserted (as shown) into the IMS DC media path and DCSF is notified, the illustrations shown in clause 4.10.2.4.2 would apply when the Remote UE is IMS DC capable. In this case, when Remote UE is the target, the DCSF (present in the Remote Network) provides the IRI-POI functions and the MF (present in the Remote Network) provides the CC-POI functions with CC-TF in IMS-AS. MF in the Remote Network is always in a DC Application Proxy mode as illustrated in clause 4.10.2.4.2.

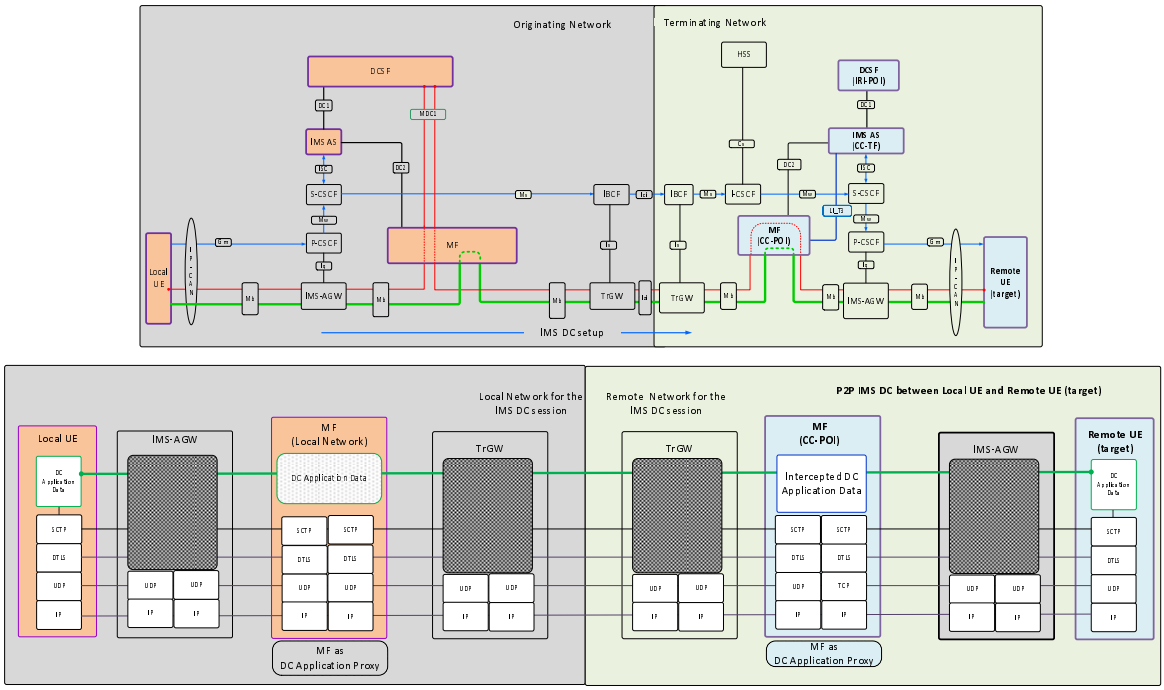


Figure 4.10.2.4.4-1: When the Remote Network has limited IMS DC support – view 1 (inserts MF, DCSF)

Figure 4.10.2.4.4-2 illustrates another possibility when the IMS network that has limited IMS DC support is presumed to not have deployed MF and DCSF. In this case, the Remote Network may disallow its locally registered IMS capable UEs from having BDC and may limit the media from P2A2P and P2P IMS DC media transferred to and from the Remote UE in a transcoded form as illustrated in clause 4.1.2.4.3. In this case, when Remote UE is the target, the S-CSCFSCSF (present in the Remote Network) provides the IRI-POI functions and the IMS-AGW (present in the Remote Network) provides the CC-POI functions with CC-TF in P-CSCF. This is similar to the illustrations shown in clause 4.10.2.4.3.

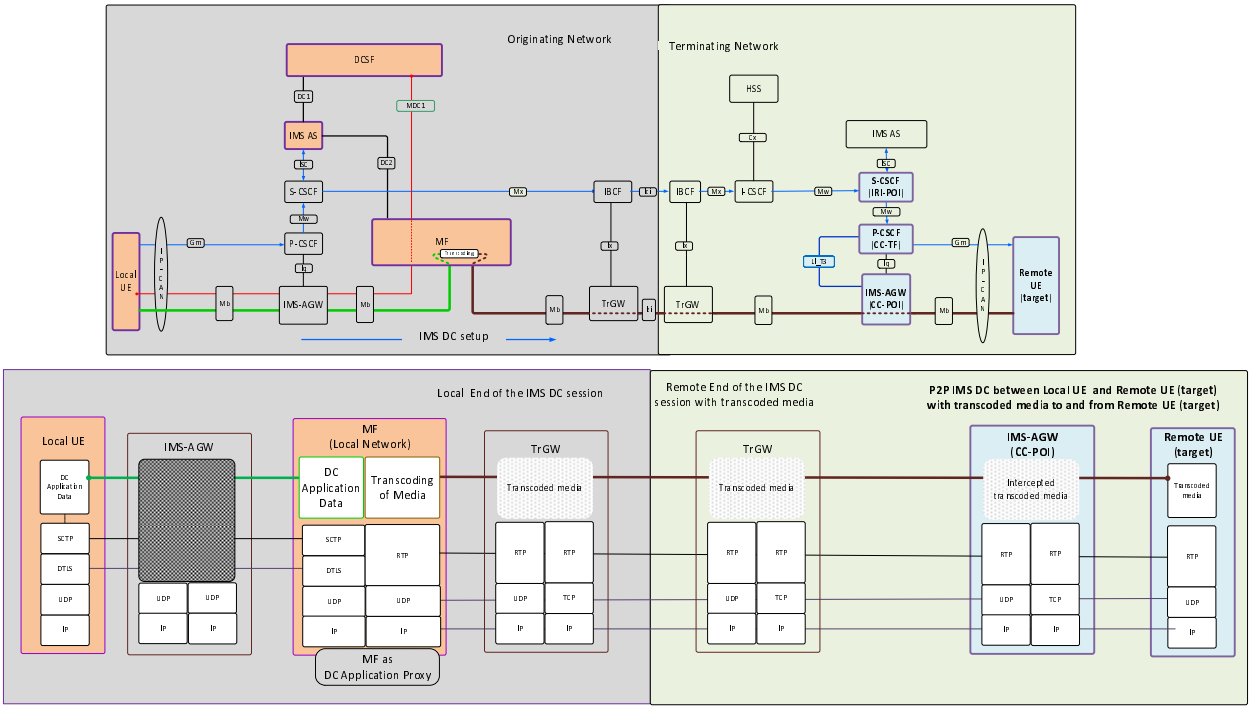


Figure 4.10.2.4.4-2: When the Remote Network has limited IMS DC support – view 2 (transcoded media to/from remote UE)

Figure 4.10.2.4.4-3 illustrates third possibility when the IMS network that has limited IMS DC support is presumed to not have deployed MF and DCSF. In this case, the Remote Network may allow its locally registered IMS capable UEs to have the IMS DCs (BDC and ADCs) set up the other UE present in the peer network.

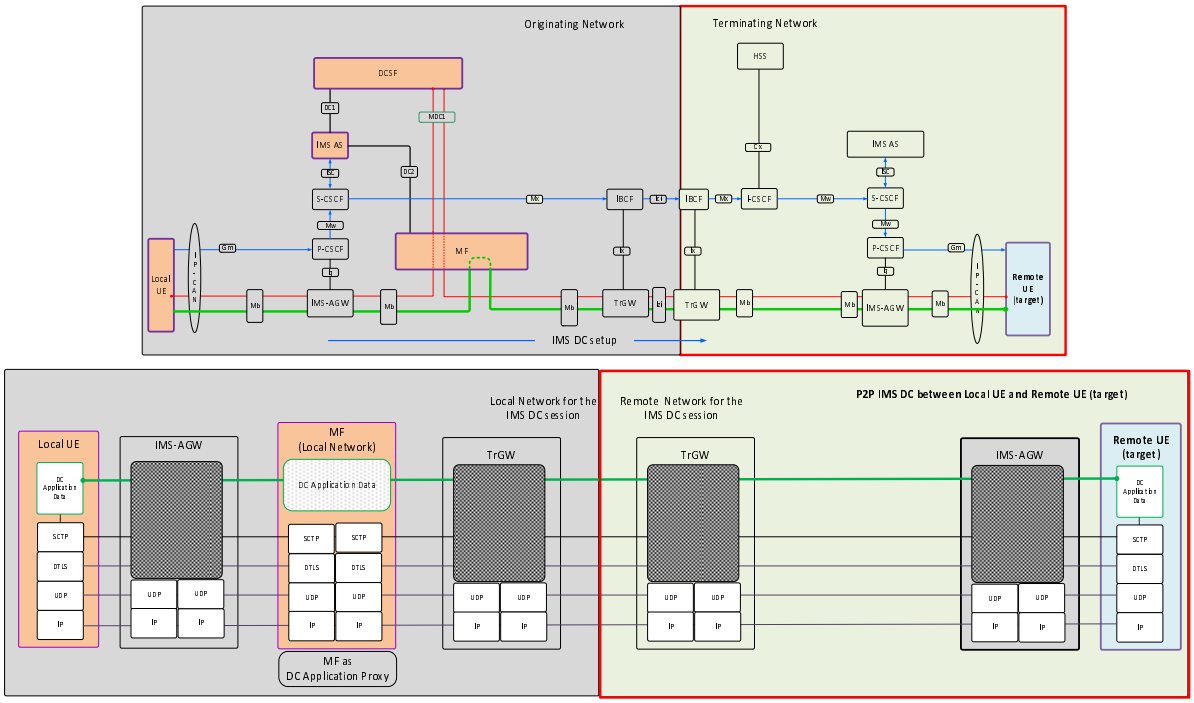


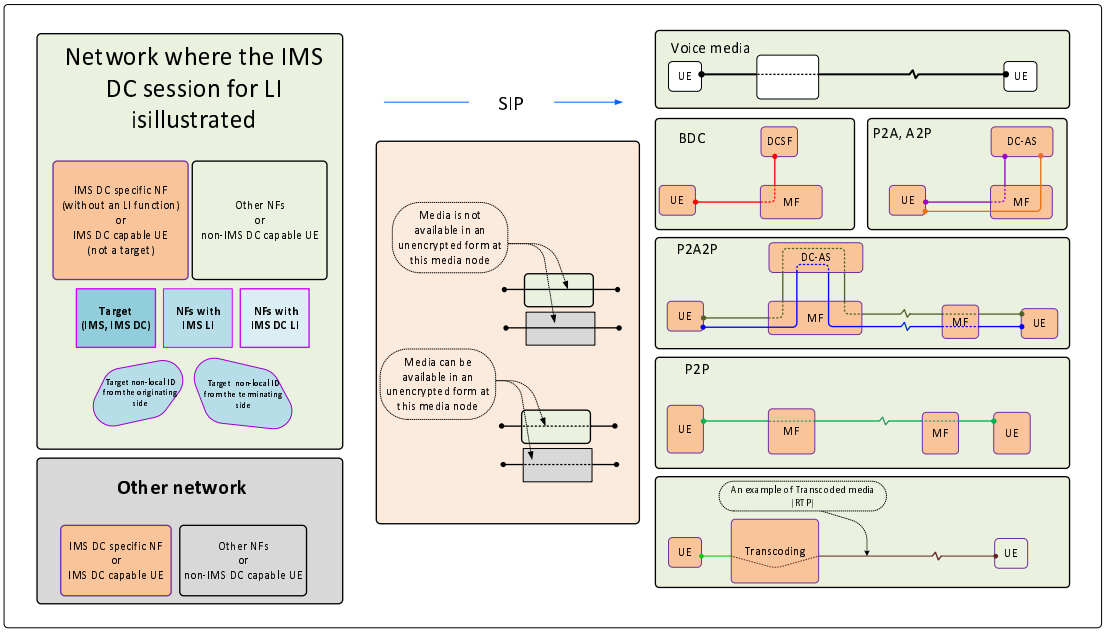
Figure 4.10.2.4.4-3: When the Remote Network has limited IMS DC support – view 3 (no insertion of MF, DCSF)

Figure 4.10.2.4.4-3 shows the LI implication if the Remote Network allows its locally registered UEs to have BDC and other ADCs setup by the other IMS DC capable UE present in the peer network. The cases where MF is not present, or not inserted, are further study.

4.10.3 LI Network topologies

4.10.3.1 Few conventions used

The following conventions are used (when not self-explanatory) within the topology view diagrams).



4.10.3.1-1: Few conventions used within the topology view diagrams

The illustrations include intra-network and inter-network scenarios. The illustrations show IMS-LI and IMS DC LI as the target UE is generally presumed to be the target for IMS-based services as well. Since the TS 33.128 [4] specifies that the concept of target non-local ID is not applicable to IMS DC, any related illustrations are presumed to be part of IMS LI.

4.10.3.2 Network topologies from local network's perspective

4.10.3.2.1 General

The subsequent clauses illustrate the network topology views for LI at the local end where the originating party is a target for IMS-based services and for IMS DC. In the illustrations, the originating party is the Local UE for IMS DC.

The illustrations also show when the terminating party is a target non-local ID in the originating network for IMS-based services when IMS DC session is present between the parties of IMS session. According to TS 33.128 [4], target non-local ID does not apply to IMS DC sessions.

4.10.3.2.2 Scenario 1: With IMS DCs to remote UE – DC-AS is in the CSP domain

Figure 4.10.3.2.2-1 and figure 4.10.3.2.2-2 illustrate topology views of IMS LI and IMS DC LI when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the terminating party (UE-B). The UE-A (Local UE) is the target of interception. The UE-B (Remote UE) is IMS DC capable, has subscribed to IMS DC and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-B. In these illustrations, DC-AS is in the CSP domain.

Figure 4.10.3.2.2-1 shows the scenario for an inter-network case. The illustration assumes that the terminating network has IMS DC capabilities and IMS DC capable UE-B has subscribed to IMS DC.

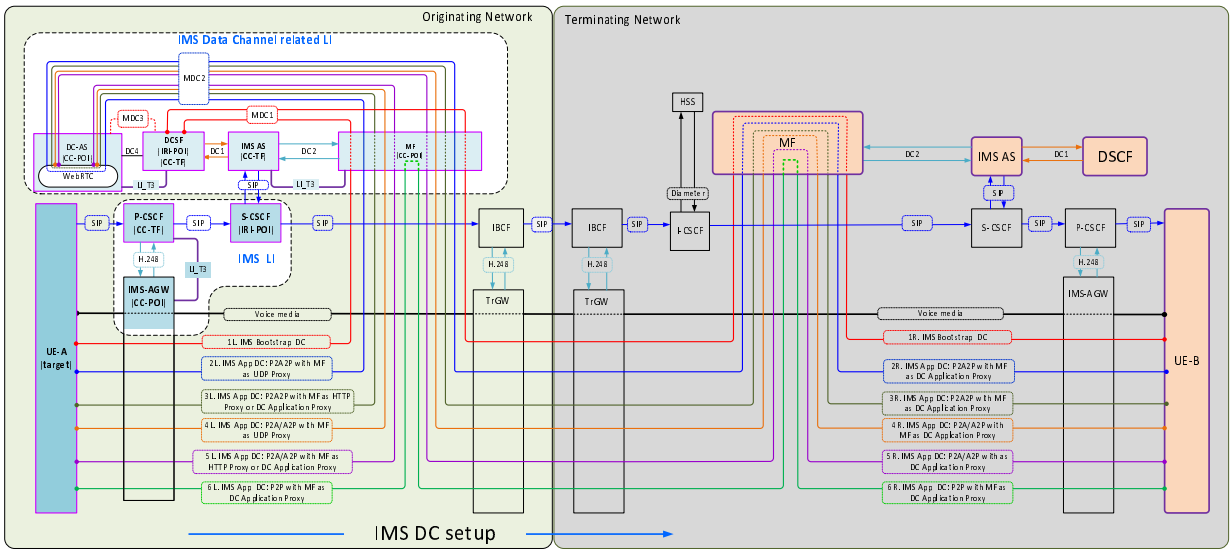


Figure 4.10.3.2.2-1: Topology view when originating party is the local UE and the target (inter-network case)

Figure 4.10.3.2.2-2 shows the scenario for an intra-network case. The illustration assumes that the IMS DC capable UE-B has subscribed to IMS DC.

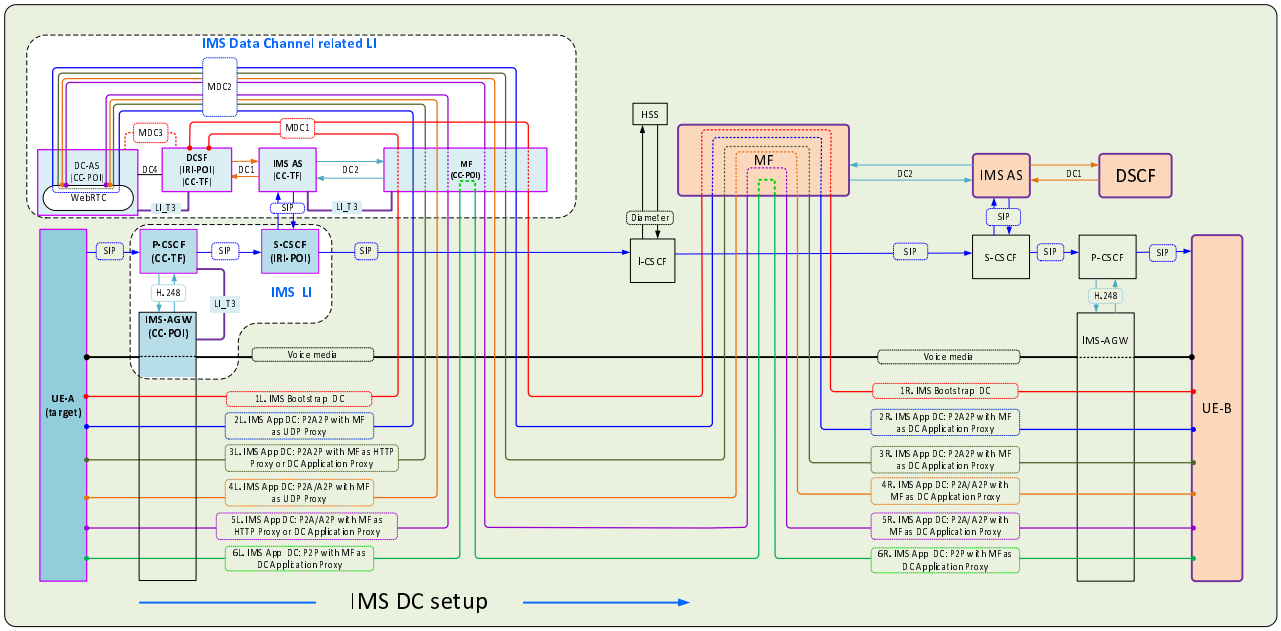


Figure 4.10.3.2.2-2: Topology view when originating party is the local UE and the target (intra-network case)

As shown in figure 4.10.3.2.2-1 and figure 4.10.3.2.2-2, the S-CSCF provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF. The IRI-POI function for IMS DC is provided by the DCSF. The CC-POI functions are provided by MF (with CC-TF at IMS AS) for all IMS DCs when the MF is operating in an HTTP proxy mode or DC Application Proxy mode. The CC-POI functions are provided by the DC-AS when the MF is operating in an UDP Proxy mode for P2A, A2P and P2A2P IMS DCs.

4.10.3.2.3 Scenario 2: With IMS DCs to remote UE – DC-AS is in the third party domain

Figure 4.10.3.2.3-1 and figure 4.10.3.2.3-2 illustrate topology views of IMS LI and IMS DC LI when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the terminating party (UE-B). The UE-A (Local UE) is the target of interception. The UE-B (Remote UE) is IMS DC capable, has subscribed to IMS DC and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-B. In these illustrations, DC-AS is in a third party domain.

Figure 4.10.3.2.3-1 shows the scenario for an inter-network case. The illustration assumes that the terminating network has IMS DC capabilities and IMS DC capable UE-B has subscribed to IMS DC.

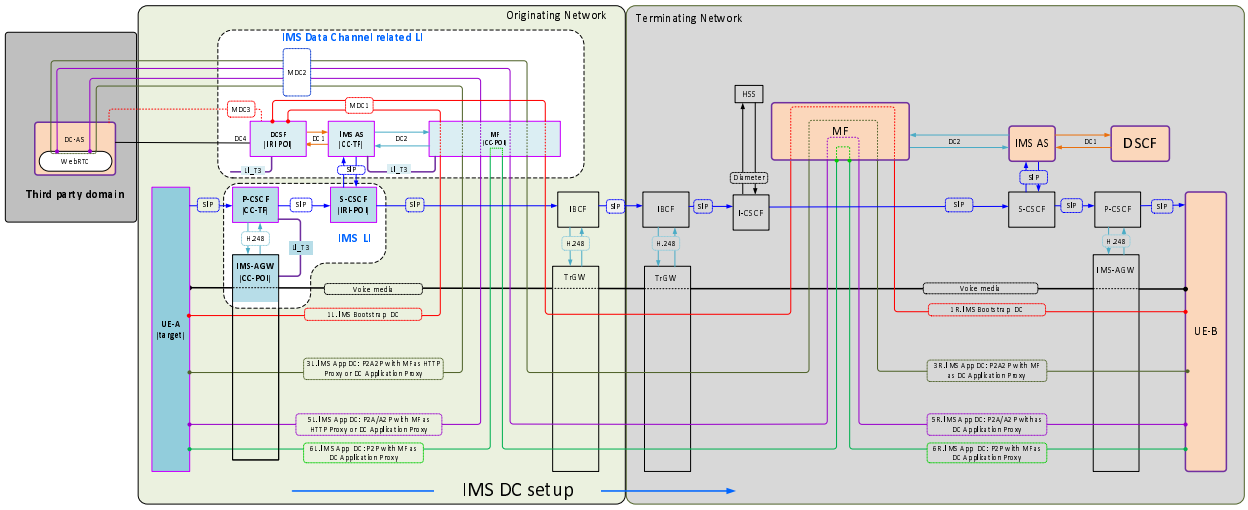


Figure 4.10.3.2.3-1: Topology view when originating party is the local UE and the target (inter-network case) – DC in third party domain

Figure 4.10.3.2.3-2 shows the scenario for an intra-network case. The illustration assumes that the IMS DC capable UE-B has subscribed to IMS DC.

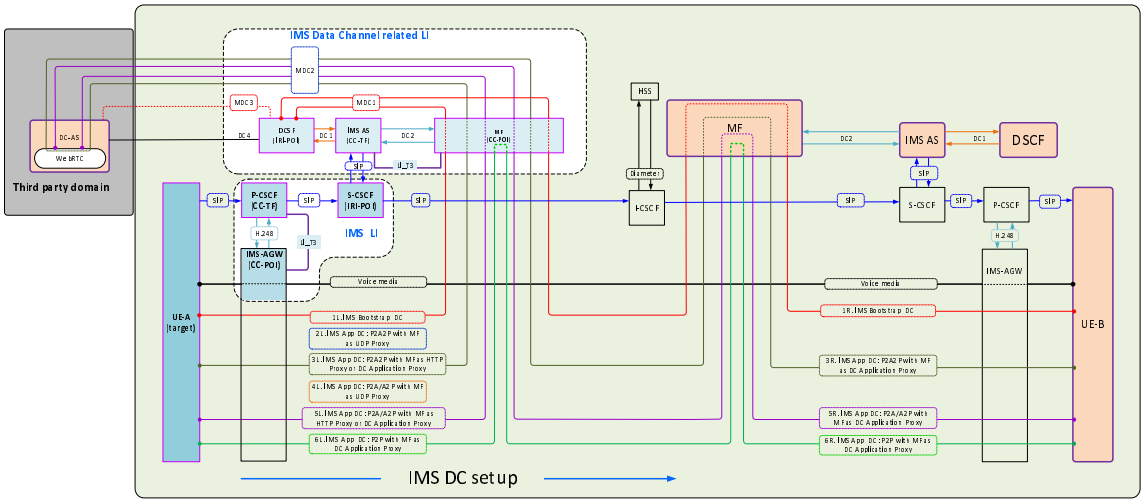


Figure 4.10.3.2.3-2: Topology view when originating party is the local UE and the target (intra-network case) – DC in third party domain

As shown in figure 4.10.3.2.3-1 and figure 4.10.3.2.3-2, the S-CSCF provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF. The IRI-POI function for IMS DC is provided by the DCSF. The CC-POI functions are provided by MF (with CC-TF at IMS AS) for all IMS DCs when the MF is operating in an HTTP proxy mode or DC Application Proxy mode. Since DC-AS is in the third party domain, the MF should not be operating in UDP proxy mode.

4.10.3.2.4 Scenario 3: Transcoded IMS DC media to remote UE - DC-AS in the CSP domain

Figure 4.10.3.2.4-1 and figure 4.10.3.2.4-2 illustrate topology views of IMS LI and IMS DC LI when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the terminating party (UE-B). The UE-A (Local UE) is target of interception. With no IMS DC capabilities at the terminating end, P2A2P and P2P IMS DC media is transferred to and from UE-B in a transcoded form. In these illustrations, the DC-AS is in the CSP domain.

Figure 4.10.3.2.4-1 shows the scenario for an inter-network case. The illustration assumes that either terminating network has no IMS DC capabilities or UE-B in the terminating network is not IMS DC capable. In the former case, basically, the terminating network does not support its locally registered IMS DC capable UEs to have Bootstrap and other application IMS DCs setup by an IMS DC capable UE present in a peer network.

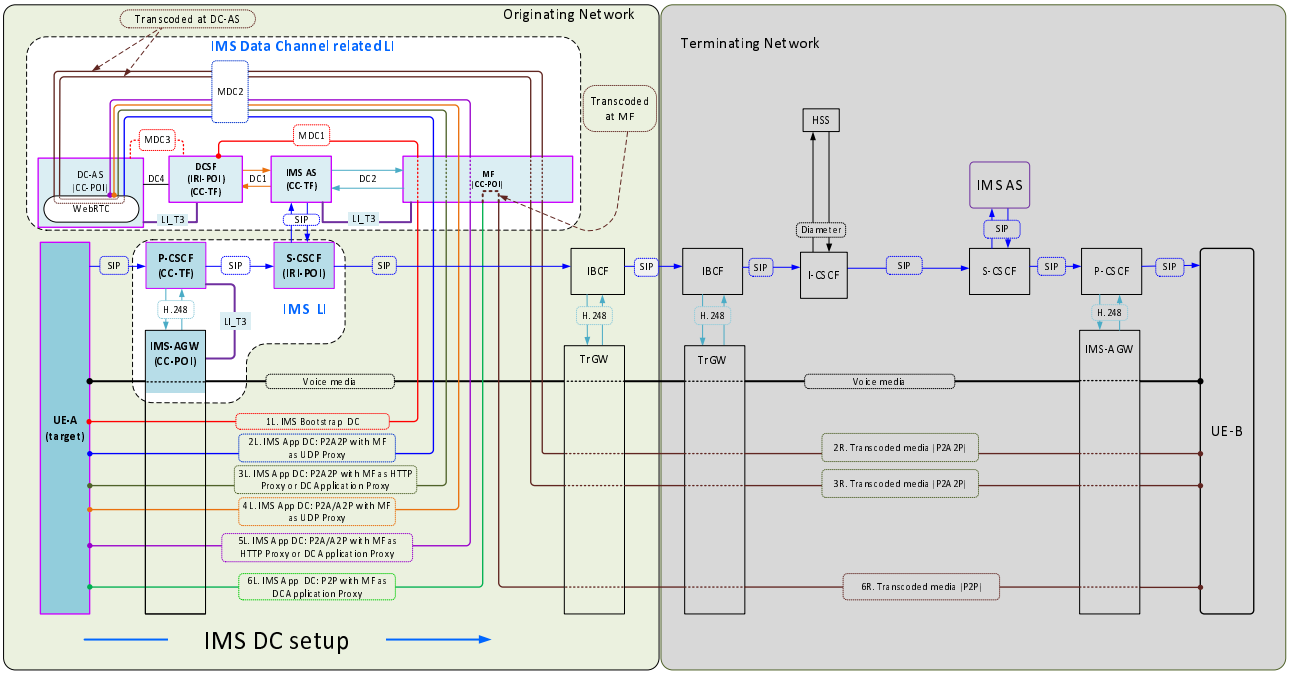


Figure 4.10.3.2.4-1: Topology view when originating party is the local UE and the target (inter-network case) – transcoded media to terminating party

Figure 4.10.3.2.4-2 shows the scenario for an intra-network case. The illustration assumes that the UE-B is not IMS DC capable.

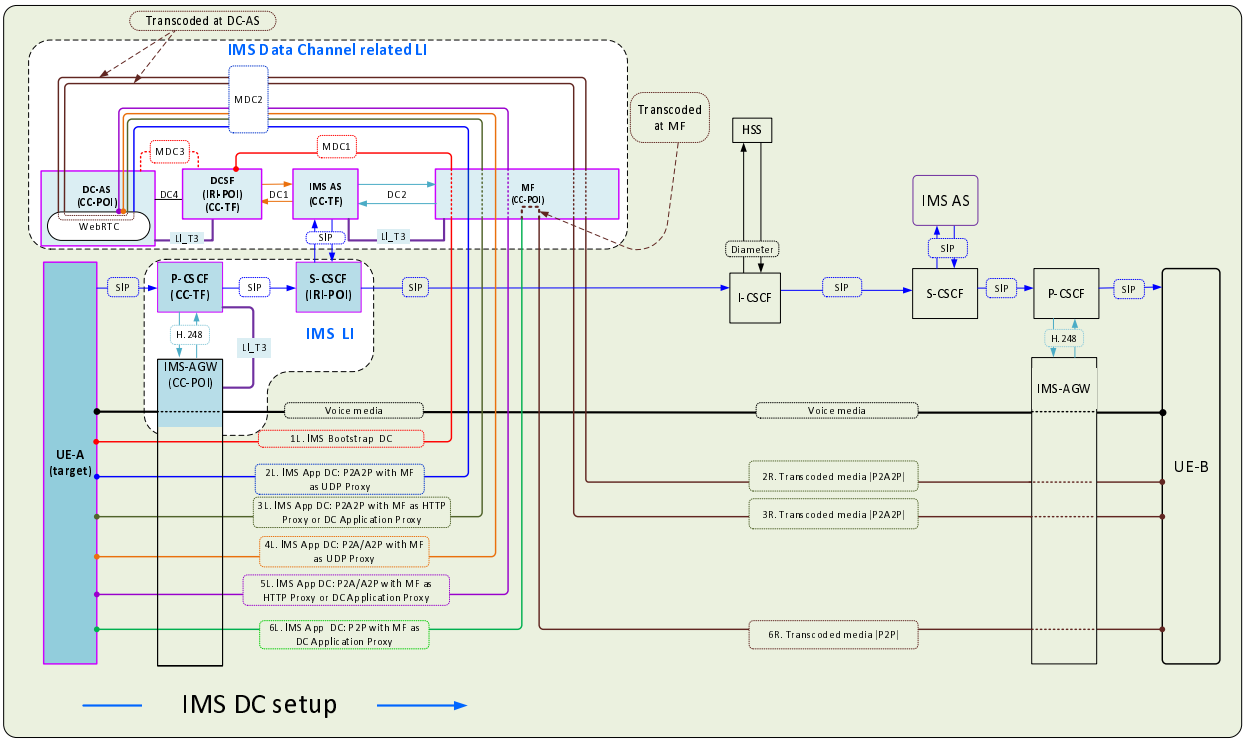


Figure 4.10.3.2.4-2: Topology view when originating party is the local UE and the target (intra-network case) – transcoded media to terminating party

As shown in figure 4.10.3.2.4-1 and 4.10.3.2-4-2, the S-CSCF provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF. The IRI-POI functions for IMS DC are provided by the DCSF. The CC-POI functions are provided by MF (with CC-TF at IMS AS) for all the IMS DCs when the MF is operating in an HTTP proxy mode or DC Application Proxy mode. The CC-POI functions are provided by the DC-AS when the MF is operating in an UDP Proxy mode for P2A, A2P and P2A2P IMS DCs.

For P2A2P IMS DC, the DC-AS provides the transcoding functions and for P2P IMS DC, the MF provides the transcoding functions. The illustrations also show that the media transfer in a transcoded form to Remote UE does not have any influence on LI when the Local UE is the target.

4.10.3.2.5 Scenario 4: Transcoded IMS DC media to the remote UE - DC-AS in the third party domain

Figure 4.10.3.2.5-1 and figure 4.10.3.2.5-2 illustrate the topology views of IMS LI and IMS DC LI when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the terminating party (UE-B). The UE-A (Local UE) is target of interception. With no IMS DC capabilities at the terminating end, P2A2P and P2P IMS DC media is transferred to and from UE-B in a transcoded form. In these illustrations, the DC-AS is in the CSP domain.

Figure 4.10.3.2.5-1 shows the scenario for an inter-network case. The illustration assumes that either terminating network has no IMS DC capabilities or UE-B in the terminating network is not IMS DC capable. In the former case, basically, the terminating network does not support its locally registered IMS DC capable UEs to have Bootstrap and other application IMS DCs setup by an IMS DC capable UE present in a peer network.

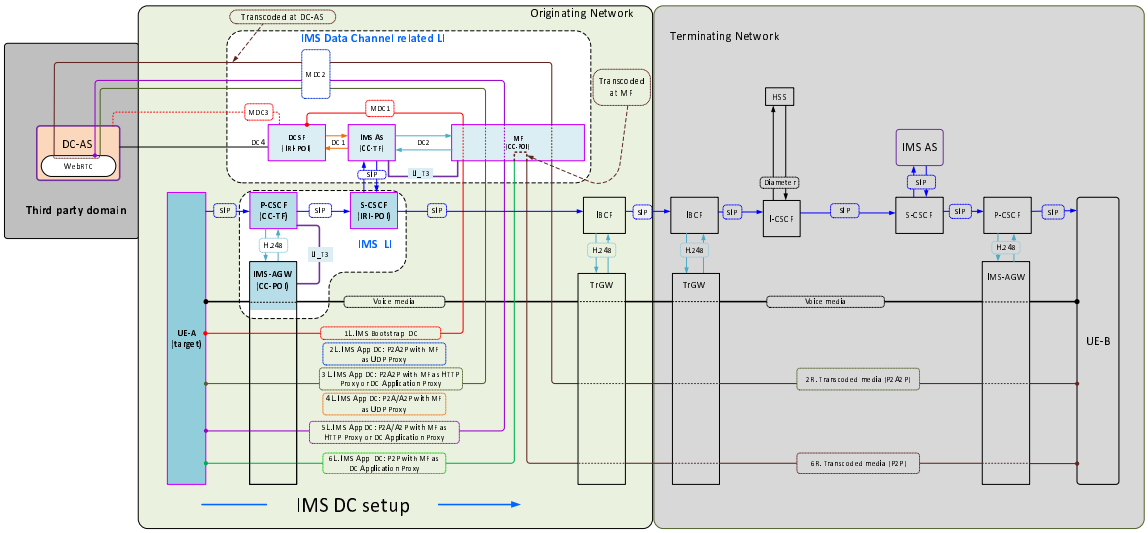


Figure 4.10.3.2.5-1: Topology view when originating party is the local UE and the target (inter-network case) – DC in third party domain

Figure 4.10.3.2.5-2 shows the scenario for an intra-network case. The illustration assumes that the UE-B is not IMS DC capable.

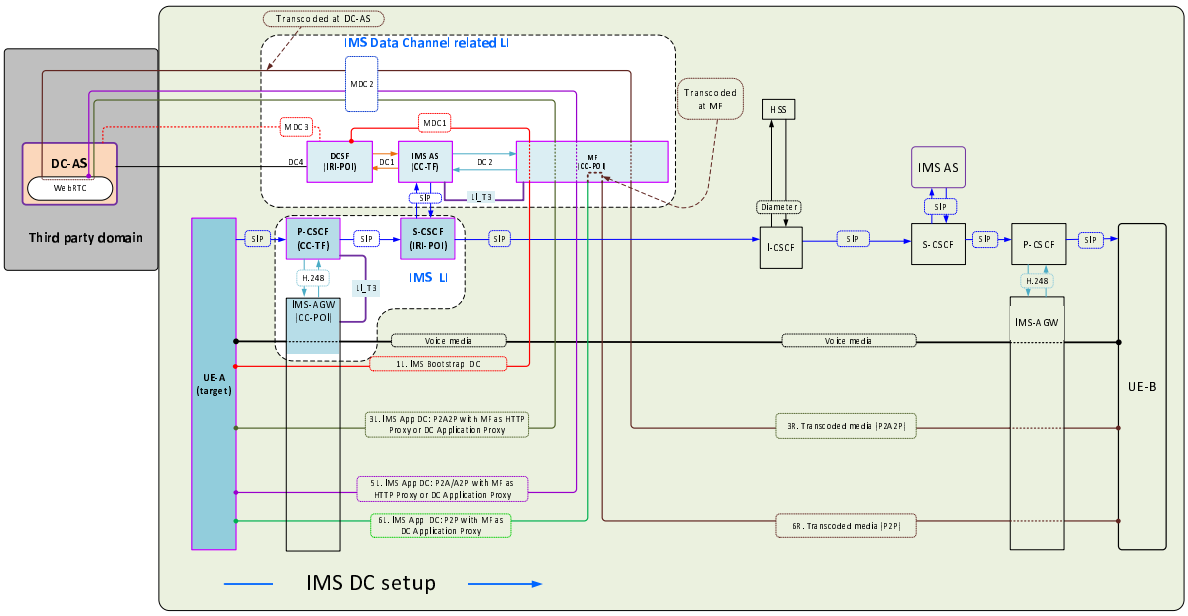


Figure 4.10.3.2.5-2: Topology view when originating party is the local UE and the target (intra-network case) – DC in third party domain

As shown in figure 4.10.3.2.5-1 and figure 4.10.3.2.5-2, the S-CSCF provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF. The IRI-POI functions for IMS DC are provided by the DCSF. The CC-POI functions are provided by MF (with CC-TF at IMS AS) for all IMS DCs when the MF is operating in an HTTP proxy mode or DC Application Proxy mode. Since DC-AS is in the third party domain, the MF should not be operating in UDP proxy mode.

For P2A2P IMS DC, the DC-AS provides the transcoding functions and for P2P IMS DC, the MF provides the transcoding functions. The illustrations also show that the media transfer in a transcoded form to Remote UE does not have any influence on LI when the Local UE is the target.

4.10.3.2.6 Scenario 5: Terminating party (Remote UE) is a target non-local ID in the originating network with IMS DCs to remote UE.

Figure 4.10.3.2.6-1 illustrates the topology view of IMS LI when the terminating party (UE-B) is a target non-local ID in the originating network for IMS-based services while the originating party (UE-A) of an IMS session has initiated the IMS DC setup with the terminating party (UE-B). The UE-B (Remote UE) is IMS DC capable, has subscribed to IMS DC and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-B.

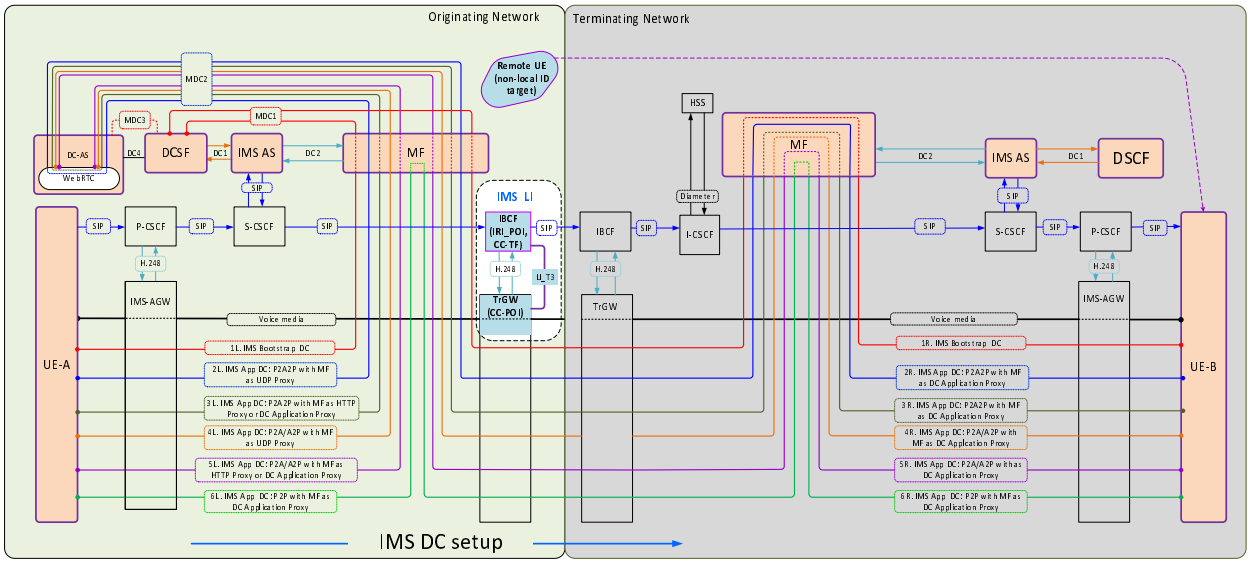


Figure 4.10.3.2.6-1: Topology view when originating party is the local UE and remote UE is a target non-local ID in the originating network

As shown in figure 4.10.3.2.6-1, the IBCF present in the originating network provides the IRI-POI functions for IMS-LI LI with respect to a target non-local ID (default option) with TrGW (present in the originating network) providing the CC-POI functions where the CC-TF is at the IBCF. As per TS 33.128 [4], the concept of target non-local ID does not apply to IMS DC and therefore, as shown the IMS DC media is not intercepted.

4.10.3.2.7 Scenario 6: Terminating party (Remote UE) is a target non-local ID in the originating network with transcoded IMS DC media to the remote UE

Figure 4.10.3.2.7-1 illustrates the topology view of IMS LI when the terminating party (UE-B) is a target non-local ID in the originating network for IMS-based services while the originating party (UE-A) of an IMS session has initiated the IMS DC setup with the terminating party (UE-B). With no IMS DC capabilities at the terminating end, P2A2P and P2P IMS DC media is transferred to and from UE-B in a transcoded form. The illustration assumes that either terminating network has no IMS DC capabilities or UE-B in the terminating network is not IMS DC capable. In the former case, basically, the terminating network does not support its locally registered IMS DC capable UEs to have Bootstrap and other application IMS DCs setup by an IMS DC capable UE present in a peer network.

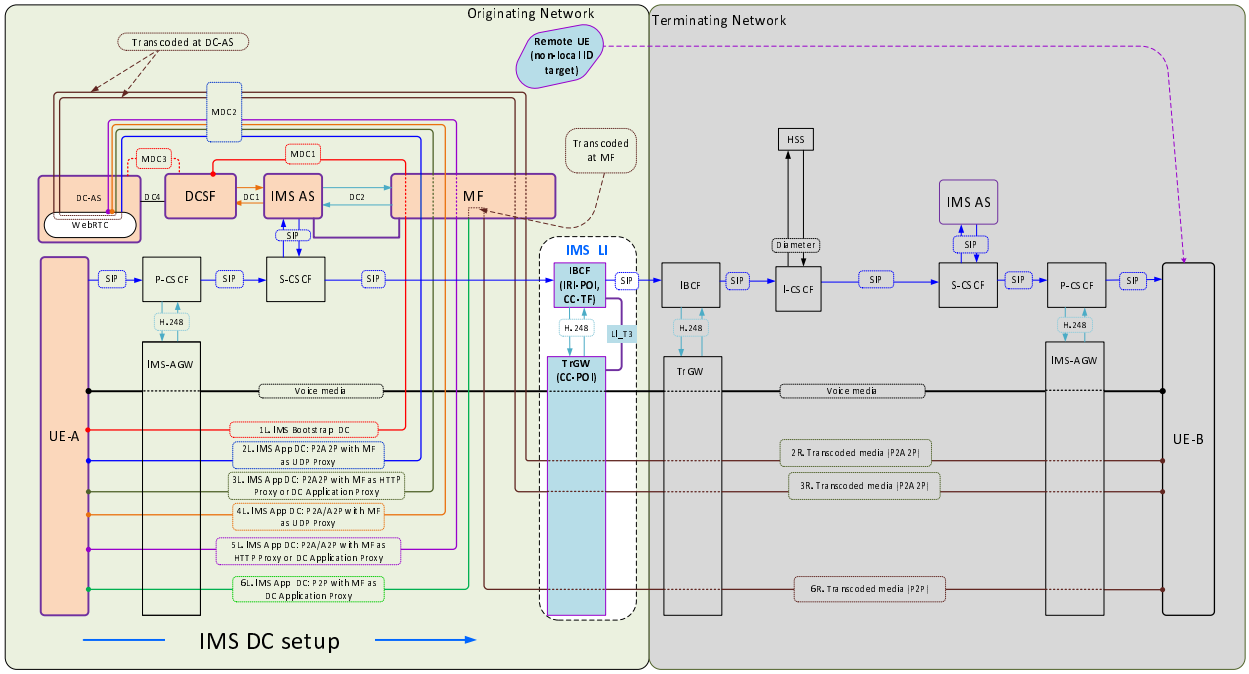


Figure 4.10.3.2.7-1: Topology view when originating party is the local UE and remote UE is a target non-local ID in the originating network – transcoded media to the remote UE

As shown in figure 4.10.3.2.7-1, the IBCF present in the originating network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with TrGW (present in the originating network) providing the CC-POI functions where the CC-TF is at the IBCF. The transcoded IMS DC media sent or received from the Remote UE (target non-local ID) is also intercepted at the CC-POI present in the TrGW.

4.10.3.3 Network topologies from remote network's perspective with IMS DC capabilities at the remote network

4.10.3.3.1 General

The subsequent clauses illustrate the network topology views where the terminating party or the redirected-to party is a target for IMS-based services and for IMS DC. In the illustrations, the originating party is the Local UE for IMS DC. In the inter-network scenario, the terminating network has IMS DC capabilities and the terminating party is IMS DC capable.

The illustrations also show the following target non-local ID cases for IMS-based services when IMS DC session is present between the parties of IMS session:

- Originating party (i.e. Local UE) is a target non-local ID in the terminating network or redirected-to network (i.e. Remote Network).
- Redirecting party is a target non-local ID in the redirected-to network (i.e. Remote Network).

According to TS 33.128 [4], target non-local ID does not apply to IMS DC sessions.

4.10.3.3.2 Scenario 1: Terminating party as a remote UE is the target

Figure 4.10.3.3.2-1 and figure 4.10.3.3.2-2 illustrate topology views of IMS LI and IMS DC LI when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the terminating party (UE-B). The UE-B (Remote UE) is the target of interception. The UE-B (Remote UE) is IMS DC capable, has subscribed to IMS DC and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-B.

Figure 4.10.3.3.2-1 shows the scenario for an inter-network case.

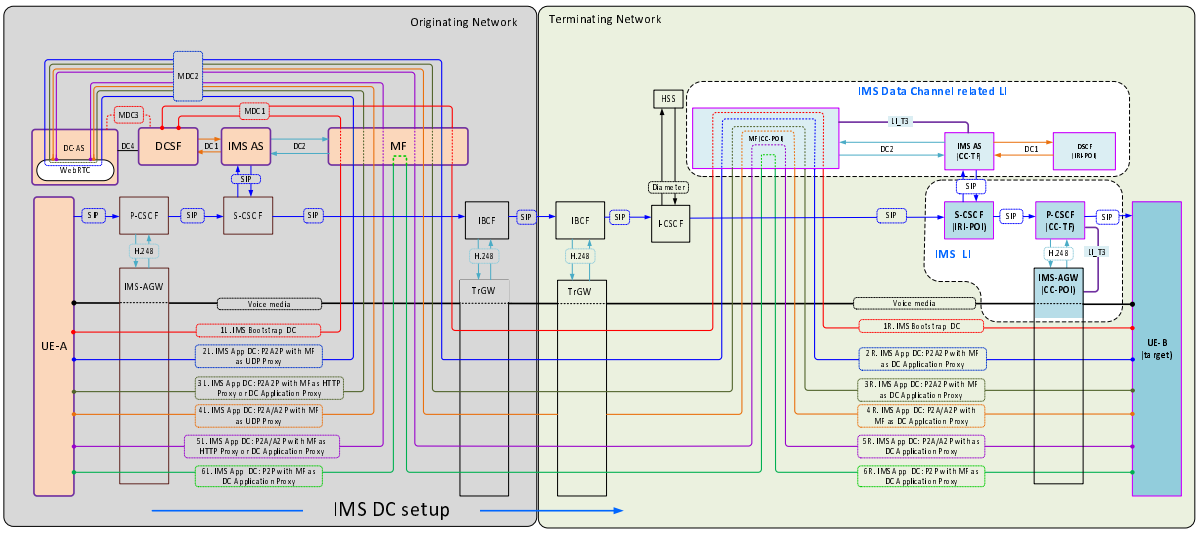


Figure 4.10.3.3.2-1: Topology view when the terminating party is the remote UE and the target (inter-network case)

Figure 4.10.3.3.2-2 shows the scenario for an intra-network case.

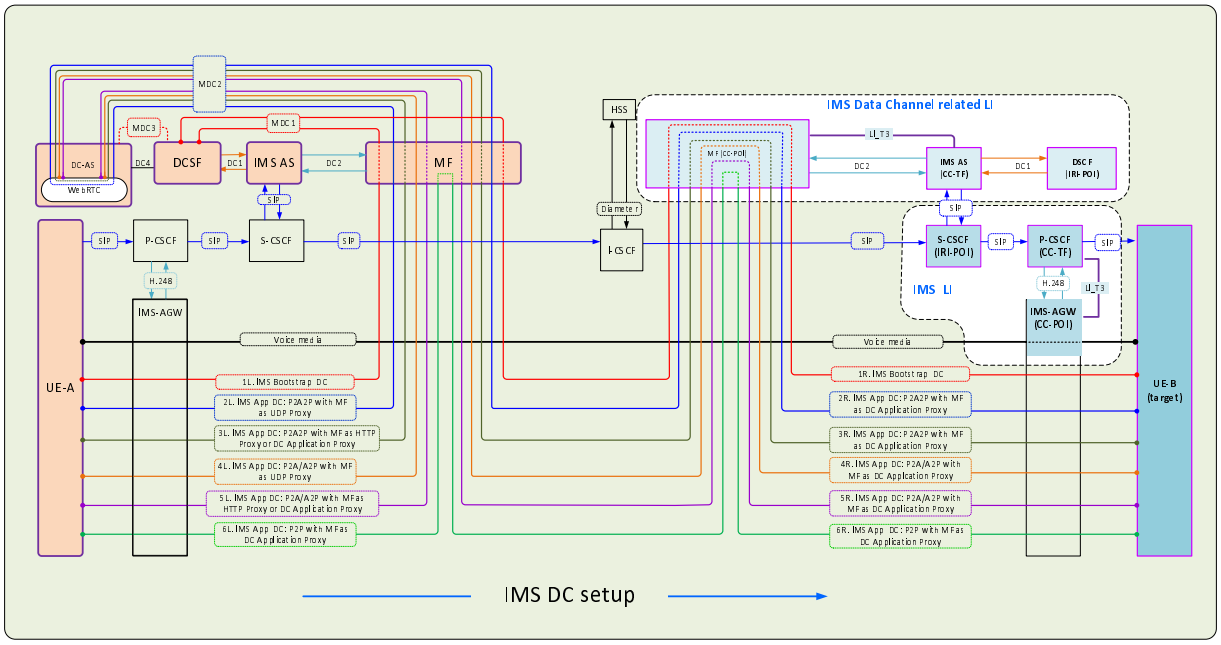


Figure 4.10.3.3.2-2: Topology view when the terminating party is the remote UE and the target (intra-network case)

As shown in figure 4.10.3.3.2-1 and figure 4.10.3.3.2-2, the S-CSCF provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF. The IRI-POI function for IMS DC is provided by the DCSF. The CC-POI functions are provided by MF (with CC-TF at IMS AS) for all IMS DCs.

4.10.3.3.3 Scenario 2: Redirected-to party as the remote UE

Figure 4.10.3.3.3-1 and figure 4.10.3.3.3-2 illustrate topology views of IMS LI and IMS DC LI when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the redirected-to party (UE-C. The UE-C (Remote UE) is the target of interception. The UE-C (Remote UE) is IMS DC capable, has subscribed to IMS DC and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-C. In this illustration, the redirecting network has the IMS DC capabilities.

Figure 4.10.3.3.3-1 shows the scenario for an inter-network case.

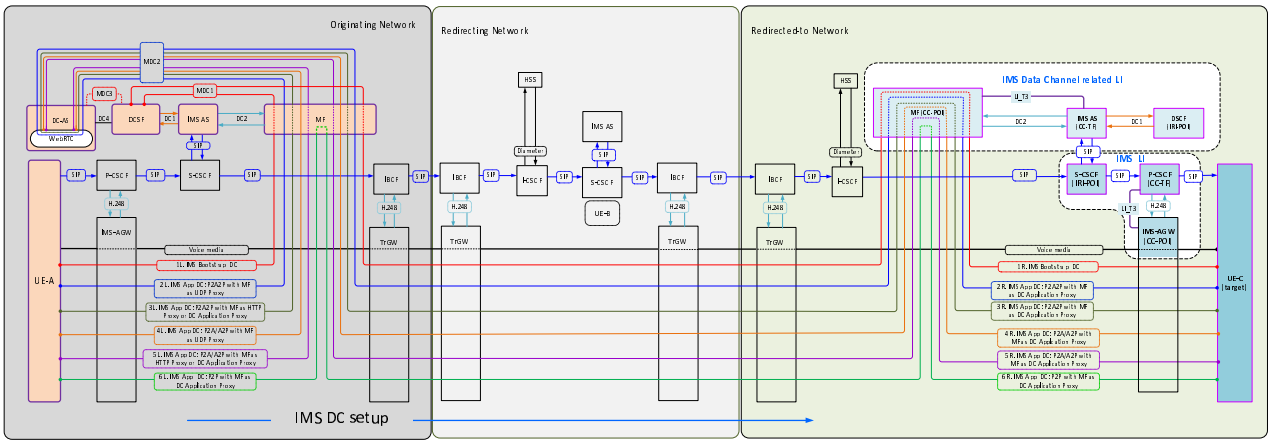


Figure 4.10.3.3.3-1: Topology view when redirected-to party is the remote UE and the target (inter-network case)

Figure 4.10.3.3.3-2 shows the scenario for an intra-network case.

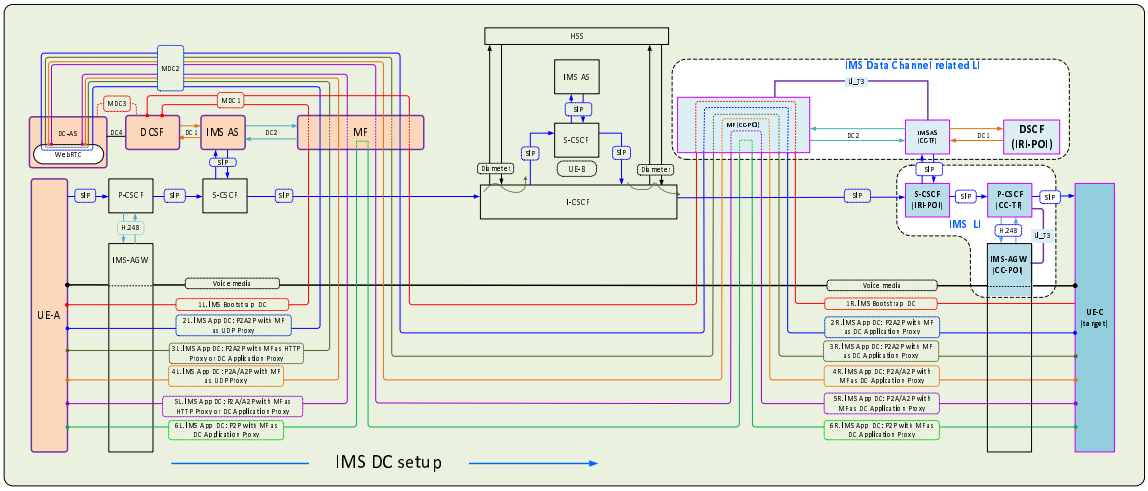


Figure 4.10.3.3.3-2: Topology view when redirected-to party is the remote UE and the target (intra-network case)

As shown in figure 4.10.3.3.3-1 and figure 4.10.3.3.3-2, the S-CSCF provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF. The IRI-POI function for IMS DC is provided by the DCSF. The CC-POI functions are provided by MF (with CC-TF at IMS AS) for all IMS DCs.

4.10.3.3.4 Scenario 3: Originating party is a target non-local ID in the terminating network

Figure 4.10.3.3.4-1 illustrates the topology view of IMS LI when originating party (UE-A) is a target non-local ID in the terminating network for IMS-based services while the originating party (UE-A, target non-local ID) of the IMS session has initiated the IMS DC setup with the terminating party (UE-B). The UE-B (Remote UE) is IMS DC capable, has subscribed to IMS DC and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-B.

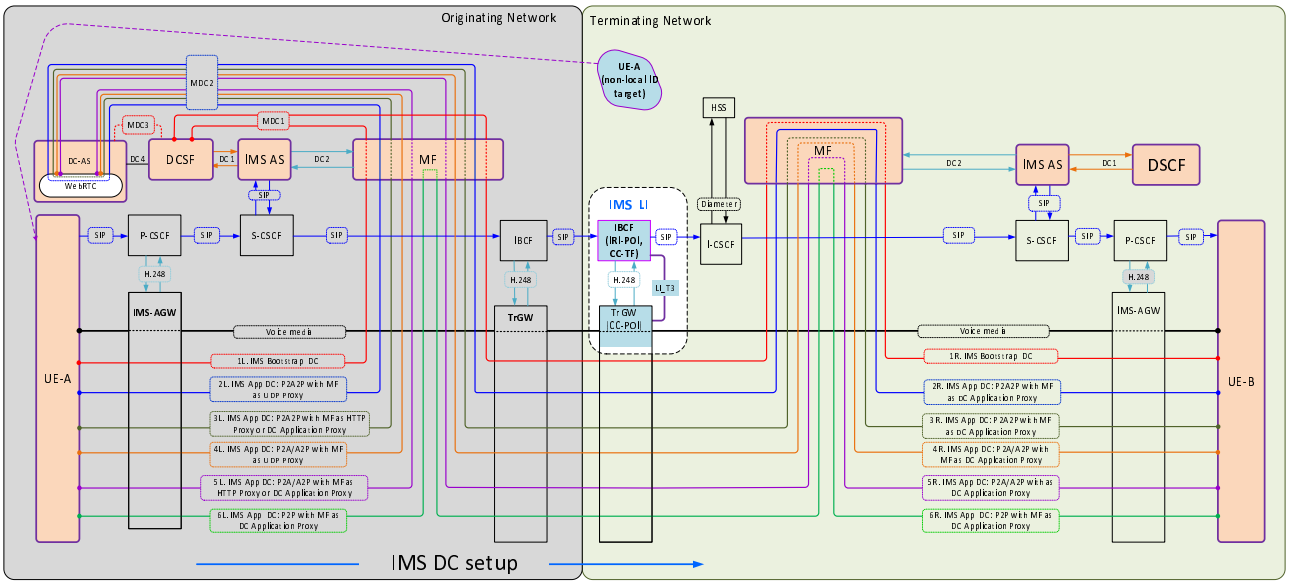


Figure 4.10.3.3.4-1: Topology view when the terminating party is the remote UE and the originating party is a target non-local ID

As shown in figure 4.10.3.3.4-1 the IBCF present in the terminating network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with TrGW (present in the terminating network) providing the CC-POI functions where the CC-TF is at the IBCF. As per TS 33.128 [4], the concept of target non-local ID does not apply to IMS DC and therefore, as shown the IMS DC media is not intercepted.

4.10.3.3.5 Scenario 4: Originating party is a target non-local ID in the redirected-to network

Figure 4.10.3.3.5-1 illustrates the topology view of IMS LI when originating party (UE-A) is a target non-local ID in the redirected-to network for IMS-based services while the originating party (UE-A, target non-local ID) of the redirected IMS session has initiated the IMS DC setup with the redirected-to party (UE-C). The Redirecting Network supports the IMS DC capabilities and the UE-C (Remote UE) is IMS DC capable, has subscribed to IMS DC in the Redirected-to Network (which obviously, has IMS DC capabilities) and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-C.

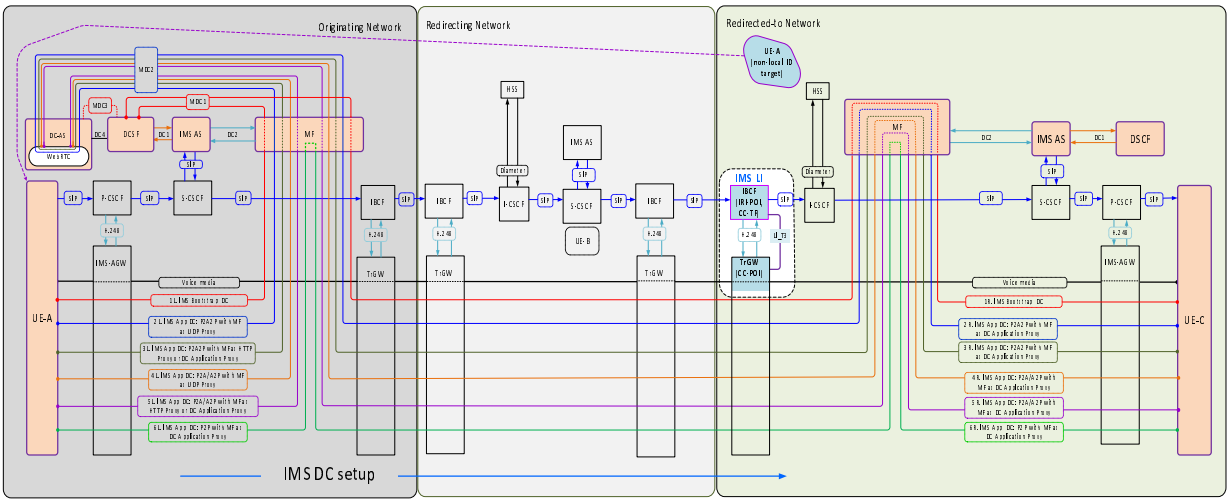


Figure 4.10.3.3.5-1: Topology view when the redirected-to party is the remote UE and the originating party is a target non-local ID in the redirected-to network

As shown in figure 4.10.3.3.5-1, the IBCF present in the redirected-to network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with TrGW (present in the redirected-to network) providing the CC-POI functions where the CC-TF is at the IBCF. As per TS 33.128 [4], the concept of target non-local ID does not apply to IMS DC and therefore, as shown the IMS DC media is not intercepted.

4.10.3.3.6 Scenario 5: Redirecting party is a target non-local ID in the redirected-to network

Figure 4.10.3.3.6-1 illustrates the topology view of IMS LI when redirecting party (UE-B) is a target non-local ID in the redirected-to network for IMS-based services while the originating party (UE-A) of the redirected IMS session has initiated the IMS DC setup with the redirected-to party (UE-C). The Redirecting Network supports the IMS DC capabilities and the UE-C (Remote UE) is IMS DC capable, has subscribed to IMS DC in the Redirected-to Network (which obviously, has IMS DC capabilities) and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-C.

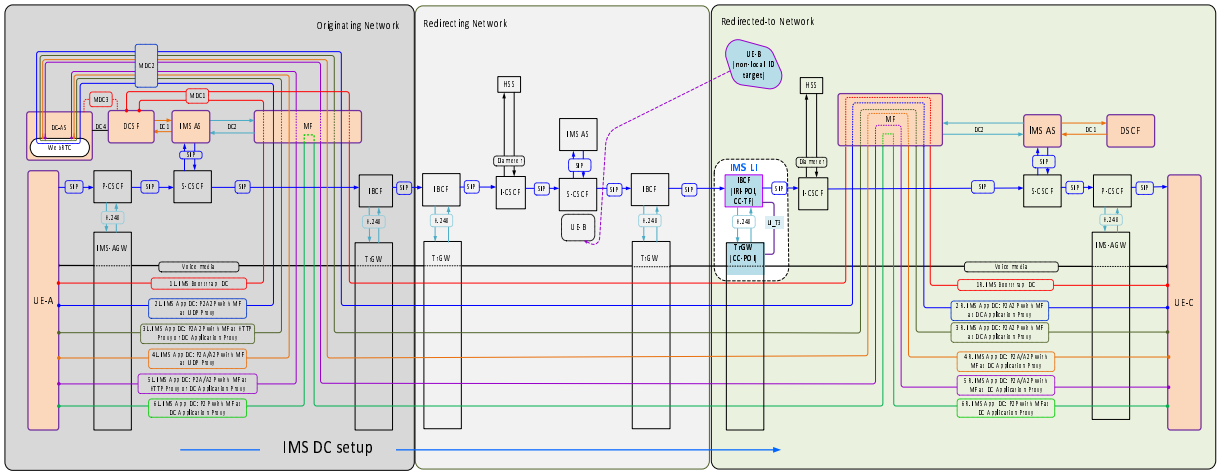


Figure 4.10.3.3.6-1: Topology view when the redirected-to party is the remote UE and the redirecting party is a target non-local ID in the redirected-to network

As shown in figure 4.10.3.3.6-1 the IBCF present in the redirected-to network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with TrGW (present in the redirected-to network) providing the CC-POI functions where the CC-TF is at the IBCF. As per TS 33.128 [4], the concept of target non-local ID does not apply to IMS DC and therefore, as shown the IMS DC media is not intercepted.

4.10.3.4 Network topologies from remote network's perspective (either remote network or remote UE does not have IMS DC capabilities)

4.10.3.4.1 General

The subsequent clauses illustrate the network topology views where the terminating party or the redirected-to party is a target for IMS-based services. In the illustrations, the originating party is the Local UE for IMS DC. In the inter-network cases, either one or both of the terminating network and the terminating party do not have IMS DC capabilities. In a redirected case, either one, or two, or all of the redirecting network, the redirected-to network and the redirected-to party do not have s IMS DC capabilities.

The illustrations also show the following target non-local ID cases for IMS-based services when IMS DC session is present between the parties of IMS session:

- Originating party (i.e. Local UE) is a target non-local ID in the terminating network or redirected-to network (i.e. Remote Network).

- Redirecting party is a target non-local ID in the redirected-to network (i.e. Remote Network).

According to TS 33.128 [4], target non-local ID does not apply to IMS DC sessions.

4.10.3.4.2 Scenario 1: Terminating party as a remote UE is the target with transcoded media

Figure 4.10.3.4.2-1 and figure 4.10.3.4.2-2 illustrate topology views of IMS LI that includes the transcoded IMS DC media when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the terminating party (UE-B). The UE-B (Remote UE) is the target of interception. IMS DC related media is sent to or received from the Remote UE (UE-B) in a transcoded form.

Figure 4.10.3.4.2-1 shows the scenario for an inter-network case. In this case, either the UE-B (Remote UE) or the terminating network does not have IMS DC capabilities.

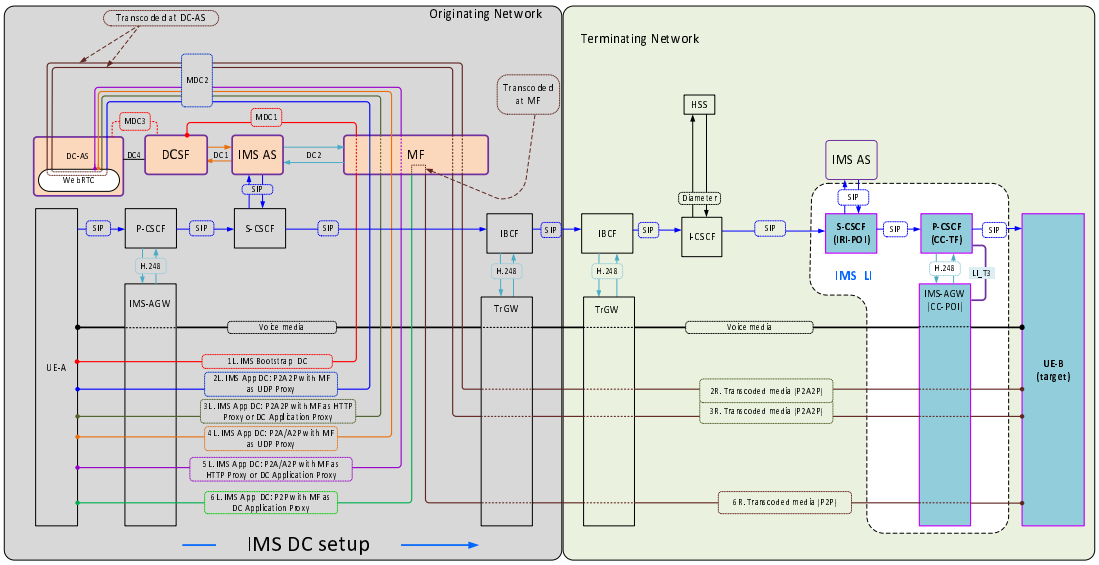


Figure 4.10.3.4.2-1: Topology view when the terminating party is the remote UE and the target (inter-network case)

Figure 4.10.3.4.2-2 shows the scenario for an intra-network case. In this case, UE-B (Remote UE) does not have IMS DC capabilities.

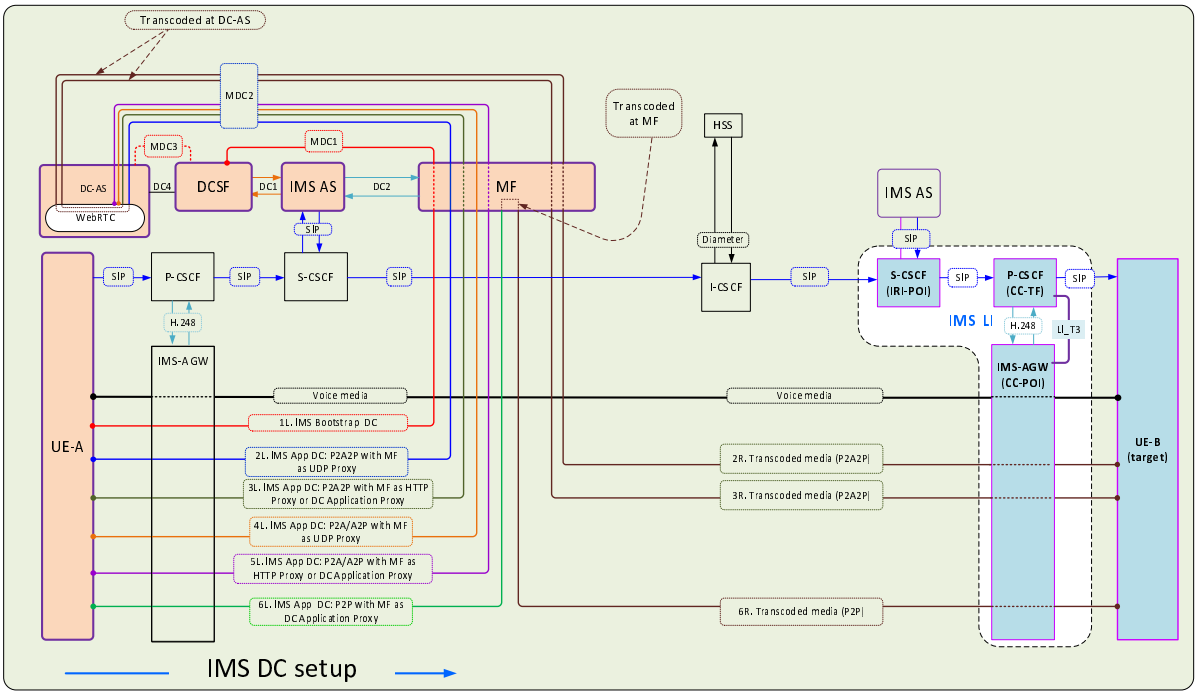


Figure 4.10.3.4.2-2: Topology view when the terminating party is the remote UE and the target (intra-network case)

As shown in figure 4.10.3.4.2-1 and figure 4.10.3.4.2-2, the S-CSCF (present in the terminating network) provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF. LI for transcoded IMS DC media is part of the IMS LI.

4.10.3.4.3 Scenario 2: Redirected-to party as the remote UE with transcoded media

Figure 4.10.3.4.3-1 and figure 4.10.3.4.3-2 illustrate topology views of IMS LI and IMS DC LI when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the redirected-to party (UE-C). The UE-C (Remote UE) is the target of interception. IMS DC related media is sent to or received from the Remote UE (UE-C) in a transcoded form.

Figure 4.10.3.4.3-1 shows the scenario for an inter-network case. In this case, either the UE-C (Remote UE) or the redirecting network or the redirected-to network does not have IMS DC capabilities.

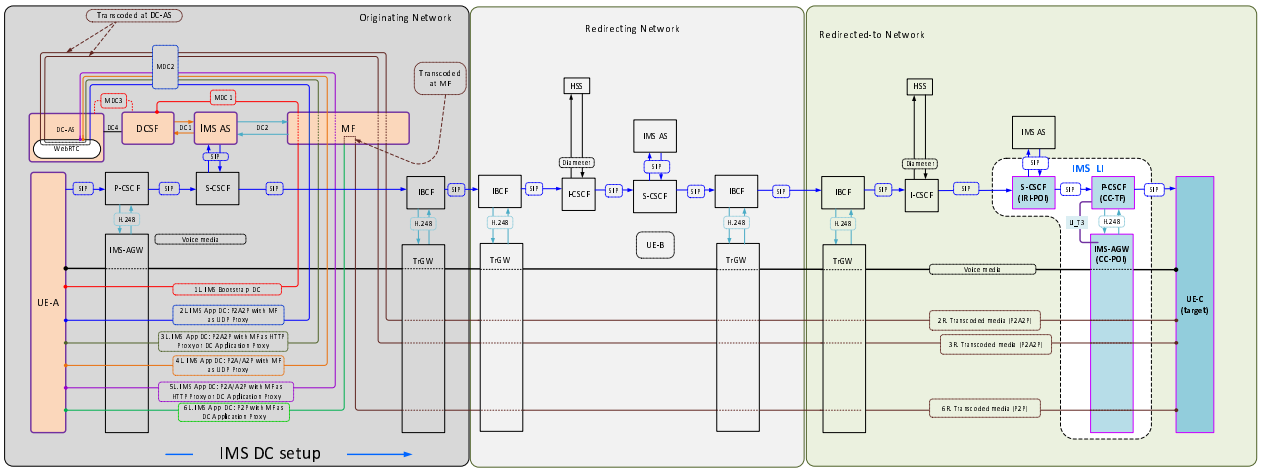


Figure 4.10.3.4.3-1: Topology view when redirected-to party is the remote UE and the target (inter-network case)

Figure 4.10.3.4.3-2 shows the scenario for an intra-network case. In this case, the redirected-to UE does not have the IMS DC capabilities.

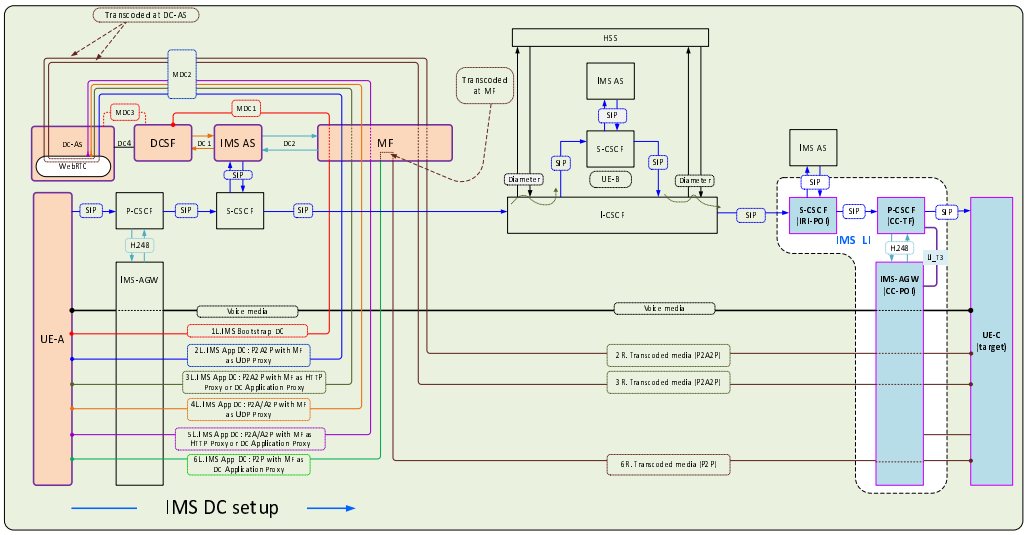


Figure 4.10.3.4.3-2: Topology view when redirected-to party is the remote UE and the target (intra-network case)

As shown in figure 4.10.3.4.3-1 and figure 4.10.3.4.3-2, the S-CSCF (present in the redirected-to network) provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW (present in the redirecting-to network) providing the CC-POI functions where the CC-TF is at the P-CSCF (present in the redirected-to). LI for transcoded IMS DC media is part of the IMS LI.

4.10.3.4.4 Scenario 3: Originating party is a target non-local ID in the terminating network with transcoded media

Figure 4.10.3.4.4-1 illustrates the topology view of IMS LI when originating party (UE-A) is a target non-local ID in the terminating network for IMS-based services while the originating party (UE-A, target non-local ID) of the IMS session has initiated the IMS DC setup with the terminating party (UE-B). IMS DC related media is sent to or received from the Remote UE (UE-B) in a transcoded form. In this case, either the UE-B (Remote UE) or the terminating network does not have IMS DC capabilities.

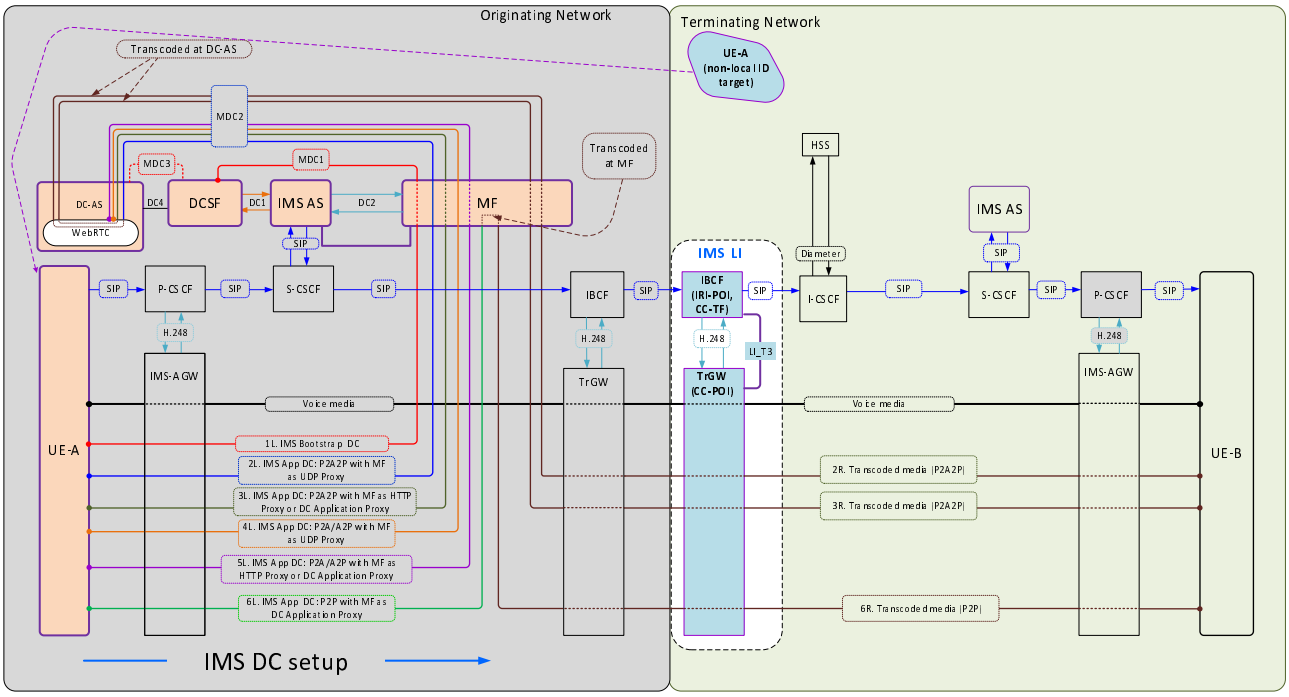


Figure 4.10.3.4.4-1: Topology view when the terminating party is the remote UE and the originating party is a target non-local ID – transcoded IMS DC media

As shown in figure 4.10.3.4.4-1 the IBCF present in the terminating network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with TrGW (present in the terminating network) providing the CC-POI functions where the CC-TF is at the IBCF. The transcoded IMS DC media sent from or received at the redirected-to network is also intercepted at the CC-POI present in the TrGW.

4.10.3.4.5 Scenario 6: Originating party is a target non-local ID in the redirected-to network

Figure 4.10.3.4.5-1 illustrates the topology view of IMS LI when originating party (UE-A) is a target non-local ID in the redirected-to network for IMS-based services while the originating party (UE-A, target non-local ID) of the redirected IMS session has initiated the IMS DC setup with the redirected-to party (UE-C). IMS DC related media is sent to or received from the Remote UE (UE-C in a transcoded form. In this case, any one or more of redirecting network, redirected-to network or the Remote UE (UE-C) do not have the IMS DC capabilities.

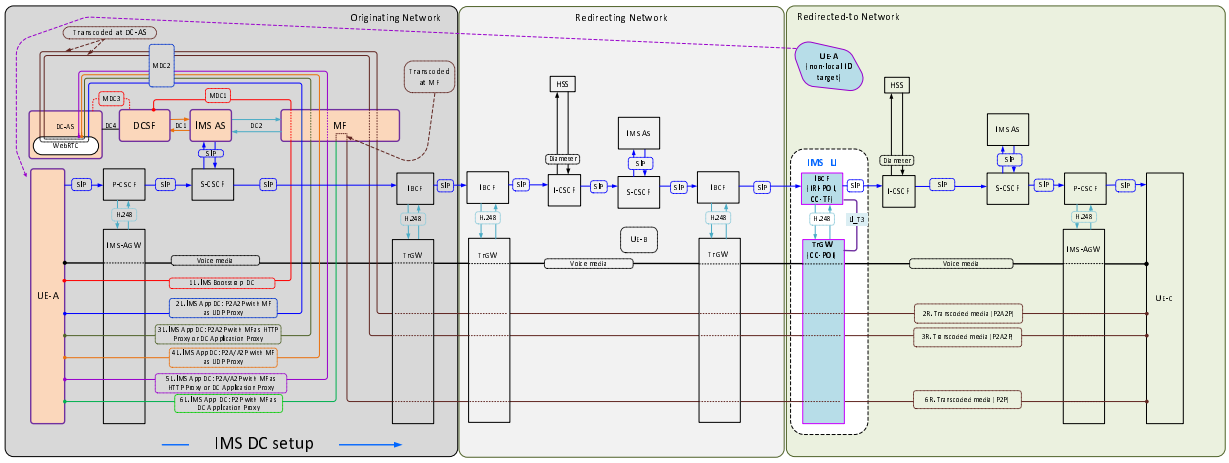


Figure 4.10.3.4.5-1: Topology view when the redirected-to party is the remote UE and the originating party is a target non-local ID in the redirected-to network – transcoded IMS DC media

As shown in figure 4.10.3.4.5-1 the IBCF present in the redirected-to network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with TrGW (present in the redirected-to network) providing the CC-POI functions where the CC-TF is at the IBCF. The transcoded IMS DC media sent from or received at redirected-to network is also intercepted at the CC-POI present in the TrGW.

4.10.3.4.6 Scenario 5: Redirecting party is a target non-local ID in the redirected-to network

Figure 4.10.3.4.6-1 illustrates the topology view of IMS LI when redirecting party (UE-B) is a target non-local ID in the redirected-to network for IMS-based services while the originating party (UE-A) of the redirected IMS session has initiated the IMS DC setup with the redirected-to party (UE-C). IMS DC related media is sent to or received from the Remote UE (UE-C in a transcoded form. In this case, any one or more of redirecting network, redirected-to network or the Remote UE (UE-C) do not have the IMS DC capabilities.

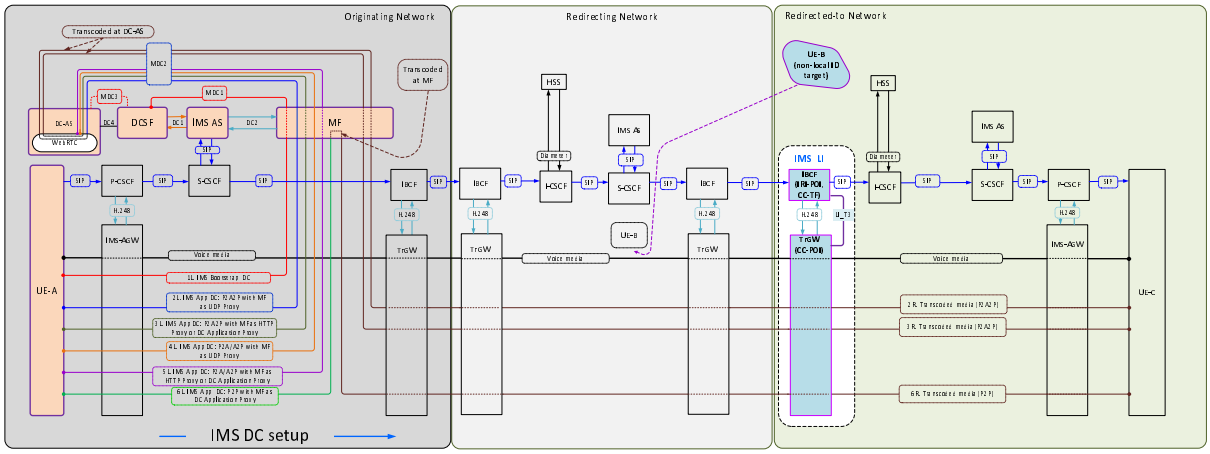


Figure 4.10.3.4.6-1: Topology view when the redirected-to party is the remote UE and the redirecting party is a target non-local ID in the redirected-to network – transcoded IMS DC media

As shown in figure 4.10.3.4.6-1 the IBCF present in the redirected-to network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with TrGW (present in the redirected-to network) providing the CC-POI functions where the CC-TF is at the IBCF. The transcoded IMS DC media sent from or received at redirected-to network is also intercepted at the CC-POI present in the TrGW.

4.10.3.5 Network topologies for special redirection cases

4.10.3.5.1 General

The subsequent clauses illustrate the network topology views where the LI in the redirecting network where redirecting party is the target of interception. In the illustrations, the originating party is the Local UE and redirected-to party is the Remote UE for IMS DC.

The illustrations also show the following target non-local ID cases for IMS-based services when IMS DC session is present between the parties of IMS session:

- Redirected-to party (i.e. Remote UE) is a target non-local ID in the redirecting network.
- Originating party (i.e. Local UE) is a target non-local ID in the redirecting network.

According to TS 33.128 [4], target non-local ID does not apply to IMS DC sessions. Note that the IMS DC capable redirected-to party may also setup the IMS DC with the originating party when the redirected-to network has the IMS DC capabilities.

4.10.3.5.2 Scenario 1: Redirecting party is the target

This clause illustrates the topology views of IMS LI and IMS DC LI when the originating party (UE-A) of an IMS session initiates the IMS DC setup with the redirected-to party (UE-C). The UE-B (redirecting party) is the target of interception. The UE-C (redirecting party) is not on the IMS DC. Therefore, the illustration shows only the regular IMS DC applies when the redirecting party is the target. IMS DC related media is not (cannot be) intercepted in the redirecting network.

Figure 4.10.3.5.2-1 shows the scenario for an inter-network case.

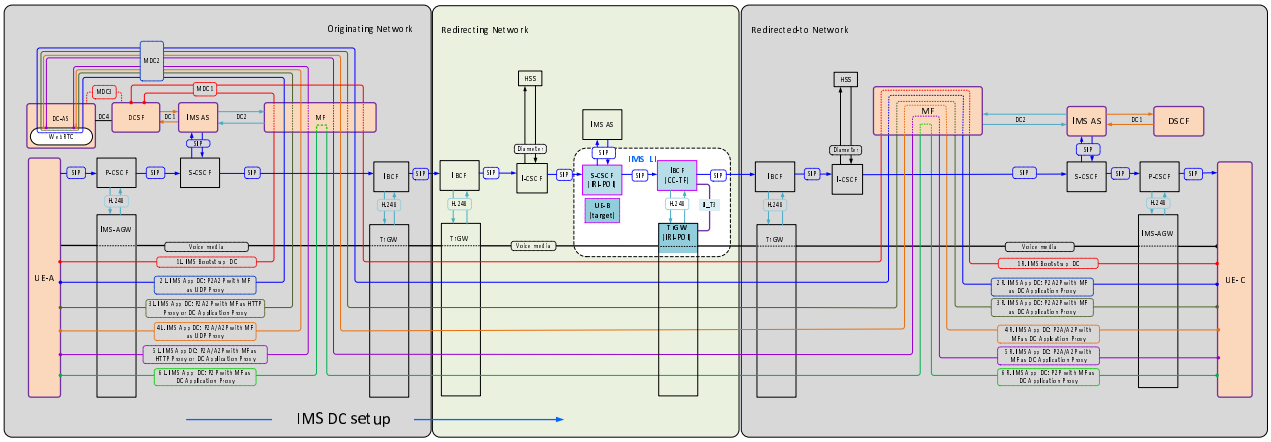


Figure 4.10.3.5.2-1: Topology view when redirecting party is the target (inter-network case)

Figure 4.10.3.5.2-2 shows the scenario for an intra-network case.

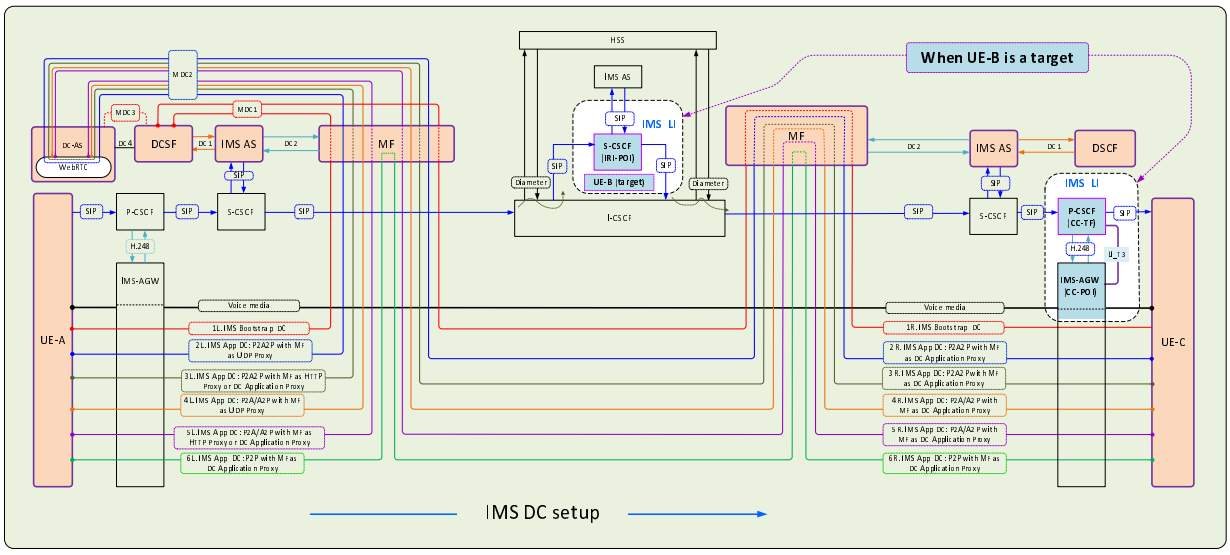


Figure 4.10.3.5.2-2: Topology view when redirecting party is the target (intra-network case)

As shown in figure 4.10.3.5.2-1 and figure 4.10.3.5.2-2, the S-CSCF (present in the redirecting network) provides the IRI-POI functions for IMS-LI (default option) with the egress TrGW (present in the redirecting) providing the CC-POI functions where the CC-TF is at the egress IBCF (present in the redirecting network). LI for IMS DC is not applicable to the redirecting network where the redirecting party is the target.

4.10.3.5.3 Scenario 2: Originating party is a target non-local ID in the redirecting network

Figure 4.10.3.5.3-1 illustrates the topology view of IMS LI when originating party (UE-A) is a target non-local ID in the redirecting network for IMS-based services while the originating party (UE-A, target non-local ID) of the redirected IMS session has initiated the IMS DC setup with the redirected-to party (UE-C). The Redirecting Network supports the IMS DC capabilities and the UE-C (Remote UE) is IMS DC capable, has subscribed to IMS DC in the Redirected-to Network (which obviously, has IMS DC capabilities) and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-C.

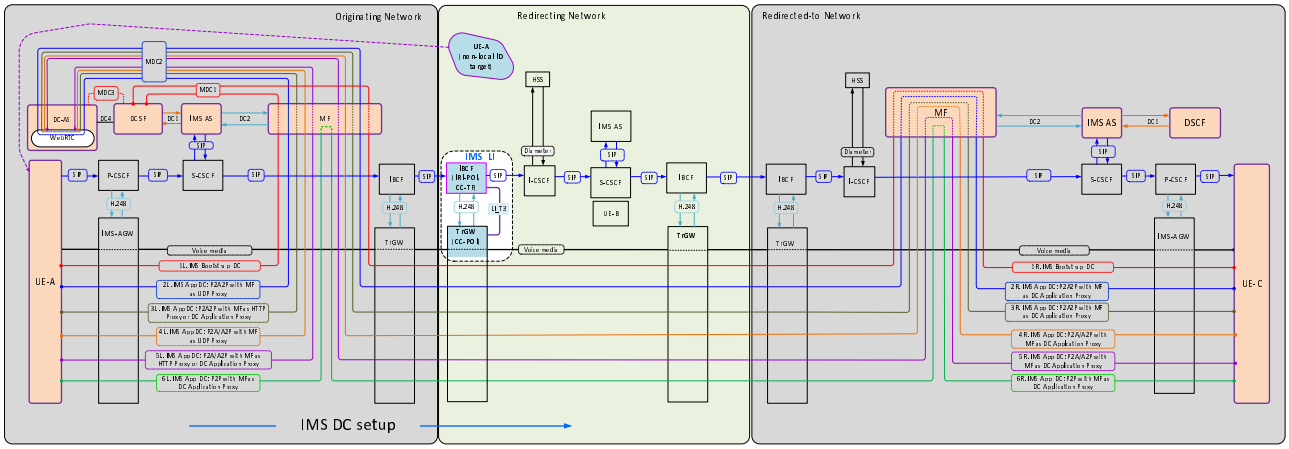


Figure 4.10.3.5.3-1: Topology view when the redirected-to party is the remote UE and the originating party is a target non-local ID in the redirecting network

As shown in figure 4.10.3.5.3-1 ingress IBCF present in the redirecting network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with ingress TrGW (present in the redirecting network) providing the CC-POI functions where the CC-TF is at the ingress IBCF. As per TS 33.128 [4], the concept of target non-local ID does not apply to IMS DC and therefore, as shown the IMS DC media is not intercepted.

4.10.3.5.4 Scenario 3: Redirecting party is a target non-local ID in the redirected-to network

Figure 4.10.3.5.4-1 illustrates the topology view of IMS LI when redirected-to party (UE-C) is a target non-local ID in the redirecting network for IMS-based services while the originating party (UE-A) of the redirected IMS session has initiated the IMS DC setup with the redirected-to party (UE-C). The Redirecting Network supports the IMS DC capabilities and the UE-C (Remote UE) is IMS DC capable, has subscribed to IMS DC in the Redirected-to Network (which obviously, has IMS DC capabilities) and hence, all the IMS DCs including the Bootstrap IMS Data Channel are setup to UE-C.

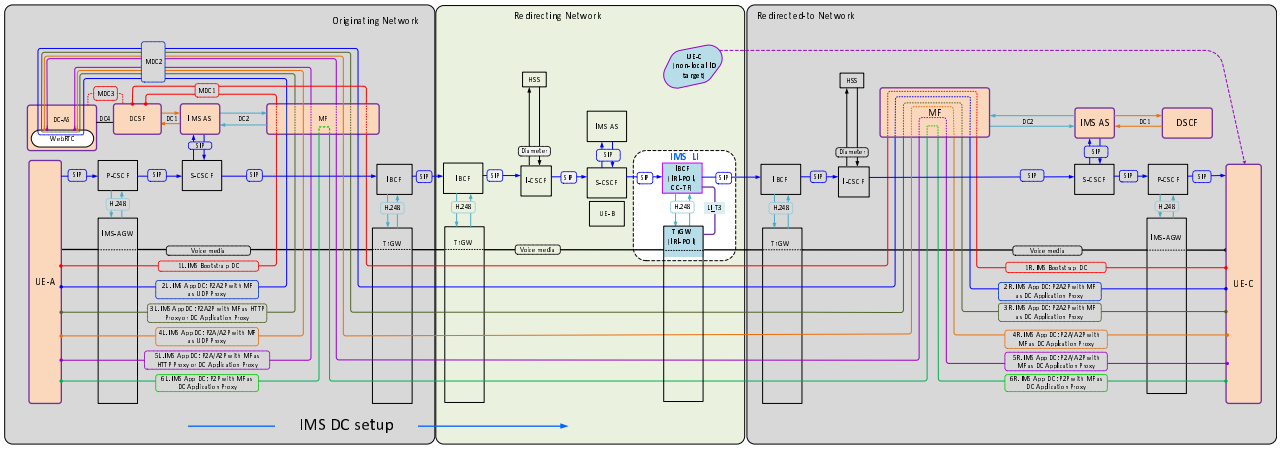


Figure 4.10.3.5.4-1: Topology view when the redirected-to party is the remote UE and the redirected-to party is a target non-local ID in the redirecting network

As shown in figure 4.10.3.5.4-1, the egress IBCF present in the redirecting network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with egress TrGW (present in the redirecting network) providing the CC-POI functions where the CC-TF is at the egress IBCF. As per TS 33.128 [4], the concept of target non-local ID does not apply to IMS DC and therefore, as shown the IMS DC media is not intercepted.

4.10.3.5.5 Scenario 4: Redirecting party is the target with transcoded IMS DC media to remote UE

Figure 4.10.3.5.5-1 and figure 4.10.3.5.5-2 illustrate topology views of IMS LI that includes the transcoded IMS DC media when the originating party (UE-A) of an IMS session initiates the IMS DC setup with redirected-to party (UE-C). The UE-B (redirecting party) is the target of interception. IMS DC related media is transferred to and from the Remote UE (UE-C) in a transcoded form.

Figure 4.10.3.5.5-1 shows the scenario for an inter-network case. In this case, any one or more of the UE-B (Remote UE) or redirecting network or the redirected-to network do not have the IMS DC capabilities.

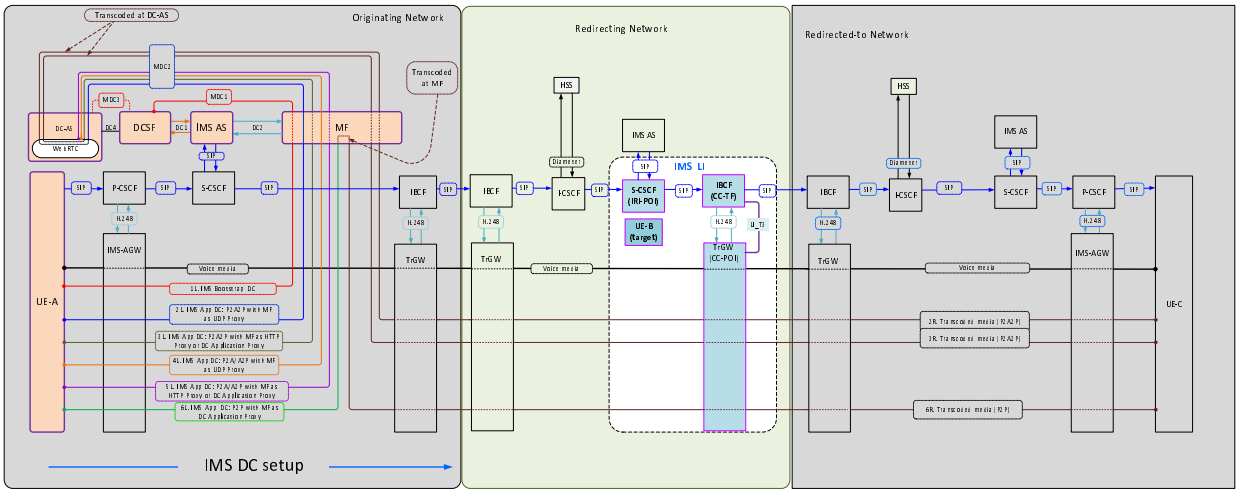


Figure 4.10.3.5.5-1: Topology view when redirecting party is the target (inter-network case)

Figure 4.10.3.5.5-2 shows the scenario for an intra-network case. In this case, the redirected-to UE does not have the IMS DC capabilities.

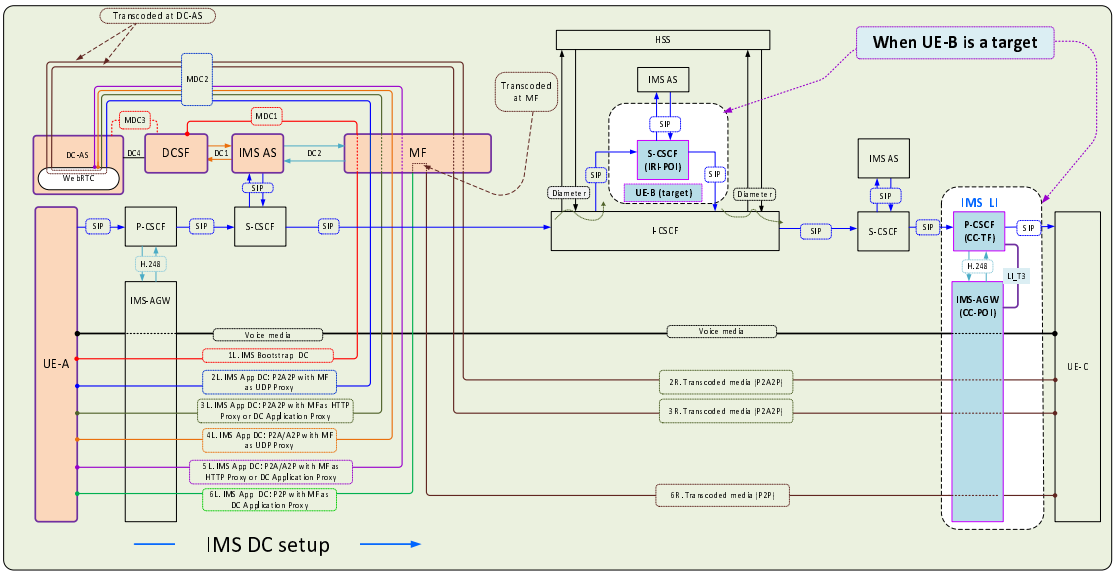


Figure 4.10.3.5.5-2: Topology view when redirecting party is the target (intra-network case)

As shown in figure 4.10.3.5.5-1 and figure 4.10.3.5.5-2, the S-CSCF (present in the redirecting network) provides the IRI-POI functions for IMS-LI (default option) with the egress TrGW (present in the redirecting network) providing the CC-POI functions where the CC-TF is at the egress IBCF (present in the redirecting network). Even though LI for IMS DC is not applicable to the redirecting network, since the IMS DC media is in transcoded form, LI for that transcoded IMS DC media is part of the IMS LI.

4.10.3.5.6 Scenario 5: Redirected-to party is a target non-local ID in the redirecting network with transcoded IMS DC media to the remote UE

Figure 4.10.3.5.6-1 illustrates the topology view of IMS LI when redirected-to party (UE-C) is a target non-local ID in the redirecting network for IMS-based services while the originating party (UE-A) of the redirected IMS session has initiated the IMS DC setup with the redirected-to party (UE-C). IMS DC related media is sent to or received from the Remote UE (UE-C in a transcoded form. In this case, any one or more of redirecting network, redirected-to network or the Remote UE (UE-C) do not have the IMS DC capabilities.

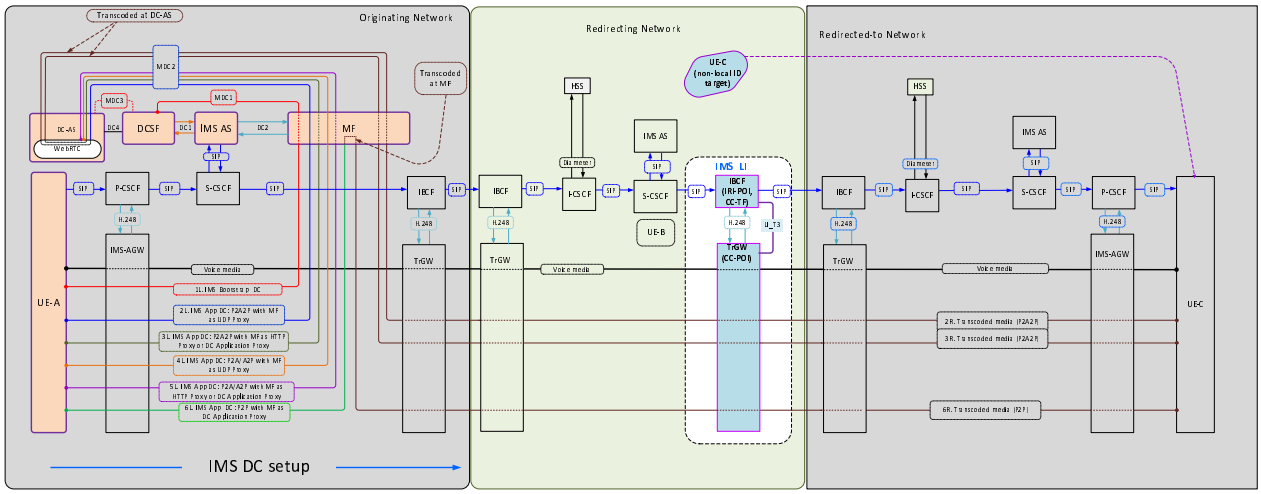


Figure 4.10.3.5.6-1: Topology view when the redirected-to party is the remote UE and the redirected-to party is a target non-local ID in the redirecting network – transcoded IMS DC media

As shown in figure 4.10.3.5.6-1 the egress IBCF present in the redirecting network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with egress TrGW (present in the redirecting network)

providing the CC-POI functions where the CC-TF is at the egress IBCF. The transcoded IMS DC media sent from or received at redirected-to network is also intercepted at the CC-POI present in the TrGW.

4.10.3.5.7 Scenario 6: Originating party is a target non-local ID in the redirecting network with transcoded IMS DC media to the remote UE

Figure 4.10.3.5.7-1 illustrates the topology view of IMS LI when originating party (UE-A) is a target non-local ID in the redirecting network for IMS-based services while the originating party (UE-A, target non-local ID) of the redirected IMS session has initiated the IMS DC setup with the redirected-to party (UE-C). IMS DC related media is sent to or received from the Remote UE (UE-C in a transcoded form. In this case, any one or more of redirecting network, redirected-to network or the Remote UE (UE-C) do not have the IMS DC capabilities.

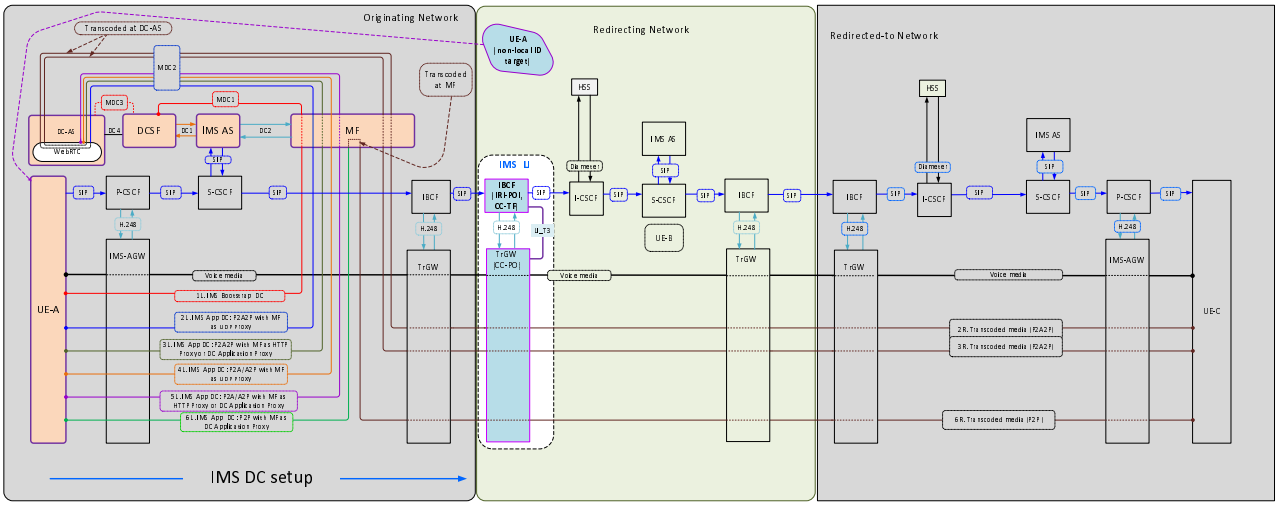


Figure 4.10.3.5. 7-1: Topology view when the redirected-to party is the remote UE and the originating party is a target non-local ID in the redirecting network – transcoded IMS DC media

As shown in figure 4.10.3.5.7-1 the ingress IBCF present in the redirecting network provides the IRI-POI functions for IMS-LI with respect to a target non-local ID (default option) with ingress TrGW (present in the redirecting network) providing the CC-POI functions where the CC-TF is at the ingress IBCF. The transcoded IMS DC media sent from or received at redirected-to network is also intercepted at the CC-POI present in the TrGW.

4.10.3.6 Topology view for two IMS DC sessions in parallel

4.10.3.6.1 General

The subsequent clauses a topology views when two IMS DC sessions are in parallel. The illustrations show that the LI when both parties of an IMS session setup IMS DC at the same time with the other party. The LI provided for the IMS DCs setup by the two parties have to be intercepted separately, in the sense, separate correlation IDs have to be used.

4.10.3.6.2 Originating party is the target

Figure 4.10.3.6.2-1 and 4.10.3.6.2-2 illustrate the topology views where the originating party (UE-A) initiates an IMS DC session with the terminating party (UE-B) and in parallel, the terminating party (UE-B) also sets up an IMS DC session with the originating party (UE-A). In both illustrations, originating party (UE-A) is the target.

For the IMS DC setup by the originating party, the illustration shows a P2P IMS DC with MF in the originating network operating in a DC Application Proxy mode. For the IMS DC setup by the terminating party, the illustration shows a P2A2P IMS DC with MF in the terminating network operating in a UDP Proxy mode.

Figure 4.10.3.6.2-1 illustrates an inter-network case.

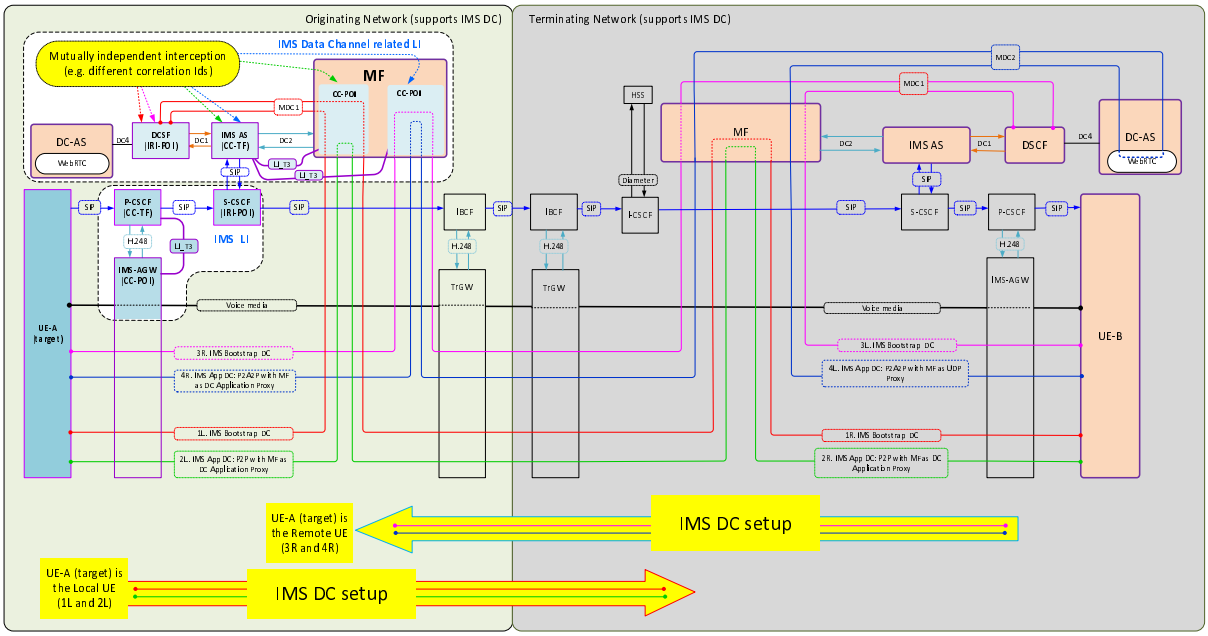


Figure 4.10.3.6.2-1: Two parallel IMS DC sessions (inter-network case) – originating party (UE-A) is the target

Figure 4.10.3.6.2-2 illustrates an intra-network case.

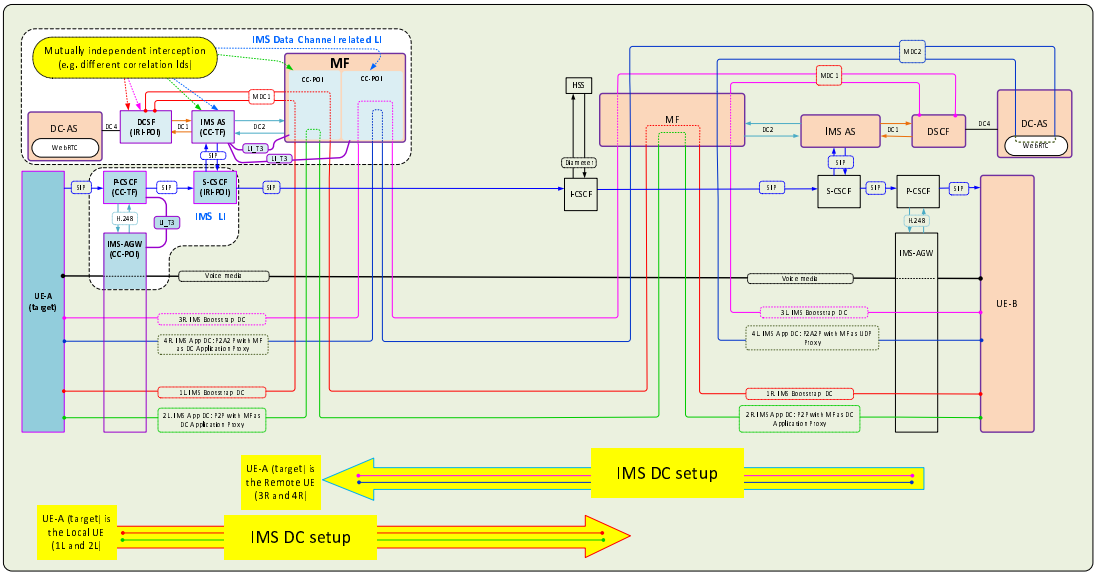


Figure 4.10.3.6.2-2: Two parallel IMS DC sessions (intra-network case) – originating party (UE-A) is the target

As shown in figures 4.10.3.6.2-1 and 4.10.3.6.2-2, the LI functions are provided at the originating end. The S-CSCF provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF.

For the IMS DC setup by the originating party (UE-A, the target), the DCSF provides the IRI-POI functions. The P2P IMS DC media is intercepted at the MF with CC-TF present in the IMS AS. Here, the target (UE-A) is the Local UE.

For the IMS DC setup by the terminating party (UE-B), the DCSF (originating end) provides the IRI-POI functions. The P2A2P IMS DC media is intercepted at the MF with CC-TF present in the IMS AS. Here, the target (UE-A) is the Remote UE.

Even though the same NFs provide the IRI-POI, CC-POI and CC-TF functions for the two independently setup IMS DC sessions, the LI for the two IMS DC sessions have to be treated separately. In the sense, they should have separate correlation IDs.

4.10.3.6.3 Terminating party is the target

Figure 4.10.3.6.3-1 and 4.10.3.6.3-2 illustrate the topology views where the originating party (UE-A) initiates an IMS DC session with the terminating party (UE-B) and in parallel, the terminating party (UE-B) also sets up an IMS DC session with the originating party (UE-A). In both illustrations, terminating party (UE-B) is the target.

For the IMS DC setup by the originating party, the illustration shows a P2P IMS DC with MF in the originating network operating in a DC Application Proxy mode. For the IMS DC setup by the terminating party, the illustration shows a P2A2P IMS DC with MF in the terminating network operating in a UDP Proxy mode.

Figure 4.10.3.6.3-1 illustrates an inter-network case.

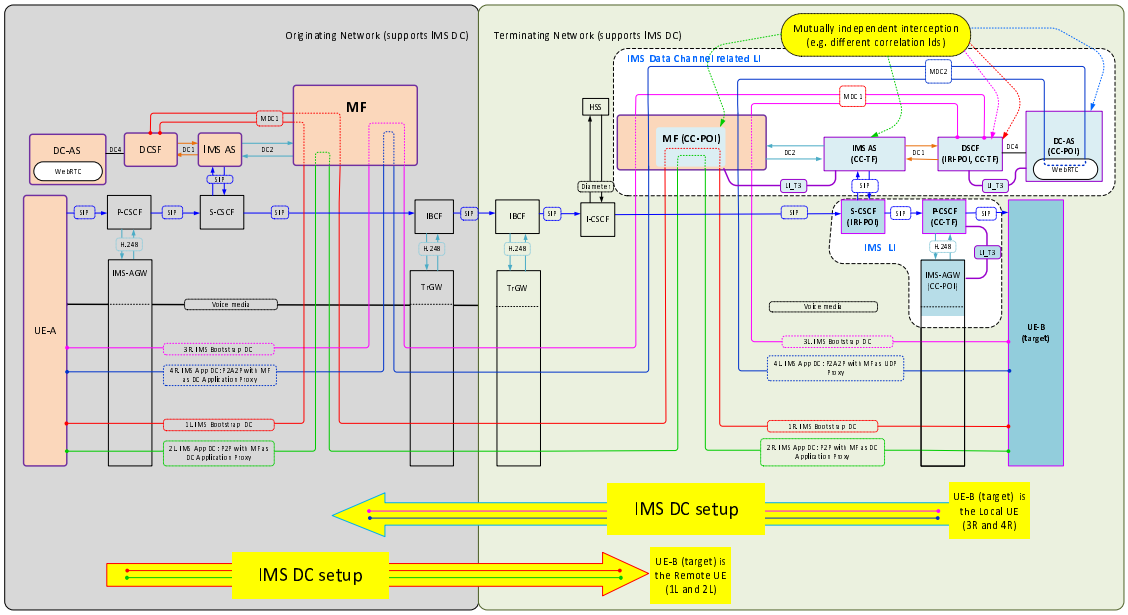


Figure 4.10.3.6.3-1: Two parallel IMS DC sessions (inter-network case) – terminating party (UE-B) is the target

Figure 4.10.3.6.3-2 illustrates an intra-network case.

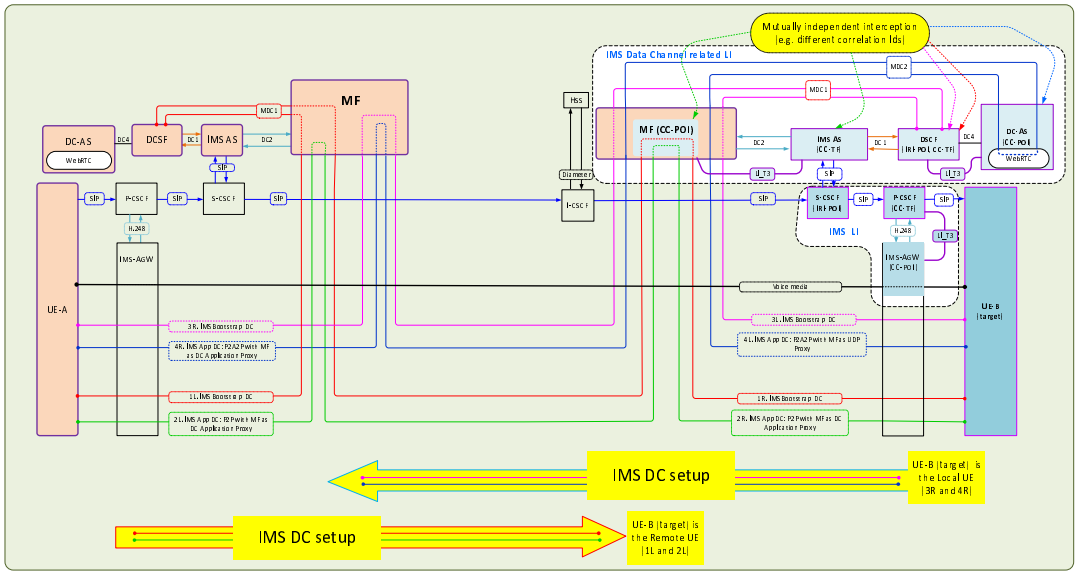


Figure 4.10.3.6.3-2: Two parallel IMS DC sessions (intra-network case) – terminating party (UE-B) is the target

As shown in figures 4.10.3.6.3-1 and 4.10.3.6.3-2, the LI functions are provided at the terminating end. The S-CSCF provides the IRI-POI functions for IMS-LI (default option) with IMS-AGW providing the CC-POI functions where the CC-TF is at the P-CSCF.

For the IMS DC setup by the originating party (UE-A), the DCSF (terminating end) provides the IRI-POI functions. The P2P IMS DC media is intercepted at the MF with CC-TF present in the IMS AS. Here, the target (UE-B) is the Remote UE.

For the IMS DC setup by the terminating party (UE-B, target), the DCSF (terminating end) provides the IRI-POI functions. The P2A2P IMS DC media is intercepted at the DC-AS (since MF at the terminating end is operating in an UDP Proxy mode) with CC-TF present in the DCSF.

The LI for the two IMS DC sessions has to be treated separately. In the sense, they should have separate correlation IDs.

5 Example call flows of LI for IMS-based services

5.1 General remarks

All the call flows illustrate that the CC delivery begins once the SDP offer and answer is completed (i.e. when the media bearer is setup). In all the call flows shown, the first reliable response is SIP 200 OK.

In all the call flows, the originating end of the IMS session sends the SDP offer and the terminating end of the IMS session gives the SDP answer. Thereafter, the first reliable response is SIP 200 OK, therefore, the SDP answer is always given in the SIP 200 OK message.

All the call flows assume the presence of an Application Server (shown as AS) that provides the services like digit translation, invoking the call forwarding, etc.

The IRI in the visited CSP with Local Break-out as the roaming architecture is intercepted by the IRI-POI present in the P-CSCF and the IRI in the home CSP is intercepted by the IRI-POI present in the S-CSCF.

The CC interception can be done at the CC-POIs present in the IMS-AGW, IM-MGW or the TrGW depending on the IMS session scenario. The option of CC interception performed at the PGW (supported for VoLTE) is not shown.

Not all the functional elements are shown in the call flows. For example, the call flows do not show I-CSCF and HSS that are involved in handling the terminating leg of an IMS session.

All the call flows show a summary of SIP messages that are delivered to the LEMF (not all SIP messages are shown).

5.2 Originating IMS sessions from the target – non-roaming

5.2.1 Introduction

This clause 5.2 describes the call flows to illustrate the scenarios of IMS sessions originated by the target. None of the parties in the sessions are presumed to be roaming.

In all call flows, the CC interception trigger from CC-TF in P-CSCF is sent to the CC-POI in IMS-AGW when the P-CSCF sends the H.248 requests to provide the SDP information related to the IMS media sessions to the IMS-AGW.

The IRI-POI in the S-CSCF generates the xIRI using the SIP messages.

5.2.2 Party_A (target) calls Party_B

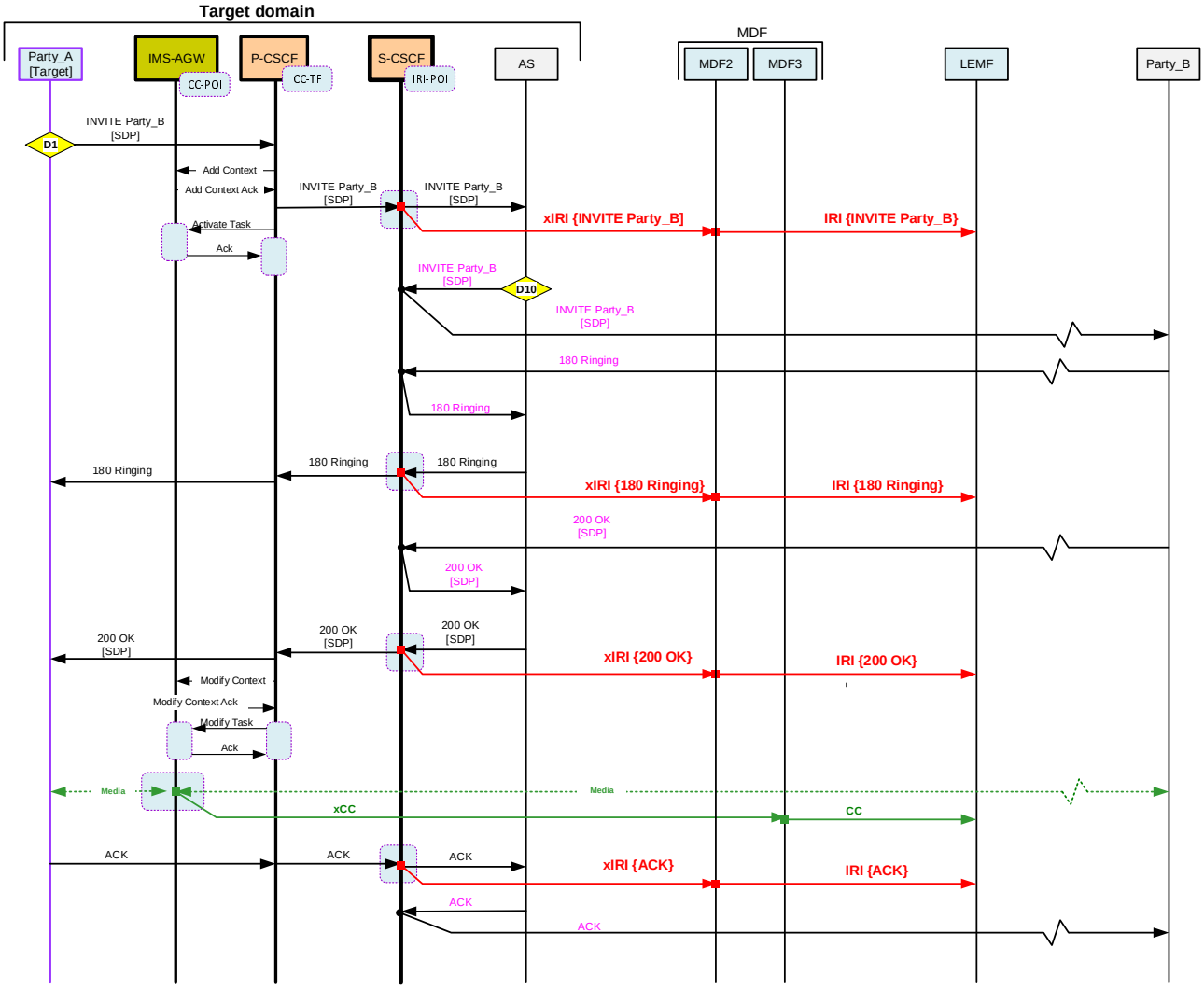


Figure 5.2.2-1: Party_A (target) calls Party_B

Figure 5.2.2-1 illustrates the case where Party_A (target) calls Party_B.

5.2.3 Party_A (target) dials a special number

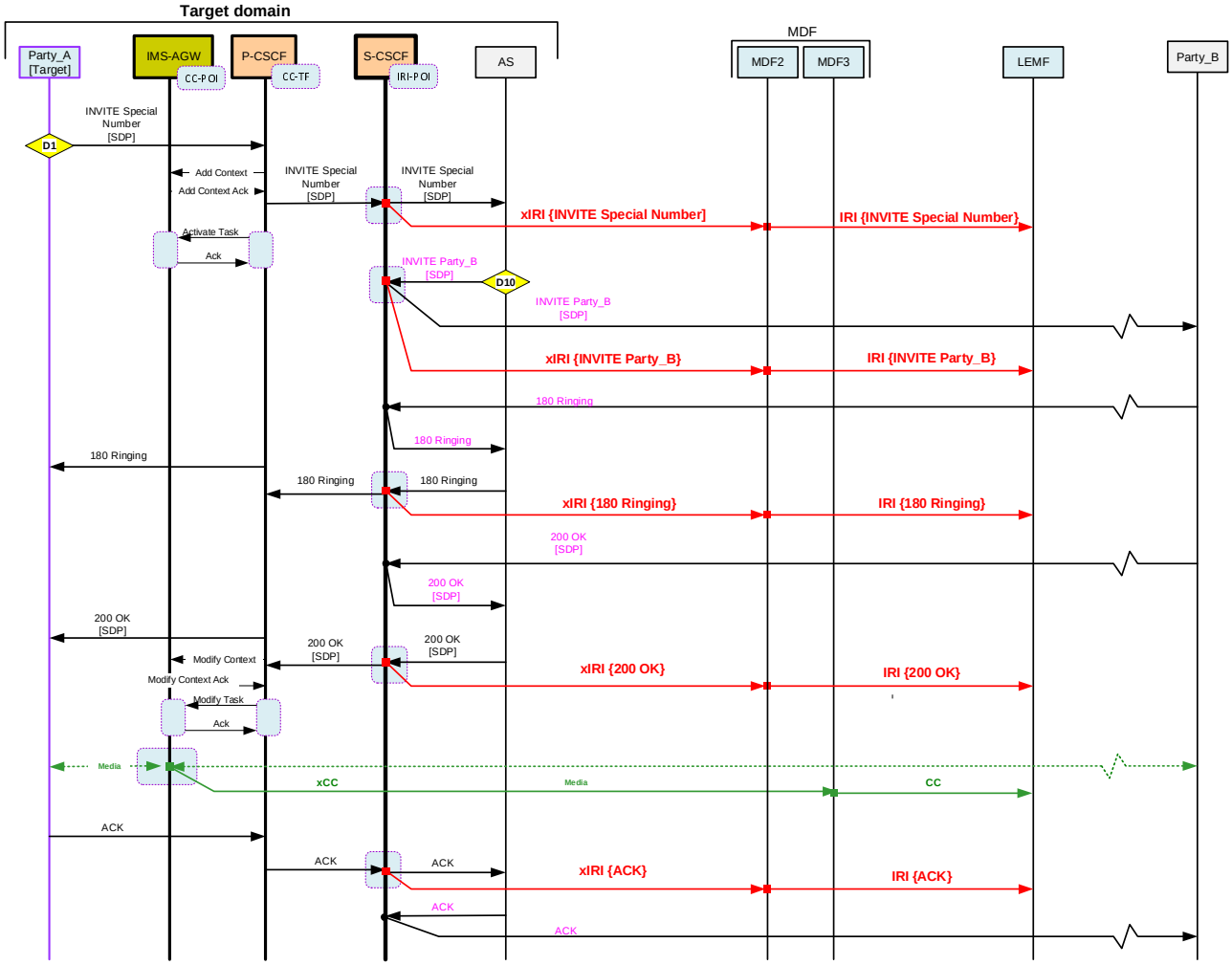


Figure 5.2.3-1: –Party_A (target) dials a special number (translated to Party_B)

Figure 5.2.3-1 illustrates the case where Party_A (target) dials a special number (e.g. a speed call number or an 800-number), which is translated to Party B by the AS.

5.3 Terminating IMS sessions to the target – non-roaming

5.3.1 Introduction

This clause 5.3 describes the call flows to illustrate the scenarios of IMS sessions terminated to the target. None of the parties in the sessions are presumed to be roaming.

In all call flows, the CC interception trigger from the CC-TF in the P-CSCF is sent to the CC-POI in the IMS-AGW as and when the P-CSCF sends the H.248 requests to provide the SDP information related to the IMS media sessions to the IMS-AGW.

The IRI-POI in S-CSCF generates the xIRI using the SIP messages.

5.3.2 Party_A calls Party_B (target)

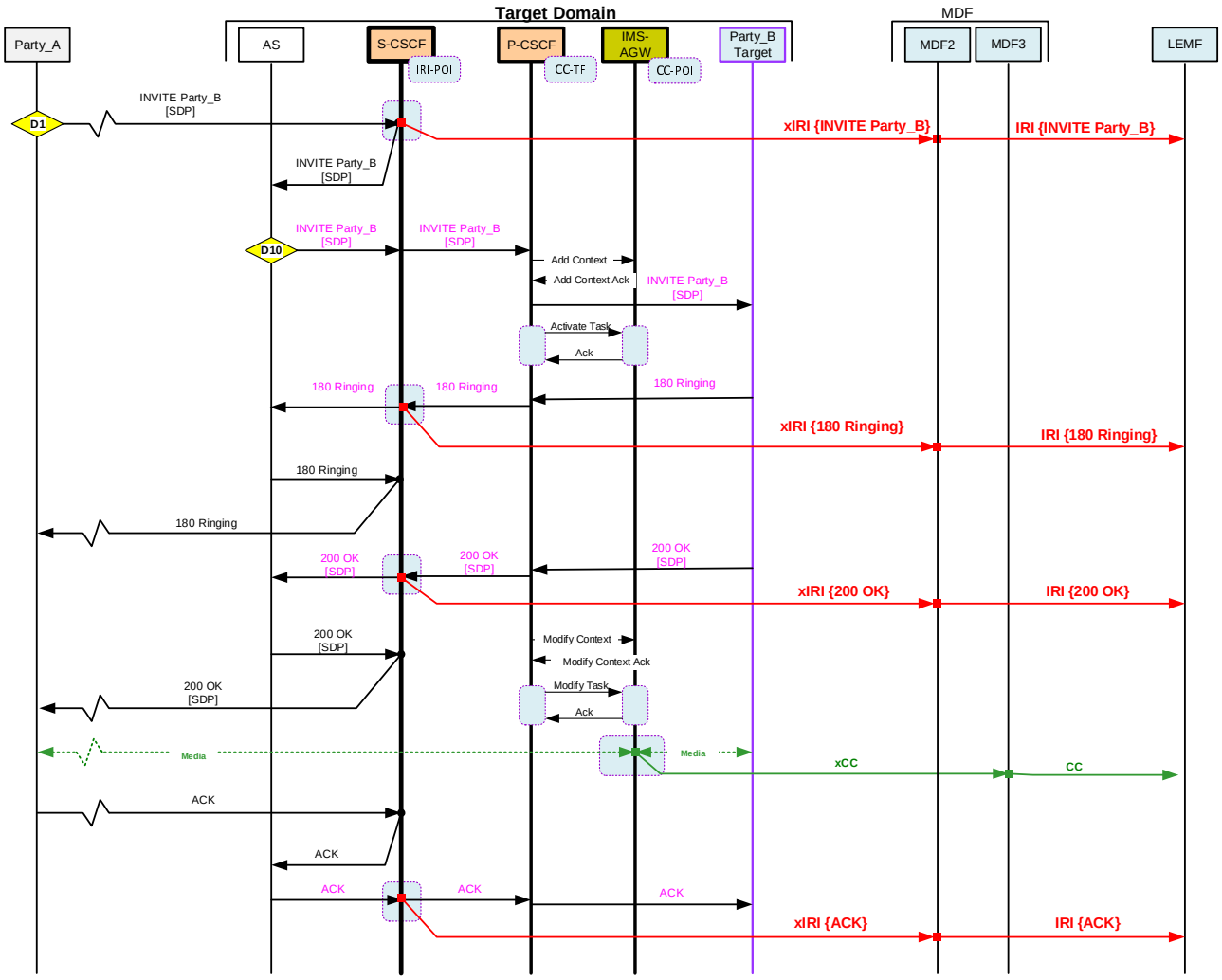


Figure 5.3.2-1: Party_A calls Party B (target)

Figure 5.3.2-1 illustrates a case where the Party_A calls Party_B (target).

5.4 Redirected IMS sessions – no roaming

5.4.1 Introduction

Clause 5.4 describes the call flows to illustrate scenarios of IMS session redirections redirection (aka call forwarding). In the illustrated call flows, the redirecting party is the target and the redirected-to-party may be served within the same CSP domain (intra-CSP redirection) as that of the redirecting party or in a different CSP domain (inter-CSP redirection).

In the intra-CSP redirections, the CC interception is done in the CC-TF/CC-POI present in the network functions (P-CSCF/IMS-AGW) that handle the redirected-to-party (Party_C). The IRI interception is done in the IRI-POI present in the network function (S-CSCF) that serves the target (Party_B).

In the case of inter-CSP redirections, the CC interception is done in the CC-POI present in the network functions (TrGW/IM-MGW) that have access to the media associated with the redirected IMS session. The CC-TF functions are provided by the IBCF (when the CC-POI is in the TrGW) and by the MGCF (when the CC-POI is in the IM-MGW). The IRI interception is done in the network function (S-CSCF) that serves the target (Party B).

5.4.2 Intra-CSP redirection unconditional – Party_A calls Party_B (target) redirected to Party_C

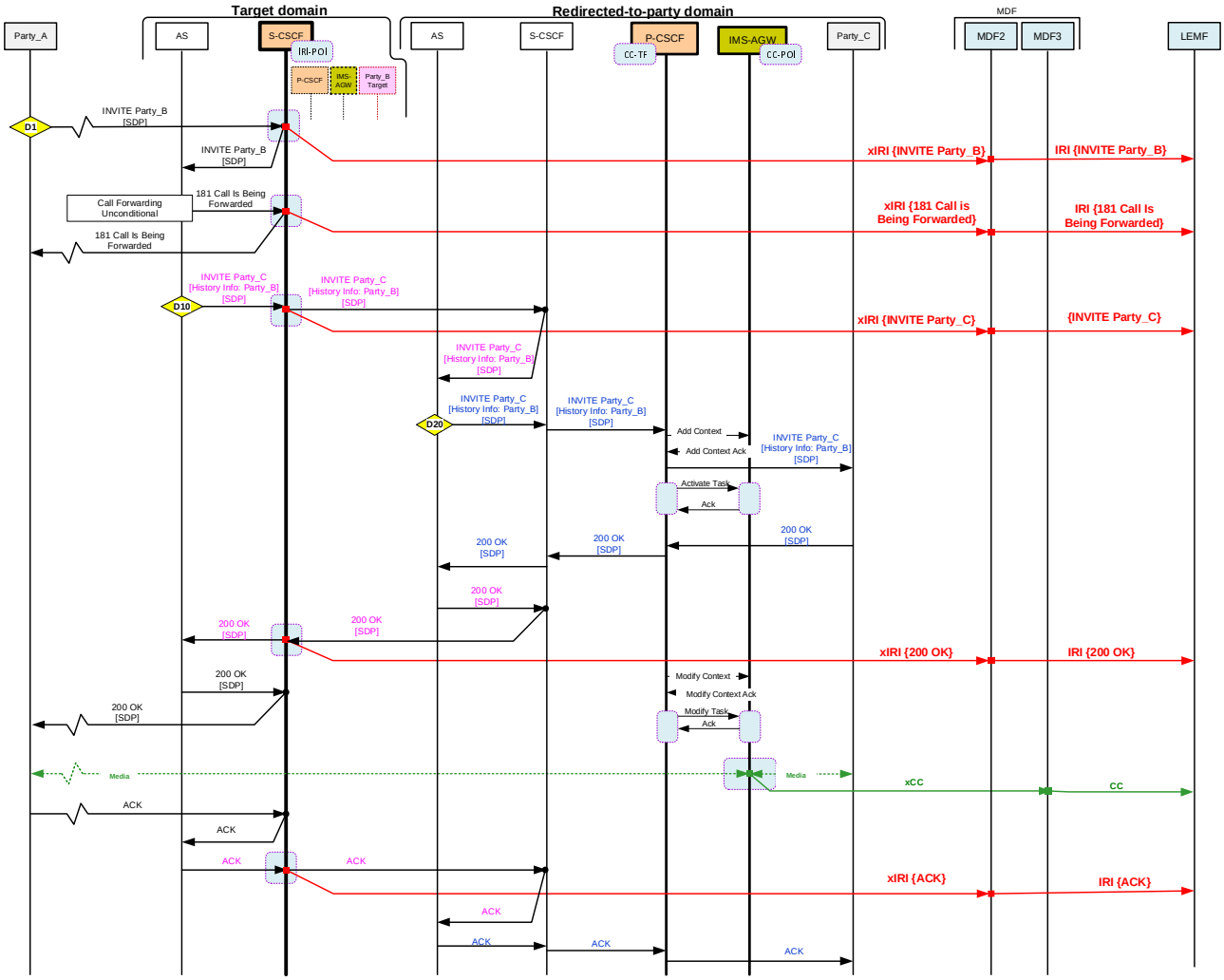


Figure 5.4.2-1: Party_A calls Party_B (target) redirected to Party_C

Figure 5.4.2-1 illustrates a case of an intra-CSP redirection unconditional. Here, Party A calls Party B (target). The AS determines that all incoming IMS sessions to the Party B (target) have to be redirected to Party C served by the same CSP.

5.4.3 Intra-CSP redirection due to no answer - Party_A calls Party_B (target) redirected to Party_C – flow 1 of 2

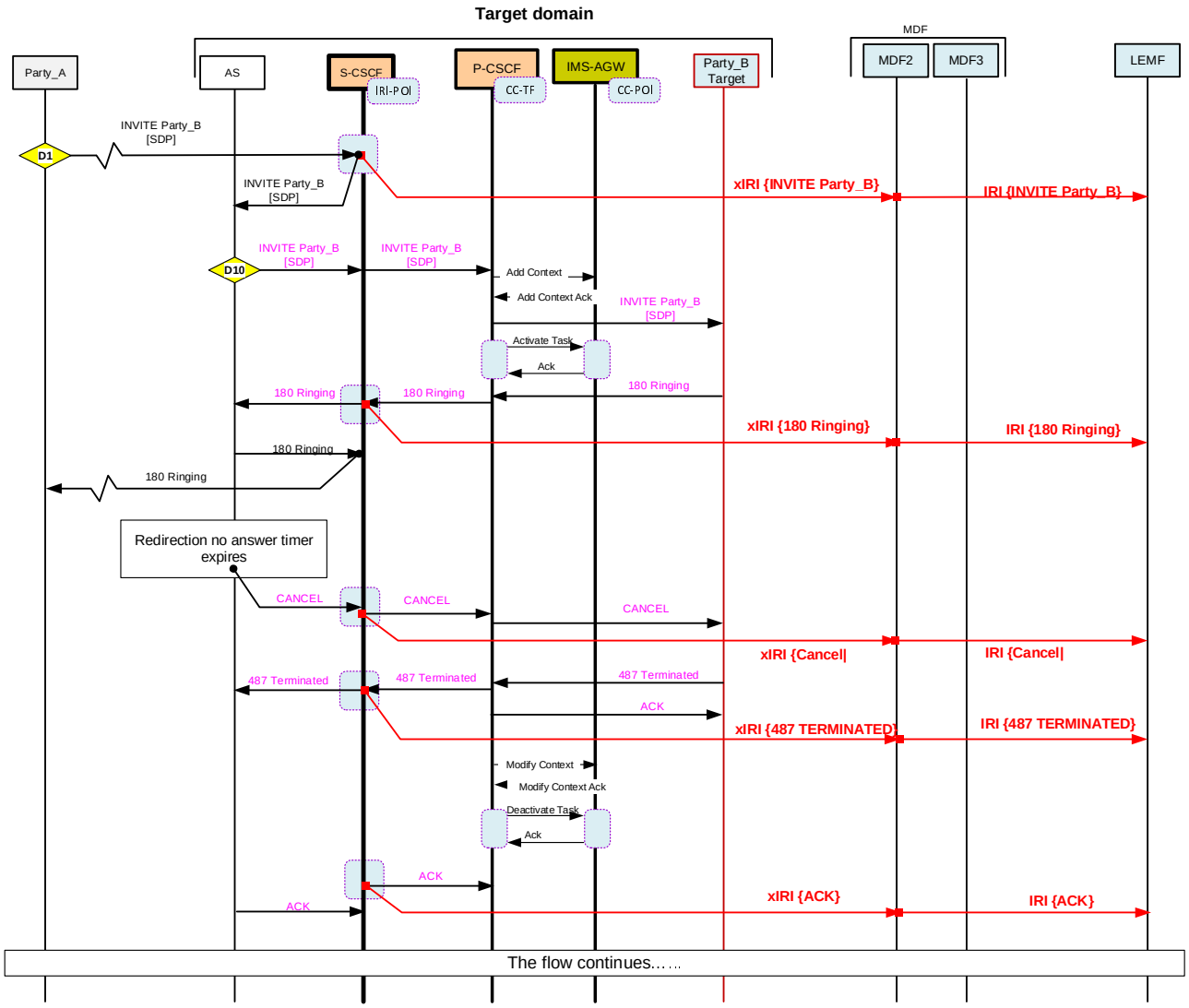


Figure 5.4.3-1: Party_A calls Party B (target) redirected to Party_C due to no answer (flow 1 of 2)

NOTE: An incoming IMS session to Party_B (target) is redirected to Party_C when Party_B fails to answer before the redirection no answer timer expires. Party_C is in the same CSP domain as that of Party_B.

5.4.4 Intra-CSP redirection due to no answer - Party_A calls Party_B (target) redirected to Party_C – flow 2 of 2

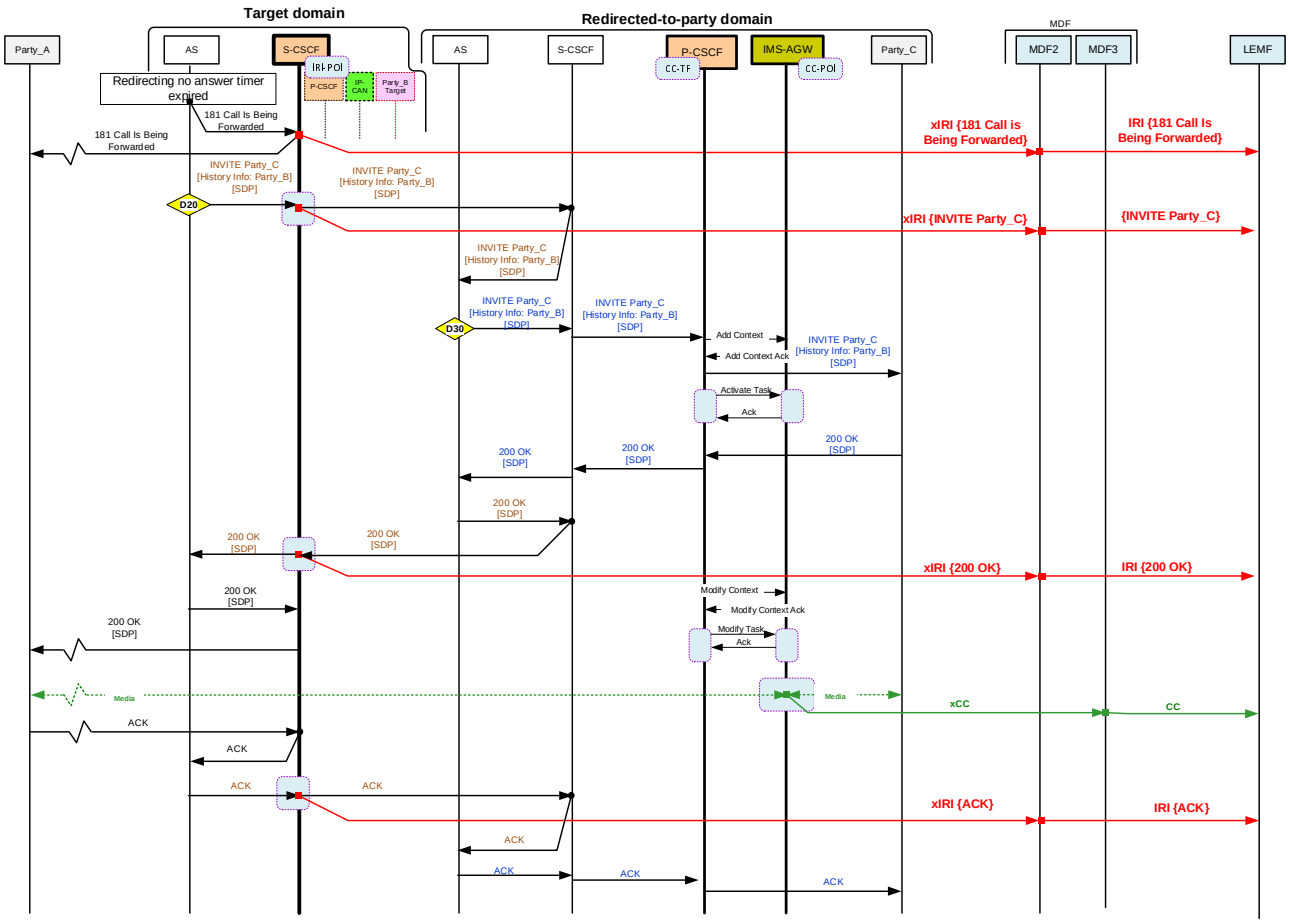


Figure 5.4.4-1: Party_A calls Party B (target) redirected to Party_C due to no answer (flow 2 of 2)

Figures 5.4.3-1 and 5.4.4-1 illustrate a case of an intra-CSP redirection no answer. Here, Party_A calls Party_B (target). The AS determines when an incoming IMS session to Party B is not answered within a timer-period, the call is to be redirected to Party_C served by the same CSP.

5.4.5 Inter-CSP redirection unconditional - Party_A calls Party_B (target) redirected to Party_C

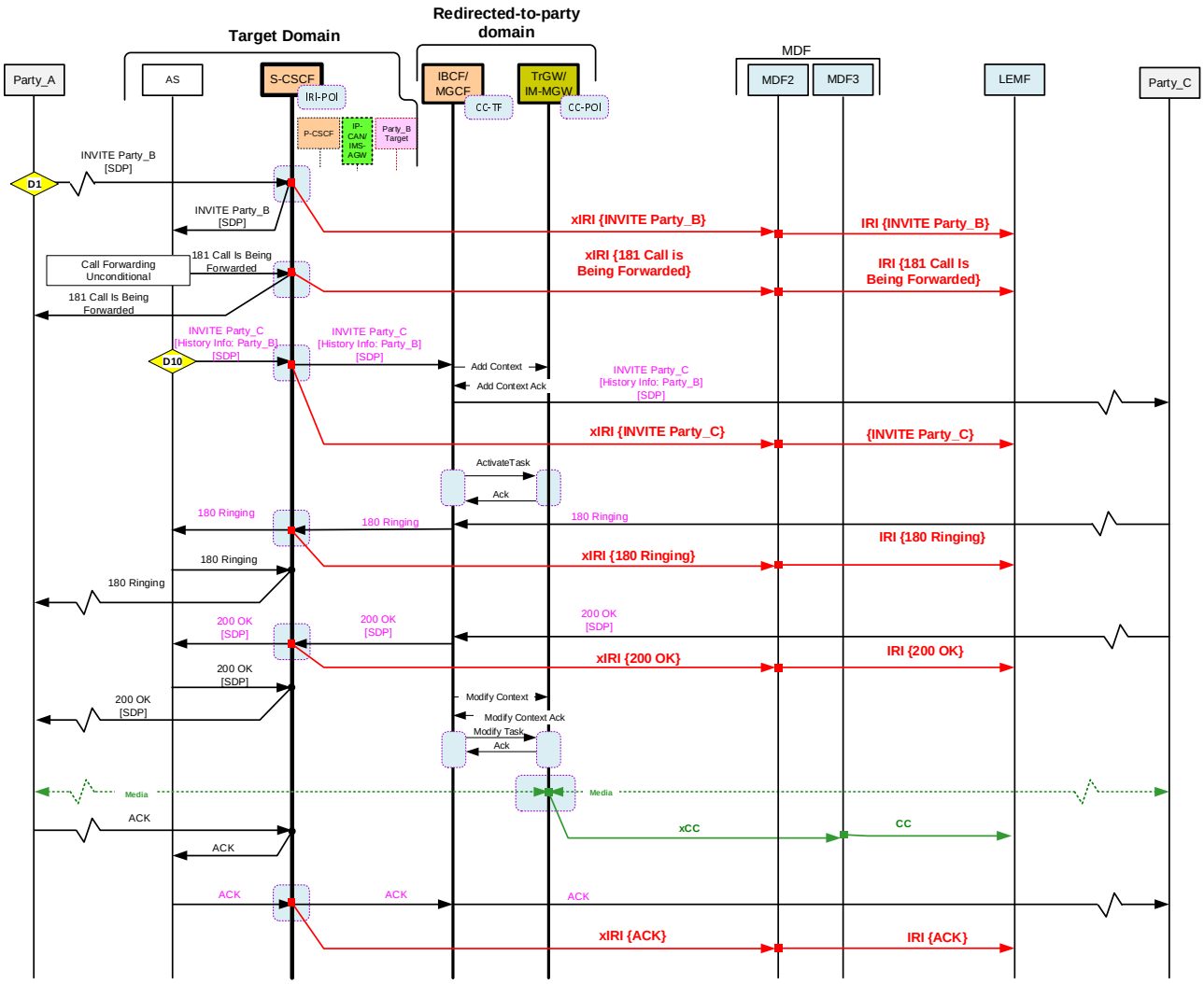


Figure 5.4.5-1: Inter-CSP direction - Party_A calls Party_B (target) redirected to Party_C

Figure 5.4.5-1 illustrates a case of an inter-CSP redirection unconditional. Here, Party_A calls Party_B (target). The AS determines that all incoming IMS sessions to the Party_B (target) have to be redirected to Party_C served in a different CSP domain.

5.5 Originating IMS sessions from the target – roaming

5.5.1 Introduction

Clause 5.5 describes the call flows to illustrate the scenarios of IMS sessions originated by a roaming target. The flows assume that the roaming architecture in use is Local Breakout (LBO).

In all flows, when the CSP domain of interest is the Home CSP, the CC interception trigger from the CC-TF in the IBCF is sent to the CC-POI in the TrGW when the IBCF sends the H.248 requests to provide the SDP information related to the IMS media sessions to the TrGW.

When the CSP domain of interest is the Visited CSP, the CC interception trigger from CC-TF in the P-CSCF is sent to the CC-POI in the IMS-AGW when the P-CSCF sends the H.248 requests to provide the SDP information related to the IMS media sessions to the IMS-AGW.

In the Home CSP domain, the IRI-POI in the S-CSCF generates the xIRI using the SIP messages. In the Visited CSP domain, the P-CSCF generates the xIRI using the SIP messages.

5.5.2 Party_A (outbound roaming target) calls Party_B

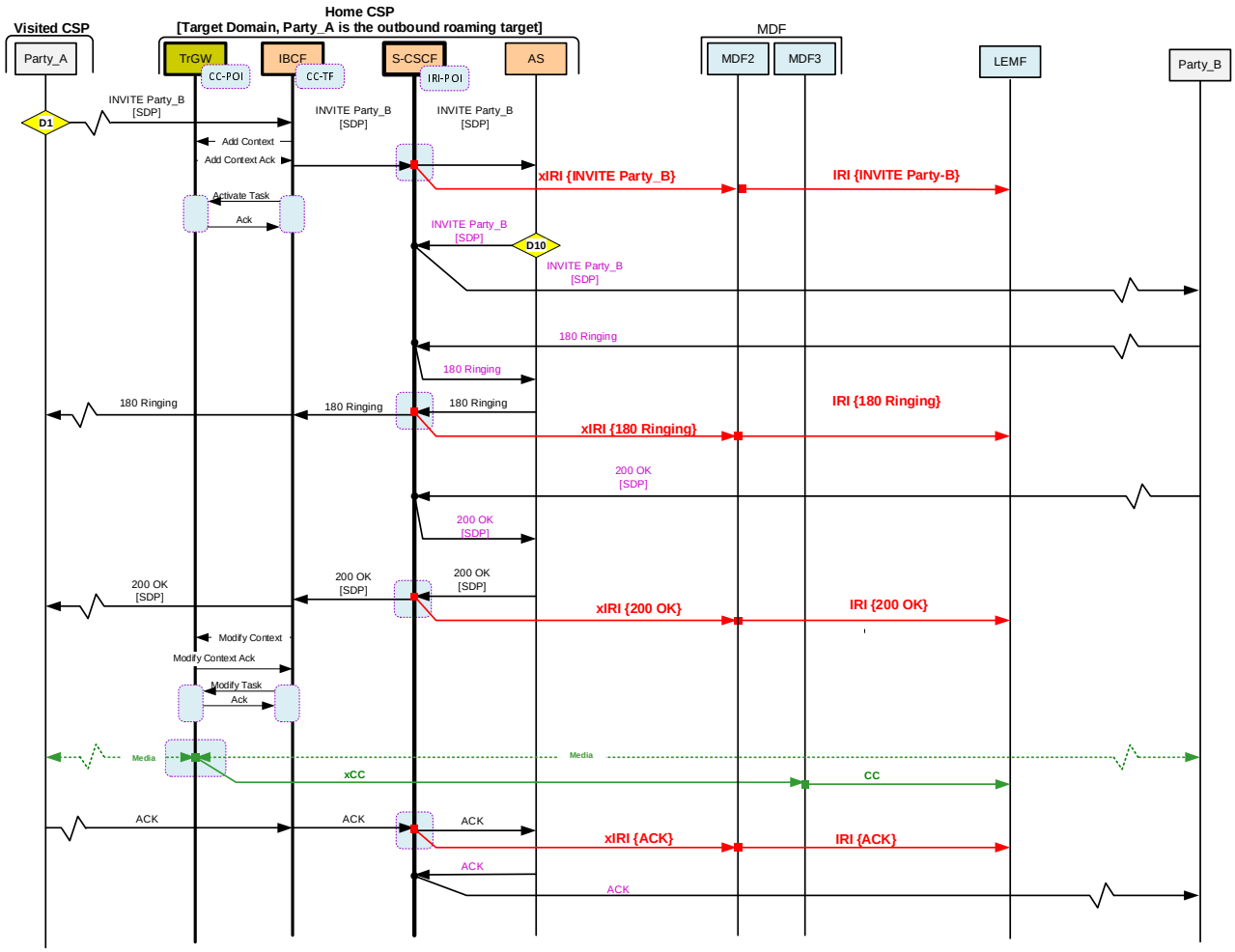


Figure 5.5.2-1: Party_A (outbound roaming target) calls Party_B

Figure 5.5.2-1 illustrates the case where Party_A (outbound roaming target) calls Party_B. The CSP domain of interest is a Home CSP.

5.5.3 Party_A (outbound roaming target) dials a special number

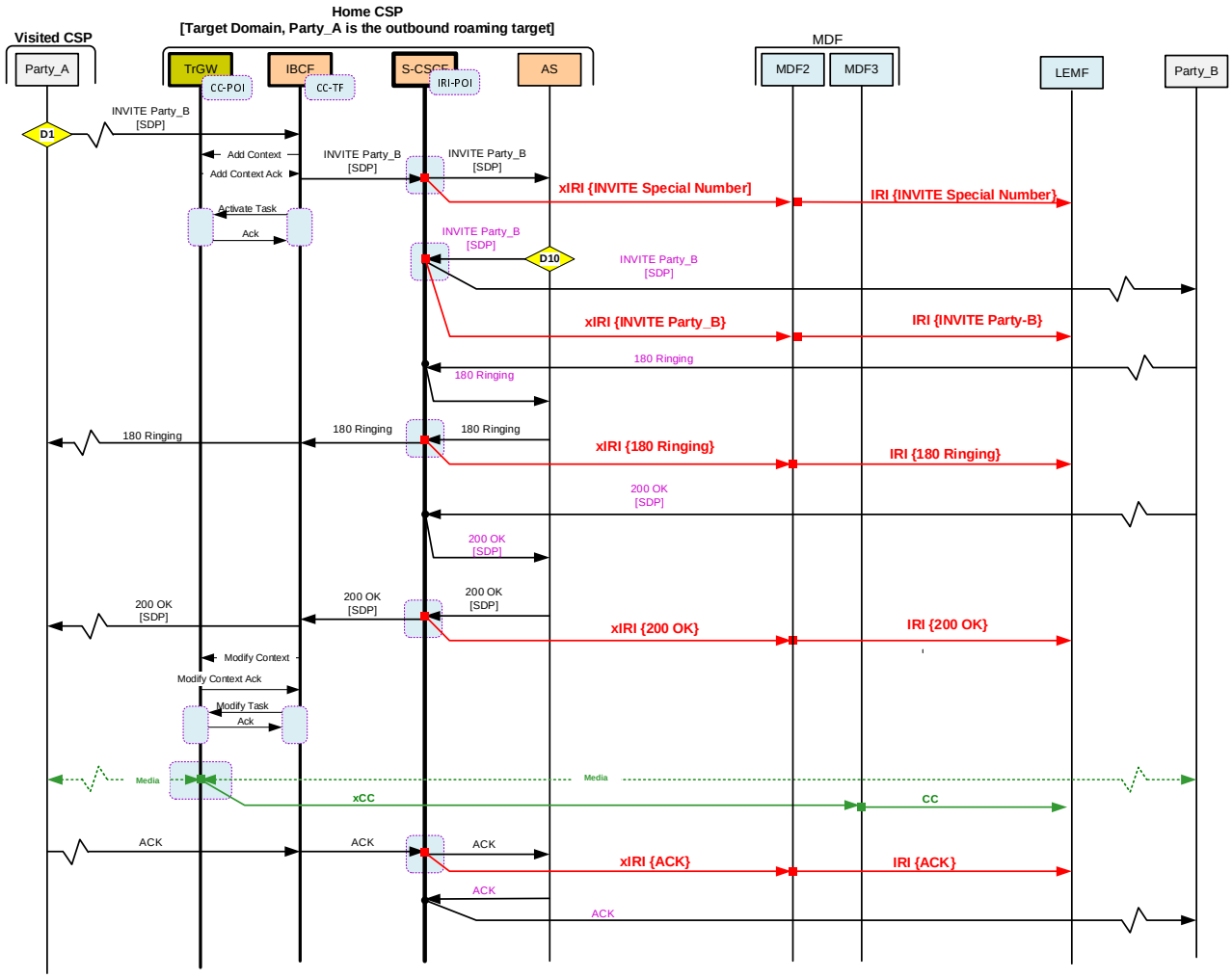


Figure 5.5.3-1: Party_A (outbound roaming target) dials a special number (translated to Party_B)

Figure 5.5.3-1 illustrates the case where Party_A (outbound roaming target) dials a Special Number (e.g. a speed call number or an 800-number), which is translated to Party_B by the AS. The CSP domain of interest is a Home CSP.

5.5.4 CC Unavailable in home CSP due to optimal media routing

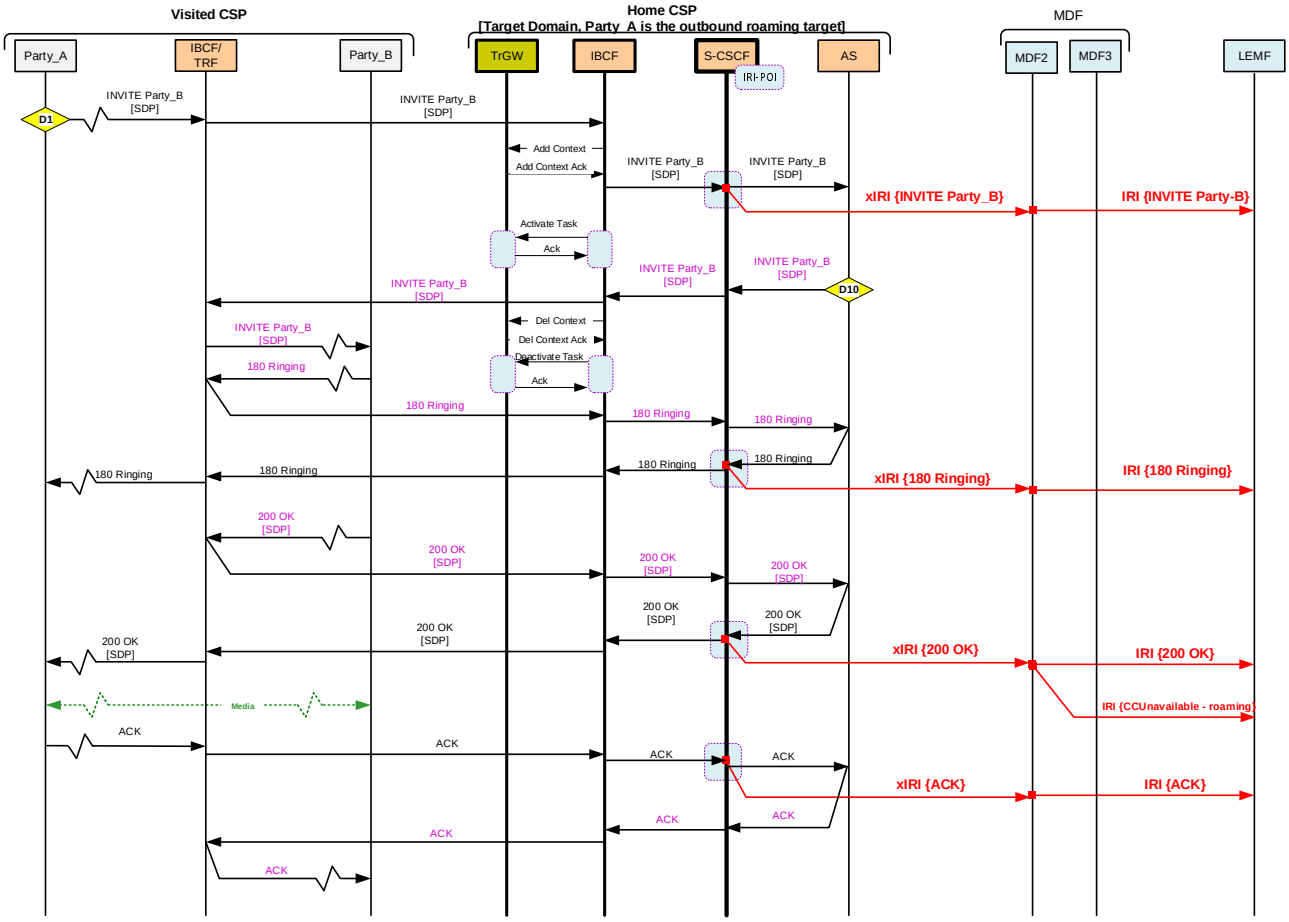


Figure 5.5.4-1: Party_A (outbound roaming target) calls Party_B, and optimal media routing is applied

Many different call scenarios exist that employ optimal media routing. In this particular example, the called party (Party_B) is served by the same CSP where the target (Party_A) is currently roaming (i.e. Home CSP of Party_B is the Visited CSP of Party_A).

NOTE: Different implementations can be employed to trigger the generation of the CCUnavailable IRI message.

5.5.5 Party_A (inbound roaming target) calls Party_B

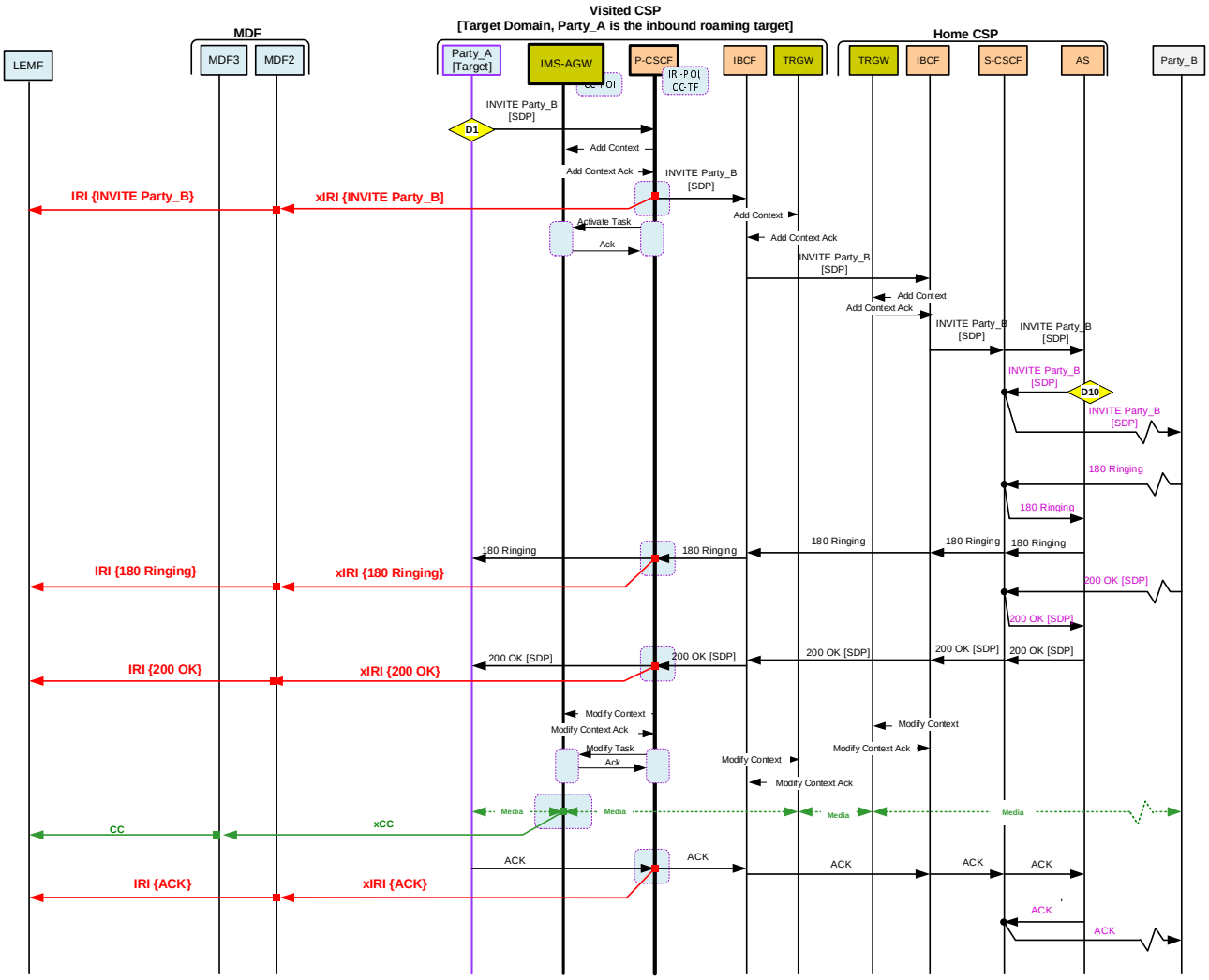


Figure 5.5.5-1: Party_A (inbound roaming target) calls Party_B

Figure 5.5.5-1 illustrates the case where Party_A (inbound roaming target) calls Party_B. The CSP domain of interest is a Visited CSP.

5.5.6 Party_A (inbound roaming target) dials a special number

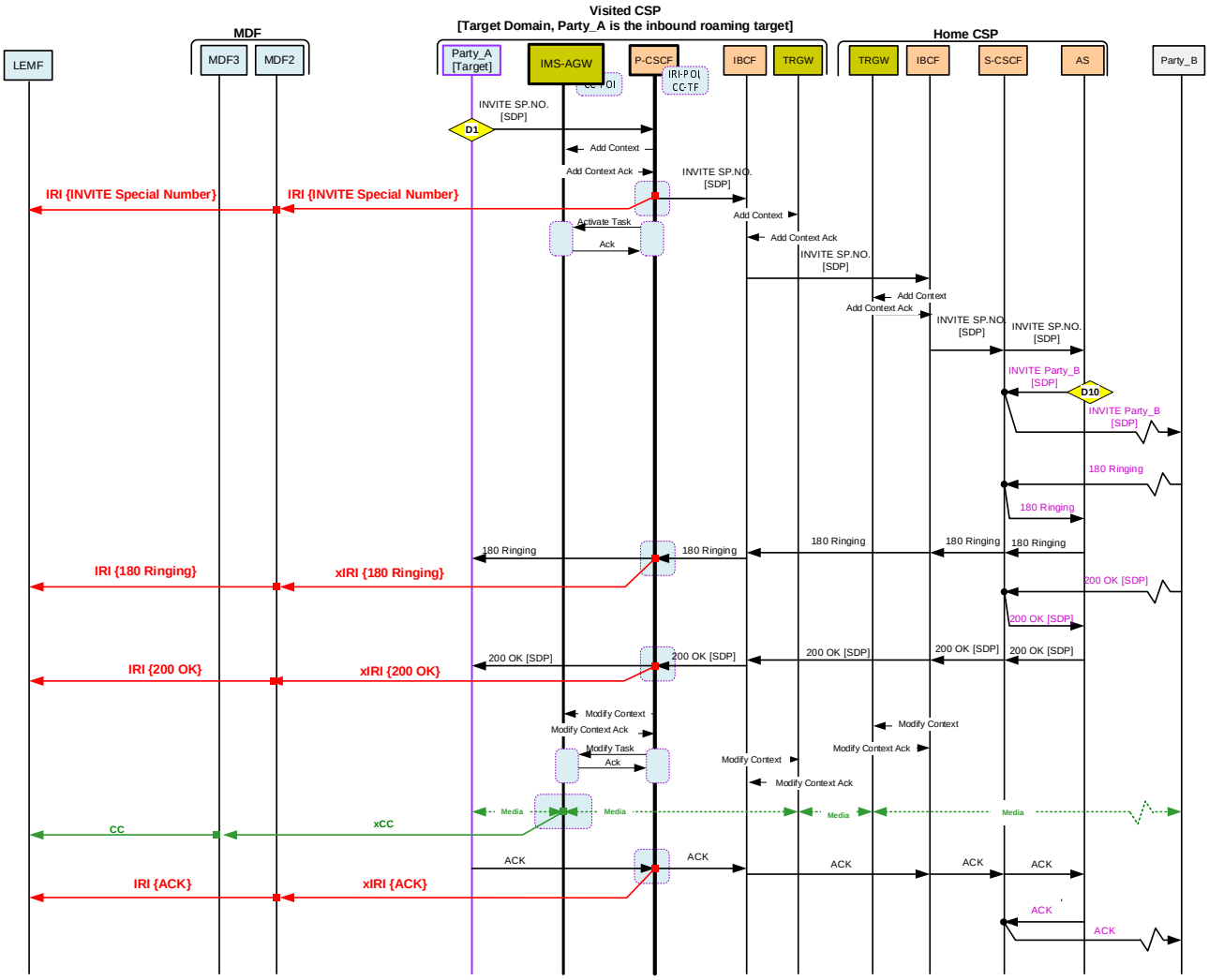


Figure 5.5.6-1: Party_A (inbound roaming target) dials a special number (translated to Party_B)

Figure 5.5.6-1 illustrates the case where Party_A (inbound roaming target) dials a Special Number (e.g. a speed call number or an 800-number), which is translated to Party B by the AS (in the Home CSP domain). The CSP domain of interest is a Visited CSP.

NOTE: Visited CSP is not aware of the fact that the Special Number is translated to Party_B in the Home CSP.

5.6 Terminating IMS sessions to the target – roaming

5.6.1 Introduction

Clause 5.6 describes the call flows to illustrate the scenarios of IMS sessions terminated to a roaming target. The flows assume that the roaming architecture in use is Local Breakout (LBO).

In all flows, when the CSP domain of interest is the Home CSP, the CC interception trigger from CC-TF in IBCF is sent to the CC-POI in TrGW when the IBCF sends the H.248 requests to provide the SDP information related to the IMS media sessions to the TrGW.

When the CSP domain of interest is the Visited CSP, the CC interception trigger from CC-TF in P-CSCF is sent to the CC-POI in the IMS-AGW when the P-CSCF sends the H.248 requests to provide the SDP information related to the IMS media sessions to the IMS-AGW.

In the Home CSP domain, the IRI-POI in the S-CSCF generates the xIRI using the SIP messages. In the Visited CSP domain, the P-CSCF generates the xIRI using the SIP messages.

5.6.2 Party_A calls Party_B (outbound roaming target)

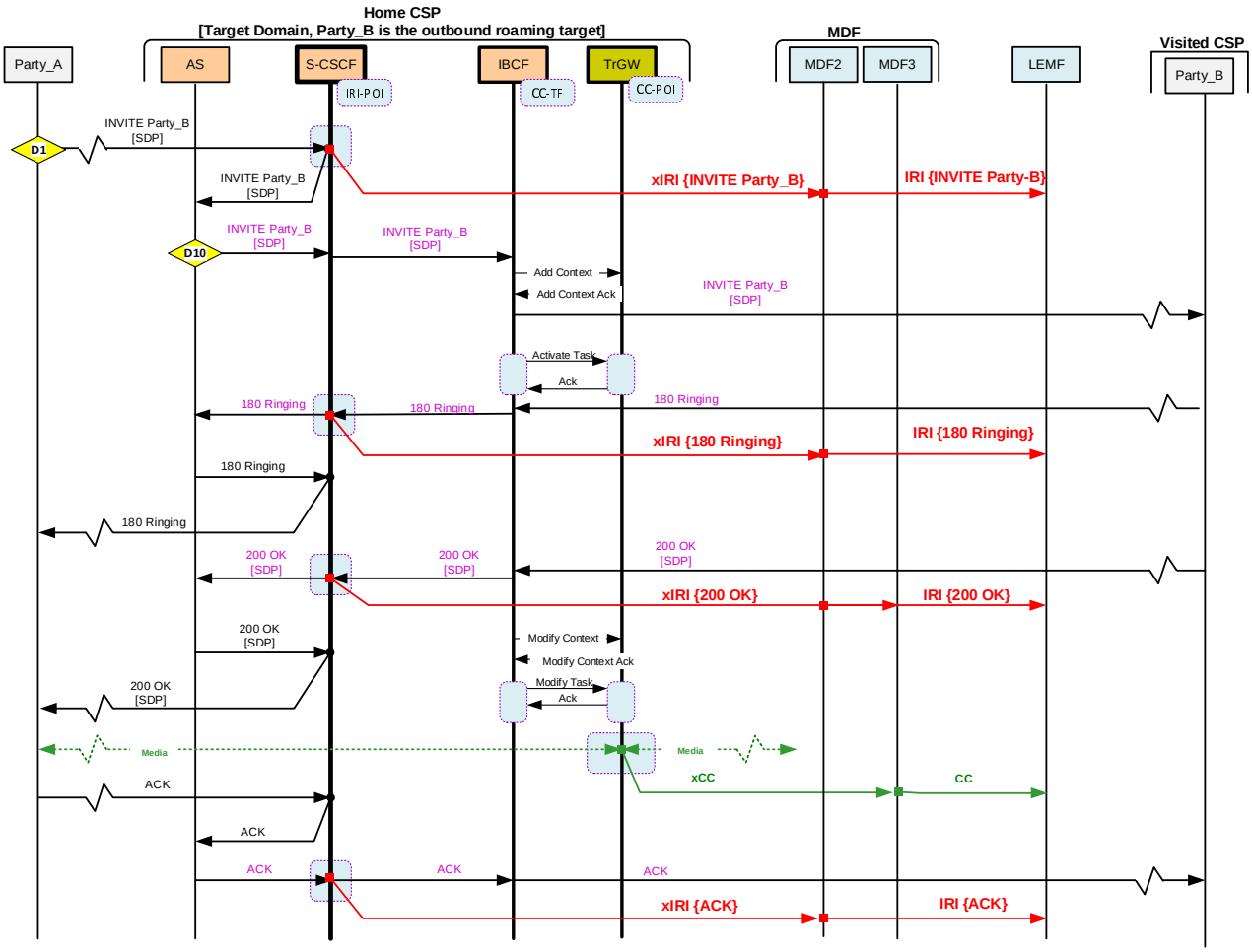


Figure 5.6.2-1: Party_A calls Party_B (outbound roaming target)

Figure 5.6.2-1 illustrates the case where Party_A calls Party_B (outbound roaming target). The CSP domain of interest is a Home CSP.

5.6.3 Party_A calls Party_B (outbound roaming target), redirected due to no answer

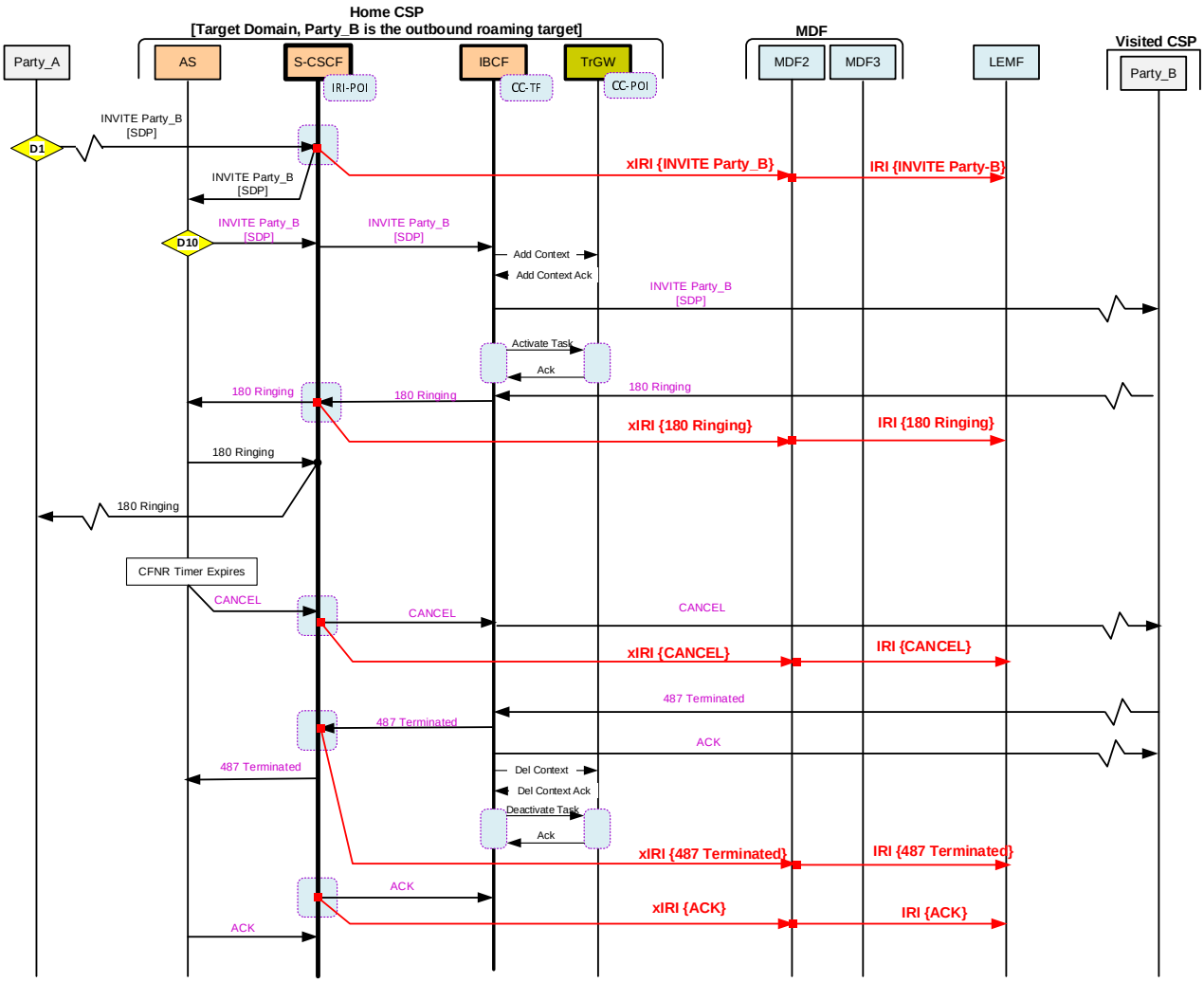


Figure 5.6.3-1: Party_A calls Party_B (outbound roaming target), redirected due to no answer

Figure 5.6.3-1 illustrates the case where an incoming IMS session to Party_B (outbound roaming target) is redirected when Party_B fails to answer before the redirection no answer timer expires.

NOTE: The redirection steps along with the LI aspects for those steps are same as shown in figure 5.4.4-1 when the redirected-to-party is in the same CSP domain as the Home CSP of Party B. The redirected-to-party may also be outbound roaming and in that case the session setup to that redirecting-to-party follows the steps shown in figure 5.6.2-1.

5.6.4 Party_A calls Party_B (inbound roaming target)

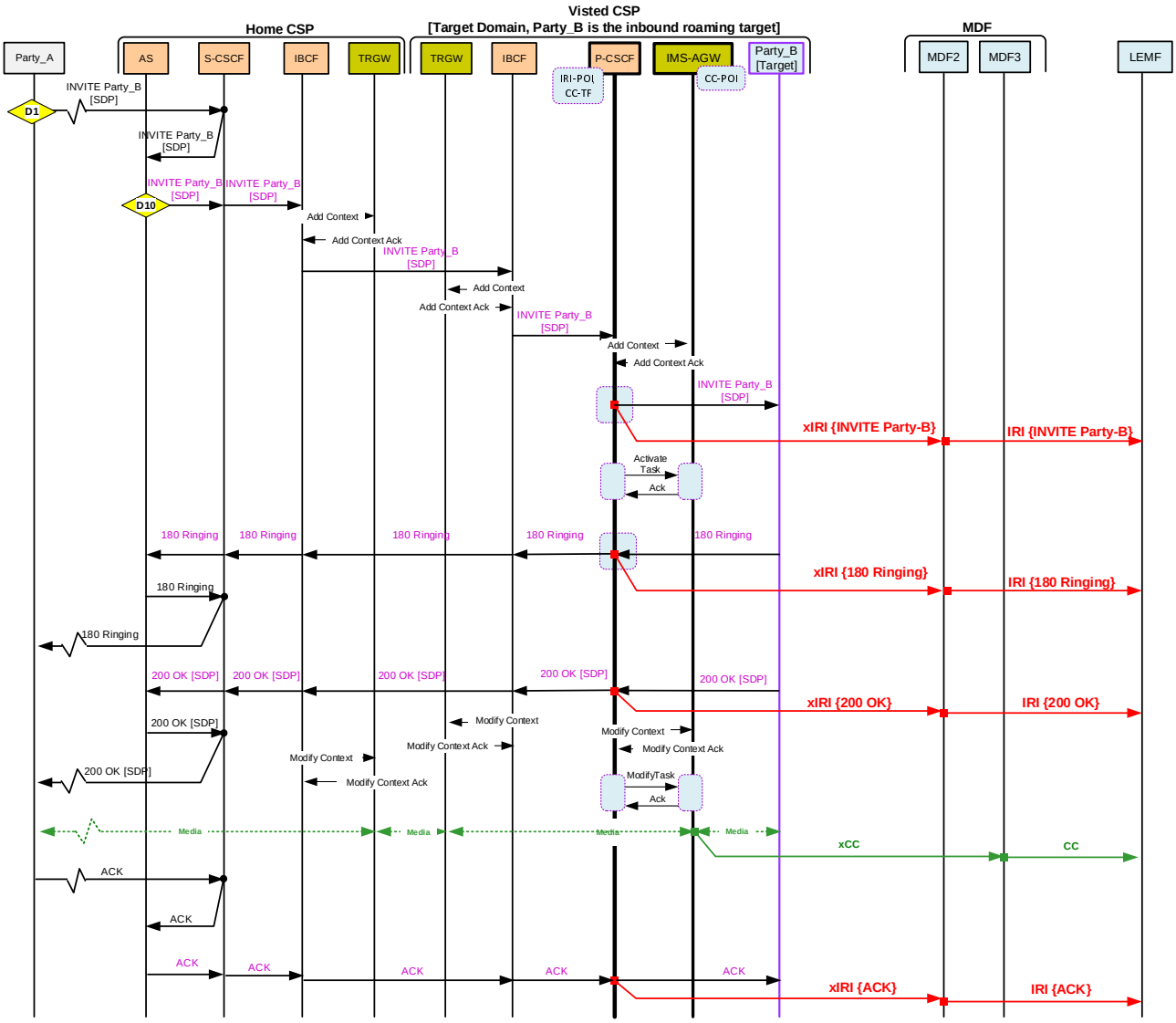


Figure 5.6.4-1: Party_A calls Party_B (inbound roaming target)

Figure 5.6.4-1 illustrates the case where Party_A calls an inbound roaming target Party B. The CSP domain of interest is a Visited CSP.

5.6.5 Party_A calls Party_B (inbound roaming target), redirected due to no answer

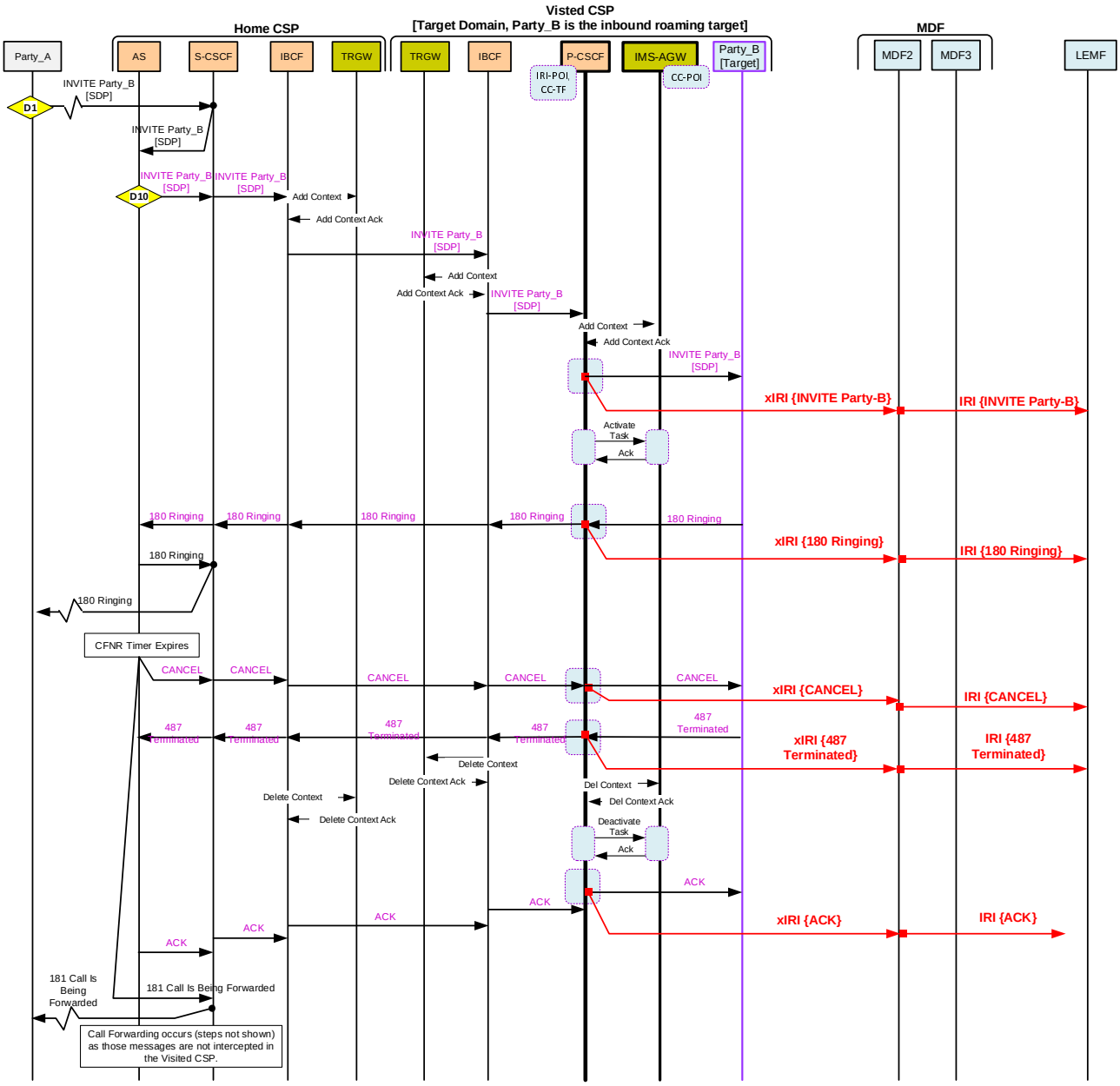


Figure 5.6.5-1: Party_A calls Party_B (inbound roaming target), redirected due to no answer

Figure 5.6.5-1 illustrates the case where an incoming IMS session to Party_B (inbound roaming target) is redirected by the Home CSP when Party_B fails to answer before the redirection no answer timer expires.

NOTE: The Visited CSP is not aware that the session is redirected. No interception occurs in the visited CSP after the redirection. Therefore, session setup to the redirected-to-party is not shown.

5.7 Ad-hoc IMS conference established by the target

5.7.1 Introduction

This clause gives the call flows to illustrate the steps related to ad-hoc IMS conference established by the target. The flows assume that the Party_A (target) has already made two sessions, one to Party_B and one to Party_C, and placed both sessions on hold so as to merge the two sessions into a conference.

Figure 5.7.2-1 illustrates the case where Party_A (target) creates the conference.

Figure 5.7.3-1 and figure 5.7.3-2 illustrate the case where Party_A (target) brings Party_C into the conference.

Figure 5.7.4-1 and figure 5.7.4-2 illustrate the case where Party_A (target) brings Party_B into the conference.

Figure 5.7.5-1 illustrates the case where Party_C drops out of the conference session.

Figure 5.7.6-1 illustrates the case where the session between two parties (Party_A (target) and Party_B) is converted back to a normal two-party session.

Figure 5.7.7-1 illustrates the case where Party_A (target) places the conference on hold.

Figure 5.7.8-1 illustrates the case where Party_A (target) retrieves the held conference from hold.

Some of the steps may be executed by the target's UE automatically (in other words, no action is required by the target). For example, when the target tries to merge the sessions into a conference, the target's UE may execute the steps shown in figure 5.7.2-1, figure 5.7.3-1, figure 5.7.3-2, figure 5.7.4-1 and figure 5.7.4-2 automatically in a serial fashion. The same way, the steps shown in figure 5.7.6-1 may be executed automatically after the steps shown in figure 5.7.5-1 when one of the conferees drops out of the conference.

Figures 5.7.7-1 and figure 5.7.8-1 are not part of the conferencing steps, however they are included here to show how the content of a held conference (a requirement in some countries) is delivered to the LEMFs.

5.7.2 Party_A (target) creates the conference

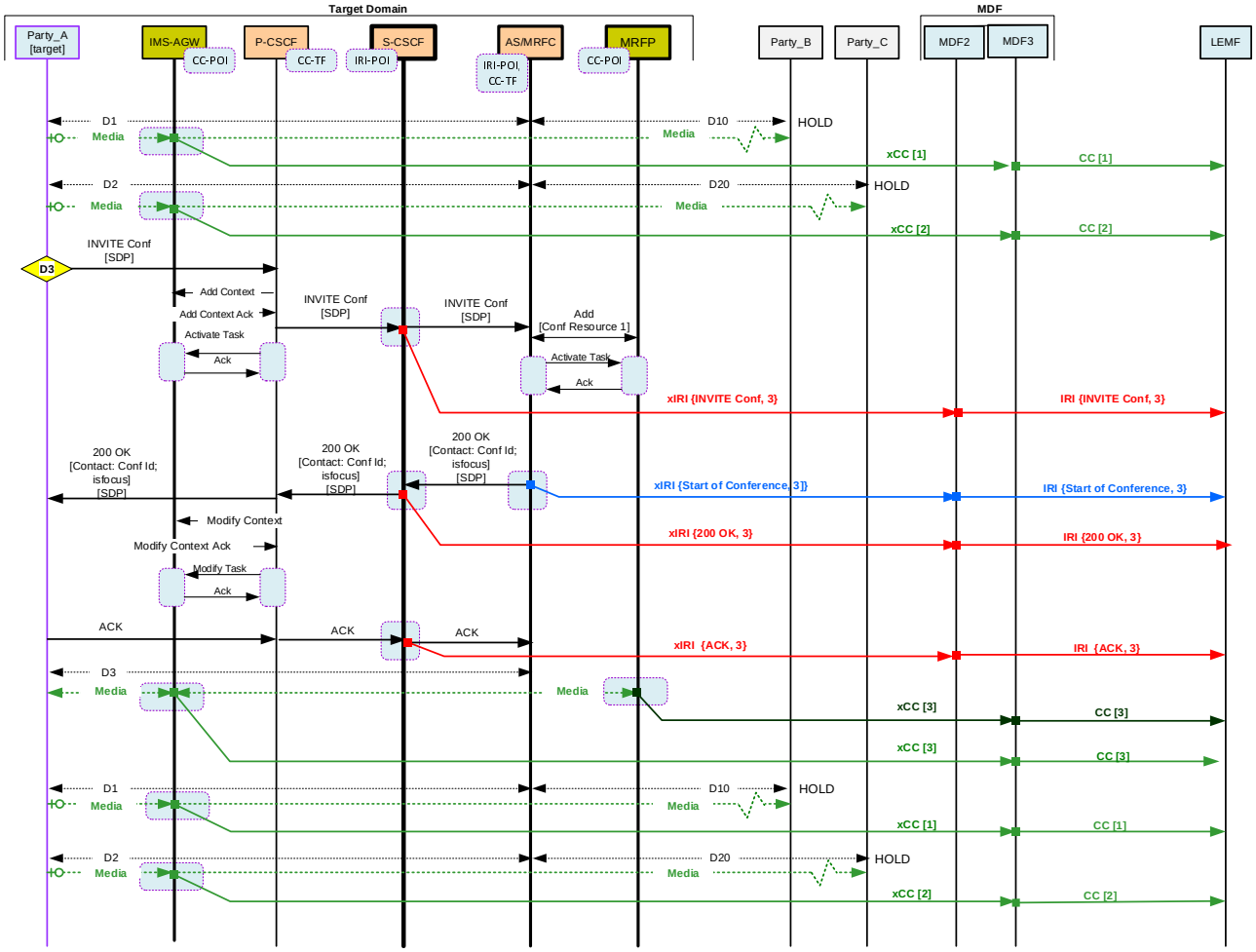


Figure 5.7.2-1: Party_A (target) creates the conference

D1 and D10 represent the dialogue of the original session between the Party_A (target) and the Party_B. D2 and D20 represent the original the dialogue of the original session between Party_A (target) and the Party_C. D3 represents the new dialogue of session between Party_A and the conference.

The IRI/CC delivered for D1 and D10 use the Correlation ID 1. The IRI/CC delivered for D2 and D20 use the Correlation ID 2. The IRI/CC delivered for the conferencing (i.e. D3) uses the Correlation ID 3.

5.7.3 Party_C (conferee) joins the conference

This flow is illustrated in two figures: Figure 5.7.3-1 and figure 5.7.3-2.

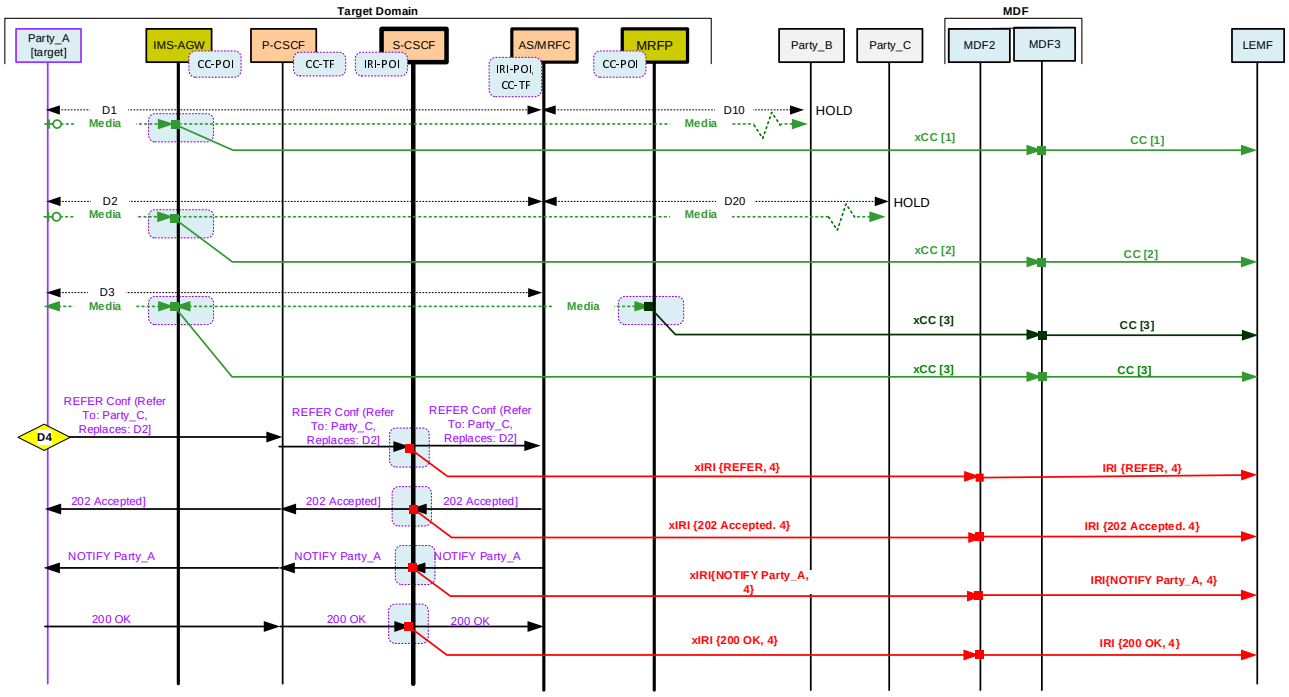


Figure 5.7.3-1: Party_C joins the conference (flow 1 of 2)

D1 and D10 represent the dialogue of the original session between Party_A (target) and Party_B. D2 and D20 represent the original the dialogue of the original session between Party_A (target) and Party_C. D3 represents the dialogue of the session between Party_A and the conference. D4 represents the dialogue that the Party_A (target) uses to refer Party_C to the conference.

NOTE: Here, REFER sent by Party_A is outside of any existing dialogues. Sending of REFER inside a dialogue is also possible.

The IRI/CC delivered for D1 and D10 use the Correlation ID 1. The IRI/CC delivered for D2 and D20 use the Correlation ID 2. The IRI/CC delivered for the conferencing (i.e. D3) uses the Correlation ID 3. The IRI for D4 uses the Correlation ID 4.

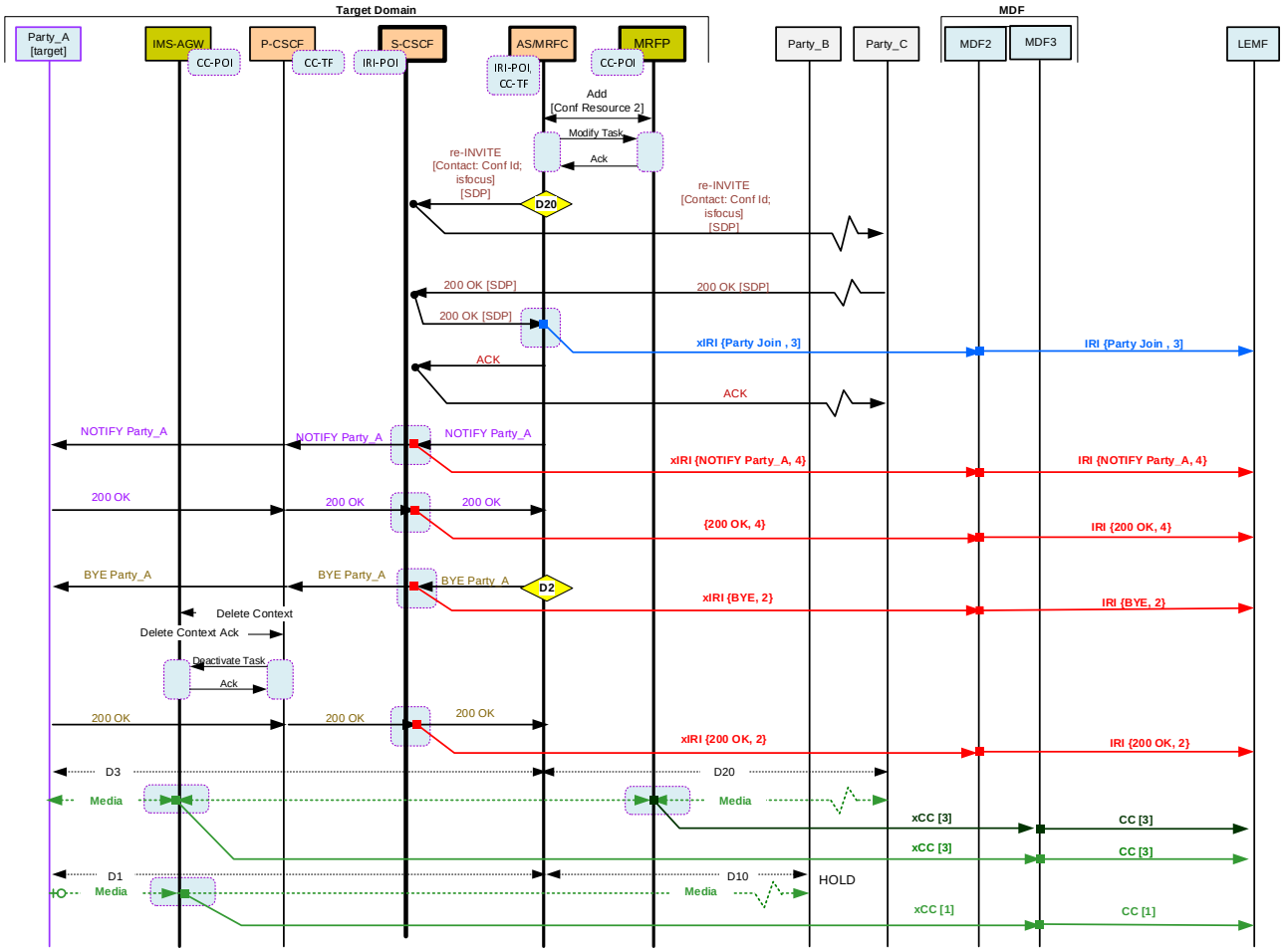


Figure 5.7.3-2: Party_C joins the conference (flow 2 of 2)

At the end of this flow, Party_A (target) and Party_C are connected via the conference. Party_B is still on hold. Part of the original session between Party_A (target) and Party_C (D2) is released with D20 now representing the session between the Party_C and the conference.

5.7.4 Party_B joins the conference

This flow is illustrated in two figures: figure 5.7.4-1 and figure 5.7.4-2.

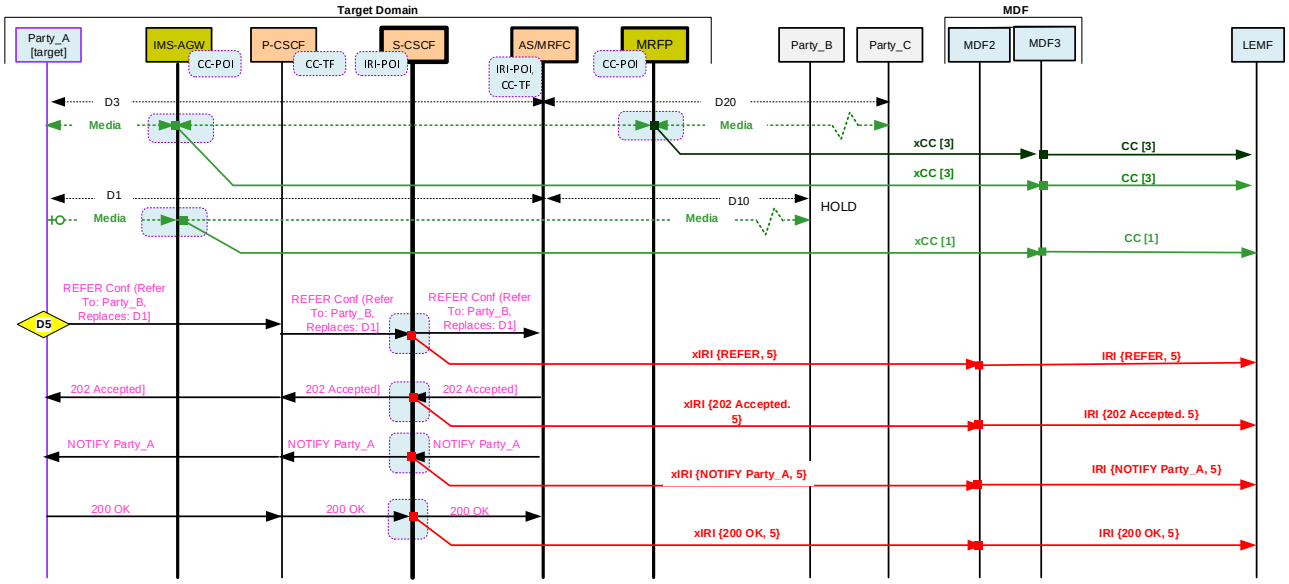


Figure 5.7.4-1: Party_B joins the conference (flow 1 of 2)

D3 represents the dialogue of the session between Party_A (target) and the conference. D20 represents the dialogue between Party_C and the conference. D1 and D10 represent the original the dialogue of the original session between Party_A (target) and Party_B. D5 represents the dialogue that the Party_A (target) uses to refer Party_B to the conference.

NOTE: Here, REFER is sent by Party_A outside of any existing dialogues. Sending of REFER inside a dialogue is also possible.

The IRI/CC delivered for D1 and D10 use the Correlation ID 1. The IRI/CC delivered for the conferencing (i.e. D3) uses the Correlation ID 3. The IRI for D5 uses the Correlation ID 5.

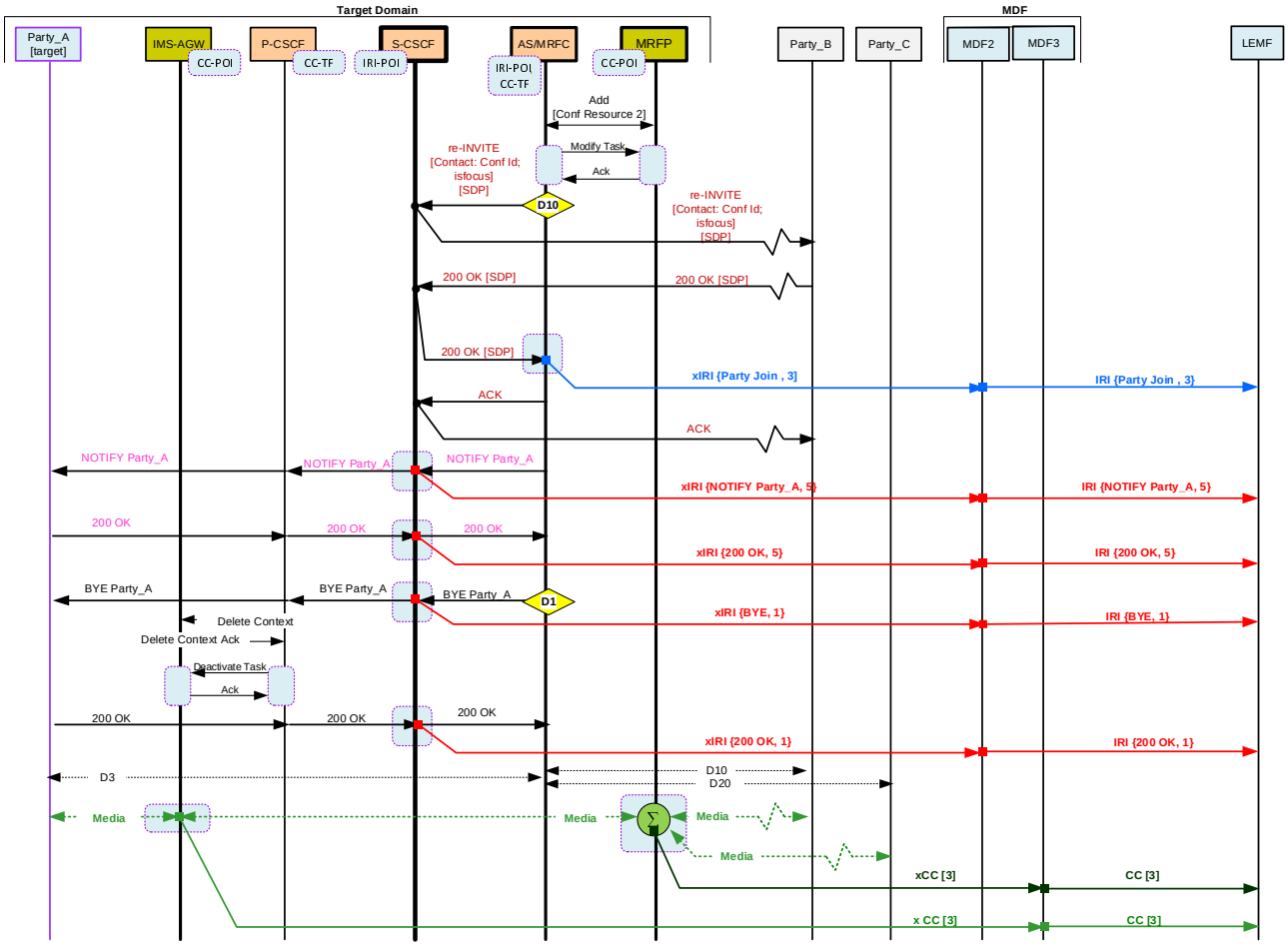


Figure 5.7.4-2: Party_B joins the conference (flow 2 of 2)

At the end of this flow, Party_A (target), Party_B and Party_C are connected via the conference. Part of the original session between Party_A (target) and Party_B (D1) is released with D10 now representing the session between Party_B and the conference.

5.7.5 Party C drops out of the conference

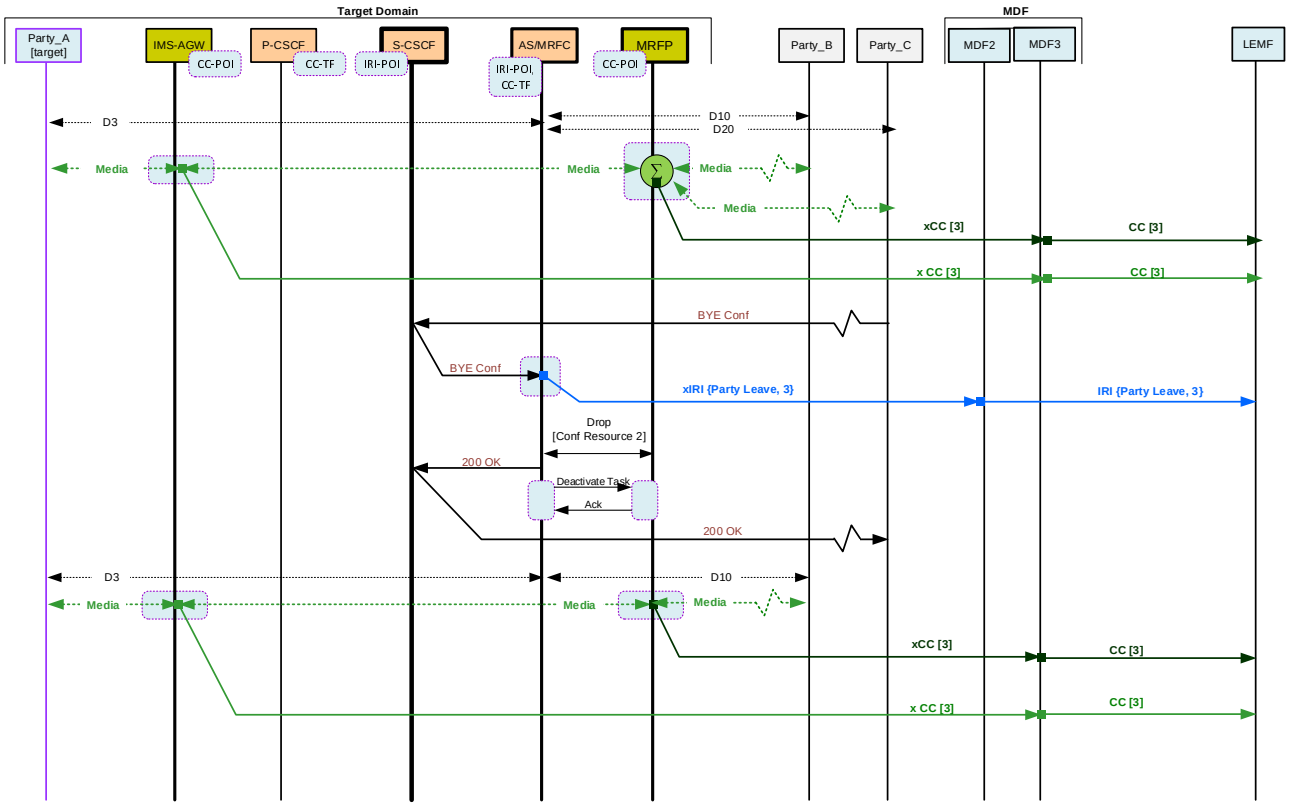


Figure 5.7.5-1: Party_C drops out of the conference

D3 represents the dialogue of the session between Party A (target) and the conference. D20 represents the dialogue between Party C and the conference. D10 represents the dialogue between Party B and the conference.

The IRI/CC delivered for the conferencing (i.e. D3) uses the Correlation ID 3.

At the end of this flow, Party A (target) and Party B are connected through the conference.

5.7.6 Reconfiguration from conference to two-party session

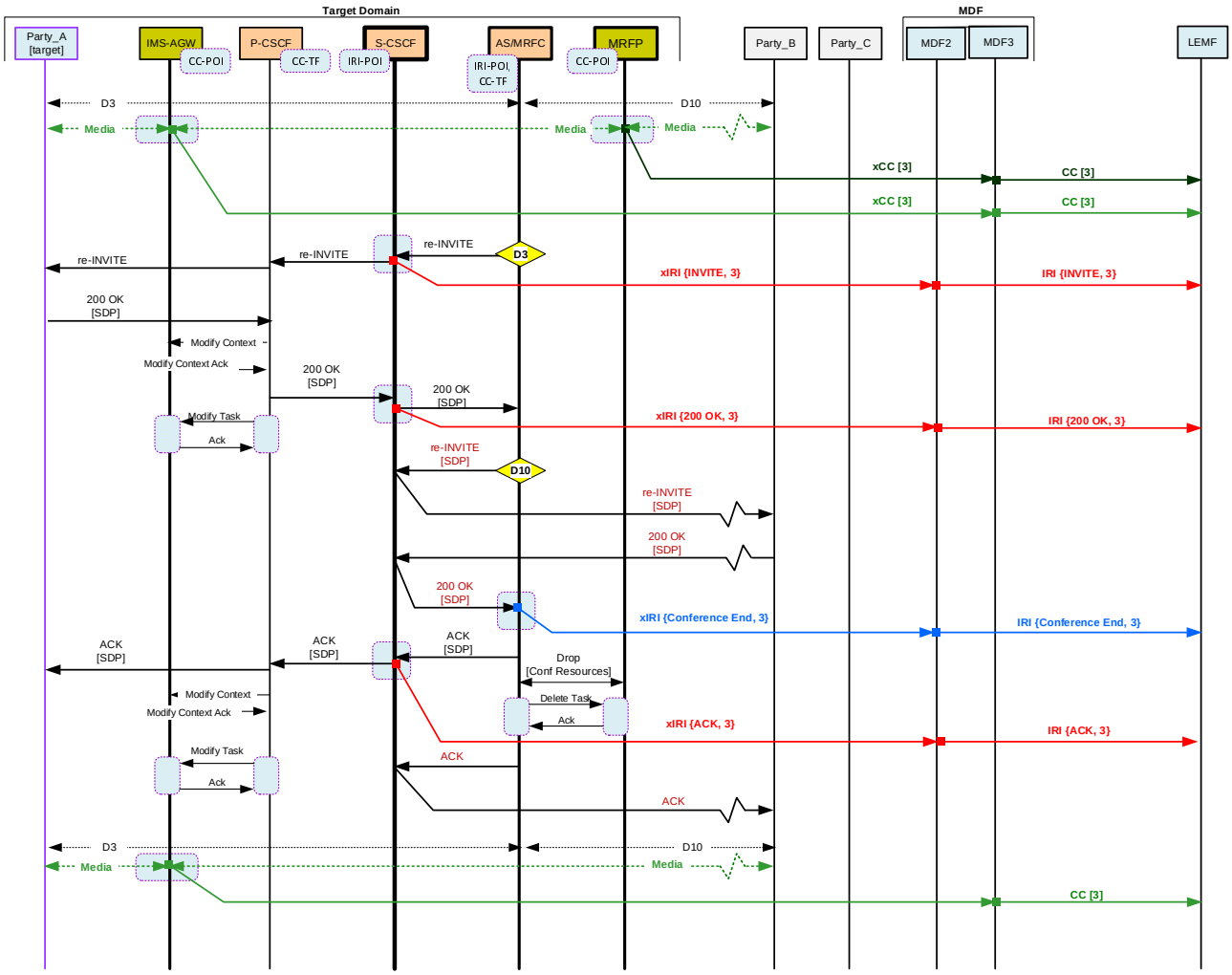


Figure 5.7.6-1: Conference is reconfigured to a two-party session

D3 represents the dialogue of the session between Party A (target) and the conference. D10 represents the dialogue between Party B and the conference.

The IRI/CC delivered for the conferencing (i.e. D3) uses the Correlation ID 3.

At the end of this flow, Party A (target) and Party B are connected directly (without the conference). The IRI/CC delivered for this session between Party A (target) and Party B (D3 and D10) uses the Correlation ID 3.

NOTE: Reconfiguration may be done using other ways (e.g. AS/MRFC sending a REFER to Party_A (target) to invite Party C). The sequence of steps and the Correlation ID used can be different with such a flow.

5.7.7 Party_A (target) places conference on hold

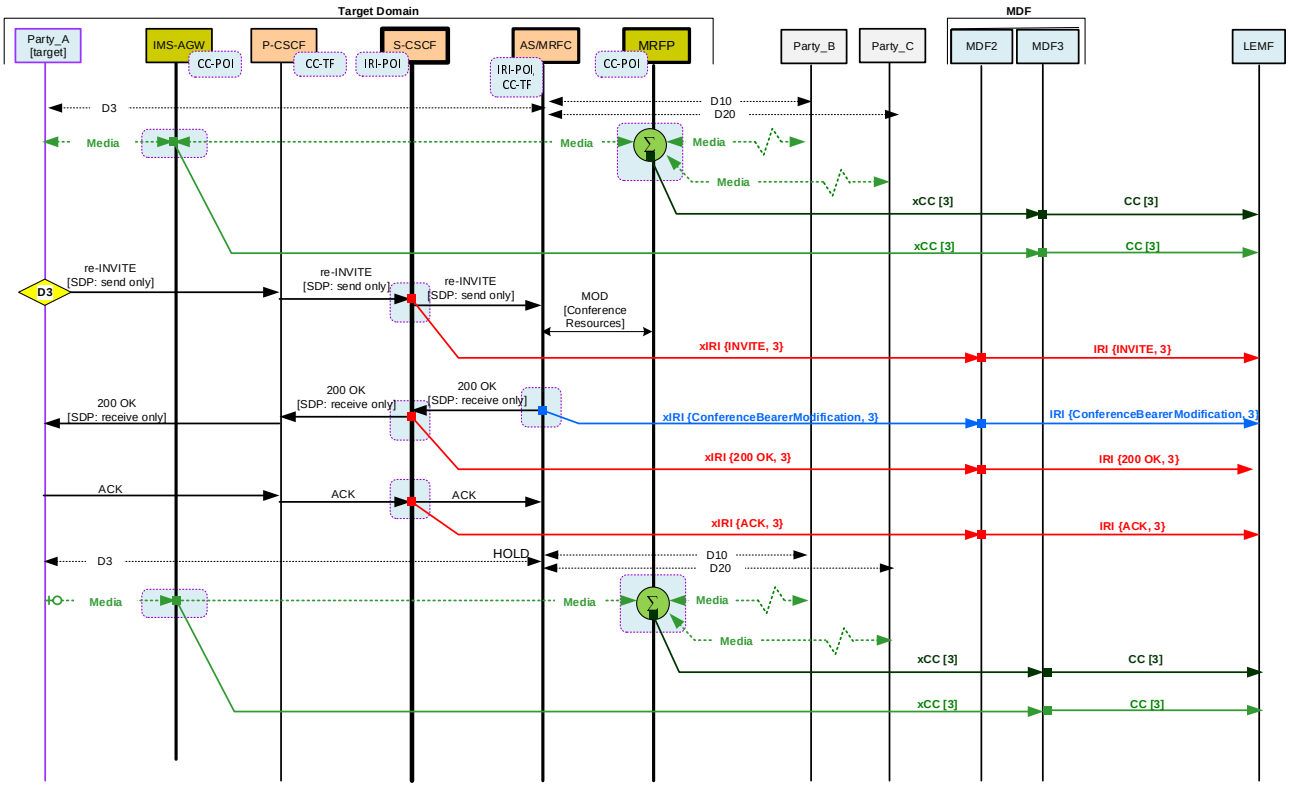


Figure 5.7.7-1: Party_A (target) places a conference on hold

D3 represents the dialogue of the session between Party A (target) and the conference. D20 represents the dialogue between Party C and the conference. D10 represents the dialogue between Party B and the conference.

The IRI/CC delivered for the conferencing (i.e. D3) uses the Correlation ID 3.

At the end of this flow, Party_B and Party_C can still communicate via the conference, but without the Party_A. The CC delivered from the MRFP contains the communication content of that conversation.

NOTE: Similar steps as shown here are used when Party_A (target) places a two-party session from hold.

5.7.8 Party_A (target) retrieves conference from hold

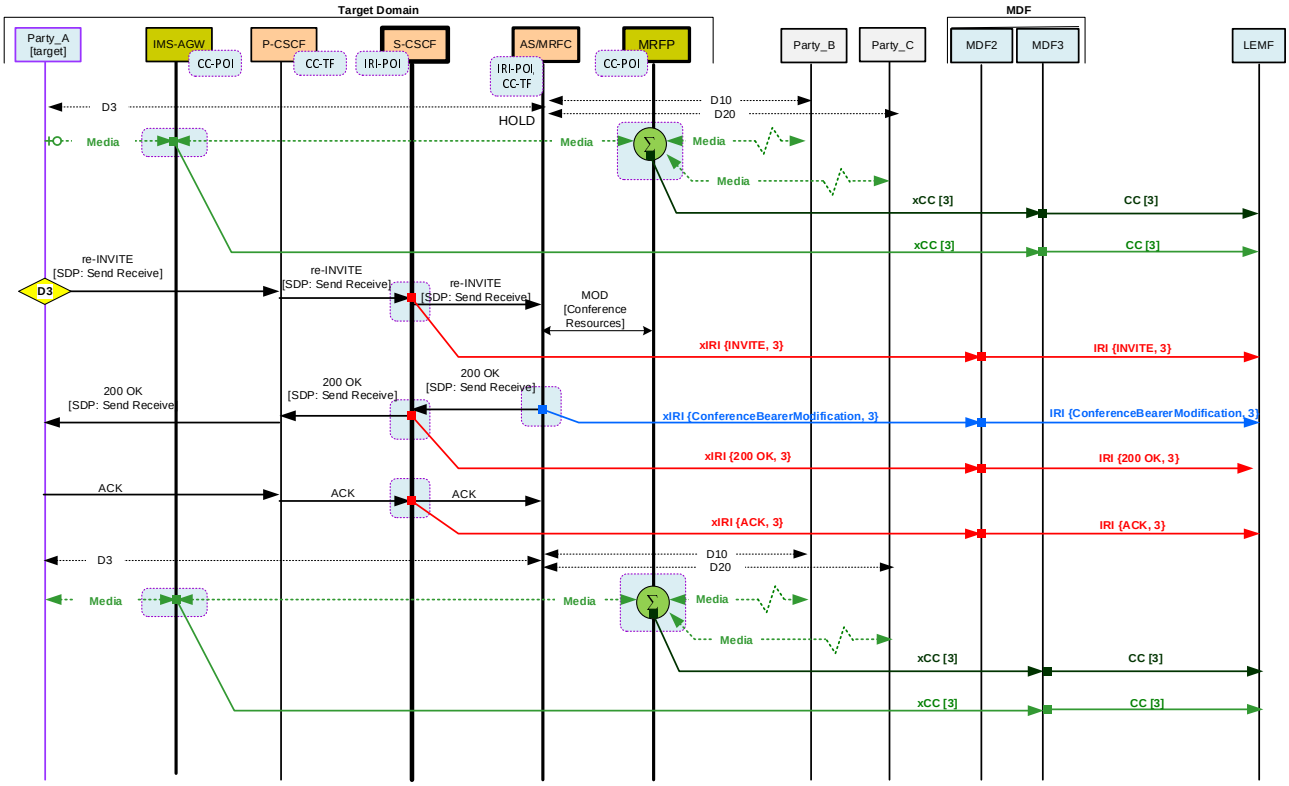


Figure 5.7.8-1: Party A (target) retrieves conference from hold

D3 represents the dialogue of the session between Party A (target) and the conference. D20 represents the dialogue between Party C and the conference. D10 represents the dialogue between Party B and the conference.

The IRI/CC delivered for the conferencing (i.e. D3) uses the Correlation ID 3.

At the end of this flow, Party A (target), Party B and Party C are communicating via the conference.

NOTE: Similar steps as shown here are used when Party A (target) retrieves a two-party session from hold.

5.8 Home routed roaming architecture

5.8.1 General

Five category call flows are presented in this clause:

- An inbound roaming target originates a voice call. The CC interception is required.
- An inbound roaming user originates a voice call. The CC interception is not required.
- A voice call is terminated to an inbound roaming target. The CC interception is required.
- An interception is activated while an inbound roaming user is active on a call (aka mid-call interception).
- A relocation occurs in the NF that provides the BBIFF functions.

In all the call flows, the target identity is the IMPU or the IMPI or the PEI (or IMEI). All the call flows assume that the SIP messages and the media are not encrypted at the NF that provides the BBIFF functions.

NOTE 1: In all the call flows, the term *channel setup* is used as a common term to represent the creation of a bearer within an EPC PDN connection or the creation of a QoS flow within a 5GC PDU session. The same term is used in TS 33.127 [2] as well.

Independently of any active intercept on a target, the BBIFF-C (or BBIFF in EPC without CUPS) notifies the LMISF-IRI whenever an IMS signalling channel or IMS media channel for a S8HR APN or a N9HR DNN is created, modified or deleted. Such notifications include the most recently known UE location information at the NF that has the BBIFF (i.e. the SGW/SGW-C receives from the MME in the case of S8HR and SMF receives from the AMF in the case of N9HR). The LMISF-IRI includes the latest UE location information in the applicable xIRI that it sends to the MDF2 for active intercepts.

NOTE 2: The BBIFF-C to BBIFF-U interactions to pass the correlation information is not specified in this TR.

NOTE 3: A voice call referenced in this clause is an IMS session within the IMS domain (in HPLMN).

5.8.2 Call origination (IRI + CC)

Figure 5.8.2-1 illustrates a call flow where an inbound roaming target originates a voice call. In the flow, Party_A (target) calls Party_B. The flow shows that Party_B is also an IMS user (SIP messages are shown), however, Party_B can also be a non-IMS user served by CS domain.

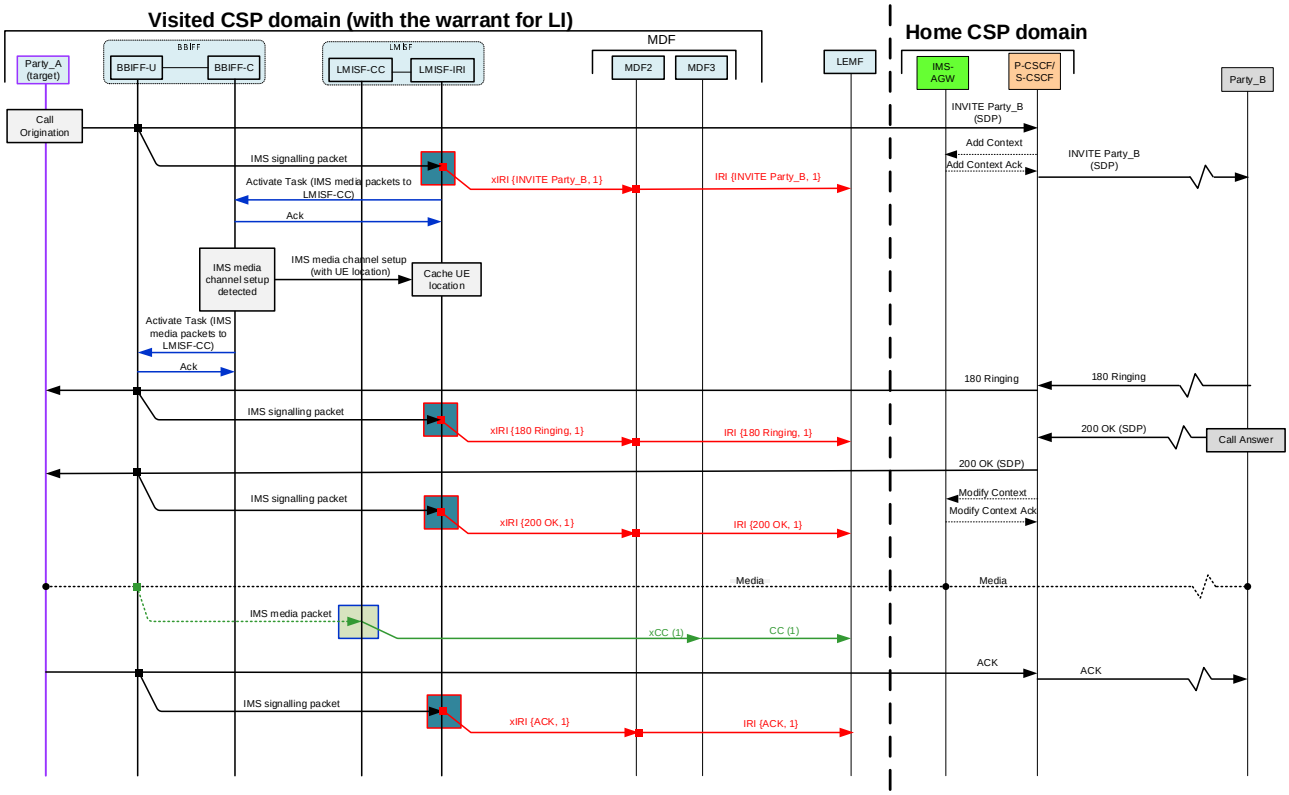


Figure 5.8.2-1: Call Origination from an inbound roaming target with HR – IRI + CC

The LMISF-IRI examines the SIP messages present within the IMS signalling packets received from the BBIFF-U to verify whether the SIP headers pointing to the calling party (e.g. P-Preferred-Identity, From, additional identities as specified in TS 24.174 [7]) is a target.

5.8.3 Call Origination (IRI only)

Figure 5.8.3-1 illustrates a call flow where an inbound roaming target originates a voice call. The lawful interception does not require CC interception. In the flow, Party_A (target) calls Party_B. The flow shows that Party_B is also an IMS user (SIP messages are shown), however, Party_B can also be a non-IMS user served by CS domain.

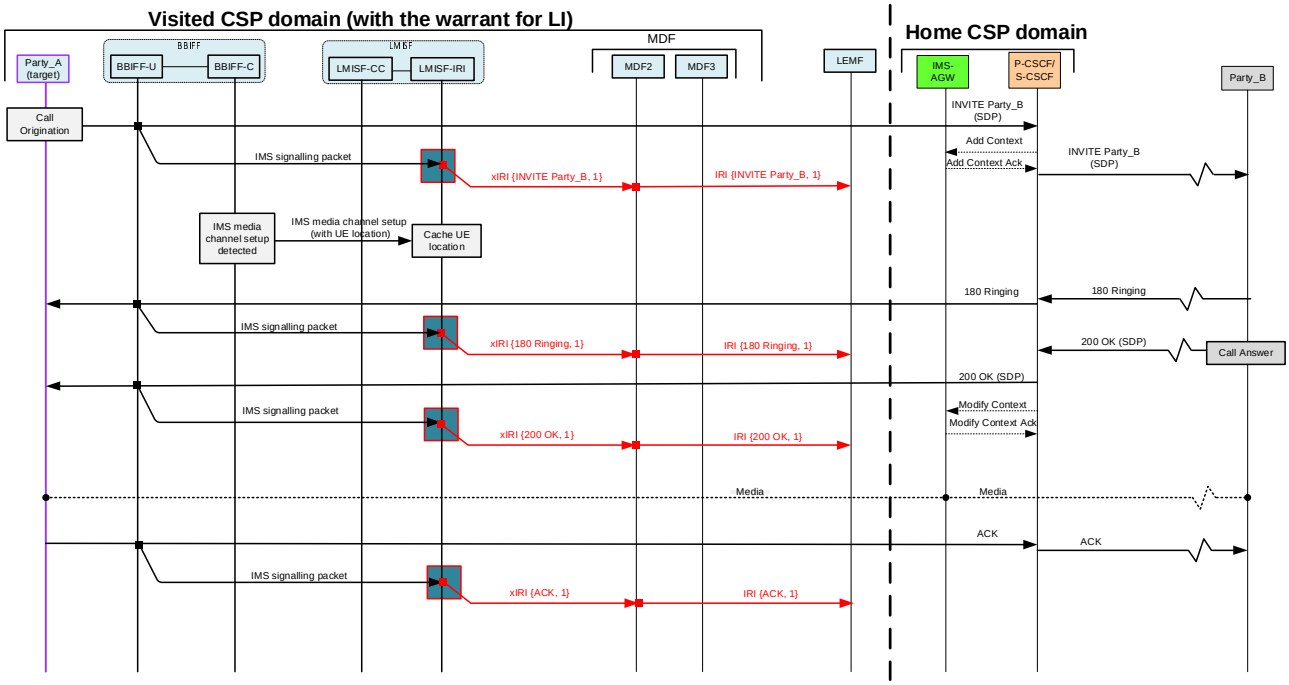


Figure 5.8.3-1: Call Origination from an inbound roaming target with HR – IRI only

As described in clause 5.8.2, for call originations from a target, the LMISF-IRI examines the SIP messages present within the IMS signalling packets received from the BBIFF-U to verify whether the SIP headers pointing to the calling party is a target.

In this flow, since no CC delivery is required for voice calls, no per-call based interaction is required between the LMISF-IRI and the BBIFF-C.

5.8.4 Call termination (IRI + CC)

Figure 5.8.4-1 illustrates a call flow where an inbound roaming target receives a voice call. In the flow, Party_A calls Party_B (target). The flow shows that Party_A is also an IMS user (SIP messages are shown), however, Party_A can also be a non-IMS user served by CS domain.

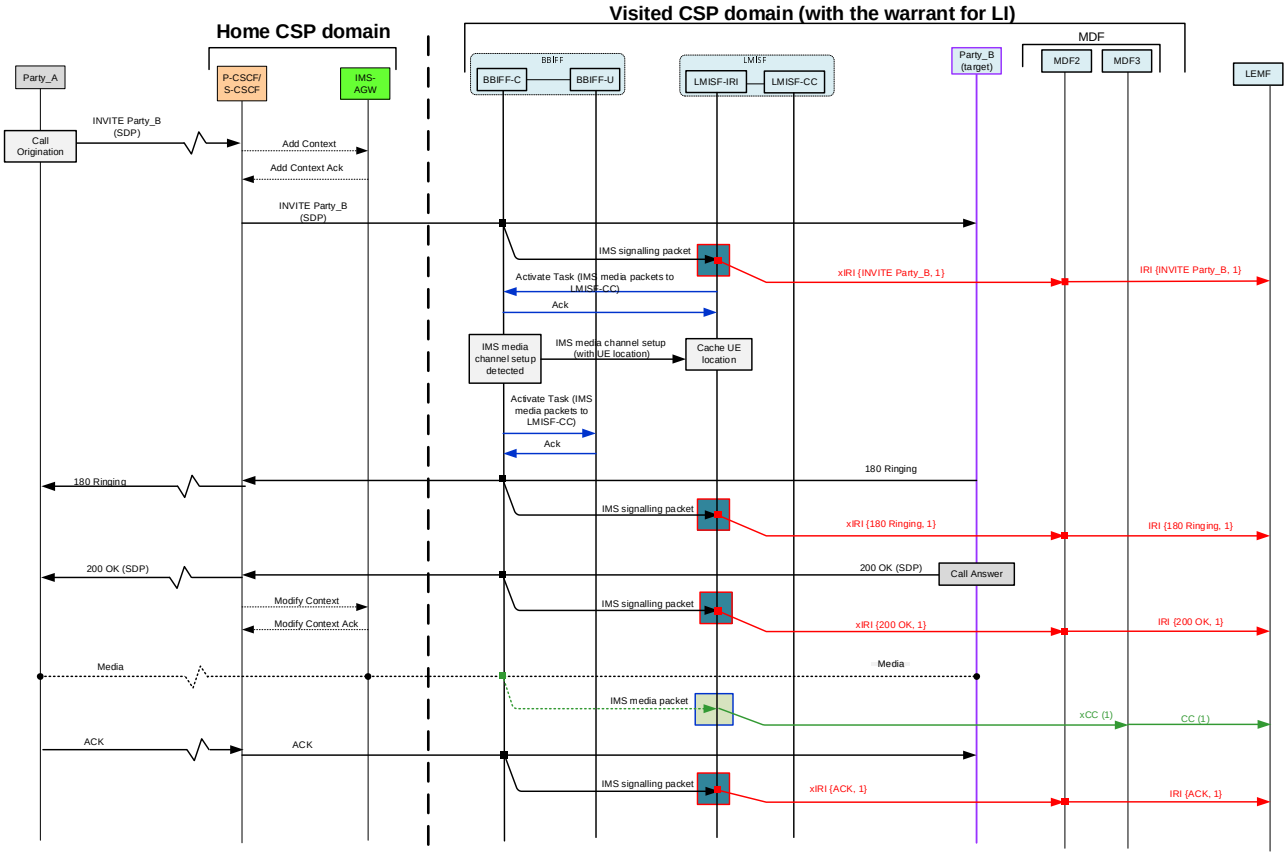


Figure 5.8.4-1: Call termination to an inbound roaming target with HR – IRI + CC

The LMISF-IRI examines the SIP messages present within the IMS signalling packets received from the BBIFF-U to verify whether the SIP headers pointing to the called party (e.g. Request URI, P-Called-Party-Id, To) is a target.

For call terminations to a target that does not require the CC delivery, no per-call based interaction is required between the LMISF-IRI and the BBIFF-C as illustrated in figure 5.8.3-1 (call origination with IRI only).

5.8.5 Mid-call interception

5.8.5.1 Before the call is completely answered

Figure 5.8.5.1-1 illustrates a call flow where a lawful interception is activated while an inbound roaming user is in the process of having an active voice call. In the flow, Party_A (target) calls Party_B. The flow shows that Party_B is also an IMS user (SIP messages are shown), however, Party_B can also be a non-IMS user served by CS domain.

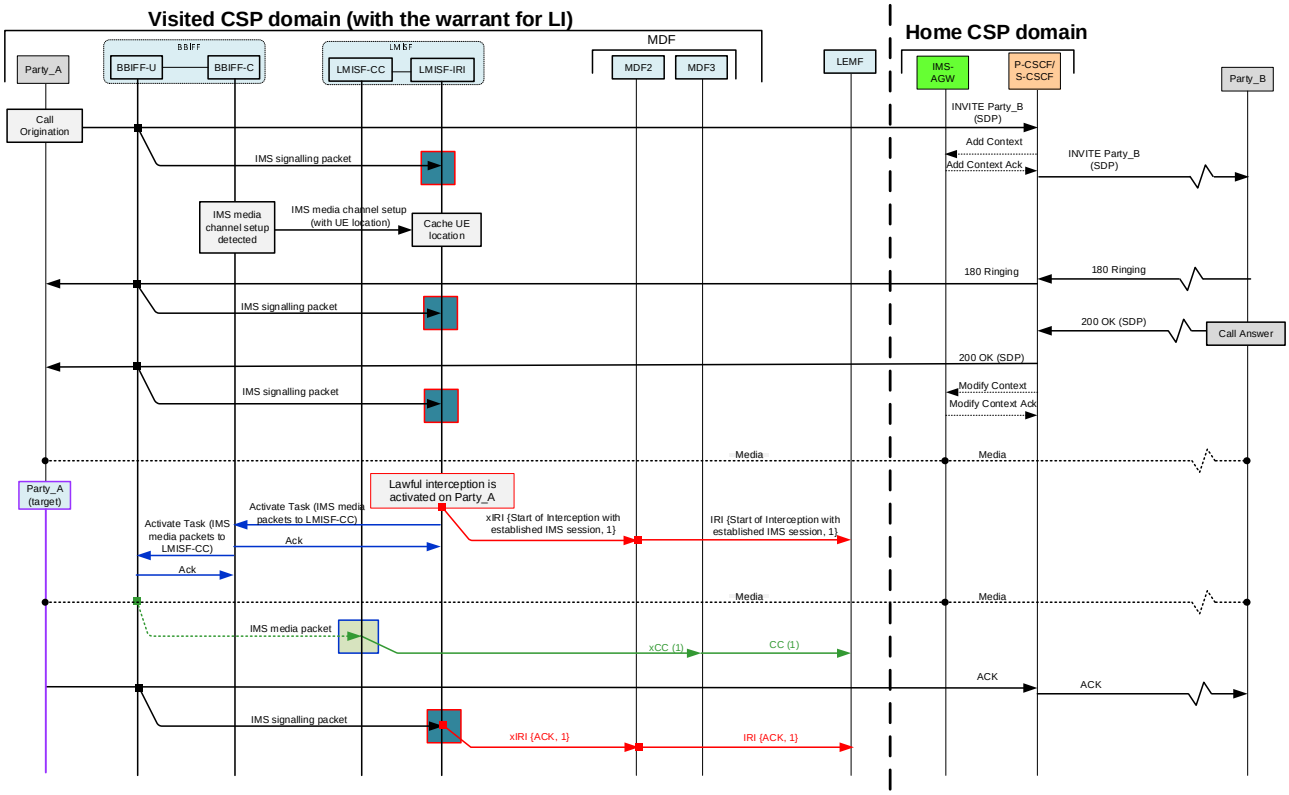


Figure 5.8.5.1-1: Mid call interception (IRI + CC) – scenario 1

In this illustration, a lawful interception is activated on the inbound roaming user right after the called party (Party_B) answers the call, but before the Party_A's (target's) UE has a chance to send the ACK message. Since the SDP offer and SDP answer related steps are already completed, the LMISF-IRI generates an xIRI with *Start Interception for established IMS session.*

NOTE: This flow assumes that no other lawful intercepts are active on the target.

In this illustration, the lawful interception requires the CC delivery. If CC delivery is not required, then no LMISF-IRI to BBIFF-C interaction is required.

5.8.5.2 After the call is completely answered

Figure 5.8.5.2-1 illustrates a call flow where a lawful interception is activated while an inbound roaming user is active on a voice call. In the flow, Party_A (target) calls Party_B. The flow shows that Party_B is also an IMS user (SIP messages are shown), however, Party_B can also be a non-IMS user served by CS domain.

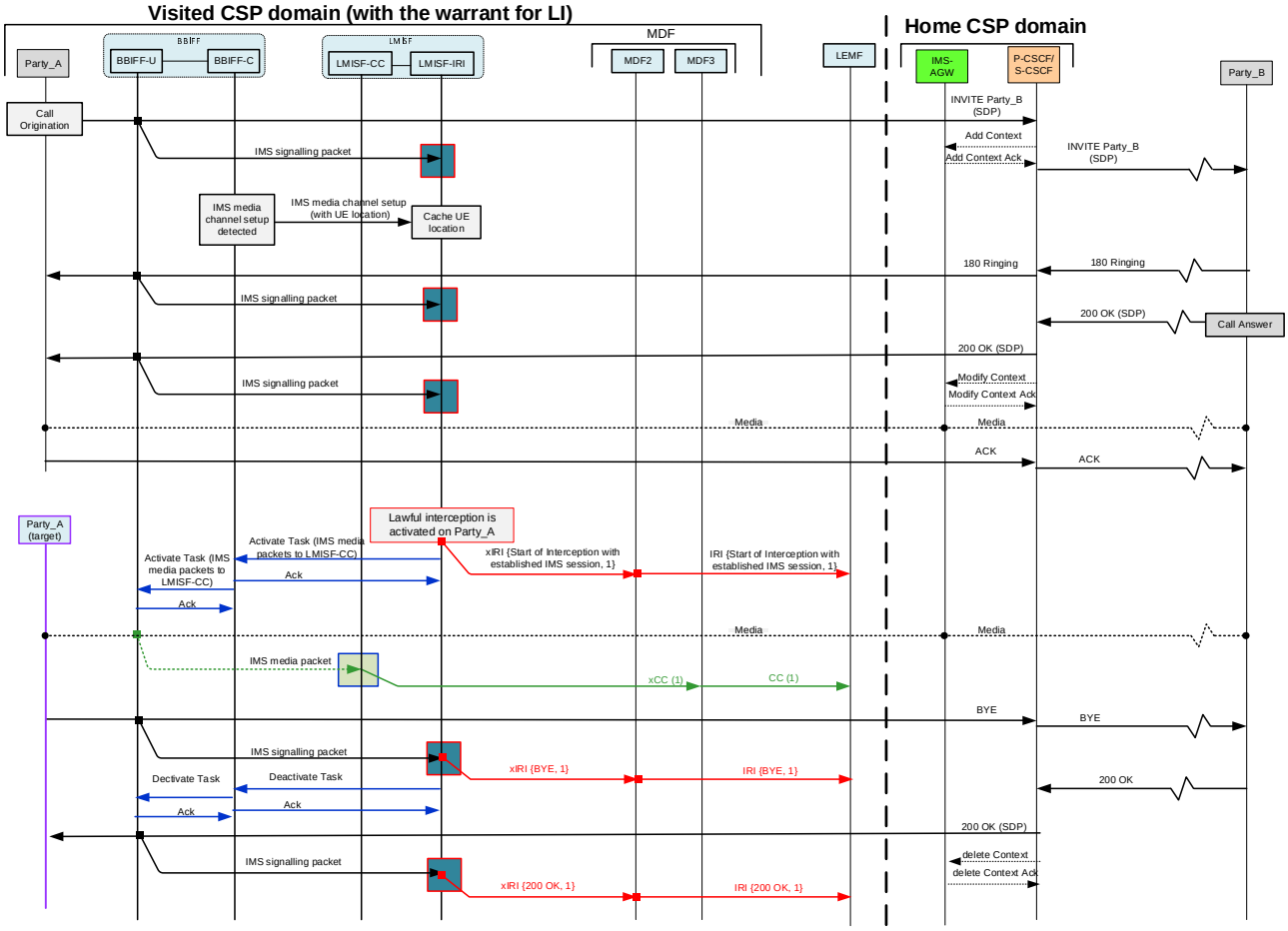


Figure 5.8.5.2-1: Mid call interception (IRI + CC) – scenario 2

In this illustration, a lawful interception is activated on the inbound roaming user well after the call is answered. The flow illustrates that in such a scenario, it is possible that the IRI may include just the SIP BYE and the SIP 200 OK. The trigger between LMISF-IRI and BBIFF-C and between BBIFF-C and BBIFF-U may encounter a case where the IMS media channel is already released by the time the BBIFF-C has a chance to send the trigger of Deactivate Task to stop the delivery of IMS media packets from BBIFF-U to LMISF-CC.

5.8.6 BBIFF relocation scenarios

5.8.6.1 BBIFF-U relocation

Figure 5.8.6.1-1 illustrates a call flow where the relocation of the parent NF that has the BBIFF-U occurs while an inbound roaming target is on a voice call. In the flow, Party_A (target) calls Party_B. The flow shows that Party_B is also an IMS user (SIP messages are shown), however, Party_B can also be a non-IMS user served by CS domain.

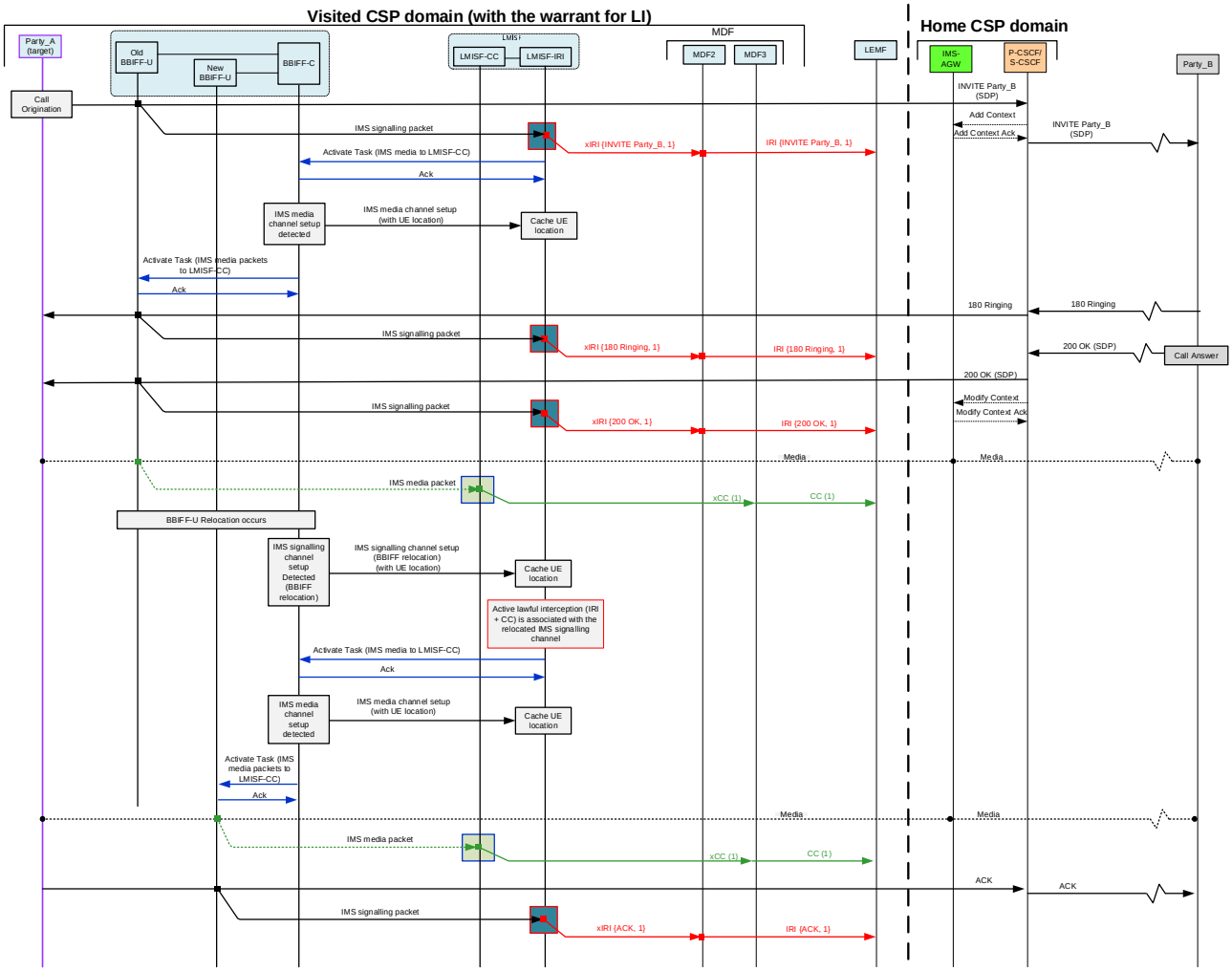


Figure 5.8.6.1-1: HR LI with BBIFF-U relocation (IRI + CC)

In this illustration, a BBIFF-U relocation occurs straight after the called party (Party_B) answers the call, but before the Party_A's (target's) UE has a chance to send the ACK message.

5.8.6.2 BBIFF-C relocation

Figure 5.8.6.2-1 illustrates a call flow where the relocation of the parent NF that has the BBIFF-C occurs while an inbound roaming target is on a voice call. The relocation of BBIFF-U is also presumed to occur. In the flow, Party_A (target) calls Party_B. The flow shows that Party_B is also an IMS user (SIP messages are shown), however, Party_B can also be a non-IMS user served by CS domain.

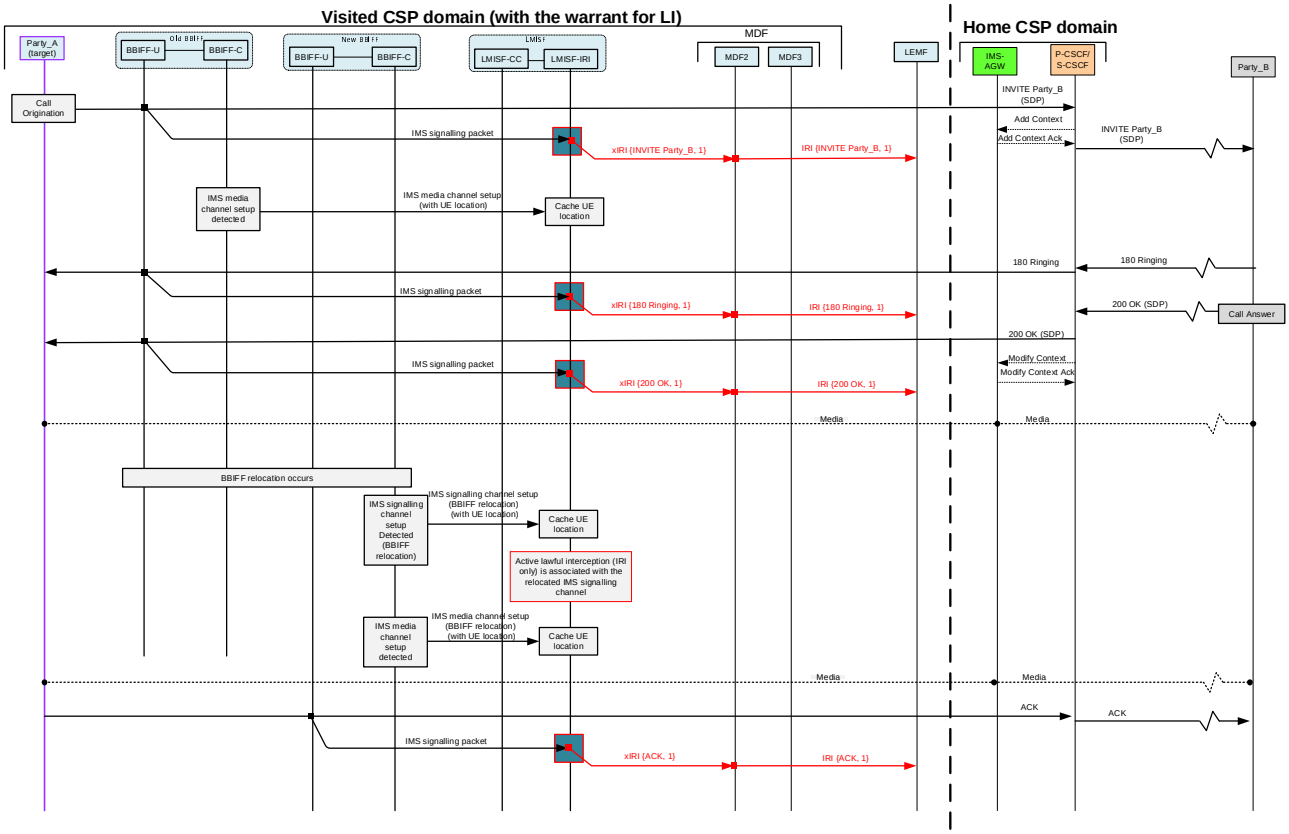


Figure 5.8.6.2-1: HR LI with BBIFF-C relocation (IRI only)

In this illustration, a BBIFF-C relocation occurs straight after the called party (Party_B) answers the call, but before the Party_A's (target's) UE has a chance to send the ACK message. In this illustration, the lawful interception is for IRI only and therefore, the only handling required in the LMISF-IRI is to have the ability to associate the IMS signalling packets received from the new BBIFF-U with the IMS session that is being intercepted already.

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3#94-LI	S3i240504				Initial version	0.0.1
2026-05	SA3LI#101	S3i260231				pCRs from SA3LI#101	0.0.12
2026-07	SA3LI#102	S3i260356				Agreed for Change Control	0.0.13
2026-08	SA#113	SP-260727				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication